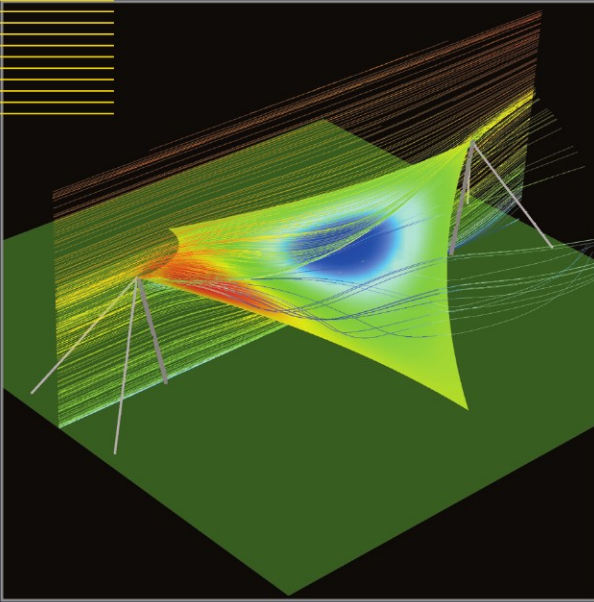


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Editorial
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Fluid-Structure Interaction

Modelling, Simulation,
Optimization

 Springer

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Fluid-Structure Interaction

Modelling, Simulation, Optimisation

With 251 Figures and 48 Tables



Springer

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Preface

The increasing accuracy requirements in many of today's simulation tasks in science and engineering more and more often entail the need to take into account more than one physical effect. Among the most important and, with respect to both modelling and computational issues, most challenging of such 'multiphysics' problems are fluid-structure interactions (FSI), i.e. interactions of some movable or deformable elastic structure with an internal or surrounding fluid flow. The variety of FSI occurrences is abundant and ranges from huge tent-roofs to tiny micropumps, from parachutes and airbags to blood flow in arteries.

Although a lot of research has been done in this thriving field, with sometimes really impressive results, and although most of today's software packages for computational fluid dynamics or computational structural mechanics offer extensions that, at least to some extent, allow for simulating certain classes of FSI scenarios, some of the key questions have not been answered yet in a satisfying way: How can the coupling itself be modelled in an appropriate way? What are the possibilities and limits of monolithic and partitioned coupling schemes or hybrid approaches? What can be said concerning the advantages and drawbacks of the various discretization schemes used on the flow and on the structure side? How reliable are the results, and what about error estimation? How can a flexible data and geometry model look like – especially against the background of large geometric or even topological changes? What can be said about the design of robust and efficient solvers? And how can sensitivity and optimization issues enter the game?

The book in hand contains the proceedings of a workshop on fluid-structure interactions held in Hohenwart, Germany, in October 2005. This 2-day workshop was organized by the Research Unit 493 'Fluid-Structure Interaction: Modelling, Simulation, Optimization' established by the Deutsche Forschungsgemeinschaft (DFG) in 2003 and bringing together researchers from seven German universities from the fields of mathematics, informatics, mechanical engineering, chemical engineering, and civil engineering. Designed as a forum for presenting the research unit's latest results as well as for exchanging ideas with leading international experts, the workshop consisted of fifteen lectures on computational aspects of fluid-structure interactions. The topics now gathered in this volume cover a broad spectrum of up-to-date FSI issues, ranging from more methodical aspects to applications.

We would like to thank the editors of Springer's Lecture Notes in Computational Science and Engineering for admitting our volume to this series and Springer Verlag and, in particular, Dr. Martin Peters, for their helpful support from the first ideas up to the final layout. Furthermore, we are obliged to Markus Brenk, who did a great job in compiling the single contributions

to a harmonic ensemble. Finally, we are grateful for the Research Unit 493 ‘Fluid-Structure Interaction: Modelling, Simulation, Optimization’ funded by the Deutsche Forschungsgemeinschaft (DFG). Without this financial support, neither many of the results presented in this book nor the book itself would have been possible.

München and Darmstadt
March 2006

Hans-Joachim Bungartz
Michael Schäfer

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Implicit Coupling of Partitioned Fluid-Structure Interaction Solvers using Reduced-Order Models

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Abstract. In this contribution a powerful technique is described which allows the strong coupling of partitioned solvers in fluid-structure interaction (FSI) problems. The method allows the use of a black box fluid and structural solver because it builds up a reduced order model of the fluid and structural problem during the coupling process. Each solution of the fluid/structural solver in the coupling process can be seen as a sensitivity response of an applied displacement/pressure mode. The applied modes and their responses are used to build up the reduced order model. The method is applied on the flow in the left ventricle during the filling and emptying phase. Two to three modes are needed, depending on the moment in the heart cycle, to reduce the residual by four orders of magnitude and to achieve a fully coupled solution at each time step.

1 Introduction

The computation of fluid-structure interaction (FSI) problems has gain a lot of interest in the past decade. The interaction can be loose or strong. For loose coupling problems (e.g. for flutter analysis [1–3]) existing fluid and structural solvers can be used as partitioned solvers. The main difficulty is the data exchange between those solvers.

When strong interaction is present, strong coupling of the fluid and structural solver can be achieved with a monolithic scheme [4]. However partitioned schemes can also be used for these applications. Vierendeels et al. [5,6] used a partitioned procedure and reached stabilization of the interaction procedure by introducing artificial compressibility in the subiterations by preconditioning the fluid solver. Recently strongly coupled partitioned methods were developed [7–10] using approximate or exact Jacobians of the fluid and structural solver. In these methods no black box fluid and/or solid solver can be used.

When existing fluid and structural solvers are used to solve strongly coupled FSI problems, a subiteration process has to be set up for every time step in order to achieve the strong coupling, but in order to obtain convergence typically quite a lot of subiterations are required. Mok et al. [11] used an Aitken-like method to enhance the convergence behaviour of this subiteration process.

In this contribution a coupling procedure is presented which outperforms the Aitken-like method for strongly coupled FSI problems. A partitioned procedure is used and implicit coupling is achieved through sensitivity analysis of the important displacement and pressure modes. These modes are detected during the subiteration procedure for each time step. The method allows the use of black box fluid and structural solver. The method is applied to a 2D axisymmetrical model of the cardiac wall which motion is computed during a complete heart cycle. The structural solver was already developed in previous work [5]. As fluid solver the commercial CFD software package Fluent 6.1 (Fluent Inc.) is used to illustrate the practical applicability of the method.

2 Methods

2.1 Fluid and Structural Solver

The black box fluid solver which is used has to fulfill some conditions. It must be possible to prescribe the movement of the boundary of the fluid domain through e.g. a user subroutine and it must be possible to extract the stress data at the moving boundaries. In our application we only need the pressure distribution at the moving boundary. The response of the flow solver can be represented by the function F :

$$p_{k+1}^{n+1} = F^{n+1}(X_{k+1}^{n+1}), \quad (1)$$

where X_{k+1}^{n+1} denotes the prescribed position of the boundary nodes obtained from the structural solver in subiteration $k+1$ when computing the solution on time level $n+1$. It is assumed that the solution on time level n is known. The superscript $n+1$ on F denotes other variables in the flow solver that are already known on time level $n+1$, such as in- and outflow boundary conditions. Starting from time level n the pressure distribution on the boundary nodes p_{k+1}^{n+1} can be computed, which is then passed to the structural solver.

The choice of the boundary conditions needs some attention. When the ventricle is filling the fluid domain has only an inlet, no outlet is present. Therefore it is impossible to specify a velocity at the inlet boundary. This would conflict with the change in volume of the ventricle which is already prescribed by the boundary position on the new time level. Moreover, also the pressure field will be undefined upto a constant value if no pressure boundary is specified. Therefore it is necessary to prescribe the pressure at the inflow boundary during the filling phase and at the outflow during the emptying phase.

The structural model which is used was already developed in previous work [5]. The structural equations are given by G :

$$G^{n+1}(X_{k+1}^{n+1}, p_k^{n+1}, \Delta p_{k+1}^{n+1}) = 0. \quad (2)$$

Since we are dealing with the cardiac cycle the function G^{n+1} incorporates the prescribed time dependency of the structural properties. In our application,

it is assumed that the volume of the ventricle is known as a function of time, therefore the structural solver does not only compute the new position of the boundary nodes, given a pressure distribution at the boundary, but it also computes a pressure shift, Δp_{k+1}^{n+1} , equal for all nodes, so that the volume corresponds with the prescribed volume at that time level. This pressure shift is used to adjust the pressure level in the fluid calculations by adjusting the pressure level of the boundary conditions. In the sequel we denote the structural equations as

$$G^{n+1}(X_{k+1}^{n+1}, p_k^{n+1}) = 0 \quad (3)$$

for a given pressure input p_k^{n+1} coming from the fluid solver, neglecting the notation for the update of the pressure boundary condition needed in the fluid solver. The structural solver can also be denoted as

$$X_{k+1}^{n+1} = S^{n+1}(p_k^{n+1}). \quad (4)$$

The superscript $n + 1$ on F, G and S are dropped from now on. Equation (3) is solved by Newton's method.

2.2 Classical Strong Coupling Methods for Partitioned Solvers

Explicit subiterations within a time step Strong coupling can be obtained by calling the fluid and structural solver subsequently during the calculation of a time step until convergence is obtained. When there is a lot of interaction between both subproblems, this approach can lead to divergence in the subiteration process. When underrelaxation is introduced with a constant underrelaxation parameter, divergence can be avoided but convergence is not really obtained as is illustrated below.

A non-constant underrelaxation parameter can be used to improve the convergence of the subiteration process. The underrelaxation parameter can be obtained with an Aitken-like acceleration method [11] as follows:

$$\omega^k = \frac{(X^k - X^{k-1}) \cdot (R(X^k) - R(X^{k-1}))}{(R(X^k) - R(X^{k-1})) \cdot (R(X^k) - R(X^{k-1}))} \quad (5)$$

where $R(X) = S \circ F(X) - X$. X^{k+1} can be obtained with

$$X^{k+1} = X^k - \omega^k R(X^k). \quad (6)$$

An initial value for ω has to be chosen. We used an initial value of 0.01.

Comparison of the different classical methods If subsequently the structural solver and the fluid solver are called within the subiterations of a time step, divergence is detected. This is shown in Fig. 1 for the first time step of the first heart cycle at the onset of filling. Even when underrelaxation

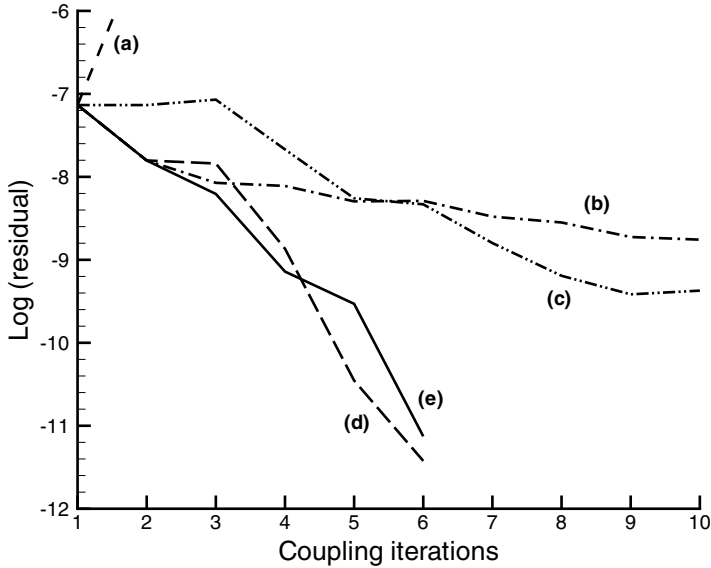


Fig. 1. Residual behaviour of the coupling method for the first time step of the first heart cycle at the onset of filling: (a) no reduced order model, without under-relaxation, (b) no reduced order model, with underrelaxation 0.05, (c) no reduced order model, with Aitken-like acceleration technique, (d) with the reduced order model for the fluid solver, (e) with the reduced order models for both the fluid and structural solver.

is used, convergence within the subiterations could not be obtained in a reasonable number of subiterations (Fig. 1). With the Aitken-like method, convergence was also not really obtained for the first time step within a reasonable number of subiterations. During the next time steps even a worse convergence behaviour was observed.

Figure 2 shows the evolution of the position of the boundary during the subiteration process of the first time step when subsequent calls of structural and fluid solver without underrelaxation are performed. One can detect that the behaviour of low frequency modes are responsible for the divergence behaviour.

From this observation, it can be expected that when implicitness is introduced in the subiteration process for a few low frequency modes, convergence could be obtained.

2.3 Coupling Method with a Reduced Order Model for the Fluid Solver (Method 1)

Since the fluid solver is a black box commercial code, it is not possible to retrieve or construct the Jacobian F_X , which is needed to solve the structural

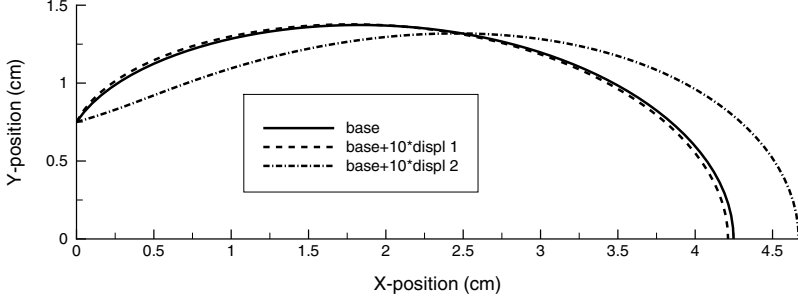


Fig. 2. Illustration of the computed displacements of the heart wall if subsequently the structural and fluid solver are called and when no underrelaxation is used.

problem in an implicit way:

$$G(X_{k+1}^{n+1}, p_{k+1}^{n+1}) = 0. \quad (7)$$

However it is possible to construct a reduced order model of the fluid solver which can be differentiated easily. Let's denote the reduced order model of the fluid solver by

$$\hat{p}_{k+1}^{n+1} = \hat{F}(X_{k+1}^{n+1}), \quad (8)$$

then the equations for the structure are written as

$$G(X_{k+1}^{n+1}, \hat{p}_{k+1}^{n+1}) = 0. \quad (9)$$

A Newton iteration method can be set up after inserting (8) into (9) as follows:

$$G(X_{k+1,s}^{n+1}, \hat{p}_{k+1,s}^{n+1}) + \left(\frac{\partial G}{\partial X} + \frac{\partial G}{\partial \hat{p}} \frac{\partial \hat{p}}{\partial X} \right) (X_{k+1,s+1}^{n+1} - X_{k+1,s}^{n+1}) \approx 0, \quad (10)$$

which is solved for $X_{k+1,s+1}^{n+1}$ upon convergence. Remark that this iteration procedure with index s involves only the solution of the structural problem.

The problem can be solved if we have an expression for the Jacobian $\frac{\partial \hat{p}}{\partial X}$ of the reduced order model for the fluid problem which we will denote by $\hat{F}_X$ in the sequel.

Construction of the reduced order model After k subiteration loops (and thus k fluid solver calls) k sets of boundary positions and corresponding pressure distributions are obtained that fulfill the flow equations (1). From the moment that minimum two sets $(X_i, p_i), i = 1 \dots k$ are available, a set of displacement modes $V_m = \{v_m, m = 1 \dots k-1\}$ is constructed with

$$v_m = X_k - X_m. \quad (11)$$

The corresponding pressure mode to v_m is denoted by $\Delta p_m = p_k - p_m$. A pressure mode matrix ΔP_{k-1} is constructed:

$$\Delta P_{k-1} = [\Delta p_1 \cdots \Delta p_{k-1}], \quad (12)$$

where the columns contain the computed pressure modes.

An arbitrary displacement ΔX can be projected onto the set of displacement modes V_m . The displacement ΔX can be written as

$$\Delta X = \sum_{m=1}^{k-1} \alpha_m v_m + \Delta X_{corr} \quad (13)$$

where α_m denotes the coordinates of ΔX in the set V_m . Note that the number of displacement modes ($k-1$) is much smaller than the dimension of ΔX , which explains the correction term. If the displacement modes are well chosen, ΔX can be approximated by $\Delta \tilde{X}$:

$$\Delta X \approx \Delta \tilde{X} = \sum_{m=1}^{k-1} \alpha_m v_m. \quad (14)$$

This is an overdetermined problem for the coordinates α_m , which can be faced with the least square approach. With this approach, the coordinates α_m can be computed as

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{k-1} \end{bmatrix} = \begin{bmatrix} \langle v_1, v_1 \rangle & \langle v_1, v_2 \rangle & \cdots & \langle v_1, v_{k-1} \rangle \\ \langle v_2, v_1 \rangle & \langle v_2, v_2 \rangle & \cdots & \langle v_2, v_{k-1} \rangle \\ \vdots & \vdots & & \vdots \\ \langle v_{k-1}, v_1 \rangle & \langle v_{k-1}, v_2 \rangle & \cdots & \langle v_{k-1}, v_{k-1} \rangle \end{bmatrix}^{-1} \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_{k-1}^T \end{bmatrix} \Delta X \quad (15)$$

The coordinates α_m denote the amount of each mode in the displacement ΔX so that the corresponding change in pressure Δp can be approximated as

$$\Delta p \approx \Delta P_{k-1} \alpha, \quad (16)$$

where $\alpha = [\alpha_1 \cdots \alpha_{k-1}]^T$. The Jacobian $\hat{F}_X$ of the reduced order model can thus be written as

$$\hat{F}_X = [\Delta p_1 \cdots \Delta p_{k-1}] \begin{bmatrix} \langle v_1, v_1 \rangle & \cdots & \langle v_1, v_{k-1} \rangle \\ \vdots & & \vdots \\ \langle v_{k-1}, v_1 \rangle & \cdots & \langle v_{k-1}, v_{k-1} \rangle \end{bmatrix}^{-1} \begin{bmatrix} v_1^T \\ \vdots \\ v_{k-1}^T \end{bmatrix} \quad (17)$$

The reduced order model, used in subiteration $k+1$ is written as

$$\hat{p}_{k+1}^{n+1} = p_k^{n+1} + \hat{F}_X (X_{k+1}^{n+1} - X_k^{n+1}). \quad (18)$$

Once eq. (9) is solved for X_{k+1}^{n+1} , p_{k+1}^{n+1} is obtained from eq. (1) and the residual

$$r_{k+1}^{n+1} = G(X_{k+1}^{n+1}, p_{k+1}^{n+1}) \quad (19)$$

is computed with the pressure from the fluid solver, i.e. not from the reduced order model.

Startup and summary of the coupling procedure During the first two subiteration loops the reduced order model can not be used since at least two sets of boundary positions and corresponding pressure distributions are needed. In the first subiteration, an initial guess for the position of the boundary X_1^{n+1} is achieved by integrating in time the position of the boundary on the previous time level X^n with an explicit forward Euler scheme:

$$X_1^{n+1} = X^n + V^n \Delta t, \quad (20)$$

where V denotes the vector of the velocities of the boundary nodes. The corresponding pressure distribution p_1^{n+1} is obtained by calling the fluid solver. In the second subiteration the displacement is computed from eq. (3) and underrelaxation with a factor 0.05 is used to obtain the position X_2^{n+1} . p_2^{n+1} is obtained from the fluid solver. From now on the reduced order model can be built up and no underrelaxation is applied anymore in the coupling process.

The subiteration process can be summarized as follows:

1. Obtain X_1^{n+1} by integrating in time eq. (20) and compute the corresponding pressure distribution p_1^{n+1} .
2. With p_1^{n+1} , compute the displacement with the structural solver
3. Obtain X_2^{n+1} by underrelaxing the displacement with a factor 0.05 and compute the corresponding pressure distribution p_2^{n+1} .
4. Start FSI loop with $k = 2$.
5. Build the reduced order model (8) with $k - 1$ modes.
6. Compute X_{k+1}^{n+1} with eq. (9).
7. Compute p_{k+1}^{n+1} with eq. (1).
8. Compute the residual r_{k+1}^{n+1} with eq. (19).
9. Repeat from step 5 with $k = k + 1$ until convergence is obtained.

Since the structure is very compliant, especially during filling, the eigenmodes of the structure are not necessarily a good choice as base modes for the FSI displacement. In the proposed coupling strategy, it is not needed to detect eigenmodes of the structure. During the subiteration process the base modes which are used are constructed from the intermediate positions that are computed from the structural solver, so they become available in a natural way.

2.4 Coupling Method with Reduced Order Models for both the Fluid and the Structural Solver (Method 2)

In the method described above it is assumed that the reduced order model for the fluid solver can be coupled with the structural solver in an implicit way. This means that the structural solver code has to be accessible enough so that the prescription of the pressure at the boundary can be done as a function of the unknown positions of those boundary nodes. This means that it must be possible to pass the Jacobian of the reduced order model of the

fluid solver to the structural solver. If the structural solver is not accessible enough the above mentioned technique cannot be applied. However it is also possible to construct a reduced order model for the structural solver as well. Both reduced order models can then be coupled in an implicit way. The latter technique is presented below in more detail.

As described above, the reduced order model for the fluid solver is built up from sets of positions (X_k^f) and corresponding pressure distributions (p_k^f). The superscript $n + 1$ is omitted here and the superscript f is introduced to distinguish between the fluid and the structural solver. From pressure distributions ($p_{k'}^s$) applied to the structural solver and the corresponding boundary positions ($X_{k'}^s$) a reduced order model for the structural solver can be built in the same way as this was done for the fluid solver. The superscript s is used here to denote the structural solver.

After k fluid solver calls and k' structural solver calls the reduced order models, respectively for the fluid and the structural solver, can be written as:

$$\hat{p}_{k+1}^f = p_k^f + \hat{F}_X \left(X_{k+1}^f - X_k^f \right), \quad (21)$$

$$\hat{X}_{k'+1}^s = X_{k'}^s + \hat{S}_p \left(p_{k'+1}^s - p_{k'}^s \right). \quad (22)$$

A solution for $\begin{bmatrix} \hat{p} & \hat{X} \end{bmatrix}^T$ is sought that fulfills both equations, i.e. $\hat{p} = \hat{p}_{k+1}^f = p_{k'+1}^s$ and $\hat{X} = \hat{X}_{k'+1}^s = X_{k+1}^f$. The solution can be found as

$$\begin{bmatrix} \hat{p} \\ \hat{X} \end{bmatrix} = \left(I - \begin{bmatrix} 0 & \hat{F}_X \\ \hat{S}_p & 0 \end{bmatrix} \right)^{-1} \left(\begin{bmatrix} p_k^f \\ X_{k'}^s \end{bmatrix} - \begin{bmatrix} 0 & \hat{F}_X \\ \hat{S}_p & 0 \end{bmatrix} \begin{bmatrix} p_{k'}^s \\ X_k^f \end{bmatrix} \right) \quad (23)$$

This solution can be obtained each time before the fluid or structural solver is called and this solution can then be used as input for these calls. However when calling the fluid solver only the solution for $\hat{X}$ is needed and when the structural solver is called only the solution for $\hat{p}$ has to be obtained. Eqs. (21) and (22) can be solved for $\hat{X}$:

$$\hat{X} = \left(I - \hat{S}_p \hat{F}_X \right)^{-1} \left[X_{k'}^s + \hat{S}_p \left(p_k^f - p_{k'}^s - \hat{F}_X X_k^f \right) \right] \quad (24)$$

or for $\hat{p}$:

$$\hat{p} = \left(I - \hat{F}_X \hat{S}_p \right)^{-1} \left[p_k^f + \hat{F}_X \left(X_{k'}^s - X_k^f - \hat{S}_p p_{k'}^s \right) \right]. \quad (25)$$

The subiteration process to obtain the solution at time level $n + 1$ can be summarized as follows. It is obvious that several variants can be constructed based on this idea.

1. Obtain X_1^f by integrating in time eq. (20) and compute the corresponding pressure distribution p_1^f with the fluid solver.

2. With $p_1^s = p_1^f$, compute the displacement with the structural solver and obtain X_1^s .
3. Obtain X_2^f by underrelaxing the displacement $X_1^s - X_1^f$ with a factor 0.05 and compute the corresponding pressure distribution p_2^f with the fluid solver.
4. With $p_2^s = p_2^f$, compute the displacement with the structural solver and obtain X_2^s .
5. Build the reduced order model for the fluid solver (21) with 1 mode.
6. Start FSI loop with $k = 2$ and $k' = 2$.
7. Build the reduced order model for the structural solver (22) with $k' - 1$ modes.
8. Compute X_{k+1}^f from the reduced order models with eq. (24).
9. Compute the corresponding pressure distribution p_{k+1}^f with the fluid solver (1).
10. Build the reduced order model for the fluid solver (21) with k modes.
11. Compute $p_{k'+1}^s$ from the reduced order models with eq. (25).
12. Compute the corresponding position of the boundary nodes $X_{k'+1}^s$ with the structural solver (3).
13. Compute the residual r_{k+1}^{n+1} with eq. (19).
14. Repeat from step 7 with $k = k + 1$ and $k' = k' + 1$ until convergence is obtained.

3 Results

The method is applied to the filling process of the left ventricle. This FSI problem has already been studied in previous work using the same structural model, but with an own written fluid solver in which artificial compressibility was used as a technique to stabilize and converge the subiteration process [5].

The geometry of the left ventricle is represented by a truncated ellipsoid in the zero stress state. At the zero stress state and with blood at rest, the transmural pressure is zero. The zero stress state is assumed to correspond with a cavity volume of 12 ml, diameter of the mitral annulus of 1.5 cm and base-to-apex distance of 4 cm. These are physiological relevant parameters for a canine heart for which the model was validated.

Away from the zero stress state, the shape of the left ventricle is computed from equilibrium equations for the left ventricular wall. These equilibrium equations involve the time dependent circumferential and longitudinal cardiac stresses, the curvature of the heart wall and the transmural pressure difference. A nonlinear extension of the thin shell equations is used. The position of the mitral valve annulus is kept fixed. We have used Meisner's lumped parameter model for the complete circulation [12] to obtain the necessary boundary conditions for our 2D axisymmetrical calculations. The velocity patterns computed from the Meisner's model at the mitral and aortic

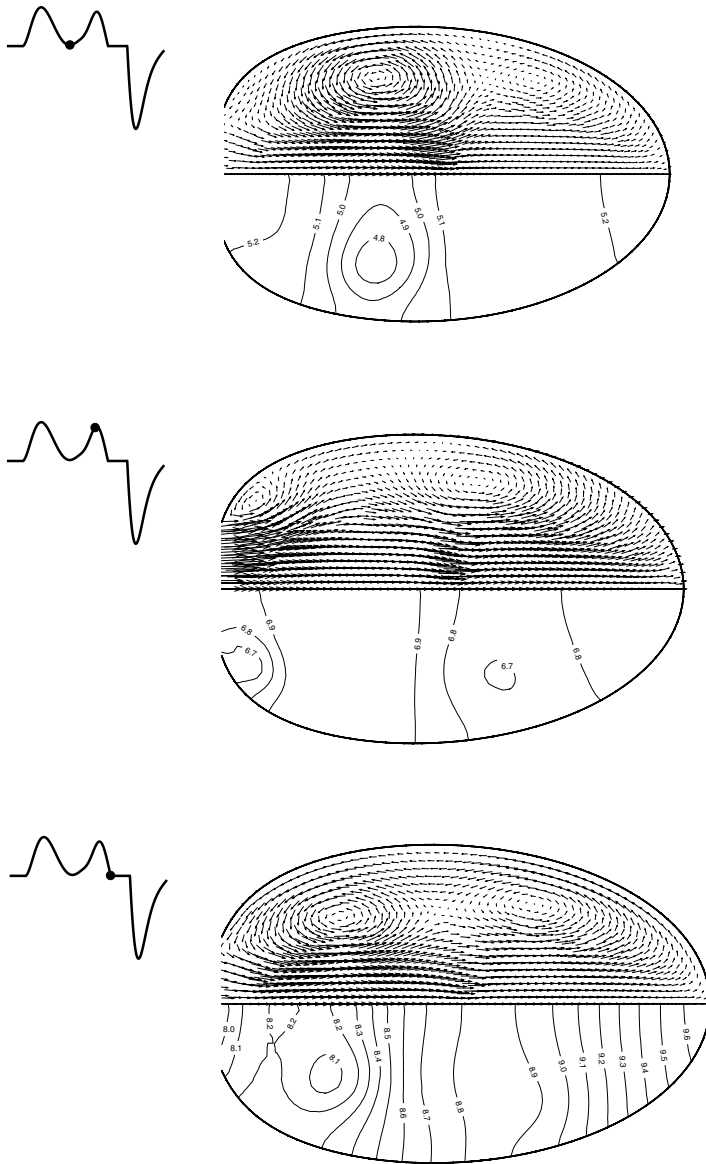


Fig. 4. Velocity vectors and pressure contours in the left ventricle during atrial contraction.

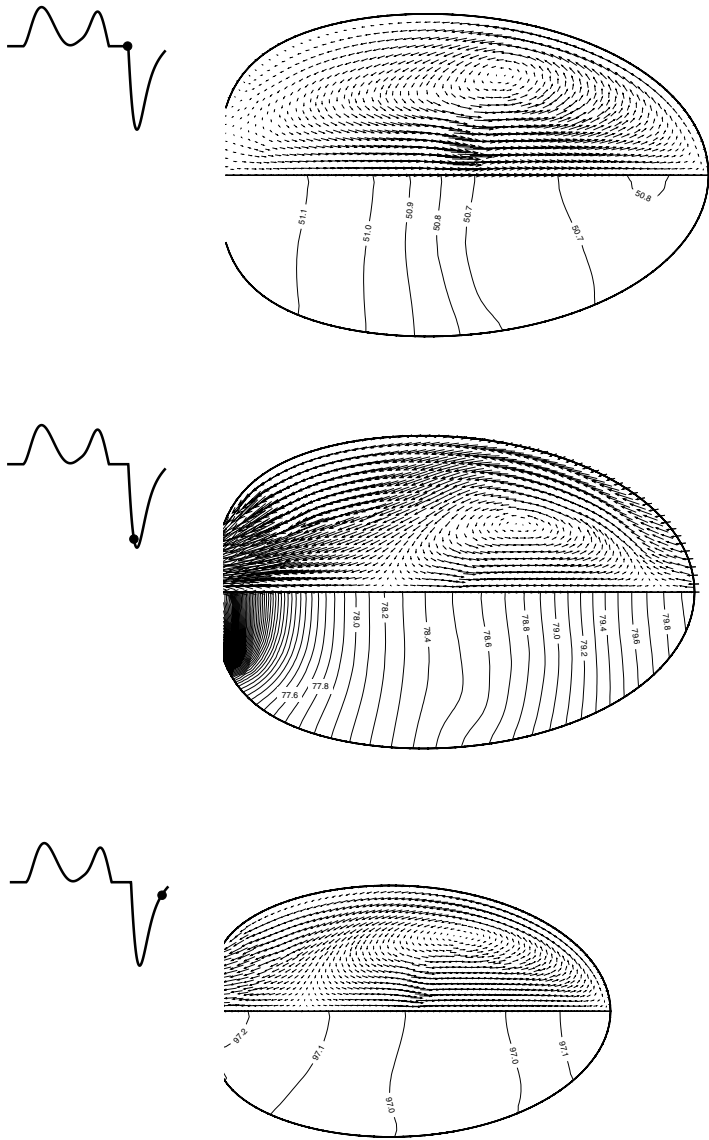


Fig. 5. Velocity vectors and pressure contours in the left ventricle during the ejection phase.

valves can be seen in Figs. 3–5. From the velocity data, the volume change of the ventricle is computed as a function of time, which is used as input for our calculations as explained above. Since the model is 2D axisymmetrical the position of the mitral valve and the aortic valve is assumed to be the same. This can be done since in normal physiological conditions at most one valve is open at the same time. All details of the model can be found in [5].

Figures 3-5 show velocity vectors and pressure contours during the third heart cycle. The time is indicated on the velocity profile diagram which shows the biphasic mitral inflow pattern and the aortic outflow pattern computed with Meisner’s model. These results correspond with results which were obtained earlier with another coupling technique presented in previous work [5,6]. Figure 6 shows the pressure-volume relationship for the three computed heart cycles. It can be seen that convergence for the cycle is already obtained during the second heart cycle. We will further not discuss physiological issues.

Figure 1 shows the convergence behaviour of the proposed methods compared to the three explicit coupling procedures during the first time step of the first heart cycle at the onset of filling.

Figure 7 shows convergence results for method 1 for two different time steps in the third heart cycle (time step 12: fast convergence and time step 162: slowest convergence). Each heart cycle is computed with 250 time steps.

Figure 8 shows convergence results for method 2 for two different time steps in the third heart cycle (time step 84: slowest convergence and time step 157: fast convergence).

Typically, two to three modes are needed to reduce the residual with three or four orders of magnitude (see Table 1).

Table 1. Mean number of modes needed to drop the residual by three or four orders of magnitude. Method 1: only reduced order model for the fluid solver, method 2: reduced order models for both the fluid and structural solver

Method	Residual drop	Mean number of modes
1	3 orders	1.95
1	4 orders	2.78
2	3 orders	2.30
2	4 orders	2.78

This is also shown in detail in Figs. 9, 10, 11 and 12 where the number of modes which are used per time step are shown for the third heart cycle.

As an example the computed displacement modes used in the reduced order model at the onset of filling is shown in Fig. 13. It can be seen that the first two modes are very low frequency modes with one node. The frequency of the third one is somewhat higher. It is also a low frequency mode but with already two nodes. During the coupling procedure in each subiteration the

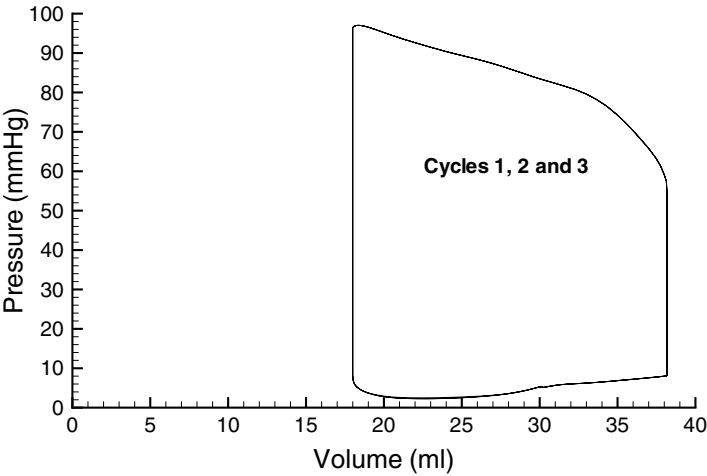


Fig. 6. Pressure-volume relationship computed during the first three heart cycles.

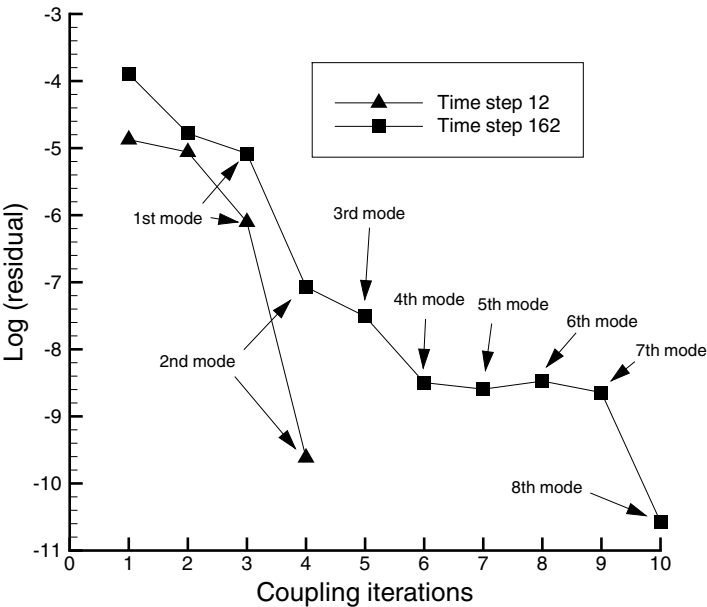


Fig. 7. Convergence behaviour of the subiteration process for method 1 for two time steps: time steps 12 and 162 in the third heart cycle.

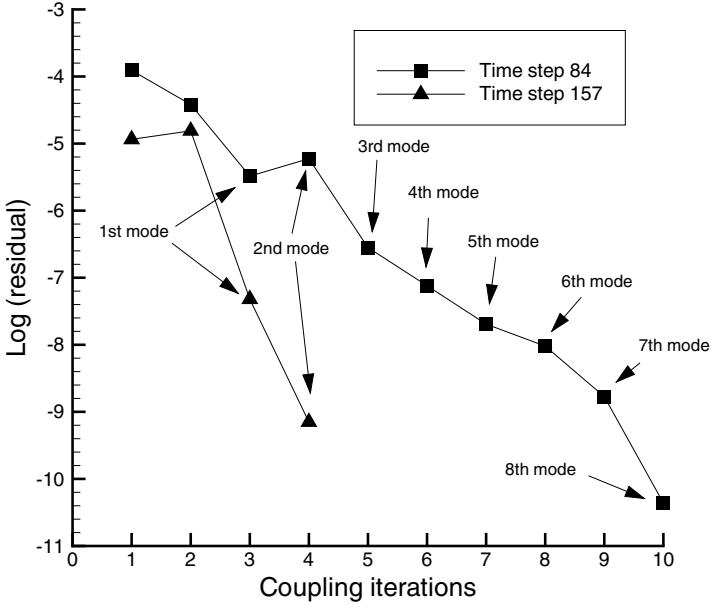


Fig. 8. Convergence behaviour of the subiteration process for method 2 for two time steps: time steps 84 and 157 in the third heart cycle.

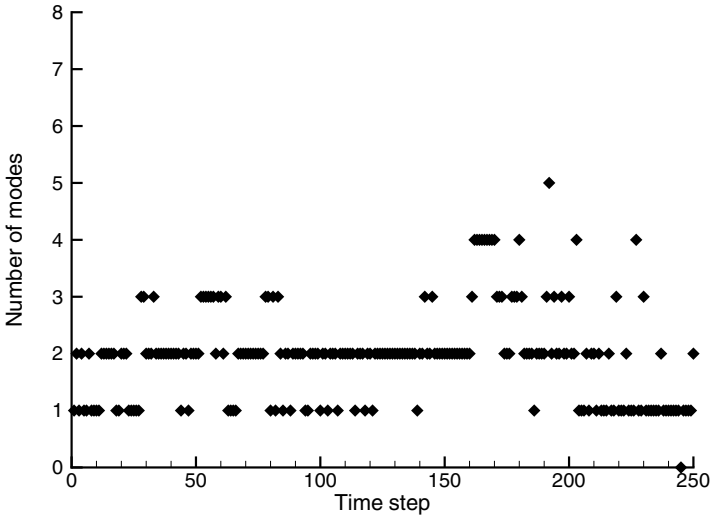


Fig. 9. Number of modes used in method 1 to reduce the residual below -8 (average of three orders of magnitude) during each time step of the third heart cycle.

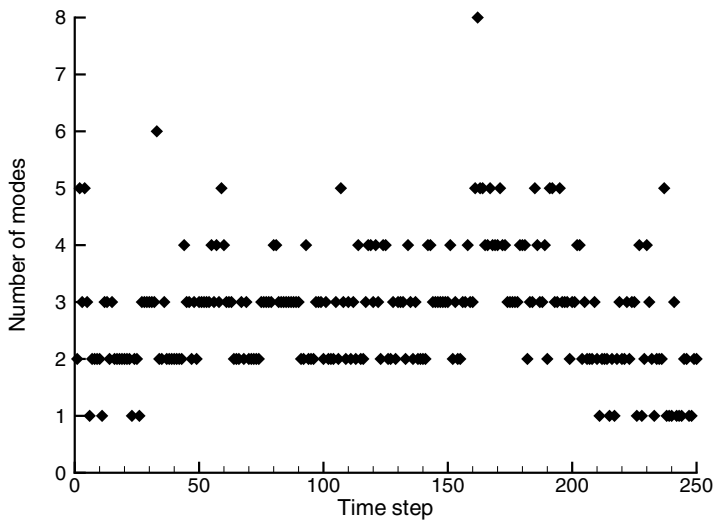


Fig. 10. Number of modes used in method 1 to reduce the residual below -9 (average of four orders of magnitude) during each time step of the third heart cycle.

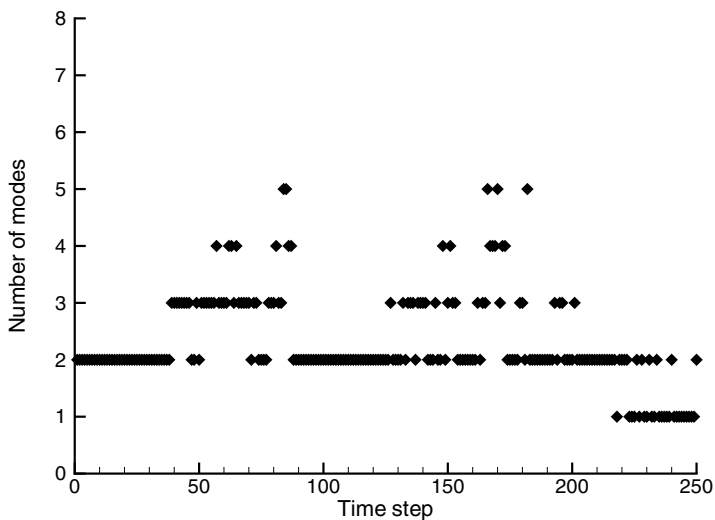


Fig. 11. Number of modes used in method 2 to reduce the residual below -8 (average of three orders of magnitude) during each time step of the third heart cycle.

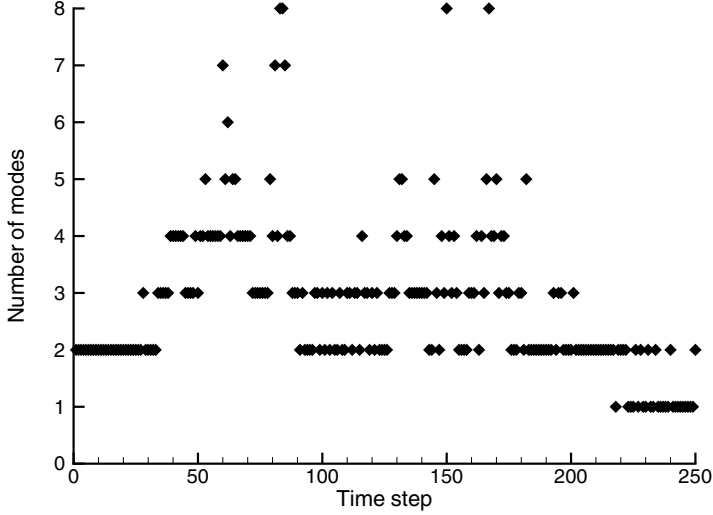


Fig. 12. Number of modes used in method 2 to reduce the residual below -9 (average of four orders of magnitude) during each time step of the third heart cycle.

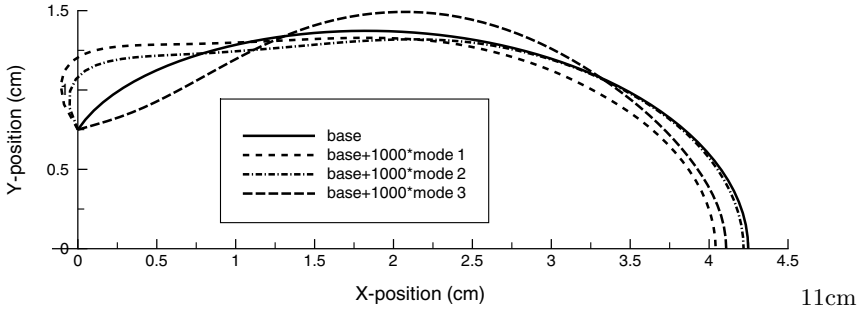


Fig. 13. Illustration of the computed modes used in the reduced order model during the coupling process at the onset of filling.

most ‘dangerous’ displacement mode is detected and treated implicitly in the subsequent subiterations.

4 Conclusion

As conclusion, it can be stated that a very efficient coupling strategy is developed and presented that allows the strong coupling of partitioned solvers. The construction of a reduced order model for the black box fluid solver is crucial to obtain very good convergence. If the structural solver is not accessible enough to implement this reduced order model of the fluid solver in

the boundary conditions then also a reduced order model for the structural solver has to be constructed. Both approaches show a very good convergence behaviour with respect to the more classical methods.

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oomph-lib – An *Object-Oriented Multi-Physics* Finite-Element *Library*

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Abstract. This paper discusses certain aspects of the design and implementation of `oomph-lib`, an object-oriented multi-physics finite-element library, available as open-source software at <http://www.oomph-lib.org>. The main aim of the library is to provide an environment that facilitates the robust, adaptive solution of multi-physics problems by monolithic discretisations, while maximising the potential for code re-use. This is achieved by the extensive use of object-oriented programming techniques, including multiple inheritance, function overloading and template (generic) programming, which allow existing objects to be (re-)used in many different ways without having to change their original implementation.

These ideas are illustrated by considering some specific issues that arise when implementing monolithic finite-element discretisations of large-displacement fluid-structure-interaction problems within an Arbitrary Lagrangian Eulerian (ALE) framework. We also discuss the development of wrapper classes that permit the generic and efficient evaluation of the so-called “shape derivatives”, the derivatives of the discretised fluid equations with respect to those solid mechanics degrees of freedom that affect the nodal positions in the fluid mesh. Finally, we apply the methodology in several examples.

1 Introduction

The development of efficient and robust methods for the numerical solution of multi-physics problems, such as large-displacement fluid-structure interactions, involves numerous challenges. One of the key issues is how best to combine existing “optimal” methodologies for the solution of the constituent single-physics problems in a coupled framework.

The two main approaches are “partitioned” and “monolithic” solvers. In a partitioned approach, existing single-physics codes are coupled via a global fixed-point (Picard) iteration and the single-physics codes are treated as “black-box” modules, whose internal data structures are regarded as inaccessible. The approach facilitates (in fact, requires) code re-use and is the only feasible approach if the source code for the single-physics solvers is unavailable, e.g. commercial software packages. The disadvantage of this approach is that the Picard iteration often converges very slowly, or not at all, even when good initial guesses are available. Under-relaxation or Aitken extrapolation may improve the convergence characteristics (see, e.g., [1] and, more recently, [2]), but in many cases (especially in time-dependent problems with strong

fluid-structure interaction) even these methods are not sufficient to ensure convergence.

Monolithic solvers are based on the fully-coupled discretisation of the governing equations, allowing (but also demanding) complete control over every aspect of the implementation. This approach allows the complete system of nonlinear algebraic equations that results from the coupled discretisation to be solved using Newton’s method. If good initial guesses for the solution are available, e.g. from continuation methods or when time-stepping, the Newton iteration converges quadratically, leading to a robust method for solving the coupled problem.

A monolithic discretisation allows direct access to the code’s internal data structures and facilitates the implementation of non-standard boundary conditions, or other “exotic” constraints. Furthermore, preconditioners for the iterative solution of the linear systems that must be solved during the Newton iteration may be derived directly from the governing equations; see, e.g., [3]. While these characteristics make monolithic solvers attractive, their implementation is often regarded as (too) labour intensive, and code re-use is perceived to be difficult to achieve.

In this paper we shall discuss the design and implementation of `oomph-lib`, an object-oriented multi-physics finite-element library, available as open-source software at <http://www.oomph-lib.org>. The main aim of the library is to provide an environment that facilitates the monolithic discretisation of multi-physics problems while maximising the potential for code re-use. This is achieved by the extensive use of object-oriented programming techniques, including multiple inheritance, function overloading and template (generic) programming, which allow existing objects to be (re-)used in many different ways without having to change their original implementation.

We shall illustrate these techniques by considering some specific issues that arise when implementing monolithic finite-element discretisations of large-displacement fluid-structure-interaction problems (and many other free boundary problems) within an Arbitrary Lagrangian Eulerian (ALE) framework:

1. It must be possible for the “load terms” in the solid mechanics finite elements to depend on unknowns in the coupled problem because the traction that the fluid exerts onto the solid must be determined as part of the overall coupled solution.
2. The solution of the equations of solid mechanics determines the shape of the fluid domain. We, therefore, require clearly defined interfaces that allow the transfer of geometric information between the solid mechanics elements and the procedures that generate the (fluid) mesh, and update its nodal positions in response to changes in the shape and position of the domain boundary.
3. The discretised fluid equations are affected by changes in the nodal positions within the fluid mesh, which are determined indirectly (via the node

update procedures referred to in 2.) by the solid mechanics degrees of freedom. A monolithic discretisation of the coupled problem requires the efficient evaluation of the so-called “shape derivatives” — the derivatives of the discretised fluid equations with respect to those solid mechanics degrees of freedom that affect the nodal positions in the fluid mesh.

In order to maximise the potential for code re-use, it is desirable to provide this functionality without having to re-implement any existing fluid or solid elements or any mesh generation/update procedures.

The outline of this paper is as follows: after a brief discussion of oomph-lib’s general design objectives in Section 2, Section 3 provides an overview of oomph-lib’s data structures and discusses the library’s fundamental objects: **Data**, **Node**, **Element**, **Mesh** and **Problem**. In Section 4 we illustrate how multiple inheritance, combining the **GeneralisedElement** and **GeomObject** base classes, is used to represent domain boundaries whose positions are determined as part of the solution. Section 5 explains the mesh generation process and illustrates how oomph-lib’s mesh adaptation procedures allow the fully-automatic spatial adaptation of meshes in domains that are bounded by curvilinear boundaries. In Section 6 we describe the joint use of template programming and multiple inheritance to create wrapper classes that “upgrade” existing elements to elements that allow the generic and efficient evaluation of the “shape-derivatives”. Finally, Section 7 presents two examples: a “toy” free-boundary problem in which the solution of a 2D Poisson equation is coupled to the shape of the domain boundary; and an unsteady large-displacement fluid-structure-interaction problem: finite-Reynolds-number flow in a rapidly oscillating elastic tube.

2 The Overall Design

2.1 General Design Objectives

The main aim of the library is to provide a general framework for the discretisation and the robust, adaptive solution of a wide range of multi- (and single-)physics problems. The library provides fully-functional elements for a wide range of “classical” partial differential equations (the Poisson, Advection-Diffusion, and the Navier–Stokes equations; the Principle of Virtual Displacements (PVD) for solid mechanics; etc.) and it is easy to formulate new elements for other, more “exotic” problems. Furthermore, it is straightforward to combine existing single-physics elements to create hybrid elements that can be used in multi-physics simulations.

“Raw speed” is regarded as less important than robustness and generality, but this is not an excuse for inefficiency. The use of appropriate data structures and “easy-to-use” spatial and temporal adaptivity are a key feature of the library.

Generic tasks such as equation numbering, the assembly and solution of the system of coupled nonlinear algebraic equations, timestepping, etc. are

fully implemented and may be executed via simple and intuitive high-level interfaces. This allows the “user” to concentrate on the problem formulation, performed by writing C++ “driver” codes that specify the discretisation of a (mathematical) problem as a `Problem` object.

2.2 The Overall Framework

Within `oomph-lib`, all problems are regarded as nonlinear and it is assumed that any continuous (sub-)problems will be discretised in time and space, i.e. the problem’s (approximate) solution must be represented by M discrete values V_j ($j = 1, \dots, M$), e.g. the nodal values in a finite-element mesh. Boundary conditions and other constraints prescribe some of these values, and so only a subset of the M values are unknown. We shall denote these unknowns by U_i ($i = 1, \dots, N$) and assume that they are determined by a system of N non-linear algebraic equations that may be written in residual form:

$$\mathcal{R}_i(U_1, \dots, U_N) = 0 \quad \text{for } i = 1, \dots, N. \quad (1)$$

By default, `oomph-lib` uses Newton’s method to solve the system (1). The method requires the provision of an initial guess for the unknowns, and the repeated solution of the linear system

$$\sum_{j=1}^N \mathcal{J}_{ij} \delta U_j = -\mathcal{R}_i \quad \text{for } i = 1, \dots, N, \quad (2)$$

where

$$\mathcal{J}_{ij} = \frac{\partial \mathcal{R}_i}{\partial U_j} \quad \text{for } i, j = 1, \dots, N \quad (3)$$

is the Jacobian matrix. The solution of the linear system is followed by an update of the unknowns,

$$U_i := U_i + \delta U_i \quad \text{for } i = 1, \dots, N. \quad (4)$$

Steps (2) and (4) are repeated until the infinity norm of the residual vector, $\|\mathcal{R}\|_\infty$, is sufficiently small. Within this framework, linear problems are special cases for which Newton’s method converges in a single iteration.

The adaptive solution of a given problem involves three main tasks:

1. THE (REPEATED) “ASSEMBLY” OF THE GLOBAL JACOBIAN MATRIX AND RESIDUAL VECTOR

`oomph-lib` employs a finite-element-type framework in which each “element” provides a contribution to the global Jacobian matrix, $\mathcal{J}$, and the global residual vector, $\mathcal{R}$, as illustrated in Fig. 1. We note that `oomph-lib`’s definition of an “element” is very general. While the elemental residual vectors and Jacobian matrices *may* arise from finite-element discretisations, they could equally well represent finite-difference stencils or algebraic constraints.

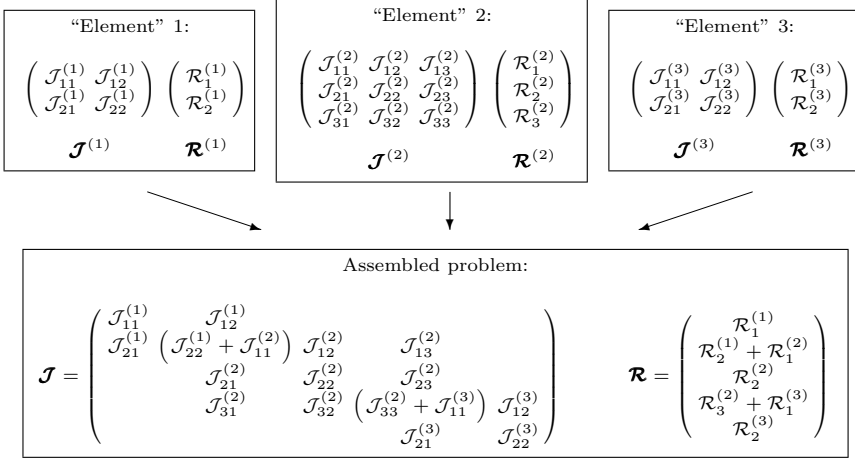


Fig. 1. Schematic illustration of the “assembly” process. Each “element” provides a contribution to the global Jacobian matrix $\mathcal{J}$ and residual vector $\mathcal{R}$. The elemental contributions may arise from a finite-element discretisation but they could equally well represent finite-difference stencils or algebraic constraints.

2. THE SOLUTION OF THE LINEAR SYSTEMS

The Newton iteration requires the repeated solution of the linear systems (2) that are (at least formally) specified by the global Jacobian matrix $\mathcal{J}$ and residual vector $\mathcal{R}$. (Not all linear solvers actually require the assembly of these objects. For instance, frontal solvers perform the LU decomposition of the Jacobian matrix “on the fly”.) In cases where the assembled Jacobian matrix is required, the assembly can be performed serially or in parallel, using an MPI-based implementation of the assembly process. The solution of the linear systems is performed by **LinearSolver** objects, most of which currently represent wrappers to state-of-the-art third-party direct solvers such as the frontal solver MA42 from the HSL library [4] or the serial and parallel versions of SuperLU [5,6]. **IterativeSolver** and **Preconditioner** classes are currently under development.

3. ERROR ESTIMATION AND PROBLEM ADAPTATION

Following the solution of the discretised problem with a given spatial discretisation, oomph-lib’s **ErrorEstimator** objects may be used to determine error estimates for each “element” in the mesh. oomph-lib provides fully automatic mesh adaptation procedures which refine (or unrefine) the mesh in regions in which the estimated error is larger (or smaller) than certain user-specified thresholds. The procedures are implemented via high-level interfaces so that a simple modification to the driver code suffices to compute a fully adaptive solution to a given problem; see the two example driver codes shown in Fig. 3 below.

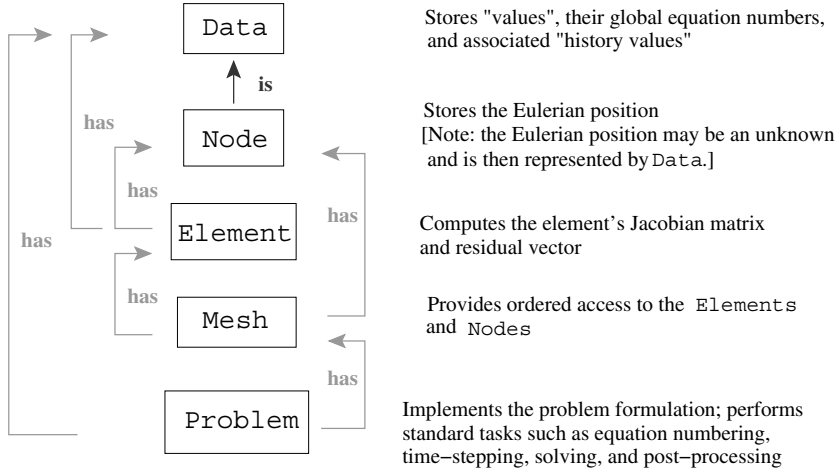


Fig. 2. Overview of the relation between `oomph-lib`'s fundamental objects.

3 The Data Structure

3.1 The Fundamental Objects

Figure 2 presents an overview of the relation between `oomph-lib`'s fundamental objects: `Data`, `Node`, `Element`, `Mesh` and `Problem`.

Data: The ultimate aim of any `oomph-lib` computation is the determination of the M values V_i ($i = 1, \dots, M$) that represent the solution to the discretised problem. These values are either prescribed ("pinned") by boundary conditions, or are unknowns. Each of the N unknown values is associated with a unique global (equation) number in the range 1 to N . `oomph-lib`'s `Data` object provides storage for values and their global equation numbers.

In many problems, the values represent components of vectors and it is often desirable to combine related values in a single object. For instance, in the finite-element discretisation of a 3D Navier-Stokes problem, each node stores three velocity components. `Data` therefore provides storage for multiple values. Furthermore, in time-dependent problems, the implicit approximation of time-derivatives requires the storage of auxiliary "history values". For instance, in a backward Euler time-discretisation, the value of the unknown at the previous timestep is required to evaluate an approximation of the value's time-derivative. `Data` provides storage for such history values, and stores a (pointer to a) `TimeStepper` object that translates the history values into approximations of the values' time-derivatives.

Nodes: `Nodes` are `Data`, i.e. they store values, but they also store the node's spatial (Eulerian) coordinates. In solid mechanics problems, the nodal coor-

<pre>// Driver code solves problem // on a fixed mesh main() { // Create the problem object ReallyHardProblem problem; // Solve the problem on // the specified mesh problem.newton_solve(); // Document the solution problem.doc_solution(); }</pre>	<pre>// Driver code solves problem // with spatial adaptivity main() { // Create the problem object ReallyHardProblem problem; // Solve, adapt the mesh, // re-solve, ... up to // three times problem.newton_solve(3); // Document the solution problem.doc_solution(); }</pre>
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Fig. 3. Two simple driver codes illustrate oomph-lib’s high-level interfaces. Note that fully-automatic spatial adaptivity is enabled by a trivial change to the driver code.

dinates can themselves be unknowns and in that case they are represented by **Data**.

Elements: The main role of **Elements** is to provide contributions to the global Jacobian matrix and the residual vector. The elemental contributions typically depend on a subset of the problem’s **Data**, which are accessed via pointers, stored in the **Element**. Within an **Element**, we distinguish between three different types of **Data**: (i) **Internal Data** contains values that are local to the element, such as the pressure in a Navier-Stokes element with a discontinuous pressure representation; (ii) **Nodal Data** is usually shared with other elements and all elements that share a given **Node** make contributions to the global equations that determine its values; (iii) **External Data** contains values that *affect* the element’s residual vector and its Jacobian matrix but are not determined *by* it. For instance, in a fluid-structure-interaction problem, the load that acts on a solid-mechanics finite element *affects* its residual but is determined *by* the adjacent fluid element(s).

Meshes: The main role of a **Mesh** is to provide ordered access to its **Nodes** and **Elements**. A **Mesh** also provides storage for (and access to) lookup schemes that identify the **Nodes** that are located on domain boundaries.

Problem: To solve a given (mathematical) problem with oomph-lib, its discretisation must be specified in a suitable **Problem** object. This usually involves the specification of the **Mesh** and the **Element** types, followed by

the application of boundary conditions. If spatial adaptivity is required, an **ErrorEstimator** object must also be specified. The error estimator is used by **oomph-lib**'s automatic mesh adaptation procedures to determine which elements should be refined or unrefined. The **Problem** base class implements generic tasks such as equation numbering, the solution of the non-linear algebraic equations by Newton's method, time-stepping, error estimation and spatial adaptation, etc. Typically, the problem specification is provided in the constructor, in which case the driver code can be as simple as the ones shown in Fig. 3. Note the trivial change required to enable spatial adaptivity.

3.2 An Example of Object Hierarchies: The Inheritance Structure for Elements

Most of **oomph-lib**'s fundamental objects are implemented in a hierarchical structure to maximise the potential for code re-use. Typically, abstract base classes are employed to (i) define interfaces for functions that all objects of this type must have, but that cannot be implemented in generality; and (ii) to implement concrete functions that perform generic tasks common to all such objects. Templating is used extensively to define families of related objects.

As an example, Fig. 4 illustrates the typical inheritance structure for finite elements. As discussed above, the minimum requirement for all elements is that they must be able to compute their contribution to the global Jacobian matrix and the residual vector. Interfaces for these tasks are defined¹ in the base class **GeneralisedElement**. For instance, the computation of the elemental Jacobian matrix must be implemented in the function **GeneralisedElement::get_jacobian(...)**. The class also provides storage for the (pointers to the) element's external and internal **Data**. (**GeneralisedElements** do not necessarily have **Nodes**; see Section 4.1 for an example). Finally, the class implements various generic tasks, such as the setup of the local/global equation numbering scheme for the values associated with the **Data** objects that affect the element.

The next level in the element hierarchy are **FiniteElements**. All **FiniteElements** have **Nodes**, and the **FiniteElement** class provides pointer-based access to these. Furthermore, all **FiniteElements** have (geometric) shape functions which are used to compute the mapping between the element's local and global (Eulerian) coordinates. The number and functional form of these shape functions depend on the specific element geometry, therefore the **FiniteElement** class only defines abstract interfaces for these functions.

Shape functions are implemented in specific "geometric" **FiniteElements**, such as the **QElement** family of 1D line, 2D quad and 3D brick elements. **QElements** are templated by the spatial dimension and the number of nodes

¹ This is achieved by implementing them as "pure virtual" C++ functions.

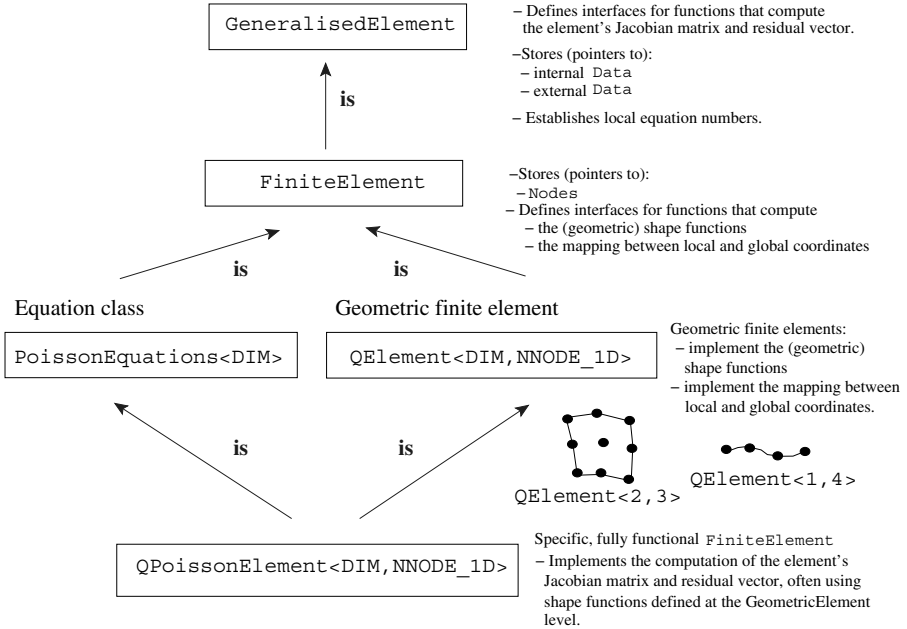


Fig. 4. Typical inheritance structure for **FiniteElements**.

along the element's 1D edges so that **QElement<1,4>** represents a four-node line element, while **QElement<3,2>** is an eight-node brick element, etc.

"Equation classes", such as **PoissonEquations**, are also derived directly from the **FiniteElement** class and implement the computation of the element's Jacobian matrix and its residual vector for a specific mathematical problem, based on the weak form of the partial differential equation (PDE). Within the equation classes, we only define the interfaces to the functions that compute the shape functions (used to represent the element geometry), the basis functions (used to represent the unknown functions) and the test functions. Their full specification is delayed until the next and final level of the element hierarchy. Templating is again used to implement equations in dimension-independent form, wherever possible. Table 1 provides a partial list of currently implemented equation classes. *oomph-lib*'s documentation provides instructions and numerous "worked examples" that illustrate how to create additional equation classes.

Finally, fully functional elements are constructed via multiple inheritance, by combining a specific geometric **FiniteElement** with a specific equation class. The (geometric) shape functions, provided by the geometric **FiniteElement** class implement the abstract shape functions defined in the equation class. For isoparametric Galerkin finite elements, the geometric

Table 1. Partial list of `oomph-lib`’s equation classes. These may be combined (by multiple inheritance) with any geometric finite element that provides sufficient inter-element continuity to form fully-functional finite elements. The presence of a template argument, `DIM`, indicates the dimension-independent implementation of the equations. The two PVD equation elements implement the principal of virtual displacements in the displacement and displacement/pressure formulations, respectively.

• <code>PoissonEquations<DIM></code>
• <code>AdvectionDiffusionEquations<DIM></code>
• <code>UnsteadyHeatEquations<DIM></code>
• <code>LinearWaveEquations<DIM></code>
• <code>NavierStokesEquations<DIM></code>
• <code>AxisymmetricNavierStokesEquations</code>
• <code>PVDEquations<DIM></code>
• <code>PVDEquationsWithPressure<DIM></code>
• <code>KirchhoffLoveBeamEquations</code>
• <code>KirchhoffLoveShellEquations</code>

shape functions are also used as the basis and test functions; for Petrov-Galerkin methods or for elements that use different interpolations for different variables (e.g. velocity and pressure in mixed Navier-Stokes elements), additional basis and test functions may be specified when the specific element is defined. Again, templating is used to create families of elements. For instance, the `QPoissonElement<DIM,NNODE_1D>` represents the family of isoparametric, Galerkin finite elements that discretise the `DIM`-dimensional Poisson equation on line, quad or brick elements with `NNODE_1DDIM` nodes.

The hierarchical implementation maximises the potential for code re-use, because any equation class may be combined with any geometric element, provided the degree of inter-element continuity of the geometric element is consistent with the differentiability requirements imposed by the weak form of the PDE represented by the equation class. The distinction between equation classes and geometric elements also facilitates the generic implementation of mesh generation and adaptation procedures, which both operate on the level of geometric `FiniteElements`.

4 GeneralisedElements and GeomObjects – How to Represent Unknown Domain Boundaries

The inheritance structure discussed in the previous section contains objects that arise naturally in the course of the finite-element discretisation of “classical” PDE problems. However, `oomph-lib` does not require the PDEs to be discretised by finite-element methods. The `GeneralisedElements`’ contributions to the global residual vector and the Jacobian matrix may equally well

represent finite-difference stencils or algebraic constraints. We shall now illustrate how this allows the representation of unknown domain boundaries in fluid-structure-interaction problems.

4.1 An Example of a GeneralisedElement

Figure 5(a) shows a very simple example of an object that may be encountered in a fluid-structure-interaction problem: a circular ring of radius R whose centre is located at (X_c, Y_c) . The ring is mounted on an elastic foundation (a spring of stiffness k), and is loaded by an external force f . The vertical displacement of the ring is governed by the algebraic equilibrium equation

$$f = k Y_c. \quad (5)$$

If the ring represents a boundary in a fluid-structure-interaction problem, f would be the (resultant) vertical force that the surrounding fluid exerts onto the ring. To allow the determination of the ring’s vertical displacement, Y_c , as part of the overall solution, the ring must be represented by a **GeneralisedElement** – a **RingOnElasticBeddingElement**, say. For this purpose we represent Y_c as the element’s internal **Data** whose single unknown value is determined by the residual equation (5). In a fluid-structure-interaction problem, the load f is an unknown. Its magnitude *affects* the element’s residual equation, but is not determined *by* the element, so we represent the load as external **Data**. If the load f is prescribed, a situation that would arise if the **RingOnElasticBeddingElement** was used in a (trivial) single-physics problem, “pinning” the value that represents f (using the **Data** member function **Data::pin(...)**) automatically excludes it from the element’s list of unknowns. Similarly, the vertical position of the ring may be fixed by “pinning” the value that represents Y_c .

The entries in the element’s residual vector contain the element’s contribution to the global equations that determine the values of its (up to) two

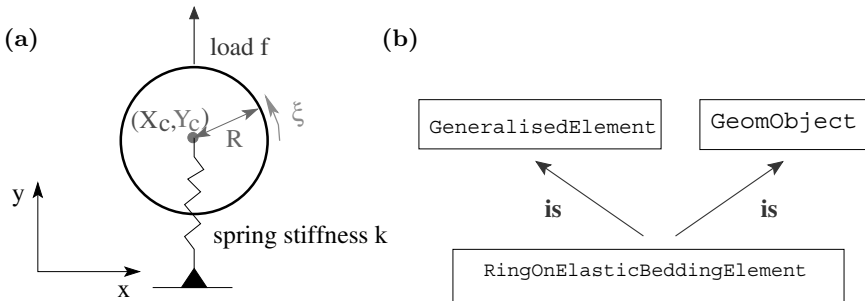


Fig. 5. A ring on an elastic foundation and its implementation as a **GeneralisedElement** and a **GeomObject**.

unknowns. If both Y_c and f are unknown, the first entry in the residual vector is given by the residual of the equilibrium equation (5). The element does not make a direct contribution to the equation that determines the external load, therefore we set the second entry to zero. The 2×2 elemental Jacobian matrix contains the derivatives of the two components of the residual vector with respect to the corresponding unknowns, so we have

$$\mathcal{R}^{(E)} = \begin{pmatrix} f - k Y_c \\ 0 \end{pmatrix} \quad \text{and} \quad \mathcal{J}^{(E)} = \begin{pmatrix} -k & 1 \\ 0 & 0 \end{pmatrix}. \quad (6)$$

If either f or Y_c are pinned, the element contains only a single unknown and its Jacobian matrix and residual vector are reduced to the appropriate 1×1 sub-blocks. If both values are pinned, the element does not make any contribution to global Jacobian matrix and residual vector.

4.2 An Example of a `GeomObject`

If used in a fluid-structure-interaction problem, the `RingOnElasticBeddingElement` defines the boundary of the fluid domain. Hence its position and shape must be accessible (via standard interfaces) to `oomph-lib`'s mesh generation and mesh update procedures. `oomph-lib` provides an abstract base class, `GeomObject`, that defines the common functionality of all objects that describe geometric features. It is assumed that the shape of a `GeomObject` may be specified explicitly by a position vector $\mathbf{R}(\xi)$, parameterised by a vector of intrinsic (Lagrangian) coordinates, ξ , where $\dim(\mathbf{R}) \geq \dim(\xi)$. For instance, the ring's shape may be represented by a 2D position vector $\mathbf{R}$, parametrised by the 1D Lagrangian coordinate ξ ;

$$\mathbf{R}(\xi) = \begin{pmatrix} X_c + R \cos(\xi) \\ Y_c + R \sin(\xi) \end{pmatrix}. \quad (7)$$

This parametrisation must be implemented in the `GeomObject`'s member function `GeomObject::position(xi,r)`, which computes the position vector $\mathbf{r}$ as a function of the vector of the intrinsic coordinates $\mathbf{xi}$.

Multiple inheritance allows the `RingOnElasticBeddingElement` to exist as both a `GeneralisedElement` and a `GeomObject`, as indicated by the inheritance diagram in Fig. 5(b). Its role as a `GeneralisedElement` allows us to determine its vertical height, Y_c , as part of the overall solution process; its role as a `GeomObject` allows us to use it for the parametrisation of the domain boundary, e.g. during mesh generation.

In fluid-structure-interaction problems, the (solid mechanics) unknowns that determine the position of the domain boundary affect the residuals and Jacobian matrices of the elements in the fluid mesh. The monolithic solution of the coupled problem via Newton's method requires the evaluation of the derivatives of the fluid mechanics residuals with respect to the (solid mechanics) unknowns that determine the shape of the fluid domain — the so-called

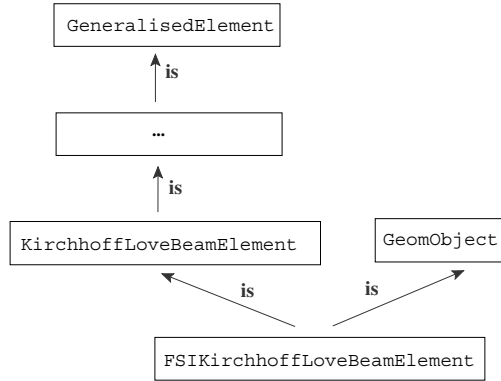


Fig. 6. Inheritance structure illustrating how a `KirchhoffLoveBeamElement` is “upgraded” to an element that may be used in fluid-structure-interaction problems.

“shape derivatives”. To facilitate such computations, the `GeomObject` class provides storage for (pointers to) those `Data` objects whose values affect the object’s shape and position. We refer to these as “geometric `Data`” and note that they should be identified and declared whenever a specific `GeomObject` is implemented. For instance, in the above example, the internal `Data` object that stores the value of Y_c represents the `RingOnElasticBeddingElement`’s only geometric `Data`.

Similar inheritance structures are implemented for “real” solid mechanics elements. For instance, in the 2D fluid-structure-interaction problem to be discussed in Section 7, the fluid domain is bounded by a thin-walled elastic ring. The ring is discretised by a surface mesh of `KirchhoffLoveBeamElements`. The shape of a deformed beam element is defined by interpolation between its nodal coordinates (represented by the `Node`’s positional `Data`), using the element’s geometric shape functions. The element’s 1D local coordinate, therefore, parametrises its 2D shape and allows it to be implemented as a `GeomObject` that can be used to define domain boundaries in fluid-structure-interaction problems. The positional `Data` stored at the element’s `Nodes` is the `GeomObject`’s geometric `Data`. Figure 6 illustrates the inheritance structure for this element.

5 Mesh Generation and Adaptation in Domains with Curvilinear Boundaries

In the previous Section we demonstrated how `GeomObjects` provide standardised interfaces for the specification of domain boundaries, and illustrated how multiple inheritance may be used to deal with domain boundaries whose positions must be determined as part of the overall solution. We now discuss

how the geometric information provided by `GeomObjects` is used to create and adapt meshes in domains with arbitrary, curvilinear boundaries. The methodology employed during the mesh generation allows a sparse update of the nodal positions in response to changes in the boundary shape — a key requirement for the efficient solution of fluid-structure-interaction problems using monolithic schemes.

5.1 Two Simple Examples

We first illustrate `oomph-lib`'s mesh adaptation capabilities in two simple single-physics problems. To begin, consider the 2D Poisson problem

$$\nabla^2 u = 1 \quad \text{in } D_{\text{fish}} \quad \text{subject to} \quad u = 0 \quad \text{on } \partial D_{\text{fish}}, \quad (8)$$

where D_{fish} is the fish-shaped domain, shown in Fig. 7(a). The “fish body” is bounded by two circular arcs of radius R , whose centres are located at $(X_c, \pm Y_c)$; the remaining domain boundaries are straight lines. The plots in Figs. 7(b-e) show contours of the solution, computed on the meshes that are generated at successive stages of the fully-automatic mesh-adaptation process. Note that `oomph-lib` requires only the provision of a very coarse initial mesh, here containing just four nine-node quad elements of type `QPoisson-Element<2,3>`.

Following the initial solution, `oomph-lib`'s mesh adaptation procedures refine the mesh, first uniformly throughout the domain, then predominantly near the inward facing corners, where the solution of Poisson's equation is singular.

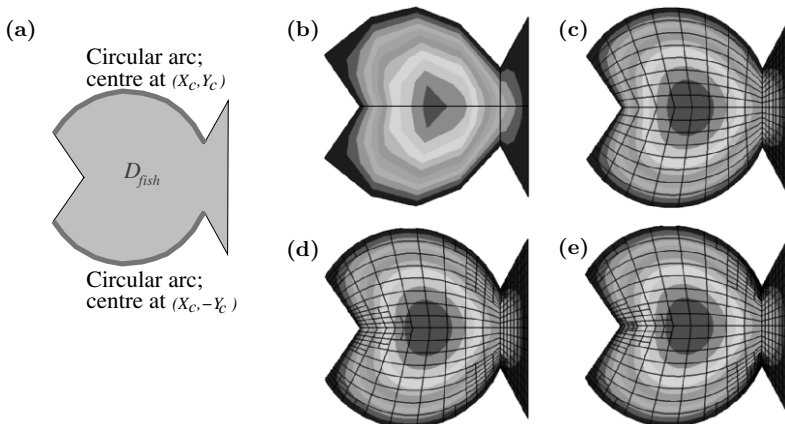


Fig. 7. The solution of a 2D Poisson equation in a fish-shaped domain. Figure (a) shows the problem sketch; Figs. (b)-(e) show contours of the computed solution, obtained on the meshes that are generated by `oomph-lib`'s automatic mesh adaptation procedures.

Figure 8 shows a 3D example: steady entry flow into a circular cylindrical tube at a Reynolds number of $Re = 200$. The axial velocity profiles illustrate how the flow develops from the entry profile $\mathbf{u} = (1 - r^{20})\mathbf{e}_z$ towards the parabolic Hagen-Poiseuille profile. This computation was started with a very coarse initial mesh, comprising six axial layers, each containing three elements. The automatic mesh adaptations then refined the mesh, most strongly near the inflow where a thin boundary layer develops on the tube wall.

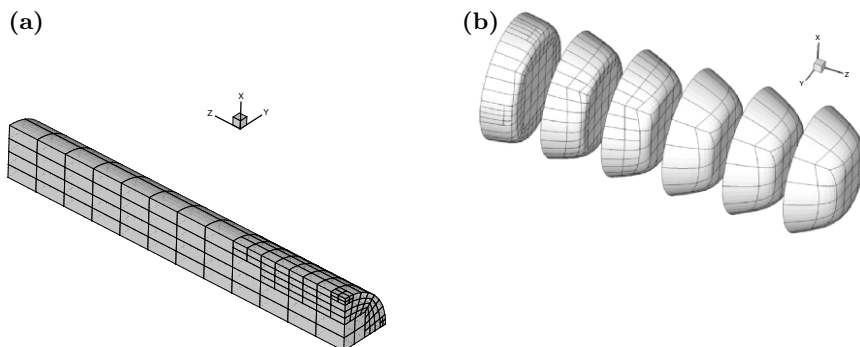


Fig. 8. Adaptive computation of finite-Reynolds-number entry flow into a circular cylindrical tube at $Re = 200$. (a) The adapted mesh (flow is from right to left). (b) Axial velocity profiles (flow is from left to right).

5.2 Some Details of the Implementation

oomph-lib’s fully-automatic mesh adaptation routines use generic high-level interfaces to the procedures that implement the adaptation for specific types of meshes (e.g. meshes consisting of quad or brick elements). The adaptation involves the following specific steps:

1. Compute an error estimate for all elements in the mesh. This task is performed by a specific **ErrorEstimator** object. Error estimation in the two examples shown above was performed with oomph-lib’s **Z2ErrorEstimator** which provides an implementation of Zhu & Zienkiewicz’s flux recovery procedure [7].
2. Select all elements whose error estimate exceeds (or falls below) certain user-specified thresholds for refinement (or unrefinement).
3. Split all elements that are scheduled for refinement into “son” elements and collapse groups of elements that are scheduled for unrefinement into their “father” element, provided *all* elements in the group are scheduled for unrefinement.

4. Delete any nodes that might have become obsolete, and create new ones where required. Interpolate the previously computed solution onto the new nodes and apply the correct boundary conditions for any newly created nodes that are located on a domain boundary.
5. Identify any “hanging nodes”, i.e. nodes on the edges or faces of elements that are not shared by adjacent (larger) elements. Inter-element continuity of the solution is ensured by constraining the nodal values and coordinates at such nodes so that they represent suitable linear combinations of these quantities at the associated “master nodes”; see Fig. 9(c). This is achieved through the implementation of the access functions to the nodal values and coordinates. For instance, at non-hanging nodes, the function `Node::value(j)` returns the j -th nodal value itself; at hanging nodes, it returns the weighted averages of the j -th values at the “master nodes”.
6. Re-generate the equation numbering scheme.

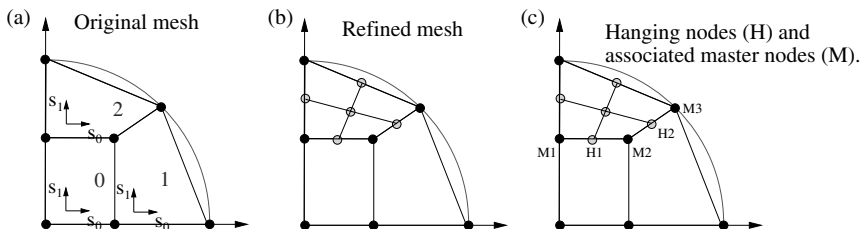


Fig. 9. (a,b): Adaptive mesh refinement without `MacroElements`. The positions of newly created nodes are determined by interpolation, using the “father” element’s geometric shape functions. Mesh refinement does not improve the representation of the curvilinear domain boundary. (c): Hanging nodes and associated master nodes: M1 and M2 are master nodes for H1; M2 and M3 are master nodes for H2.

Provided an `ErrorEstimator` object has been specified, the above steps are performed completely automatically by a call to the function `Problem::adapt()`. On return from this function, the adapted problem may be re-solved immediately.

At present, `oomph-lib` provides implementations of these procedures for meshes that contain 2D quad and 3D brick elements. Generalised quadtree and octree data structures are used to store the refinement pattern, and to identify efficiently the elements’ edge and face neighbours during the determination of the hanging nodes and their associated “master nodes”.

5.3 How to Resolve Curvilinear Boundaries: Domains and MacroElements

One particular aspect of the implementation requires a more detailed discussion. How do oomph-lib’s mesh adaptation procedures determine the position of newly created nodes in domains with curvilinear boundaries? Figure 9 illustrates the potential problem. A quarter-circle domain has been discretised with a very coarse initial mesh, containing three four-node quad elements. Assume that the error estimation indicates that element 2 should be refined. During the refinement, four new “son” elements and five new nodes are created. By default, the nodal values and coordinates of newly created nodes are obtained by interpolation from the corresponding quantities in the “father” element, using the “father” elements shape and basis functions. This procedure is perfectly adequate for meshes in domains with polygonal boundaries. However, in problems with curvilinear domain boundaries, we must ensure that the refined meshes provide a progressively more accurate representation of the exact boundary shape.

In order to achieve this, oomph-lib requires domains with curvilinear boundaries to be represented by objects that are derived from the abstract base class `Domain`. All specific `Domain` objects decompose an “exact” domain into a number of macro elements which must have the same topology and be parametrised by the same set of local coordinates as the geometric finite elements in the coarse initial mesh, as illustrated in Fig. 10. A `Domain` object defines the boundaries of its constituent macro elements, given by either the exact curvilinear boundaries (typically represented by `GeomObjects`), or arbitrary (usually straight/planar) internal edges/faces. Common interfaces for macro elements are defined in the abstract base class `MacroElement`. All macro elements must implement the member function `MacroElement::macro_map(S,r)` which establishes the mapping between the macro element’s vector of local coordinates, S , and the global (Eulerian) position vector, r , to a point inside it. The `QMacroElement<DIM>` family provides an implementation of this mapping for 2D quad- and 3D brick-shaped macro

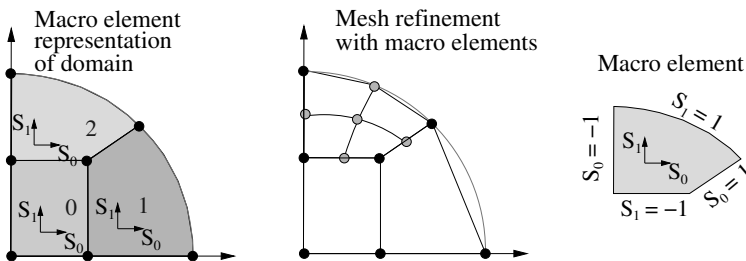


Fig. 10. `MacroElements` decompose a `Domain` into a number of subdomains which have the same topology as the corresponding `FiniteElements` in the coarse initial mesh.

elements and may be used with geometric finite elements from the `QElement` family.

The only non-trivial task to be performed when creating a new `Domain` object is the parametrisation of its macro element boundaries; not an overly onerous task, given that a domain may (and indeed should) be parametrised by very few macro elements. Once a physical domain is represented by a `Domain` object, each `FiniteElement` in the coarse initial mesh is associated with one of the `Domain`'s macro elements. The `FiniteElement`'s macro element representation is then employed (i) to determine the position of the nodes in the coarse initial mesh, and (ii) to determine the position of newly created nodes during mesh refinement.

6 Evaluation of “Shape Derivatives”

6.1 Macro-Element-Based Node Updates

The macro-element-based representation of the domain may also be used to update the nodal positions in response to changes in the domain boundary. The update may be performed on a node-by-node basis, if we allow each `Node` to store (i) a pointer to the macro element in which it is located², and (ii) its local coordinates in that macro element. Thus each `Node` is able to determine (or update) “its own” position by a call to the macro-element mapping `MacroElement::macro_map(S,r)`. (To avoid the allocation of unnecessary storage in problems that do not involve moving boundaries, the storage for these quantities is provided in the class `MacroElementUpdateNode`, derived from `oomph-lib`'s `Node` class.) Once `Nodes` can “update their own positions”, the generic and efficient evaluation of shape derivatives in fluid-structure interaction (or any other free-boundary) problems is possible.

6.2 The Generic Evaluation of “Shape Derivatives”

For simplicity we shall illustrate the methodology by considering a “toy” free-boundary problem: the solution of a 2D Poisson problem, coupled to the position of the boundary. Recall that `oomph-lib`'s `QPoissonElement<DIM, NNODE_1D>` is a single-physics element that implements the discretisation of the `DIM`-dimensional Poisson equation $\nabla^2 u(x_i) = g(x_i)$, via an isoparametric Galerkin approach in which the element's $N_u^{(E)}$ geometric shape functions $\psi_i(x_j)$ ($i = 1, \dots, N_u^{(E)}$) are also used as test and basis functions. The $N_u^{(E)}$

² The macro-element mappings of adjacent `MacroElements` are continuous, therefore `Nodes` that are located at the interface between two `MacroElements` may be associated with either one.

components of the element’s residual vector are given by

$$\mathcal{R}_i^{(E)} = \int_E \left(\sum_{j=1}^{N_u^{(E)}} U_j^{(E)} \sum_{k=1}^{\text{DIM}} \frac{\partial \psi_j}{\partial x_k} \frac{\partial \psi_i}{\partial x_k} + g(x_1, \dots, x_{\text{DIM}}) \psi_i \right) dV \quad \text{for } i = 1, \dots, N_u^{(E)}, \quad (9)$$

and depend on the element’s $N_u^{(E)}$ unknown nodal values, $U_j^{(E)}$ ($j = 1, \dots, N_u^{(E)}$). The element’s $N_u^{(E)} \times N_u^{(E)}$ Jacobian matrix contains the derivatives of the residual vector with respect to these unknowns,

$$\mathcal{J}_{ij}^{(E)} = \frac{\partial \mathcal{R}_i^{(E)}}{\partial U_j^{(E)}} = \int_E \left(\sum_{k=1}^{\text{DIM}} \frac{\partial \psi_j}{\partial x_k} \frac{\partial \psi_i}{\partial x_k} \right) dV \quad \text{for } i, j = 1, \dots, N_u^{(E)}. \quad (10)$$

In a free-boundary problem, the residual also depends on the nodal positions which are determined (via the element’s macro element representation) by the position of the domain boundary. As discussed above, unknown domain boundaries are represented by **GeomObjects**, whose shape and position is specified by their geometric **Data**. We shall denote the set of $N_G^{(E)}$ geometric unknowns, represented by the geometric **Data** that affect the nodal positions in an element, by $G_i^{(E)}$ ($i = 1, \dots, N_G^{(E)}$).

In order to use an existing **FiniteElement** in a free-boundary problem, the geometric unknowns $G_i^{(E)}$ ($i = 1, \dots, N_G^{(E)}$) that determine the element’s nodal positions must be added to the list of unknowns that affect the element’s residual vector. This requires the extension of the element’s Jacobian matrix and residual vector to

$$\mathcal{J}^{(E)} = \begin{pmatrix} \mathcal{J}_{[DD]}^{(E)} & \mathcal{J}_{[DB]}^{(E)} \\ \mathcal{J}_{[BD]}^{(E)} & \mathcal{J}_{[BB]}^{(E)} \end{pmatrix} \quad \text{and} \quad \mathcal{R}^{(E)} = \begin{pmatrix} \mathcal{R}_{[D]}^{(E)} \\ \mathcal{R}_{[B]}^{(E)} \end{pmatrix}, \quad (11)$$

where the subscripts “D” and “B” indicate entries that correspond to equations/unknowns that are associated with the equation being solved inside the domain and those that determine the shape of its boundary, respectively.

oomph-lib provides a templated wrapper class,

```
template<class ELEMENT>
class MacroElementNodeUpdateElement<ELEMENT> : public virtual ELEMENT
```

that computes the augmented quantities in (11) in complete generality. Given any existing finite element, specified by the template parameter **ELEMENT**, the **MacroElementNodeUpdateElement** class automatically incorporates the dependence of the element’s residual vector on the geometric unknowns involved in its macro-element-based node update functions into the computation of the element’s Jacobian matrix.

To explain the implementation, we consider the origin of the various terms in (11) for our “free boundary” Poisson element. The vector $\mathcal{R}_{[D]}^{(E)}$ contains the residuals of the discretised Poisson equation, evaluated for the current values of nodal unknowns, $U_i^{(E)}$ ($i = 1, \dots, N_U^{(E)}$), and the geometric unknowns $G_i^{(E)}$ ($i = 1, \dots, N_G^{(E)}$). The main diagonal block $\mathcal{J}_{[DD]}^{(E)}$ contains the derivatives of $\mathcal{R}_{[D]}^{(E)}$ with respect to the element’s nodal values. $\mathcal{J}_{[DD]}^{(E)}$ and $\mathcal{R}_{[D]}^{(E)}$ are therefore given by the Jacobian matrix and the residual vector of the single-physics element, as specified in (9) and (10). In the `MacroElementNodeUpdateElement` these may be obtained directly by calling `ELEMENT::get_jacobian(...)`. The Poisson element does not make a direct contribution to the equations that determine the shape of the domain boundary, therefore we set $\mathcal{R}_{[B]}^{(E)} = \mathbf{0}$, which implies that $\mathcal{J}_{[BD]}^{(E)} = \mathcal{J}_{[BB]}^{(E)} = \mathbf{0}$. Hence, the only non-trivial entry in the augmented element’s Jacobian matrix is the off-diagonal block $\mathcal{J}_{[DB]}^{(E)}$. It contains the derivatives of the residual vector of the underlying element with respect to the geometric unknowns $G_i^{(E)}$ ($i = 1, \dots, N_G^{(E)}$) — the “shape derivatives”. For our “free boundary” Poisson element these are given by

$$\mathcal{J}_{[DB]}^{(E)}{}_{ij} = \frac{\partial}{\partial G_j} \int_E \left(\sum_{l=1}^{N_u^{(E)}} U_l^{(E)} \sum_{k=1}^{\text{DIM}} \frac{\partial \psi_l}{\partial \underline{x}_k} \frac{\partial \psi_i}{\partial \underline{x}_k} + g(\underline{x}_1, \dots, \underline{x}_{\text{DIM}}) \psi_i \right) dV$$

for $i = 1, \dots, N_U^{(E)}$, $j = 1, \dots, N_G^{(E)}$. (12)

In Eqn. (12) the underlined quantities are affected by a change in the element’s nodal positions, and hence by a change in the geometric unknowns $G_i^{(E)}$ ($i = 1, \dots, N_G^{(E)}$). A change in the geometric unknowns affects the Jacobian of the mapping between local and global coordinates, contained in the differential dV ; the derivatives of the shape functions with respect to the global coordinates; and the argument to the source function, $g(\underline{x}_i)$. The analytical evaluation of the derivatives of these quantities with respect to the geometric unknowns would result in extremely lengthy algebraic expressions. Furthermore, the precise form of the derivatives is element-specific and would also depend on the macro-element mapping. To permit the evaluation of these terms for any template argument, `ELEMENT`, and any `MacroElement`, `oomph-lib` approximates the derivatives using finite differences,

$$\begin{aligned} \mathcal{J}_{[DB]}^{(E)}{}_{ij} &= \frac{\partial \mathcal{R}_i^{(E)} \left(U_1, \dots, U_{N_U^{(E)}}; G_1, \dots, G_{N_G^{(E)}} \right)}{\partial G_j} \\ &\approx \frac{\mathcal{R}_i^{(E)}(\dots, G_j + \epsilon_{FD}, \dots) - \mathcal{R}_i^{(E)}(\dots, G_j, \dots)}{\epsilon_{FD}}, \end{aligned} \quad (13)$$

where $\epsilon_{FD} \ll 1$. The evaluation of the finite-difference expressions is a sparse operation because the element’s list of geometric unknowns includes only

unknowns that actually change the position of at least one of its nodes, implying that only non-zero entries in the Jacobian matrix are computed.

The implementation of the above steps is completely generic, allowing the wrapper class to be used with *any* of oomph-lib’s existing finite elements and macro elements. In addition, because the adaptive solution of problems in domains with curvilinear boundaries already requires a macro-element-based representation of the domain, “upgrading” an existing fixed-domain problem to a free-boundary problem is trivial. In fact, it is only necessary to pass a list of (the pointers to) those **GeomObjects** that determine the boundaries of a given macro element to the associated **FiniteElement** when the coarse initial mesh is created. oomph-lib automatically extracts the geometric unknowns from the **GeomObject**’s geometric **Data** and includes them in the list of the element’s unknowns. Moreover, during mesh refinement, the relevant information is automatically passed to the “son” elements when a coarse element is split.

6.3 Other Node Update Approaches in oomph-lib

The generic implementation of the **MacroElementNodeUpdateElement** as a templated wrapper class is only possible because **MacroElementUpdateNodes** are able to update their own positions in response to changes in the shape of the domain boundaries. oomph-lib provides a number of alternative node update strategies and associated wrapper elements:

SpineElement<ELEMENT>: A generalisation of Kistler & Scriven’s “Method of Spines” [8], often used for free-surface fluids problems.

AlgebraicElement<ELEMENT>: A generalisation of the **MacroElementNodeUpdateElement** class, discussed above. These elements increase the sparsity of the node update operations in cases where a domain is bounded by many **GeomObjects** and are explained in more detail in Section 7.2. These elements are more efficient than **MacroElementNodeUpdateElement** but require more “user” input.

All elements discussed so far update the nodal positions based on algebraic update functions. oomph-lib also provides the

PseudoElasticNodeUpdateElement<ELEMENT,SOLID_ELEMENT>: A doubly-templated wrapper class that uses the equations of solid mechanics (discretised by the solid mechanics element specified by the second template parameter) to update the nodal positions. This element is easiest to use because it requires neither **MacroElements** nor any algebraic update functions. However, it is much more computationally expensive than the other wrapper classes, because it introduces a large number of additional unknowns into the problem.

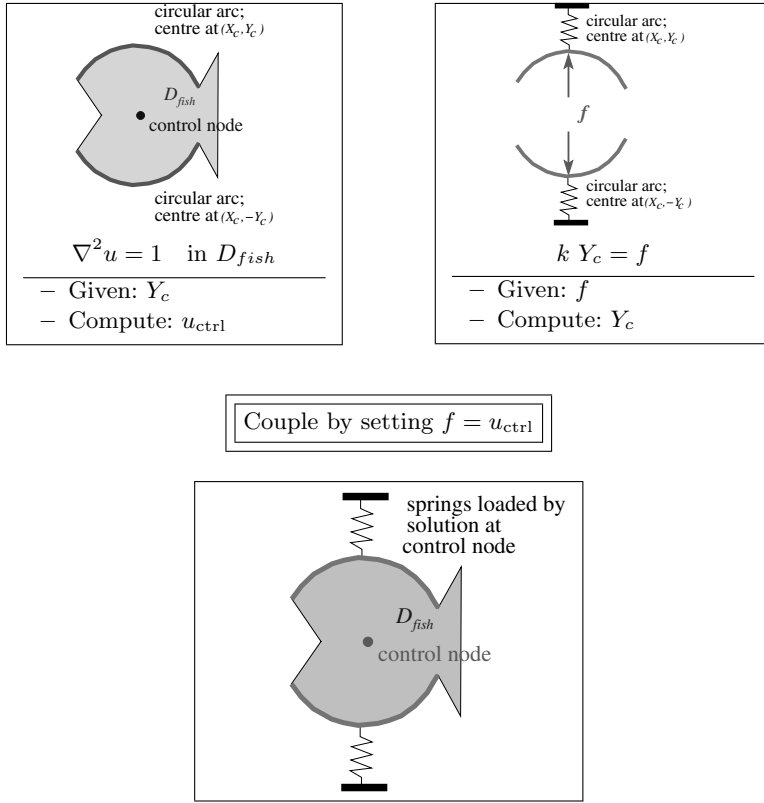


Fig. 11. Sketch illustrating the combination of two simple single-physics problems into a coupled "free boundary" Poisson problem.

7 Examples

We present several examples that illustrate the application of the methodologies discussed in the previous sections. Fully-documented demo codes for the solution of the example problems are available from <http://www.oomph-lib.org>.

7.1 A "Toy" Free-Boundary Problem: The Solution of Poisson's Equation, Coupled to the Position of the Domain Boundary

In our first example we combine the two simple single-physics problems of Section 4.1 and Section 5.1 into a "toy" free-boundary problem: Two rings on elastic foundations define the upper and lower curvilinear boundaries of the fish-shaped domain, while u_{ctrl} , the solution of Poisson's equation at a

pre-selected control node, specifies the load that drives the rings’ vertical displacements, as shown in Fig. 11

In *oomph-lib*, the solution of the coupled problem only requires a few trivial changes to the single-physics (Poisson) code:

- Replace the element type, `QPoissonElement<3,2>`, by `MacroElementNodeUpdateElement<QPoissonElement<3,2> >`.
- Pass the pointers to the `RingOnElasticBeddingElements` (which are already used during the macro-element-based mesh generation in the single-physics code) to the `MacroElementNodeUpdateElements` to indicate that their geometric `Data` affects their nodal positions. During this step, the `RingOnElasticBeddingElements` are used in their role as `GeomObjects`.
- Pass the pointer to the control Node in the fish mesh to the `RingOnElasticBeddingElements` to specify the “load”. During this step, the control Node is used in its role as `Data`.
- Add the `RingOnElasticBeddingElements` to the fish mesh. During this step, the `RingOnElasticBeddingElements` are used in their role as `GeneralisedElements`.

Figure 12 compares the results of a sequence of single-physics computations in which Y_c is prescribed, to the solution of the coupled problem. An increase in Y_c increases the distance between the two `RingOnElasticBeddingElements` that define the upper and lower curvilinear boundaries of the fish-shaped domain. Figures 12(a-d) shows that this increases the amplitude of the solution of Poisson’s equation, causing $u_{\text{ctrl}}(Y_c)$ to increase with Y_c , as shown by the solid line in Fig. 12(e).

For a spring stiffness of $k = 1$, the solution of the coupled problem should be located at the intersection of $u_{\text{ctrl}}(Y_c)$ with the line $u_{\text{ctrl}} = Y_c$ (the dashed

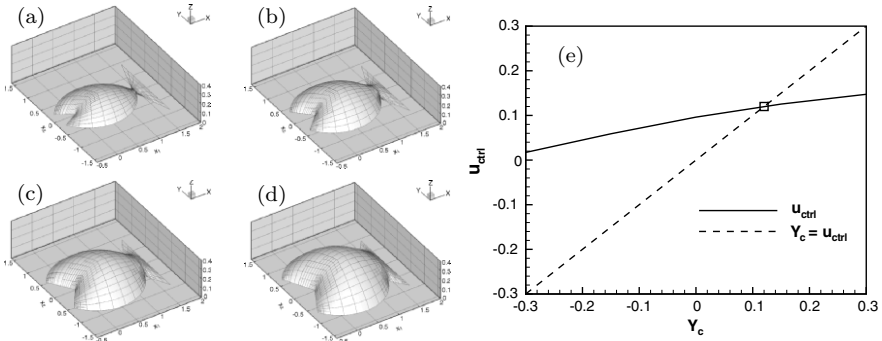


Fig. 12. (a-d) The single-physics solutions of Poisson’s equation for various values of Y_c . (e) The solution of Poisson’s equation at a control node, u_{ctrl} as a function of Y_c (solid line), and the solution of the coupled problem for $k = 1$ (square marker). The solution is located at the intersection of $u_{\text{ctrl}}(Y_c)$ with the line $u_{\text{ctrl}} = Y_c$ (the dashed line).

line). This is in perfect agreement with `oomph-lib`'s solution of the coupled problem, represented by the square marker.

7.2 A Fluid-Structure-Interaction Problem: Flow in Collapsible Tubes

I. BACKGROUND

Our next example is concerned with the classical biomechanical fluid-structure-interaction problem of flow in collapsible tubes. Many physiological flow problems (e.g. blood flow in the veins and arteries) involve finite-Reynolds-number flows in elastic vessels (see, e.g., [9] for a recent review). Experimental studies of such flows, reviewed in [10], are often performed using a Starling Resistor, a thin-walled elastic tube mounted on two rigid tubes and enclosed in a pressure chamber. Viscous fluid is pumped through the tube, while the external pressure in the chamber is kept constant. The tube wall is loaded by the external pressure and the fluid traction. When the compressive load exceeds a critical threshold, the tube buckles non-axisymmetrically, undergoing large deflections and inducing strong fluid-structure interaction. The non-axisymmetric collapse is often followed by the development of large-amplitude, self-excited oscillations. The mechanism responsible for the development of these oscillations is not fully understood. This is partly because the theoretical or computational analysis of the problem involves the solution of the 3D unsteady Navier–Stokes equations, coupled to the equations of large-displacement thin-shell theory, a formidable task. However, scaling arguments may be used to simplify the problem in particular regions of parameter space. Here we consider the case in which the Reynolds number associated with the mean flow through the tube is large, and the tube wall performs high-frequency, small-amplitude oscillations. If the 3D unsteady flow $\mathbf{u}(\mathbf{x}, t)$ in the tube is decomposed into a steady and a time-periodic unsteady component, so that $\mathbf{u}(\mathbf{x}, t) = \bar{\mathbf{u}}(\mathbf{x}) + \hat{\mathbf{u}}(\mathbf{x}, t)$, it may be shown that the unsteady component $\hat{\mathbf{u}}(\mathbf{x}, t)$ uncouples from the mean flow (see [11] for details). Furthermore, because the amplitude of the wall deformation is small, the oscillation causes small changes in the tube's cross-sectional area and only drives small axial flows. This implies that $\hat{\mathbf{u}}(\mathbf{x}, t)$ is dominated by its transverse components, allowing $\hat{\mathbf{u}}(\mathbf{x}, t)$ to be determined by computing the 2D flows that develop within the tube's individual cross-sections, as indicated by the sketch in Fig. 13.

II. PRESCRIBED WALL MOTION – `MacroElement`-BASED NODE UPDATE

We start by analysing the 2D internal fluid flows generated by the non-axisymmetric deflections of a circular ring performing a prescribed high-frequency oscillation that resembles the *in vacuo* oscillations of an elastic ring in its N -th fundamental mode. If we denote the amplitude of the oscillations by ϵ , the time-dependent wall shape, parametrised by the Lagrangian

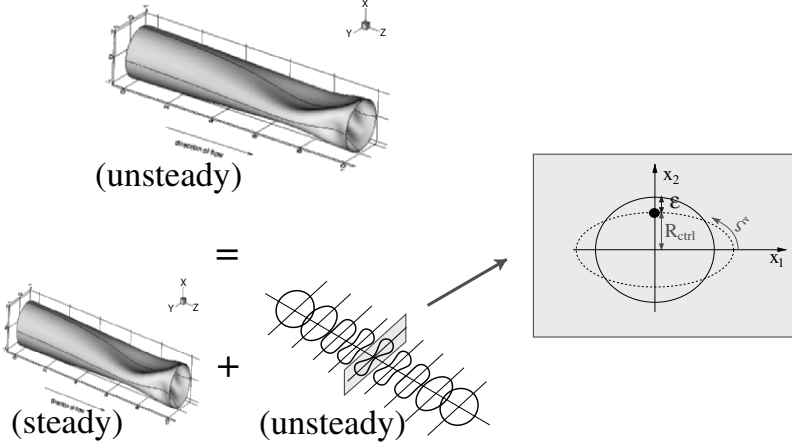


Fig. 13. Large-Reynolds-number flows in collapsible tubes that perform high-frequency, small-amplitude oscillations may be decomposed into steady and time-periodic unsteady components. The time-periodic unsteady flows are dominated by their transverse velocity components and may be determined independently in the cross-sections of the tube.

coordinate ξ , shown in Fig. 13, is given by

$$\mathbf{R}(\xi, t) = R_0 \begin{pmatrix} \cos(\xi) \\ \sin(\xi) \end{pmatrix} + \epsilon \mathbf{V}_N(\xi) \sin(\Omega t), \quad (14)$$

where the wall displacement field, $\mathbf{V}_N(\xi)$, has the form

$$\mathbf{V}_N(\xi) = \begin{pmatrix} \cos(N\xi) \cos(\xi) - A \sin(N\xi) \sin(\xi) \\ \cos(N\xi) \sin(\xi) + A \sin(N\xi) \cos(\xi) \end{pmatrix}, \quad (15)$$

see [12]. To investigate this problem with **oomph-lib**, we represent the wall shape by a **GeomObject** and use it to create the coarse initial fluid mesh, shown in the leftmost plot in Fig. 14. The fluid mesh initially contains only three Crozier-Raviart (Q2Q-1) Navier-Stokes elements. Before starting the computation, we perform three uniform refinements, using the **Problem** member function **Problem::refine_uniformly()**, and assign the initial conditions on the resulting mesh, shown in the second mesh plot. The remaining mesh plots in Fig. 14 illustrate how **oomph-lib**'s automatic adaptation procedures adjust the fluid mesh throughout the simulation.

Because the wall shape is prescribed by equations (14) and (15), the current problem does not involve any “proper” fluid-structure interaction. However, the small but finite change in the ring’s cross-sectional area, induced by the prescribed wall displacement field (15), would violate the discrete mass conservation enforced by the discretised continuity equation. It is therefore necessary to adjust the shape of the boundary so that the total area of the

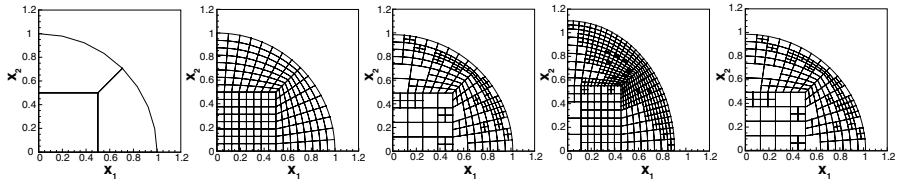


Fig. 14. Mesh adaptation during the simulation of 2D flows that are driven by the motion of the oscillating ring. The simulation is started with the uniform mesh that is obtained by performing three levels of uniform refinement of the coarse initial mesh. The remaining plots illustrate how `oomph-lib`'s automatic mesh adaptation procedures adjust the mesh throughout the simulation.

computational domain is maintained, e.g. by allowing the ring's mean radius, R_0 , to vary. In principle, R_0 could be determined via the constraint on the area of the computational domain. However, a more elegant (and easier-to-implement) approach is to give the ring some nominal elasticity, so that its mean radius is determined by the “equilibrium equation”

$$R_0 - 1 = p_{\text{ctrl}} \quad (16)$$

where p_{ctrl} is the fluid pressure at a certain fixed position, e.g. at the origin. In this approach, variations in the fluid pressure (which, in an incompressible fluid, is only determined up to an arbitrary constant) adjust R_0 so that the area of the computational domain is conserved.

The implementation of this approach is straightforward. We treat R_0 as the `GeomObject`'s geometric `Data` whose single unknown value is determined by the “equilibrium equation” (16). The resulting free-boundary problem may be solved exactly as the free-boundary Poisson problem considered in the previous example.

Figure 15 compares the computational results (instantaneous streamlines and pressure contours at two characteristic phases during the oscillation) with Heil & Waters' [11] asymptotic predictions. The two upper plots show the velocity field at an instant when the moving wall approaches its undeformed, axisymmetric configuration. The fluid is accelerated and the velocity field resembles an unsteady stagnation point flow. A thin boundary layer exists near the wall but it has little effect on the overall flow field. The two lower plots show the flow during the second half of the periodic oscillation, when the wall approaches its most strongly deformed configuration. A strong adverse pressure gradient decelerates the fluid and an area of “flow separation” appears to form near the wall.

The flow is characterised by the amplitude of the wall oscillation, ϵ , and the Womersley number, α^2 , an unsteady Reynolds number. The computations were performed for $\epsilon = 0.1$ and $\alpha^2 = 100$. Given that the asymptotic

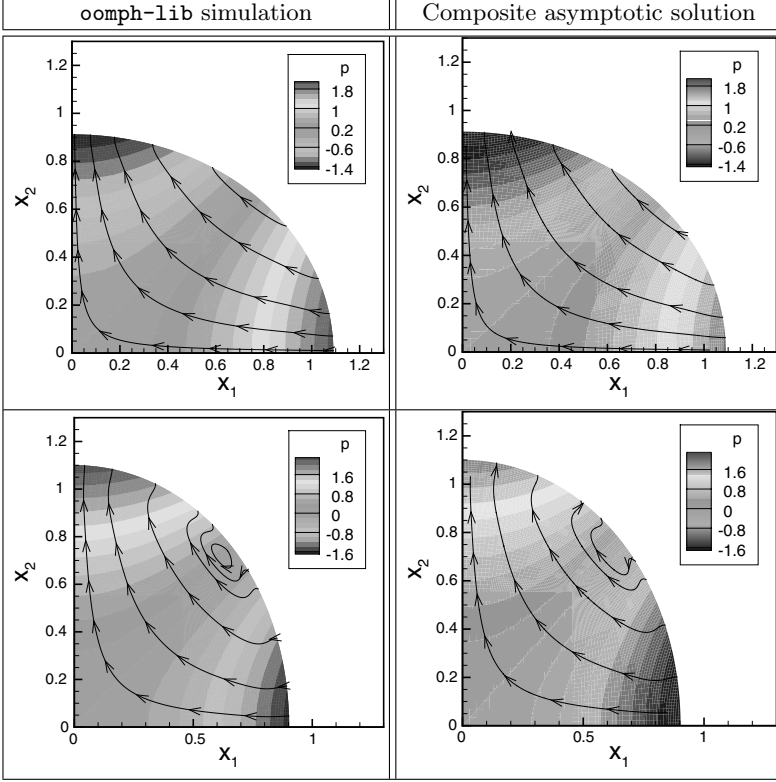


Fig. 15. Instantaneous streamlines and contours of the pressure at two characteristic phases of the oscillation. The plots on the left show the computed results, those on the right show the analytical predictions of reference [11].

predictions are only valid in the limit of small amplitude, $\epsilon \ll 1$, and large Womersley number, $\alpha^2 \gg 1$, the agreement between the two sets of results is very satisfying.

III. FULL FLUID-STRUCTURE INTERACTION

III.A NODE UPDATE WITH `AlgebraicNodes`

Next, we consider the problem with “full” fluid-structure interaction. For this purpose, we replace the `GeomObject` that prescribes the wall motion by a surface mesh of `FSIKirchhoffLoveBeamElements`, loaded by the traction exerted by the adjacent fluid elements. The implementation of the fluid-mesh update in response to changes in the shape of the domain boundary could, in principle, be performed by the macro-element-based mesh update used in all previous examples. For this purpose, we would have to combine the individual `FSIKirchhoffLoveBeamElements` into a single `GeomObject` whose

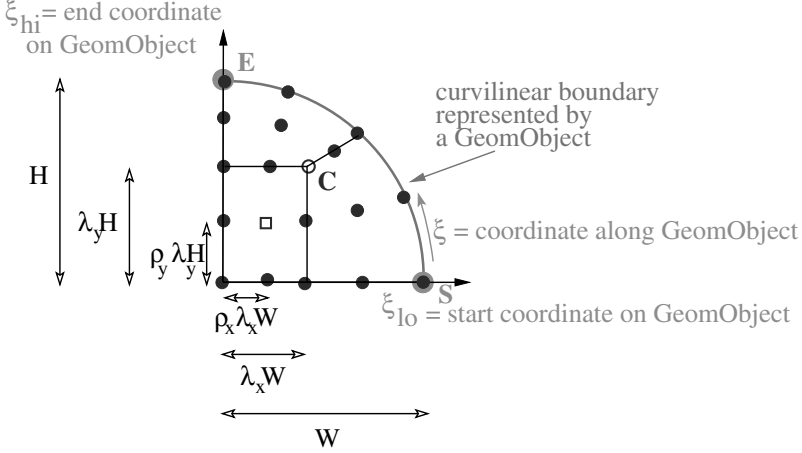


Fig. 16. Sketch illustrating the sparse node update procedures for a “quarter circle” mesh.

geometric `Data` contains the positional `Data` of *all* nodes in the wall mesh. Using this representation, the change in the nodal positions of *any* node in the wall mesh potentially induces a change in the position of *all* fluid nodes.

To avoid this undesirable feature, `oomph-lib` provides an alternative mesh update strategy that allows node updates to be performed much more sparsely. The idea is illustrated in Fig. 16. Consider the coarse three-element mesh in a “quarter circle domain” whose curved boundary is represented by one (or possibly more) `GeomObjects`. One strategy for distributing the nodes is to place the central node, “C”, at fixed fractions, λ_x and λ_y , of the domain’s width W and height H , respectively. This defines the boundary of the central element. Its constituent nodes can then be located at fixed fractions, ρ_x and ρ_y , of its width and height. Similarly, the nodes in the two elements adjacent to the curvilinear boundary may be placed on straight lines that connect the central element to reference points on the curvilinear boundary, identified by their intrinsic coordinate ξ_{ref} on the `GeomObject`. If the curvilinear boundary is represented by multiple `GeomObjects` (as in the fluid-structure-interaction problem), the reference points on the curvilinear boundary may be identified by a pointer to one of those `GeomObjects`, and the reference point’s intrinsic coordinate, ξ_{ref} , within it.

To update the nodal positions in response to changes in the domain boundary, each `Node` therefore requires (at most) two types of data: (i) Pointer(s) to the `GeomObject`(s) that affect its position, and (ii) a certain number of parameters such as λ_x , λ_y , ρ_x , ρ_y and ξ_{ref} , that identify reference points on the `GeomObjects`, and the `Node`’s relative position to these. Storage for this “node update data” is provided in the `AlgebraicNode` class, which is derived from the `Node` class.

Using this mesh-update strategy, the position of each fluid node in the “quarter circle domain” depends on no more than three `FSIKirchhoffLoveBeamElements`. As a result, the shape-derivative sub-matrix in the global Jacobian matrix is much sparser than that generated by the macro-element-based node update.

It is important to note that, as in the case of the macro-element-based node updates, `oomph-lib` only requires the specification of the “node update data” on a coarse initial mesh. Once created, the mesh may be refined with `oomph-lib`’s mesh adaptation procedures which automatically determine the “node update data” for any newly created `AlgebraicNodes`, based on the data stored at the previously existing nodes.

The computation of the “shape derivatives” is again performed fully-automatically by a templated wrapper class, `AlgebraicElement<ELEMENT>`, which determines the geometric unknowns that affect the element’s nodal positions by extracting the geometric `Data` from the `GeomObjects` that are stored in the “node update data” of its constituent `AlgebraicNodes`.

III.B RESULTS

Figure 17 shows a result from the numerical simulation of the fully-coupled fluid-structure-interaction problem. The computations were performed with the `AlgebraicElement`-version of the 2D Crozier-Raviart Navier-Stokes element used in the previous example. The simulations were started from an initial configuration in which the ring and the fluid are at rest. The oscillation

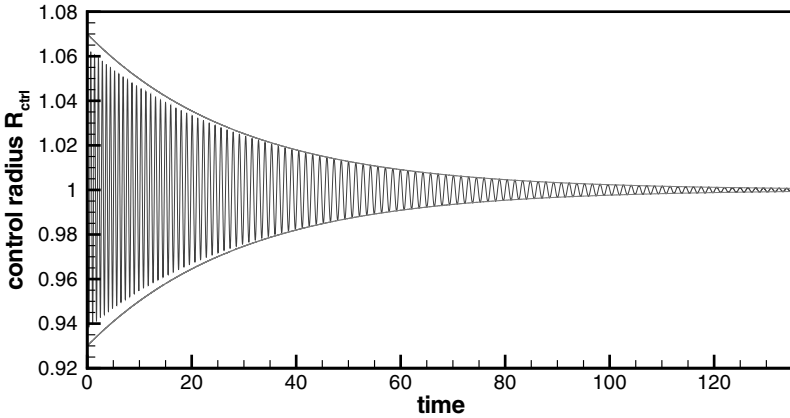


Fig. 17. Time history of the control radius, $R_{ctrl}(t)$ for the fully-coupled fluid-structure-interaction problem at $\alpha^2 = 200$. At large times, the amplitude of the oscillation, $\widehat{R_{ctrl}}$, decays exponentially, i.e. $\widehat{R_{ctrl}} \sim \exp(-\lambda t)$, as shown by the envelope.

was initiated by subjecting the ring to a short transient load perturbation of the form

$$\mathbf{f}_{\text{transient}} = \begin{cases} \mathbf{0} & \text{for } t < 0 \\ p_{\cos} \cos(N\xi) \mathbf{N} & \text{for } 0 \leq t \leq 0.3 \\ \mathbf{0} & \text{for } t > 0.3, \end{cases} \quad (17)$$

where $\mathbf{N}$ is the unit normal vector on the ring. Figure 17 shows a plot of the control radius R_{ctrl} (identified in the sketch in Fig. 13) as a function of time. The transient perturbation deforms the ring non-axisymmetrically, with a maximum amplitude of approximately 6% of its undeformed radius. Subsequently, the ring performs slowly decaying oscillations about its axisymmetric equilibrium state. Heil & Waters' theoretical analysis [11] demonstrates that the period of the oscillations is controlled by a dynamic balance between fluid inertia and the wall's elastic restoring forces, while viscous dissipation causes the oscillations to decay over a timescale that is much larger than the period of the oscillations. The frequency of the oscillations decreases slightly and ultimately approaches a constant value. At this stage, the system performs damped harmonic oscillations whose amplitude decays exponentially, as shown by the envelope in Fig. 17. The period and decay rate of the oscillation observed in the computations is in excellent agreement with Heil & Waters' theoretical predictions, obtained from a multiple-scales analysis of the problem.

8 Acknowledgements

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Modeling of Fluid-Structure Interactions with the Space-Time Techniques

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Abstract. We provide an overview of the space-time finite element techniques developed by the Team for Advanced Flow Simulation and Modeling (T★AFSM) for modeling of fluid–structure interaction problems. The core method is the Deforming-Spatial-Domain/Stabilized Space-Time formulation, complemented with the mesh update methods, including the Solid-Extension Mesh Moving Technique and Move-Reconnect-Renode Mesh Update Method. Also complementing the core method are the block-iterative, quasi-direct and direct coupling methods for the solution of the fully-discretized, coupled fluid and structural mechanics equations. Additionally, the Surface-Edge-Node Contact Tracking technique is introduced as a contact algorithm for the purpose of protecting the quality of the fluid mechanics mesh between the structural surfaces coming into contact. We present mesh-moving tests and numerical examples with incompressible flows and membrane and cable structures.

1 Introduction

Modeling of fluid–structure interactions (FSI) offers a number of challenges in computational engineering. The spatial domain occupied by the fluid changes in time as the interface moves, and the mathematical model will need to handle that. Accurate representation of the flow field near the fluid–structure interface requires that the mesh be updated to track the interface, and this requires special attention in 3D problems with complex geometries. Solution of the coupled fluid and structural mechanics equations offers additional challenges, and what technique should be used depends on how sensitive the structure is to the variations in the fluid dynamics forces and how much modularity is desired for the fluid and structural mechanics solvers. Structural surfaces coming into contact create difficulties for the generation and update of the fluid mechanics mesh, and this needs to be addressed with a contact algorithm that can guard the quality of the mesh. In this paper, we focus on the stabilized space–time finite element techniques, with emphasis on mesh update, light structures and contact between structural surfaces.

In our FSI modeling we prefer to use an interface-tracking technique. In this category of techniques, as the structure moves and the spatial domain

occupied by the fluid changes its shape, the mesh moves to accommodate this shape change and to follow (i.e. “track”) the fluid–structure interface. Moving the fluid mesh to track the interface enables us to control the mesh resolution near that interface and obtain more accurate solutions in such critical flow regions. One of the most well known examples of the interface-tracking techniques is the Arbitrary Lagrangian–Eulerian (ALE) finite element formulation [1]. The interface-tracking technique we use for discretizing the fluid dynamics equations is the Deforming-Spatial-Domain/Stabilized Space–Time (DSD/SST) formulation [2–4]. The stabilization is based on the Streamline-Upwind/Petrov-Galerkin (SUPG) [5,6] and Pressure-Stabilizing/Petrov-Galerkin (PSPG) [2,7] formulations. The SUPG formulation prevents numerical instabilities that might be encountered when we have high Reynolds number and strong boundary layers. With the PSPG formulation, we can use, without numerical instabilities, equal-order interpolation functions for velocity and pressure. An earlier version of the pressure stabilization, for Stokes flows, was introduced in [8].

The DSD/SST formulation was originally introduced in [2–4] as a general-purpose interface-tracking technique for simulation of problems involving moving boundaries or interfaces, whether fluid–solid or fluid–fluid. The stabilized space–time formulations were introduced and tested earlier by other researchers in the context of problems with fixed spatial domains (see for example [9]), mainly because of the superior stability and accuracy characteristics of these formulations. The space–time computations are carried out for one space–time “slab” at a time, where the “slab” is the slice of the space–time domain between the time levels n and $n + 1$. This spares a 3D computational problem from becoming a 4D problem including the time dimension. Some additional special features are exploited in the Special DSD/SST (S-DSD/SST) formulation [10,11] to make the calculation of the element-level vectors and matrices more efficient.

In the mesh update strategy originally envisioned with the DSD/SST formulation, the updating is based on moving it for as many time steps as we can and remeshing (generating fully or partially new set of nodes or elements) only as frequently as we need to. The mesh moving algorithm is essentially based on the one introduced in [12,13], where the motion of the nodes is governed by the equations of elasticity and the mesh deformation is dealt with selectively based on the sizes of the elements (see also [14]). The Jacobian of the transformation from the element domain to the physical domain is dropped in the finite element formulation of the elasticity equations. This “Jacobian-Based Stiffening” is equivalent to dividing the elastic modulus by the element Jacobian and results in an increase in the stiffness of the smaller elements, which are typically placed near the fluid–structure interfaces. Mesh moving techniques with functionally comparable features were later introduced in [15].

The DSD/SST formulation was first applied to FSI problems in [16] for 2D flow computation of vortex-induced vibrations of a cylinder (0D struc-

ture), and in [17] for 3D computation of flow in a flexible, cantilevered pipe (1D structure). The earliest application of the DSD/SST formulation to axisymmetric FSI computations was reported in [18]. Applications of the DSD/SST formulation to 3D FSI computations with incompressible flows and membranes and cables were first reported, in the context of parachute simulations, in [19–21]. More parachute applications were reported in [22–25]. In our FSI modeling approach, the structural mechanics equations are solved using a semi-discrete, finite element formulation. We see no compelling reason to use a space–time formulation for those equations.

The Team for Advanced Flow Simulation and Modeling (T★AFSM) introduced a number of enhancements to the general mesh update technique originally introduced in [12]. A mesh-moving optimization study based on using different ratios of the Lamé parameters of the elasticity equations was reported in [13]. A “stiffening exponent” was introduced in [26] for the Jacobian-Based Stiffening, together with a mesh-moving optimization study based on using different values of this exponent. The Solid-Extension Mesh Moving Technique (SEMMT) [27–30] addresses the challenges involved in moving a mesh with very thin fluid elements near the solid surfaces. In the Move-Reconnect-Renode Mesh Update Method (MRRMUM) [31], two remeshing options are defined, with each one proposed to be used when it is most effective to do so.

In modeling of FSI problems with the DSD/SST formulation (or any other interface-tracking technique), at each time step, we need to solve the fully-discretized, coupled fluid and structural mechanics and mesh-moving equations. What technique to use for the solution of these equations should, to a large extent, depend on the nature of the application problem. With that in mind, the T★AFSM has developed block-iterative [17,18,21], quasi-direct [10,11,32], and direct coupling techniques [10,11,32]. The direct coupling approach is based on the mixed analytical/numerical element-vector-based (AEVB/NEVB) computation technique introduced in [14,33]. The block-iterative technique gives us more flexibility in terms of algorithmic modularity and independence of the fluid and structural mechanics solvers and also better parallel efficiency. The quasi-direct and direct coupling techniques give us more robust algorithms for FSI computations where the structure is light and therefore more sensitive to the variations in the fluid dynamics forces. Various aspects of FSI modeling, including the coupling between the equations of fluid and structural mechanics and mesh motion, have also been addressed by many other researchers (see [34–41]) in recent years.

As we discussed in an earlier paragraph of this section, the core methods the T★AFSM developed for moving boundaries and interfaces have been applied to a wide range of FSI problems, some more than a decade ago (see, for example, [16,17]), and some with very challenging complexities (see, for example, [21,24,31,42]). The enhancements we describe in this paper, which include the advanced mesh update methods and the robust solution techniques for the coupled fluid and structure equations, are also applicable to a

wide range of FSI problems. In this paper, however, we explain these enhancements in the context of FSI problems where the fluid mechanics is governed by the Navier–Stokes equations of incompressible flows and the structural mechanics, where the structures are light, is governed by the membrane and cable equations. We also introduce in this paper a contact algorithm that we call the Surface-Edge-Node Contact Tracking (SENCT) technique. This algorithm, which is in early stages of its development and testing, is introduced for the purpose of protecting the quality of the fluid mechanics mesh between the structural surfaces coming into contact.

The governing equations are reviewed in Section 2. The finite element formulations, including the DSD/SST formulation, are described in Section 3. In Section 4 we review the mesh update techniques, including the SEMMT and MRRMUM. In Section 5 we describe the block-iterative, quasi-direct, and direct coupling techniques. The SENCT contact algorithm is described in Section 6. Numerical examples are presented in Section 7.

2 Governing Equations

2.1 Fluid Mechanics

Let $\Omega_t \subset \mathbb{R}^{n_{sd}}$ be the spatial domain with boundary Γ_t at time $t \in (0, T)$. The subscript t indicates the time-dependence of the domain. The Navier–Stokes equations of incompressible flows are written on Ω_t and $\forall t \in (0, T)$ as

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - \mathbf{f} \right) - \nabla \cdot \boldsymbol{\sigma} = \mathbf{0} , \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 , \quad (2)$$

where ρ , $\mathbf{u}$ and $\mathbf{f}$ are the density, velocity and the external force, respectively. The stress tensor $\boldsymbol{\sigma}$ is defined as $\boldsymbol{\sigma}(p, \mathbf{u}) = -p\mathbf{I} + 2\mu\boldsymbol{\varepsilon}(\mathbf{u})$, with $\boldsymbol{\varepsilon}(\mathbf{u}) = ((\nabla \mathbf{u}) + (\nabla \mathbf{u})^T)/2$. Here p is the pressure, $\mathbf{I}$ is the identity tensor, $\mu = \rho\nu$ is the viscosity, ν is the kinematic viscosity, and $\boldsymbol{\varepsilon}(\mathbf{u})$ is the strain-rate tensor. The essential and natural boundary conditions for Eq. (1) are represented as $\mathbf{u} = \mathbf{g}$ on $(\Gamma_t)_g$ and $\mathbf{n} \cdot \boldsymbol{\sigma} = \mathbf{h}$ on $(\Gamma_t)_h$, where $(\Gamma_t)_g$ and $(\Gamma_t)_h$ are complementary subsets of the boundary Γ_t , $\mathbf{n}$ is the unit normal vector, and $\mathbf{g}$ and $\mathbf{h}$ are given functions. A divergence-free velocity field $\mathbf{u}_0(\mathbf{x})$ is specified as the initial condition.

2.2 Structural Mechanics

Let $\Omega_t^s \subset \mathbb{R}^{n_{xd}}$ be the spatial domain with boundary Γ_t^s , where $n_{xd} = 2$ for membranes and $n_{xd} = 1$ for cables. The parts of Γ_t^s corresponding to the essential and natural boundary conditions are represented by $(\Gamma_t^s)_g$ and

$(\Gamma_t^s)_h$. The superscript “s” indicates the structure. The equations of motion are written as

$$\rho^s \left(\frac{d^2 \mathbf{y}}{dt^2} + \eta \frac{d\mathbf{y}}{dt} - \mathbf{f}^s \right) - \nabla \cdot \boldsymbol{\sigma}^s = \mathbf{0} , \quad (3)$$

where ρ^s , $\mathbf{y}$, $\mathbf{f}^s$ and $\boldsymbol{\sigma}^s$ are the material density, structural displacement, external force and the Cauchy stress tensor [43,44], respectively. Here η is the mass-proportional damping coefficient. The damping provides additional stability and can be used where time-accuracy is not required, such as in determining the deformed shape of the structure for specified fluid mechanics forces acting on it. The stresses are expressed in terms of the 2nd Piola–Kirchhoff stress tensor $\mathbf{S}$, which is related to the Cauchy stress tensor through a kinematic transformation. Under the assumption of large displacements and rotations, small strains, and no material damping, the membranes and cables are treated as Hookean materials with linear elastic properties. For membranes, under the assumption of plane stress, $\mathbf{S}$ becomes (see [45]):

$$S^{ij} = (\bar{\lambda}_m G^{ij} G^{kl} + \mu_m [G^{il} G^{jk} + G^{ik} G^{jl}]) E_{kl} , \quad (4)$$

where for the case of isotropic plane stress $\bar{\lambda}_m = 2\lambda_m\mu_m/(\lambda_m + 2\mu_m)$. Here, E_{kl} are the components of the Cauchy–Green strain tensor, G^{ij} are the components of the contravariant metric tensor in the original configuration, and λ_m and μ_m are the Lamé constants. For cables, under the assumption of uniaxial tension, $\mathbf{S}$ becomes $S^{11} = E_c G^{11} G^{11} E_{11}$, where E_c is the Young’s modulus for the cable. To account for stiffness-proportional material damping, the Hookean stress–strain relationships defined by Eq. (4) and its version for cables are modified, and E_{kl} is replaced by $\hat{E}_{kl}$, where $\hat{E}_{kl} = E_{kl} + \zeta \dot{E}_{kl}$. Here ζ is the stiffness-proportional damping coefficient and $\dot{E}_{kl}$ is the time derivative of E_{kl} .

3 Finite Element Formulations

3.1 DSD/SST Formulation of Fluid Mechanics

In the DSD/SST method [2], the finite element formulation is written over a sequence of N space–time slabs Q_n , where Q_n is the slice of the space–time domain between the time levels t_n and t_{n+1} . At each time step, the integrations are performed over Q_n . The space–time finite element interpolation functions are continuous within a space–time slab, but discontinuous from one space–time slab to another. The notation $(\cdot)_n^-$ and $(\cdot)_n^+$ denotes the function values at t_n as approached from below and above. Each Q_n is decomposed into elements Q_n^e , where $e = 1, 2, \dots, (n_{el})_n$. The subscript n used with n_{el} is for the general case in which the number of space–time elements may change from one space–time slab to another. The essential and

natural boundary conditions are enforced over $(P_n)_g$ and $(P_n)_h$, the complementary subsets of the lateral boundary of the space-time slab. The finite element trial function spaces $(\mathcal{S}_u^h)_n$ for velocity and $(\mathcal{S}_p^h)_n$ for pressure, and the test function spaces $(\mathcal{V}_u^h)_n$ and $(\mathcal{V}_p^h)_n = (\mathcal{S}_p^h)_n$ are defined by using, over Q_n , first-order polynomials in space and time. The DSD/SST formulation is written as follows: given $(\mathbf{u}^h)_n^-$, find $\mathbf{u}^h \in (\mathcal{S}_u^h)_n$ and $p^h \in (\mathcal{S}_p^h)_n$ such that $\forall \mathbf{w}^h \in (\mathcal{V}_u^h)_n$ and $\forall q^h \in (\mathcal{V}_p^h)_n$:

$$\begin{aligned}
 & \int_{Q_n} \mathbf{w}^h \cdot \rho \left(\frac{\partial \mathbf{u}^h}{\partial t} + \mathbf{u}^h \cdot \nabla \mathbf{u}^h - \mathbf{f}^h \right) dQ + \int_{Q_n} \boldsymbol{\varepsilon}(\mathbf{w}^h) : \boldsymbol{\sigma}(p^h, \mathbf{u}^h) dQ \\
 & - \int_{(P_n)_h} \mathbf{w}^h \cdot \mathbf{h}^h dP + \int_{Q_n} q^h \nabla \cdot \mathbf{u}^h dQ + \int_{\Omega_n} (\mathbf{w}^h)_n^+ \cdot \rho ((\mathbf{u}^h)_n^+ - (\mathbf{u}^h)_n^-) d\Omega \\
 & + \sum_{e=1}^{(n_{el})_n} \int_{Q_n^e} \frac{1}{\rho} \left[\tau_{\text{SUPG}} \rho \left(\frac{\partial \mathbf{w}^h}{\partial t} + \mathbf{u}^h \cdot \nabla \mathbf{w}^h \right) + \tau_{\text{PSPG}} \nabla q^h \right] \\
 & \quad \cdot [\mathbf{L}(p^h, \mathbf{u}^h) - \rho \mathbf{f}^h] dQ \\
 & + \sum_{e=1}^{(n_{el})_n} \int_{Q_n^e} \nu_{\text{LSIC}} \nabla \cdot \mathbf{w}^h \rho \nabla \cdot \mathbf{u}^h dQ = 0, \tag{5}
 \end{aligned}$$

where

$$\mathbf{L}(q^h, \mathbf{w}^h) = \rho \left(\frac{\partial \mathbf{w}^h}{\partial t} + \mathbf{u}^h \cdot \nabla \mathbf{w}^h \right) - \nabla \cdot \boldsymbol{\sigma}(q^h, \mathbf{w}^h). \tag{6}$$

Here τ_{SUPG} , τ_{PSPG} and ν_{LSIC} are the SUPG, PSPG and LSIC (least-squares on incompressibility constraint) stabilization parameters. For ways of calculating τ_{SUPG} , τ_{PSPG} and ν_{LSIC} , see [46,47,33]. This formulation is applied to all space-time slabs $Q_0, Q_1, Q_2, \dots, Q_{N-1}$, starting with $(\mathbf{u}^h)_0^- = \mathbf{u}_0$. For an earlier, detailed reference on the formulation see [2].

3.2 Semi-Discrete Formulation of Structural Mechanics

With $\mathbf{y}^h$ and $\mathbf{w}^h$ coming from appropriately defined trial and test function spaces, respectively, the semi-discrete finite element formulation of the structural mechanics equations are written as

$$\begin{aligned}
 & \int_{\Omega_0^s} \mathbf{w}^h \cdot \rho^s \frac{d^2 \mathbf{y}^h}{dt^2} d\Omega^s + \int_{\Omega_0^s} \mathbf{w}^h \cdot \eta \rho^s \frac{d\mathbf{y}^h}{dt} d\Omega^s + \int_{\Omega_0^s} \delta \mathbf{E}^h : \mathbf{S}^h d\Omega^s \\
 & = \int_{\Omega_t^s} \mathbf{w}^h \cdot (\mathbf{t}^h + \rho^s \mathbf{f}^s) d\Omega^s. \tag{7}
 \end{aligned}$$

The fluid mechanics forces acting on the structure are represented by vector $\mathbf{t}^h$. This force term is geometrically nonlinear and thus increases the overall nonlinearity of the formulation. The left-hand-side terms of Eq. (7) are referred to in the original configuration and the right-hand-side terms in the

deformed configuration at time t . From this formulation at each time step we obtain a nonlinear system of equations. In solving that nonlinear system with an iterative method, we use the following incremental form:

$$\left[\frac{\mathbf{M}}{\beta \Delta t^2} + \frac{(1-\alpha)\gamma \mathbf{C}}{\beta \Delta t} + (1-\alpha)\mathbf{K} \right] \Delta \mathbf{d}^i = \mathbf{R}^i, \quad (8)$$

where $\mathbf{C} = \eta \mathbf{M} + \zeta \mathbf{K}$. Here $\mathbf{M}$ is the mass matrix, $\mathbf{K}$ is the consistent tangent matrix associated with the internal elastic forces, $\mathbf{C}$ is a damping matrix, $\mathbf{R}^i$ is the residual vector at the i^{th} iteration, and $\Delta \mathbf{d}^i$ is the i^{th} increment in the nodal displacements vector $\mathbf{d}$. The damping matrix $\mathbf{C}$ is used only in stand-alone structural mechanics computations with specified fluid mechanics forces while establishing a starting shape for the FSI computations. In Eq. (8), all of the terms known from the previous iteration are lumped into the residual vector $\mathbf{R}^i$. The parameters α, β, γ are part of the Hilber–Hughes–Taylor [48] scheme, which is used here for time-integration.

4 Mesh Update Methods

How the mesh is updated depends on several factors, including the complexity of the fluid–structure interface and overall geometry, how unsteady the interface is, and how the starting mesh was generated. In general, the mesh update has two components: moving the mesh for as long as it is possible, and full or partial remeshing (i.e., generating a new set of elements, and sometimes also a new set of nodes) when the element distortion becomes too high. In mesh moving, the only rule to follow is that at the interface the normal velocities of the mesh and the fluid boundary have to match. With that constraint satisfied, the mesh can be moved in ways to reduce the frequency of remeshing. In most 3D applications, remeshing requires calling an automatic, unstructured-mesh generator. Reducing that cost becomes a major incentive for reducing the frequency of remeshing. Maintaining the parallel efficiency of the computations is another major incentive for reducing the frequency of remeshing, because parallel efficiency of most automatic mesh generators is substantially lower than that of most flow solvers. For example, reducing the frequency of remeshing to every ten time steps or less would sufficiently reduce the influence of remeshing in terms of its added cost and lack of parallel efficiency. In most of the complex flow problems the T★AFSM computed in the past, the frequency of remeshing was far less than every ten time steps. In our current parallel computations on a PC cluster, typically we perform remeshing on one of the nodes, which, with its 2 GigaBytes of memory, is powerful enough to generate large meshes. If remeshing does not consist of (full or partial) regeneration of just the element connectivities but also involves (full or partial) node regeneration, we need to project the solution from the old mesh to the new one. This involves a search process, which can be carried out in parallel. Still, the computational cost involved in this, and the

projection errors introduced by remeshing, add more incentives for reducing the frequency of remeshing.

4.1 Automatic Mesh Moving Technique

In the automatic mesh moving technique introduced in [12], the motion of the internal nodes is determined by solving the equations of elasticity. As the boundary condition, the motion of the nodes at the interfaces is specified to match the normal velocity of the fluid. Similar mesh moving techniques were used earlier by other researchers (see for example [49]). In [12] the mesh deformation is dealt with selectively based on the sizes of the elements. Mesh moving techniques with comparable features were later introduced in [15]. In the technique introduced in [12], selective treatment based on element sizes is attained by altering the way we account for the Jacobian of the transformation from the element domain to the physical domain. The objective is to stiffen the smaller elements, which are typically placed near solid surfaces, more than the larger ones. When this technique was first introduced in [12], it consisted of simply dropping the Jacobian from the finite element formulation of the mesh moving (elasticity) equations. This results in the smaller elements being stiffened more than the larger ones. The method described in [12] was augmented in [26] to a more extensive kind by introducing a stiffening power that determines the degree by which the smaller elements are rendered stiffer than the larger ones. This approach, when the stiffening power is set to 1.0, would be identical to the one first introduced in [12].

4.2 Solid-Extension Mesh Moving Technique (SEMMT)

In dealing with fluid–solid interfaces, sometimes we need to generate structured layers of elements around the solid objects to fully control the mesh resolution there and have more accurate representation of the boundary layers. In the mesh moving technique introduced in [12], such structured layers of elements move “glued” to the solid objects, undergoing a rigid-body motion. No equations are solved for the motion of the nodes in these layers, because these nodal motions are not governed by the equations of elasticity. This results in some cost reduction. But more importantly, the user has full control of the mesh resolution in these layers. For early examples of automatic mesh moving combined with structured layers of elements undergoing rigid-body motion with solid objects, see [50]. Earlier examples of element layers undergoing rigid-body motion, in combination with deforming structured meshes, can be found in [2].

In computation of flows with fluid–solid interfaces where the solid is deforming, the motion of the fluid mesh near the interface cannot be represented by a rigid-body motion. Depending on the deformation mode of the solid, the automatic mesh moving technique described above may need to be used. In such cases, the thin fluid elements near the solid surface becomes a challenge

for the automatic mesh moving technique. In the Solid-Extension Mesh Moving Technique (SEMMT) [27,28], it was proposed to treat those thin fluid elements almost like an extension of the solid elements. In the SEMMT, in solving the equations of elasticity governing the motion of the fluid nodes, higher rigidity is assigned to these thin elements compared to the other fluid elements. Two ways of accomplishing this were proposed in [27,28]: solving the elasticity equations for the nodes connected to the thin elements separate from the elasticity equations for the other nodes, or together. If they are solved separately, for the thin elements, as boundary conditions at the interface with the other elements, traction-free conditions would be used. The separate solution option is referred to as “SEMMT – Multiple Domain (SEMMT–MD)” and the unified solution option as “SEMMT – Single Domain (SEMMT–SD)”. In [51,30], test computations were presented to demonstrate how the SEMMT functions as part of the T★AFSM mesh update method. Both SEMMT options described above were employed. The test computations included mesh deformation tests [51,30] and a 2D FSI model problem [30]. In Subsection 4.4, we provide a brief description of the 2D FSI test computations.

4.3 General Test Conditions and Mesh Quality Measures

The tests reported here are carried out with the standard technique (where stiffening power is set to 1.0 for all the elements, and all the nodes are moved together) and SEMMT–SD (with stiffening power set to 2.0 for the inner elements and 1.0 for the outer elements). The mesh over which the elasticity equations are solved is updated at each increment. This update is based on the displacements calculated over the current mesh that has been selectively stiffened. That way, the element Jacobians used in stiffening are updated every time the mesh deforms. As a result, the most current size of an element is used in determining how much it is stiffened. Also as a result, as an element approaches a tangled state, its Jacobian approaches zero, and its stiffening becomes very large. As a measure of mesh quality, we define, similar to the way done in [52], the Element Shape Change: $f_{AR}^e = |\log (AR^e/AR_o^e)|$. Here subscript “o” refers to the undeformed mesh (i.e., the mesh obtained after the last remesh), and AR^e is the element aspect ratio, defined as $AR^e = (\ell_{max}^e)^2/A^e$, where ℓ_{max}^e is the maximum edge length for element e .

4.4 2D FSI Model Problem with SEMMT

The model represents a parachute-like structure. The “canopy” is modeled with 50 membrane elements and the “suspension lines” with 22 cable elements. Two layers of elements extend outward from the upper and lower surfaces of the canopy. In other words, the membrane elements are between the two two-layer inner fluid meshes. Detailed information on the flow conditions, structural parameters, and the solution steps can be found in [30].

In this test computation the $\frac{\partial \mathbf{w}^h}{\partial t}$ term in Eq. (5) has been dropped. Figure 1 shows the deformed meshes for the standard mesh moving technique and the SEMMT-SD. The orthogonality of the mesh lines at the canopy surface is much better preserved with the SEMMT-SD. For more on this test computation, see [30].

4.5 Move-Reconnect-Renode Mesh Update Method (MRRMUM)

The MRRMUM was proposed in [31]. In the MRRMUM, two remeshing options were defined, with each one proposed to be used when it is most effective to do so. In the “reconnect” option, only the way the nodes are connected is changed and thus only the elements are replaced (fully or partially) with a new set of elements. The mesh generator developed in [53] provides the reconnect option. In the “renode” option, the existing nodes are replaced (fully or partially) with a new set of nodes. This, of course, results in also replacing the existing elements with a new set of elements. Because the reconnect option is simpler and involves less projection errors, it is preferable to the renode option. In the MRRMUM, we move the mesh for as many time steps as we can, reconnect only as frequently as we need to, and renode only when doing so is the only remaining option.

In [31], for the prescribed rigid-body rotation of a parachute, the performances of the two remeshing options described above were compared. By examining the aerodynamical forces acting on the parachute in all three directions, performances of remeshing with the “reconnect” and “renode” options were evaluated. The evaluations showed that the force oscillations seen immediately after the remeshing are reduced substantially with the “reconnect” option.

5 Solution of Fully-Discretized Equations

Full discretizations of the formulations described in Subsections 3.1 and 3.2 lead to coupled, nonlinear equation systems that need to be solved at every time step. In a form that is partitioned with respect to the models represented, these nonlinear equations can be written as follows:

$$\mathbf{N}_1(\mathbf{d}_1, \mathbf{d}_2) = \mathbf{F}_1, \quad (9)$$

$$\mathbf{N}_2(\mathbf{d}_1, \mathbf{d}_2) = \mathbf{F}_2, \quad (10)$$

where $\mathbf{d}_1$ and $\mathbf{d}_2$ are the vectors of nodal unknowns corresponding to unknown functions $\mathbf{u}_1$ and $\mathbf{u}_2$, respectively. In the context of a coupled FSI problem, $\mathbf{u}_1$ and $\mathbf{u}_2$ represent the fluid and structure unknowns, respectively. For the space-time formulation of the fluid mechanics problem, $\mathbf{d}_1$ represents unknowns associated with the finite element formulation written for

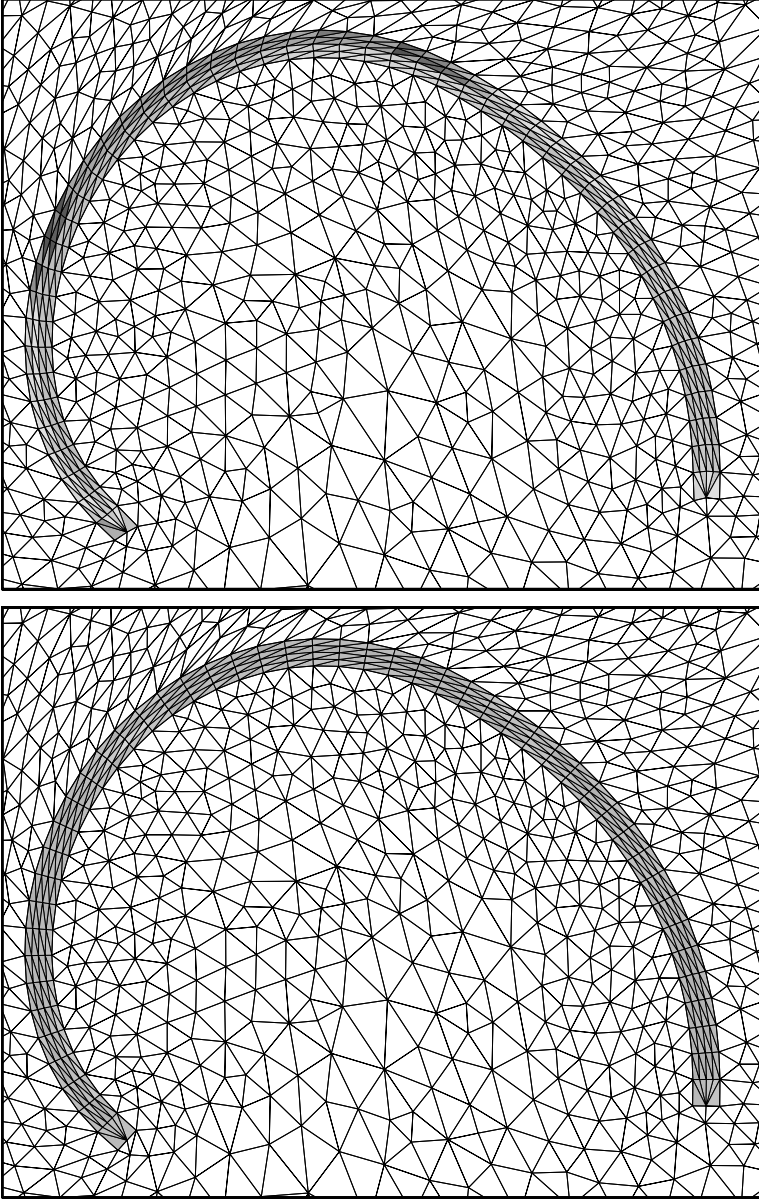


Fig. 1. Mesh deformations of the FSI model problem for the standard mesh moving technique (top) and SEMMT-SD (bottom). Colors of the inner elements indicate element distortions as measured by the aspect ratio change. The color range from light blue to light red corresponds to the range $0 \leq f_{AR}^e \leq 0.25$.

the space-time slab between the time levels n to $n + 1$ (see [2–4]). Solution of these equations with the Newton–Raphson method would necessitate at every Newton–Raphson step solution of the following linear equation system:

$$\mathbf{A}_{11}\mathbf{x}_1 + \mathbf{A}_{12}\mathbf{x}_2 = \mathbf{b}_1 , \quad (11)$$

$$\mathbf{A}_{21}\mathbf{x}_1 + \mathbf{A}_{22}\mathbf{x}_2 = \mathbf{b}_2 , \quad (12)$$

where $\mathbf{b}_1$ and $\mathbf{b}_2$ are the residuals of the nonlinear equations, $\mathbf{x}_1$ and $\mathbf{x}_2$ are the correction increments for $\mathbf{d}_1$ and $\mathbf{d}_2$, and $\mathbf{A}_{\beta\gamma} = \partial\mathbf{N}_\beta/\partial\mathbf{d}_\gamma$. Keeping the coupling matrices $\mathbf{A}_{12}$ and $\mathbf{A}_{21}$ in the picture requires taking into account the dependence of Eq. (9) on the mesh motion. In Subsections 5.1–5.3 we describe different ways of handling the coupling between Eqs. (9) and (10).

5.1 Block-Iterative Coupling

In the block-iterative coupling, the coupling matrices $\mathbf{A}_{12}$ and $\mathbf{A}_{21}$ are not kept in the picture. In an iteration step taking us from iterative solution i to $i + 1$, the following set of equations are solved:

$$\left. \frac{\partial\mathbf{N}_1}{\partial\mathbf{d}_1} \right|_{(\mathbf{d}_1^i, \mathbf{d}_2^i)} (\Delta\mathbf{d}_1^i) = \mathbf{F}_1 - \mathbf{N}_1(\mathbf{d}_1^i, \mathbf{d}_2^i) , \quad (13)$$

$$\left. \frac{\partial\mathbf{N}_2}{\partial\mathbf{d}_2} \right|_{(\mathbf{d}_1^{i+1}, \mathbf{d}_2^i)} (\Delta\mathbf{d}_2^i) = \mathbf{F}_2 - \mathbf{N}_2(\mathbf{d}_1^{i+1}, \mathbf{d}_2^i) . \quad (14)$$

The linear equations systems given by Eqs. (13)–(14) are also solved iteratively, using the GMRES search technique [54].

Because the matrix blocks representing the coupling between the fluid and structural mechanics systems are not in the picture, in computations where the structure is light, structural response becomes very sensitive to small changes in the fluid mechanics forces and convergence becomes difficult to achieve. In Subsections 5.2 and 5.3 we describe ways of keeping the coupling matrix blocks in the picture. In the absence of keeping the coupling matrices $\mathbf{A}_{12}$ and $\mathbf{A}_{21}$, a shortcut approach was proposed in [33,55] for improving the convergence of the block iterations. In this approach, to reduce “over-correcting” (i.e. “over-incrementing”) the structural displacements during the block iterations, the mass matrix contribution to $\mathbf{A}_{22}$ is increased. This is achieved without altering $\mathbf{b}_1$ or $\mathbf{b}_2$ (i.e. $\mathbf{F}_1 - \mathbf{N}_1(\mathbf{d}_1, \mathbf{d}_2)$ or $\mathbf{F}_2 - \mathbf{N}_2(\mathbf{d}_1, \mathbf{d}_2)$), and therefore when the block iterations converge, they converge to the solution of the problem with the correct structural mass.

5.2 Quasi-Direct Coupling

In the quasi-direct coupling approach, the coupling matrices $\mathbf{A}_{12}$ and $\mathbf{A}_{21}$ are kept in the picture partially, without taking into account the dependence of

$\mathbf{A}_{12}$ on the mesh motion. In describing this approach, we re-write the finite element formulations given by Eqs. (5) and (7), with a slight change of notation, and with a clarification of how the fluid–structure interface conditions are handled:

$$\begin{aligned}
& \int_{Q_n} \mathbf{w}_{1E}^h \cdot \rho \left(\frac{\partial \mathbf{u}^h}{\partial t} + \mathbf{u}^h \cdot \nabla \mathbf{u}^h - \mathbf{f}^h \right) dQ + \int_{Q_n} \boldsymbol{\varepsilon}(\mathbf{w}_{1E}^h) : \boldsymbol{\sigma}(p^h, \mathbf{u}^h) dQ \\
& - \int_{(P_n)_h} \mathbf{w}_{1E}^h \cdot \mathbf{h}_{1E}^h dP + \int_{Q_n} q_{1E}^h \nabla \cdot \mathbf{u}^h dQ \\
& + \int_{\Omega_n} (\mathbf{w}_{1E}^h)_n^+ \cdot \rho ((\mathbf{u}^h)_n^+ - (\mathbf{u}^h)_n^-) d\Omega \\
& + \sum_{e=1}^{(n_{el})_n} \int_{Q_n^e} \frac{1}{\rho} \left[\tau_{\text{SUPG}} \rho \left(\frac{\partial \mathbf{w}_{1E}^h}{\partial t} + \mathbf{u}^h \cdot \nabla \mathbf{w}_{1E}^h \right) + \tau_{\text{PSPG}} \nabla q_{1E}^h \right] \\
& \quad \cdot [\mathbf{L}(p^h, \mathbf{u}^h) - \rho \mathbf{f}^h] dQ \\
& + \sum_{e=1}^{(n_{el})_n} \int_{Q_n^e} \nu_{\text{LSIC}} \nabla \cdot \mathbf{w}_{1E}^h \rho \nabla \cdot \mathbf{u}^h dQ = 0 , \tag{15}
\end{aligned}$$

$$\begin{aligned}
& \int_{Q_n} q_{1I}^h \nabla \cdot \mathbf{u}^h dQ \\
& + \sum_{e=1}^{(n_{el})_n} \int_{Q_n^e} \frac{1}{\rho} [\tau_{\text{PSPG}} \nabla q_{1I}^h] \cdot [\mathbf{L}(p^h, \mathbf{u}^h) - \rho \mathbf{f}^h] dQ = 0 , \tag{16}
\end{aligned}$$

$$\int_{\Gamma_{1I}} (\mathbf{w}_{1I}^h)_{n+1}^- \cdot ((\mathbf{u}_{1I}^h)_{n+1}^- - \mathbf{u}_{2I}^h) d\Gamma = 0 , \tag{17}$$

$$\begin{aligned}
& \int_{Q_n} (\mathbf{w}_{1I}^h)_{n+1}^- \cdot \rho \left(\frac{\partial \mathbf{u}^h}{\partial t} + \mathbf{u}^h \cdot \nabla \mathbf{u}^h - \mathbf{f}^h \right) dQ \\
& + \int_{Q_n} \boldsymbol{\varepsilon}((\mathbf{w}_{1I}^h)_{n+1}^-) : \boldsymbol{\sigma}(p^h, \mathbf{u}^h) dQ - \int_{(P_n)_h} (\mathbf{w}_{1I}^h)_{n+1}^- \cdot \mathbf{h}_{1I}^h dP \\
& + \sum_{e=1}^{(n_{el})_n} \int_{Q_n^e} \frac{1}{\rho} \left[\tau_{\text{SUPG}} \rho \left(\frac{\partial (\mathbf{w}_{1I}^h)_{n+1}^-}{\partial t} + \mathbf{u}^h \cdot \nabla (\mathbf{w}_{1I}^h)_{n+1}^- \right) \right] \\
& \quad \cdot [\mathbf{L}(p^h, \mathbf{u}^h) - \rho \mathbf{f}^h] dQ \\
& + \sum_{e=1}^{(n_{el})_n} \int_{Q_n^e} \nu_{\text{LSIC}} \nabla \cdot (\mathbf{w}_{1I}^h)_{n+1}^- \rho \nabla \cdot \mathbf{u}^h dQ = 0 , \tag{18}
\end{aligned}$$

$$\begin{aligned}
& \int_{(\Omega_2)_0} \mathbf{w}_2^h \cdot \rho_2 \frac{d^2 \mathbf{y}^h}{dt^2} d\Omega + \int_{(\Omega_2)_0} \mathbf{w}_2^h \cdot \eta \rho_2 \frac{d\mathbf{y}^h}{dt} d\Omega + \int_{(\Omega_2)_0} \delta \mathbf{E}^h : \mathbf{S}^h d\Omega \\
&= \int_{\Omega_2} \mathbf{w}_2^h \cdot \rho_2 \mathbf{f}_2^h d\Omega + \int_{\Omega_{2E}} \mathbf{w}_{2E}^h \cdot \mathbf{h}_{2E}^h d\Omega - \int_{\Omega_{2I}} \mathbf{w}_{2I}^h \cdot \mathbf{h}_{1I}^h d\Omega . \quad (19)
\end{aligned}$$

While the subscript I refers to the fluid–structure interface, the subscript E refers to “elsewhere” in the fluid and structure domains or boundaries. In reconciling the slightly modified notation used here with the notation we used in Eqs. (5) and (7), we note that $\rho_2 = \rho^s$, $\mathbf{f}_2^h = \mathbf{f}^s$, $(\Omega_2)_0 = \Omega_0^s$, $\Omega_2 = \Omega_2^s$, and Ω_{2I} and Ω_{2E} indicate the partitions of Ω_2 corresponding to the interface and “elsewhere”. We also note that $\mathbf{h}_{1I}^h = -\mathbf{t}^h$, and $\mathbf{h}_{2E}^h$ denotes the prescribed external forces acting on the structure in Ω_{2E} , which is separate from $\mathbf{f}_2^h$. In this formulation, $(\mathbf{u}_{1I}^h)_{n+1}^-$ and $\mathbf{h}_{1I}^h$ (the fluid velocity and stress at the interface) are treated as separate unknowns, and Eqs. (17) and (18) can be seen as equations corresponding to these two unknowns, respectively. The structural displacement rate at the interface, $\mathbf{u}_{2I}^h$, is derived from $\mathbf{y}^h$. We note that Eq. (18) represents the stabilized space–time finite element formulation of the fluid mechanics momentum equation, where the test functions are limited to the fluid–structure interface.

The formulation above is based on allowing for cases when the fluid and structure meshes at the interface are not identical. If they are identical, the same formulation can still be used, but one can also use its reduced version where Eq. (17) is no longer needed and $\mathbf{h}_{1I}^h$ is no longer treated as a separate unknown. If the structure is represented by a 3D continuum model instead of a membrane model, the formulation above would still be applicable if the the domain integrations over Ω_{2E} and Ω_{2I} in the last two terms of Eq. (19) are converted to boundary integrations over Γ_{2E} and Γ_{2I} . In such cases, $\mathbf{h}_{2E}^h$ would represent the prescribed forces acting “elsewhere” on the surface of the structure.

In a slightly altered version of the formulation given by Eqs. (15)–(19), the SUPG and LSIC stabilizations are suppressed in Eq. (18), and the following equation is used:

$$\begin{aligned}
& \int_{Q_n} (\mathbf{w}_{1I}^h)_{n+1}^- \cdot \rho \left(\frac{\partial \mathbf{u}^h}{\partial t} + \mathbf{u}^h \cdot \nabla \mathbf{u}^h - \mathbf{f}^h \right) dQ \\
& + \int_{Q_n} \boldsymbol{\varepsilon}((\mathbf{w}_{1I}^h)_{n+1}^-) : \boldsymbol{\sigma}(p^h, \mathbf{u}^h) dQ - \int_{(P_n)_h} (\mathbf{w}_{1I}^h)_{n+1}^- \cdot \mathbf{h}_{1I}^h dP = 0 . \quad (20)
\end{aligned}$$

More recent experiences indicate that Eq. (20) leads to a more robust solution algorithm. Although this may not be so surprising when Eq. (20) is considered in conjunction with Eq. (17), it is something that needs further investigation.

5.3 Direct Coupling

The mixed analytical/numerical element-vector-based (AEVB/NEVB) computation technique introduced in [14,33] can be employed to keep the coupling

Let us suppose that the nonlinear vector function $\mathbf{N}_\beta$ corresponds to a finite element integral form $\mathbf{B}_\beta(\mathbf{W}_\beta, \mathbf{u}_1, \dots, \mathbf{u}_N)$. Here $\mathbf{W}_\beta$ represents the vector of nodal values associated with the weighting function $\mathbf{w}_\beta$, which generates the nonlinear equation block β . Let us also suppose that we are able to, without major difficulty, derive the first-order terms in the expansion of $\mathbf{B}_\beta(\mathbf{W}_\beta, \mathbf{u}_1, \dots, \mathbf{u}_N)$ in $\mathbf{u}_\gamma$. We represent those first-order terms in $\Delta \mathbf{u}_\gamma$ with the finite element integral form $\mathbf{G}_{\beta\gamma}(\mathbf{W}_\beta, \mathbf{u}_1, \dots, \mathbf{u}_N, \Delta \mathbf{u}_\gamma)$. For example, $\mathbf{G}_{11}(\mathbf{W}_1, \mathbf{u}_1, \dots, \mathbf{u}_N, \Delta \mathbf{u}_1)$ represents the first-order terms obtained by expanding the finite element formulation of the fluid mechanics equations (i.e. momentum equation and incompressibility constraint) in fluid mechanics unknowns (i.e. fluid velocity and pressure). We note that the integral form $\mathbf{G}_{\beta\gamma}$ will generate $\frac{\partial \mathbf{N}_\beta}{\partial \mathbf{d}_\gamma}$. Consequently, the product $\mathbf{A}_{\beta\gamma} \mathbf{x}_\gamma$ can be evaluated as [14,33]

$$\mathbf{A}_{\beta\gamma} \mathbf{x}_\gamma = \frac{\partial \mathbf{N}_\beta}{\partial \mathbf{d}_\gamma} \mathbf{x}_\gamma = \bigwedge_{e=1}^{n_{el}} \mathbf{G}_{\beta\gamma}(\mathbf{W}_\beta, \mathbf{u}_1, \dots, \mathbf{u}_N, \mathbf{v}_\gamma), \quad (27)$$

where, $\mathbf{v}_\gamma$ is a function interpolated from $\mathbf{x}_\gamma$ in the same way $\mathbf{u}_\gamma$ is interpolated from $\mathbf{d}_\gamma$. This EVB technique allows us to evaluate matrix–vector products without dealing with numerical evaluation of directional derivatives. To differentiate between the EVB techniques defined by Eqs. (26) and (27), we call them, respectively, numerical EVB (NEVB) and analytical EVB (AEVB) techniques.

Two ways of using the mixed AEVB/NEVB computation technique were proposed in [10,11,32] to take into account the dependence of the coupling matrices on the mesh motion. In the first way, it was proposed to use the NEVB technique to compute $\mathbf{A}_{12} \mathbf{x}_2$ while including the dependence of $\mathbf{A}_{12}$ on the mesh motion. This would be done as seen in Eq. (23). In the second way proposed, the use of the NEVB technique is limited to evaluation of the matrix–vector products involving coupling matrices representing the dependence on the mesh motion. This would be done by considering a three-block version of the nonlinear equation system given by Eqs. (9)–(10), where $\mathbf{d}_3$ is the vector of nodal unknowns representing the mesh motion, and the third block of equations represents the mesh-moving equations. The three-block version of Eq. (11)–(12) can be solved iteratively with the three-block version of Eqs. (21)–(22), which is written as follows:

$$\mathbf{P}_{11} \Delta \mathbf{y}_1 + \mathbf{P}_{12} \Delta \mathbf{y}_2 + \mathbf{P}_{13} \Delta \mathbf{y}_3 = \mathbf{b}_1 - (\mathbf{A}_{11} \mathbf{x}_1 + \mathbf{A}_{12} \mathbf{x}_2 + \mathbf{A}_{13} \mathbf{x}_3), \quad (28)$$

$$\mathbf{P}_{21} \Delta \mathbf{y}_1 + \mathbf{P}_{22} \Delta \mathbf{y}_2 + \mathbf{P}_{23} \Delta \mathbf{y}_3 = \mathbf{b}_2 - (\mathbf{A}_{21} \mathbf{x}_1 + \mathbf{A}_{22} \mathbf{x}_2 + \mathbf{A}_{23} \mathbf{x}_3), \quad (29)$$

$$\mathbf{P}_{31} \Delta \mathbf{y}_1 + \mathbf{P}_{32} \Delta \mathbf{y}_2 + \mathbf{P}_{33} \Delta \mathbf{y}_3 = \mathbf{b}_3 - (\mathbf{A}_{31} \mathbf{x}_1 + \mathbf{A}_{32} \mathbf{x}_2 + \mathbf{A}_{33} \mathbf{x}_3). \quad (30)$$

The NEVB technique can be used for computing $\mathbf{A}_{13} \mathbf{x}_3$ as follows:

$$\mathbf{A}_{13} \mathbf{x}_3 = \bigwedge_{e=1}^{n_{el}} \lim_{\epsilon_1 \rightarrow 0} \left[\frac{\mathbf{N}_1^e(\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3 + \epsilon_1 \mathbf{x}_3) - \mathbf{N}_1^e(\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3)}{\epsilon_1} \right]. \quad (31)$$

6 Surface-Edge-Node Contact Tracking (SENCT)

In this section we propose the Surface-Edge-Node Contact Tracking (SENCT) technique as a contact algorithm. In this technique, which is in early stages of its development and testing, the objective is to prevent the structural surfaces from coming closer than a predetermined minimum distance we would like to maintain to protect the quality of the fluid mechanics mesh between the structural surfaces.

The contact detection is based on searching and calculating for each node the projection distance from that node to all the structural surface elements (“surfaces”), edges and nodes. During this search, we check to see if that projection distance is smaller than the minimum distance we would like to maintain between the structural surfaces. If it is, then we declare that surface or edge or node to be a contact surface or edge or node for the node we are conducting the search for. We note that for each node we are conducting a search for, the predetermined minimum distance between the structural surfaces is calculated locally (i.e. based on the local length scales related to that node).

The search algorithm involves some exclusion criteria. For example, edges belonging to contact surfaces are excluded, and the nodes belonging to contact surfaces or edges are excluded. The search algorithm also involves some more obvious exclusion criteria. For example, we exclude the surfaces containing the node we are conducting the search for, and exclude the edges and nodes that belong to the surface that also contains the node we are conducting the search for. The surfaces, edges and nodes beyond a certain distance from the node we are conducting the search for are excluded from the projection distance calculations, because they are not expected to be candidates for a contact surface or edge or node. The contact detection search is conducted at every n_{tsbcd} time steps, a parameter specified by the user. Once all the contact surfaces, edges and nodes are identified for the “contacted node” (i.e. the node we were conducting the search for), the nodes belonging to those surfaces, edges and nodes are declared to be the “contact-node set” for the contacted node.

We propose two variations of the SENCT technique. In the SENCT-Force (SENCT-F) technique, the contacted node is subjected to penalty forces that are inversely proportional to the projection distances to the contacting surfaces, edges and nodes. In the SENCT-Displacement (SENCT-D) technique, the displacement of the contacted node is adjusted to correlate with the motion of the contacting surfaces, edges and nodes. There are various ways of accomplishing that. For example, at every time step, the contacted node can be allowed to move for a certain number of nonlinear iterations without any contact restrictions, followed by some more nonlinear iterations where the motion of the contacted node is set to the mean displacement of the contact-node set. If the displacement of the contacted node at the end of the first set of nonlinear iterations shows that it no longer qualifies as a contacted node,

then during the next set of nonlinear iterations the node is allowed to move without any contact restrictions.

7 Numerical Examples

All computations were carried out in a parallel computing environment, using PC clusters.

7.1 Flow Past a “Flag”

In this test problem (see [56]), the FSI involved in the flapping of a “flag” is computed. Results from a very similar problem were presented in [57]. The quasi-direct coupling approach, as described by Eqs. (15)–(17) and Eqs. (19)–(20), is used. The dimensions of the flag are 1.5 m in the flow direction and 1.0 m in the span-wise direction. The fluid velocity, density and kinematic viscosity are 2 m/s, 1 kg/m³ and 0.008 m²/s, respectively. The flag is modeled as a membrane with density 1000 kg/m³, thickness 0.2 mm, and Young’s modulus 40,000 N/m². The leading edge of the flag is held fixed and the lateral edges of the flag are constrained to move only in a normal plane. The structural mechanics mesh consists of 2,809 nodes and 5,416 three-node triangular membrane elements. The fluid mechanics mesh, generated with the mesh generator developed in [53], contains approximately 95,000 nodes and 560,000 four-node tetrahedral elements. The FSI computations are carried out until a nearly cyclic pattern of flapping is reached. Figure 2 shows a

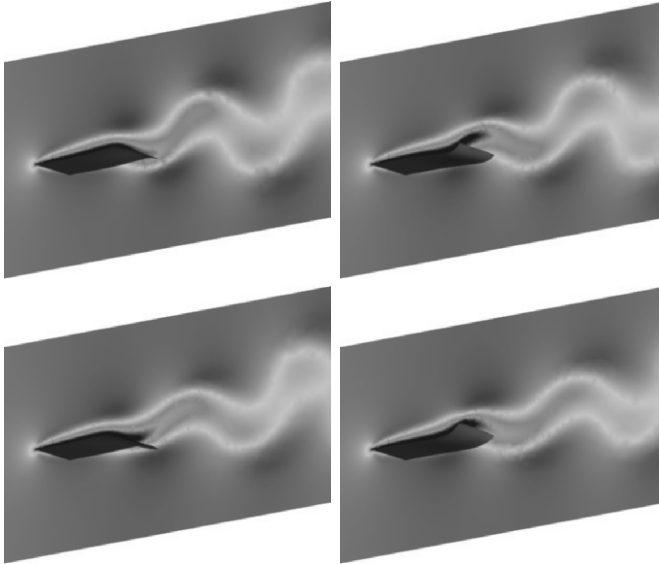


Fig. 2. Time history (left to right, top to bottom) of the flag motion and the horizontal velocity on a normal plane.

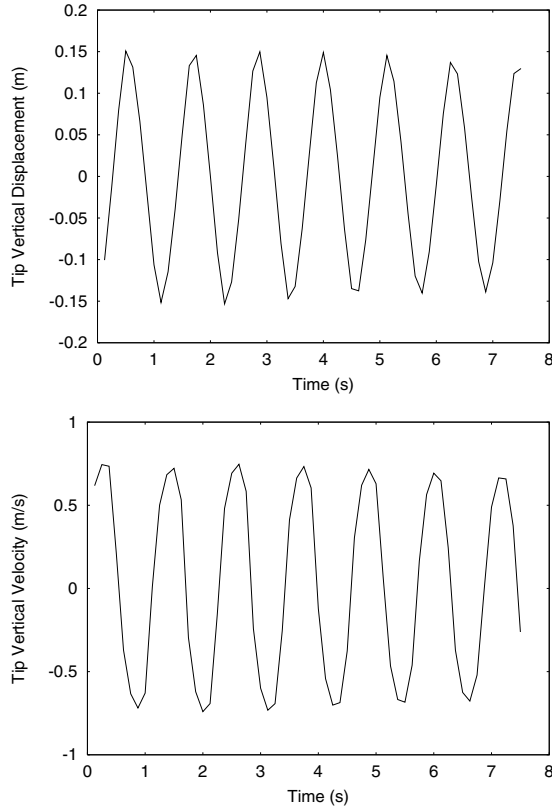


Fig. 3. Vertical displacement (top) and velocity (bottom) for the midpoint of the free edge of the flag.

sequence of snapshots of the flag and the horizontal velocity on a normal plane. Figure 3 shows the displacement and velocity for the midpoint of the free edge of the flag. We note that the results are very similar to those reported in [10,11], using the quasi-direct coupling technique described by Eqs. (15)–(19).

7.2 Soft Landing of a G-12 Parachute

In this test problem (see [56]), the soft-landing of a G-12 parachute is computed using a direct coupling technique based on Eqs. (15)–(19) and Eq. (31). This is a 64-ft diameter parachute with 64 suspension lines, each about 51 ft long, and 4 risers, where each riser is connected to 16 suspension lines. The risers meet at a single confluence point, which is connected to the payload with 4 pneumatic muscle actuators (PMAs), each about 15 ft long. The payload is 2,200 lb. The soft landing is accomplished by the retraction of these

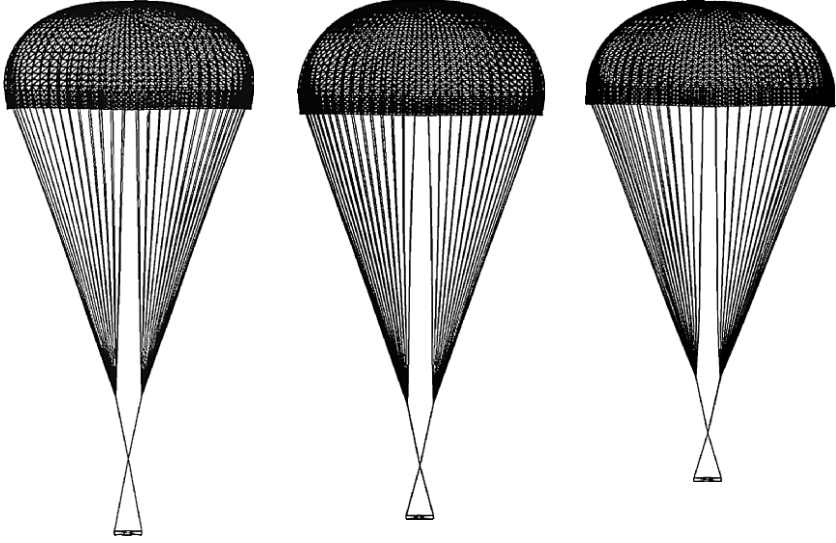


Fig. 4. Soft landing of a G-12 parachute. Dynamics of the parachute, represented by images at three instants during and after the retraction. For more on this computation, see [32].

PMAs prior to landing. In this test case the PMAs are retracted 7.1 ft in 0.23 s while the parachute is descending at 28 ft/s. In the computation the $\frac{\partial \mathbf{w}^h}{\partial t}$ term in Eqs. (15) and (18) has been dropped. The structural mechanics mesh consists of 5,261 nodes, and 8,832 three-node triangular membrane elements, 2,650 two-node cable elements, and 8 one-node (point-mass) payload elements. The fluid mechanics mesh contains approximately 140,000 nodes and 850,000 four-node tetrahedral elements. Figure 4 shows the dynamics of the parachute during and after the retraction. The payload descent speed becomes as low as 10 ft/s, and we see a large increase in drag. Figure 5 shows the flow field during and after the retraction. For more details on this computation, see [32].

7.3 Disreefing of a G-12 Parachute

A parachute can be initially deployed in a reefed configuration and with a lower drag area so that higher descent speeds are achieved during most of its descent. Prior to landing, it can be disreefed to increase the drag area to its full level and thus reduce the landing speed to its normal value. The challenge involved in mesh update increases significantly in this type of FSI problems. The test problem is used to show how such challenges are addressed. The additional geometric complexities causing the increase in mesh-update difficulties are limited to certain periods of the FSI simulation. Therefore it

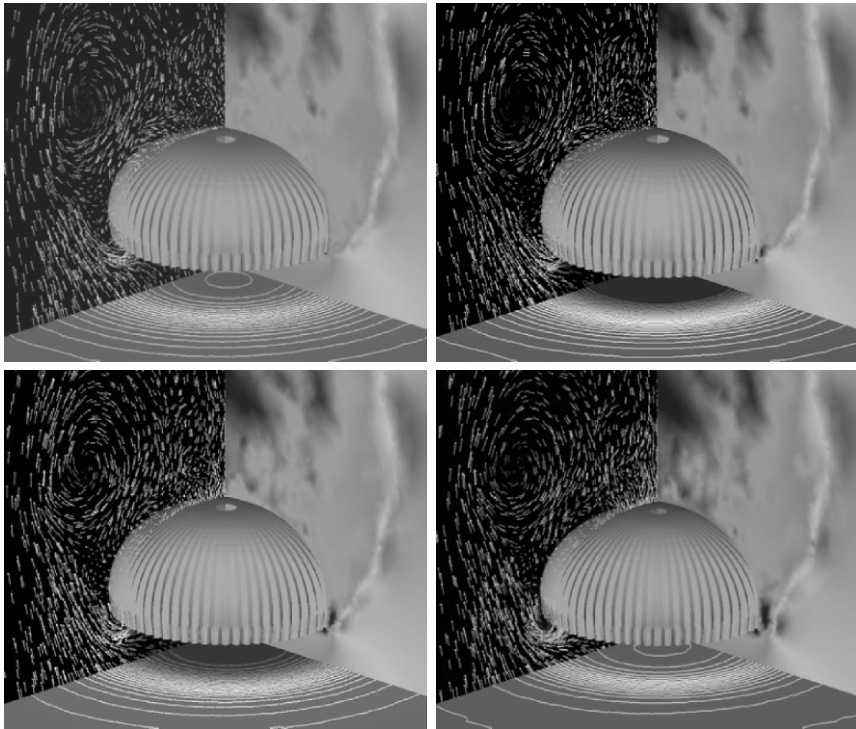


Fig. 5. Soft landing of a G-12 parachute. Flow field at four instants during and after the retraction (left to right and top to bottom). Left plane: velocity vectors; right plane: magnitude of vertical velocity; bottom plane: pressure. For more on this computation, see [32].

was proposed in [31] to overcome those difficulties by selectively decreasing the time-step size or increasing the remeshing frequency during those periods. The simulation starts with a reefed parachute with a diameter 25% smaller than the diameter of a fully-inflated parachute. Figure 6 shows the fully-inflated and reefed parachutes. Figure 7 shows the surface mesh for the reefed parachute. The structural mechanics mesh consists of 5,261 nodes, and 8,832 three-node triangular membrane elements, 2,714 two-node cable elements, and 8 one-node payload elements. The fluid mechanics mesh, generated with the mesh generator developed in [53], contains approximately 120,000 nodes and 730,000 four-node tetrahedral elements. The FSI computation is carried out with the quasi-direct coupling approach, as described by Eqs. (15)–(17) and Eqs. (19)–(20). The computation is carried out for a total of 2,625 time steps, with a total of 21 remeshes. Figure 8 shows the parachute at some instants during the disreefing.

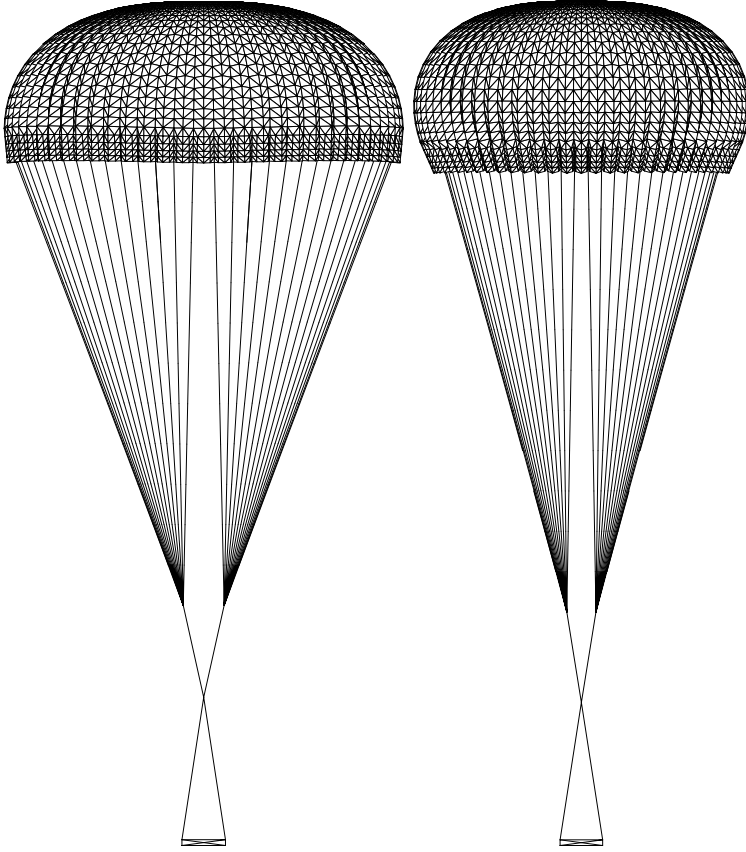


Fig. 6. Disreefing of a G-12 parachute. Fully inflated (left) and reefed (right) parachutes. The reduction in diameter is 25%.

7.4 A Cloth Piece Falling Over a Rigid Rod

A $1.0 \text{ m} \times 1.0 \text{ m}$ piece of cloth is dropped in the air over a rigid rod that is 0.1 m thick and 1.2 m long, from a height of 0.02 m above the surface of the rod. The thickness, density and stiffness of the cloth are 0.002 m , 100 kg/m^3 and $1.0 \times 10^4 \text{ N/m}^2$, respectively. The structural mechanics meshes for the cloth and rod are shown in Figure 9. The mesh for the cloth consists of 2,601 nodes and 5,000 three-node triangular membrane elements. The initial configuration is shown by the top picture in Figure 10. The fluid mechanics mesh, generated with the mesh generator developed in [53], contains approximately 52,000 nodes and 310,000 four-node tetrahedral elements. The rigid rod is not modeled in this fluid mechanics mesh. The FSI computation is carried out with the quasi-direct coupling approach, as described by Eqs. (15)–(17) and Eqs. (19)–(20). The SENCT-D technique is used as the contact algorithm.

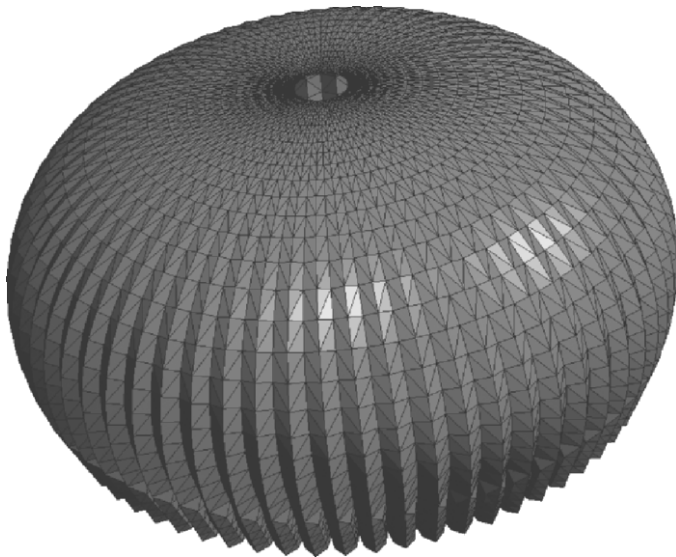


Fig. 7. Disreefing of a G-12 parachute. Surface mesh for the reefed parachute.

The computation is carried out for a total of 500 time steps, with a total of 4 remeshes. Figures 10–12 show the cloth at various instants during the simulation.

8 Concluding Remarks

We provided an overview of the space–time finite element techniques developed by the Team for Advanced Flow Simulation and Modeling (T★AFSM) to address the computational challenges involved in modeling of fluid–structure interactions. The core method is the Deforming-Spatial-Domain/Stabilized Space–Time (DSD/SST) formulation, which has been proven to be a powerful and comprehensive method for solving flow problems with moving boundaries and interfaces. The core method is enhanced by a number of additional methods developed by the T★AFSM to address more specific challenges. The mesh update methods developed, including the Solid-Extension Mesh Moving Technique (SEMMT) and Move-Reconnect-Renode Mesh Update Method (MRRMUM), help us update the mesh effectively as the spatial domain occupied by the fluid changes in time. What technique to use for the solution of the fully-discretized, coupled fluid and structural mechanics equations should, to a large extent, depend on the nature of the application problem. Keeping that in mind, the T★AFSM developed the block-iterative, quasi-direct, and direct coupling techniques. The block-iterative technique gives us more flexibility in terms of algorithmic modularity and independence of the fluid and

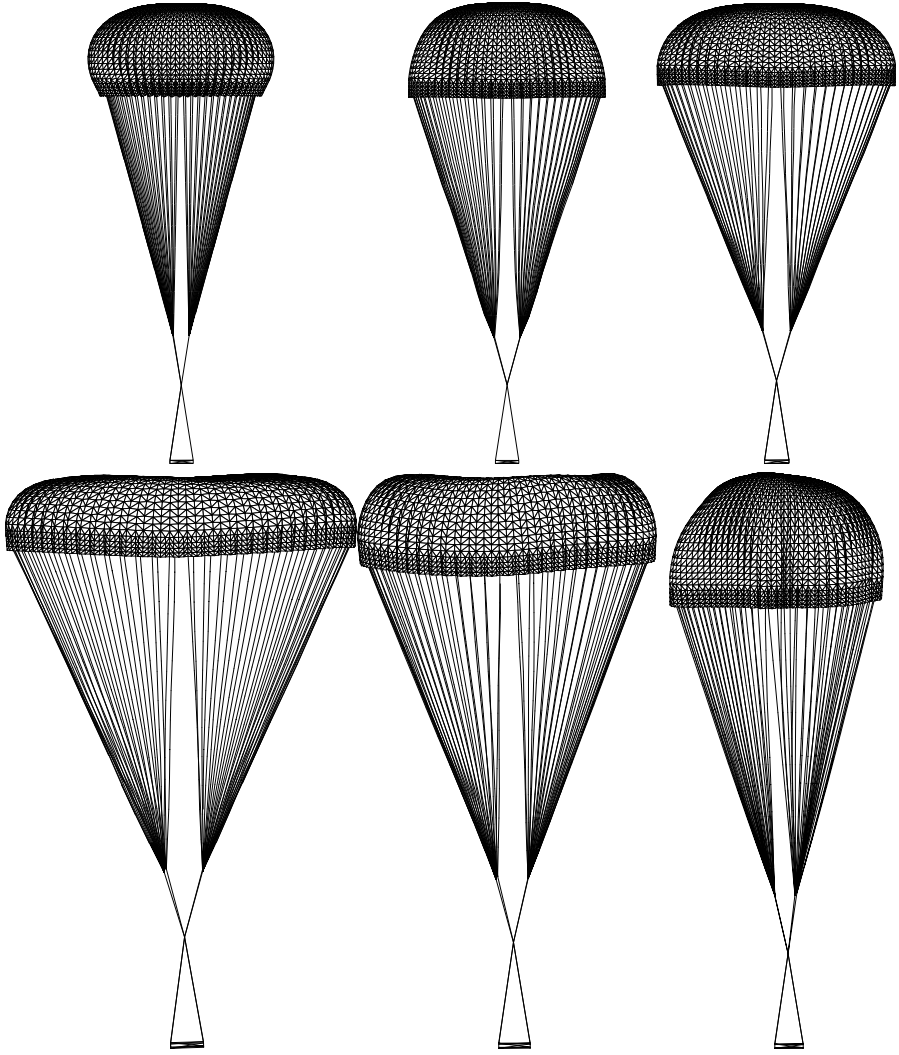


Fig. 8. Disreefing of a G-12 parachute. Parachute at various instants during disreefing (left to right and top to bottom).

structural mechanics solvers. The quasi-direct and direct coupling techniques give us more robust algorithms for FSI computations where the structure is light. Additionally, in this paper, the Surface-Edge-Node Contact Tracking (SENCT) technique has been introduced as a contact algorithm for the purpose of protecting the quality of the fluid mechanics mesh between the structural surfaces coming into contact. The algorithm is still in early stages of its development and testing but clearly offers a good potential for over-

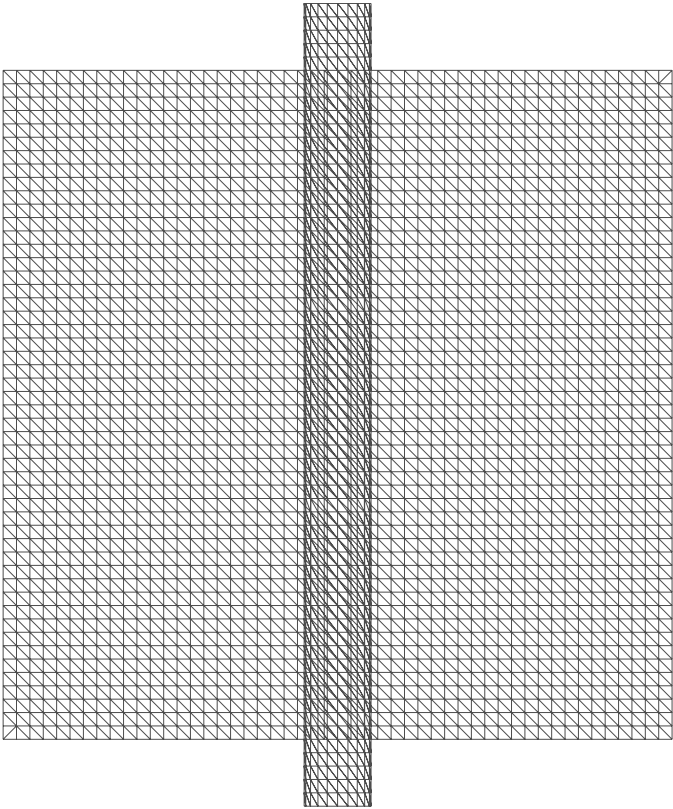


Fig. 9. A cloth piece falling over a rigid rod. Meshes for the cloth and rod.

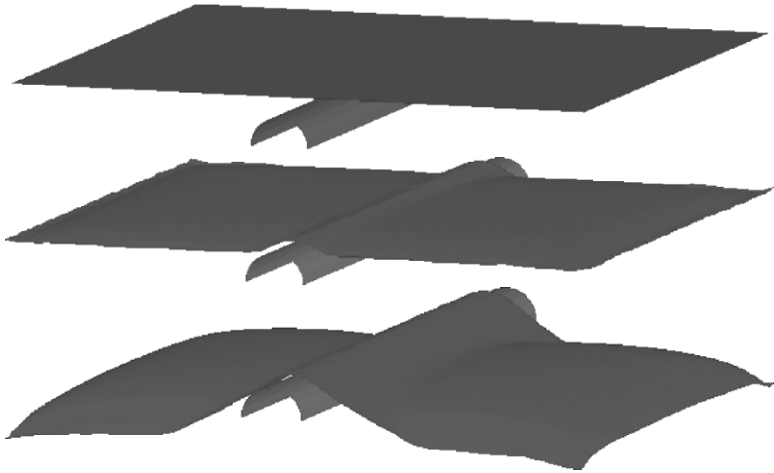


Fig. 10. A cloth piece falling over a rigid rod, at $t = 0, 0.1$ and 0.2 s.



Fig. 11. A cloth piece falling over a rigid rod, at $t = 0.3, 0.4$ and 0.5 s.



Fig. 12. A cloth piece falling over a rigid rod, at $t = 0.6, 0.7$ and 0.8 s.

coming the mesh update challenges involved in FSI problems with contact. Our overview included test computations for the mesh moving techniques we developed, as well as test computations for FSI applications with light structures. We believe that in this article we demonstrated that the core method and its enhancements are significantly increasing the scope and accuracy of FSI computations.

Acknowledgments

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Extending the Range and Applicability of the Loose Coupling Approach for FSI Simulations

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Abstract. Several algorithms for fluid-structure interaction are described. All of them are useful for the loose coupling of fluid and structural dynamics codes. The first class of algorithms considers the loose coupling of implicit time-marching codes. Of these, a predictor-corrector algorithm that may be interpreted as a Jacobi iteration with block-diagonal terms was found to be a good compromise of simplicity, generality and speed. The second class of algorithms treats the displacement of the surface of the structure that is in contact with the fluid. It is shown that a straightforward treatment of the displacements for arbitrary choice of timesteps can lead to instabilities. For optimal stability, at each timestep the ending time of the fluid should be just beyond the ending time of the structure. The third class of algorithms treats the movement of the flow mesh in an ALE setting. The use of a projective prediction of mesh velocities, as well as linelet preconditioning for the resulting PCG system can reduce significantly the effort required. Examples are included that show the effectiveness of the proposed procedures.

1 Introduction

Over the last two decades, the trend in each of the individual disciplines that are required in order to predict the behaviour of processes or products - fluid dynamics, structural mechanics, combustion, heat transfer, control, acoustics, electromagnetics, etc. - has followed the typical bottom-up direction. Starting from sufficiently simple geometries and equations to have an impact and be identified as ‘computational’, more and more realism was added at the geometrical and physics level. While the engineering process (Fig. 1) follows the line: project, objectives, optimization goals, discipline, problem definition, gridding, solution of the PDE and evaluation, the developments (in particular of software) in the Computational Sciences tend to run in the opposite direction: solvers, mesh generators, pre-processors, multi-disciplinary links, complete database.

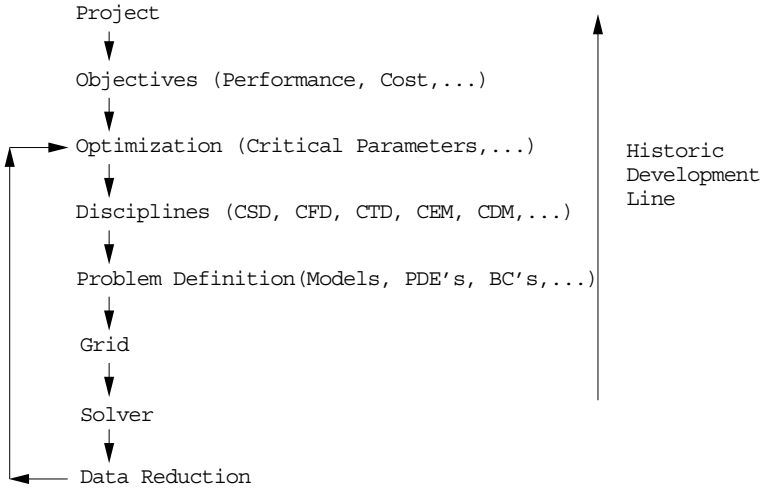


Fig. 1. Design and Analysis Process in Engineering

With the advancement of numerical techniques and the advent first of affordable 3-D graphics workstations and scalable compute servers, and more recently of personal computers with sufficiently large memory and 3-D graphics cards, public domain and commercial software for each of the ‘computational core disciplines’ has matured rapidly and received wide acceptance in the design and analysis process. Most of these packages [7] are now at the threshold mesh generator: pre-processor. This has prompted the development of the next logical step: multi-disciplinary links of codes, a trend that is clearly documented by the growing number of publications and software releases in this area.

The desire to solve multidisciplinary problems can not only be explained by the maturity of discipline codes. Indeed, the biggest payoffs expected from multidisciplinary analysis are increased **insight** into complex phenomena and industrial processes, as well as a more comprehensive **optimization** of products and processes.

The space of possible approximations for coupled fluid, structure and thermal analysis is shown in Fig. 2. Note that for each discipline, different levels of physical approximations and realism are possible. The CPU cost, as well as model preparation times, can vary by orders of magnitude as one moves away from the origin. Application areas associated with particular locations in this fluid- structure- thermal- analysis space have been compiled in Fig. 3. These only represent the better known application classes of what is a fast-growing range of possibilities.

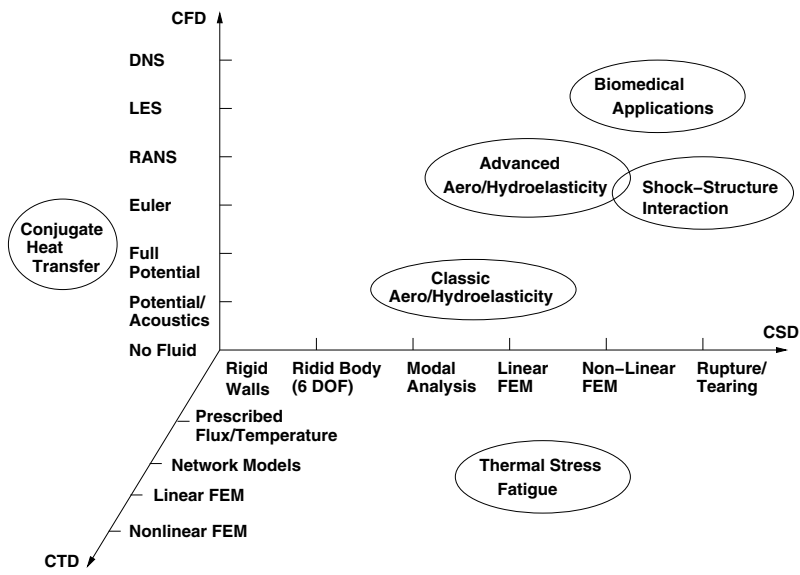


Fig. 2. CFD/CSD/CTD Space

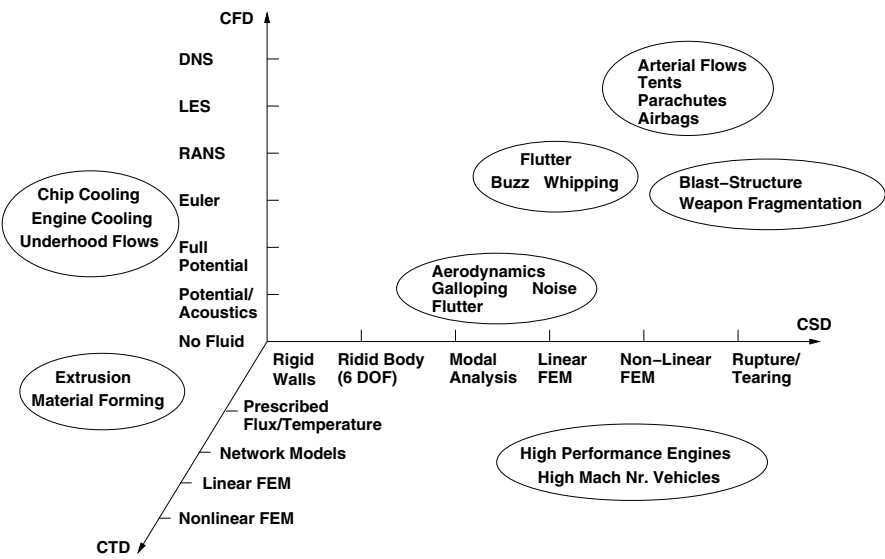


Fig. 3. CFD/CSD/CTD Application Areas

The present paper first reviews the requirements for general FSI capabilities. The loose coupling approach appears as a good candidate for such a problem class. The attention then turns to algorithms that extend the range and applicability of the loose coupling approach.

The first class of algorithms considers the loose coupling of implicit time-marching codes. Of these, a predictor-corrector algorithm that may be interpreted as a Jacobi iteration with block-diagonal terms was found to be a good compromise of simplicity, generality and speed.

The second class of algorithms treats the displacement of the surface of a structure that is in contact with the fluid. It is shown that a straightforward treatment of the displacements for arbitrary choice of timesteps can lead to instabilities. For optimal stability, at each timestep the ending time of the fluid should be just beyond the ending time of the structure.

The third class of algorithms treats the movement of the flow mesh in an ALE setting. The use of a projective prediction of mesh velocities, as well as linelet preconditioning for the resulting PCG system can reduce significantly the effort required.

The fourth class of algorithms treats the detection and tracking of surfaces embedded in an Eulerian fluid mesh. This technique has shown great promise for problems with severe contact, as well as cracking, rupture and topology change. Examples are included that show the effectiveness of the proposed procedures.

2 Basic Requirements

Considering the fluid-structure-thermal interaction problem as an example, we see from the list of possibilities displayed in Figs. 2,3 that any multidisciplinary capability must satisfy the following requirements:

- a) Optimal Methods/Grids: As long as CPU requirements matter (and indications are this will be the case for decades to come), each discipline will attempt to obtain the required results with the least amount of effort. This implies the use of an adequate enough physical/mathematical model, as well as a mesh that is optimally suited for the problem at hand. For the FSI problem, this calls for the possibility to exchange data across different grids and even different abstractions (e.g. 3-D fluid surface to 3-D solid beam).
- b) Modularity of Codes: Codes and solution methods evolve in time. What seemed impossibly expensive can suddenly be common practice. New codes appear, some disappear. Moreover, in each of the disciplines mature ‘legacy codes’ exist. Some of them are from commercial providers, others are open source, others associated with institutions. All of them encompass hundreds of man-years of coding, debugging and benchmarking. Therefore, a general FSI capability must be able to exchange different CFD/CSD/CTD codes in a simple way, allowing the unhindered development of the individual sub-discipline codes.

c) Extendability: After solving a typical FSI problem, in many cases the analyst will be asked to take the next step: perform FSI optimization, increase the realism of physics into the molecular/atomic level, link the FSI problem to a control loop, or add further options into the codes used. Therefore, a general FSI capability must be easily extendable.

d) Fast Multidisciplinary Problem Definition: FSI analysis will only become routine if the setup can be simplified to such an extent that solving a multidisciplinary problem requires a similar input of man-hours as the most expensive subdiscipline problem. Experience in the area of shock-structure interaction calculations indicates that the structure is, in the overwhelming majority of cases, by far the most demanding subdiscipline for problem setup. The demand in this case would be that once the structural problem is ready, setting up the coupled fluid-structure problem should only add a small fraction of extra man-hours.

e) Insightful Visualization: Increased insight and understanding of multidisciplinary problems can only occur if the analyst has the ability to see all relevant phenomena concurrently. This implies that for FSI applications the post-processing tools must allow the visualization of all relevant sub-discipline quantities.

3 Coupling Schemes

The question of how to couple CSD and CFD codes has been treated extensively in the literature [27,14,12,15,4,5,17,2], [10,28,11]. Two main approaches have been pursued to date: strong coupling and loose coupling. The strong (or tight) coupling technique solves the discrete system of coupled, nonlinear equations resulting from the CFD, CSD, CTD and interface conditions **in a single step**. For an extreme example of the tight coupling approach, where even the discretization on the surfaces was forced to be the same, see Thornton [27] and Huebner et al. [10,28,11]. At each timestep, the resulting matrix system is of the form:

$$\begin{bmatrix} \mathbf{K}_{ss} & \mathbf{K}_{sf} & \mathbf{K}_{st} \\ \mathbf{K}_{fs} & \mathbf{K}_{ff} & \mathbf{K}_{ft} \\ \mathbf{K}_{ts} & \mathbf{K}_{tf} & \mathbf{K}_{tt} \end{bmatrix} \cdot \begin{pmatrix} \Delta \mathbf{u}_s \\ \Delta \mathbf{u}_f \\ \Delta \mathbf{u}_t \end{pmatrix} = \begin{pmatrix} \mathbf{r}_s \\ \mathbf{r}_f \\ \mathbf{r}_t \end{pmatrix}, \quad (1)$$

where the sub-indexes s, f, t stand for structure, fluid and thermal fields, $\mathbf{u}$ are the unknowns, $\mathbf{r}$ the right-hand sides (sum of internal and external forces/fluxes), the diagonal sub-matrices are the ones usually obtained for each sub-discipline, and the off-diagonal sub-matrices represent the coupling between disciplines. If we focus on the fluid-structure interaction problem, this reduces to:

One may also interpret this predictor-corrector scheme as a Jacobi iteration. If we consider the linearized implicit scheme for the complete fluid-

structure system:

$$\begin{bmatrix} \mathbf{K}_{ss} & \mathbf{K}_{sf} \\ \mathbf{K}_{fs} & \mathbf{K}_{ff} \end{bmatrix} \cdot \begin{pmatrix} \Delta \mathbf{u}_s \\ \Delta \mathbf{u}_f \end{pmatrix} = \begin{pmatrix} \mathbf{r}_s \\ \mathbf{r}_f \end{pmatrix}, \quad (2)$$

a Jacobi iteration can be written as:

$$\mathbf{K}_{ss} \Delta \mathbf{u}_s^i = \mathbf{r}_s - \mathbf{K}_{sf} \Delta \mathbf{u}_f^{i-1}, \quad (3a)$$

$$\mathbf{K}_{ff} \Delta \mathbf{u}_f^i = \mathbf{r}_f - \mathbf{K}_{fs} \Delta \mathbf{u}_s^{i-1}. \quad (3b)$$

The steps taken in each iteration may also be interpreted as follows:

- Obtain loads from fluid and apply to structure ($\mathbf{K}_{sf} \Delta \mathbf{u}_f^{i-1}$);
- Obtain new displacements ($\Delta \mathbf{u}_s^i$);
- Obtain mesh velocities for the fluid boundary from the structure ($\mathbf{K}_{fs} \Delta \mathbf{u}_s^{i-1}$); and
- Obtain new flow variables ($\Delta \mathbf{u}_f^i$).

This interpretation is not exact, as the mesh motion of the flow solver, and the displacement field of the structure are linked beyond nearest neighbours in $\mathbf{K}_{sf}, \mathbf{K}_{fs}$. The interpretation would only be exact for explicit timestepping schemes. However, it is useful in deriving the so-called loose coupling technique, which solves the complete FSI system given by Eqn. (2) by using an iterative strategy of repeated ‘CFD solution followed by CTD solution followed by CSD solution’ until convergence is achieved (see Fig. 4). In this case, the coupling matrices in Eqns. (3a,b) contain only the direct load and displacement transfer terms.

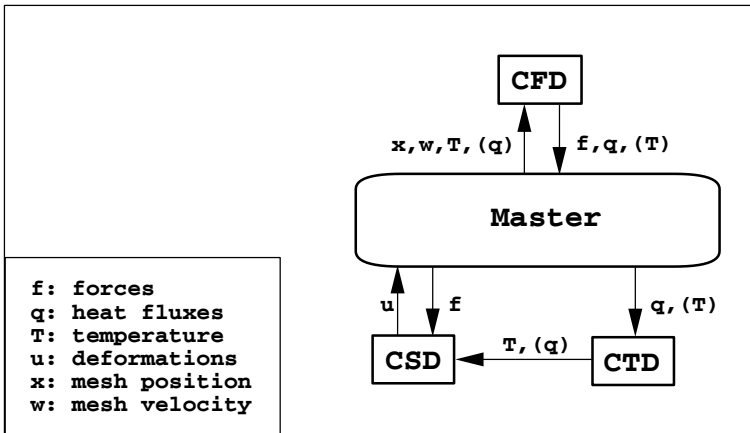


Fig. 4. Loose Coupling for Fluid/Structure/Thermal Simulations

Special cases of the loose coupling approach include the direct coupling in time of explicit CFD and CSD codes and the incremental load approach of steady aero- and hydro-elasticity. The variables on the boundaries are transferred back and forth between the different codes by a master code that directs the multi-disciplinary run. Each code (CFD, CSD, CTD, ..) is seen as a subroutine, or object, that is called by the master code, or as a series of processes that communicate via message passing. This implies that the transfer of geometrical and physical information is performed between the different codes without affecting their efficiency, layout, basic functionality, and coding styles. At the same time, different CSD, CTD or CFD codes may be replaced, making this a very modular approach. This allows for a straightforward re-use of existing codes and the choice of the ‘best model’ for a given application. The information transfer software may be developed, to a large extent, independently from the CSD, CTD and CFD codes involved, again leading to modularity and software reuse. For this reason, this approach is favoured for industrialization. Indeed, considerable effort has been devoted to develop general, scalable information transfer libraries [15,23,3,6,9].

4 Coupling of Implicit Time-Marching Codes

The loose coupling of explicit CFD and CSD codes has been used extensively and with considerable success over the last decade [15,1,4,5,17,2,21]. For some classes of problems, explicit time-marching is required to capture all transient phenomena. On the other hand, if the relevant physical phenomena are not linked to the highest eigenmodes of the system, explicit time-marching can be a prohibitively expensive proposition. For such problems, which include low-frequency aeroelasticity, subsonic parachute unfolding, haemodynamics with responding vessel walls and windshield defrosting, implicit time-marching is required. The question is then whether the loose coupling approach can be extended to solve multidisciplinary problems with codes that employ implicit time-marching schemes.

An under-relaxed predictor-corrector scheme was found to offer a good compromise between simplicity and stability. The scheme is illustrated for the case of fluid- structure interaction. Denoting by i the iteration step, α the under-relaxation factor, $\mathbf{x}_s$ the position of the surface of the structure wetted by the fluid, σ_f the stresses exerted by the fluid on the structure, $f(\sigma_f)$ the surface deformation due to fluid loads and $g(x_s)$ the change of fluid stresses due to surface deformation, the predictor-corrector scheme for each timestep takes the following form:

```
while: not converged:
  update structure with fluid loads:
     $x_s^i = (1 - \alpha)x_s^{i-1} + \alpha f(\sigma_f^i)$ 
  update fluid with structure position/velocity:
     $\sigma_f^i = (1 - \alpha)\sigma_f^{i-1} + \alpha g(x_s^i)$ 
endwhile
```

Typical under-relaxation factors are in the range $0.5 \leq \alpha \leq 0.9$. Note that the flow of information is the same as in the case of explicit/explicit code coupling. Current research in this area is focused on convergence criteria and acceleration techniques.

5 Calculation of Displacements

A loose coupling between the CFD and CSD codes implies that the ending times of the respective codes can be different. The fluid surface is imposed by the CSD surface, which is moving according to fluid forces. If the time of the CFD code t_f at the end of the CFD timestep lies beyond the time of the CSD code t_s , the CSD surface is extrapolated. This can be done in a variety of ways (linear [4], cubic [5], etc.), but in all probability the assumed position of the CSD surface and the one calculated by the CSD code in the next timestep will not coincide. This implies that some form of correction will be required at the beginning of the next CFD timestep. An exaggerated situation where this can happen has been sketched in Fig. 5. Note the jump in surface positions at the beginning of the next CFD timestep. Experience with coupled explicit codes indicates that for isotropic (Euler) grids, this jump does not affect the results. However, for highly stretched grids that resolve boundary layers in the fluid (RANS grids), a jump in the surface can easily lead to elements with negative volumes and a breakdown of the simulation.

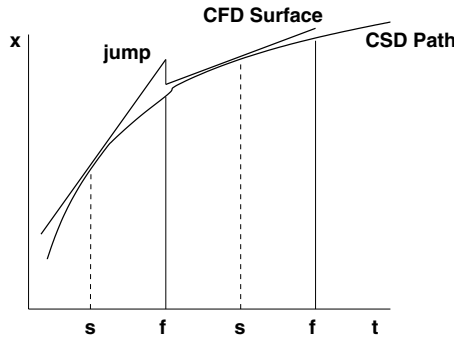


Fig. 5. Loose Coupling: Surface Matching

An alternative is to ignore the new surface velocities at the end of the CSD timestep, and to compute velocities directly from positions. This can be easily done if the surface mesh of the CSD domain remains intact (e.g. no changes in topology).

The surface velocity of the CFD domain is then continued from the previous position in such a way that the CSD surface seen by the CFD code

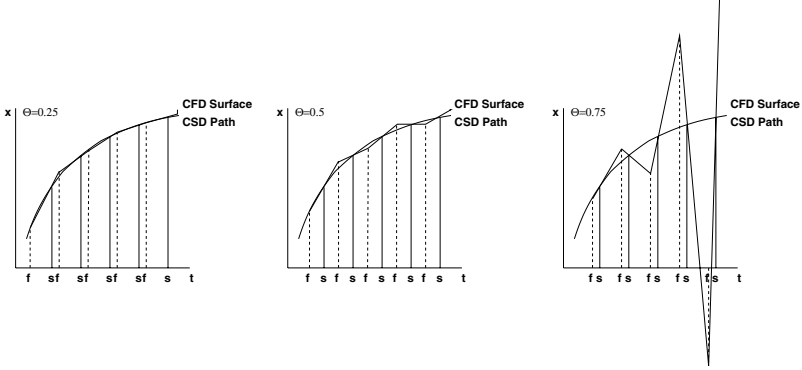


Fig. 6. Loose Coupling: Mesh Velocity Instability

coincides exactly with the position of the CSD surface seen by the CSD code at $t = t_s$. Although intuitively simple, the method is unstable if the ending time of the fluid is larger than the ending time of the structure by half the timestep taken. This behaviour is shown in Fig. 6. For optimal stability, the ending time of the fluid should be just beyond the ending time of the structure: $t_f = t_s + \epsilon$.

6 Projective Prediction of Mesh Velocities

The movement of the CFD mesh as the surfaces of bodies immersed or surrounding the flowfield move or deform can have a profound effect on overall code performance. A bad choice of mesh movement leads to frequent remeshings with all associated negative effects (artificial viscosity due to loss of information during reinterpolation, degradation of performance on parallel machines, etc.). Many good mesh movement techniques have been proposed [16,24,26]. We have used a nonlinear Laplacian-based technique [16,19] that solves the system:

$$\nabla k \nabla \mathbf{w} = 0 \quad , \quad \mathbf{w}_\Gamma = \mathbf{w}_b \quad , \quad (3)$$

where $\mathbf{w}$ denotes the velocity of the mesh, $\mathbf{w}_b$ the (imposed) velocity at the surface of the bodies immersed or surrounding the flowfield, and k is a diffusivity that depends on the distance from the body. The discretization of Eqn. (3) via finite elements results in a discrete system of the form:

$$\mathbf{K} \cdot \mathbf{w} = \mathbf{r} \quad . \quad (4)$$

The solution of this system is performed using a preconditioned conjugate gradient solver. Diagonal preconditioning is employed for isotropic grids, while for RANS grids linelet preconditioning [25] is preferred.

In what follows, the **basic assumption** is that $\mathbf{K}$ does not change significantly in time. For many situations this is indeed the case. The mesh does not move significantly between timesteps, and the distance from the bodies for individual gridpoints does not change considerably either.

If we denote by $\mathbf{w}^i, \mathbf{r}^i, i = 1, l$ the values of the mesh velocity and right-hand sides at previous timesteps $n - i$, we know that:

$$\mathbf{K} \cdot \mathbf{w}^i = \mathbf{r}^i . \quad (5)$$

Given the new right-hand side $\mathbf{r}$, we can perform a least-squares approximation to it in the basis $\mathbf{r}^i, i = 1, l$:

$$(\mathbf{r} - \alpha_i \mathbf{r}^i)^2 \rightarrow \min , \quad (6)$$

which results in

$$\mathbf{A}\boldsymbol{\alpha} = \mathbf{s} , \quad A^{ij} = \mathbf{r}^i \cdot \mathbf{r}^j , \quad s^i = \mathbf{r}^i \cdot \mathbf{r} . \quad (7)$$

Having solved for the approximation coefficients α_i , we can estimate the start value $\mathbf{w}$ from:

$$\mathbf{w} = \alpha_i \mathbf{w}^i . \quad (8)$$

We remark that, in principle, the use of the right-hand sides $\mathbf{r}^i, i = 1, l$ as a basis may be numerically dangerous. After all, if any of these vectors are parallel, the matrix $\mathbf{A}$ is singular. One could perform a Gram-Schmidt orthogonalization instead. This option was invoked by Fischer [8] who looked at a number of possible schemes to accelerate the convergence of iterative solvers using successive right-hand sides within the context of incompressible flow solvers based on spectral elements. However, we have not found this to be a problem for any of the cases tried to date. The advantage of using simply the original right-hand sides is that the update of the basis is straightforward. We keep an index for the last entry in the basis, and simply insert the new entries at the end of the timestep in the position of the oldest basis vector. The storage requirements for this projective predictor scheme are rather modest: $2 \cdot \text{ndim} \cdot \text{npoin} \cdot \text{nvec}$. We typically use 1-4 basis vectors, i.e. the storage is at most $24 \cdot \text{npoin}$.

The effect of using even 2 search directions is dramatic: for the cylinder and bridge-cases shown below, the number of mesh velocity iterations per timestep drops from $O(20)$ to $O(2)$.

7 Examples

We include two examples that demonstrate the proposed coupling strategy for implicit time-marching schemes.

7.1 Tacoma Narrows Bridge Section

The failure of the Tacoma Narrows Bridge in 1940 may be considered as the classic case of aeroelastic coupling for large civil engineering structures. In this first case, we consider a section of the bridge. The main dimensions of the bridge section (in meters), which is taken to be $dz = 0.5\text{ m}$ are shown in Fig. 7(a). The incompressible flow equations are solved in an ALE frame of reference. The solver is implicit in the pressure and viscous terms, and explicit in the advection terms, which are advanced using a 5-stage Runge-Kutta scheme [22] with a Courant-number of $C = 1.0$.

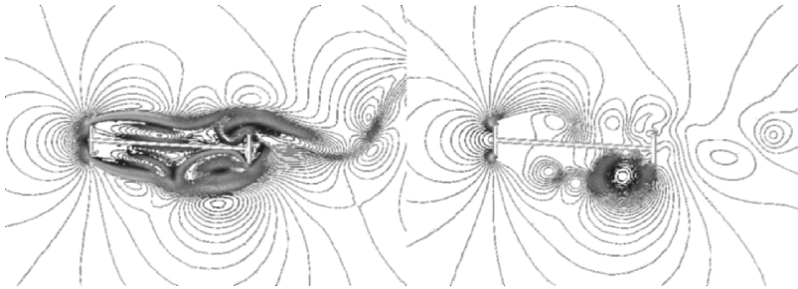
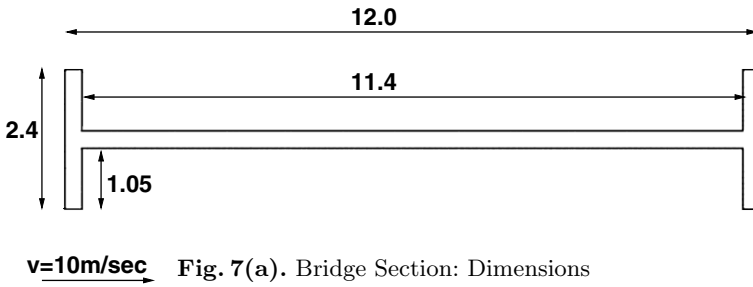


Fig. 7(b). Bridge Section: Typical Flowfield (Abs(vel), Pressure)

The flow mesh has approximately 381K elements. The structure is computed with an implicit eigenmode integrator that uses the second-order Newmark scheme [29]. In the present case, only two eigenmodes are considered: heave and torsion [10]. The incoming flow conditions are set to $\rho = 1.25\text{ kg/m}^3$, $\mathbf{v}_\infty = (10.0, 0.0, 0.0)\text{ m/sec}$, $\mu = 0.1\text{ kg/m/sec}$. The coupled system is advanced in time using the explicit loose coupling strategy, but making sure that for each timestep the ending time of the fluid is just beyond the ending time of the structure ($t_f = t_s + \epsilon$). Figure 7(b) shows a typical result after the structure has started to vibrate. Note the low pressure zone below

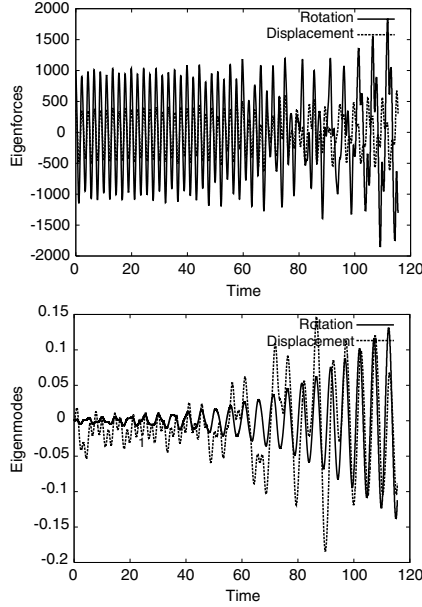


Fig. 7(c,d). Bridge Section: Eigenforces and Eigenmodes

the aft-zone of the bridge at the core of the large vortex. This portion of the bridge is actually moving down, i.e. the flow is pumping energy into the structure. The time-histories of the eigenforces and eigenmodes can be seen in Figs. 7(c,d). Note the onset of the torsional instability after approximately 2 minutes of real time. Even though the mesh was moved through many cycles, and torsion was considerable, no remeshing was required.

7.2 Tacoma Narrows Bridge

This example extends the previous one to a full 3-D situation. The eigenmodes are approximated by a parabolic profile along the span, assuming that one end is clamped. Due to symmetry, only half of the bridge (i.e. 50 m) is computed. The solvers and coupling strategy used are the same as in the last example. However, in this case the flow mesh had approximately 8.5M elements. Figures 8(a,b) show a typical result after the structure has started to vibrate. Note the difference in the vortex structure for the planes where the bridge is clamped (left) and free to vibrate (right). The time-histories of the eigenforces and eigenmodes can be seen in Figs. 8(c,d). Note the onset of the torsional instability after approximately 4 minutes of real time. Even though the mesh was moved through many cycles, and torsion was considerable, no remeshing was required. Figure 8(e) depicts the exchange of work between the fluid and the structure. Note that with the onset of the instability, the

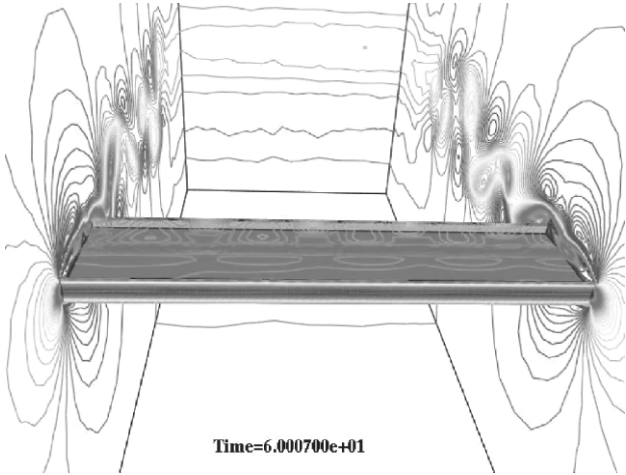


Fig. 8(a). Bridge: Typical Flowfield

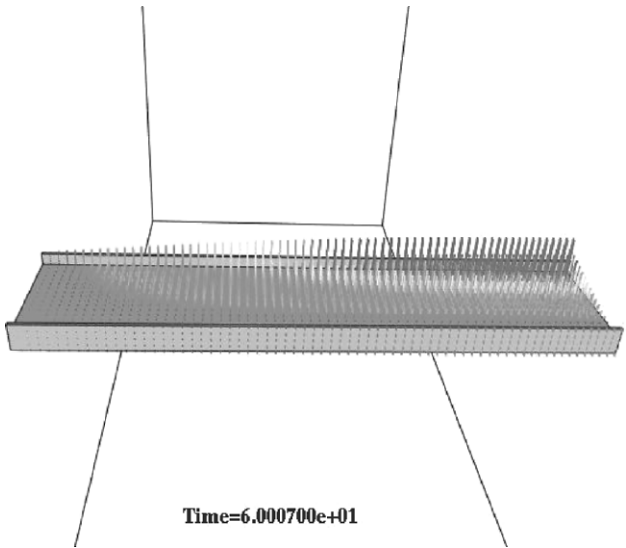


Fig. 8(b). Velocity of Structure

work exchange is no longer balanced, but shifts to a net flow of work (energy) from the fluid to the structure.

7.3 Ship in Heavy Sea State

This example shows the use of the loose coupling methodology to predict the effects of large waves on ships. The problem definition is given in Fig. 9(a). The ship is a generic LNG tanker that is allowed to move rigidly and elasti-

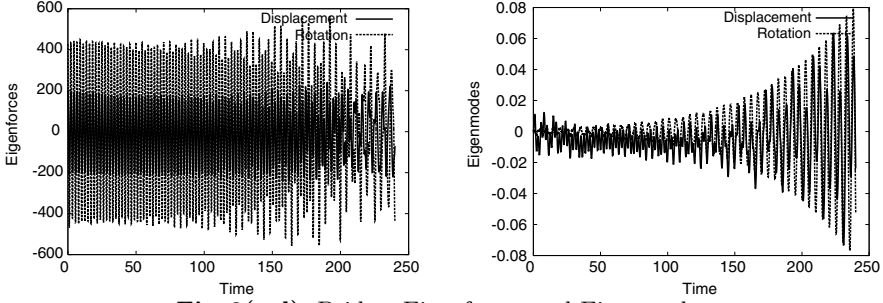


Fig. 8(c,d). Bridge: Eigenforces and Eigenmodes

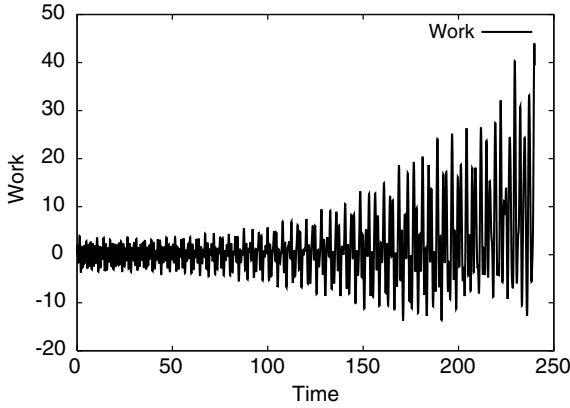


Fig. 8(e). Bridge: Work Exchanged Between Fluid and Solid

cally in the y -direction, and to rotate rigidly in the z direction. The three main bending eigenmodes were computed using the ABAQUS CSD code. These modes, together with their stiffnesses and masses were then used as input to FEEIGEN, an implicit eigenmode integrator. The waves are generated by moving the left wall of the domain. A large element size was specified at the far end of the domain in order to dampen the waves. The mesh at the ‘wave-maker plane’ is moved using a sinusoidal excitation. The surface nodes of the ship move according to the 6 DOF integration of the rigid body motion equations, as well as the eigenmodes. Approximately 30 layers of elements close to the ‘wave-maker plane’ and the ship are moved, and the Navier-Stokes/VOF equations are integrated using the arbitrary Lagrangean- Eulerian frame of reference. The mesh had approximately 1.25M elements, and the integration to 3 minutes of real time took 2.5 hours on a Dell laptop PC (Intel P4 CPU, 2Gbyte RAM, Suse Linux OS, Intel compiler). Figure 9(b) shows the evolution of the flowfield. Note the change in position for the ship, as well as the roll. The position of the center of gravity may be discerned from Fig. 9(c), and the evolution of the first eigenmode from Fig. 9(d). Note the

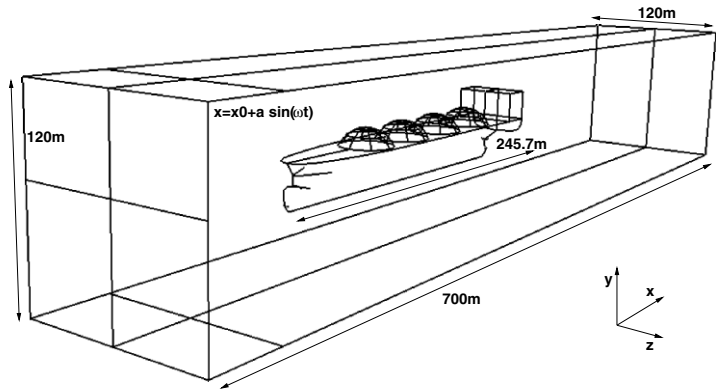


Fig. 9(a). Ship in Heavy Sea State: Problem Definition

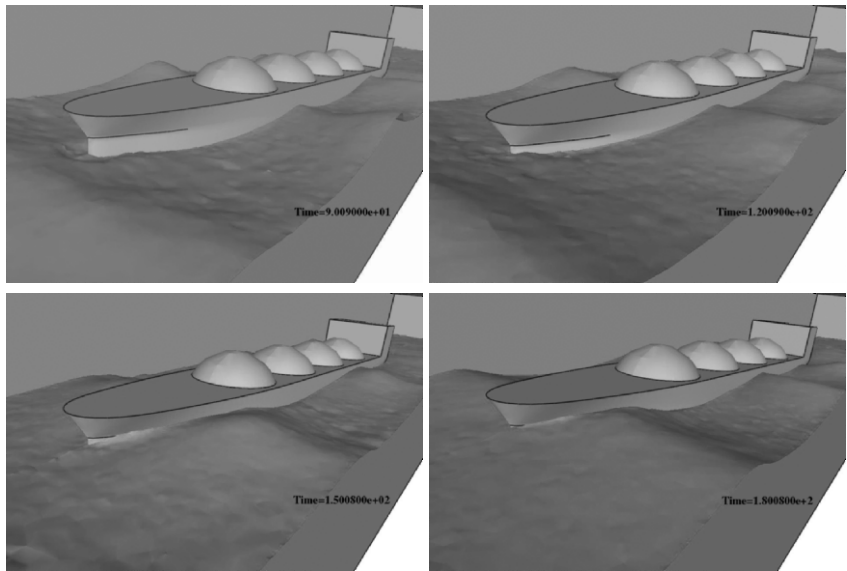


Fig. 9(b). Evolution of the Free Surface/Ship Motion

initial ‘surfacing’ of the ship due to the imbalance of weight and floating forces, as well as the eigenmode values, which lead to approximately 50 cm of whipping at the stern of the ship.

7.4 Heated Cylinder

This classic case considers a cylinder of unit diameter that is heated uniformly with a source. The surrounding flow is cold, and transports the heat away. The geometry is shown in Fig. 10(a). A planar cut through the plane $z = 0$ shows that the grids employed for the flow domain and the solid domain are

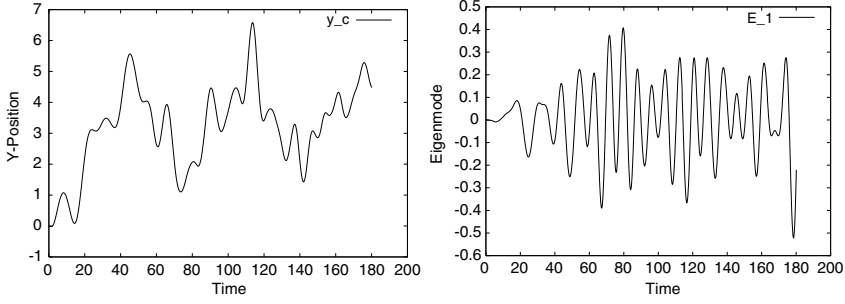


Fig. 9(c,d). Evolution of Center of Gravity and 1st Eigenmode

very different indeed (Fig. 10(b)). The flow domain requires a mesh suited for the boundary layer, with highly stretched elements. The solid domain is discretized with isotropic elements. The material data were set as follows:

- Fluid: $\rho = c_p = 1.0, \mathbf{v} = (1.0, 0.0, 0.0), \mu = k = 0.054$
- Solid: $\rho = c_p = k = 1.0, s = 1.0$

The velocity and temperature fields in the same plane are shown in Figs. 10(c,d) for a given time. Note the transport of heat into the fluid. The under-relaxation factor for this case was set to $\alpha = 0.2$ and 5 iterations between the solid and fluid were required per timestep.

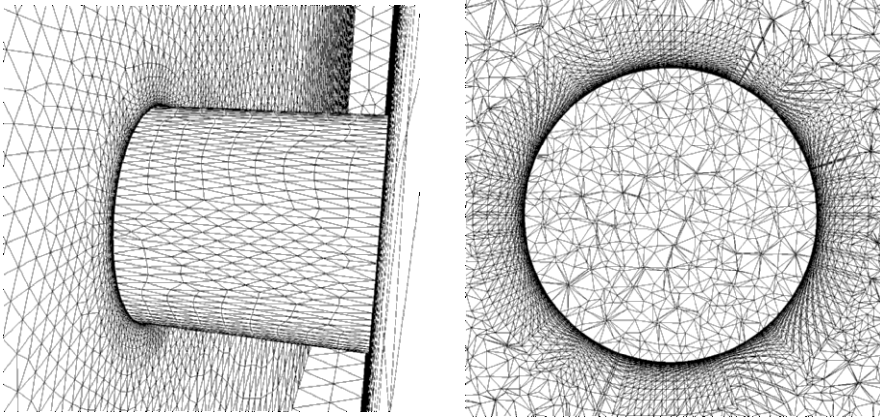


Fig. 10(a,b). Heated Cylinder: Fluid Domain and Plane $z = 0.0$

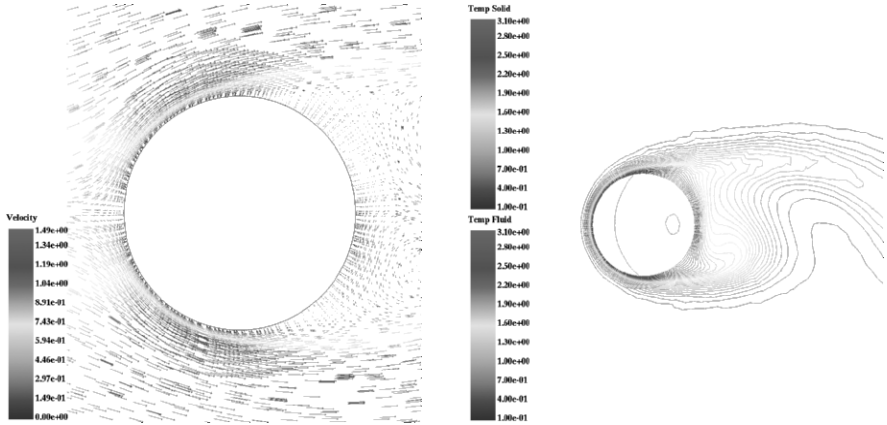


Fig. 10(c,d). Heated Cylinder: Velocity and Temperature ($z = 0.0$)

8 Conclusions and Outlook

Several algorithms for fluid-structure interaction have been described. All of them are useful for the loose coupling of fluid and structural dynamics codes.

The first class of algorithms considers the loose coupling of implicit time-marching codes. Of these, a predictor-corrector algorithm that may be interpreted as a Jacobi iteration with block-diagonal terms was found to be a good compromise of simplicity, generality and speed.

The second class of algorithms treats the displacement of the surface of the structure that is in contact with the fluid. It is the motion of this surface that determines the movement of the fluid domain. Given that CSD codes return positions and velocities, the question is how to treat these. It is shown that a naive treatment of the displacements for arbitrary choice of timesteps can lead to instabilities. For optimal stability, at each timestep the ending time of the fluid should be just beyond the ending time of the structure.

The third class of algorithms treats the movement of the flow mesh in an ALE setting. The use of a projective prediction of mesh velocities, as well as linelet preconditioning for the resulting PCG system can reduce significantly the effort required.

Examples are included that show the effectiveness of the proposed procedures.

9 Acknowledgements

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A New Fluid Structure Coupling Method for Airbag OOP

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Abstract. The deployment of an airbag is most fatal and dangerous to a passenger when they are in an out of position (OOP) situation, with the airbag making contact before it is fully inflated. This can lead to severe, if not life threatening, injuries to the passenger. This situation is more commonly associated with small females and children who are positioned near to the airbag module, i.e. in an OOP load cases. The aim of this research is to assess the response of a Hybrid III 5th Percentile female anthropomorphic dummy positioned in a FMVSS 208 low risk static airbag deployment OOP load cases using a transient dynamic finite element program called LS-DYNA. The simulation considers the standard procedures utilised in the LS-DYNA, where assumptions such as uniform airbag pressure and temperature are made, along with a more recently developed procedure that takes into account the fluid-structure interaction between the inflating gas source and the airbag fabric, referred to as Arbitrary Lagrangian Eulerian (ALE) theory. Both simulations were compared to test data received by Jaguar, indicating satisfactory results in terms of correlation, with the more recently developed procedure, ALE theory, showing the greatest accuracy, both in terms of graphical and schematic comparison, especially in the very early stages of the inflation process. As a result, the new simulation procedure model was utilised to research into the effects of changing the designs of the airbag module.

1 Nomenclature

M	Capsule Mass	div	Divergence
F	Vector field	dS	Small surface area of the control volume
dV	Small volume of control volume	k	C_p/C_v
C_p	Constant pressure specific heat of gas	C_v	Constant volume specific heat of gas
e	specific internal energy	v	airbag volume
P	Pressure applied to the airbag	ρ	density of the gas
P_2	Airbag Pressure	V_2	Airbag Volume
m_2	Airbag Mass	R	Gas Constant
T_2	Airbag Temperature	$u\Delta t$	Transport volume.f
OOP	Out of Position	FMVSS	Federal Motor Vehicle Safety Standards

2 Control Volume Method

This is the standard simulation method of deploying an airbag in LS-DYNA, based upon the works of Wang and Nefske [1,2]. It is particularly appropriate for intricate, highly nonlinear systems with deploying airbags that have complex contact conditions from the initial folded position. The CVM assumes that is no heat transfer into the control volume (adiabatic), the inflating gas is assumed to behave as an ideal gas with constant specific heats, and also, that the temperature and pressure are uniform within the control volume (lumped parameter approach).

At any time step the volume of the airbag is calculated utilising Gauss' Theorem, which related the surface area of the airbag to its volume.

$$\int_V div F dV = \int_S F dS$$

This calculation is possible since the airbag material is made up of shell or membrane elements, which exact position, orientation and current surface area is calculated and stored at each time step.

Since the volume at any time step is known, using the specific internal energy evolution equation:

$$e_2 = e_1 \left(\frac{v_2}{v_1} \right)^{(1-k)}$$

The internal energy at the next time step can be calculated. In a similar way, the density at the next time step can be calculated since it is related to the volume. With this information, it possible to calculate the pressure at the next time step using the gamma law ideal gas equation:

$$P = \rho(1 - k)e$$

This pressure is then applied to the internal surface of the airbag fabric as a load normal and the whole above process then repeats its self until the simulation time is reached.

The more complex models developed by Wang and Nefske involve taking into account the mass flow in and out of the airbag. With this method, the mass flow rate, $\dot{m}_{in}$, and temperature, T_1 , from the inflator are tabulated as a function of time that are determined by tank tests, whereas the mass flowing out is due to airbag vents, $\dot{m}_{23}$, and airbag porosity, are calculated using equations developed by Wang and Nefske (see Fig. 1).

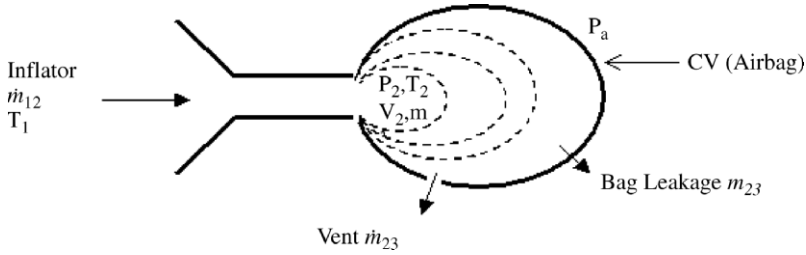


Fig. 1. Control Volume Inflation process

Using the above information, the first law of thermodynamics is applied to the above model, resulting in an equation that expresses the increase in the internal energy

$$E = \frac{d}{dt} (m_2 c_v T_2) = c_p T_1 \dot{m}_{in} - c_p T_2 \dot{m}_{out} - P_2 \frac{dV_2}{dt}.$$

By utilising the equations for the mass flow rate out of the airbag and the conservation of energy equation above, it is possible to solve these equations iteratively by function evaluation and the using the gamma law gas equation of state as well as gauss' theorem to find the volume, the pressure at the next time step can be calculated via a half index-shift method. All other variations of the CVM, from the hybrid model that takes into account gas mixture and models that include jetting points, are all based upon this above procedure.

For IP situations, where the airbag is near full inflation, this type of modelling is accurate enough, with assumptions not affecting the model accuracy. However, due to the assumptions in the model, this type of simulation is not appropriate for OOP load cases, where an accurate modelling of the gas flow is required due to the quick impact with the passenger. It therefore required to build a model that considers the fluid flow and the coupling between the inflating gas source and the airbag fabric, along with the surrounding air.

3 ALE Method

In lagrangian systems [3], mainly used for structural mechanics, all the nodes of the computational mesh follow the movement of the associated material nodes, since the mesh nodes are embedded in the material nodes. This type of modelling allows materials with history-dependent constitutive relations to be treated, the facile tracking of free surface and the interfaces between the different types of material. However, it has a major draw back, since large distortions of the computational domain are not possible without remeshing frequently.

In Eulerian systems, which are most commonly used in fluid dynamic modelling, the computational mesh and associated nodes are fixed in position, but the material nodes are free to move with respect to the eulerian grid. This enables the ability to handle large distortions in the modelling, but at the cost of flow detail resolution and the accurate definition of the interface.

ALE [4,5] is a technique that has been developed to combine the positive aspects of the two above procedures, where the computational nodes of the mesh can move with the material nodes as in lagrangian systems, remain fixed like in eulerian systems, or move arbitrarily of the material nodes. This enables greater distortions to be handled by the model than if it were a lagrangian model, with greater accuracy than if it were a eulerian system (see Fig. 2).

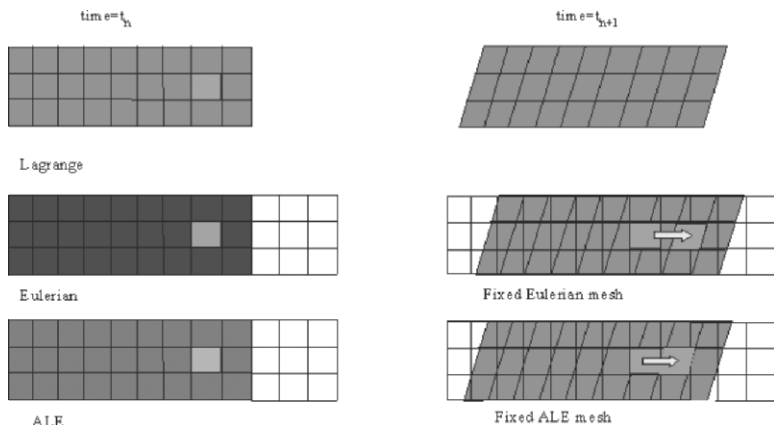


Fig. 2. Lagrangian, Eulerian and ALE Comparison

The general procedure for ALE formulation consists of a Lagrangian time-step being performed, followed by an advection step. This is when the mesh is mapped from a distorted Lagrangian mesh to an undistorted mesh [6].

Advection is the transport of the mass between the elements. For airbag simulation purposes, a second order accurate monotonic advection algorithm

is used based upon the Van Leer procedure. This assumes that the mesh topology is fixed and the Courant number, C , is less than one, which means that the motion of the mesh during a time step is less than the surrounding elements characteristic lengths.

$$C = \frac{u\Delta t}{\Delta x} = \frac{f}{V} \leq 1$$

The factor that links the Lagrangian parts to the Eulerian parts is the coupling method. Here, this enables the interaction between the airbag fabric and the gas source. There are two coupling methods available, the penalty-based method and the constraint-based method. The penalty-based coupling method is the preferred procedure, cause unlike the constraint-based coupling method, energy is conserved, even though there may be certain problems with stability. The diagram below shows the penalty-based coupling method concept.

The method works by calculating a vector force, F , which denotes the inflating gas source flowing into the surrounding air, and so moving through the airbag fabric. This is the penalty-force that is applied to push the inflating gas back into the airbag fabric. It is proportional to the displacement, d , between the Lagrangian node and the location of the fluid material in the Eulerian mesh.

The accuracy of the coupling also depends upon the number of coupling points utilised, besides the element nodes. Additional coupling points can be positioned on the airbag structure, as can be seen in Fig. 3. A compromise has to be made between too many additional coupling points, leading long computational times, or a few if no additional coupling points, resulting in the effect of artificial leakage through the airbag fabric. The leakage is caused by the inflating gas source not connecting with the coupling points and passing through the airbag fabric structure. Another factor is the direction of inflating gas source flow differing from the direction of advection, which results in mass being advected through the airbag fabric (see Fig. 4).

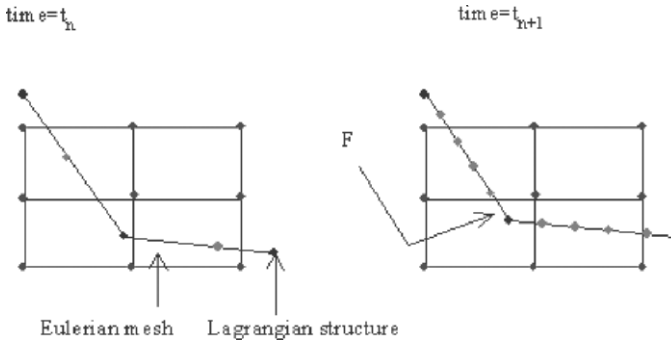


Fig. 3. Penalty-based coupling method

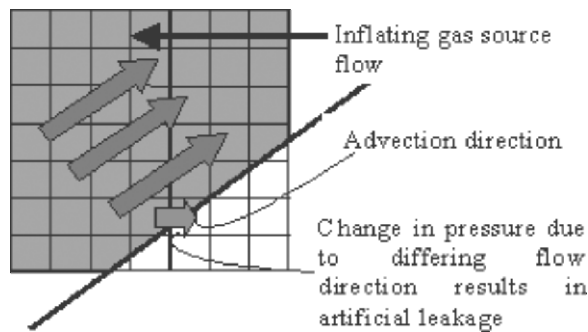


Fig. 4. Airbag leakage

4 Simulation

The simulation example below of the dummy in an OOP load case is for the control volume. This type of modelling situation is utilised by literally all automotive companies and suppliers since the set up of the dummy is in accordance with the FMVSS 208 load cases for OOP dummy airbag deployment analysis for low risk deployment testing.

The graph in Fig. 5 shows the pressure on the airbag during the inflation process. The initial peak is due to the uniform pressure being applied which is an inaccurate depiction of what actually happens, this leads to the volume of the airbag being inflated at a rate much quicker than expected, which can

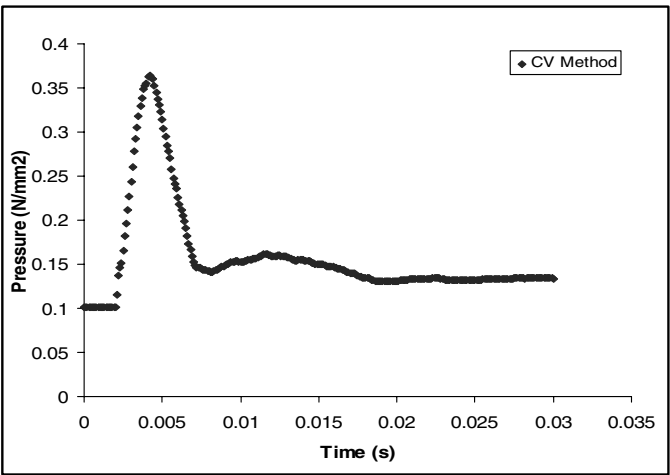


Fig. 5. Airbag Pressure

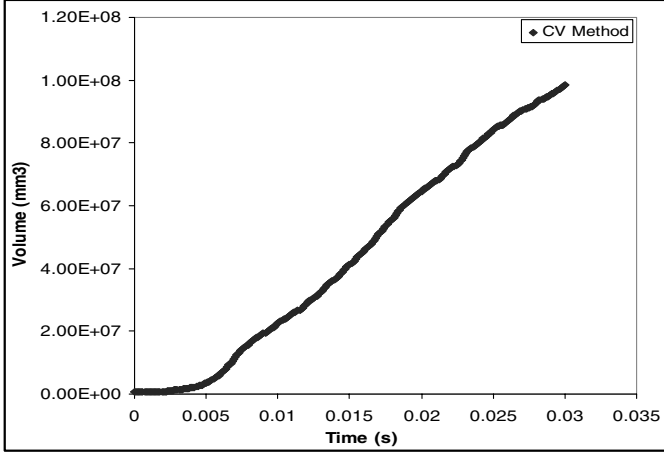


Fig. 6. Airbag Volume

be seen in the graph Fig. 6. Figure 7 gives a kinematic description of the whole simulation process, which lasted 0.3 ms.

5 Summary

An outline of the currently present and newly developed simulation procedures for airbag deployment modelling has been detailed and explained in this paper. The mostly commonly used methodology of airbag simulation is the Control Volume Method, where the airbag has a uniform pressure applied to it internally and several assumptions have been made about the inflating gas source. This has been compared to the latest simulation procedure of Arbitrary Lagrangian Eulerian (ALE) modelling, a fluid-structure method that couples the benefits of Lagrangian and Eulerian theory.

The greatest advantage of ALE is its ability to accurately model the inflating gas source, the effects of the fluid flow and the interaction with structural parts of the model (e.g. airbag fabric). This is especially useful in the early stages of the inflation process when dealing with OOP load cases, since the CV Method is not accurate enough to model for these situations, ALE can be utilised to accurately depict the initial pressure distribution in the airbag.

At present, simulations for the above model are being run utilising the ALE methodology described in this paper and will be published at a later date, comparing the results with the CV Method.

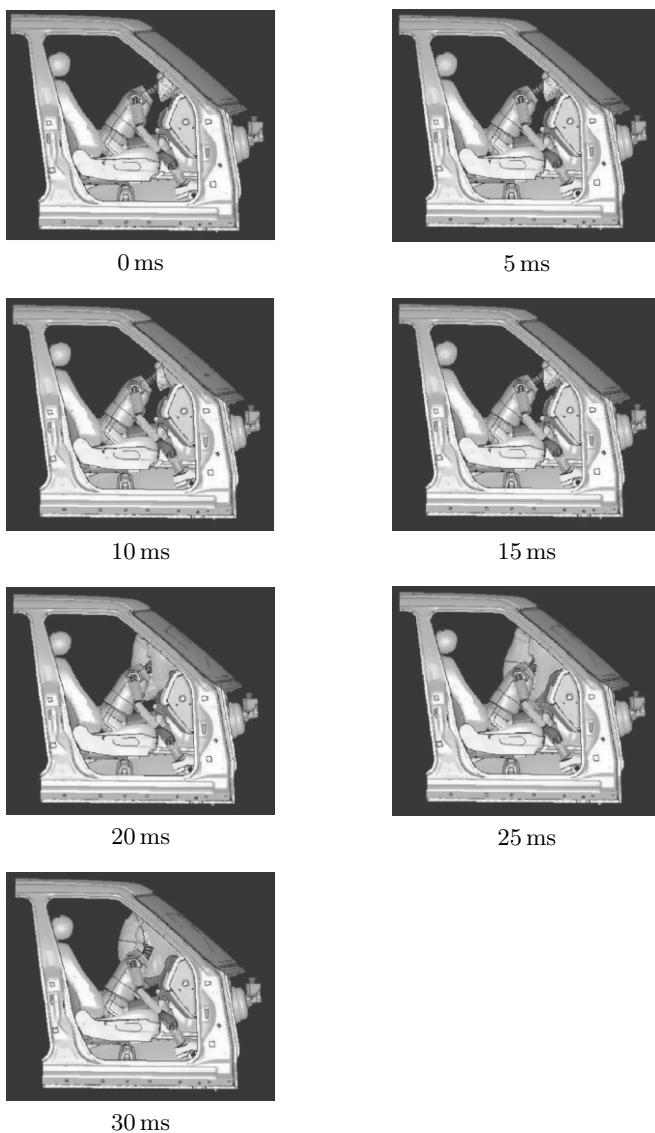


Fig. 7. OOP load case for CV Method

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Adaptive Finite Element Approximation of Fluid-Structure Interaction Based on an Eulerian Variational Formulation

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Abstract. We propose a general variational framework for the adaptive finite element approximation of fluid-structure interaction problems. The modeling is based on an Eulerian description of the (incompressible) fluid as well as the (elastic) structure dynamics. This is achieved by tracking the movement of the initial positions of all ‘material’ points. In this approach the deformation appears as a primary variable in an Eulerian framework. Our approach uses a technique which is similar to the *Level Set* method in so far that it also tracks initial data, in our case the set of *Initial Positions*, and from this determines to which ‘phase’ a point belongs. To avoid the need for reinitialization of the initial position set, we employ the harmonic continuation of the structure velocity field into the fluid domain. Based on this monolithic model of the fluid-structure interaction we apply the *dual weighted residual method* for goal-oriented a posteriori error estimation and mesh adaptation to fluid-structure interaction problems. Several stationary as well as nonstationary examples are presented.

1 Introduction

Computational fluid dynamics and computational structure mechanics are two major areas of numerical simulation of physical systems. With the introduction of high performance computing it has become possible to tackle systems with a coupling of fluid and structure dynamics. General examples of such fluid-structure interaction (FSI) problems are flow transporting elastic particles (particulate flow), flow around elastic structures (airplanes, submarines) and flow in elastic structures (haemodynamics, transport of fluids in closed containers). In all these settings the dilemma in modeling the coupled dynamics is that the fluid model is normally based on an Eulerian perspective in contrast to the usual Lagrangian approach for the solid model. This makes the setup of a common variational description difficult. However, such a variational formulation of FSI is needed as the basis of a consistent approach to residual-based a posteriori error estimation and mesh adaptation as well as to the solution of optimal control problems by the Euler-Lagrange method. This is the subject of the present paper.

Combining the Eulerian and the Lagrangian setting for describing FSI involves conceptional difficulties. On the one hand the fluid domain itself is

time-dependent and depends on the deformation of the structure domain. On the other hand, for the structure the fluid boundary values (velocity and the normal stress) are needed. In both cases values from the one problem are used for the other, which is costly and can lead to a drastic loss of accuracy. A common approach to dealing with this problem is to separate the two models, solve each separately, and so converge iteratively to a solution which satisfies both together with the interface conditions (Figure 1). Solving the separated problems serially multiple times is referred to as a ‘partitioned approach’.

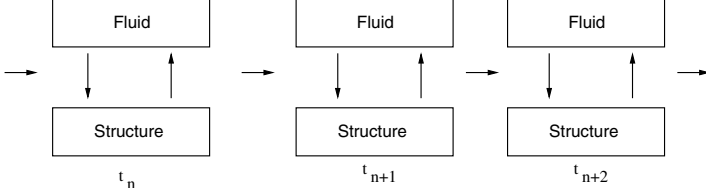


Fig. 1. Partitioned approach, Lagrangian and Eulerian frameworks coupled.

A partitioned approach does not contain a variational equation for the fluid-structure interface. To achieve this, usually an auxiliary unknown coordinate transformation function ζ_f is introduced for the fluid domain. With its help the fluid problem is rewritten as one on the transformed domain which is fixed in time. Then, all computations are done on the fixed reference domain and as part of the computation the auxiliary transformation function ζ_f has to be determined at each time step. Figure 2 illustrates this approach for the driven cavity problem considered in Section 5, below. Such, so-called ‘arbitrary Lagrangian-Eulerian’ (ALE) methods are used in [17,31,16], and corresponding transformed space-time finite element formulations in [27,28].

Both, the partitioned and the transformation approach to overcome the Euler-Lagrange discrepancy explicitly track the fluid-structure interface by

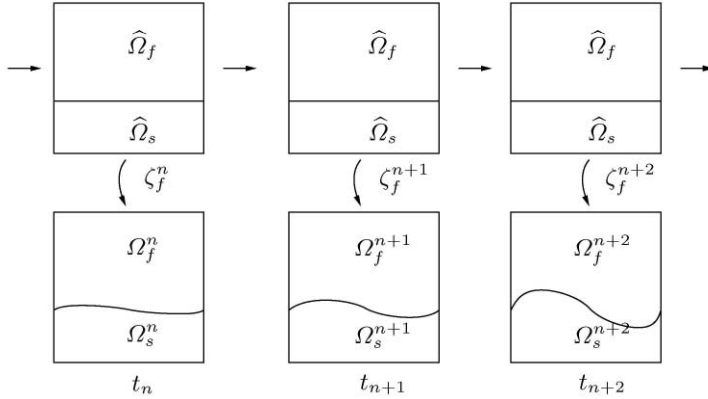


Fig. 2. Transformation approach, both frameworks Lagrangian

mesh adjustment and are generally referred to as ‘interface tracking’ methods. Both methods leave the structure problem in its natural Lagrangian setting.

In this paper, we follow the alternative way of posing the fluid as well as the structure problem in a fully Eulerian framework. A similar approach has been used by Lui and Walkington [21] in the context of the transport of visco-elastic bodies in a fluid. In the Eulerian setting a phase variable is employed on the fixed mesh to distinguish between the different phases, liquid and solid. This approach to identifying the fluid-structure interface is generally referred to as ‘interface capturing’, a method commonly used in the simulation of multiphase flows, [18,19]. Examples for the use of such a phase variable are the Volume of Fluid (VoF) method [14] and the Level Set (LS) method [9,22,26]. In the classical LS approach the distance function has to continually be reinitialized, due to the smearing effect by the convection velocity in the fluid domain. This makes the use of the LS method delicate for modeling FSI problems particularly in the presence of cornered structures. To cope with this difficulty, we propose a variant of the LS method that makes reinitialization unnecessary and which easily copes with cornered structures.

The method we describe does not depend on the specific structure model. The key variable in structure dynamics is the deformation, and since this depends on the deflection, it is understandable why structure dynamics is preferably described in the Lagrangian frame. To be able to describe the deformations in the Eulerian frame, we introduce the set of ‘initial positions’ (IP set) of all structure points. This set is then transported with the structure velocity in each time step. Based on the IP set points and their Eulerian coordinates the displacement is now available in an Eulerian sense. Also its gradient has to be rewritten appropriately, which will be explained later in Section 3.2. Since the fluid-structure interface will be crossing through cells, we will have to also transport the IP set in the fluid domain.

If we were to use the fluid velocity for the convection of the IP set, this would lead to entanglement of the respective displacements, which would ‘wreak havoc’ on the interface cells. This is a known problem with LS approaches. A common way for fixing this problem has been to occasionally fix the LS field between the time steps. The problem with this approach is that the variational formulation is no longer consistent. As an alternative, we harmonically continue the structure velocity into the fluid domain. In the fluid domain we then use this velocity for the convection of the IP set. Since an IP set is available in both domains, we can always at each point determine if it belongs to the fluid or solid part of the model.

Again this approach is similar to the LS approach. Actually, it is possible to also develop a model for FSI using the level set approach, [20]. But when developing a complete variational formulation the two key characteristics of the LS approach also become the main cause of concern: reinitialization and the signed distance function. Although the problem of reinitialization here can also be avoided by using an harmonically extended velocity, the trouble concerning corner approximation persists. In contrast to this, by using an

initial position set, we are deforming a virtual mesh of the structure which is extended into the whole domain.

The equations we use are based on the momentum and mass conservation equations for the flow of an incompressible Newtonian fluid and the deformation of a compressible St. Venant-Kirchhoff or likewise incompressible neo-Hookean solid. The spatial discretization is by a second-order finite element method with conforming equal-order (bilinear) trial functions using ‘local projection stabilization’ as introduced by Becker and Braack [2,3]. The time discretization uses the second-order ‘Fractional-Step- θ ’ scheme originally proposed by Bristeau, Glowinski, and Periaux [7]. This method has the same complexity as the Crank–Nicolson scheme but better stability properties, [23].

Based on the Eulerian variational formulation of the FSI system, we use the ‘dual weighted residual’ (DWR) method, described in [4,5,1], to derive ‘goal-oriented’ a posteriori error estimates. The evaluation of these error estimates requires the approximate solution of a linear dual variational problem. The resulting a posteriori error indicators are then used for automatic local mesh adaption. The full application of the DWR method to FSI problems requires a Galerkin discretization in space as well as in time. Due to the use of a difference scheme in time, in this paper we are limited to ‘goal-oriented’ mesh adaptation in computing steady states or (somewhat heuristically) to quasi-steady states within the time stepping process. The incorporation of automatic time-step control will be the subject of forthcoming work.

The method for computing FSI described in this paper is validated at a stationary model problem that is a lid-driven cavity involving the interaction of an incompressible Stokes fluid with a linearized incompressible neo-Hookean solid. Then, as a more challenging test the self-induced oscillation of a thin elastic bar immersed in an incompressible fluid is treated (FLUSTRUK-A benchmark described in [15]). For this test problem, our method is also compared against a standard ‘arbitrary Lagrange Eulerian’ (ALE) approach. The possible potential of the fully Eulerian formulation of the FSI problems is indicated by its good behavior for large structure deformations. All computations and visualizations are done using the flow-solver package GASCOIGNE [33] and the graphics package VISUSIMPLE [32]. More details on the software implementation can be found in [10].

The outline of this paper is as follows. Section 2 introduces the basic notation for the Eulerian formulation of the FSI problem which is then described in Section 3. Section 4 presents the discretization in space and time and the derivation of a posteriori error estimates and strategies for mesh adaptation. In Section 5 the proposed method is validated at a stationary test problem ‘elastic flow cavity’. Finally, Section 6 contains the results obtained for the nonstationary benchmark problem FLUSTRUK-A (oscillations of a thin elastic bar) for various combinations of material models and flow conditions.

2 Notation

We begin with introducing some notation which will be used throughout the paper. By $\Omega \subset \mathbb{R}^d$ ($d = 2$ or $d = 3$), we denote the domain of definition of the FSI problem. The domain Ω is supposed to be *time independent* but to consist of two possibly time-dependent subdomains, the fluid domain $\Omega_f(t)$ and the structure domain $\Omega_s(t)$. Unless needed, the explicit time dependency will be skipped in this notation. The boundaries of Ω , Ω_f , and Ω_s are denoted by $\partial\Omega$, $\partial\Omega_f$, and $\partial\Omega_s$, respectively. The common interface between Ω_f and Ω_s is $\Gamma_i(t)$, or simply Γ_i .

The initial structure domain is denoted by $\hat{\Omega}_s$. Spaces, domains, coordinates, values (such as pressure, displacement, velocity) and operators associated to $\hat{\Omega}_s$ (or $\hat{\Omega}_f$) will likewise be indicated by a ‘hat’.

Partial derivatives of a function f with respect to the i -th coordinate are denoted by $\partial_i f$, and the total time-derivative by $d_t f$. The divergences of vectors and tensors are written as $\operatorname{div} f = \sum_i \partial_i f_i$ and $(\operatorname{div} F)_i = \sum_j \partial_j F_{ij}$. The gradient of a vector valued function v is the tensor $(\nabla v)_{ij} = \partial_j v_i$.

By $[f]$, we denote the jump of a (possibly discontinuous) function f across an interior boundary, where n is always the unit vector n at points on that boundary.

For a set X , we denote by $L^2(X)$ the Lebesgue space of square-integrable functions on X equipped with the usual inner product and norm

$$(f, g)_X := \int_X f g \, dx, \quad \|f\|_X^2 = (f, f)_X,$$

respectively, and correspondingly for vector- and matrix-valued functions. Mostly the domain X will be Ω , in which case we will skip the domain index in products and norms. For Ω_f and Ω_s , we similarly indicate the associated spaces, products, and norms by a corresponding index ‘f’ or ‘s’.

Let $L_X := L^2(X)$ and $L_X^0 := L^2(X)/\mathbb{R}$. The functions in L_X (with $X = \Omega$, $X = \Omega_f(t)$, or $X = \Omega_s(t)$) with first-order distributional derivatives in L_X make up the Sobolev space $H^1(X)$. Further, $H_0^1(X) = \{v \in H^1(X) : v|_{\partial X_D} = 0\}$, where ∂X_D is that part of the boundary ∂X at which Dirichlet boundary conditions are imposed. Further, we will use the function spaces $V_X := H^1(X)^d$, $V_X^0 := H_0^1(X)^d$, and for time-dependent functions

$$\begin{aligned} \mathcal{L}_X &:= L^2[0, T; L_X], & \mathcal{V}_X &:= L^2[0, T; V_X] \cap H^1[0, T; V_X^*], \\ \mathcal{L}_X^0 &:= L^2[0, T; L_X^0], & \mathcal{V}_X^0 &:= L^2[0, T; V_X^0] \cap H^1[0, T; V_X^*], \end{aligned}$$

where V_X^* is the dual of V_X^0 . Again, the X -index will be skipped in the case of $X = \Omega$, and for $X = \Omega_f$ and $X = \Omega_s$ a corresponding index ‘f’ or ‘s’ will be used.

3 Formulation

In this section, we introduce the Eulerian formulation of the FSI problem.

3.1 Fluid

For the liquid part, we assume a Newtonian incompressible fluid governed by the usual Navier-Stokes equations, i.e., the equations describing conservation of mass and momentum. The (constant) density and kinematic viscosity of the fluid are ρ_f and ν_f , respectively. The equations are written in an Eulerian framework in the time-dependent domain $\Omega_f(t)$. The physical unknowns are the scalar pressure field $p_f \in \mathcal{L}_f$ and the vector velocity field $v_f \in v_f^D + \mathcal{V}_f$. Here, v_f^D is a suitable extension of the prescribed Dirichlet data on the boundaries (both moving or stationary) of Ω_f , and g_1 is a suitable extension to all of $\partial\Omega_f$ of the Neumann data for $\sigma_f \cdot n$ on the boundaries. We have ‘hidden’ the fluid-structure interface conditions of steadiness of velocity and normal stress in parts of the boundary conditions v_f^D and g_1 . These are addressed in Section 3.3, below. We write the fluid equations in variational form: *Find* $\{v_f, p_f\} \in \{v_f^D + \mathcal{V}_f^0\} \times \mathcal{L}_f$, *such that* $v_f(0) = v_f^0$, *and*

$$\begin{aligned} (\rho_f(\partial_t + v_f \cdot \nabla)v_f, \psi^v)_f + (\sigma_f, \epsilon(\psi^v))_f &= (g_1, \psi^v)_{\partial\Omega_f} + (f_1, \psi^v)_f, \\ (\operatorname{div} v_f, \psi^p)_f &= 0, \end{aligned} \quad (1)$$

for all $\{\psi^v, \psi^p\} \in V_f^0 \times L_f$, where

$$\sigma_f := -p_f I + 2\rho_f \nu_f \epsilon(v_f), \quad \epsilon(v) := \frac{1}{2}(\nabla v + \nabla v^T).$$

3.2 Structure

In the examples in Sections 5 and 6, below, we consider two different types of materials, an ‘incompressible neo-Hookean’ (INH) material and a compressible elastic material described by the ‘St. Venant-Kirchhoff’ (STVK) model. These two models will be described in the next two subsections.

The density of the structure is ρ_s . The material elasticity is usually described by a set of two parameters, the Poisson ration ν_s and the Young modulus E_s , or alternatively, the Lamé coefficients λ_s and μ_s . These parameters satisfy the following relations:

$$\begin{aligned} \nu_s &= \frac{\lambda_s}{2(\lambda_s + \mu_s)}, & E_s &= \mu_s \frac{3\lambda_s + 2\mu_s}{\lambda_s + \mu_s}, \\ \mu_s &= \frac{E_s}{2(1 + \nu_s)}, & \lambda_s &= \frac{\nu_s E_s}{(1 + \nu_s)(1 - 2\nu_s)}, \end{aligned}$$

where $\nu_s = \frac{1}{2}$ for incompressible and $\nu_s < \frac{1}{2}$ for compressible material.

Incompressible Neo-Hookean (INH) material: We consider the case of an incompressible neo-Hookean (INH) elastic medium governed by the equations based on conservation of mass and momentum. Usually these equations are formulated in Lagrangian coordinates in the domain $\hat{\Omega}_s$ with the scalar pressure field $\hat{p}_s \in \hat{\mathcal{L}}_f$ and the vector displacement and velocity fields $\hat{u}_s \in \hat{u}^D + \hat{\mathcal{V}}_s^0$, $\hat{v}_s \in \hat{v}_s^D + \hat{\mathcal{V}}_s^0$. Here, $\hat{u}^D$ and $\hat{v}_s^D$ are suitable extensions of the prescribed Dirichlet data on the boundaries of $\hat{\Omega}_s$, and $\hat{g}_2$ is a suitable extension to all of $\partial\hat{\Omega}_s$ of the Neumann data for $\hat{\sigma}_s \cdot n$ on the boundaries. Again, similarly as for the fluid problem, we have ‘hidden’ the fluid-structure interface conditions of steadiness of velocity and normal stress in parts of the boundary conditions $\hat{v}_s^D$ and $\hat{g}_2$. These are addressed in Section 3.3, below.

For the sake of simplicity, we assume that the only boundary displacements that take place are on $\hat{\Gamma}_i$, i.e.,

$$\hat{u}^D = \hat{v}_s^D = 0 \quad \text{on } \partial\hat{\Omega}_s \setminus \hat{\Gamma}_i.$$

We write these equations in the following variational form: *Find* $\{\hat{u}_s, \hat{v}_s, \hat{p}_s\} \in \{\hat{u}^D + \hat{\mathcal{V}}_s^0\} \times \{\hat{v}_s^D + \hat{\mathcal{V}}_s^0\} \times \hat{\mathcal{L}}_s$, *such that* $\hat{u}_s(0) = \hat{u}_s^0$, $\hat{v}_s(0) = \hat{v}_s^0$, *and*

$$\begin{aligned} (\rho_s d_t \hat{v}_s, \hat{\psi}^u)_{\hat{s}} + (\hat{\sigma}_s \hat{F}^{-T}, \hat{\epsilon}(\hat{\psi}^u))_{\hat{s}} &= (\hat{g}_2, \hat{\psi}^u)_{\partial\hat{\Omega}_s} + (\hat{f}_2, \hat{\psi}^u)_{\hat{s}}, \\ (d_t \hat{u}_s - \hat{v}_s, \hat{\psi}^v)_{\hat{s}} &= 0, \\ (\det \hat{F}, \hat{\psi}^p)_{\hat{s}} &= (1, \hat{\psi}^p)_{\hat{s}}, \end{aligned} \quad (2)$$

for all $\{\hat{\psi}^u, \hat{\psi}^v, \hat{\psi}^p\} \in \hat{V}_s^0 \times \hat{V}_s^0 \times \hat{L}_s$, where

$$\hat{F} := I + \hat{\nabla} \hat{u}_s, \quad \hat{\sigma}_s := -\hat{p}_s I + \mu_s (\hat{F} \hat{F}^T - I), \quad \hat{\epsilon}(\hat{\psi}^u) := \frac{1}{2} (\hat{\nabla} \hat{\psi}^u + \hat{\nabla} \hat{\psi}^{uT}).$$

St. Venant-Kirchhoff (STVK) material: We consider a compressible elastic material described by the St. Venant-Kirchhoff (STVK) model. As in the case of the incompressible neo-Hookean material, this model is governed by the equations for conservation of mass and momentum. Again these equations are written in Lagrangian form using the same notation as above. *Find* $\{\hat{u}_s, \hat{v}_s\} \in \{\hat{u}^D + \hat{\mathcal{V}}_s^0\} \times \{\hat{v}_s^D + \hat{\mathcal{V}}_s^0\}$, *such that* $\hat{u}_s(0) = \hat{u}_s^0$, $\hat{v}_s(0) = \hat{v}_s^0$, *and*

$$\begin{aligned} (\rho_s d_t \hat{v}_s, \hat{\psi}^u)_{\hat{s}} + (\hat{J} \hat{\sigma}_s \hat{F}^{-T}, \hat{\epsilon}(\hat{\psi}^u))_{\hat{s}} &= (\hat{g}_2, \hat{\psi}^u)_{\partial\hat{\Omega}_s} + (\hat{f}_2, \hat{\psi}^u)_{\hat{s}}, \\ (d_t \hat{u}_s - \hat{v}_s, \hat{\psi}^v)_{\hat{s}} &= 0, \end{aligned} \quad (3)$$

for all $\{\hat{\psi}^u, \hat{\psi}^v\} \in \hat{V}_s^0 \times \hat{V}_s^0$, where

$$\begin{aligned} \hat{F} &:= I + \hat{\nabla} \hat{u}_s, \quad \hat{J} := \det \hat{F}, \quad \hat{E} := \frac{1}{2} (\hat{F}^T \hat{F} - I), \\ \hat{\epsilon}(\hat{\psi}^u) &:= \frac{1}{2} (\hat{\nabla} \hat{\psi}^u + \hat{\nabla} \hat{\psi}^{uT}), \quad \hat{\sigma}_s := \hat{J}^{-1} \hat{F} (\lambda_s (\text{tr} \hat{E}) I + 2\mu_s \hat{E}) \hat{F}^T. \end{aligned}$$

Conversion to Eulerian frame (the IP set approach): To rewrite the above conservation equations in an Eulerian frame, we need the pressure $\hat{p}_s$, displacement $\hat{u}_s$, and its gradient $\hat{\nabla}\hat{u}_s$ expressed in the Eulerian sense, which are denoted by p_s , u_s , and ∇u_s , respectively. There holds $p_s(x) = \hat{p}_s(\hat{x})$ and $u_s(x) = \hat{u}_s(\hat{x})$, or more precisely,

$$p_s(x) = \hat{p}_s(D(x)) = \hat{p}_s(\hat{x}), \quad u_s(x) = \hat{u}_s(D(x)) = \hat{u}_s(\hat{x}), \quad (4)$$

where $D(x)$ is the (inverse) displacement function of points in the deformed domain Ω_s back to points in the initial domain $\hat{\Omega}_s$. The corresponding displacement function is $\hat{D}(\hat{x})$,

$$\begin{aligned} \hat{D} : \hat{\Omega}_s &\rightarrow \Omega_s, & \hat{D}(\hat{x}) &= \hat{x} + \hat{u}_s = x, \\ D : \Omega_s &\rightarrow \hat{\Omega}_s, & D(x) &= x - u_s = \hat{x}. \end{aligned} \quad (5)$$

Since $\det \hat{\nabla} \hat{D} = \det \hat{F} \neq 0$ the displacements D and $\hat{D}$ are well defined.

The immediate difficulty with the relations (4) is that u_s is only implicitly determined by $\hat{u}_s$, since $D(x)$ also depends on u_s . This is unpractical, and we therefore need a direct way of determining the displacement $u(x)$ of a ‘material’ point located at x with respect to its initial position at point $\hat{x}$. To achieve this, we introduce the so-called ‘set of initial positions’ (IP set) $\phi(\Omega)$ of all points of Ω at time t . If we look at a given ‘material’ point at the position $x \in \Omega$ and the time $t \in (0, T]$, then the value $\phi(t, x)$ will tell us what the initial position of this point was at time $t = 0$. These points are transported in the full domain with a certain velocity w . The convection velocity in the structure will be the structure velocity itself, $w|_{\Omega_s} = v_s$. If the fluid velocity were to be used for convection in the fluid domain, then the displacements there would eventually become very entangled. For this reason we use an alternative velocity. We explain this in more detail in Subsection 3.3, below. With this notation, the mapping ϕ is determined by the following variational problem: *Find $\phi \in \phi_0 + \mathcal{V}^0$, such that*

$$(\partial_t \phi + w \cdot \nabla \phi, \psi) = 0 \quad \forall \psi \in V^0. \quad (6)$$

where ϕ_0 is a suitable extension of the Dirichlet data along the boundaries,

$$\begin{aligned} \phi(0, x) &= x, & x &\in \Omega, \\ \phi(t, x) &= x, & \{t, x\} &\in (0, T] \times \partial\Omega. \end{aligned}$$

This means that $\hat{x} + \hat{u}(t, \hat{x}) = x$, for any point with the initial position $\hat{x}$ and the position x later at time t . Since $\hat{x} = \phi(0, \hat{x}) = \phi(t, x)$ and $\hat{u}(t, \hat{x}) = u(t, x)$ it follows that

$$x = \phi + u. \quad (7)$$

Using this in the IP set equation (6) yields

$$(\partial_t u - w + w \cdot \nabla u, \psi) = 0 \quad \forall \psi \in V^0.$$

To access the Eulerian deformation, we use an identity of the displacement functions from (5) $D(\hat{D}(\hat{x})) = \hat{x}$. Differentiating this yields

$$(I - \nabla u)(I + \hat{\nabla} \hat{u}) = I \quad \Leftrightarrow \quad \hat{\nabla} \hat{u} = (I - \nabla u)^{-1} - I.$$

Thus, the Cauchy stress tensor σ_s can be written for INH and STVK materials in an Eulerian framework as follows:

$$\begin{aligned} \sigma_s &= \begin{cases} -p_s I + \mu_s (F F^T - I) & \text{(INH material),} \\ J^{-1} F (\lambda_s (\text{tr} E) I + 2\mu_s E) F^T & \text{(STVK material),} \end{cases} \\ F &= I + \hat{\nabla} \hat{u} = (I - \nabla u)^{-1}, \quad J = \det F, \quad E = \frac{1}{2}(F^T F - I). \end{aligned}$$

Finally, we write the structure equations in the Eulerian framework for both types of materials, INH and STVK.

INH material in Eulerian formulation: We write the structure equations in Eulerian form in the time-dependent domain $\Omega_s(t)$ with the corresponding scalar pressure field p_s and the vector displacement and velocity fields u_s and v_s , respectively. Here, the incompressibility condition $\det(F) = 1$ is replaced by the divergence condition for the velocity. Find $\{u_s, v_s, p_s\} \in \{u^D + \mathcal{V}_s^0\} \times \{v_s^D + \mathcal{V}_s^0\} \times \mathcal{L}_s$, such that $u_s(0) = u_s^0$, $v_s(0) = v_s^0$, and

$$\begin{aligned} (\rho_s(\partial_t v_s + v_s \cdot \nabla v_s), \psi^u)_s + (\sigma_s, \epsilon(\psi^u))_s &= (g_2, \psi^v)_{\partial\Omega_s} + (f_2, \psi^u)_s, \\ (\partial_t u_s - v_s + v_s \cdot \nabla u_s, \psi^v)_s &= 0, \\ (\text{div} v_s, \psi^p)_s &= 0, \end{aligned} \tag{8}$$

for all $\{\psi^u, \psi^v, \psi^p\} \in V_s^0 \times V_s^0 \times L_s$, where

$$\sigma_s := -p_s I + \mu_s (F F^T - I), \quad F := (I - \nabla u)^{-1}, \quad \epsilon(\psi^u) = \frac{1}{2}(\nabla \psi^u + \nabla \psi^{uT}).$$

STVK material in Eulerian formulation: We write the structure equations in Eulerian form in the time-dependent domain $\Omega_s(t)$ with the corresponding displacement and velocity fields u_s and v_s , respectively. Find $\{u_s, v_s\} \in \{u^D + \mathcal{V}_s^0\} \times \{v_s^D + \mathcal{V}_s^0\}$, such that $u_s(0) = u_s^0$, $v_s(0) = v_s^0$, and

$$\begin{aligned} (\rho_s(\partial_t v_s + v_s \cdot \nabla v_s), \psi^u)_s + (\sigma_s, \epsilon(\psi^u))_s &= (g_2, \psi^u)_{\partial\Omega_s} + (f_2, \psi^u)_s, \\ (\partial_t u_s - v_s + v_s \cdot \nabla u_s, \psi^v)_s &= 0, \end{aligned} \tag{9}$$

for all $\{\psi^u, \psi^v\} \in V_s^0 \times V_s^0$, where

$$\begin{aligned} \sigma_s &:= J^{-1} F (\lambda_s (\text{tr} E) I + 2\mu_s E) F^T, \\ F &:= (I - \nabla u)^{-1}, \quad J := \det F, \quad E := \frac{1}{2}(F^T F - I). \end{aligned}$$

3.3 The FSI Problem in Eulerian Formulation

Since now both the fluid and the structure equations (1), (8), and (9) for conservation of mass and momentum are written in an Eulerian frame, they can be combined into one unified formulation. Here, the steadiness of velocity across the fluid-structure interface Γ_i is strongly enforced by requiring one common continuous field for the velocity on Ω . The Dirichlet boundary data v_f^D and v_s^D on parts of $\partial\Omega$ are merged into a suitable velocity field $v^D \in V$. Finally, the force balance condition $\sigma_f \cdot n = \sigma_s \cdot n$ on Γ_i now appears as a boundary integral of the jump $[\sigma \cdot n]$ on the right hand side,

$$([\sigma \cdot n], \psi^v)_{\Gamma_i} = \int_{\Gamma_i} (\sigma_f - \sigma_s) \cdot n_f \psi^v \, do.$$

By omitting this boundary integral the (weak) continuity of $\sigma \cdot n$ becomes an implicit condition of the combined variational formulation. The remaining parts of the Neumann data g_1 and g_2 now form the Neumann boundary data on $\partial\Omega$ and are combined to g_3 . We write the Cauchy stress tensor for the whole domain as follows:

$$\sigma := \chi_f \sigma_f + \chi_s \sigma_s.$$

Here, χ_f and χ_s are the characteristic functions of Ω_f and Ω_s , respectively, which are determined by the values of the IP-set function ϕ :

$$\chi_f(t, x) = \begin{cases} 1, & \phi(t, x) \in \hat{\Omega}_f \setminus \hat{\Gamma}_i, \\ 0, & \phi(t, x) \in \hat{\Omega}_s, \end{cases} \quad \chi_s = 1 - \chi_f. \quad (10)$$

The interface Γ_i will usually intersect mesh cells. Due to this, we need a reasonable continuation of the displacement and its gradient from the structure domain into the fluid domain. The value of u in the fluid domain will be determined by the choice of the convection velocity w . If we were to use the fluid velocity this would eventually lead to increasing entanglement, which would necessitate a continual reinitialization of the IP set. As an alternative, we use the harmonic continuation of the structure velocity to the whole domain Ω , which is likewise denoted by w and satisfies

$$(\chi_s(w - v), \psi) + (\chi_f \alpha_w \nabla w, \nabla \psi) = 0, \quad \forall \psi \in V^0, \quad (11)$$

where α_w is a small positive parameter. By this construction, the deflection u_f in the fluid domain becomes an artificial quantity without any real physical meaning, i.e., $d_t u_s = v_s$, but generally $d_t u_f \neq v_f$.

Now, we can combine the formulations (1), (8), (9), (11), and (10) with (7), to obtain a complete variational formulation of the FSI problem in Eulerian formulation. In the case of STVK material the (non-physical) pressure p_s in the structure subdomain is determined as harmonic extension of the flow pressure p_f .

Eulerian formulation of the FSI problem: Find fields $\{v, w, u, p\} \in \{v^D + \mathcal{V}^0\} \times \mathcal{V}^0 \times \mathcal{V}^0 \times \mathcal{L}$, such that $v(0) = v^0$, $u(0) = u^0$, and

$$\begin{aligned} (\rho(\partial_t v + v \cdot \nabla v), \psi) + (\sigma, \epsilon(\psi)) &= (g_3, \psi)_{\partial\Omega} + (f_3, \psi) \quad \forall \psi \in V^0, \\ (\operatorname{div} v, \chi) &= 0 \quad \forall \chi \in L \quad (\text{INH material}), \\ (\chi_f \operatorname{div} v, \chi) + (\chi_s \alpha_p \nabla p, \nabla \chi) &= 0 \quad \forall \chi \in L \quad (\text{STVK material}), \\ (\partial_t u - w + w \cdot \nabla u, \psi) &= 0 \quad \forall \psi \in V^0, \\ (\chi_s(w - v), \psi) + (\chi_f \alpha_w \nabla w, \nabla \psi) &= 0 \quad \forall \psi \in V^0, \end{aligned} \quad (12)$$

where α_p is a small positive constant, $\rho := \chi_f \rho_f + \chi_s \rho_s$ and $\sigma := \chi_f \sigma_f + \chi_s \sigma_s$, with

$$\chi_f := \begin{cases} 1, & x - u \in \hat{\Omega}_f \setminus \hat{\Gamma}_i, \\ 0, & x - u \in \hat{\Omega}_s, \end{cases} \quad \chi_s = 1 - \chi_f,$$

and

$$\begin{aligned} \sigma_f &:= -pI + 2\rho_f \nu_f \epsilon(v), \\ \sigma_s &:= \begin{cases} -pI + \mu_s (FF^T - I) & (\text{INH material}), \\ J^{-1} F(\lambda_s (\operatorname{tr} E)I + 2\mu_s E)F^T & (\text{STVK material}), \end{cases} \\ F &:= (I - \nabla u)^{-1}, \quad J := \det F, \quad E := \frac{1}{2}(F^T F - I). \end{aligned}$$

In this variational formulation the position of the fluid structure interface Γ_i is implicitly given by the displacement u and the characteristic function χ_s ,

$$\Gamma_i(t) = \{x \in \Omega, x - u(x, t) \in \hat{\Gamma}_i\}. \quad (13)$$

Notice that the system (12) is *nonlinear* even if the two subproblems are linear, e.g., for a Stokes fluid interacting with a linear elastic structure.

In some situations the solution of an FSI problem may tend to a ‘steady state’ as $t \rightarrow \infty$. For later purposes, we derive the set of equations determining such a steady state solution $\{v^*, w^*, u^*, p^*\} \in \{v^D + V^0\} \times V^0 \times V^0 \times L$. The corresponding limits of the characteristic functions and subdomains are denoted by χ_f^*, χ_s^* and Ω_f^*, Ω_s^* , respectively. Further, the fluid velocity becomes constant in time, $v_f^* := \lim_{t \rightarrow \infty} v|_{\Omega_f}$, and the structure velocity vanishes, $v_s^* \equiv 0$, which in turn implies $w^* \equiv 0$. The steady state structure displacement u_s^* is likewise well defined, but the corresponding (‘non-physical’) fluid displacement is merely defined by $u_f^* = u_f^{\lim} := \lim_{t \rightarrow \infty} u|_{\Omega_f}$ and therefore depends on the chosen construction of $w|_{\Omega_f}$ as harmonic extension of $w|_{\Omega_s}$. Actually, it could be defined by any suitable continuation of u_s^* to all of Ω , e.g., by harmonic continuation. On the other hand the steady state pressure p^* is to be determined from the limiting equations. Then, with suitable extensions u^D and v^D of the prescribed Dirichlet data on $\partial\Omega$, the FSI system (12) reduces to the following ‘stationary’ form (dropping for simplicity the stars):

Eulerian formulation of the ‘stationary’ FSI problem: Find $\{u, v, p\} \in \{u^D + V^0\} \times \{v^D + V^0\} \times L$, such that

$$\begin{aligned} (\rho v \cdot \nabla v, \psi) + (\sigma, \epsilon(\psi)) &= 0 \quad \forall \psi \in V^0, \\ (\operatorname{div} v, \chi) &= 0 \quad \forall \chi \in L \quad (\text{INH material}), \\ (\chi_f \operatorname{div} v, \chi) + (\chi_s \alpha_p \nabla p, \nabla \chi) &= 0 \quad \forall \chi \in L \quad (\text{STVK material}), \\ (\chi_f(u - u_f^{\lim}), \psi) + (\chi_s v, \psi) &= 0 \quad \forall \psi \in V^0, \end{aligned} \tag{14}$$

where $\rho := \chi_f \rho_f + \chi_s \rho_s$ and $\sigma := \chi_f \sigma_f + \chi_s \sigma_s$, with

$$\chi_f := \begin{cases} 1, & x - u \in \hat{\Omega}_f \setminus \hat{\Gamma}_i, \\ 0, & x - u \in \hat{\Omega}_s, \end{cases} \quad \chi_s = 1 - \chi_f,$$

and

$$\begin{aligned} \sigma_f &:= -pI + 2\rho_f \nu_f \epsilon(v), \\ \sigma_s &:= \begin{cases} -pI + \mu_s(FF^T - I) & (\text{INH material}), \\ J^{-1}F(\lambda_s(\operatorname{tr} E)I + 2\mu_s E)F^T & (\text{STVK material}), \end{cases} \\ F &:= (I - \nabla u)^{-1}, \quad J := \det F, \quad E := \frac{1}{2}(F^T F - I). \end{aligned}$$

4 Discretization

In this section, we detail the discretization in space and time of the FSI problem based on its Eulerian variational formulation (12).

4.1 Mesh Notation

The spatial discretization is by a conforming finite element Galerkin method on meshes $\mathbb{T}_h$ consisting of cells denoted by K , which are (convex) quadrilaterals in 2d or hexaedra in 3d. The mesh parameter h is a scalar cell-wise constant function defined by $h|_K := h_K = \operatorname{diam}(K)$. ‘Refinement’ of cells is always by bisection, i.e., by joining opposite midpoints of sides or faces. ‘Coarsening’ of a cell is possible if it has been generated by prior refinement of some ‘parent cell’. The ‘finest level’ of cells of a mesh $\mathbb{T}_h$ consists of all cells that can be removed by coarsening in one sweep. The resulting coarsened mesh is referred to as $\mathbb{T}_{2h}$. To facilitate mesh refinement and coarsening, we allow the cells to have a certain number of nodes that are at the midpoint of sides or faces of neighboring cells. These ‘hanging nodes’ do not carry degrees of freedom and the corresponding function values are determined by linear or bilinear interpolation of neighboring ‘regular’ nodal points. For more details on this approach see [8] or [1].

4.2 Galerkin Formulation

For arguments $U = \{v, w, u, p\}$ and $\Psi = \{\psi^v, \psi^w, \psi^u, \psi^p\} \in \mathcal{W} := \mathcal{V} \times \mathcal{V} \times \mathcal{V} \times \mathcal{V}$, we introduce the space-time semilinear form

$$\begin{aligned} A(U)(\Psi) := & \int_0^T \left\{ (\rho(\partial_t v + v \cdot \nabla v), \psi^v) + (\sigma(U), \epsilon(\psi^v)) \right. \\ & + \begin{cases} (\operatorname{div} v, \psi^p) & \text{(INH material)} \\ (\chi_f \operatorname{div} v, \psi^p) + (\chi_s \alpha_p \nabla p, \nabla \psi^p) & \text{(STVK material)} \end{cases} \\ & - (g_3, \psi^v)_{\partial\Omega} - (f_3, \psi^v) + (\partial_t u - w + w \cdot \nabla u, \psi^u) \\ & \left. + (\chi_s(w - v), \psi^w) + (\chi_f \alpha_w \nabla w, \nabla \psi^w) \right\} dt. \end{aligned}$$

With this notation, we can write the variational problem (12) in compact form: *Find* $U \in U^D + \mathcal{W}^0$, *such that*

$$A(U)(\Psi) = 0 \quad \forall \Psi \in \mathcal{W}^0, \quad (15)$$

where U^D is an appropriate extension of the Dirichlet boundary and initial data and the space $\mathcal{W}^0$ is defined by

$$\mathcal{W}^0 := \{\Psi \in \mathcal{V}^0 \times \mathcal{V}^0 \times \mathcal{V}^0 \times \mathcal{V}^0, \psi^u(0) = \psi^v(0) = 0\}.$$

For discretizing this problem in space, we use equal-order Q_1 finite elements (d-linear shape functions) for all unknowns, where the corresponding finite element spaces are denoted by $L_h \subset L$, $V_h \subset V$, $W_h \subset W$, etc.. Within the present abstract setting the discretization in time is likewise thought as by a Galerkin method, such as the dG(r) ('discontinuous' Galerkin) or the cG(r) ('continuous' Galerkin) method. Here, the dG(0) method is closely related to the backward Euler scheme and the dG(1) method to the Crank–Nicolson scheme. However, in the test computations described below, we have used a Galerkin method only in space but finite difference schemes in time. The full space-time Galerkin framework is mainly introduced as basis for a systematic approach to residual-based a posteriori error estimation as described below.

The spatial discretization by 'equal-order' finite elements for velocity and pressure needs stabilization in order to compensate for the missing 'inf-sup stability'. We use the so-called '*local projection stabilization*' (LPS) introduced by Becker and Braack [2,3]. An analogous approach is also employed for stabilizing the convection in the flow model as well as in the transport equation for the displacement u . We define the mesh-dependent bilinear form

$$\begin{aligned} (\varphi, \psi)_\delta &:= \sum_{K \in \mathbb{T}_h} \delta_K (\varphi, \psi)_K, \\ \delta_K &:= \alpha (\chi_f \rho_f \nu_f h_K^{-2} + \chi_s \mu_s h_K^{-2} + \beta \rho |v_h|_{\infty;K} h_K^{-1} + \gamma |w_h|_{\infty;K} h_K^{-1})^{-1}. \end{aligned}$$

Further, we introduce the ‘fluctuation operator’ $\pi_h : V_h \rightarrow V_{2h}$ on the finest mesh level $\mathbb{T}_h$ by $\pi_h = I - P_{2h}$, where $P_{2h} : V_h \rightarrow V_{2h}$ is the L^2 -projection. The operator π_h measures the fluctuation of a function in V_h with respect to its projection into the next coarser space V_{2h} . With this notation, we define the stabilization form

$$S_\delta(U_h)(\Psi_h) := \int_0^T \left\{ (\nabla \pi_h p_h, \nabla \pi_h \psi_h^p)_\delta + (\rho v_h \cdot \nabla \pi_h v_h, v_h \cdot \nabla \pi_h \psi_h^v)_\delta + (w_h \cdot \nabla \pi_h u_h, w_h \cdot \nabla \pi_h \psi_h^u)_\delta \right\} dt,$$

where the first term stabilizes the fluid pressure, the second one the INH structure pressure, the third one the transport in the flow model, and the fourth one the transport of the displacement u_h . Then, the stabilized Galerkin approximation of problem (15) reads: Find $U_h \in U_h^D + \mathcal{W}_h^0$, such that

$$A_\delta(U_h)(\Psi_h) := A(U_h, \Psi_h) + S_\delta(U_h)(\Psi_h) = 0, \quad \forall \Psi_h \in \mathcal{W}_h^0. \quad (16)$$

The LPS has the important property that it acts only on the diagonal terms of the coupled system and that it does not contain any second-order derivatives. However, it is only ‘weakly’ consistent, as it does not vanish for the continuous solution, but it tends to zero with the right order as $h \rightarrow 0$. The choice of the numbers α, β, γ in the stabilization parameter δ_K is, based on practical experience, in our computations $\alpha = 1/2$, and $\beta = \gamma = 1/6$.

4.3 Time Discretization

The discretization in time is by the so-called ‘fractional-step- θ scheme’ in which each time step $t_{n-1} \rightarrow t_n$ is splitted into three substeps $t_{n-1} \rightarrow t_{n-1+\theta} \rightarrow t_{n-\theta} \rightarrow t_n$. For brevity, we formulate this time stepping method for an abstract differential-algebraic equation (DAE)

$$\begin{bmatrix} M & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{v}(t) \\ \dot{p}(t) \end{bmatrix} + \begin{bmatrix} A(v(t)) & B \\ -B^T & C \end{bmatrix} \begin{bmatrix} v(t) \\ p(t) \end{bmatrix} = \begin{bmatrix} b(t) \\ c(t) \end{bmatrix}, \quad (17)$$

which resembles the operator form of the spatially discretized incompressible Navier-Stokes equations with pressure stabilization. With the parameters $\theta = 1 - \sqrt{2}/2 = 0.292893\dots$, $\theta' = 1 - 2\theta$, $\alpha \in (1/2, 1]$, and $\beta = 1 - \alpha$, the fractional-step- θ scheme reads:

$$\begin{aligned} \begin{bmatrix} M + \alpha \theta k A^{n-1+\theta} & \theta k B \\ -B^T & C \end{bmatrix} \begin{bmatrix} v^{n-1+\theta} \\ p^{n-1+\theta} \end{bmatrix} &= \begin{bmatrix} [M - \beta \theta k A^{n-1}] v^{n-1} + \theta k b^{n-1} \\ c^{n-1+\theta} \end{bmatrix} \\ \begin{bmatrix} M + \beta \theta' k A^{n-\theta} & \theta' k B \\ -B^T & C \end{bmatrix} \begin{bmatrix} v^{n-\theta} \\ p^{n-\theta} \end{bmatrix} &= \begin{bmatrix} [M - \alpha \theta' k A^{n-1+\theta}] v^{n-1+\theta} + \theta' k b^{n-\theta} \\ c^{n-\theta} \end{bmatrix} \\ \begin{bmatrix} M + \alpha \theta k A^n & \theta k B \\ -B^T & C \end{bmatrix} \begin{bmatrix} v^n \\ p^n \end{bmatrix} &= \begin{bmatrix} [M - \beta \theta k A^{n-\theta}] v^{n-\theta} + \theta k b_h^{n-\theta} \\ c^n \end{bmatrix}, \end{aligned}$$

where $A^{n-1+\theta} := A(x^{n-1+\theta})$, $b^{n-1} := b(t_{n-1})$, etc.. This scheme is of second order and has a similar work complexity as the well-known Crank–Nicolson scheme (case $\alpha = 1/2$). The fractional-step- θ scheme was originally proposed in form of an operator splitting scheme separating the two complications ‘non-linearity’ and ‘incompressibility’ within each cycle $t_{n-1} \rightarrow t_{n-1+\theta} \rightarrow t_{n-\theta} \rightarrow t_n$. However, it has also very attractive features as a pure time-stepping method. Being *strongly* A-stable, for any choice of $\alpha \in (1/2, 1]$, it possesses the full smoothing property in the case of rough initial data, in contrast to the Crank–Nicolson scheme which is only conditionally smoothing (for $k \sim h^2$). Furthermore, it is less dissipative than most of the other second-order implicit schemes and therefore suitable for computing oscillatory solutions; for more details, we refer to [23], [24], and [11].

For computing steady state solutions, we use a pseudo-time stepping techniques based on the simple (first-order) backward Euler scheme, which in the notation from before reads

$$\begin{bmatrix} M + kA^n & kB \\ -B^T & C \end{bmatrix} \begin{bmatrix} v^n \\ p^n \end{bmatrix} = \begin{bmatrix} Mv^{n-1} + kb_h^{n-1} \\ c^n \end{bmatrix},$$

4.4 Solution of the Algebraic Systems

After time and space discretization, in each substep of the fractional-step- θ scheme (or any other fully implicit time-stepping scheme) a quasi-stationary nonlinear algebraic system has to be solved. This is done by a standard Newton-type method with adaptive step-length selection, in which the transport terms are correctly linearized. Only the stabilization terms and the terms involving the characteristic function χ_f , determining the position of the interface, are treated by a simple functional iteration. In all cases the iteration starts from the values at the preceding time level. The resulting linear subproblems are then solved by the GMRES method with preconditioning by a geometric multigrid method with block-ILU smoothing. Since such an approach is rather standard nowadays, we omit its details and refer to the relevant literature, e.g., [29], [23], or [16].

4.5 Mesh Adaptation

Now, we come to the main issues of this paper, namely the automatic mesh adaptation within the finite element solution of the FSI problem. The computations shown in Sections 5 and 6, below, have been done on three different types of meshes:

- globally refined meshes obtained using several steps of uniform (edge) bisection of a coarse initial mesh,
- locally refined meshes obtained using a purely geometry-based criterion by marking all cells for refinement which have certain prescribed distances from the fluid-structure interface,

- locally refined meshes obtained using a systematic residual-based criteria by marking all cells for refinement which have error indicators above a certain threshold.

The main goal of this project is to employ the ‘*dual weighted residual method*’ (DWR method) for the adaptive solution of FSI problems. This method has been developed in [4] (see also [5] and [1]) as an extension of the duality technique for a posteriori error estimation described in [12]. The DWR method provides a general framework for the derivation of ‘goal-oriented’ a posteriori error estimates together with criteria of mesh adaptation for the Galerkin discretization of general linear and nonlinear variational problems, including optimization problems. It is based on a complete *variational* formulation of the problem, such as (15) for the FSI problem. In fact, this was one of the driving factors for deriving the Eulerian formulation underlying (15). In order to incorporate also the time discretization into this framework, we have to use a fully space-time Galerkin method, i.e., a standard finite element method in space combined with the dG(r) or cG(r) (‘discontinuous’ Galerkin or ‘continuous’ Galerkin) method in time. The following discussion assumes such a space-time Galerkin discretization, though in our test computations, we have used the fractional-step- θ scheme which is a difference scheme. Accordingly, in this paper the DWR method is used only in its stationary form in computing either steady states or intermediate quasi-steady states within the time stepping process.

We begin with the description of the DWR method for the special case of an FSI problem governed by an abstract variational equation such as (15). For notational simplicity, we think the nonhomogeneous boundary and initial data U^D to be incorporated into a linear forcing term $F(\cdot)$, or to be exactly representable in the approximating space $\mathcal{W}_h$. Then, the problem reads as follows: *Find* $U \in U^D + \mathcal{W}^0$, *such that*

$$A(U)(\Psi) = F(\Psi) \quad \forall \Psi \in \mathcal{W}^0. \quad (18)$$

The corresponding (stabilized) Galerkin approximation reads: *Find* $U_h \in U_h^D + \mathcal{W}_h$, *such that*

$$A(U_h)(\Psi_h) + S_\delta(U_h)(\Psi_h) = F(\Psi) \quad \forall \Psi_h \in \mathcal{W}_h^0. \quad (19)$$

Suppose now that the goal of the computation is the evaluation of the value $J(U)$ for some functional $J(\cdot)$ which is defined on $\mathcal{W}$ and (for notational simplicity only) assumed as linear. We want to control the quality of the discretization in terms of the error $J(U - U_h)$. To this end, we introduce the directional derivative

$$A'(U)(\Phi, \Psi) := \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \{A(U + \epsilon\Phi)(\Psi) - A(U)(\Psi)\}, \quad \Phi, \Psi \in \mathcal{W}^0,$$

the existence of which is assumed.

Remark 1. The above assumption of differentiability may cause concerns in treating the FSI problem since the dependence of the characteristic function $\chi_f(u)$ in (12) on the deflection u is generically not differentiable (only Lipschitzian). However, this non-differentiability is confined to the interface between fluid and structure which can be assumed to form a lower dimensional manifold. Hence, for practical applications, after discretization along the interface, the directional derivative can safely be replaced by a mesh-size dependent difference quotient. This pragmatic approach has proven very successful in similar situation, e.g., for Hencky elasto-plasticity [25].

With the above notation, we introduce the bilinear form

$$L(U, U_h)(\Phi, \Psi) := \int_0^1 A'(U_h + s(U - U_h))(\Phi, \Psi) ds,$$

and formulate the ‘dual problem’

$$L(U, U_h)(\Phi, Z) = J(\Phi) \quad \forall \Phi \in \mathcal{W}^0. \quad (20)$$

In the present abstract setting the existence of a solution $Z \in \mathcal{W}^0$ of the dual problem (20) has to be assumed. Now, taking $\Phi = U - U_h \in \mathcal{W}^0$ in (20) and using the Galerkin orthogonality property

$$A(U)(\Psi_h) - A(U_h)(\Psi_h) = S_\delta(U_h)(\Psi_h), \quad \Psi \in \mathcal{W}_h^0,$$

yields the error representation

$$\begin{aligned} J(U - U_h) &= L(U, U_h)(U - U_h, Z) \\ &= \int_0^1 A'(U_h + s(U - U_h))(U - U_h, Z) ds \\ &= A(U)(Z) - A(U_h)(Z) \\ &= F(Z - \Psi_h) - A(U_h)(Z - \Psi_h) - S_\delta(U_h)(\Psi_h) \\ &=: \rho(U_h)(Z - \Psi_h) - S_\delta(U_h)(\Psi_h), \end{aligned}$$

where $\Psi_h \in \mathcal{W}^0$ is an arbitrary element, usually taken as the generic nodal interpolant $I_h Z \in \mathcal{W}_h^0$ of Z . For the evaluation of the terms on the right-hand side, we split the integrals in the residual term $\rho(U_h)(Z - \Psi_h)$ into their contribution from the single mesh cells $K \in \mathbb{T}_h$ and integrate by parts. This results in an estimate of the error $|J(U - U_h)|$ in terms of computable local residual terms $\rho_K(U_h)$ multiplied by certain weight factors $\omega_K(Z)$ which depend on the dual solution Z ,

$$|J(U - U_h)| \leq \sum_{K \in \mathbb{T}_h} \rho_K(U_h) \omega_K(Z) + |S_\delta(U_h)(\Psi_h)|. \quad (21)$$

The explicit form of the terms in the sum on the right-hand side will be stated for a special situation below. The second term due to the regularization is usually neglected.

Since the dual solution Z is unknown, the evaluation of the weights $\omega_K(Z)$ requires further approximation. We linearize by assuming

$$L(U, U_h)(\Phi, \Psi) \approx L(U_h, U_h)(\Phi, \Psi) = A'(U_h)(\Phi, \Psi)$$

and use the approximate ‘discrete’ dual solution $Z_h \in \mathcal{W}_h^0$ defined by

$$A'(U_h)(\Phi, Z_h) = J(\Phi_h) \quad \forall \Phi_h \in \mathcal{W}_h^0. \quad (22)$$

From Z_h , we generate improved approximations to Z in a post-processing step by patchwise higher-order interpolation. For example in 2d, on 2×2 -patches of cells in $\mathbb{T}_h$ the 9 nodal values of the piecewise bilinear Z_h are used to construct a patchwise biquadratic function $\tilde{Z}$. This is then used to obtain the approximate error estimate

$$|J(U - U_h)| \approx \eta := \sum_{K \in \mathbb{T}_h} \rho_K(U_h) \omega_K(\tilde{Z}) \quad (23)$$

which is the basis of automatic mesh adaptation, [6,1].

Mesh adaptation algorithm. The approach we use for the adaptive refinement of the spatial mesh is straightforward. Particularly, for the refinement criteria there exist much more sophisticated versions, which are not used in this paper for sake of simplicity. Let an error tolerance TOL be give. Then, on the basis of the (approximate) a posteriori error estimate (23), the mesh adaptation proceeds as follows:

1. Compute the primal solution U_h from (19) on the current mesh, starting from some initial state, e.g., that with zero deformation.
2. Compute the solution $\tilde{Z}_h$ of the approximate discrete dual problem (22).
3. Evaluate the cell-error indicators $\eta_K := \rho_K(U_h) \omega_K(\tilde{Z}_h)$.
4. If $\eta < TOL$ then accept U_h and evaluate $J(U_h)$, otherwise proceed to the next step.
5. Determine the 30% cells with largest and the 10% cells with smallest values of η_K . The cells of the first group are refined and those of the second group coarsened. Then, continue with Step 1. (Coarsening usually means canceling of an earlier refinement. Further refinement may be necessary to prevent the occurrence of too many hanging nodes. In two dimensions this strategy leads to about a doubling of the number of cells in each refinement cycle. By a similar strategy it can be achieved that the number of cells stays about constant during the adaptation process within a time stepping procedure.)

Remark 2. The error representation (21) has been derived assuming the error functional $J(\cdot)$ as linear. In many applications nonlinear, most often quadratic, error functionals occur. Examples are the spatial L^2 -norm error

$J(U_h) := \|(U - U_h)(T)\|^2$ at the end time T of a nonstationary computation or the least-squares cost-functional $\|U_h - \bar{U}\|^2$ of a stationary optimal control problem. For nonlinear (differentiable) error functionals the DWR approach can be extended to yield an error representation of the form (21); see [5,1].

Remark 3. The DWR method can also be applied to optimization problems of the form

$$\min_{q \in Q} J(u, q) \quad a(u, q)(\psi) = f(\psi) \quad \forall \psi \in V,$$

for instance in the context of an FSI setting. To this end we introduce the Lagrangian functional $\mathcal{L}(u, q, \lambda) := J(u, q) + f(\lambda) - a(u, q)(\lambda)$, with the adjoint variable $\lambda \in V$. Its stationary points $\{u, q, \lambda\}$ are possible solutions of the optimization problem. These are determined by the nonlinear variational equation (so-called KKT system)

$$\mathcal{L}'(u, q, \lambda)(\phi, \chi, \psi) = 0 \quad \forall \{\phi, \chi, \psi\} \in V \times Q \times V,$$

which has saddle-point character. For the solutions $\{u_h, q_h, \lambda_h\}$ of the corresponding finite element Galerkin approximation, there are residual-based error estimates available similar to (23); see [5,1].

A stationary special case. We will develop the explicit form of the error representation (21) and the approximate dual problem (22) for the *stationary* FSI model with an ‘incompressible neo-Hookean’ (INH) solid. We assume the system as being driven only by Dirichlet boundary conditions, i.e., volume and surface forces are zero, $f_3 \equiv 0$ and $g_3 \equiv 0$. Let $\{v, w, u, p\} \in \{v^D + V^0\} \times V^0 \times V^0 \times L$ be a steady state solution of the corresponding FSI model (12) determined by the system (14). In order to simplify the formulation of the corresponding dual problem, we omit higher order terms, e.g., in the Cauchy stress tensor for the structure. Since there is no movement in the structure domain, the mass conservation condition $\operatorname{div} v_s = 0$ is not practical for the sensitivity analysis. For this reason the conservation condition in the structure domain will be $\det(I - \nabla u) = 1$, from which we again omit the higher order terms by approximating $\det(I - \nabla u) - 1 \approx \operatorname{div} u = 0$. Then, the variational formulation of the stationary FSI problem further reduces to

$$A(U)(\Psi) = F(\Psi) \quad \forall \Psi \in W^0, \quad (24)$$

with the (time-independent) semilinear form

$$\begin{aligned} A(U)(\Psi) := & (\rho v \cdot \nabla v, \psi^v) + (\sigma(U), \epsilon(\Psi)) + (\chi_f \operatorname{div} v + \chi_s \operatorname{div} u, \psi^p) \\ & + (\chi_f u + \chi_s v, \psi^u), \end{aligned}$$

and the linear functional $F(\Psi) := (\chi_f u_f, \psi^u)$. Here, the stress-strain relation is given by

$$\sigma(U) = \begin{cases} -pI + 2\rho_f \nu_f \epsilon(v), & \text{in } \Omega_f, \\ -pI + 2\mu_s \epsilon(u), & \text{in } \Omega_s. \end{cases}$$

where the (small-strain) approximation $FF^T - I \approx 2\epsilon(\psi^u)$ has been used.

Suppose now that the discretization error is to be controlled with respect to some linear functional on W of the form $J(\Phi) = \{j^u(\phi^u), j^v(\phi^v), j^p(\phi^p)\}$. In order to correctly set up the corresponding dual problem, we would have to differentiate the semi-linear form $A(U)(\Psi)$ with respect to all components of U . However, this is not directly possible since the unknown position of the interface Γ_i depends on the displacement function u in a non-differentiable way. Because of this difficulty, we will adopt a more heuristic approach which is rather common in solving problems with free, i.e., only implicitly determined, boundaries. We assume that the interface obtained on the current mesh is already in good agreement with the correct one, $\Gamma_{ih} \approx \Gamma_i$, and set up the dual problem formally with Γ_{ih} as a fixed interface.

Adopting these simplifications and dropping all stabilization terms leads us to the following approximate dual problem: *Find* $Z = \{z^u, z^v, z^p\} \in W^0$, *such that*

$$\tilde{A}'(U_h)(\Phi, Z) = J(\Phi) \quad \forall \Phi \in W^0. \quad (25)$$

for all $\Phi = \{\phi^u, \phi^v, \phi^p\} \in W^0$, with the bilinear form

$$\begin{aligned} \tilde{A}'(U_h)(\Phi, Z) = & (\rho v_h \cdot \nabla \phi^v, z^v) + (\rho \phi^v \cdot \nabla v_h, z^v) + (\sigma'(U_h) \epsilon(\Phi), \epsilon(Z)) \\ & + (\chi_f \operatorname{div} \phi^v + \chi_s \operatorname{div} \phi^u, z^p) + (\chi_f \phi^u + \chi_s \phi^v, z^u), \end{aligned}$$

where

$$\sigma'(U_h) \epsilon(U) := \begin{cases} -pI + 2\rho_f \nu_f \epsilon(v), & \text{in } \Omega_{f,h}, \\ -pI + 2\mu_s \epsilon(u), & \text{in } \Omega_{s,h}. \end{cases}$$

Remark 4. The solvability of the dual problem (25) is not for granted and has to be established by exploiting the particular properties of the primal problem (24). This is a difficult task in view of the rather few existence results in the literature for general FSI problems. However, in our test calculation, we have never encountered difficulties in obtaining the ‘discrete’ dual solution. In the test example considered in Section 5, below, the discrete dual problem has been set up simply by transposing the Newton matrix at the final solution.

Let Z be a solution of (25) and Z_h its finite element approximation. To evaluate the approximate error estimate

$$J(E) \approx F(Z - Z_h) - A(U_h)(Z - Z_h),$$

for $E := U - U_h$, we introduce two modified submeshes

$$\mathbb{T}_{hs} := \{K \cap \Omega_{s,h}, K \in \mathbb{T}_h\}, \quad \mathbb{T}_{hf} := \{K \cap \Omega_{f,h}, K \in \mathbb{T}_h\},$$

and their union $\tilde{\mathbb{T}}_h := \mathbb{T}_{hs} \cup \mathbb{T}_{hf}$. The mesh $\tilde{\mathbb{T}}_h$ differs only from $\mathbb{T}_h$ in so far that the cells that contain the fluid-structure interface are subdivided into fluid domain part and structure domain part. Now, by cellwise integration by parts and rearranging boundary terms, we obtain

$$J(E) = \sum_{K \in \tilde{\mathbb{T}}_h} \left\{ (\operatorname{div} \sigma(U_h) - \rho v_h \cdot \nabla v_h, z^v - z_h^v)_K - \left(\frac{1}{2} [\sigma(U_h) \cdot n], z^v - z_h^v \right)_{\partial K} \right. \\ \left. - (\chi_f \operatorname{div} v_h + \chi_s \operatorname{div} u_h, z^p - z_h^p)_K - (\chi_f(u_h - u_f) + \chi_s v_h, z^u - z_h^u)_K \right\},$$

where $[\cdot]$ denotes the jump across intercell boundaries Γ . If Γ is part of the boundary $\partial\Omega$ the ‘jump’ is assigned the value $[\sigma \cdot n] = 2\sigma \cdot n$. We note that in this error representation the ‘cell residuals’ $\{\operatorname{div} \sigma(U_h) - \rho v_h \cdot \nabla v_h\}_K$, $\{\chi_f \operatorname{div} v_h + \chi_s \operatorname{div} u_h\}_K$, and $\{\chi_f(u_h - u_f) + \chi_s v_h\}_K$ represent the degree of consistency of the approximate solution U_h , while the ‘edge term’ $\frac{1}{2}[\sigma(U_h) \cdot n]$ measures its ‘discrete’ smoothness. These residual terms are multiplied by the weights (sensitivity factors) $z^v - z_h^v$, $z^p - z_h^p$, and $z^u - z_h^u$, respectively. From this error representation, we can deduce the following error estimate

$$|J(E)| \approx \sum_{K \in \tilde{\mathbb{T}}_h} \eta_K, \quad \eta_K := \sum_{i=1}^4 \rho_K^{(i)} \omega_K^{(i)}, \quad (26)$$

with the residual terms and weights

$$\begin{aligned} \rho_K^{(1)} &:= \|\operatorname{div} \sigma(U_h) - \rho v_h \cdot \nabla v_h\|_K, & \omega_K^{(1)} &:= \|Z - Z_h\|_K, \\ \rho_K^{(2)} &:= \frac{1}{2} h_K^{-1/2} \|[\sigma(U_h) \cdot n]\|_{\partial K}, & \omega_K^{(2)} &:= h_K^{1/2} \|z^v - z_h^v\|_{\partial K}, \\ \rho_K^{(3)} &:= \|\chi_f \operatorname{div} v_h + \chi_s \operatorname{div} u_h\|_K, & \omega_K^{(3)} &:= \|z^p - z_h^p\|_K, \\ \rho_K^{(4)} &:= \|\chi_f(u_h - u_f) + \chi_s v_h\|_K, & \omega_K^{(4)} &:= \|z^u - z_h^u\|_K. \end{aligned}$$

The weights $\omega^{(i)}$ are approximated, for instance by post-processing the discrete dual solution Z_h . Then, the cellwise error indicators η_K can be used for the mesh adaptation process as described above.

5 Numerical Test 1: Elastic Flow Cavity

For validating the numerical method described in the preceding section, we use a simple stationary test example, the lid-driven cavity with an elastic bottom wall, as shown in Figure 3. For simplicity, for modeling the fluid the linear Stokes equations are used and the material of the bottom wall is assumed to be linear neo-Hookean and incompressible. The structure material is taken as very soft such that a visible deformation of the fluid-structure interface can be expected. Then, the other material parameters are chosen such that flow and solid deformation velocity are small enough to allow for a

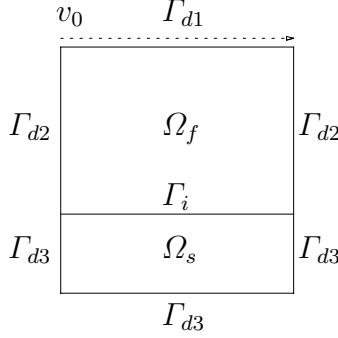


Fig. 3. Configuration of the ‘elastic’ lid-driven cavity.

stationary solution of the coupled linear systems. This solution is computed by a pseudo-time stepping method employing the implicit Euler scheme. A steady state is reached once the kinetic energy of the structure is below a prescribed small tolerance, here $\|v_s\|^2 \leq 10^{-8}$.

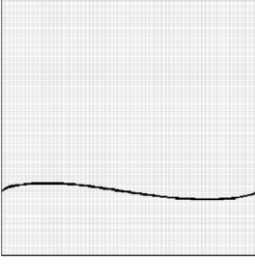
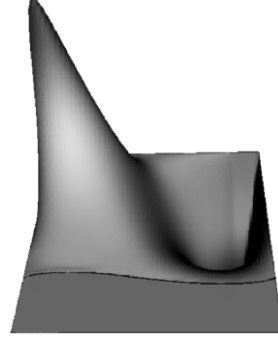
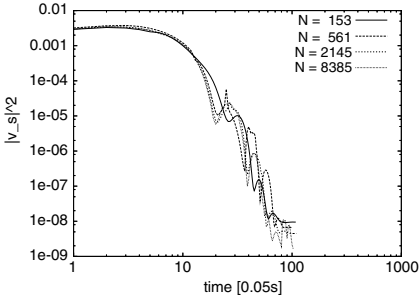
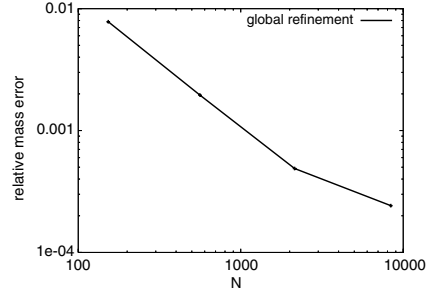
The cavity has a size of 2×2 , and its elastic part has a height of 0.5. The material constants are $\rho_f = \rho_s = 1$, $\nu_f = 0.2$, and $\mu_s = 2.0$. At the top boundary Γ_{d1} the regularized tangential flow profile

$$v_0 = 0.5 \begin{cases} 4x, & x \in [0.0, 0.25], \\ 1, & x \in (0.25, 1.75), \\ 4(2 - x), & x \in [1.75, 2.0], \end{cases}$$

is prescribed, in order to avoid problems due to pressure singularities.

5.1 Computations on Globally Refined Meshes

Figure 6 shows the development of $\|v_s\|^2$ during the pseudo-time stepping process depending on the number of cells of the mesh. As expected the kinetic energy tends to zero. The multiple ‘bumps’ occur due to the way the elastic structure reaches its stationary state by ‘swinging’ back and forth a few times. At the extreme point of each swing the kinetic energy has a local minimum. Figures 4 and 5 show the final stationary state computed on globally uniform meshes. In Figure 7, we monitor the mass error of the structure at the stationary state and find that it is actually of the expected order $O(h^2)$.

**Fig. 4.** Final position of interface.**Fig. 5.** Vertical velocity.**Fig. 6.** Variation of $\|v_s\|^2$ in time for different numbers N of mesh cells.**Fig. 7.** Relative error of mass conservation in the steady state on globally and locally refined meshes.

5.2 Computations on Locally Adapted Meshes

Next, we apply the simplified stationary version of the DWR method as described in Subsection 4.5 for local mesh adaptation in the present test problem. For the ‘goal-oriented’ a posteriori error estimation, we take the value of the pressure at the point $A = (0.5, 1.0)^T$ which is located in the flow region. To avoid sharp singularities in the corresponding dual solution, the associated functional is regularized to

$$J(u, p) = |K_A|^{-1} \int_{K_A} p \, dx \approx p(A),$$

where $K_A \in \mathbb{T}_h$ is a cell containing the point A . As reference value of $p(A)$, we use the result obtained on a very fine uniform mesh.

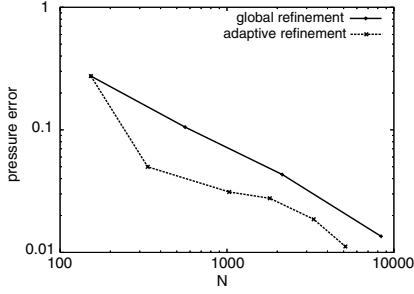


Fig. 8. Reduction of the pressure value error.

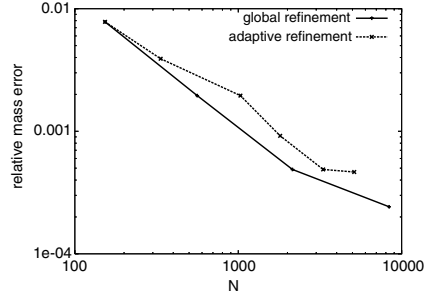


Fig. 9. Relative error of mass conservation.

In Figures 8 and 9 the resulting pressure error and the relative error in mass conservation is displayed as a function of the number of mesh cells. Figure 10 shows a sequence of adapted meshes. As expected two effects can be seen. There is local refinement around the point of interest and since the position of the fluid-structure interface is a decisive factor for the pressure field, local refinement also occurs along the interface.

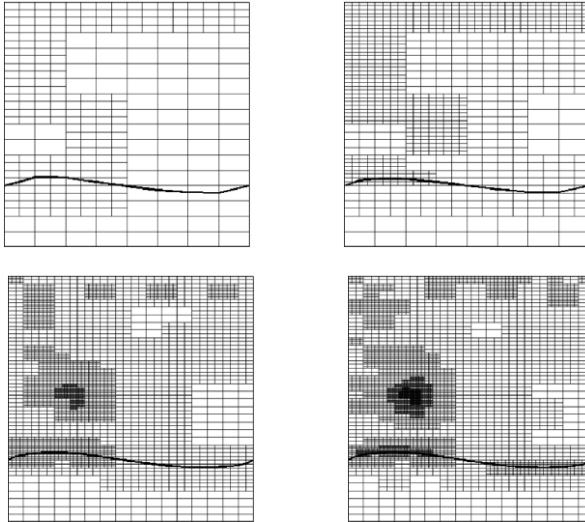


Fig. 10. Locally adapted meshes with $N = 335, 1031, 3309, 5123$ cells.

It may seem surprising that in Figure 9 there is no reduction of the mass error in the last iteration. This is due to the approach we are using here. After each step of mesh adaption a new primal solution is calculated, starting with the initial state of no deformation. The sensitivity analysis though does not take the initial state into account. Mesh adaption takes place around the

final state of the interface, it does not reflect its initial state. An easy way of alleviating the mass error problem is to explicitly move a certain amount of local refinements with the interface from one time step to the next. Doing that though in this example would have made it unclear if the local refinement at the final interface position was due to the sensitivity analysis or the explicit movement of interface-bound refinement.

6 Numerical Test 2: FSI Benchmark FLUSTRUK-A

The second example is the FSI benchmark FLUSTRUK-A described in [15]. A thin elastic bar immersed in an incompressible fluid develops self-induced time-periodic oscillations of different amplitude depending on the material properties assumed. This benchmark has been defined to validate and compare the different computational approaches and software implementations for solving FSI problems. In order to have a fair comparison of our Eulerian-based method with the traditional Eulerian-Lagrangian approach, we have also implemented an ALE method for this benchmark problem.

The configuration of this benchmark shown in Figure 11 is based on the successful CFD benchmark ‘flow around a cylinder’, [30].

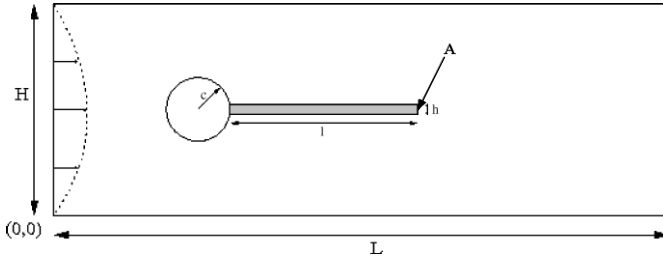


Fig. 11. Configuration of the FSI benchmark ‘FLUSTRUK-A’.

Configuration: The computational domain has length $L = 2.5$, height $H = 0.41$, and left bottom corner at $(0, 0)$. The center of the circle is positioned at $C = (0.2, 0.2)$ with radius $r = 0.05$. The elastic bar has length $l = 0.35$ and height $h = 0.02$. Its right lower end is positioned at $(0.6, 0.19)$ and its left end is clamped to the circle. Control points are $A(t)$ fixed at the trailing edge of the structure with $A(0) = (0.6, 0.20)$, and $B = (0.15, 0.2)$ fixed at the cylinder (stagnation point).

Boundary and initial conditions: The boundary conditions are as follows: Along the upper and lower boundary the usual ‘no-slip’ condition is used for

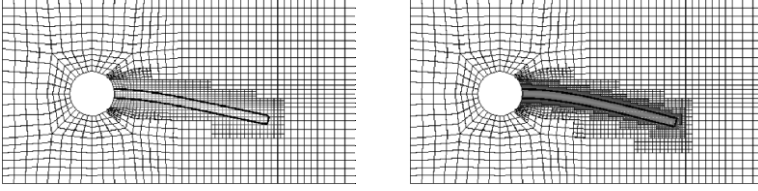


Fig. 12. CSM test: Stationary position of the control point A on heuristically refined meshes with $N = 1952$ and $N = 7604$ cells.

the velocity. At the (left) inlet a constant parabolic inflow profile,

$$v(0, y) = 1.5 \bar{U} \frac{4y(H - y)}{H^2},$$

is prescribed which drives the flow, and at the (right) outlet zero-stress $\sigma \cdot n = 0$ is realized by using the ‘do-nothing’ approach in the variational formulation, [13,23]. This implicitly forces the pressure to have zero mean-value at the outlet. The initial condition is zero flow velocity and structure displacement.

Material properties: The fluid is assumed as incompressible and Newtonian, the cylinder as fixed and rigid, and the structure as (compressible) St. Venant-Kirchhoff (STVK) type.

Discretization: The first set of computations is done on globally refined meshes for validating the proposed method and its software implementation. Then, for the same configuration adaptive meshes are used where the refinement criteria are either purely heuristic, i.e., based on the cell distance from the interface, or are based on a simplified stationary version of the DWR approach (at every tenth time step) as already used before for the cavity example. In all cases a uniform time-step size of 0.005 s is used. The curved cylinder boundary is approximated to second order by polygonal mesh boundaries as can be seen in Figure 12.

The following four different test cases are considered:

- *Computational fluid dynamics test (CFD Test):* The structure is made very stiff, to the effect that we can compare the computed drag and lift coefficients with those obtained for a pure CFD test (with rigid structure).
- *Computational structure mechanics test (CSM Test):* The fluid is set to be initially in rest around the bar. The deformation of the bar under a vertical gravitational force is compared to the deformation of the same bar in a pure CSM test.
- *FSI tests:* Three configurations are treated corresponding to different inflow velocities and material stiffness parameters, and the Eulerian approach is compared to the standard ALE method.

- *FSI with large deflections*: The fluid is set to be initially in rest around the bar. The gravitational force on the bar is very large, causing a large deformation of the bar and eventually it reaching and running up against the channel wall. This case is difficult for the ALE method but can easily be handled by the Eulerian approach.

6.1 CFD Test

Here, the structure is set to be very stiff, to the effect that we can compare derived drag and lift values with those obtained with a pure CFD approach. The forces are calculated based on the closed path S around the whole structure, cylinder and bar,

$$J(u, p) := \int_S \sigma_f \cdot n \, do. \tag{27}$$

The CFD test has been done with the parameters listed in Table 1.

Table 1. Parameters for the CFD test.

Parameters	CFD test
$\rho_f [10^3 kg \, m^{-3}]$	1
$\nu_f [10^{-3} m^2 s^{-1}]$	1
ν_s	0.4
$\rho_s [10^6 kg \, m^{-3}]$	1
$\mu_s [10^{12} kg \, m^{-1} s^{-2}]$	1
$U [m \, s^{-1}]$	1

For the chosen parameters there is a steady state solution. The reference values for the drag and lift forces are calculated using a pure CFD approach on globally refined meshes (see also [15]). The results are shown in Table 2. Using the Eulerian FSI approach, we calculate the same forces again. As a method of mesh adaption we use a heuristic approach as described above.

Table 2. CFD test: Results of CFD computation on uniform meshes (left), and by the Eulerian FSI approach on heuristically adapted meshes (right).

N	dof	drag	lift
1278	3834	145.75	10.042
4892	14676	133.91	10.239
19128	57384	136.00	10.373
75632	226896	136.54	10.366
300768	902304	136.67	10.369
∞	∞	136.70	10.530

N	dof	drag	lift
1300	9100	122.66	12.68
2334	16338	126.13	11.71
9204	64428	131.77	10.53
36680	256760	134.47	10.45
∞	∞	136.70	10.530

Table 3. Parameters for the CSM test.

parameter	CSM test
$\rho_f [10^3 kg m^{-3}]$	1
$\nu_f [10^{-3} m^2 s^{-1}]$	1
ν_s	0.4
$\rho_s [10^3 kg m^{-3}]$	1
$\mu_s [10^6 kg m^{-1} s^{-2}]$	0.5
$\bar{U} [m s^{-1}]$	0
$g [m s^{-2}]$	2

Table 4. CSM test: Displacement of the control point A for three levels of heuristic mesh adaption.

N	dof	$u_x(A) [10^{-3}m]$	$u_y(A) [10^{-3}m]$
1952	13664	-5.57	-59.3
3672	25704	-6.53	-63.4
7604	53228	-6.74	-64.6
∞	∞	-7.187	-66.10

6.2 CSM Test

Here, the inflow velocity is set to zero and the fluid is initially at rest. A vertical gravitational force is applied, which causes the bar to slowly sink in the fluid filled volume. Due to the viscous effect of the fluid the bar will eventually come to rest. The value of final displacement can be compared to the results calculated with a pure CSM approach in a Lagrangian framework. The quantity of interest is the displacement of the point A at the middle of the trailing tip. The corresponding reference values are taken from [15]. The CSM test has been done with the parameters listed in Table 3. Using the Eulerian FSI approach, we calculate the displacements with mesh adaption by the heuristic approach described above. The final stationary positions and the heuristically adapted meshes can be seen in Figure 12.

Next, we apply the DWR method as described above to the CSM test case. In the dual problem, we use the Jacobi matrix of the model as presented in Section 4. In the first example the DWR method was always applied to the final stationary state. The results were used for mesh adaption. The generated mesh was then used with the initially unperturbed problem to determine a new final stationary state. In contrast to that approach, we now apply the DWR method at periodic intervals without restarting. To control the resulting mesh adaption at each interval we try to keep the number of nodes N below a certain threshold N_t . This is achieved by reducing refinement and/or increasing coarsening at each interval. As an example we calculate the point-value of the component sum of $u(A)$ at the control point A . The

position x_A is determined from $x_A - u(x_A) = A(0) = (0.6, 0.2)^T$. As a error control functional, we use a regularized delta function at x_A applied to $(e_1 + e_2)^T u$,

$$J(u) = |K_A|^{-1} \int_{K_A} (e_1 + e_2)^T u(x) dx ,$$

where K_A is the cell in the Mesh $\mathbb{T}_h$ containing the point A .

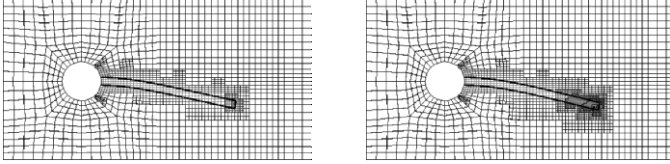


Fig. 13. CSM Test: Stationary position of the bar computed on locally refined meshes (DWR method) with $N = 2016$ and $N = 4368$ cells.

Table 5. CSM Test: Displacements of the control point A for three levels of locally refined meshes (DWR method).

N_t	N	dof	$u_x(A)$ [$10^{-3}m$]	$u_y(A)$ [$10^{-3}m$]
2000	2016	14112	-5.73	-59.8
3000	2614	18298	-6.54	-63.2
4500	4368	30576	-6.88	-64.6
	∞	∞	-7.187	-66.10

6.3 FSI Tests

Three test cases, FSI-2, FSI-3, and FSI-3*, are treated with different inflow velocities and material stiffness values as stated in Table 6. The parameters are chosen such that a visible transient behavior of the bar can be seen. To ensure a ‘fair’ comparison of results, we calculate the comparison values using the ALE method. Using the Eulerian FSI approach, we calculate the displacements on three mesh levels, where the heuristic approach as described above is used for mesh refinement.

We begin with the FSI-2 and FSI-3 test cases. Some snapshots of the results of these simulations are shown in Figures 14 and 16. The time-dependent behavior of the displacements for the tests are shown in Figures 15 and 17.

The FSI-3* test case is used to illustrate some special features of the Eulerian solution approach. Figure 18 illustrates the treatment of corners in the structure by the IP set approach compared to the LS approach. In the LS method the interface is identified by all points for which $\phi = 0$, while in the

Table 6. Parameter settings for the FSI test cases.

parameter	FSI-2	FSI-2*	FSI-3	FSI-3*
structure model	STVK	STVK	STVK	INH
$\rho_f [10^3 kg m^{-3}]$	1	1	1	1
$\nu_f [10^{-3} m^2 s^{-1}]$	1	1	1	1
ν_s	0.4	0.4	0.4	0.5
$\rho_s [10^3 kg m^{-3}]$	10	20	1	1
$\mu_s [10^6 kg m^{-1} s^{-2}]$	0.5	0.5	2	2
$\bar{U} [m s^{-1}]$	1	0	2	2

IP set method the interface is identified by all points which are on one of the

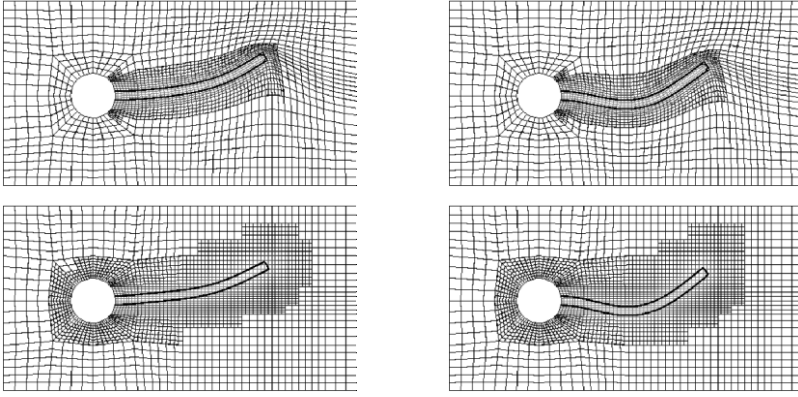


Fig. 14. FSI-2: Snapshots of results obtained by the ALE (top two) and by the Eulerian (bottom two) approaches.

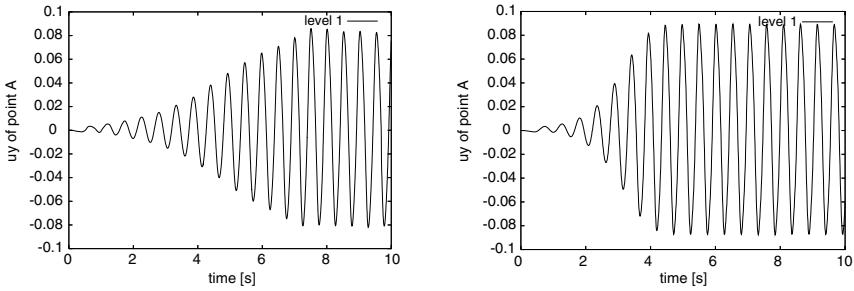


Fig. 15. FSI-2: Vertical displacement of the control point A , obtained by the Eulerian approach (left, $N = 2082$ cells) with maximum amplitude $2.226 \cdot 10^{-2}$ and frequency $1.92 s^{-1}$, and by the ALE approach (right, $N = 2784$ cells) with maximum amplitude $2.68 \cdot 10^{-2}$ and frequency $1.953 s^{-1}$.

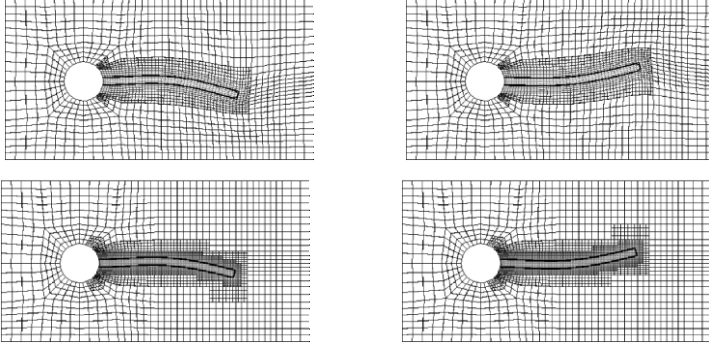


Fig. 16. FSI-3 Test: Some snapshots of results obtained by the ALE (top two) and the Eulerian (bottom two) approaches.

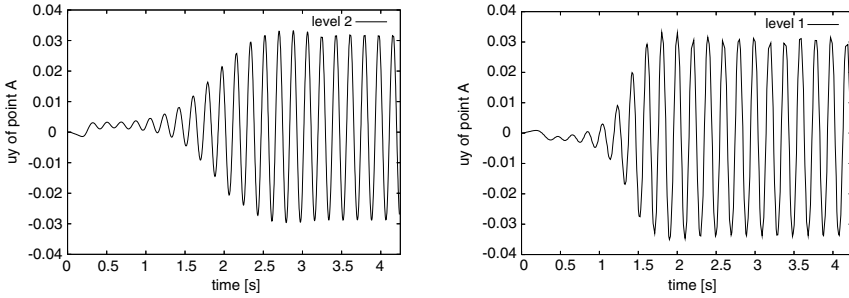


Fig. 17. FSI-3 Test: Vertical Displacement of the control point A , obtained by the Eulerian approach (left, $N = 3876$ cells) with maximum amplitude $6.01 \cdot 10^{-2}$ and frequency 5.48 s^{-1} , and by the ALE approach (right, $N = 2082$ cells) with maximum amplitude $6.37 \cdot 10^{-2}$ and frequency 5.04 s^{-1} .

respective isoline segments belonging to the edges of the bar. The differences are visible in the cells that contain the corners.

Since in the Eulerian approach the structure deformations are not in a Lagrangian framework, it is not immediately clear, due to the coupling with the fluid, how well the mass of the structure is conserved in an Eulerian approach, especially in the course of an instationary simulation comprising hundreds of time steps. In Figure 19, we display the bar's relative mass error as a function of time. Except for certain initial jitters, the relative error is less than 1%.

Finally, Figure 20 illustrates the time dynamics of the structure and the adapted meshes over the time interval $[0, T]$. More detailed properties of this dynamics is shown in Figure 20. For both approaches, we obtain a periodic

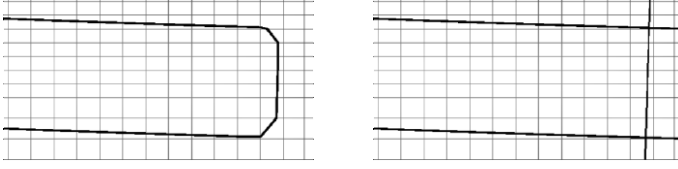


Fig. 18. FSI-3*: Treatment of corners by the LS method (left) and by the IP set method (right).

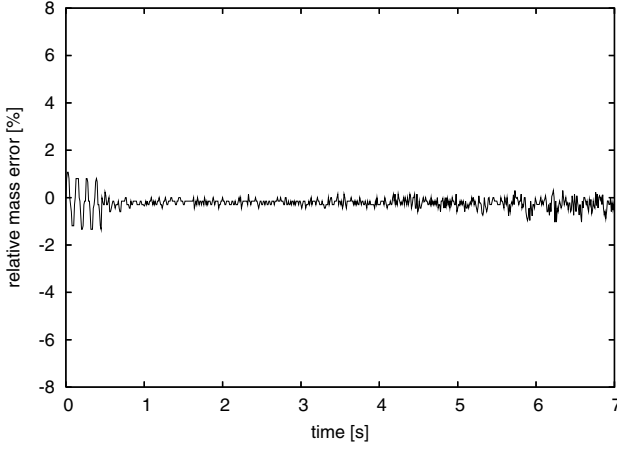


Fig. 19. FSI-3*: Relative mass error of the bar.

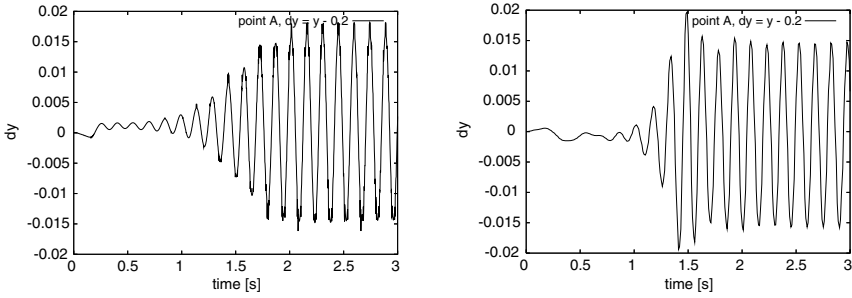


Fig. 20. FSI-3*: Vertical displacement of the control point A , obtained by the Eulerian approach (left) with maximum amplitude $1.6 \cdot 10^{-2}$ and frequency 6.86 s^{-1} , and by the ALE approach (right) with maximum amplitude $1.51 \cdot 10^{-2}$ and frequency 6.70 s^{-1} .

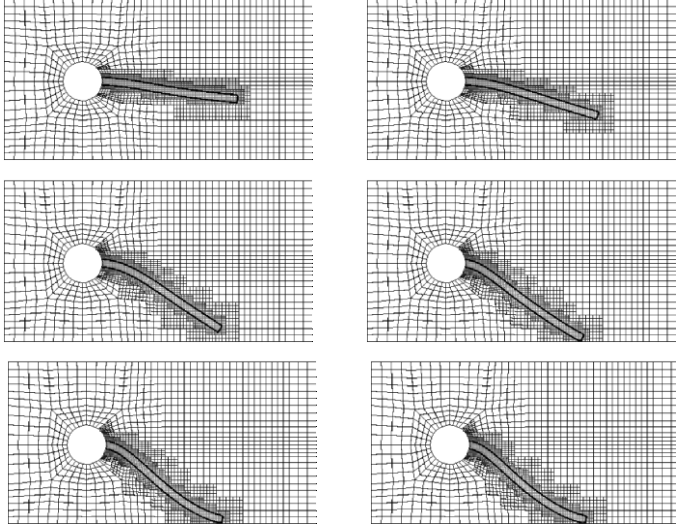


Fig. 21. A sequence of snap-shots of the bar’s large deformation under gravitational loading obtained by the Eulerian approach.

oscillation. For the Eulerian approach we obtain an amplitude of $1.6\text{e-}2$ with an oscillation frequency of 6.86s^{-1} . In comparison to that, based on the ALE approach, we obtain an amplitude of $1.51\text{e-}2$ with an oscillation frequency of 6.70s^{-1} .

6.4 FSI Test with Large Deformations

In the test case FSI-2* (see Table 6) the fluid is initially in rest and the bar is subjected to a vertical force. This causes the bar to bend downward until it touches the bottom wall. A sequence of snapshots of the transition to steady state obtained by the Eulerian approach for this problem is shown in Figure 21. The position of the trailing tip A is shown in Figure 22.

7 Summary and Future Development

In this paper we presented a fully Eulerian variational formulation for ‘fluid-structure interaction’ (FSI) problems. This approach uses the ‘initial position’ set (IP set) method for interface capturing, which is similar to the ‘level set’ (LS) method, but preserves sharp corners of the structure. The harmonic continuation of the structure velocity avoids the need of reinitialization of the IP set. This approach allows us to treat FSI problems with free bodies and large deformations. This is the main advantage of this method compared to interface tracking methods such as the arbitrary Lagrangian-Eulerian (ALE)

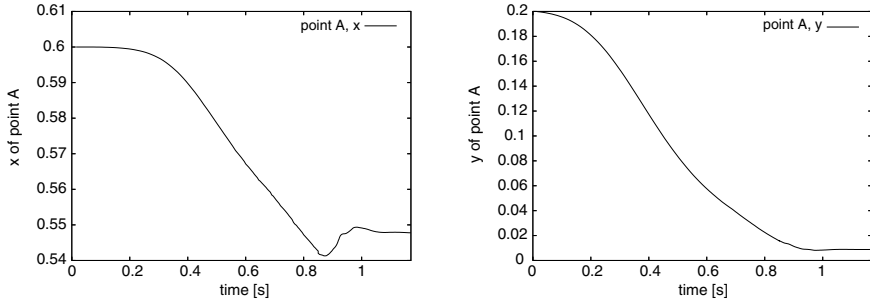


Fig. 22. FSI-2*: Position of the control point A during the deformation of the bar.

method. At several examples the Eulerian approach turns out to yield results which are in good agreement with those obtained by the ALE approach. In order to have a ‘fair’ comparison both methods have been implemented using the same numerical components and software library GASCOIGNE [33]. The method based on the Eulerian approach is inherently more expensive than the ALE method, by about a factor of two, but it allows to treat also large deformations and topology changes.

The full variational formulation of the FSI problem provides the basis for the application of the ‘dual weighted residual’ (DWR) method for ‘goal-oriented’ a posteriori error estimation and mesh adaptation. In this method inherent sensitivities of the FSI problem are utilized by solving linear ‘dual’ problems, similar as in the Euler-Lagrange approach to solving optimal control problems. The feasibility of the DWR method for FSI problems has, in a first step, been demonstrated for the computation of steady state solutions.

In the next step, we will employ the DWR method also for nonstationary FSI problems, particularly for the simultaneous adaptation of spatial mesh and time step size. Another goal is the development of the Eulerian approach for three-dimensional FSI problems and to explore its potential for FSI problems with large deformations and topology changes. Finally, we intend to extend the variational Eulerian approach and the DWR method to the optimal control of FSI systems.

8 Acknowledgement

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A Monolithic FEM/Multigrid Solver for an ALE Formulation of Fluid-Structure Interaction with Applications in Biomechanics

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Abstract. We investigate a new method of solving the problem of fluid-structure interaction of an incompressible elastic object in laminar incompressible viscous flow. Our proposed method is based on a fully implicit, monolithic formulation of the problem in the arbitrary Lagrangian-Eulerian framework. High order FEM is used to obtain the discrete approximation of the problem. In order to solve the resulting systems a quasi-Newton method is applied with the linearized systems being approximated by the divided differences approach. The linear problems of saddle-point type are solved by a standard geometric multigrid with local multilevel pressure Schur complement smoothers.

1 Overview

We consider the problem of viscous fluid flow interacting with an elastic body which is being deformed by the fluid action. Such a problem is encountered in many real life applications of great importance. Typical examples of this type of problem are the areas of aero-elasticity, biomechanics or material processing. For example, a good mathematical model for biological tissue could be used in such areas as early recognition or prediction of heart muscle failure, advanced design of new treatments and operative procedures, and the understanding of atherosclerosis and associated problems. Other possible applications include the development of virtual reality programs for training new surgeons or designing new operative procedures (see [1]).

1.1 Fluid Structure Models in Biomechanics

There have been several different approaches to the problem of fluid-structure interaction. Most notably these include the work of [2–5] where an immersed boundary method was developed and applied to a three-dimensional model of the heart. In this model they consider a set of one-dimensional elastic fibers immersed in a three-dimensional fluid region and using a parallel supercomputer they were able to model the pulse of the heart ventricle.

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A fluid-structure model with the wall modelled as a thin shell was used to model the left heart ventricle in [6,7] and [8,9]. In [10,11] a similar approach was used to model the flow in a collapsible tube. In these models the wall is modelled by two-dimensional thin shells which can be modified to capture the anisotropy of the muscle. In reality the thickness of the wall can be significant and very important. For example in arteries the wall thickness can be up to 30% of the diameter and its local thickening can lead to the creation of an aneurysm. In the case of heart ventricle the thickness of the wall is also significant and also the direction of the muscle fibers changes through the wall.

1.2 Theoretical Results

The theoretical investigation of fluid structure interaction problems is complicated by the need of mixed description. While for the solid part the natural view is the material (Lagrangian) description, for the fluid it is the spatial (Eulerian) description. In the case of their combination some kind of mixed description (usually referred to as the arbitrary Lagrangian-Eulerian description or ALE) has to be used which brings additional nonlinearity into the resulting equations.

In [12] a time dependent, linearized model of interaction between a viscous fluid and an elastic shell in small displacement approximation and its discretization is analyzed. The problem is further simplified by neglecting all changes in the geometry configuration. Under these simplifications by using energy estimates they are able to show that the proposed formulation is well posed and a global weak solution exists. Further they show that an independent discretization by standard mixed finite elements for the fluid and by nonconforming discrete Kirchhoff triangle finite elements for the shell together with backward or central difference approximation of the time derivatives converges to the solution of the continuous problem.

In [13] a steady problem of equilibrium of an elastic fixed obstacle surrounded by a viscous fluid is studied. Existence of an equilibrium state is shown with the displacement and velocity in $C^{2,\alpha}$ and pressure in $C^{1,\alpha}$ under assumption of small data in $C^{2,\alpha}$ and domain boundaries of class C^3 .

A numerical solution of the resulting equations of the fluid structure interaction problem poses a great challenge since it includes the features of nonlinear elasticity, fluid mechanics and their coupling. The easiest solution strategy, mostly used in the available software packages, is to decouple the problem into the fluid part and solid part, for each of those parts to use some well established method of solution then the interaction is introduced as external boundary conditions in each of the subproblems. This has an advantage that there are many well tested finite element based numerical methods for separate problems of fluid flow and elastic deformation, on the other hand the treatment of the interface and the interaction is problematic. The approach presented here treats the problem as a single continuum with

the coupling automatically taken care of as internal interface, which in our formulation does not require any special treatment.

2 Continuum Description

Let $\Omega \subset \mathbb{R}^3$ be a reference configuration of a given body. Let $\Omega_t \subset \mathbb{R}^3$ be a configuration of this body at time t . Then a one-to-one, sufficiently smooth mapping χ_Ω of the reference configuration Ω to the current configuration

$$\chi_\Omega : \Omega \times [0, T] \mapsto \mathbb{R}^3, \quad (1)$$

describes the motion of the body, see Figure 1. The mapping χ_Ω depends on the choice of the reference configuration Ω which can be fixed in a various ways. Here we think of Ω to be the initial (stress-free) configuration Ω_0 . Thus, if not emphasized, we mean by χ exactly $\chi_\Omega = \chi_{\Omega_0}$.

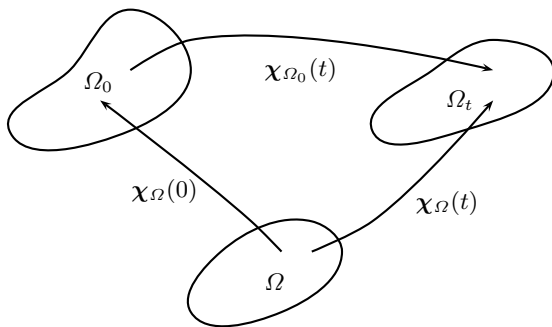


Fig. 1. The referential domain Ω , initial Ω_0 and current state Ω_t and relations between them. The identification $\Omega \equiv \Omega_0$ is adopted in this text.

If we denote by $\mathbf{X}$ a material point in the reference configuration Ω then the position of this point at time t is given by

$$\mathbf{x} = \chi(\mathbf{X}, t). \quad (2)$$

Next, the mechanical fields describing the deformation are defined in a standard manner. The displacement field, the velocity field, deformation gradient and its determinant are

$$\mathbf{u}(\mathbf{X}, t) = \chi(\mathbf{X}, t) - \mathbf{X}, \quad \mathbf{v} = \frac{\partial \chi}{\partial t}, \quad \mathbf{F} = \frac{\partial \chi}{\partial \mathbf{X}}, \quad J = \det \mathbf{F}. \quad (3)$$

Let us adopt the following useful notations for some derivatives. Any field quantity φ with values in some vector space Y (i.e. scalar, vector or tensor valued) can be expressed in the Eulerian description as a function of the spatial position $\mathbf{x} \in \mathbb{R}^3$

$$\varphi = \tilde{\varphi}(\mathbf{x}, t) : \Omega_t \times [0, T] \mapsto Y.$$

Then we define following notations for the derivatives of the field φ

$$\frac{\partial \varphi}{\partial t} := \frac{\partial \tilde{\varphi}}{\partial t}, \quad \nabla \varphi = \frac{\partial \varphi}{\partial \mathbf{x}} := \frac{\partial \tilde{\varphi}}{\partial \mathbf{x}}, \quad \operatorname{div} \varphi := \operatorname{tr} \nabla \varphi. \quad (4)$$

In the case of Lagrangian description we consider the quantity φ to be defined on the reference configuration Ω , then for any $\mathbf{X} \in \Omega$ we can express the quantity φ as

$$\varphi = \bar{\varphi}(\mathbf{X}, t) : \Omega \times [0, T] \mapsto Y,$$

and we define the derivatives of the field φ as

$$\frac{d\varphi}{dt} := \frac{\partial \bar{\varphi}}{\partial t}, \quad \operatorname{Grad} \varphi = \frac{\partial \varphi}{\partial \mathbf{X}} := \frac{\partial \bar{\varphi}}{\partial \mathbf{X}}, \quad \operatorname{Div} \varphi := \operatorname{tr} \operatorname{Grad} \varphi. \quad (5)$$

These two descriptions can be related to each other through following relations

$$\bar{\varphi}(\mathbf{X}, t) = \tilde{\varphi}(\chi(\mathbf{X}, t), t), \quad (6)$$

$$\frac{d\varphi}{dt} = \frac{\partial \varphi}{\partial t} + (\nabla \varphi) \mathbf{v}, \quad \operatorname{Grad} \varphi = (\nabla \varphi) \mathbf{F}, \quad \int_{\Omega_t} \varphi dv = \int_{\Omega} \varphi J dV \quad (7)$$

$$\frac{d\mathbf{F}}{dt} = \operatorname{Grad} \mathbf{v}, \quad \frac{\partial J}{\partial \mathbf{F}} = J \mathbf{F}^{-T}, \quad \frac{dJ}{dt} = J \operatorname{div} \mathbf{v}. \quad (8)$$

For the formulation of the balance laws we will need to express a time derivatives of some integrals. The following series of equalities obtained by using the previously stated relations will be useful

$$\begin{aligned} \frac{d}{dt} \int_{\Omega_t} \varphi dv &= \frac{d}{dt} \int_{\Omega} \varphi J dV = \int_{\Omega} \frac{d}{dt} (\varphi J) dV = \int_{\Omega_t} \left(\frac{d\varphi}{dt} + \varphi \operatorname{div} \mathbf{v} \right) dv \\ &= \int_{\Omega_t} \left(\frac{\partial \varphi}{\partial t} + \operatorname{div} (\varphi \mathbf{v}) \right) dv = \int_{\Omega_t} \frac{\partial \varphi}{\partial t} dv + \int_{\partial \Omega_t} \varphi \mathbf{v} \cdot \mathbf{n} da \\ &= \frac{\partial}{\partial t} \int_{\Omega_t} \varphi dv + \int_{\partial \Omega_t} \varphi \mathbf{v} \cdot \mathbf{n} da. \end{aligned} \quad (9)$$

And also the Piola identity will be used, $\operatorname{Div}(J \mathbf{F}^{-T}) = \mathbf{0}$, which can be checked by differentiating the left hand side and using (8) together with an identity obtained by differentiating the relation $\mathbf{F} \mathbf{F}^{-1} = \mathbf{I}$.

2.1 Balance Laws

In this section we will formulate the balance relations for mass and momentum in three forms: the Eulerian, the Lagrangian and the arbitrary Eulerian-Lagrangian (ALE) description.

The Eulerian (or spatial) description is well suited for a problem of fluid flowing through some spatially fixed region. In such a case the material particles can enter and leave the region of interest. The fundamental quantity describing the motion is the velocity vector.

On the other hand the Lagrangian (or referential) description is well suited for a problem of deforming a given body consisting of a fixed set of material particles. In this case the actual boundary of the body can change its shape. The fundamental quantity describing the motion in this case is the vector of displacement from the referential state.

In the case of fluid-structure interaction problems we can still use the Lagrangian description for the deformation of the solid part. The fluid flow now takes place in a domain with boundary given by the deformation of the structure which can change in time and is influenced back by the fluid flow. The mixed ALE description of the fluid has to be used in this case. The fundamental quantity describing the motion of the fluid is still the velocity vector but the description is accompanied by a certain displacement field which describes the change of the fluid domain. This displacement field has no connection to the fluid velocity field and the purpose of its introduction is to provide a transformation of the current fluid domain and corresponding governing equations to some fixed reference domain. This method is sometimes called a pseudo-solid mapping method, see [14].

Let $\mathcal{P} \subset \mathbb{R}^3$ be a fixed region in space (a control volume) with the boundary $\partial\mathcal{P}$ and unit outward normal vector $\mathbf{n}_{\mathcal{P}}$, such that

$$\mathcal{P} \subset \Omega_t \quad \text{for all } t \in [0, T].$$

Let ϱ denote the mass density of the material. Then the balance of mass in the region $\mathcal{P}$ can be written as

$$\frac{\partial}{\partial t} \int_{\mathcal{P}} \varrho dv + \int_{\partial\mathcal{P}} \varrho \mathbf{v} \cdot \mathbf{n}_{\mathcal{P}} da = 0. \quad (10)$$

If all the fields are sufficiently smooth this equation can be written in local form with respect to the current configuration as

$$\frac{\partial \varrho}{\partial t} + \operatorname{div}(\varrho \mathbf{v}) = 0. \quad (11)$$

It will be useful to derive the mass balance equation from the Lagrangian point of view. Let $\mathcal{Q} \subset \Omega$ be a fixed set of particles. Then $\chi(\mathcal{Q}, t) \subset \Omega_t$ is a region occupied by these particles at the time t , and the balance of mass can be expressed as

$$\frac{d}{dt} \int_{\chi(\mathcal{Q}, t)} \varrho dv = 0, \quad (12)$$

which in local form w.r.t. the reference configuration can be written as

$$\frac{d}{dt}(\varrho J) = 0. \quad (13)$$

In the case of an arbitrary Lagrangian-Eulerian description we take a region $\mathcal{Z} \subset \mathbb{R}^3$ which is itself moving independently of the motion of the body. Let the motion of the control region $\mathcal{Z}$ be described by a given mapping

$$\zeta_{\mathcal{Z}} : \mathcal{Z} \times [0, T] \mapsto \mathbb{R}^3, \quad \mathcal{Z}_t \subset \Omega_t \quad \forall t \in [0, T],$$

with the corresponding velocity $\mathbf{v}_{\mathcal{Z}} = \frac{\partial \zeta_{\mathcal{Z}}}{\partial t}$, deformation gradient $\mathbf{F}_{\mathcal{Z}} = \frac{\partial \zeta_{\mathcal{Z}}}{\partial \mathbf{X}}$ and its determinant $J_{\mathcal{Z}} = \det \mathbf{F}_{\mathcal{Z}}$. The mass balance equation can be written as

$$\frac{\partial}{\partial t} \int_{\mathcal{Z}_t} \varrho dv + \int_{\partial \mathcal{Z}_t} \varrho (\mathbf{v} - \mathbf{v}_{\mathcal{Z}}) \cdot \mathbf{n}_{\mathcal{Z}_t} da = 0, \quad (14)$$

this can be viewed as an Eulerian description with a moving spatial coordinate system or as a grid deformation in the context of the finite element method. In order to obtain a local form of the balance relation we need to transform the integration to the fixed spatial region $\mathcal{Z}$

$$\frac{\partial}{\partial t} \int_{\mathcal{Z}} \varrho J_{\mathcal{Z}} dv + \int_{\partial \mathcal{Z}} \varrho (\mathbf{v} - \mathbf{v}_{\mathcal{Z}}) \cdot \mathbf{F}_{\mathcal{Z}}^{-T} \mathbf{n}_{\mathcal{Z}} J_{\mathcal{Z}} da = 0, \quad (15)$$

then the local form is

$$\frac{\partial}{\partial t} (\varrho J_{\mathcal{Z}}) + \operatorname{div} (\varrho J_{\mathcal{Z}} (\mathbf{v} - \mathbf{v}_{\mathcal{Z}}) \cdot \mathbf{F}_{\mathcal{Z}}^{-T}) = 0. \quad (16)$$

The two previous special formulations can be now recovered. If the region $\mathcal{Z}$ is not moving in space, i.e. $\mathcal{Z} = \mathcal{Z}_t, \forall t \in [0, T]$, then $\zeta_{\mathcal{Z}}$ is the identity mapping, $\mathbf{F}_{\mathcal{Z}} = \mathbf{I}$, $J_{\mathcal{Z}} = 1$, $\mathbf{v}_{\mathcal{Z}} = \mathbf{0}$ and (16) reduces to (11). If the region $\mathcal{Z}$ moves exactly with the material, i.e. $\zeta_{\mathcal{Z}} = \chi|_{\mathcal{Z}}$ then $\mathbf{F}_{\mathcal{Z}} = \mathbf{F}$, $J_{\mathcal{Z}} = J$, $\mathbf{v}_{\mathcal{Z}} = \mathbf{v}$ and (16) reduces to (13).

The balance of linear momentum is postulated in a similar way. Let $\boldsymbol{\sigma}$ denote the Cauchy stress tensor field, representing the surface forces per unit area, $\mathbf{f}$ be the body forces acting on the material per unit mass. Then the balance of linear momentum in the Eulerian description is stated as

$$\frac{\partial \varrho \mathbf{v}}{\partial t} + \operatorname{div} (\varrho \mathbf{v} \otimes \mathbf{v}) = \operatorname{div} \boldsymbol{\sigma}^T + \varrho \mathbf{f}, \quad (17)$$

or with the use of (11) we can write

$$\varrho \frac{\partial \mathbf{v}}{\partial t} + \varrho (\nabla \mathbf{v}) \mathbf{v} = \operatorname{div} \boldsymbol{\sigma}^T + \varrho \mathbf{f}. \quad (18)$$

From the Lagrangian point of view the momentum balance relation is

$$\frac{d}{dt} (\varrho J \mathbf{v}) = \operatorname{Div} (J \boldsymbol{\sigma}^T \mathbf{F}^{-T}) + \varrho J \mathbf{f}, \quad (19)$$

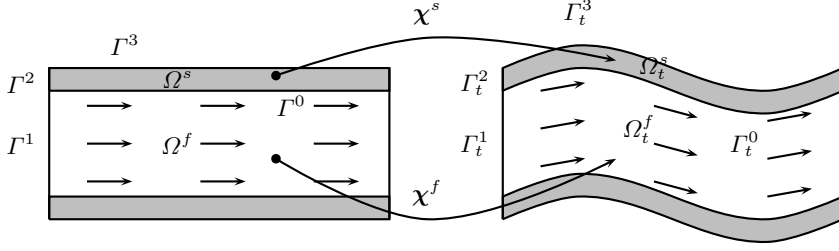


Fig. 2. Undeformed (original) and deformed (current) configurations.

or using (13) we can write

$$\varrho J \frac{d\mathbf{v}}{dt} = \text{Div} (J \boldsymbol{\sigma}^T \mathbf{F}^{-T}) + \varrho J \mathbf{f}. \quad (20)$$

In the arbitrary Lagrangian-Eulerian formulation we obtain in the local form

$$\frac{\partial \varrho J_{\mathcal{Z}} \mathbf{v}}{\partial t} + \text{div} (\varrho J_{\mathcal{Z}} \mathbf{v} \otimes (\mathbf{v} - \mathbf{v}_{\mathcal{Z}}) \mathbf{F}_{\mathcal{Z}}^{-T}) = \text{div} (J_{\mathcal{Z}} \boldsymbol{\sigma}^T \mathbf{F}_{\mathcal{Z}}^{-T}) + \varrho J_{\mathcal{Z}} \mathbf{f}, \quad (21)$$

or with the use of (16) we can write

$$\varrho J_{\mathcal{Z}} \frac{\partial \mathbf{v}}{\partial t} + \varrho J_{\mathcal{Z}} (\nabla \mathbf{v}) \mathbf{F}_{\mathcal{Z}}^{-T} (\mathbf{v} - \mathbf{v}_{\mathcal{Z}}) = \text{div} (J_{\mathcal{Z}} \boldsymbol{\sigma}^T \mathbf{F}_{\mathcal{Z}}^{-T}) + \varrho J_{\mathcal{Z}} \mathbf{f}. \quad (22)$$

In the case of angular momentum balance we assume that there are no external or internal sources of angular momentum. It then follows that the Cauchy stress tensor has to be symmetric, i.e. $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$. Assuming isothermal conditions the energy balance is satisfied if the choice of the constitutive relation for the materials is compatible with the balance of entropy.

3 Fluid Structure Interaction Problem Formulation

At this point we make a few assumptions that allow us to deal with the task of setting up a tractable problem. Let us consider a flow between thick elastic walls as shown in Figure 2. We will use the superscripts s and f to denote the quantities connected with the solid and fluid. Let us assume that both materials are incompressible and all the processes are isothermal, which is a well accepted approximation in biomechanics, and let us denote the constant densities of each material by ϱ^f, ϱ^s .

3.1 Monolithic Description

We denote by Ω_t^f the domain occupied by the fluid and Ω_t^s by the solid at time $t \in [0, T]$. Let $\Gamma_t^0 = \bar{\Omega}_t^f \cap \bar{\Omega}_t^s$ be the part of the boundary where the solid interacts with the fluid and $\Gamma_t^i, i = 1, 2, 3$ be the remaining external boundaries of the solid and the fluid as depicted in Figure 2.

Let the deformation of the solid part be described by the mapping χ^s

$$\chi^s : \Omega^s \times [0, T] \mapsto \mathbb{R}^3, \quad (23)$$

with the corresponding displacement $\mathbf{u}^s$ and the velocity $\mathbf{v}^s$ given by

$$\mathbf{u}^s(\mathbf{X}, t) = \chi^s(\mathbf{X}, t) - \mathbf{X}, \quad \mathbf{v}^s(\mathbf{X}, t) = \frac{\partial \chi^s}{\partial t}(\mathbf{X}, t). \quad (24)$$

The fluid flow is described by the velocity field $\mathbf{v}^f$ defined on the fluid domain Ω_t^f

$$\mathbf{v}^f(\mathbf{x}, t) : \Omega_t^f \times [0, T] \mapsto \mathbb{R}^3. \quad (25)$$

Further we define the auxiliary mapping, denoted by ζ^f , to describe the change of the fluid domain and corresponding displacement $\mathbf{u}^f$ by

$$\zeta^f : \Omega^f \times [0, T] \mapsto \mathbb{R}^3, \quad \mathbf{u}^f(\mathbf{X}, t) = \zeta^f(\mathbf{X}, t) - \mathbf{X}. \quad (26)$$

We require that the mapping ζ^f is sufficiently smooth, one to one and has to satisfy

$$\zeta^f(\mathbf{X}, t) = \chi^s(\mathbf{X}, t), \quad \forall (\mathbf{X}, t) \in \Gamma^0 \times [0, T]. \quad (27)$$

In the context of the finite element method this will describe the artificial mesh deformation inside the fluid region and it will be constructed as a solution to a suitable boundary value problem with (27) as the boundary condition.

The momentum and mass balance of the fluid in the time dependent fluid domain according to (16) and (21) are

$$\varrho^f \frac{\partial \mathbf{v}^f}{\partial t} + \varrho^f (\nabla \mathbf{v}^f) (\mathbf{v}^f - \frac{\partial \mathbf{u}^f}{\partial t}) = \text{div } \boldsymbol{\sigma}^f \quad \text{in } \Omega_t^f, \quad (28)$$

$$\text{div } \mathbf{v}^f = 0 \quad \text{in } \Omega_t^f, \quad (29)$$

together with the momentum (17) and mass (11) balance of the solid in the solid domain

$$\varrho^s \frac{\partial \mathbf{v}^s}{\partial t} + \varrho^s (\nabla \mathbf{v}^s) \mathbf{v}^s = \text{div } \boldsymbol{\sigma}^s \quad \text{in } \Omega_t^s, \quad (30)$$

$$\text{div } \mathbf{v}^s = 0 \quad \text{in } \Omega_t^s. \quad (31)$$

The interaction is due to the exchange of momentum through the common part of the boundary Γ_t^0 . On this part we require that the forces are in balance and simultaneously the no slip boundary condition holds for the fluid, i.e.

$$\boldsymbol{\sigma}^f \mathbf{n} = \boldsymbol{\sigma}^s \mathbf{n} \quad \text{on } \Gamma_t^0, \quad \mathbf{v}^f = \mathbf{v}^s \quad \text{on } \Gamma_t^0. \quad (32)$$

The remaining external boundary conditions can be of the following kind. A natural boundary condition on the fluid inflow and outflow part Γ_t^1

$$\boldsymbol{\sigma}^f \mathbf{n} = p_B \mathbf{n} \text{ on } \Gamma_t^1, \quad (33)$$

with p_B given value. Alternatively we can prescribe a Dirichlet type boundary condition on the inflow or outflow part Γ_t^1

$$\mathbf{v}^f = \mathbf{v}_B \text{ on } \Gamma_t^1, \quad (34)$$

where $\mathbf{v}_B$ is given. The Dirichlet boundary condition is prescribed for the solid displacement at the part Γ_t^2

$$\mathbf{u}^s = \mathbf{0} \text{ on } \Gamma_t^2, \quad (35)$$

and the stress free boundary condition for the solid is applied at the part Γ_t^3

$$\boldsymbol{\sigma}^s \mathbf{n} = \mathbf{0} \text{ on } \Gamma_t^3. \quad (36)$$

We introduce the domain $\Omega = \Omega^f \cup \Omega^s$, where Ω^f, Ω^s are the domains occupied by the fluid and solid in the initial undeformed state, and two fields defined on this domain as

$$\mathbf{u} : \Omega \times [0, T] \rightarrow \mathbb{R}^3, \quad \mathbf{v} : \Omega \times [0, T] \rightarrow \mathbb{R}^3,$$

such that the field $\mathbf{v}$ represents the velocity at the given point and $\mathbf{u}$ the displacement on the solid part and the artificial displacement in the fluid part, taking care of the fact that the fluid domain is changing with time,

$$\mathbf{v} = \begin{cases} \mathbf{v}^s & \text{on } \Omega^s, \\ \mathbf{v}^f & \text{on } \Omega^f, \end{cases} \quad \mathbf{u} = \begin{cases} \mathbf{u}^s & \text{on } \Omega^s, \\ \mathbf{u}^f & \text{on } \Omega^f. \end{cases} \quad (37)$$

Due to the conditions (27) and (32) both fields are continuous across the interface Γ_t^0 and we can define global quantities on Ω as the deformation gradient and its determinant

$$\mathbf{F} = \mathbf{I} + \text{Grad } \mathbf{u}, \quad J = \det \mathbf{F}. \quad (38)$$

Using this notation the solid balance laws (30) and (31) can be expressed in the Lagrangian formulation with the initial configuration Ω^s as reference, cf. (19),

$$J \varrho^s \frac{d\mathbf{v}}{dt} = \text{Div } \mathbf{P}^s \quad \text{in } \Omega^s, \quad (39)$$

$$J = 1 \quad \text{in } \Omega^s. \quad (40)$$

The fluid equations (28) and (29) are already expressed in the arbitrary Lagrangian-Eulerian formulation with respect to the time dependent region

Ω_t^f , now we transform the equations to the fixed initial region Ω^f by the mapping ζ^f defined by (26)

$$\varrho^f \frac{\partial \mathbf{v}}{\partial t} + \varrho^f (\text{Grad } \mathbf{v}) \mathbf{F}^{-1} (\mathbf{v} - \frac{\partial \mathbf{u}}{\partial t}) = J^{-1} \text{Div}(J \boldsymbol{\sigma}^f \mathbf{F}^{-T}) \quad \text{in } \Omega^f, \quad (41)$$

$$\text{Div}(J \mathbf{v} \mathbf{F}^{-T}) = 0 \quad \text{in } \Omega^f. \quad (42)$$

It remains to prescribe some relation for the mapping ζ^f . In terms of the corresponding displacement $\mathbf{u}^f$ we formulate some simple relation together with the Dirichlet boundary conditions required by (27), for example

$$\frac{\partial \mathbf{u}}{\partial t} = \Delta \mathbf{u} \quad \text{in } \Omega^f, \quad \mathbf{u} = \mathbf{u}^s \quad \text{on } \Gamma^0, \quad \mathbf{u} = \mathbf{0} \quad \text{on } \Gamma^1. \quad (43)$$

Other choices are possible. For example, the mapping $\mathbf{u}^f$ can be realized as a solution of the elasticity problem with the same Dirichlet boundary conditions, see [14].

The complete set of the equations can be written as

$$\frac{\partial \mathbf{u}}{\partial t} = \begin{cases} \mathbf{v} & \text{in } \Omega^s, \\ \Delta \mathbf{u} & \text{in } \Omega^f, \end{cases} \quad (44)$$

$$\frac{\partial \mathbf{v}}{\partial t} = \begin{cases} \frac{1}{J \varrho^s} \text{Div } \mathbf{P}^s & \text{in } \Omega^s, \\ -(\text{Grad } \mathbf{v}) \mathbf{F}^{-1} (\mathbf{v} - \frac{\partial \mathbf{u}}{\partial t}) + \frac{1}{J \varrho^f} \text{Div}(J \boldsymbol{\sigma}^f \mathbf{F}^{-T}) & \text{in } \Omega^f, \end{cases} \quad (45)$$

$$0 = \begin{cases} J - 1 & \text{in } \Omega^s, \\ \text{Div}(J \mathbf{v} \mathbf{F}^{-T}) & \text{in } \Omega^f, \end{cases} \quad (46)$$

with the initial conditions

$$\mathbf{u}(0) = \mathbf{0} \quad \text{in } \Omega, \quad \mathbf{v}(0) = \mathbf{v}_0 \quad \text{in } \Omega, \quad (47)$$

and boundary conditions

$$\mathbf{u} = \mathbf{0}, \quad \mathbf{v} = \mathbf{v}_B \quad \text{on } \Gamma^1, \quad \mathbf{u} = \mathbf{0} \quad \text{on } \Gamma^2, \quad \boldsymbol{\sigma}^s \mathbf{n} = \mathbf{0} \quad \text{on } \Gamma^3. \quad (48)$$

3.2 Constitutive Equations

In order to solve the balance equations we need to specify the constitutive relations for the stress tensors. For the fluid we use the incompressible Newtonian relation

$$\boldsymbol{\sigma}^f = -p^f \mathbf{I} + \mu (\nabla \mathbf{v}^f + (\nabla \mathbf{v}^f)^T), \quad (49)$$

where μ represents the viscosity of the fluid and p^f is the Lagrange multiplier corresponding to the incompressibility constraint (29).

For the solid part we assume that it can be described by an incompressible hyper-elastic material. We specify the Helmholtz potential Ψ and the solid stress is given by

$$\boldsymbol{\sigma}^s = -p^s \mathbf{I} + \varrho^s \frac{\partial \Psi}{\partial \mathbf{F}} \mathbf{F}^T, \quad (50)$$

the first Piola-Kirchhoff stress tensor is then given by

$$\mathbf{P}^s = -J p^s \mathbf{F}^{-T} + J \varrho^s \frac{\partial \Psi}{\partial \mathbf{F}}, \quad (51)$$

where p^s is the Lagrange multiplier corresponding to the incompressibility constraint (40).

Then the material is specified by prescribing the Helmholtz potential as a function of the deformation

$$\Psi = \hat{\Psi}(\mathbf{F}) = \tilde{\Psi}(\mathbf{C}), \quad (52)$$

where $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ is the right Cauchy-Green deformation tensor. Typical examples for the Helmholtz potential used for isotropic materials like rubber is the Mooney-Rivlin material

$$\tilde{\Psi} = c_1(I_C - 3) + c_2(II_C - 3), \quad (53)$$

where $I_C = \text{tr } \mathbf{C}$, $II_C = \text{tr } \mathbf{C}^2 - \text{tr}^2 \mathbf{C}$, $III_C = \det \mathbf{C}$ are the invariants of the right Cauchy-Green deformation tensor $\mathbf{C}$ and c_i are some material constants. A special case of neo-Hookean material is obtained for $c_2 = 0$. With a suitable choice of the material parameters the entropy inequality and the balance of energy are automatically satisfied.

3.3 Weak Formulation

We non-dimensionalize all the quantities by a given characteristic length L and speed V as follows

$$\begin{aligned} \hat{t} &= t \frac{V}{L}, & \hat{\mathbf{x}} &= \frac{\mathbf{x}}{L}, & \hat{\mathbf{u}} &= \frac{\mathbf{u}}{L}, & \hat{\mathbf{v}} &= \frac{\mathbf{v}}{V}, \\ \hat{\boldsymbol{\sigma}}^s &= \boldsymbol{\sigma}^s \frac{L}{\varrho^f V^2}, & \hat{\boldsymbol{\sigma}}^f &= \boldsymbol{\sigma}^f \frac{L}{\varrho^f V^2}, & \hat{\mu} &= \frac{\mu}{\varrho^f V L}, & \hat{\Psi} &= \Psi \frac{L}{\varrho^f V^2}, \end{aligned}$$

further using the same symbols, without the hat, for the non-dimensional quantities. The non-dimensionalized system with the choice of material rela-

tions, (49) for viscous fluid and (51) for the hyper-elastic solid is

$$\frac{\partial \mathbf{u}}{\partial t} = \begin{cases} \mathbf{v} & \text{in } \Omega^s, \\ \Delta \mathbf{u} & \text{in } \Omega^f, \end{cases} \quad (54)$$

$$\frac{\partial \mathbf{v}}{\partial t} = \begin{cases} \frac{1}{\beta} \text{Div} \left(-J p^s \mathbf{F}^{-T} + \frac{\partial \Psi}{\partial \mathbf{F}} \right) & \text{in } \Omega^s, \\ -(\text{Grad } \mathbf{v}) \mathbf{F}^{-1} \left(\mathbf{v} - \frac{\partial \mathbf{u}}{\partial t} \right) & \text{in } \Omega^f, \\ + \text{Div} \left(-J p^f \mathbf{F}^{-T} + J \mu \text{Grad } \mathbf{v} \mathbf{F}^{-1} \mathbf{F}^{-T} \right) & \end{cases} \quad (55)$$

$$0 = \begin{cases} J - 1 & \text{in } \Omega^s, \\ \text{Div}(J \mathbf{v} \mathbf{F}^{-T}) & \text{in } \Omega^f, \end{cases} \quad (56)$$

and the boundary conditions

$$\boldsymbol{\sigma}^f \mathbf{n} = \boldsymbol{\sigma}^s \mathbf{n} \quad \text{on } \Gamma_t^0, \quad \mathbf{v} = \mathbf{v}_B \quad \text{on } \Gamma_t^1, \quad (57)$$

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_t^2, \quad \boldsymbol{\sigma}^f \mathbf{n} = \mathbf{0} \quad \text{on } \Gamma_t^3. \quad (58)$$

Let $I = [0, T]$ denote the time interval of interest. We multiply the equations (54)–(56) by the test functions $\boldsymbol{\zeta}, \boldsymbol{\xi}, \gamma$ such that $\boldsymbol{\zeta} = \mathbf{0}$ on Γ^2 , $\boldsymbol{\xi} = \mathbf{0}$ on Γ^1 and integrate over the space domain Ω and the time interval I . Using integration by parts on some of the terms and the boundary conditions we obtain

$$\int_0^T \int_{\Omega} \frac{\partial \mathbf{u}}{\partial t} \cdot \boldsymbol{\zeta} dV dt = \int_0^T \int_{\Omega^s} \mathbf{v} \cdot \boldsymbol{\zeta} dV dt - \int_0^T \int_{\Omega^f} \text{Grad } \mathbf{u} \cdot \text{Grad } \boldsymbol{\zeta} dV dt, \quad (59)$$

$$\begin{aligned} & \int_0^T \int_{\Omega^f} J \frac{\partial \mathbf{v}}{\partial t} \cdot \boldsymbol{\xi} dV dt + \int_0^T \int_{\Omega^s} \beta J \frac{\partial \mathbf{v}}{\partial t} \cdot \boldsymbol{\xi} dV dt \\ &= - \int_0^T \int_{\Omega^f} J \text{Grad } \mathbf{v} \mathbf{F}^{-1} \left(\mathbf{v} - \frac{\partial \mathbf{u}}{\partial t} \right) \cdot \boldsymbol{\xi} dV dt \\ &+ \int_0^T \int_{\Omega} J p \mathbf{F}^{-T} \cdot \text{Grad } \boldsymbol{\xi} dV dt \\ &- \int_0^T \int_{\Omega^s} \frac{\partial \Psi}{\partial \mathbf{F}} \cdot \text{Grad } \boldsymbol{\xi} dV dt \\ &- \int_0^T \int_{\Omega^f} J \mu \text{Grad } \mathbf{v} \mathbf{F}^{-1} \mathbf{F}^{-T} \cdot \text{Grad } \boldsymbol{\xi} dV dt, \end{aligned} \quad (60)$$

$$0 = \int_0^T \int_{\Omega^s} (J - 1) \gamma dV dt + \int_0^T \int_{\Omega^f} \text{Div}(J \mathbf{v} \mathbf{F}^{-T}) \gamma dV dt. \quad (61)$$

Let us define the following spaces

$$\begin{aligned} U &= \{ \mathbf{u} \in L^\infty(I, [W^{1,2}(\Omega)]^3), \mathbf{u} = \mathbf{0} \text{ on } \Gamma^2 \}, \\ V &= \{ \mathbf{v} \in L^2(I, [W^{1,2}(\Omega_t)]^3) \cap L^\infty(I, [L^2(\Omega_t)]^3), \mathbf{v} = \mathbf{0} \text{ on } \Gamma^1 \}, \\ P &= \{ p \in L^2(I, L^2(\Omega)) \}, \end{aligned}$$

then the variational formulation of the fluid-structure interaction problem is to find $(\mathbf{u}, \mathbf{v} - \mathbf{v}_B, p) \in U \times V \times P$ such that equations (59), (60) and (61) are satisfied for all $(\boldsymbol{\zeta}, \boldsymbol{\xi}, \gamma) \in U \times V \times P$ including appropriate initial conditions.

3.4 Discretization

In the following, we restrict ourselves to two dimensions which allows systematic tests of the proposed methods in a very efficient way, particularly in view of grid-independent solutions. The time discretization is done by the Crank-Nicholson scheme which is only conditionally stable but which has better conservation properties than for example the implicit Euler scheme, see [15,16]. The Crank-Nicholson scheme can be obtained by dividing the time interval I into the series of time steps $[t^n, t^{n+1}]$ with step length $k_n = t^{n+1} - t^n$. Assuming that the test functions are piecewise constant on each time step $[t^n, t^{n+1}]$, writing the weak formulation (59)–(60) for the time interval $[t^n, t^{n+1}]$, approximating the time derivatives by the central differences

$$\frac{\partial f}{\partial t} \approx \frac{f(t^{n+1}) - f(t^n)}{k_n} \quad (62)$$

and approximating the time integration for the remaining terms by the trapezoidal quadrature rule as

$$\int_{t^n}^{t^{n+1}} f(t) dt \approx \frac{k_n}{2} (f(t^n) + f(t^{n+1})), \quad (63)$$

we obtain the time discretized system. The last equation corresponding to the incompressibility constraint is taken implicitly for the time t^{n+1} and the corresponding term with the Lagrange multiplier p_h^{n+1} in the equation (60) is also taken implicitly.

The discretization in space is done by the finite element method. We approximate the domain Ω by a domain Ω_h with polygonal boundary and by $\mathcal{T}_h$ we denote a set of quadrilaterals covering the domain Ω_h . We assume that $\mathcal{T}_h$ is regular in the sense that any two quadrilateral are disjoint or have a common vertex or a common edge. By $\bar{T} = [-1, 1]^2$ we denote the reference quadrilateral.

Our treatment of the problem as one system suggests that we use the same finite elements on both, the solid part and the fluid region. Since both materials are incompressible we have to choose a pair of finite element spaces known to be stable for the problems with incompressibility constraint. One possible choice is the conforming biquadratic, discontinuous linear Q_2, P_1 pair, see Figure 3 for the location of the degrees of freedom. This choice results in 39 degrees of freedom per element in the case of our displacement, velocity, pressure formulation in two dimensions and 112 degrees of freedom per element in three dimensions.

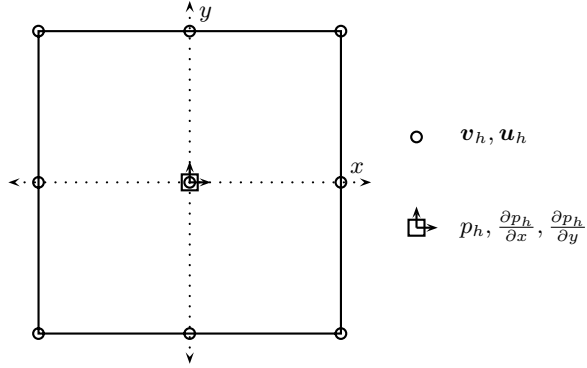


Fig. 3. Location of the degrees of freedom for the Q_2, P_1 element.

The spaces U, V, P on an interval $[t^n, t^{n+1}]$ would be approximated in the case of the Q_2, P_1 pair as

$$\begin{aligned} U_h &= \{\mathbf{u}_h \in [C(\Omega_h)]^2, \mathbf{u}_h|_T \in [Q_2(T)]^2 \quad \forall T \in \mathcal{T}_h, \mathbf{u}_h = \mathbf{0} \text{ on } \Gamma_2\}, \\ V_h &= \{\mathbf{v}_h \in [C(\Omega_h)]^2, \mathbf{v}_h|_T \in [Q_2(T)]^2 \quad \forall T \in \mathcal{T}_h, \mathbf{v}_h = 0 \text{ on } \Gamma_1\}, \\ P_h &= \{p_h \in L^2(\Omega_h), p_h|_T \in P_1(T) \quad \forall T \in \mathcal{T}_h\}. \end{aligned}$$

Let us denote by $\mathbf{u}_h^n$ the approximation of $\mathbf{u}(t^n)$, $\mathbf{v}_h^n$ the approximation of $\mathbf{v}(t^n)$ and p_h^n the approximation of $p(t^n)$. Further we will use following shorthand notation

$$\begin{aligned} \mathbf{F}^n &= \mathbf{I} + \text{Grad } \mathbf{u}_h^n, \quad J^n = \det \mathbf{F}^n \quad J^{n+\frac{1}{2}} = \frac{1}{2}(J^n + J^{n+1}), \\ (f, g) &= \int_{\Omega} f \cdot g dV, \quad (f, g)_s = \int_{\Omega^s} f \cdot g dV, \quad (f, g)_f = \int_{\Omega^f} f \cdot g dV, \end{aligned}$$

f, g being scalars, vectors or tensors.

Writing down the discrete equivalent of the equations (59)–(61) yields

$$\begin{aligned} (\mathbf{u}_h^{n+1}, \boldsymbol{\eta}) - \frac{k_n}{2} \left\{ (\mathbf{v}_h^{n+1}, \boldsymbol{\eta})_s + (\nabla \mathbf{u}_h^{n+1}, \nabla \boldsymbol{\eta})_f \right\} \\ - (\mathbf{u}_h^n, \boldsymbol{\eta}) - \frac{k_n}{2} \left\{ (\mathbf{v}_h^n, \boldsymbol{\eta})_s + (\nabla \mathbf{u}_h^n, \nabla \boldsymbol{\eta})_f \right\} = 0, \end{aligned} \quad (64)$$

$$\begin{aligned}
& \left(J^{n+\frac{1}{2}} \mathbf{v}_h^{n+1}, \boldsymbol{\xi} \right)_f + \beta \left(\mathbf{v}_h^{n+1}, \boldsymbol{\xi} \right)_s - k_n \left(J^{n+1} p_h^{n+1} (\mathbf{F}^{n+1})^{-T}, \text{Grad } \boldsymbol{\xi} \right)_s \\
& + \frac{k_n}{2} \left\{ \left(\frac{\partial \Psi}{\partial \mathbf{F}} (\text{Grad } \mathbf{u}_h^{n+1}), \text{Grad } \boldsymbol{\xi} \right)_s \right. \\
& + \mu \left(J^{n+1} \text{Grad } \mathbf{v}_h^{n+1} (\mathbf{F}^{n+1})^{-1}, \text{Grad } \boldsymbol{\xi} (\mathbf{F}^{n+1})^{-1} \right)_f \\
& + \left. \left(J^{n+1} \text{Grad } \mathbf{v}_h^{n+1} (\mathbf{F}^{n+1})^{-1} \mathbf{v}_h^{n+1}, \boldsymbol{\xi} \right)_f \right\} \\
& - \frac{1}{2} \left(J^{n+1} \text{Grad } \mathbf{v}_h^{n+1} (\mathbf{F}^{n+1})^{-1} (\mathbf{u}_h^{n+1} - \mathbf{u}_h^n), \boldsymbol{\xi} \right)_f \\
& - \left(J^{n+\frac{1}{2}} \mathbf{v}_h^n, \boldsymbol{\xi} \right)_f - \beta \left(\mathbf{v}_h^n, \boldsymbol{\xi} \right)_s \\
& + \frac{k_n}{2} \left\{ \left(\frac{\partial \Psi}{\partial \mathbf{F}} (\text{Grad } \mathbf{u}_h^n), \text{Grad } \boldsymbol{\xi} \right)_s \right. \\
& + \mu \left(J^n \text{Grad } \mathbf{v}_h^n (\mathbf{F}^n)^{-1}, \text{Grad } \boldsymbol{\xi} (\mathbf{F}^n)^{-1} \right)_f \\
& + \left. \left(J^n \text{Grad } \mathbf{v}_h^n (\mathbf{F}^n)^{-1} \mathbf{v}_h^n, \boldsymbol{\xi} \right)_f \right\} \\
& + \frac{1}{2} \left(J^n \text{Grad } \mathbf{v}_h^n (\mathbf{F}^n)^{-1} (\mathbf{u}_h^{n+1} - \mathbf{u}_h^n), \boldsymbol{\xi} \right)_f = 0, \\
& (J^{n+1} - 1, \gamma)_s + (J^{n+1} \text{Grad } \mathbf{v}_h^{n+1} (\mathbf{F}^{n+1})^{-1}, \gamma)_f = 0. \tag{65}
\end{aligned}$$

Using the basis of the spaces U_h, V_h, P_h as the test functions $\boldsymbol{\zeta}, \boldsymbol{\xi}, \gamma$ we obtain a nonlinear algebraic set of equations. In each time step we have to find $\mathbf{X} = (\mathbf{u}_h^{n+1}, \mathbf{v}_h^{n+1}, p_h^{n+1}) \in U_h \times V_h \times P_h$ such that

$$\mathcal{F}(\mathbf{X}) = 0, \tag{66}$$

where $\mathcal{F}$ represents the system (64–66).

3.5 Solution Algorithm

The system (67) of nonlinear algebraic equations is solved using Newton method as the basic iteration. One step of the Newton iteration can be written as

$$\mathbf{X}^{n+1} = \mathbf{X}^n - \left[\frac{\partial \mathcal{F}}{\partial \mathbf{X}} (\mathbf{X}^n) \right]^{-1} \mathcal{F}(\mathbf{X}^n). \tag{68}$$

This basic iteration can exhibit quadratic convergence provided that the initial guess is sufficiently close to the solution. To ensure the convergence globally, some improvements of this basic iteration are used.

The damped Newton method with line search improves the chance of convergence by adaptively changing the length of the correction vector. The solution update step in the Newton method (68) is replaced by

$$\mathbf{X}^{n+1} = \mathbf{X}^n + \omega \delta \mathbf{X}, \tag{69}$$

where the parameter ω is determined such that a certain error measure decreases. One of the possible choices for the quantity to decrease is

$$f(\omega) = \mathcal{F}(\mathbf{X}^n + \omega \delta \mathbf{X}) \cdot \delta \mathbf{X}. \quad (70)$$

Since we know

$$f(0) = \mathcal{F}(\mathbf{X}^n) \cdot \delta \mathbf{X}, \quad (71)$$

and

$$f'(0) = \left[\frac{\partial \mathcal{F}}{\partial \mathbf{X}}(\mathbf{X}^n) \right] \delta \mathbf{X} \cdot \delta \mathbf{X} = \mathcal{F}(\mathbf{X}^n) \cdot \delta \mathbf{X}, \quad (72)$$

and computing $f(\omega_0)$ for $\omega_0 = -1$ or ω_0 determined adaptively from previous iterations, we can approximate $f(\omega)$ by a quadratic function

$$f(\omega) = \frac{f(\omega_0) - f(0)(\omega_0 + 1)}{\omega_0^2} \omega^2 + f(0)(\omega + 1). \quad (73)$$

Then setting

$$\tilde{\omega} = \frac{f(0)\omega_0^2}{f(\omega_0) - f(0)(\omega_0 + 1)}, \quad (74)$$

the new optimal step length $\omega \in [-1, 0]$ is

$$\omega = \begin{cases} -\frac{\tilde{\omega}}{2} & \text{if } \frac{f(0)}{f(\omega_0)} > 0, \\ -\frac{\tilde{\omega}}{2} - \sqrt{\frac{\tilde{\omega}^2}{4} - \tilde{\omega}} & \text{if } \frac{f(0)}{f(\omega_0)} \leq 0. \end{cases} \quad (75)$$

This line search can be repeated with ω_0 taken as the last ω until, for example, $f(\omega) \leq \frac{1}{2}f(0)$. By this we can enforce a monotone convergence of the approximation $\mathbf{X}^n$.

An adaptive time-step selection was found to help in the nonlinear convergence. A heuristic algorithm was used to correct the time-step length

-
1. Let $\mathbf{X}^n$ be some starting guess.
 2. Set the residuum vector $\mathbf{R}^n = \mathcal{F}(\mathbf{X}^n)$ and the tangent matrix $\mathbf{A} = \frac{\partial \mathcal{F}}{\partial \mathbf{X}}(\mathbf{X}^n)$.
 3. Solve for the correction $\delta \mathbf{X}$

$$\mathbf{A} \delta \mathbf{X} = \mathbf{R}^n.$$
 4. Find optimal step length ω .
 5. Update the solution $\mathbf{X}^{n+1} = \mathbf{X}^n - \omega \delta \mathbf{X}$.
-

Fig. 4. One step of the Newton method with the line search.

according to the convergence of the nonlinear iterations in the previous time-step. If the convergence was close to quadratic, i.e. only up to three Newton steps were needed to obtain the required precision, the time step could be slightly increased, otherwise the time-step length was reduced.

The structure of the Jacobian matrix $\frac{\partial \mathcal{F}}{\partial \mathbf{X}}$ is

$$\frac{\partial \mathcal{F}}{\partial \mathbf{X}}(\mathbf{X}) = \begin{pmatrix} S_{uu} & S_{uv} & 0 \\ S_{vu} & S_{vv} & B_u + B_v \\ B_u^T & B_v^T & 0 \end{pmatrix}, \quad (76)$$

and it can be computed by finite differences from the residual vector $\mathcal{F}(\mathbf{X})$

$$\left[\frac{\partial \mathcal{F}}{\partial \mathbf{X}} \right]_{ij}(\mathbf{X}^n) \approx \frac{[\mathcal{F}]_i(\mathbf{X}^n + \alpha_j \mathbf{e}_j) - [\mathcal{F}]_i(\mathbf{X}^n - \alpha_j \mathbf{e}_j)}{2\alpha_j}, \quad (77)$$

where $\mathbf{e}_j$ are the unit basis vectors in $\mathbb{R}^n$ and the coefficients α_j are adaptively taken according to the change in the solution in the previous time step. Since we know the sparsity pattern of the Jacobian matrix in advance, it is given by the used finite element method, this computation can be done in an efficient way so that the linear solver remains the dominant part in terms of the CPU time. However, as Table 1 shows, the resulting nonlinear and linear solution behavior is quite sensitive w.r.t. the parameters.

Table 1. nonlinear solver it. / avg. linear solver it. [CPU time] for BiCGStab(ILU(0)): *TOL* denotes the stopping criterion of the linear problems.

α/TOL	10^{-8}	10^{-4}	10^{-2}	10^{-1}
10^{-8}	7 /107 [21.52]	12 /57 [26.52]	12 /47 [23.75]	17 /33 [27.38]
10^{-4}	7 /108 [24.57]	8 /62 [17.77]	10 /42 [18.95]	18 /31 [29.05]
10^{-2}	16 /109 [51.65]	20 /47 [38.28]	25 /29 [38.58]	56 /16 [73.83]
10^{-1}	44 /116 [141.30]	48 /35 [81.72]	49 /17 [65.77]	–

3.6 Multigrid Solver

The solution of the linear problems is the most time consuming part of the solution process. A good candidate seems to be a direct solver for sparse systems like UMFPACK(see [17]); while this choice provides very robust linear solvers, its memory and CPU time requirements are too high for larger systems (i.e. more than 20000 unknowns). Large linear problems can be solved by Krylov space methods (BiCGStab, GMRes, see[18]) with suitable preconditioners. One possibility is the ILU preconditioner with special treatment of the saddle point character of our system, where we allow certain fill-in for the zero diagonal blocks, see [19]. The alternative option for larger systems is the multigrid method presented in this section.

We utilize the standard geometric multigrid approach based on a hierarchy of grids obtained by successive regular refinement of a given coarse mesh. The complete multigrid iteration is performed in the standard defect-correction setup with the V or F-type cycle. While a direct sparse solver [17] is used for the coarse grid solution, on finer levels a fixed number (2 or 4) of iterations by local MPSC schemes (Vanka-like smoother) [20,21] is performed. Such iteration can be written as

$$\begin{bmatrix} \mathbf{u}^{l+1} \\ \mathbf{v}^{l+1} \\ p^{l+1} \end{bmatrix} = \begin{bmatrix} \mathbf{u}^l \\ \mathbf{v}^l \\ p^l \end{bmatrix} - \omega \sum_{\text{Patch } \Omega_i} \begin{bmatrix} S_{uu}|\Omega_i & S_{uv}|\Omega_i & 0 \\ S_{vu}|\Omega_i & S_{vv}|\Omega_i & kB|\Omega_i \\ c_u B_s^T|\Omega_i & c_v B_f^T|\Omega_i & 0 \end{bmatrix}^{-1} \begin{bmatrix} \text{def}_u^l \\ \text{def}_v^l \\ \text{def}_p^l \end{bmatrix}.$$

The inverse of the local systems (39×39) can be done by hardware optimized direct solvers.

The full nodal interpolation is used as the prolongation operator $\mathbf{P}$ with its transposed operator used as the restriction $\mathbf{R} = \mathbf{P}^T$.

In Table 2 we compare the performance of the multigrid solver with the Krylov space based iterative solvers with ILU preconditioner. The comparison is presented for two different sizes of the timestep to illustrate the behavior of the solvers for different level of nonlinearity involved.

Table 2. Comparison of different solvers (nonlinear solver it. / avg. linear solver it. [CPU time]).

timestep 10^{-2}

Level	ndof	MG(2)	MG(4)	BiCGStab(ILU(1))	GMRES(ILU(1),200)
1	12760	2/8 [66]	2/8 [92]	2/51 [32]	2/50 [27]
2	50144	2/8 [190]	2/5 [198]	2/120 [200]	2/117 [151]
3	198784	2/9 [744]	2/6 [852]	2/311 [1646]	2/358 [1432]
4	791552	2/13 [3803]	2/7 [3924]	MEM.	MEM.

timestep 10^0

Level	ndof	MG(2)	MG(4)	BiCGStab(ILU(1))	GMRES(ILU(1),200)
1	12760	4/12 [118]	4/11 [177]	20/160 [631]	20/801 [1579]
2	50144	4/12 [466]	4/7 [470]	2/800 [] diverg.	13/801 [] diverg.
3	198784	4/13 [1898]	4/7 [2057]	2/800 [] diverg.	4/801 [] diverg.
4	791552	4/15 [8678]	4/8 [9069]	MEM.	MEM.

4 Exemplary Applications in Biomechanics

In this section we present a few example applications to demonstrate the described methods. As a motivation we consider the numerical simulation of some problems encountered in the area of cardiovascular hemodynamics, namely flow interaction with thick-walled deformable material, which can

become a useful tool for deeper understanding of the onset of diseases of the human circulatory system, as for example blood cell and intima damages in stenosis, aneurysm rupture, evaluation of the new surgery techniques of heart, arteries and veins.

In order to test the proposed numerical methods, simplified two-dimensional examples which include interaction of flow with deformable material are computed. The first example is a flow in an ellipsoidal cavity and the second is a flow through a channel with elastic walls. In both cases the flow is driven by changing the fluid pressure at the inflow part of the boundary while the elastic part of the boundary is either fixed or stress free.

The constitutive relations used for the materials are the incompressible Newtonian model (49) for the fluid and the hyper-elastic neo-Hookean material (53) with $c_2 = 0$ for the solid. This choice includes all the main difficulties the numerical method has to deal with, namely the incompressibility and significant deformations.

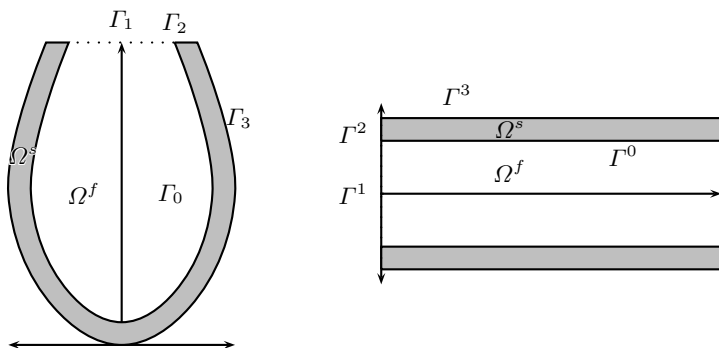


Fig. 5. Schematic view of the ventricle and elastic tube geometries.

4.1 Flow in an Ellipsoidal Cavity

The motivation for our first test is the left heart ventricle which is an approximately ellipsoidal chamber surrounded by the heart muscle. In our two-dimensional computations we use an ellipsoidal cavity, see Figure 5, with prescribed time-dependent natural boundary condition at the fluid boundary part Γ^1

$$p(t) = \sin t \quad \text{on } \Gamma^1. \quad (78)$$

The material of the solid wall is modelled by the simple neo-Hookean constitutive relation (53) with $c_2 = 0$.

The Figures 6 and 7 show the computational grid for the maximal and minimal volume configuration of the cavity and the velocity field of the fluid

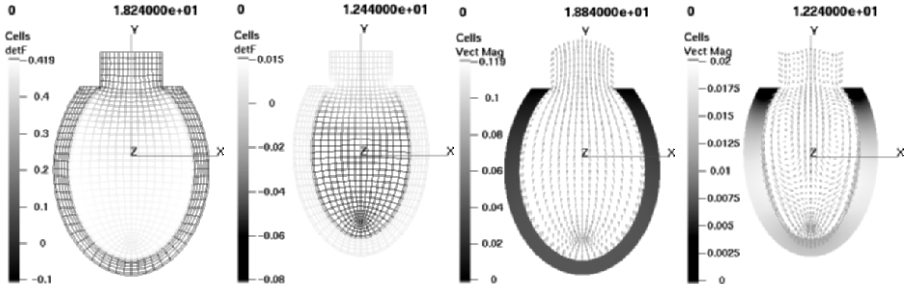


Fig. 6. Maximum and minimum volume configuration with the fluid flow.

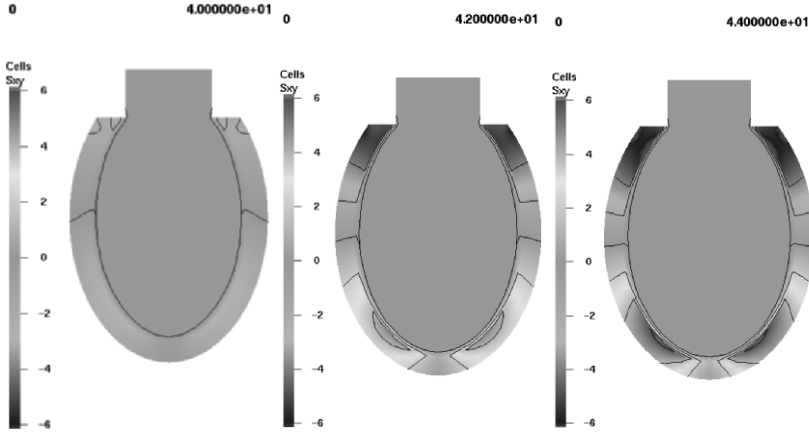


Fig. 7. Shear stress distribution in the wall during the period.

for the same configurations. One of the important characteristics is the shear stress exerted by the fluid flow on the wall material. This Figure 7 shows the distribution of the shear stress in the domain for three different times. In Figures 8 and 9 the volume change of the cavity as a function of the time and the average pressure inside the cavity vs. the volume of the cavity is shown together with the trajectory and velocity of a material point on the solid-fluid interface. We can see that after the initial cycle which was started from the undeformed configuration the system comes to a time periodic solution.

4.2 Flow in an Elastic Channel

The second application is the simulation of a flow in an elastic tube or in our 2 dimensional case a flow between elastic plates. The flow is driven by a time-dependent pressure difference between the ends of the channel of the form (78). Such flow is also interesting to investigate in the presence of some constriction as a stenosis, which is shown in Figure 13.

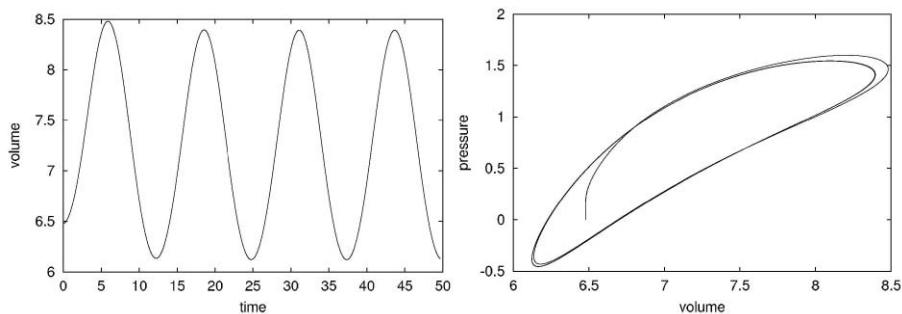


Fig. 8. Volume of the fluid inside and the pressure-volume diagram for the ellipsoidal cavity test.

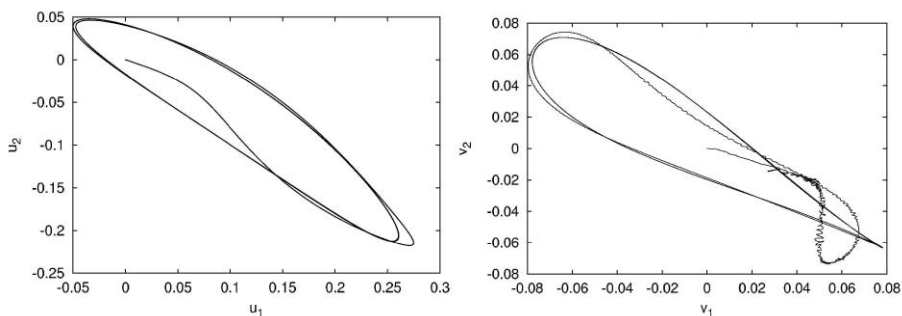


Fig. 9. The displacement trajectory and velocity of a point at the fluid-solid interface (inner side of the wall) for the ellipsoidal cavity test.

For the flow in the channel without any constriction the time dependence of the fluid volume inside the channel is shown together with the pressure volume diagram in the figure and the trajectory and velocity of a material point on the solid fluid interface in the Figures 11 and 12. The velocity field is shown in Figure 10 at different stages of the pulse.

Finally in Figure 13 the velocity field in the fluid and the pressure distribution throughout the wall is shown for the computation of the flow in a channel with elastic obstruction. In this example the elastic obstruction is modelled by the same material as the walls of the channel and is fixed to the elastic walls. Both ends of the walls are fixed at the inflow and outflow and the flow is again driven by a periodic change of the pressure at the left end.

5 Summary and Future Development

In this paper we presented a general formulation of dynamic fluid-structure interaction problem suitable for applications with finite deformations and laminar flows. While the presented example calculations are simplified to

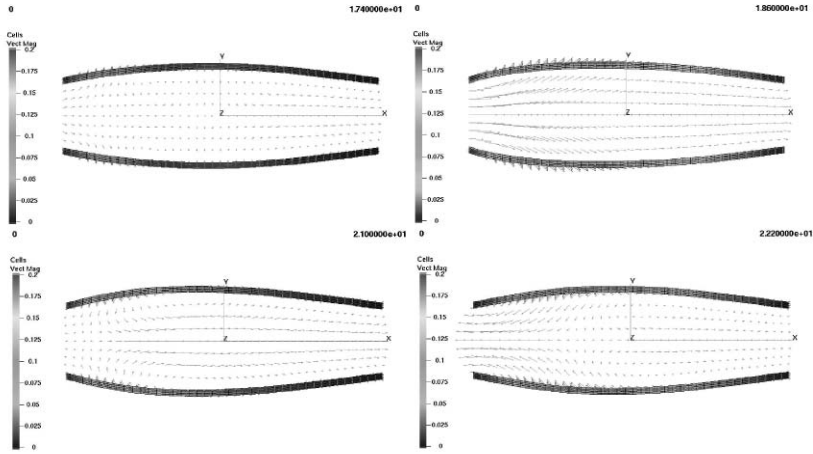


Fig. 10. Velocity field during one pulse in a channel without an obstacle.

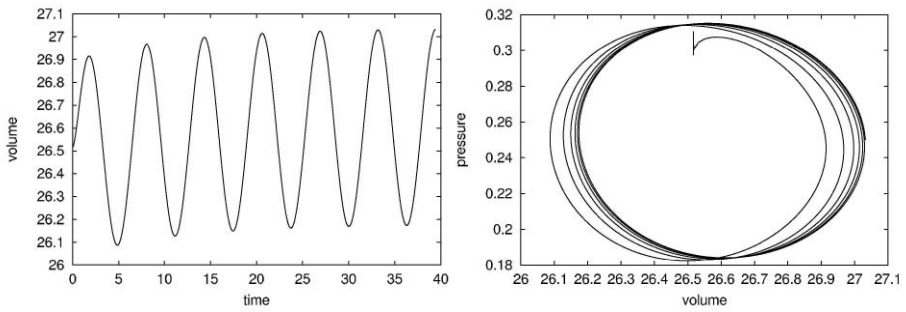


Fig. 11. Volume of the fluid in the channel and the pressure-volume diagram.

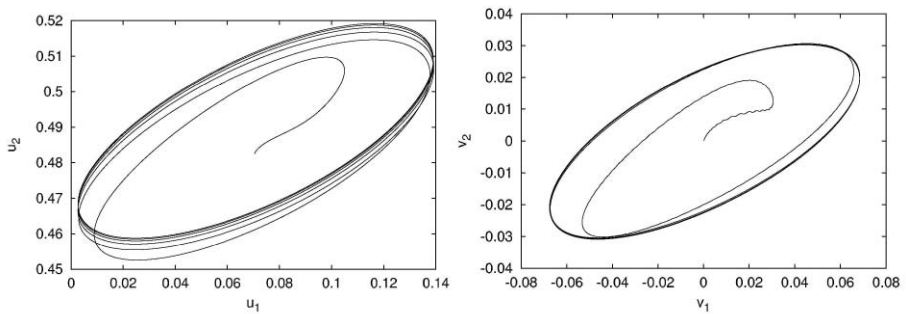


Fig. 12. Displacement trajectory and velocity of a point at the fluid solid interface (inner side of the wall).

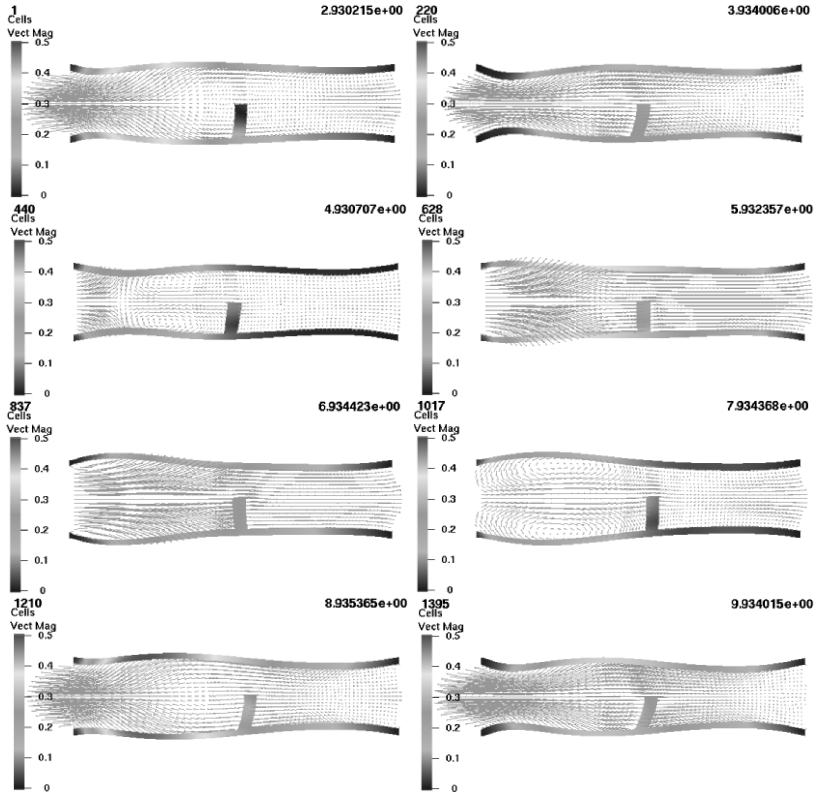


Fig. 13. Fluid flow and pressure distribution in the wall during one pulse for the example flow in a channel with constriction.

allow initial testing of the numerical methods (see [22]) the formulation is general enough to allow immediate extension to more realistic material models. For example in the case of material anisotropy one can consider

$$\tilde{\Psi} = c_1(I_C - 3) + c_2(II_C - 3) + c_3(|\mathbf{F}\mathbf{a}| - 1)^2,$$

with $\mathbf{a}$ being the preferred material direction. The term $|\mathbf{F}\mathbf{a}|$ represents the extension in the direction $\mathbf{a}$. The system can be coupled with additional models of chemical and electric activation of the active response of the biological material, see [23]. In the same manner the constitutive relation for the fluid can be directly extended to the power law models used to describe the shear thinning property of blood. Further extension to viscoelastic models and coupling with the mixture based model for soft tissues together with models for chemical and electric processes involved in biomechanical problems would allow to perform realistic simulation for real applications.

To obtain the solution approximation the discrete systems resulting from the finite element discretization of the governing equations need to be solved which requires sophisticated solvers of nonlinear systems and fast solvers for very large linear systems. The computational complexity increases tremendously for full 3D problems and with more complicated models like viscoelastic materials for the fluid or solid components. The main advantage of the presented numerical method is its accuracy and robustness with respect to the constitutive models. Possible directions of increasing the efficiency of the solvers include the development of improved multigrid solvers, for instance of global pressure Schur complement type [21], and the combination with parallel high performance computing techniques.

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An Implicit Partitioned Method for the Numerical Simulation of Fluid-Structure Interaction

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Abstract. The paper deals with an implicit partitioned solution approach for the numerical simulation of fluid-structure interaction problems. The method is realized on the basis of the finite-volume flow solver FASTEST, the finite-element structural solver FEAP, and the quasi-standard coupling interface MpCCI. Multigrid methods can be involved at different levels. The method is verified by comparisons with benchmark results. Investigations concerning the influence of the grid movement technique and an underrelaxation are presented. Results of a combined numerical-experimental study are given for validation. For the corresponding configuration also basic fluid-structure interaction mechanism are investigated.

1 Introduction

Coupled fluid-solid problems, which are characterized by the interaction of fluid forces and structural deformations occur in many applications in industry and science. For a realistic numerical simulation of such kind of problems one of the crucial issues is the algorithmic realization of the coupling mechanisms. These can be invoked at different levels in diverse manners within the numerical scheme resulting either in a more weakly or more strongly coupled solution procedure (e.g., [10]).

In the present paper we consider an implicit partitioned solution approach, which tries to combine the advantages of weakly and strongly coupled schemes in a complementary way. For each time step the (implicit) solution procedure consists in the application of different nested iteration processes for linearization, pressure-velocity coupling, and linear system solving, which are linked by a special predictor-corrector iteration for the fluid-structure coupling (following [7]). Within the predictor-corrector iteration an underrelaxation is employed to stabilize the solution procedure. In addition, a multigrid technique for accelerating the computations can be involved at the subproblem level (e.g., [8]) or as a global solver (e.g., [11]). The method is realized on the basis of the finite-volume flow solver FASTEST [2], the finite-element structural solver FEAP [14], and the quasi-standard coupling interface MpCCI [9].

On the one hand the considered approach gives a great deal of flexibility due to its modularity, and on the other hand the implicit predictor-corrector scheme ensures a strong numerical coupling which can be controlled by the underrelaxation and can be further improved by the global multigrid approach.

By comparison with benchmark cases the components of the considered approach are verified. Further studies concern the influences of the underrelaxation and the grid adaption for which an algebraic and an elliptic approach are investigated with respect to robustness, accuracy, and efficiency. Validation results related to an experimental reference configuration are presented also allowing an insight to the corresponding basic fluid-structure interaction mechanisms.

2 Governing Equations

We consider a problem domain Ω consisting of a fluid part Ω_f and a solid part Ω_s , which regarding the shape as well as the location of fluid and solid parts can be arbitrary.

For the fluid domain part Ω_f we assume a flow of an incompressible Newtonian fluid. In this case the basic conservation equations governing transport of mass and momentum for a fluid control volume V_f with surface S_f are given by:

$$\frac{d}{dt} \int_{V_f} \rho_f dV_f + \int_{S_f} \rho_f (v_j - v_j^g) n_j dS_f = 0, \quad (1)$$

$$\frac{d}{dt} \int_{V_f} \rho_f v_i dV_f + \int_{S_f} [\rho_f v_j (v_i - v_i^g) n_j - T_{ij}] dS_f = \int_{V_f} \rho_f f_{fi} dV_f, \quad (2)$$

where v_i is the velocity vector with respect to Cartesian coordinates x_i , t is the time, ρ_f is the fluid density, and f_{fi} are external body forces (e.g., buoyancy forces). v_i^g is the velocity with which S_f may move (grid velocity) due to displacements of solid parts. The stress tensor T_{ij} for incompressible Newtonian fluids is defined by

$$T_{ij} = \mu_f \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right) - p \delta_{ij} \quad (3)$$

with the pressure p and the dynamic viscosity μ_f .

The model equations for the solid domain Ω_s may take rather different forms depending on the concrete problem and coupling mechanisms involved. The models range from a simple rigid body motion without any deformation of the solid up to strongly nonlinear (physically and/or geometrically) deformations. The basic balance equation for momentum for the solid domain Ω_s

can be written as

$$\rho_s \ddot{u}_i - \frac{\partial \sigma_{ij}}{\partial x_j} = \rho_s f_{si}, \quad (4)$$

where u_i is the displacement, σ_{ij} denotes the Cauchy stress tensor, ρ_s is the density of the solid, and f_{si} are external volume forces acting on the solid (e.g., gravitational forces). The solid model equations are completed by a suitable (problem dependent) constitutive equation relating the stresses with strains. Here, for the considered test cases the St. Venant-Kirchhoff material is employed (e.g., [1]).

The problem formulation has to be closed by prescribing suitable boundary and interface conditions. On solid and fluid boundaries Γ_s and Γ_f standard conditions as for individual solid and fluid problems can be prescribed. For the velocities and the stresses on a fluid-solid interface Γ_i we have the conditions

$$v_i = \dot{u}_i^b \quad \text{and} \quad \sigma_{ij} n_j = T_{ij} n_j, \quad (5)$$

where $\dot{u}_i^b$ is the velocity of the interface.

3 Numerical Techniques

For the discretization of the problem domain we follow a block-structuring technique. Domains with different fluid or solid parts can be handled straightforwardly by assigning the different parts to different blocks. Solid blocks are treated by the finite-element solver FEAP (see [14]). For the fluid blocks the parallel multigrid finite-volume flow solver FASTEST is employed (see [2,12]). Both solvers involve second-order spatial discretizations and fully implicit second-order time discretizations.

3.1 Fluid-Structure Coupling

For the fluid-structure coupling an implicit partitioned approach is employed (see also [13]). In Fig. 1 a schematic view of the iteration process, which is performed for each time step, is given. After the initializations the flow field is determined in the actual flow geometry. From this the friction and pressure forces on the interacting walls are computed, which are passed to the structural solver as boundary conditions. The structural solver computes the deformations, with which then the fluid mesh is modified, before the flow solver is started again.

For the mesh deformation algebraic and elliptic approaches are employed which are described in more detail in Sect. 3.3. In the fluid solver a discrete form of the space conservation law

$$\frac{d}{dt} \int_{V_f} dV = \int_{S_f} v_j^g n_j dS \quad (6)$$

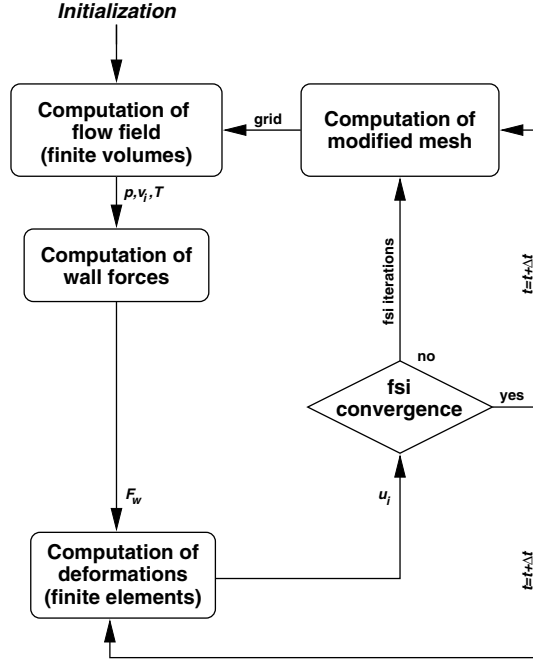


Fig. 1. Flow chart of coupled solution procedure

is taken into account in order to compute the additional convective fluxes in (1)–(2). This is done via the swept volumes δV_c of the control volume faces for which one has the relation (see [3]):

$$\sum_c \frac{\delta V_c^n}{\Delta t_n} = \frac{V_f^n - V_f^{n-1}}{\Delta t_n} = \sum_c (v_j^g n_j S_f)_c^n, \quad (7)$$

where the summation index c runs over the faces of the control volume, the index n denotes the time level t_n and Δt_n is the time step size. By this way interface displacements enter the fluid problem part in a manner strictly ensuring mass conservation.

The fluid-structure interaction (FSI) iteration loop is repeated until a convergence criterion is reached, which is defined by the change of the mean displacements:

$$\mathfrak{R}^{\text{FSI}} = \max_{i=1,2,3} \frac{\sum_{k=1}^N |u_i^{k,m-1} - u_i^{k,m}|}{N} < \varepsilon,$$

where m is the FSI iteration counter and N is the number of interface nodes. Note that an explicit coupling method would be obtained, if only one FSI iteration is performed.

The data transfer between the flow and solid solvers within the partitioned solution procedure is performed via an interface realized by the coupling library MpCCI (see [9]). In Fig. 2 the corresponding information flow is represented schematically. MpCCI is used for controlling the data communication as well as for carrying out the interpolations of the data from the fluid and solid grids.

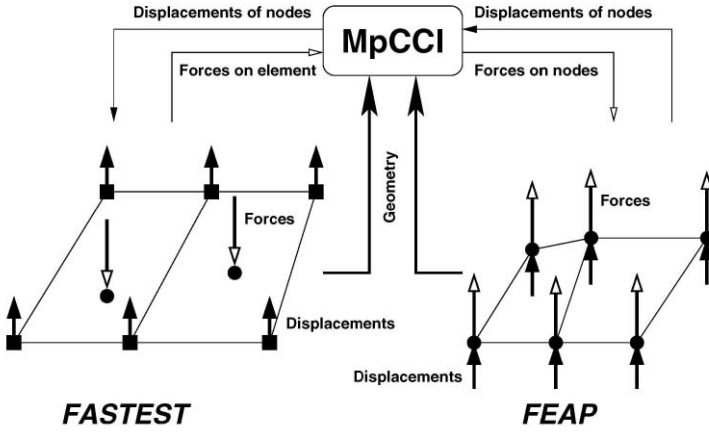


Fig. 2. Schematic representation of information flow between fluid and solid solvers via the MpCCI interface

After the initialization MpCCI is provided with the geometry information at the fluid-solid interface for both grids. From the flow solver these are the coordinates of the control volume vertices and centers at the interface. From the structural solver only the node coordinates are required. With these geometry informations the forces at the nodes of the structural grid are interpolated and passed to the structural solver. The displacements from the structural solver at the nodes are transferred to MpCCI, which interpolates the displacements to the control volume vertices of the fluid grid interface. Afterwards the complete fluid grid is adapted and the corresponding coordinates of the control volume centers are computed. Finally, the new geometry informations are passed to MpCCI for the next iteration. An update of the geometry information of the solid grid is not necessary, since the structural finite-element computations always relate to the original solid grid. Note that with the considered approach nearly arbitrary discretizations for the fluid and solid subproblems can be used, i.e., there is no need for matching grids. For the necessary interpolations MpCCI provides several alternatives. For our test cases a conservative linear interpolation is employed (see [9]).

Various test computations have shown that the coupling scheme is rather sensitive with respect to the deformations in the first FSI iterations. Here,

situations that are far away from the physical equilibrium can arise, which may lead to instabilities or even the divergence of the FSI iterations. In order to counteract this effect an underrelaxation is employed. The actually computed displacements u_i^{act} are (linearly) weighted with the values u_i^{old} from the preceding iteration to give the new displacements u_i^{new} :

$$u_i^{\text{new}} = \alpha_{\text{FSI}} u_i^{\text{act}} + (1 - \alpha_{\text{FSI}}) u_i^{\text{old}},$$

where $0 < \alpha_{\text{FSI}} \leq 1$. Note that the underrelaxation does not change the final converged result.

3.2 Multigrid Method

In order to reduce the computational effort multigrid methods can be invoked into the coupled solution procedure. This can be done at the subproblem level by solving the fluid and/or solid subproblem with a multigrid technique (see Fig. 3). An alternative approach is to employ the partitioned iteration scheme as a smoother in a global multigrid approach as it is indicated schematically in Fig. 4. This may improve the robustness of the fluid-structure coupling.

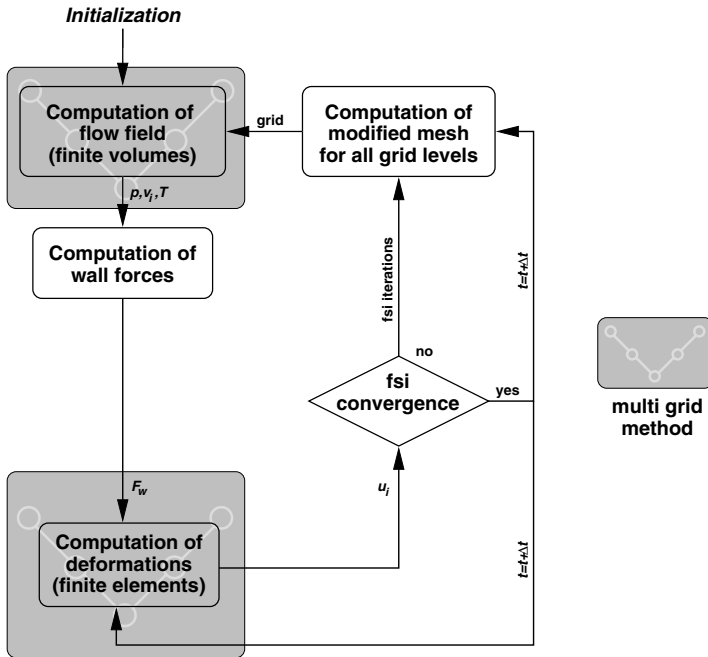


Fig. 3. Flow chart of coupled solution procedure with multigrid subproblem solving

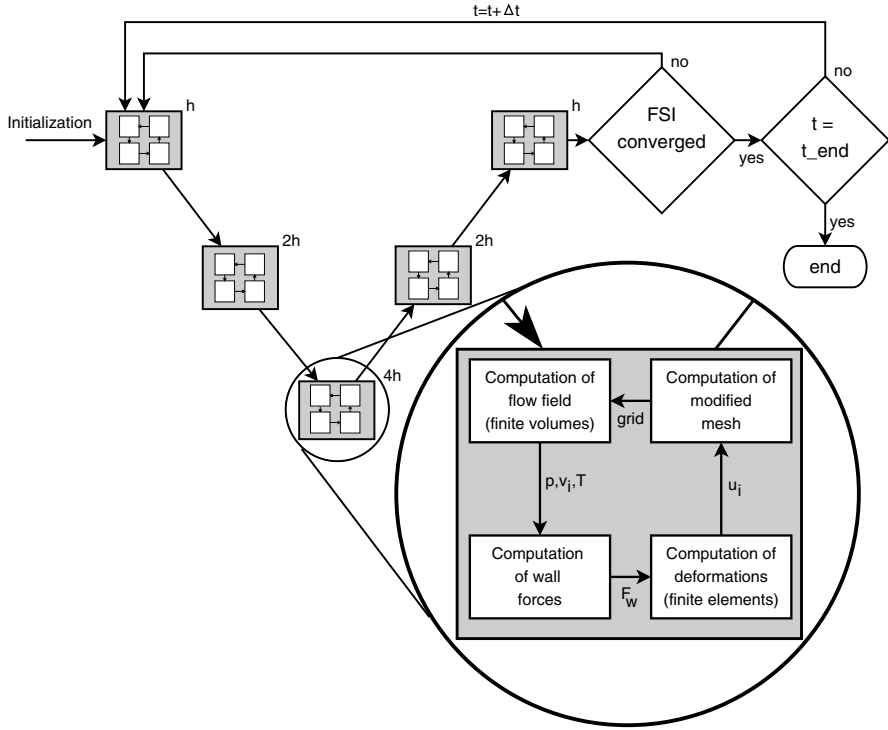


Fig. 4. Flow chart of global multigrid method for fluid-structure coupling

Concerning the multigrid components a standard geometric approach can be employed, i.e., a full approximation storage (FAS) scheme for the treatment of nonlinearities, a V-cycle strategy for the movement through the grid levels, and bilinear interpolation for prolongation and restriction. For steady cases the multigrid method can be combined with the nested iteration technique (full multigrid) to improve the initial guesses on the finer grids. Details can be found, for instance, in [5].

3.3 Grid Movement Techniques

The method for moving the grid in the fluid domain constitutes an important component of the coupled solution procedure, in particular in the case of larger structural deformations. Besides the requirements that no grid folding occurs and that the mesh exactly fits the moving boundaries one has to take care that distortions of control volumes are kept to a minimum in order not to deteriorate the discretization accuracy and the efficiency of the solver.

We consider algebraic and elliptic mesh generation techniques for the grid movement. To simplify the presentation we describe the approaches for

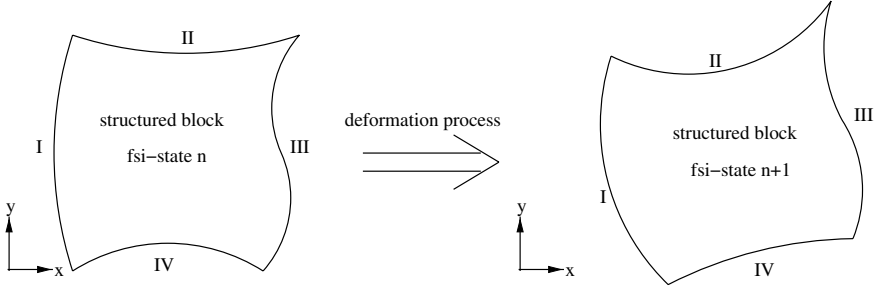


Fig. 5. Deformation of structured two-dimensional block

a single two-dimensional structured block surrounded by 4 boundary curves I to IV (see Fig. 5). The generalization to the three-dimensional case and to multiple blocks is straightforward.

For a structured two-dimensional block there is a one-to-one mapping $\mathbf{x}(\xi, \eta) = (x(\xi, \eta), y(\xi, \eta))$ of the physical coordinates $\mathbf{x} = (x, y)$ to computational coordinates (ξ, η) where without loss of generality $0 \leq \xi \leq 1$ and $0 \leq \eta \leq 1$. In each FSI iteration the coordinates of the interior grid points have to be computed from the (given) boundary points distribution.

As an algebraic approach we consider the linear transfinite interpolation (TFI) where the interior grid points are computed by (e.g., [15])

$$\begin{aligned} \mathbf{x}(\xi, \eta) = & (1 - \eta)\mathbf{x}(\xi, 0) + \eta\mathbf{x}(\xi, 1) + (1 - \xi)\mathbf{x}(0, \eta) + \xi\mathbf{x}(1, \eta) \\ & - \xi [\eta\mathbf{x}(1, 1) + (1 - \eta)\mathbf{x}(1, 0)] \\ & - (1 - \xi) [\eta\mathbf{x}(0, 1) + (1 - \eta)\mathbf{x}(0, 0)] \end{aligned} \quad (8)$$

The method is simple, but deteriorated grid cells and even grid folding may occur.

For an elliptic grid movement we adopt an approach described in [15] which is based on the following (elliptic) Poisson equation for the physical coordinates:

$$\begin{aligned} a\mathbf{x}_{\xi\xi} - 2b\mathbf{x}_{\xi\eta} + c\mathbf{x}_{\eta\eta} + (aP_{11}^1 - 2bP_{12}^1 + cP_{13}^1)\mathbf{x}_{\xi} \\ + (aP_{11}^2 - 2bP_{12}^2 + cP_{13}^2)\mathbf{x}_{\eta} = \mathbf{0} \end{aligned} \quad (9)$$

with the control functions

$$\mathbf{P}_{11} = \frac{1}{s_{\eta}t_{\xi} - s_{\xi}t_{\eta}} \begin{bmatrix} t_{\eta} & -s_{\eta} \\ -t_{\xi} & s_{\xi} \end{bmatrix} \begin{bmatrix} s_{\xi\xi} \\ t_{\xi\xi} \end{bmatrix} \quad (10)$$

$$\mathbf{P}_{12} = \frac{1}{s_{\eta}t_{\xi} - s_{\xi}t_{\eta}} \begin{bmatrix} t_{\eta} & -s_{\eta} \\ -t_{\xi} & s_{\xi} \end{bmatrix} \begin{bmatrix} s_{\xi\eta} \\ t_{\xi\eta} \end{bmatrix} \quad (11)$$

$$\mathbf{P}_{13} = \frac{1}{s_{\eta}t_{\xi} - s_{\xi}t_{\eta}} \begin{bmatrix} t_{\eta} & -s_{\eta} \\ -t_{\xi} & s_{\xi} \end{bmatrix} \begin{bmatrix} s_{\eta\eta} \\ t_{\eta\eta} \end{bmatrix} \quad (12)$$

and the abbreviations

$$a = x_\eta x_\eta + y_\eta y_\eta, \quad b = x_\xi x_\eta + y_\xi y_\eta, \quad c = x_\xi x_\xi + y_\xi y_\xi. \quad (13)$$

An index ξ and η denotes the corresponding derivatives. Figure 6 shows how the mapping between the computational space and the physical domain is performed through the parameter space (s, t) that can be used to control the quality of the mesh.

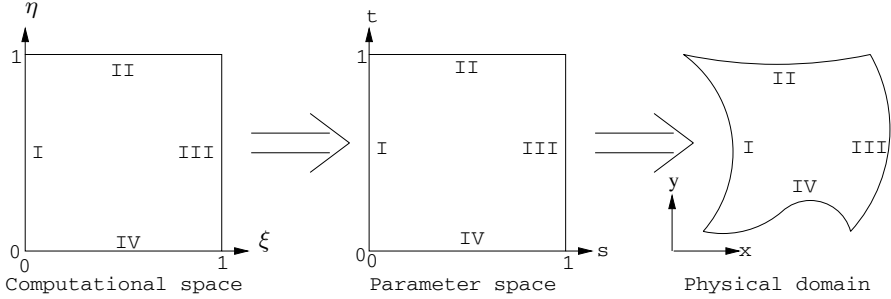


Fig. 6. Mapping strategy for elliptic grid movement

We consider a parameter space with $s(\text{I}) = 0$, $s(\text{III}) = 1$, $t(\text{IV}) = 0$, and $t(\text{II}) = 1$. The point distribution along $s(\text{II})$, $s(\text{IV})$, $t(\text{I})$, and $t(\text{III})$ arises from linear interpolation along these boundaries involving the normalized arc length. The inner parameter domain is adapted by solving simultaneously:

$$s = s(\text{IV})(1 - t) + s(\text{II})t, \quad (14)$$

$$t = t(\text{I})(1 - s) + t(\text{III})s. \quad (15)$$

Next, the control functions P_{11}^i , P_{12}^i , P_{13}^i , $i = 1, 2$ can be computed according to (10) to (12) and remain unchanged during the solution of (9). The derivatives in equation (9) are approximated by central differences and a Picard iteration process is used for linearization:

$$\begin{aligned} & a^{k-1} \mathbf{x}_{\xi\xi}^k - 2b^{k-1} \mathbf{x}_{\xi\eta}^k + c^{k-1} \mathbf{x}_{\eta\eta}^k \\ & + (a^{k-1} P_{11}^1 - 2b^{k-1} P_{12}^1 + c^{k-1} P_{13}^1) \mathbf{x}_\xi^k \\ & + (a^{k-1} P_{11}^2 - 2b^{k-1} P_{12}^2 + c^{k-1} P_{13}^2) \mathbf{x}_\eta^k = \mathbf{0}. \end{aligned} \quad (16)$$

In each step this equation system is solved by the Gauß-Seidel algorithm (the computation time for this step is negligible compared to the total time) for the unknown grid coordinates $x_{i,j}^k$ and $y_{i,j}^k$, $i = 1 \dots N - 1$, $j = 1 \dots M - 1$. The Picard iteration process is repeated as long as a certain convergence criterion is satisfied. Within the FSI iteration process the coordinates of the previous iteration are applied as initial values. The whole solution algorithm, which is summarized schematically in Fig. 7 operates like a mesh smoother. In any case a boundary conforming mesh without grid folding results.

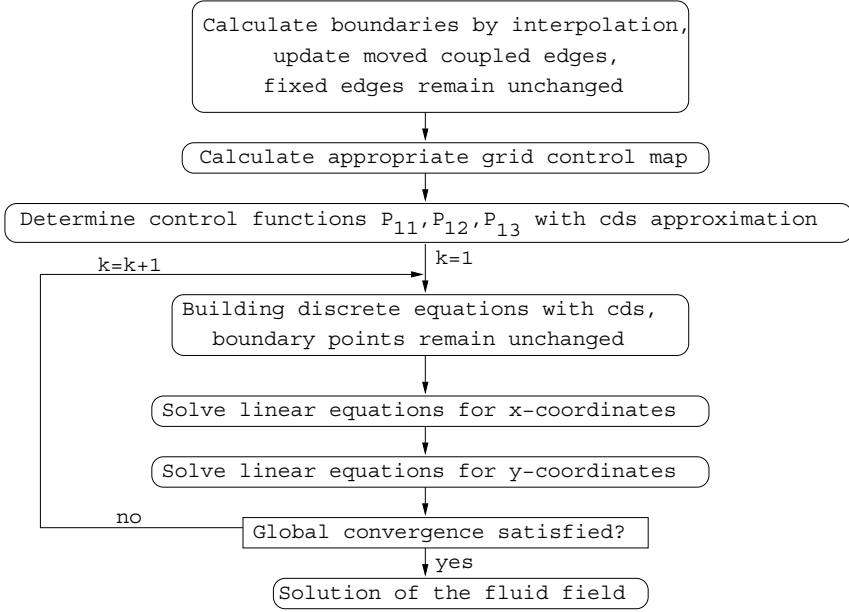


Fig. 7. Flow chart of elliptic mesh movement method

4 Numerical Investigations

In this section we present numerical results for the verification of the coupled solution procedure and for the investigation of the influence of the grid movement technique and the underrelaxation.

4.1 Benchmark Results

For verifying the functionality of the subproblem solvers and the coupled solution procedure we consider the benchmark configuration proposed in [6] for comparing pure fluid, pure structural, and fluid-structure interaction computations for both steady and unsteady cases. For details of the geometry and the physical parameters we refer to [6]. Although the benchmarks are two-dimensional, in order to test the full functionality of the present approaches, we consider all cases in a fully three-dimensional setting with symmetry boundary conditions in the third (z -)direction (see Fig. 8).

For the spatial discretizations of the flow domain a coarse and a fine grid (39 936 and 159 744 control volumes, respectively) for the x - y -plane are considered. On the structure side only the flexible part (flag) is taken into account. This is discretized with 21×4 trilinear 8-node hexahedral solid elements for the x - y -plane, i.e., the thickness of the flag is resolved by 4 elements. The rigid cylinder is not discretized. In the z -direction 2 layers of

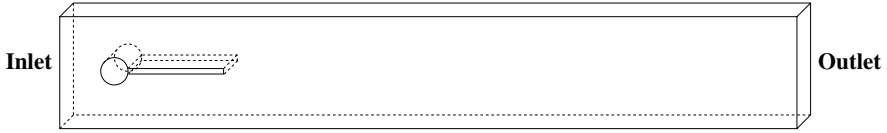


Fig. 8. Geometric configuration of benchmark test case

control volumes and elements, respectively, are employed. Remark that the fluid and structure meshes do not match, which deliberately are selected in order to verify this functionality.

The results for the pure flow test cases CFD1, CFD2, and CFD3 are given in Table 1 comparing the drag and lift forces and frequencies with the corresponding reference values given in [6]. The cases CFD1 and CFD2 corresponding to Reynolds numbers of $Re = 20$ and $Re = 100$, respectively, yield steady flows. The time step size for the transient case CFD3 at Reynolds number $Re = 200$ is $\Delta t = 0.005$ s. In Fig. 9 the temporal development of the drag force for CFD3 is shown. For the steady cases CFD1 and CFD2 already on the relatively coarse fluid grid the results agree well with the reference values given in [6]. For CFD3 the frequencies match well already on the coarse grid, but the amplitudes are not well captured. For the fine grid the latter are close to the benchmark values. The deviation in the mean lift value for CFD3 is due to the fact that the minimum and maximum lift values slightly vary from period to period, i.e., depending on which actual period is taken for the evaluation differences in the observed range can occur.

Table 1. Results for flow benchmark cases

	CFD1	CFD2	CFD3
Drag [kg/ms ²]			
present (39 936 CVs)	14.25	138.8	459.41 ± 4.25
present (159 744 CVs)	14.29	137.0	440.99 ± 5.12
from [6]	14.29	136.7	439.45 ± 5.61
Lift [kg/ms ²]			
present (39 936 CVs)	1.106	9.873	-42.60 ± 406.0
present (159 744 CVs)	1.119	10.50	-5.61 ± 436.1
from [6]	1.119	10.53	-11.89 ± 437.8
Frequency [1/s]			
present (39 936 CVs)	–	–	4.38
present (159 744 CVs)	–	–	4.43
from [6]	–	–	4.39

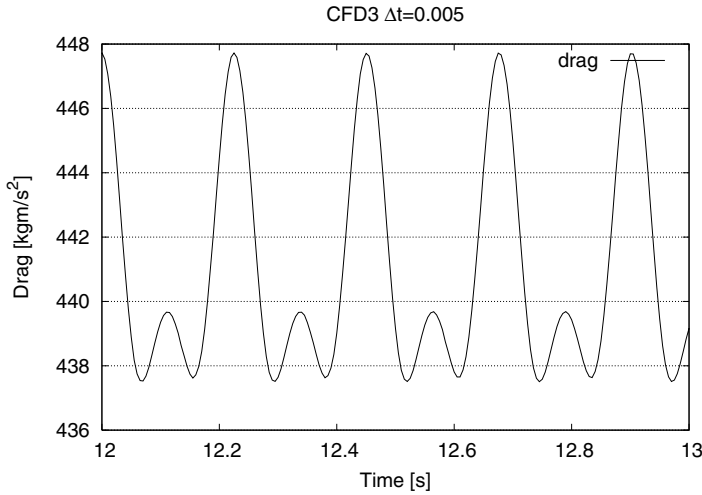


Fig. 9. Temporal behavior of drag force for flow test case CFD3

The results for the pure structural mechanics test cases CSM1, CSM2, and CSM3 are summarized in Table 2, where $ux(A)$ and $uy(A)$ denote the displacements in x - and y -direction of the endpoint of the flexible beam. In Fig. 10 the temporal development of $uy(A)$ for the transient case CSM3 is indicated. Already with the relatively coarse element subdivision the results are in good agreement with the reference values.

Table 2. Results for structure benchmark cases

	CSM1	CSM2	CSM3
$ux(A)$ [m]			
present	$-7.011 \cdot 10^{-3}$	$-4.570 \cdot 10^{-4}$	$-14.21 \cdot 10^{-3} \pm 14.21 \cdot 10^{-3}$
from [6]	$-7.187 \cdot 10^{-3}$	$-4.690 \cdot 10^{-4}$	$-14.30 \cdot 10^{-3} \pm 14.30 \cdot 10^{-3}$
$uy(A)$ [m]			
present	$-6.516 \cdot 10^{-2}$	$-1.672 \cdot 10^{-2}$	$-63.09 \cdot 10^{-3} \pm 61.61 \cdot 10^{-3}$
from [6]	$-6.610 \cdot 10^{-2}$	$-1.697 \cdot 10^{-2}$	$-63.61 \cdot 10^{-3} \pm 65.16 \cdot 10^{-3}$
Frequency [1/s]			
present	–	–	1.098
from [6]	–	–	1.099

The results for the fluid-structure interaction test cases FSI1 and FSI3 for both fluid grids are summarized in Table 3. For the transient case FSI3, that is characterized by periodic oscillations of the flag, a time step size of $\Delta t = 0.01$ s is used. In Fig. 11 the temporal development of the drag force

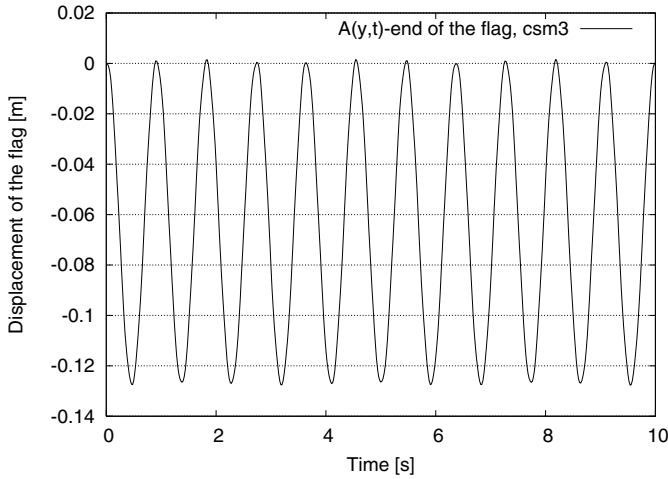


Fig. 10. Temporal behavior of displacement of endpoint of flag in y -direction for test case CSM3

for FSI3 obtained with the fine fluid grid is shown. Also in the coupled case the results are in good agreement with the reference values. The most critical value is the lift force for FSI3 which obviously requires a relatively high spatial and temporal resolution for an accurate prediction.

4.2 Influence of Grid Movement

Next, we investigate the influence of the grid movement and compare the algebraic and elliptic approaches described in Sect. 3.3. As a test case we consider

Table 3. Results for fluid-structure interaction benchmark cases

	FSI1	FSI3
Drag [kg/ms ²]		
present (39 936 CVs)	14.277	466.8 ± 24.31
present (159 744 CVs)	14.289	457.7 ± 23.75
from [6]	14.295	457.3 ± 22.66
Lift [kg/ms ²]		
present (39 936 CVs)	0.772	0.9 ± 168.3
present (159 744 CVs)	0.769	13.5 ± 156.9
from [6]	0.764	2.2 ± 149.8
Frequency [1/s]		
present (39 936 CVs)	–	5.6
present (159 744 CVs)	–	5.2
from [6]	–	5.3

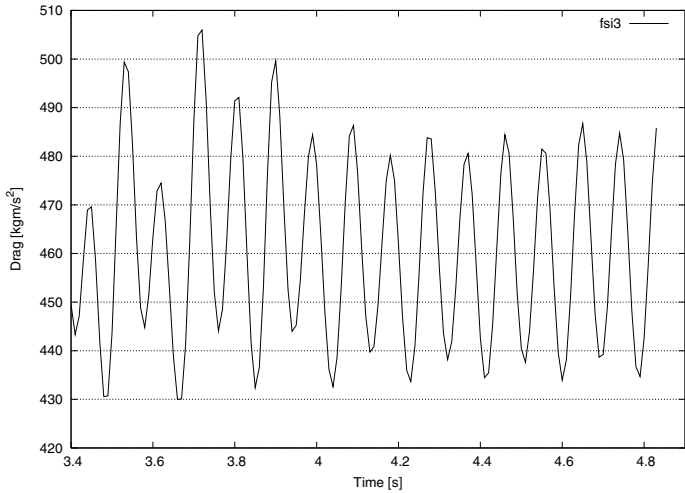


Fig. 11. Temporal behavior of drag force for fluid-structure interaction case FSI3

a laminar flow around a rotating cylinder fixed in a channel which allows for a systematic variation of the amount of grid movement. Figure 12 shows the geometry and the corresponding block structure consisting of two fixed blocks at the inlet and the outlet and four moving blocks belonging to an O-grid around the cylinder. The kinematic viscosity is defined as $\nu_f = 10^{-3} \text{ m}^2/\text{s}$ and the fluid density is $\rho_f = 1.0 \text{ kg/m}^3$. At the inflow a parabolic velocity profile with mean velocity $\bar{v} = 0.2 \text{ m/s}$ which with the cylinder diameter of $D = 0.1 \text{ m}$ gives the Reynolds number $Re = 20$. To get a physically two-dimensional problem a symmetry boundary condition is applied in the third direction.

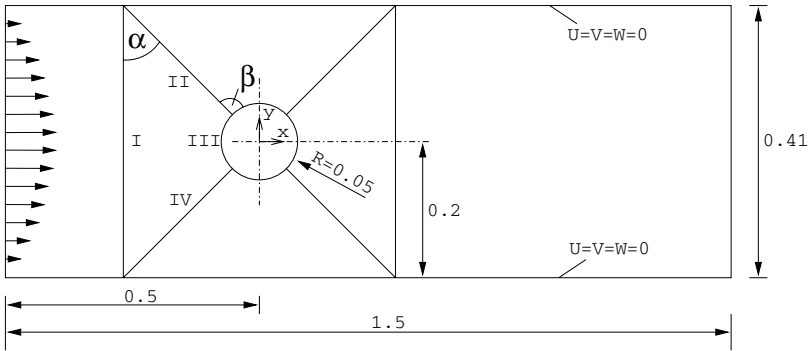


Fig. 12. Test configuration for grid movement techniques (units in m)

The cylinder is rotated in steps of 5 degrees. The grid adaption according to the rotating cylinder is done by the algebraic TFI method and the elliptic approach as outlined in Sect. 3.3. After each step a steady flow computation is performed until grid folding or divergence of the iterations occur. 125 000 control volumes and exactly the same parameters are used for both cases and for all angles of torsion. For comparison a reference solution obtained by a computation with 450 000 control volumes on an undeformed mesh is considered.

In Fig. 13 the grids with the maximum angle of torsion for which the last converged solution could be achieved are illustrated. While the elliptic approach allows for an angle of 95° the algebraic method only works until 70° . It should be mentioned that the boundaries II and IV in the elliptic case are approximated by a cubic spline interpolation to get angles of about $\alpha = 45^\circ$ and $\beta = 90^\circ$ (see Fig. 12). One can recognize that even for high distortions with the elliptic approach the grid is nearly orthogonal at the cylinder boundary.

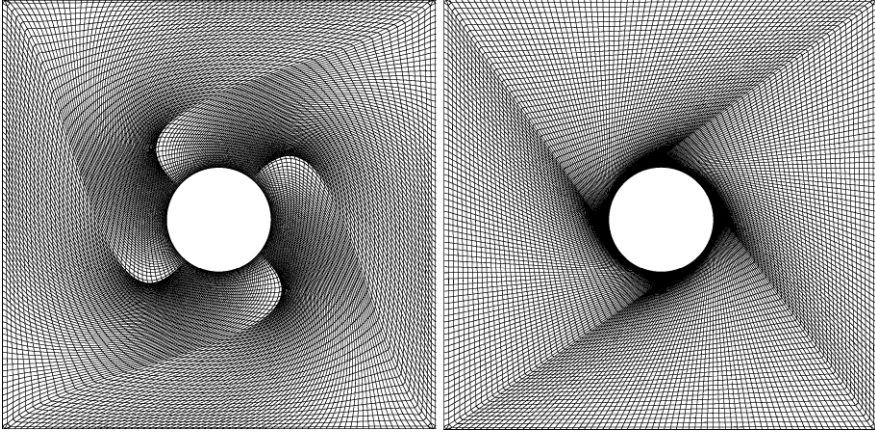


Fig. 13. Maximum possible grid distortions with elliptic (left, 95°) and algebraic (right, 70°) grid movements

In Fig. 14 a comparison of the lift coefficients with varying rotation angle is shown. One can see the increasing deviation from the reference value with increasing rotation angle. However, in particular for larger angles, the error with the elliptic method is much smaller than with the algebraic method.

In Fig. 15 the numbers of iterations required to obtain the converged solution for varying rotation angle is indicated for both grid deformations. The iteration numbers increase with increasing angle. While for small angles the algebraic approach is slightly faster, the situation changes for larger angles

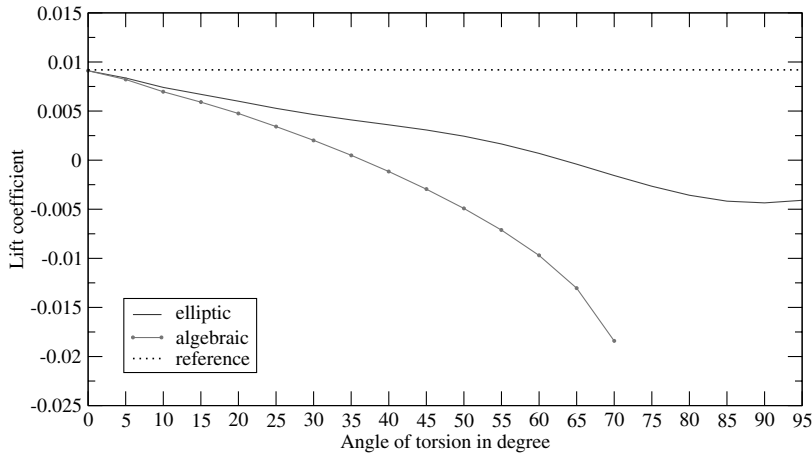


Fig. 14. Comparison of lift coefficients with algebraic and elliptic grid movements

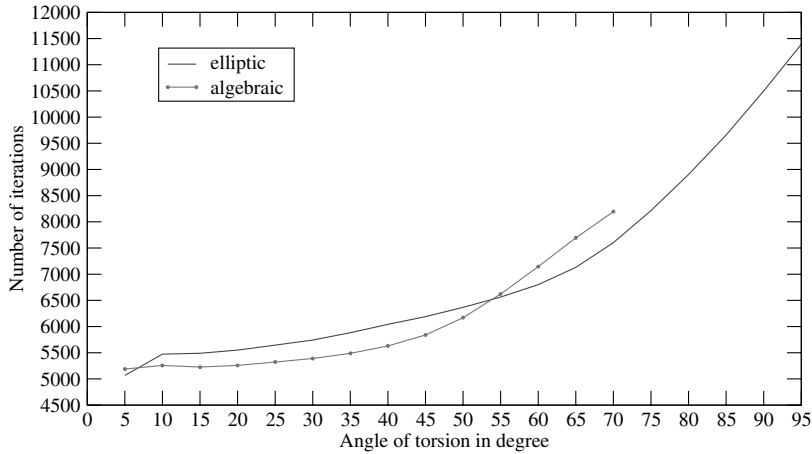


Fig. 15. Comparison of computational effort with algebraic and elliptic grid movements

due to the heavy distortions of control volumes near the boundary with the algebraic approach.

In summary one can state that for small distortions, i.e., with angles of torsion $\approx 0^\circ - 20^\circ$, the algebraic method works satisfactorily, but for larger angles with the elliptic approach significantly more accurate results can be obtained. The latter even still works in cases where the algebraic approach leads to grid folding.

4.3 Structural Underrelaxation

To study the effect of the underrelaxation we consider the transient fluid-structure interaction test case FSI3 with the coarse fluid grid (see Sect. 4.1). The influence of the underrelaxation can be seen in Fig. 16 where the total number of iterations (which are directly proportional to the total CPU time) with varying underrelaxation factor is shown for different time step sizes. In each case the same period of time is computed (1 second in the periodic state).

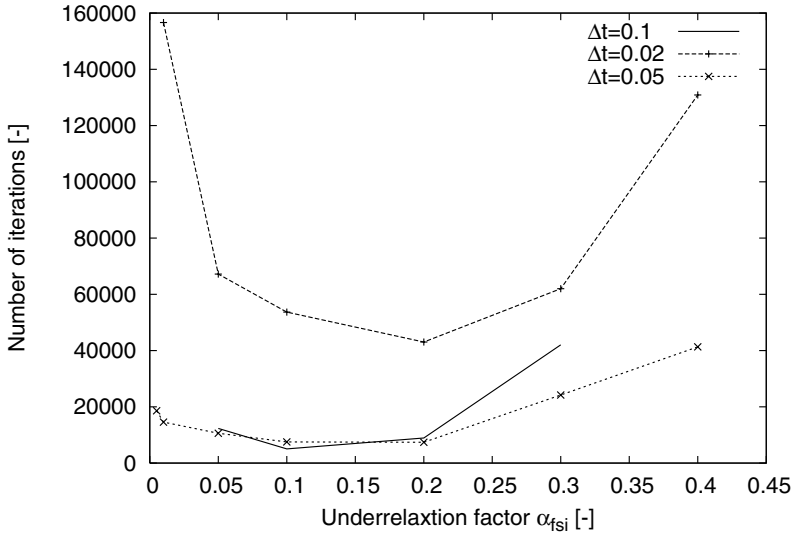


Fig. 16. Total number of iterations for varying underrelaxation factor α_{FSI} and different time step sizes

It can be seen that there is an optimal value for α_{FSI} that increases with decreasing time step size. For the large time step size $\Delta t = 0.1$ s already above $\alpha_{FSI} = 0.3$ the FSI iterations diverge, for the smaller ones above $\alpha_{FSI} = 0.4$. The results indicate that the underrelaxation – besides the stabilizing effect – can also have an effect on the efficiency of the computation. For instance, for the small time step size $\Delta t = 0.05$ s with $\alpha_{FSI} = 0.2$ the computational effort is significantly lower than with higher values with which the FSI iterations still converge. The incorporation of a strategy for an adaptive determination of α_{FSI} in the context of the multigrid method is a topic of further research.

5 Comparison with Experiment

For model validation and for studying basic fluid-structure interaction effects a corresponding reference configuration has been set up in cooperation with the Institute of Fluid Mechanics at University of Erlangen-Nürnberg (see [4]). A sketch of the test channel is shown in Fig. 17. Note that for all simulations the actual inflow boundary (with uniform inflow profile) is chosen that far away from the inlet of the test channel, where the profiles are measured, such that the flow is already fully developed there.

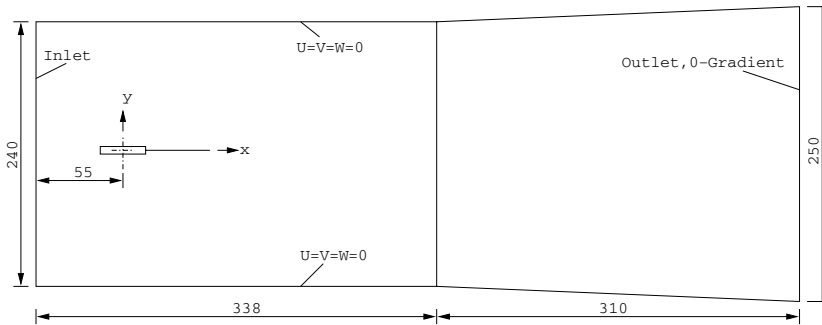


Fig. 17. Geometric properties of test channel (units in mm)

Initial experiments have been carried out for the structural configuration presented in Fig. 18 (left) with water as test fluid. It was found that a rotational degree of freedom for the front plate is necessary to obtain a swiveling motion. Changing the fluid to polyethylene glycol to get into a flow laminar regime, resulted in a totally different structural behavior. In this case for any combination of velocity, viscosity, and membrane thickness neither in the experiment nor in the simulation periodic vibrations could be achieved with this structural configuration.

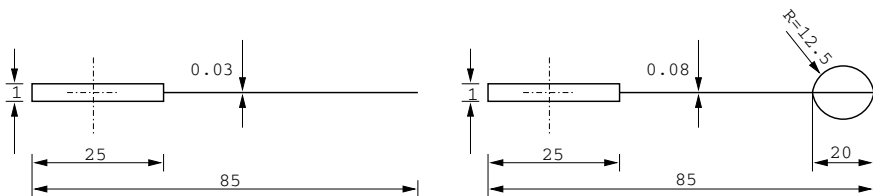


Fig. 18. Structural configurations for preliminary investigations

By numerical simulations it was found that a way to get the desired vibrations with real materials was to put a weight at the trailing edge of the membrane to increase the mass. This finally resulted in the configuration shown in Fig. 18 (right). According to a linear material behavior with slight bending the assumptions for St. Venant-Kirchhoff's material law are assumed to be valid. The parameters are: inflow velocity $\bar{v} = 2 \text{ m/s}$, fluid density $\rho_f = 1075 \text{ kg/m}^3$, kinematic viscosity $\nu_f = 5 \cdot 10^{-4} \text{ m}^2/\text{s}$, structural density $\rho_s = 7850 \text{ kg/m}^3$, and Young's modulus $E = 2 \cdot 10^{11} \text{ N/m}^2$. The resulting Reynolds number is $Re = 340$.

For the structural discretizations linear solid hexahedrons (1 840 elements) are applied in combination with enhanced strain formulations allowing for very large aspect ratios. The fluid domain is discretized by 53 776 control volumes. For mesh adaption the elliptic method is applied (the algebraic approach fails here in most cases).

Simulation results for the adapted preliminary configuration are shown in Fig. 19. Given are a snapshot of the velocity component in x -direction (left) and the temporal developments of the displacements of the front and rear ends of the structure in y -direction (right) illustrating the periodic oscillations of the configuration.

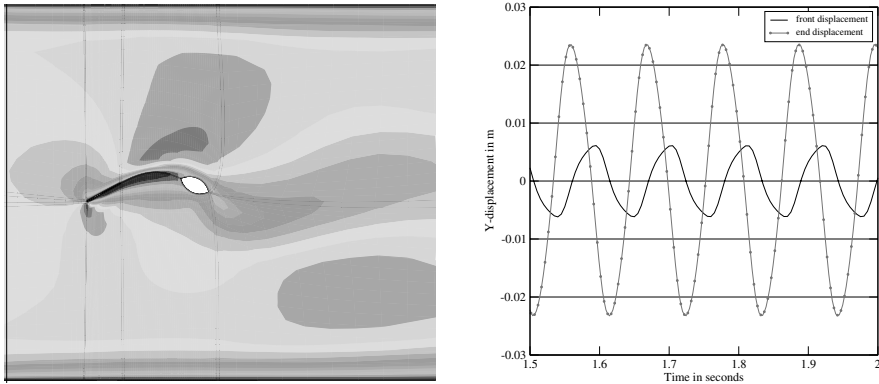


Fig. 19. Numerical results for adapted preliminary configuration: instantaneous velocity component in x -direction (left) and displacements of front end rear ends of structure in y -direction (right)

After complementary numerical-experimental fine tuning a further modified configuration as shown in Fig. 20 could be set up. It consists of a cylindrical aluminum front body ($\rho_s = 2828 \text{ kg/m}^3$, $E = 7 \cdot 10^{10} \text{ N/m}^2$), a thin membrane of stainless steel ($\rho_s = 7855 \text{ kg/m}^3$, $E = 2 \cdot 10^{11} \text{ N/m}^2$), and a rectangular fluid polyethylene glycol is chosen with density $\rho_f = 1050 \text{ kg/m}^3$

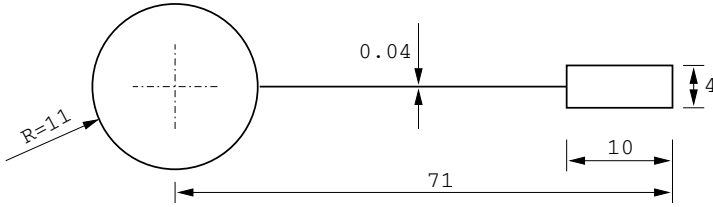


Fig. 20. Final structural configuration

and kinematic viscosity $\nu_f = 1.64 \cdot 10^{-4} \text{ m}^2/\text{s}$. According to the experiment a uniform inlet velocity of $\bar{v} = 1.37 \text{ m/s}$ is applied. The configuration is fixed in the channel with one rotational degree of freedom positioned at the center of the cylinder, where no friction is assumed at the fixing point. The gravity force is aligned with the x -axis. The physical problem can be considered in very good approximation as two-dimensional what was confirmed by several measurements. In the simulations this is represented by symmetry boundary conditions in z -direction.

For a better understanding of the physics of the fluid-structure interaction problem first an eigenfrequency analysis for the structure is carried out by a pure structural simulation. The first two eigenmodes are shown in Fig. 21 (the trivial rigid body motion with 0.048 Hz is not considered). In the first mode with frequency 6.12 Hz the front body moves in the same direction as the rear mass, while in the second mode with frequency 30.07 Hz the moving direction of the front and the end mass is opposite.

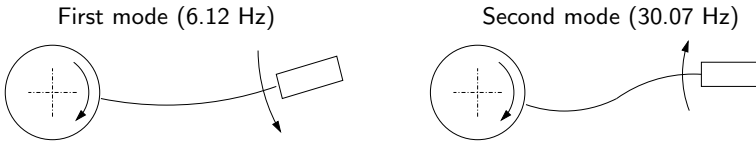


Fig. 21. Eigenmodes of final structural configuration

Now, the fluid-structure interaction case is considered. In the computation as well as in the experiment the structure starts vibrating by itself until it swivels. A snapshot of the velocity component in x -direction is given in Fig. 22. In both cases the structure vibrates in a state corresponding to the second mode, i.e. the front body moves in the opposite direction as the rear mass. The oscillation is periodic but not harmonic what is expected for non-linear vibrations. The measured experimental frequency is $13 \text{ Hz} \pm 1.5\%$ and the corresponding computation yields 12.7 Hz , i.e. the agreement is very good. The spectral representation of the rear mass and front body oscillations

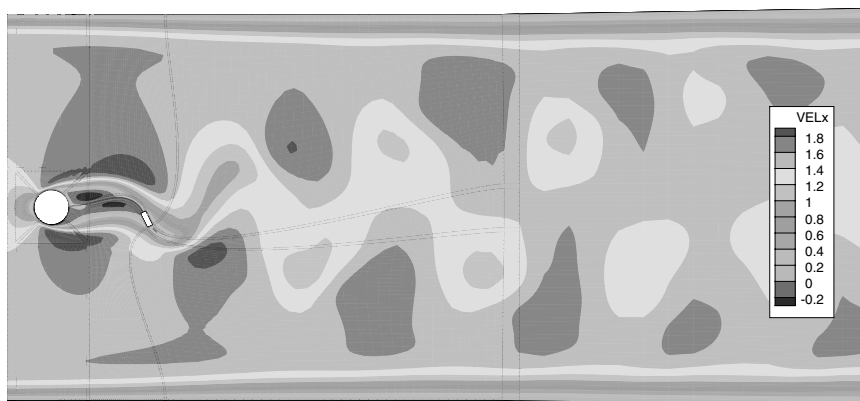


Fig. 22. Velocity in x -direction for final configuration

obtained from the simulation results is shown in Fig. 23. It indicates that the other occurring frequencies are negligibly small. One can also observe that the fluid-structure interaction swiveling motion frequencies do not coincide with the corresponding eigenfrequencies of the structure.

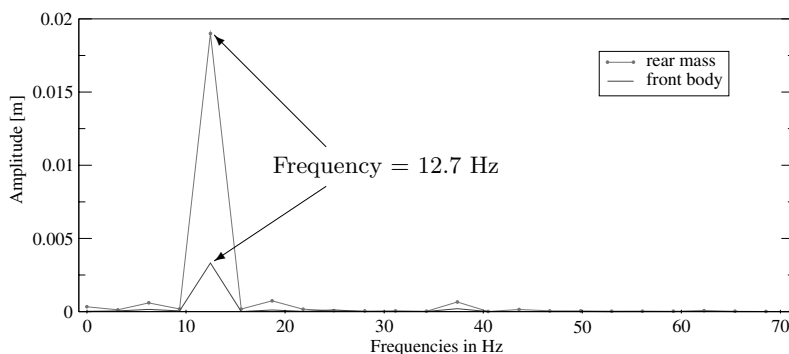


Fig. 23. Spectral representation of rear mass and front body oscillations (from simulation results)

In Fig. 24 a comparison between experiment and simulation for the displacement of the trailing edge in the x - y -plane in one period is shown. Again the experimental and numerical results are in good agreement. The maximum y -amplitude in both cases is between 18 and 20 mm. The reason for the slight difference of the x - y -displacements of the trailing edge, besides the usual disturbances in experiments (no perfect materials, measurement tolerances etc.) seems to be due to the assumption of a fully two-dimensional

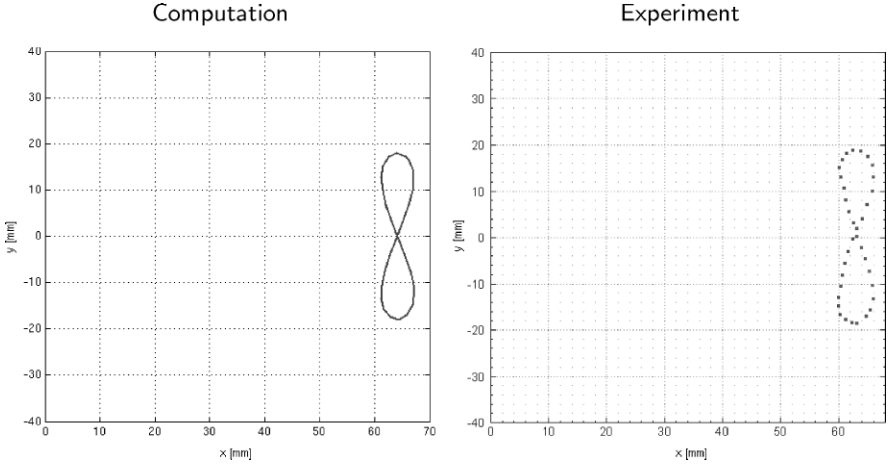


Fig. 24. Comparison of experimental and numerical trailing edge displacements for final configuration

physical problem. In the experiment the side walls and the small gap between them and the structure have a damping effect which is not represented in the simulation with symmetric boundary conditions. The influence of this effect will be investigated in future studies.

6 Conclusions

An implicit partitioned approach for the numerical simulation of fluid-structure interaction has been presented. Flexibility and modularity is achieved by employing the coupling interface MpCCI. For mesh movement an algebraic method employing linear transfinite interpolation and an elliptic approach are employed.

The approach has been verified by comparisons with benchmark test cases for which already on relatively coarse grids a fairly good agreement has been obtained. Numerical investigations with respect to the grid movement have illustrated the advantages of an elliptic approach, in particular in the case of large mesh deformations. The stabilizing effect of an underrelaxation has been studied which also may lead to an improved numerical efficiency of the coupled scheme.

In comparative studies with experimental test configurations good agreement could be achieved and the numerical simulation results could be employed as a guideline for setting up a reference configuration for validating basic fluid-structure interaction mechanisms.

In forthcoming work, in particular, further investigations are foreseen concerning the performance of the global multigrid algorithm as it already has been successfully realized for thermally coupled fluid-solid problems (see [11]).

In addition, fluid-structure interaction problems involving turbulent flows and their adequate numerical treatment will be a matter of interest.

Acknowledgement

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Large Deformation Fluid-Structure Interaction – Advances in ALE Methods and New Fixed Grid Approaches

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Abstract. This contribution focusses on computational approaches for fluid structure interaction problems from several perspectives. Common driving force is the desire to handle even the large deformation case in a robust, efficient and straightforward way. In order to meet these requirements main subjects are on the one hand necessary improvements on coupling issues as well as on Arbitrary Lagrangian Eulerian (ALE) approaches. On the other hand, we discuss pros and cons of available fixed grid approaches and start the development of new such approaches. Some numerical examples are provided along the paper.

1 Introduction

Fluid structure interaction is of great relevance in many fields of engineering as well as in the applied sciences. Hence, the development and application of respective simulation approaches has gained great attention over the past decades. Some current endeavors in this field are: the advancement from special purpose or special problem to quite general approaches; the desire to even capture very general and complex systems; and the exigent need of robust high quality approaches even for such complex cases, *i.e.* approaches that have the potential to turn over from being a challenging and fascinating research topic to a development tool with real predictive capabilities. Often when interaction effects are essential this comes along with large structural deformations. So it is of great relevance of being able to adequately deal with this case. However, many available approaches (both in research as well as in commercial codes) lack robustness especially in this situation.

In this contribution we report on different advances to improve both robustness and quality of the simulation of large deformation fluid structure interaction (FSI) problems. For this undertaken we have in mind the problem of incompressible flows interacting with flexible, often thin-walled, structures exhibiting large deformations. However, the discussed issues are not restricted to this case. We also want no restriction on the structural model, *i.e.* the approaches should for example not only work with degenerated structural

models but should also work when the real volume is taken into account even in the thin-walled case. The paper first focusses on necessary recent advances on coupling issues as well as on Arbitrary Lagrangian Eulerian (ALE) approaches. Another main focus is on fixed grid approaches, where we discuss pros and cons of some available fixed grid approaches in order to motivate the development of new such approaches, which is also started in the second part of this paper.

Obviously, key points in the simulation of coupled problems are the coupling issues itself. Surprisingly, the importance of these issues has often been disrespected in the past. Coupling issues are both the transfer of quantities at the interface and the computational coupling algorithm. Among others it will be shown, that some commonly used coupling schemes are instable in certain frequent and important cases. This is independent of the underlying discretization schemes as well as of the underlying principle approach (*i.e.* ALE or fixed grid).

Most research and commercial codes that are available for simulations of the interaction of flows and flexible, often thin-walled, structures are based on the Arbitrary Lagrangian Eulerian method. These approaches go back to early works like [1–6]. A sketch of the general problem of the interaction of a flow field and a flexible structure, in this case an embedded structure, is shown in Figure 1(a). The conjoined interface Γ^{fsi} separates the structural

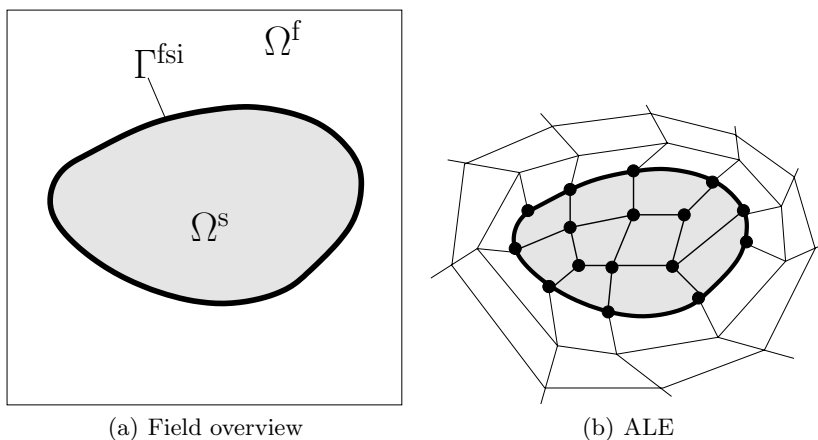


Fig. 1. Figure 1(a) shows the involved fluid field Ω^f , structural field Ω^s and the conjoined interface Γ^{fsi} . In the ALE approach, depicted in Figure 1(b), the fluid domain ends at the structural surface and the boundary fluid nodes move with the structure. The interior fluid mesh is deformed or smoothed by mesh smoothing/deforming algorithms.

domain Ω^s from the fluid domain Ω^f . The essential feature here is that the fluid field is formulated and solved on a deforming grid. This grid deforms with the structure at the interface and then the grid deformation is extended into the fluid field.

As said such ALE based approaches are by far the most wide spread schemes. However, if one is interested in large deformation and long duration simulations with high accuracy, many currently used algorithms suffer from one or more shortcomings that hamper such simulations. In the paper, we sketch some recent advances that improve this situation. These developments are with respect to the geometric conservation question, the accuracy of a flow solver on deforming grids and on mesh motion schemes.

But even the most advanced and best cultured ALE based scheme once comes to its limits where only re-meshing helps. At the latest in such situations one might be tempted to turn over to approaches that work with a fixed grid. But these schemes might also in other cases be an attractive alternative. However, also none (to the knowledge of the authors) of the currently available fixed-grid methods meet the high requirements stated above. Since a complete review on fixed grid methods would go beyond the scope of this paper, features and problems of a few existing methods are elaborated exemplary to develop a list of ‘minimum requirements’ for fixed grid methods. These requirements have to be fulfilled in order to match or surpass the ALE approach with respect to accuracy and numerical stability.

The Immersed Boundary (IB) method [7,8] is one of the most prominent and also earliest attempts to simulate the interaction of elastic structures with viscous flows. Mittal & Iaccarino [9] give an exhaustive overview about developments descending from the original formulation. In its original form, the IB method uses an elastic, spring-like fiber structure that is completely immersed into a viscous fluid, which could also be used to weave a net of fibers to represent a three-dimensional volume. More recently, the IB formulation was extended for 3d finite element formulations for fluid and structures by the Extended Immersed Boundary Method (EIBM) [10] and the Immersed Finite Element Method (IFEM) [11].

For fully fledged structures, *i.e.* no degenerated beam or shell descriptions, the fluid extends over the physical fluid domain Ω^f and into the structural domain Ω^s , which, from the fluid point of view, is a ‘background’ or ‘fictitious’ fluid domain (both names are used alternatively). The structure is deformed by interpolating the fluid velocity from surrounding fluid nodes onto Lagrangian structural nodes – at the surface and inside the structure. The structural deformation causes elastic forces computed from spring- or FE-formulations, which are subsequently interpolated back onto the fluid nodes. This results in a fluid momentum equation with an additional body force term ultimately depending on the fluid velocity. Consequently, the solid degrees of freedom are completely removed from the system of equation and the structural influence on the flow is entirely represented by the additional

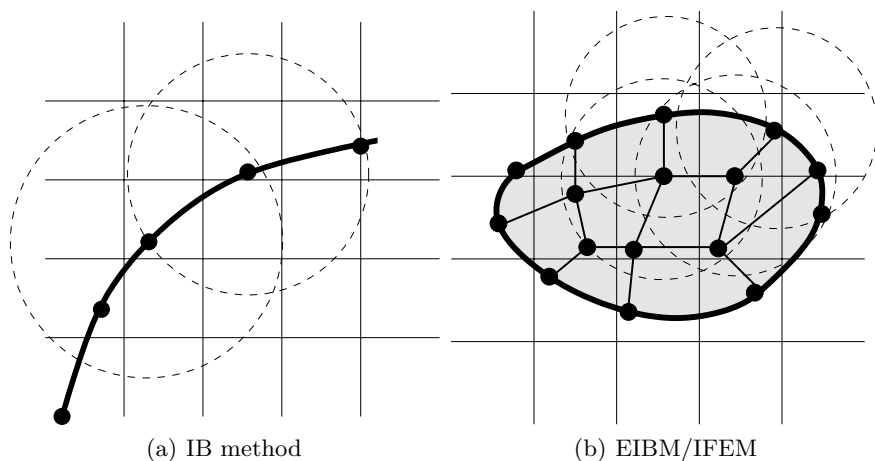


Fig. 2. Illustration of IB-type methods: The interpolation functions (dashed circles) centered at the structural Lagrangian nodes (•) maps the elastic forces onto the background fluid mesh.

force terms. A sketch of the both the IB method and the IFEM are depicted in Figure 2.

Many of the original shortcomings have meanwhile been removed or overcome (see overviews given in References [12,9,8]). However, the remaining drawbacks still limit the range of application of this approach. Much effort has been made to remove the loss of accuracy due to interpolation and resulting smoothing of the velocity field and traction discontinuity along the surface (see *e.g.* [13]). A further restriction is the mutual mesh-size dependency between fluid and solid mesh to achieve accurate kinematic matching of fluid and solid velocity at the surface and within the structure. The smoothed properties and interpolation function hidden in the nonlinear force term hamper the development of fully implicit algorithms, although many improvements towards half and fully implicit formulations have been made. Despite of these limitations, the IB method has been proven to be a useful tool in simulating biomechanic problems, where soft tissues don't limit the usage of an explicit coupling algorithm and a relative low Reynolds number reduces the impact of a smoothed boundary layer near the immersed interface. The efficient implementation of the fluid solver using a Cartesian grid did allow complex simulations such as the simulation of flow inside a beating human heart [7].

An approach with many similarities to the IB method is the so called Distributed Lagrange Multiplier / Fictitious Domain (DLM/FD) method [14,15]. Originally, the approach was developed for rigid particles. The DLM/FD

methods have since been extended to simulate thin, deformable structural surfaces [16–18] as well as to flexible and fully fledged structures [19].

In this method, the structural degrees of freedom are not removed, instead, (usually) a monolithic system is formed with both fluid and structural degrees of freedom. The kinematic matching of the background fluid velocity and the structural nodes is achieved by using Lagrange multiplier techniques to constrain the background flow field in Ω^s . In other words, for rigid or flexible, volume-occupying bodies, the background fluid is forced to perform the same motion as the overlaying structure. Figure 3 illustrates possible Lagrange multiplier discretization along a surface and within the structure. Again, the methods has no inherent way to accurately represent a stress jump over the fluid structure interface. This problem has been addressed, for example, by adapting the mesh locally [20] or using the eXtended Finite Element Method [21] (XFEM) to allow for a proper representation of the stress discontinuity.

Other fixed-grid or Cartesian-grid methods that won't be explored here in more detail are, *e.g.* the volume of fluid method [22] or level-set method [23], which are mostly used for two-phase flow problems. Methods for fluid structure interaction are, among others, Fedkiw *et al.* [24] and Cirak & Radovitzky [25]. While efficient due to the use of fixed Cartesian grids, most of them lack one or more item of the feature list below, most prominently, a conserving, accurate interface description.

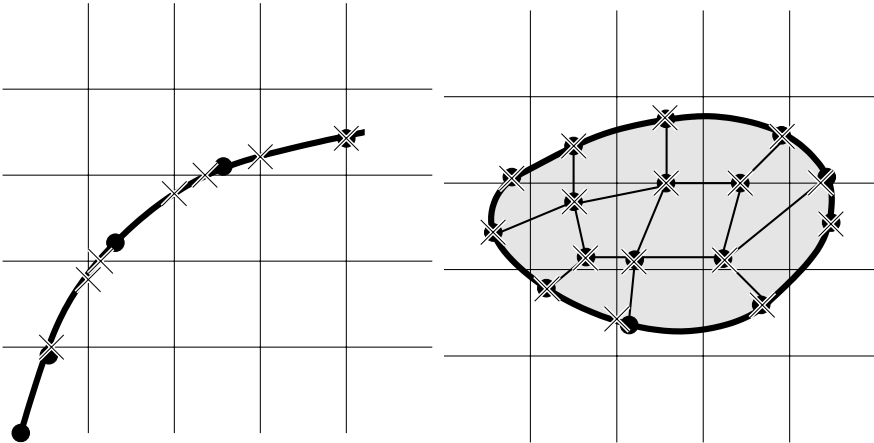


Fig. 3. Illustration of DLM/FD-type methods: Several discretization types exist for the Lagrange multiplier field (Lagrange multiplier nodes marked with $\times$, Lagrangian structure nodes with $\bullet$). Several proposals exist on the proper discretization of the Lagrange multiplier field.

Summarizing the shortcomings of currently available methods (also of methods that have not been discussed in detail above), we see the following list of necessary improvements or requirements for future fixed-grid approaches.

- *Accurate representation of the (stress) discontinuity* Fixed grid methods can only become serious tools, if the interface and the transfer of variables is represented accurately and consistently.
- *Removing the influence of the fictitious fluid on the simulation* This includes several important points as
 - *Removing the incompressibility constraint from the structural domain* If all Lagrangian points are ‘tied’ to the fictitious fluid by the IB interpolation or a Lagrangian multiplier technique, the structural deformation has to be divergence free [19,11]. While this might not be a limitation for applications in biomechanics, where usually water-filled and therefore incompressible tissues are considered, it certainly is a limitation in many engineering applications, since most materials and structures deform in a compressible way. Allowing the structure to deform in a compressible way seems crucial to make fixed grid methods generally applicable.
 - *Artificial viscosity* Despite of the incompressibility problem, the artificial fluid viscosity that is present in the discussed methods for deformable structures has to be removed. For long-time simulation, even a small viscosity will have an impact on the accuracy of the method.
 - *Removing fluid degrees of freedom from within of the structural domain* From an efficiency point of view, it would be advisable to remove the extra fluid unknowns within the structural domain as far as possible. If the structure occupies a significant space of the computational domain, the computational cost of the redundant calculations cannot be ignored.
 - *Empty structures* Removing fluid degrees of freedom from inside the solid can increase the flexibility of the method for another reason. Looking at potential applications, there might be the wish to simulate e.g. a shell structure separating two fluids.
- *Mesh size independence* Methods like the IB methods require a certain ratio between fluid and solid mesh size at the interface to simulate impermeable structural surfaces. Future methods should allow an independent mesh size for each of the simulated fields.

This list served as a guideline for the development of new fixed-grid approaches that promise to overcome the addressed shortcomings. First steps of this development are sketched in this paper. The first approach is based on an overlapping domain decomposition idea and is denoted as a Chimera-like method. The second approach extends the DLM/FD method by using XFEM

ideas to sharply and accurately separate the fluid solution outside and within the structure.

The paper is structured as follows: First the general problem is stated in Chapter 2. Chapter 3 covers the addressed coupling issues while the ALE-advances are given in Chapter 4. First development steps of the new fixed-grid methods are presented in Chapter 5 followed by some numerical examples and concluding remarks.

2 General Fluid Structure Interaction Problem

A general fluid structure interaction problem consists of the description of the fluid and solid fields, appropriate fluid structure interface conditions at the conjoined interface and conditions for the remaining boundaries, respectively. For the presentation in this paper, we restrict ourselves to incompressible flows, which is a reasonable choice for many engineering applications.

In the following, the fields and interface conditions are introduced. Furthermore, a brief sketch of the solution procedure for each of the fields is presented.

2.1 Fluid

Without consideration of the specific reference system, the conservation of momentum is stated as

$$\rho^f \frac{D\mathbf{u}}{Dt} - \nabla \cdot \boldsymbol{\sigma} = \rho^f \mathbf{b} \quad (1)$$

Here, the material time derivative of the velocity $\mathbf{u}$ times the fluid density ρ^f is balanced by the gradient of the Cauchy stress tensor $\boldsymbol{\sigma}$ and external, velocity independent volumetric forces $\mathbf{b}$. Mass conservation for an incompressible fluid is stated as

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

The Newtonian material law defines the internal stress tensor $\boldsymbol{\sigma}$ as

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\mu\boldsymbol{\varepsilon}(\mathbf{u}) \quad (3)$$

with the strain rate tensor $\boldsymbol{\varepsilon}$

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \nabla \mathbf{u} + (\nabla \mathbf{u})^T \quad (4)$$

and the pressure p and dynamic viscosity μ .

The material time derivative depends on the choice of the reference system. There are basically 3 alternative reference systems: the Eulerian, the Lagrangian and the Arbitrary Lagrangian Eulerian formulation. Traditionally, for flows without moving boundaries, the Eulerian system is used, where the momentum equation reads as

$$\left. \frac{\partial \mathbf{u}}{\partial t} \right|_{\mathbf{x}} + \nabla (\mathbf{u} \otimes \mathbf{u}) + \nabla p - 2\nu \nabla \cdot \boldsymbol{\varepsilon}(\mathbf{u}) = \mathbf{b} \quad \text{in } \Omega_f \times (0, T) \quad (5)$$

Fluid structure interaction problems require to deal with moving boundaries for the fluid domain. The most commonly used description, namely the ALE description, will be discussed in Section 4. If the moving boundary is described on a fixed grid, the Eulerian description from Eq. (5) can be used. However, additional effort goes into the description of the interface on the fixed grid. Finally, even Lagrangian flow descriptions are used, e.g. to describe the flow of fluid particles, which allows to track the interface in a simple manner [26].

For both the Eulerian and ALE formulation presented in this article, we use direct time integration using one-step- θ method or BDF2 time discretization. The spatial discretization of the semi-discrete equations is done by a stabilized (Galerkin least square like) finite element method using either bi-linear or quadratic shape functions and equal-order ansatz functions for velocity and pressure unknowns. The solution of the pre-conditioned, linearized system is usually done using the BiCGSTAB or GMRES method within the AZTEC solver package.

2.2 Structure

Most commonly, the structure is described using a Lagrangian description, where the material derivative becomes a partial derivative with respect to time, such that

$$\rho^s \frac{\partial^2 \mathbf{d}}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma} + \rho^s \mathbf{f} \quad (6)$$

with the displacement $\mathbf{d}$ defined as the difference of the current position $\mathbf{x}$ and the initial position $\mathbf{X}$.

In the large deformation case it is common to describe the constitutive equation using a stress-strain relation based on the Green-Lagrangian strain tensor $\mathbf{E}$ and the 2. Piola-Kirchhoff stress tensor $\mathbf{S}(\mathbf{E})$ as a function of $\mathbf{E}$. The 2. Piola-Kirchhoff stress can be obtained from the Cauchy stress $\boldsymbol{\sigma}$ as

$$\mathbf{S} = J \mathbf{F}^{-1} \cdot \boldsymbol{\sigma} \cdot \mathbf{F}^{-T} \quad (7)$$

and the Green-Lagrangian strain tensor $\mathbf{E}$ as

$$\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I}) \quad (8)$$

J denotes the determinant of the deformation gradient tensor $\mathbf{F}$, which itself is defined as

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} \quad (9)$$

For the numerical results presented in this article we use the St.-Venant-Kirchhoff material law for simplicity. Different structural models, mainly 3d-shell as well as solid-shell models, are used along with sophisticated element technology (see [27], [28]). The used solvers for the set of linear equations are usually the direct UMFPACK solver and, for 2d- and 3d-shell structures, also iterative methods, e.g. based on aggregation multigrid ideas [29].

2.3 Interface Conditions

The main conditions at the interface are the dynamic and kinematic coupling conditions. The force equilibrium requires the stress vectors to be equal as

$$\boldsymbol{\sigma}^f \cdot \mathbf{n} = \boldsymbol{\sigma}^s \cdot \mathbf{n} \quad \forall \mathbf{x} \in \Gamma^{\text{fsi}} \quad (10)$$

For abbreviation, the fluid surface forces at the interface are also referred to as $\mathbf{h}_\Gamma^f = \boldsymbol{\sigma}^f \cdot \mathbf{n}$

We assume no mass flow across the interface. Consequently, also the normal velocities at the interface have to match as

$$\mathbf{u} \cdot \mathbf{n} = \frac{\partial d}{\partial t} \cdot \mathbf{n} \quad \forall \mathbf{x} \in \Gamma^{\text{fsi}} \quad (11)$$

If viscous fluids are considered, there is also a matching condition for the tangential velocities, which can be combined with the equation above to obtain the ‘no slip’ boundary conditions as

$$\mathbf{u} = \frac{\partial d}{\partial t} \quad \forall \mathbf{x} \in \Gamma^{\text{fsi}} \quad (12)$$

2.4 Partitioned Analysis

In general, one can choose to describe the whole coupled system in a monolithic way and solve all fields together or separate the fields and couple them in the sense of a partitioned analysis. In the latter case either sequential (staggered) or iterative algorithms can be used. The monolithic approach is straightforward and allows to solve the resulting system of equations with a complete tangent stiffness matrix (if - in an ALE setting - fluid, structure and mesh degrees of freedom are included). However, such monolithic approaches have a number of obvious severe drawbacks like loss of software modularity, limitations with respect to the application of different sophisticated solvers in the different fields and challenges with respect to the problem size and conditioning of the overall system matrix. Hence they are generally considered not very well suited for application to real world problems, where often not only specific solution approaches but also specific codes should be used in the single fields.

For this and a number of additional reasons we prefer to use a partitioned approach. The trade-off is an incomplete tangent stiffness for the overall problem. The consequences are discussed in the following section.

3 Coupling of Partitioned Fields

The algorithmic implementation of the coupled problem of fluid structure interaction is based on a field-wise partitioned solution approach. This modular

software concept allows for complex and specifically designed approaches for the single fields of the structure, the fluid and, for the ALE approach, the mesh.

The wet structural surface is the natural interface or coupling surface Γ^{fsi} . For brevity, in Section 3 and 4, Γ^{fsi} will be simply referred to as Γ . Complete kinematic and dynamic continuity at Γ would ensure conservation of mass, momentum and energy at the interface. The boundary conditions are chosen such that forces generated from the fluid pressure and viscous friction are exerted on the structural interface as Neumann boundary conditions. The structural displacement at Γ is transferred into velocities and used as a Dirichlet condition for the fluid field and it is also used to update the mesh position in the ALE approach. For the description of the fluid mesh movement, the fluid mesh velocity $\mathbf{u}^g$ is used, which will be introduced in more detail in Section 4.

The following developments were made using a partitioned ALE approach for FSI. However, since the fundamental properties of the partitioned approach are independent of the fluid or structural discretization and the interface description, the results serve also as a foundation for the development of partitioned fixed-grid schemes and can directly be applied in this case too.

3.1 Instability of Coupling Schemes

A particularly appealing way to solve the coupled problem is the classical and most frequently used approach of a sequentially staggered formulation. While the fluid and structural domain are solved implicitly, the coupling information is exchanged once per time step.

The time step from time level n to $n + 1$ proceeds as follows, where a variety of schemes differ by the specific accuracy of the particular steps [30]:

1. Calculate an explicit predictor of the structural interface displacement at the new time level $\mathbf{d}_{\Gamma,P}^{n+1}$.
2. Solve for new nodal positions and compute fluid mesh velocity $\mathbf{u}^g$.
3. Compute the fluid velocity at Γ , which serves as a Dirichlet boundary condition: $\mathbf{u}_{\Gamma,P}^{n+1}(\mathbf{d}_{\Gamma,P}^{n+1})$.
4. Solve the fluid equations to obtain $\mathbf{u}^{n+1}$ and $\mathbf{p}^{n+1}$.
5. Obtain the fluid boundary traction along Γ : $\mathbf{h}_P^{\text{f},n+1}$.
6. Solve the structural field for the new displacements $\mathbf{d}^{n+1}$ under consideration of the fluid load $\mathbf{h}_P^{\text{f},n+1}$.
7. Proceed to the next time step.

While being very promising with respect to efficiency, sequentially staggered algorithms exhibit an inherent instability, which increases with decreasing time step [31,32]. Since the error in the transferred fluid forces act as an extra mass on the structural interface degrees of freedom, and the instability highly depends on the mass ratio between fluid and structural density, it has been called *artificial added mass effect*. An analysis shows that the instability

is caused by too large eigenvalues of the amplification operator of the explicit step [33,30].

A number of simplifying assumptions allow to summarize steps 2, 3 and 4 and identify the dimensionless added mass operator $\mathcal{M}_{\mathcal{A}}$. This operator directly transfers the predicted nodal accelerations $\dot{\mathbf{u}}_I$ at the interface Γ into the fluid forces $\mathbf{h}_\Gamma$ exerted on the structure by

$$\mathbf{h}_\Gamma = m^f \mathcal{M}_{\mathcal{A}} \dot{\mathbf{u}}_I \quad (13)$$

where m^F denotes a characteristic fluid mass. Introducing Eq. (13) into the discrete structural equations yields

$$\begin{bmatrix} \mathbf{M}_{II}^s & \mathbf{M}_{I\Gamma}^s \\ \mathbf{M}_{\Gamma I}^s & \mathbf{M}_{\Gamma\Gamma}^s \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{d}}_I \\ \ddot{\mathbf{d}}_\Gamma \end{bmatrix} + \begin{bmatrix} \mathbf{K}_{II}^s & \mathbf{K}_{I\Gamma}^s \\ \mathbf{K}_{\Gamma I}^s & \mathbf{K}_{\Gamma\Gamma}^s \end{bmatrix} \begin{bmatrix} \mathbf{d}_I \\ \mathbf{d}_\Gamma \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ -m^f \mathcal{M}_{\mathcal{A}} \dot{\mathbf{u}}_I \end{bmatrix}, \quad (14)$$

where the structural system of equations has been split into internal (subscript I) and interface (subscript Γ) degrees of freedom. Inserting the particular representation of the predictor and Dirichlet boundary condition allows to analyze the eigenvalue of the operator that transfers the interface displacement from time level n to $n+1$. Here, two different cases have to be distinguished: $\mathbf{d}_\Gamma^{n+1}$ is either a function of a finite number of old interface positions (*e.g.* $\mathbf{d}_\Gamma^{n+1} = f(\mathbf{d}_\Gamma^n, \mathbf{d}_\Gamma^{n-1}, \mathbf{d}_\Gamma^{n-2})$), or it depends on all previously calculated interface positions $\mathbf{d}_\Gamma^{n+1} = f(\mathbf{d}_\Gamma^n, \mathbf{d}_\Gamma^{n-1}, \dots, \mathbf{d}_\Gamma^0)$. In the first case an ‘instability condition’ of the form

$$\frac{m^f}{m^s} \max \mu_i > C_1$$

can be obtained, where m^s represents a characteristic structural mass and μ_i denotes the i -th eigenvalue of the added mass operator $\mathcal{M}_{\mathcal{A}}$. The limit C_1 depends upon the particular details of the temporal discretization and decreases (!) with increasing accuracy [30]. Thus, the more accurate the scheme is the earlier it becomes unstable with respect to the density ratio of fluid and structure. The instability limits obtained for different structural predictors and backward Euler (BE) or second-order backward differencing (BDF2) time discretization of the fluid equations are given in Table 1.

The problem gets even worse in the second case, which yields an instability condition of the form

$$\frac{m^f}{m^s} \max \mu_i > \frac{C_2}{n}$$

where n is the number of time steps. Thus, the instability constant C_2/n decreases during the simulation and regardless of the density ratio there will be a step at which the problem becomes unstable.

When stabilized fluid elements are considered the analysis becomes more complicated and the simple instability limits given in Table 1 are not directly applicable any more. However, it can be proven that for every sequentially

Table 1. Instability limit C_1 for sequentially staggered fluid structure interaction schemes depending on the structural predictor and fluid time discretization scheme.

predictor	BE	BDF2
0th order	3	$\frac{3}{2}$
1st order	$\frac{3}{5}$	$\frac{3}{10}$
2nd order	$\frac{1}{3}$	$\frac{1}{6}$

staggered scheme there exists a density ratio ρ^f/ρ^s at which the scheme becomes unstable [30]. Numerical investigations show that the instability limits are very restrictive when incompressible fluids are considered and stabilized finite elements are employed. Therefore, the instability limits effectively prevent stable computations by means of staggered algorithms and an iterative coupling is necessary.

3.2 Iterative Staggered Coupling

To avoid the instabilities inherent in sequential staggered schemes, an iterative staggered Dirichlet-Neumann sub-structuring scheme is formulated which is based on the approach given in [34]. Every time step from time level n to $n + 1$ requires an iteration over the fields until convergence of the respective interface displacements or residuals is obtained. The algorithm can be summarized as follows:

- Compute an explicit predictor of the structural interface displacement at the new time level $\mathbf{d}_{\Gamma,0}^{n+1}$.
- Fluid mesh:
 - Solve for new nodal positions and compute fluid mesh velocity $\mathbf{u}^g$
- Fluid:
 - Compute fluid velocity at Γ from mesh position to serve as Dirichlet boundary condition $\mathbf{u}_{\Gamma,i+1}^{n+1}$ ($\mathbf{d}_{\Gamma,i+1}^{n+1}$).
 - Solve fluid equations on the new mesh to obtain $\mathbf{u}_{i+1}^{n+1}$ and $\mathbf{p}_{i+1}^{n+1}$.
 - Obtain fluid boundary traction along Γ $\mathbf{h}_{\Gamma,i+1}^{f,n+1}$.
- Structure:
 - Solve the structural field for the new displacements $\mathbf{d}_{i+1}^{n+1}$ including the interface displacements $\tilde{\mathbf{d}}_{i+1}^{n+1}$ under consideration of the fluid load $\mathbf{h}_{\Gamma,i+1}^{f,n+1}$.
 - Relaxation of the interface displacements $\tilde{\mathbf{d}}_{i+1}^{n+1}$ by using $\mathbf{d}_{i+1}^{n+1} = \omega_i \tilde{\mathbf{d}}_{i+1}^{n+1} + (1 - \omega_i) \mathbf{d}_i^{n+1}$
- Check convergence,
- Restart iteration with new mesh calculation if not yet converged.

For fixed-grid methods, obviously a calculation of new fluid node positions is not necessary.

The approach can be formulated with overall second-order accuracy in time by using second-order accurate schemes to advance the structural and the fluid field in time and interpolating the mesh velocity with sufficiently high order, as

$$\mathbf{u}_{i+1}^{g,n+1} = \frac{3\mathbf{r}_{i+1}^{n+1} - 4\mathbf{r}^n + \mathbf{r}^{n-1}}{\Delta t}$$

where $\mathbf{r}$ denote displacements of the mesh field.

3.3 Automatic Estimation of Relaxation Parameter ω

The properties causing the instability for the sequentially staggered scheme transfers to the iteratively staggered algorithm and demands relaxation [31], [32], [35], and especially a relaxation parameter $\omega_i < 1$ to ensure convergence [33]. Two stable and robust ways of obtaining the relaxation parameter have been developed in our group. It is worth mentioning that in both cases no user dependent (input) parameter is required and ‘optimal’ relaxation parameter are obtained in every iteration step thus beating even the best ω found by trial and error. Not to speak about the hopelessness of finding an appropriate relaxation parameter by trial and error in a strongly nonlinear time-dependent problem. The first way yields an *iterative sub-structuring scheme accelerated via the gradient method*. This technique is an acceleration via the application of the gradient method (method of steepest descent) to the iterative sub-structuring scheme, and it also guarantees convergence. In each iteration, a relaxation parameter ω_i is computed by

$$\omega_i = \frac{\mathbf{g}_i^T \mathbf{g}_i}{\mathbf{g}_i^T \mathbf{S}_s^{-1} (\mathbf{S}_f + \mathbf{S}_s) \mathbf{g}_i} = \frac{\mathbf{g}_i^T \mathbf{g}_i}{\mathbf{g}_i^T (\mathbf{S}_s^{-1} \mathbf{S}_f \mathbf{g}_i + \mathbf{g}_i)} \quad (15)$$

which is locally optimal with respect to the actual search direction, *i.e.* the residual

$$\mathbf{g}_i = \mathbf{S}_s^{-1} \left(\mathbf{f}_{\Gamma_{ext}}^{mod,n+1} - (\mathbf{S}_f + \mathbf{S}_s) \mathbf{d}_{\Gamma,i}^{n+1} \right) = \tilde{\mathbf{d}}_{\Gamma,i+1}^{n+1} - \mathbf{d}_{\Gamma,i}^{n+1}$$

A procedure for evaluating Eq. (15) without explicitly computing and storing the Schur complements $\mathbf{S}_f$ and $\mathbf{S}_s$ has been proposed in [35].

Within a second approach, the coupling is formulated as an *iterative sub-structuring scheme accelerated via the Aitken method*. This technique for explicitly calculating a suitable relaxation parameter is the application of Aitken’s acceleration scheme for vector sequences according to Irons *et al.* [36]. To obtain ω_i , the interfacial displacement difference is computed as

$$\Delta \mathbf{d}_{\Gamma,i+1}^{n+1} := \mathbf{d}_{\Gamma,i}^{n+1} - \tilde{\mathbf{d}}_{\Gamma,i+1}^{n+1}$$

The Aitken factor is obtained from

$$\mu_i^{n+1} = \mu_{i-1}^{n+1} + (\mu_{i-1}^{n+1} - 1) \frac{\left(\Delta \mathbf{d}_{\Gamma,i}^{n+1} - \Delta \mathbf{d}_{\Gamma,i+1}^{n+1} \right)^T \Delta \mathbf{d}_{\Gamma,i+1}^{n+1}}{\left(\Delta \mathbf{d}_{\Gamma,i}^{n+1} - \Delta \mathbf{d}_{\Gamma,i+1}^{n+1} \right)^2}$$

and yields the relaxation parameter

$$\omega_i = 1 - \mu_i^{n+1}.$$

Even though a rigorous analysis of its convergence properties does not exist, numerical studies have shown that the Aitken acceleration applied to the FSI problems considered here shows a performance, which is often very good and sometimes even better than acceleration via the gradient method. Furthermore, the evaluation of the relaxation parameters via the Aitken method is extremely cheap (in terms of both CPU and memory) and simple to implement.

3.4 Conserving Transfer of Quantities on the Interface

Another important ingredient to obtain long-time accuracy for simulations over long periods of time is the conservation of transfer quantities at the interface. Geometric conservation demands that the Dirichlet boundary condition for the fluid domain at Γ is obtained consistent with the time discretization scheme employed on the fluid domain. Along with one-step- θ or BDF2 the interface velocity is obtained from

$$\mathbf{u}_{\Gamma,i+1}^{n+1} = 2 \frac{\mathbf{r}_{\Gamma,i+1}^{n+1} - \mathbf{r}_{\Gamma}^n}{\Delta t} - \mathbf{u}_{\Gamma}^n$$

ensuring a correct mass balance within the incompressible fluid flow.

Particular care has also to be taken in obtaining the fluid forces applied to the structure. The fluid force along the coupling interface can be obtained from

$$\mathbf{F}_{\sigma} = \int (2\mu \boldsymbol{\varepsilon}(\mathbf{u}) - \bar{p} \mathbf{I}) \cdot \mathbf{n} \, d\Gamma \quad (16)$$

A discrete version of Eq. (16) is usually used to obtain lift and drag or FSI coupling forces (*e.g.* [37–39]). However, a more accurate and also more efficient way to obtain the coupling forces is to employ consistent nodal forces [40]. Sharing the order of accuracy of the primary variables and naturally containing the viscous forces, consistent nodal fluid forces are a key ingredient for accurate coupling on moderately fine meshes. Consistent with the discretization used here, the nodal force component i at node I at the end of

the time step $n + 1$ is given by

$$\begin{aligned}
 \mathbf{F}_{Ii}^{n+1} = & \left(-\frac{\rho}{\delta} \mathbf{u}^{n+1} - \rho (\mathbf{u}^{n+1} - \mathbf{u}^{g,n+1}) \cdot \nabla \mathbf{u}^{n+1} + \frac{\rho}{\delta} \mathbf{r}^{n+1}, N_I \mathbf{e}_i \right) \\
 & - (2\mu \boldsymbol{\varepsilon}(\mathbf{u}^{n+1}), \boldsymbol{\varepsilon}(N_I \mathbf{e}_i)) + (\bar{p}^{n+1}, \nabla \cdot N_I \mathbf{e}_i) \\
 & + \sum_k \tau_k^m \rho (\mathcal{R}_M(\mathbf{u}^{n+1}, p), (\mathbf{u}^{n+1} - \mathbf{u}^{g,n+1}) \cdot \nabla N_I \mathbf{e}_i + 2\mu \nabla \cdot \boldsymbol{\varepsilon}(N_I \mathbf{e}_i))_k \\
 & - \sum_k \tau_k^c \rho (\mathcal{R}_C(\mathbf{u}^{n+1}), \nabla \cdot N_I \mathbf{e}_i)_k
 \end{aligned} \tag{17}$$

where N_I is the shape function associated with node I defined on the actual configuration $\Omega^{f,n+1}$ while $\mathbf{e}_i$ is the Cartesian basis vector in direction i . δ and $\mathbf{r}^{n+1}$ stem from the temporal discretization. In contrast to the physical forces obtained from Eq. (16), Eq. (17) evaluates accumulated nodal forces, which are fully consistent with the chosen discretization and thus ensures conservation [41]. Consistent nodal forces fit into a nodal-based data structure and are thus easy and efficiently to implement. They further allow to incorporate the effect of stabilization terms, preserving consistency in the stabilized case [40].

3.5 Further Important Comments on Coupling and Modeling of Coupled Problems

FSI with high accuracy requires new modeling techniques not only for the interface but also for each of the single fields. Small inaccuracies in a solution of a field might not significantly influence the overall solution of this field. However, if such inaccuracies occur near a fluid structure interface, the inaccurate local solution might severely influence the local transfer of information between fluid and structure, which can substantially change the result of the overall simulation. Consequently, an accurate solution near the interface is crucial for the overall exactness and performance of the coupling [40].

For example, a wrong velocity or pressure gradient due to a local boundary oscillation in the fluid solution results in wrong surface forces. Thus, such local oscillations, often not doing any harm on pure fluid simulations, can be dangerous in the coupled case. One source of such local oscillations can stem from small time steps as studied in Ref. [42]. A sketch for such local oscillations using `quad4` and `quad9` elements is presented in Figure 4.

Another important aspect to obtain reliable solutions of coupled problems already starts before any numerical scheme is applied. It has to do with the modeling of such problems. In many cases one has to re-think the modeling of a problem in a completely new way when coupling is taken into account. Standard modeling assumptions, appropriate in the single field case, might lead to defective results in the coupled case. Some first remarks in this respect can be found in [43] and [40].

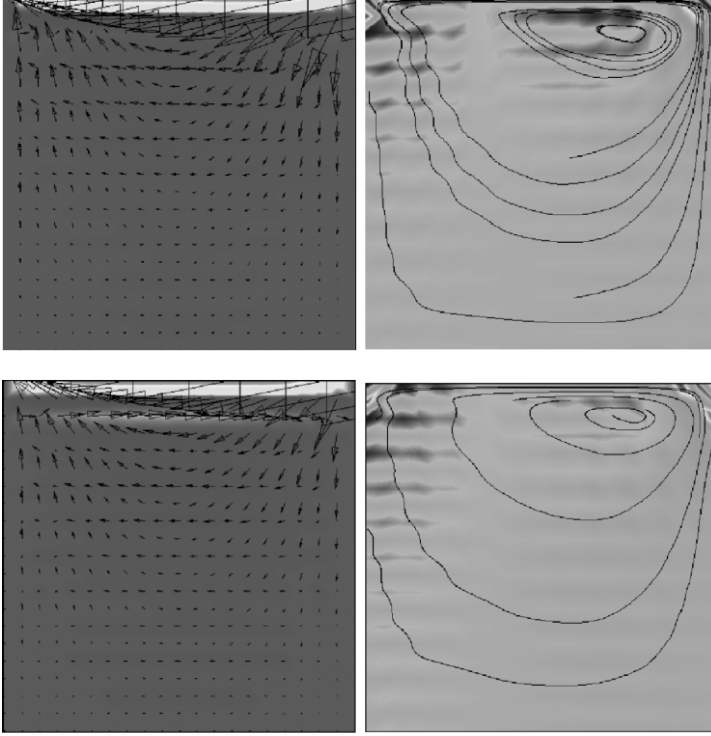


Fig. 4. Local velocity (left column) and pressure oscillations (right column) caused by small time steps for a driven cavity problem using **quad4** (upper row) and **quad9** (lower row) elements.

4 ALE Based Formulation for Fluid Structure Interaction

4.1 ALE and Geometric Conservation Revisited

ALE-based methods are the most commonly used methods for dealing with moving surfaces and a vast number of publications exist on this topic. In the following, we discuss a number of aspects, which become increasingly important when dealing with large deformations of the fluid boundary.

The ALE equation of motion can be derived from the Eulerian formulation of the balance of linear momentum in Eq. (5) by introducing the deforming reference system χ , which follows the motion of the respective boundaries and deforms, e.g. according to a mesh smoothing, in the remaining domain. The reference system deformation is described with the unique mapping $\mathbf{x} = \varphi(\chi, t)$ as depicted in Figure 5. By means of the Reynolds transport theorem,

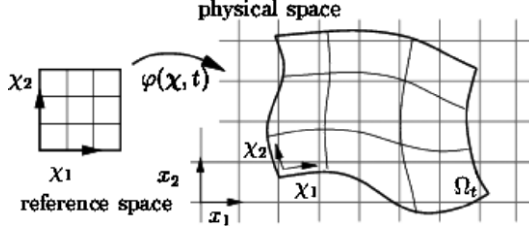


Fig. 5. Sketch of ALE system of reference

Eq. (5) can be reformulated on moving grids as

$$\left. \frac{\partial(uJ_t)}{\partial t} \right|_{\chi} + \{ \nabla \cdot (u \otimes (u - u^g)) - 2\nu \nabla \cdot \varepsilon(u) + \nabla p \} J_t = bJ_t, \quad (18)$$

where $J_t = \det(\partial \mathbf{x} / \partial \boldsymbol{\chi})$ denotes the time-dependent Jacobian of the mapping, and $\mathbf{u}^g = \partial \mathbf{x} / \partial t|_{\chi}$ represents the velocity of the reference system, which is to be identified with the grid velocity after discretization. As well as Eq. (5), Eq. (18) is in divergence form and thus the point of departure for a finite volume discretization in space [44–46]. However, in the weak form the first term of Eq. (18) yields an integral over a temporally changing domain and its improper representation in a discrete scheme may result in stability problems.

Geometric conservation demands for the correct representation of the mass term in a deforming domain by requiring that a temporally and spatially constant solution can be represented exactly by the discrete scheme. Thus geometric conservation in conjunction with a discretized Eq. (18) yields the need for temporal averaging of either geometries or fluxes within a time step [45,47,48,46,49]. Depending on the temporal discretization scheme different versions of the geometric conservation law, *i.e.* Discrete Geometric Conservation Laws (DGCL), and thus averaging schemes have to be used [46].

In a strong form the geometric conservation equation is given by

$$\frac{\partial J_t}{\partial t} = J_t \nabla \cdot \mathbf{u}^g \quad (19)$$

which links the temporal change of the domain to the domain velocity. Equation (19) can be incorporated into (18)

$$\left. \frac{\partial \mathbf{u}}{\partial t} \right|_{\chi} + (\mathbf{u} - \mathbf{u}^g) \cdot \nabla \mathbf{u} - 2\nu \nabla \cdot \varepsilon(\mathbf{u}) + \nabla p = \mathbf{b}, \quad (20)$$

yielding an ALE form of the momentum balance which does no longer require integrals over temporally deforming domains. Discretization of Eq. (20) in

time and space is straightforward, while the stability and accuracy properties of the discretization scheme can directly transfer to the deforming domain formulation. The order k of temporal accuracy is preserved provided that the mesh velocity $\mathbf{u}^g$ is interpolated with order k or higher and satisfies geometric conservation (19) which can be achieved with almost no extra effort [50]. While Eq. (20) is no longer in divergence form and thus inappropriate for the discretization with finite volumes it works perfectly with finite elements. It avoids the need for different discrete versions of the geometric conservation law or temporal averaging.

4.2 Distorted Meshes

Dealing with large deformations can result in severe deformations of the moving fluid mesh. Consequently, such meshes represents an important problem which an ALE formulation has to cope with. Thus ALE methods demand for formulations which yield reliable results also on significantly distorted meshes. Additionally, boundary adapted fluid meshes may result in highly stretched elements or complicated geometries require unfavorably shaped elements to be adequately represented. In the context of fluid structure interaction, accurate coupling information, even on distorted meshes, is of particular interest.

Numerical investigations show that the stabilized ALE formulation can be designed such that the scheme is reliable and insensitive to mesh distortion. To highlight the solution quality the errors obtained for the Kim-Moin flow on the three different meshes depicted in Figure 6 are considered. The present investigation compares linear `quad4` and quadratic `quad9` elements in 2d. A selection of the results is given in Table 2. The results highlight that the formulation is very robust and insensitive to mesh distortion, in particular, when higher order elements are employed, for which the stabilization terms are fully consistent. The high quality of the quadratic elements even on a mesh with an immense angular distortion as the one depicted in Figure 6(c) relies on the fact that the edge and face nodes are perfectly placed at the respective centers. However, as the mesh motion is managed by a specifically

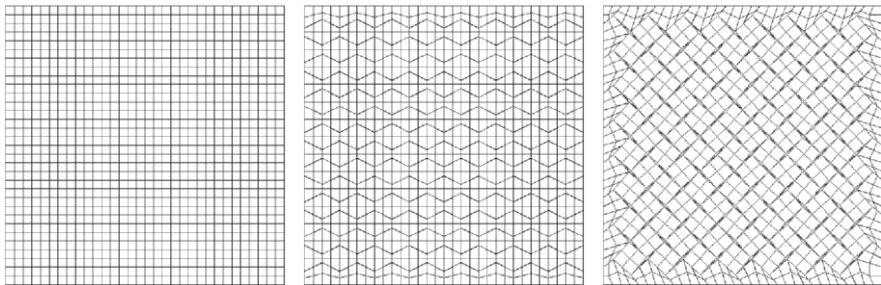


Fig. 6. Meshes used for error evaluation. From left to right: undistorted mesh for reference, distortion mode 1 and distortion mode 2

Table 2. L^2 error in velocity and pressure on differently distorted meshes of linear (quad4) and quadratic (quad9) elements.

mesh	err_u quad4	err_p quad4	err_u quad9	err_p quad9
mode 0	0.005130	0.002877	0.002624	0.002560
mode 1	0.007613	0.005044	0.002638	0.002581
mode 2	0.232357	0.185340	0.006558	0.009451

designed mesh motion scheme, this can at least inside the domain easily be guaranteed.

Further investigations confirm these results showing that distortion insensitive stabilized formulations can be constructed allowing for reliable computations on deforming domains [51].

4.3 ALE Mesh Motion Schemes

Even though ALE schemes are designed to reduce the element distortion caused by pure Lagrangian formulations, a certain portion of element distortion is unavoidable. Hence, for robust simulations of general complex systems the mesh motion scheme is a key issue. A number of mesh motion techniques aiming at equally distributing the elemental distortion in order to maximize the remaining element quality have been developed [52]. Most of these schemes are in one way or the other based on a pseudo structural approach, given as

$$\mathbf{K}^g \mathbf{r} = \mathbf{F} \quad (21)$$

where the ‘mesh stiffness’ matrix $\mathbf{K}^g$ is designed such that modes which could yield element failure are stiffened compared to other distortion modes.

However, at some point the best mesh moving algorithm cannot avoid an unacceptable fluid mesh deformation and a partial or complete re-meshing of the fluid domain becomes necessary. Such scenarios include problems with topology changes, *e.g.* when a fluid penetrates an opening crack or when a valve closes, or simply if the structure moves too far in the domain or rotates. The re-meshing involves an interpolation between the old and the new mesh, which always introduces errors. Consequently, when frequent re-meshing would be required, a fixed-grid method might be an attractive alternative.

5 Fixed (Background) Grid Approaches for FSI

5.1 An Overlapping Domain Decomposition / Chimera-Like Formulation

Chimera schemes were introduced into computational fluid dynamics in 1983 by Steger *et al.* [53]. Briefly, when used for the proposed fluid structure interaction scheme, a fixed Eulerian fluid background mesh that does not extend into the structural domain Ω^s , is overlapped by a moving, deformable fluid mesh using the ALE formulation. The moving mesh is connected to the structure and deforms according to the structural surface deformation.

For the fluid structure coupling, we use the iterative, partitioned approach described in Section 3.2. The solution procedure for FSI problems is displayed in Table 3. Due to the modularity of the partitioned coupling, the FSI coupling scheme can be kept unchanged, while the Chimera approach is used to couple the moving grid with the background fluid.

Table 3. Iterative staggered FSI coupling scheme using a Chimera-like approach for the fluid solution. Due to the modularity, the changes are ‘local’.

time loop		
repeat until STOP FSI (iteration over fields)		
solve for mesh dynamics on deforming subdomains		
solve fluid		
		Chimera procedure ... (Table 4)
solve structure		
if converged, STOP FSI		

The advantage of such a configuration is that the ALE domain does not extend over the entire fluid domain, instead, it moves and deforms with the structure, while the majority of the background mesh remains fixed. The structure always has a surface fitted ALE fluid mesh, which allows accurate resolution of the boundary layers around the structure. No adaptive meshing is required for the background fluid mesh to resolve the boundary layer. Additionally, the structure can now travel large distances through the fluid domain or can rotate multiple times without any deformation of the global fluid mesh, therefore extending the range of applications, where no re-meshing is required. The price to pay is an additional iteration over the overlapping fluid grids to obtain the fluid solution in the entire fluid domain Ω^f .

In general, Chimera schemes for the solution of the fluid problem are based on the decomposition of the fluid domain Ω^f into several overlapping

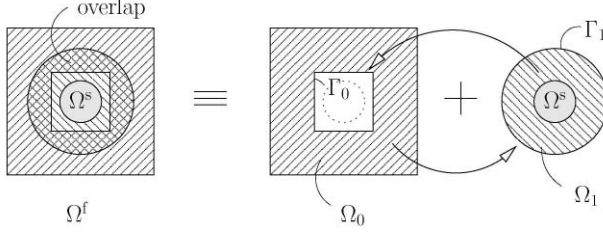


Fig. 7. Chimera-like approach for FSI: the computational fluid domain is divided into a fixed Eulerian background domain Ω_0 and an overlapping, moving ALE domain with the ‘inner’ boundaries Γ_0 and Γ_1 . The arrows indicate the transfer of information between the subdomains during the alternating solution process.

subdomains Ω_i , which are solved in an iterative manner. For simplicity, we will refer to only two fluid subdomains, the fixed Eulerian fluid mesh Ω_0 and the moving ALE fluid mesh Ω_1 around one structure (Ω^s) as depicted in Figure 7. Extension to multiple structures with multiple communication between these subdomains is obviously possible.

For the fixed background fluid domain Ω_0 , which has an inner Dirichlet boundary Γ_0 beneath the sub-domain Ω_1 , the Eulerian fluid equations are solved as

$$\left. \frac{\partial \mathbf{u}_0}{\partial t} \right|_{\mathbf{x}} + \mathbf{u}_0 \cdot \nabla \mathbf{u}_0 - 2\nu \nabla \cdot \varepsilon(\mathbf{u}_0) + \nabla p_0 = \mathbf{b} \quad \text{in } \Omega_0 \quad (22)$$

$$\nabla \cdot \mathbf{u}_0 = 0 \quad \text{in } \Omega_0 \quad (23)$$

with the ‘standard’ boundary conditions on $\partial\Omega \cap \Omega_0$ plus the boundary conditions on Γ_0 , which depend on the solution on domain Ω_1 from the previous Chimera iteration step:

$$\mathbf{u}_0(\mathbf{x}) = \mathbf{I}(\mathbf{u}_1^{prev}) \quad (24)$$

For the moving fluid domain Ω_1 , the ALE formulation from Section 4 is used as

$$\left. \frac{\partial \mathbf{u}_1}{\partial t} \right|_{\mathbf{x}} + (\mathbf{u}_1 - \mathbf{u}_1^g) \cdot \nabla \mathbf{u}_1 - 2\nu \nabla \cdot \varepsilon(\mathbf{u}_1) + \nabla p_1 = \mathbf{b} \quad \text{in } \Omega_1 \quad (25)$$

$$\nabla \cdot \mathbf{u}_1 = 0 \quad \text{in } \Omega_1 \quad (26)$$

The boundary values on Γ_1 are set by interpolation from the previous solution on Ω_0 .

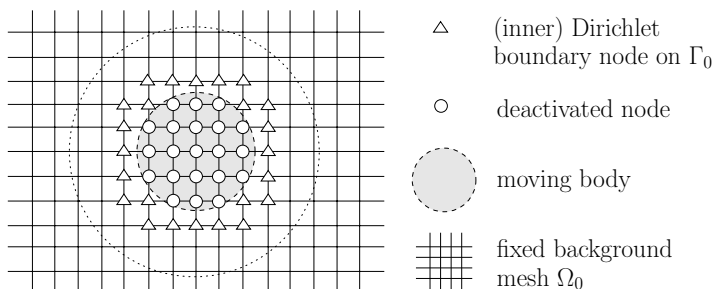
Depending on the boundary condition imposed on Γ_i (Dirichlet, Neumann or Robin), values for velocities or tractions are set by interpolation from the corresponding sub-domain. The algorithmic structure of the Chimera solution process is described in Table 4. The background mesh initially covers the fluid and solid domain, however, all nodes below the moving bodies are

Table 4. The solution procedure for the alternating Chimera approach.

solve fluid
determine the inactive nodes on Ω_0 and the interpolation nodes on Γ_0
repeat until STOP
loop all subdomains Ω_i
update boundary conditions on Γ_i by interpolation from the previous solution of the overlapping domain.
solve the fluid problem on Ω_i .
STOP if the boundary values on $\bigcup \Gamma_i$ do not change anymore during one loop over all Ω_i

deactivated and the internal boundary Γ_0 is formed for each iteration step as illustrated in Figures 8 and 9. The node search is efficiently implemented by a quadtree(2d)/octree(3d) search algorithm. The boundary condition on Γ_0 of the background is chosen to be of Dirichlet type, the boundary condition on Γ_i of the moving subdomains is usually of Neumann or Robin type. Imposing Dirichlet boundary conditions on Γ_i would prescribe the influx into the moving sub-domain Ω_i . Due to interpolation errors during the setting of the boundary conditions on Γ_i , the net influx is nonzero causing a violation of the incompressibility constraint on Ω_i . The consequence is an ill posed subproblem on Ω_i (see [54] for further details and ideas to diminish the problems by imposing additional transmission constraints).

The interpolation process corresponding to the interpolation operator $\mathbf{I}$ will be described exemplary for the interpolation of the velocity $\mathbf{u}_0^n$ at a node n (coordinates $\mathbf{x}^n$) on the Dirichlet boundary Γ_0 below the overlapping do-

**Fig. 8.** The generation of the background mesh for moving body problems is done automatically.

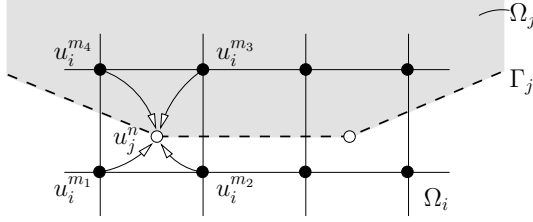


Fig. 9. The boundary value on a node of Γ_j is interpolated from the recent solution on Ω_i .

main Ω_1 (the upper index refers to the node, the lower to the sub-domain). To determine the value of $\mathbf{u}_0^n$, a search is performed in domain Ω_1 to determine a ‘parent’ element containing the point corresponding to the coordinates $\mathbf{x}^n$. The value $\mathbf{u}_0^n$ is now determined by Lagrange interpolation from the values $\mathbf{u}_1^m$ of the nodes m of the parent element, see Figure 9. For the displayed example, the values, which are calculated using the standard finite element shape functions N_{m_i} evaluated at the local coordinates $\boldsymbol{\xi}$ of the point $\mathbf{x}^n$ on the parent element are

$$\mathbf{u}_0^n = \sum N_{m_i}(\boldsymbol{\xi}) \mathbf{u}_1^{m_i}$$

For Neumann boundary conditions on Γ_1 , the gradients (*e.g.* tractions) are interpolated at the Gauss points along Γ_1 . For this purpose, a least square smoothing of the gradients in Ω_0 is recommended.

Once a converged fluid solution is obtained, the loads at Γ^{fsi} are transferred to the structure and the FSI iteration is continued.

As already pointed out in [54], the overlapping length of the subdomains plays a crucial role for the convergence of this sub-domain iteration. Also from our experience it usually pays to choose the overlapping length as large as possible. An additional important influence on the convergence speed is the adopted boundary condition, where Neumann and especially Robin type conditions have clear advantages.

Up to now, the fluid problem on the moving subdomains is solved in a moving (for example rotating) frame of reference. This allows the computation of multiple rigid bodies in relative motion (see ([55] or [56] for related approaches). This implies the solution of a modified equation, which contains the (underlined) non-inertial terms:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + \underline{2\boldsymbol{\omega} \times \mathbf{u}} - 2\nu \varepsilon(\mathbf{u}) + \nabla p = \mathbf{b} - \underline{\mathbf{a} - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{x}) - \dot{\boldsymbol{\omega}} \times \mathbf{x}} \quad (27)$$

To be able to exchange velocities and tractions between the subdomains, the data has to be transformed by means of a tensorial transformation.

In summary, for the proposed method, instead of a rotating frame of reference with additional forces, we use the moving mesh capabilities inherent to the ALE approach to solve the fluid problem on the moving subdomains.

Using an appropriate mesh moving scheme, the mesh velocity and position is known and the ALE-equation is solved for the moving and deforming fluid domain. The modified equation (27) is replaced by the ALE formulation. The grid velocities already include the movement of the sub-domain, hence no extra terms are necessary anymore. From the standard Chimera approach we adopt the iterative solution procedure and the interpolation techniques used to transfer information between the subdomains.

5.2 XFEM-Based Approach for Fluid Structure Interaction

Motivated by introductory listed drawbacks of current fixed-grid methods and inspired by the list of requirements also given above, we developed another fixed-grid approach. The discussion in the introduction shows among others the need of removing the influence of the ‘fictitious’ fluid on the structure movement. Therefore, our second proposed approach uses the properties of the eXtended Finite Element Method (XFEM) to separate the outside and inside fluid domain (fictitious domain) with respect to the structure. In Ref. [21], a similar idea was explored and implemented for compressible flow interacting with beam structures.

As throughout the paper, we employ an iterative, partitioned approach with the structures surface velocity as a Dirichlet condition on the fluid domain and the fluids force vector as a Neumann load on the structural surface. Consequently, no changes are to be made to the structural solution. The extension of the Eulerian fluid formulation Eq. (5) with the XFEM is presented below.

The eXtended Finite Element Method [57,58] was proposed for the representation of crack growth in solids on unchanging meshes and has since been used for *e.g.* the simulation of cracks [59–61], 2-phase flows [62] and the interaction of rigid particles with Stokes flow [63].

The principle features of the proposed approach are depicted in Figure 10. The mapping of the structural surface on the fluid grid defines a surface Γ^{fsi} on the fluid mesh, that is not coincident with any element boundaries. The position of the interface is always explicitly known through the structures position and no additional fields like the level-set, as used in crack problems and 2-phase flow problems, are needed. For easier reference, the ‘physical’ fluid domain outside the structure is named Ω^+ , the ‘fictitious’ fluid domain inside the structure is named Ω^- , which corresponds to a positive or negative value of an enrichment function $H(\mathbf{x}, t)$:

$$H(\mathbf{x}, t) = \begin{cases} +1 & \forall \mathbf{x} \in \Omega^+ \\ -1 & \forall \mathbf{x} \in \Omega^- \end{cases} \quad (28)$$

Similarly, one can define two additional names for the projected boundary Γ^{fsi} , namely Γ^+ and Γ^- , depending whether the enrichment function is evaluated approaching Γ^{fsi} from Ω^+ or Ω^- .

In the spirit of the original crack enrichment, the jump function $H(\mathbf{x}, t)$ is applied to include a jump of arbitrary height for the velocity and the pressure

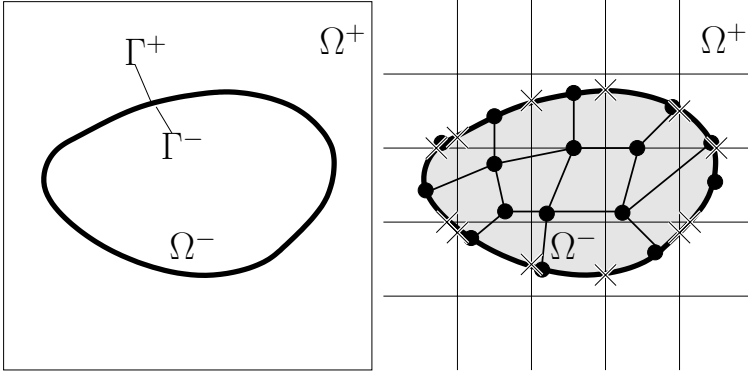


Fig. 10. An XFEM approach on FSI: On the left, the pure fluid problem is shown including the mapped surface of the structure. The structural domain Ω^s corresponds to the fluid domain Ω^- . The kinematics and kinetics are completely decoupled along this line and Dirichlet boundary conditions are applied at Γ^+ via a Lagrange multiplier technique. On the right picture, the discretized problem including the structure is sketched. Crosses mark the position of Lagrange multiplier nodes as also used in Ref. [21]

along the given interface Γ^{fsi} as

$$\mathbf{u}^h(\mathbf{x}, t) = \sum_I N_I(\mathbf{x}) \bar{\mathbf{u}}_I(t) + \sum_J N_J(\mathbf{x}) H(\mathbf{x}, t) \hat{\mathbf{u}}_J(t) \quad (29)$$

$$p^h(\mathbf{x}, t) = \sum_I N_I(\mathbf{x}) \bar{p}_I(t) + \sum_J N_J(\mathbf{x}) H(\mathbf{x}, t) \hat{p}_J(t) \quad (30)$$

The corresponding test functions are discretized accordingly as

$$\mathbf{v}^h(\mathbf{x}, t) = \sum_I N_I(\mathbf{x}) \bar{\mathbf{v}}_I(t) + \sum_J N_J(\mathbf{x}) H(\mathbf{x}, t) \hat{\mathbf{v}}_J(t) \quad (31)$$

$$q^h(\mathbf{x}, t) = \sum_I N_I(\mathbf{x}) \bar{q}_I(t) + \sum_J N_J(\mathbf{x}) H(\mathbf{x}, t) \hat{q}_J(t) \quad (32)$$

In addition to the standard finite element nodal unknowns $\bar{\mathbf{u}}_I(t)$, one has to solve for the additional unknowns $\hat{\mathbf{u}}_J(t)$. For the additional unknowns, the standard shape functions $N_J(\mathbf{x})$ are multiplied by the enrichment function. Choosing a jump function enrichment for the velocity field differs from the approach used in [21]. Figure 11 shows a 1d-solution, where the standard and the enriched degrees of freedom multiplied by the shape functions are shown. For illustration purpose, the standard and enriched degrees of freedom are already multiplied by the shape functions to represent the discretized functions $\bar{\mathbf{u}}^h = \sum_I N_I(\mathbf{x}) \bar{\mathbf{u}}_I(t)$ and $\hat{\mathbf{u}}^h = N_J(\mathbf{x}) \hat{\mathbf{u}}_J(t)$. The assembled function $\mathbf{u}^h$ shows that enriching with a jump function H allows independent physical solutions and solution derivatives on each side of the discontinuity.

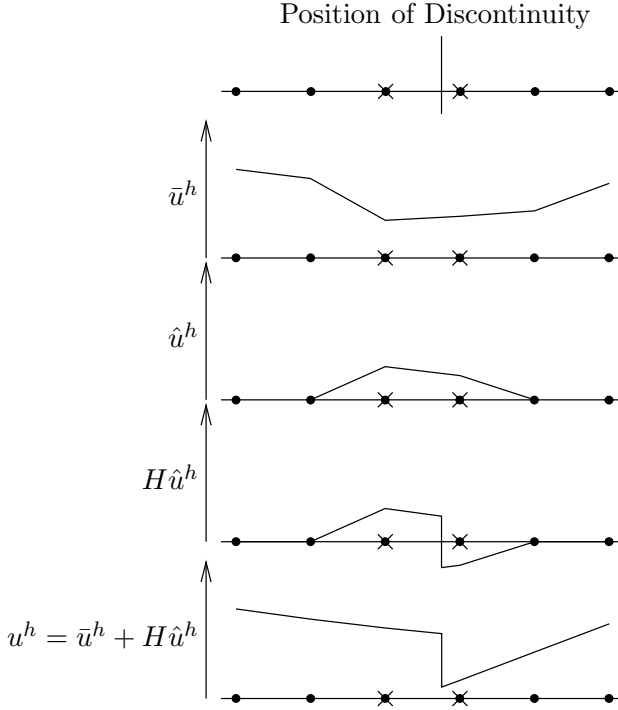


Fig. 11. 1d solution composition across a XFEM interface: Nodes with standard degrees of freedom are marked with $\bullet$, enriched nodes with $\times$. Enriching with the jump function H allows arbitrary functions and derivatives on each side of the discontinuity.

To illustrate the feature of kinematic and kinetic decoupling for flow problems, Figure 12 shows a channel, where ‘do nothing’ boundary conditions are applied on the sloped boundary Γ^+ and Γ^- . On the left end of the channel, an inflow is prescribed, while on the right inclined side, ‘do nothing’ boundary conditions are applied. It can clearly be seen, both the right and the left channel part do not interfere with each other. The flow enters the domain and, at the XFEM Neumann boundary, leaves Ω^+ without any disturbances. In absence of any prescribed inflow and surface traction, the velocity field equals zero in Ω^- . For further illustration purposes we also display the solution in such a way that only the standard shape functions are used in the post-processing step. In this case the solution would look like Figure 13. The nodal solution represents the correct solution, however, the post-processing using standard linear elements shows a gradient for the standard solution $\mathbf{u}^h$. Nevertheless, the integration using $\mathbf{u}^h$ is performed exactly using Eqs. (29) and (30). In summary, for the FSI problem, the jump function enrichment

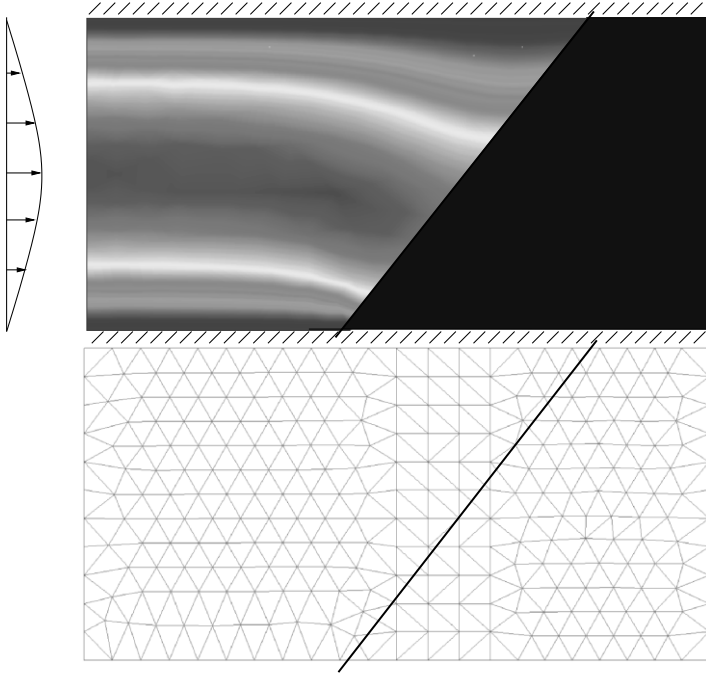


Fig. 12. Demonstration of the ‘do nothing’ BC in a flow field: The prescribed inflow on the left develops to a channel flow and exits Ω^+ at the enriched interface Γ^+ .

allows the domain enclosed by Γ^{fsi} to change its volume regardless of the enclosed, decoupled, fictitious fluid field Ω^- .

The chosen partitioned FSI coupling scheme requires the application of Dirichlet BC for the velocity along Γ^+ , which can be stated as

$$0 = \mathbf{u} - \mathbf{u}^{\text{fsi}} = \bar{\mathbf{u}} + \hat{\mathbf{u}}H - \mathbf{u}^{\text{fsi}} = \bar{\mathbf{u}} + \hat{\mathbf{u}} - \mathbf{u}^{\text{fsi}} \quad \text{at } \Gamma^+ \quad (33)$$

where the structural movement defines the prescribed velocity at Γ^+ . Since this Dirichlet condition cannot be applied to the nodes directly, it is enforced on Γ^+ with a Lagrange multiplier technique as

$$\lambda(\mathbf{u} - \mathbf{u}^{\text{fsi}}) \quad \text{at } \Gamma^+ \quad (34)$$

Various discretizations and proper mesh densities for the Lagrange multiplier can be found in the literature for DLM/FD methods, *e.g.* in Refs. [16–18] or in Ref. [21].

On Γ^- , ‘do nothing’ boundary conditions are employed. Consequently, the fictitious flow in Ω^- can cross Γ^- . The advantage of this approach is that Ω^- can change its volume as the structure does on top of it. No additional forces appear and influence the flow in Ω^+ and on Γ^+ . The weak form of the

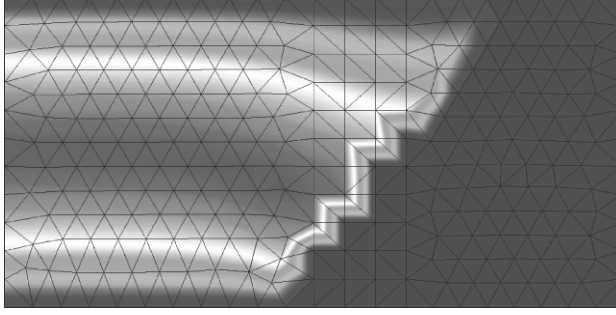


Fig. 13. Demonstration of the ‘do nothing’ BC in a flow field: The flow field with incomplete postprocessing step. The ‘perturbations’ across the interface are a post-processing artefact – the integration is performed using the exact value of $\mathbf{u}^h(\mathbf{x})$.

Eulerian Navier-Stokes equation including the Dirichlet interface conditions at Γ^+ enforced by the Lagrange multiplier λ and the ‘do nothing’ BC on Γ^- is stated as

$$\begin{aligned} & \left(\mathbf{v}, \mathbf{u}_{,t}^{n+1} \right)_{\Omega^f} + \left(\mathbf{v}, \mathbf{u} \cdot \nabla \mathbf{u} \right)_{\Omega^f} + \left(\nabla \cdot \mathbf{v}, \boldsymbol{\sigma}^f \right)_{\Omega^f} - \left(\mathbf{v}, \boldsymbol{\sigma}^f \cdot \mathbf{n} \right)_{\Gamma^f} \\ & - \left(q, \nabla \cdot \mathbf{u} \right)_{\Omega^f} - \left(\delta \lambda, \mathbf{u} - \mathbf{u}^{\text{fsi}} \right)_{\Gamma^{f+}} - \left(\mathbf{v}, \lambda \right)_{\Gamma^{f+}} \end{aligned} \quad (35)$$

For brevity, the standard Neumann conditions along non-moving fluid boundaries have been omitted. The stabilization can be applied as usual, however, further investigation is required on how the cutting of elements interferes with the calculation of the stabilization parameters.

Having described the fluid problem in this way, the iterative FSI coupling procedure as described in Section 3.2 is given as

- Compute an explicit predictor of the structural interface displacement at the new time level $\mathbf{d}_{\Gamma,0}^{n+1}$.
- Start iteration over fields
 - Fluid:
 - * Compute fluid velocity at Γ^{fsi} from structural displacements to serve as Dirichlet boundary condition $\mathbf{u}_{\Gamma,i+1}^{n+1}$ ($\mathbf{d}_{\Gamma,i+1}^{n+1}$).
 - * Solve fluid equations with updated position of Γ^{fsi} to obtain $\mathbf{u}_{i+1}^{n+1}$ and $\mathbf{p}_{i+1}^{n+1}$.
 - * Obtain fluid boundary traction along Γ^{fsi} : $\mathbf{h}_{\Gamma^{\text{fsi}},i+1}^{\text{f},n+1}$.
 - Structure:
 - * Solve the structural field for the new displacements $\mathbf{d}_{i+1}^{n+1}$ including the interface displacements $\tilde{\mathbf{d}}_{i+1}^{n+1}$ under consideration of the fluid load $\mathbf{h}_{\Gamma^{\text{fsi}},i+1}^{\text{f},n+1}$.
 - * Relaxation of the interface displacements $\tilde{\mathbf{d}}_{i+1}^{n+1}$

- * Perform relaxation by $\mathbf{d}_{i+1}^{n+1} = \omega_i \tilde{\mathbf{d}}_{i+1}^{n+1} + (1 - \omega_i) \mathbf{d}_i^{n+1}$
- Check convergence,
- Restart iteration with new mesh calculation if not yet converged.

Summarizing this section, the main goals could be attained. The influence of the fictitious fluid on the physical result could be removed. The structure can deform in an arbitrary, *i.e.* also in a compressible, way and no additional artificial ‘background viscosity’ influences the structural deformation. And very important the kinematic and the traction discontinuity is represented with the order of the spatial discretization.

6 Numerical Examples

6.1 Stirring Rotor in a Cavity

To illustrate the Chimera-type method described in Section 5.1, we present an example of a cavity flow stirred by a rigid agitator. The setup is depicted in Figure 14. The problem is solved on two Dirichlet/Neumann coupled domains. The governing equation on the subdomain fixed to the rotating agitator is the Navier Stokes equation including the additional inertia terms. Each of the subproblems is solved with a finite element code (bilinear elements, with GLS stabilization). Time discretization is done by a one-step- θ rule.

The overlapping area of the subdomains was chosen as large as possible in order to get the best convergence behavior of the iteration between the fluid subdomains. The ‘costs’ of the domain decomposition in this example are up to 20 solutions of the subproblems in one time step. The time required for one subdomain solution depends strongly on the state of convergence of the subdomain iteration. A nearly converged subdomain iteration leads to

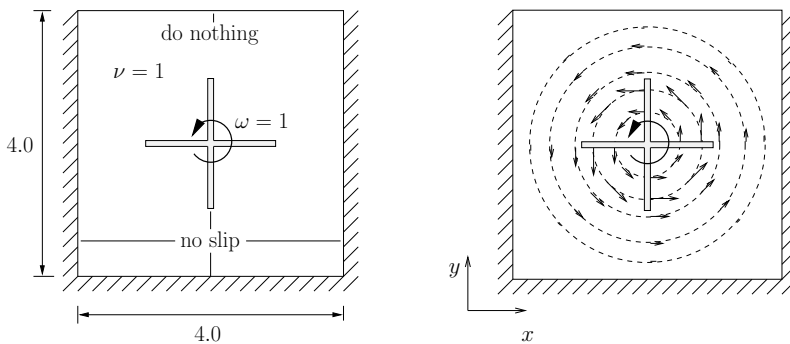


Fig. 14. Problem definition for the Chimera example problem (left). All input values are presented in dimensionless form – the Reynolds number associated with the problem is very low, hence we are expecting a primary flow pattern close to the one displayed on the right.

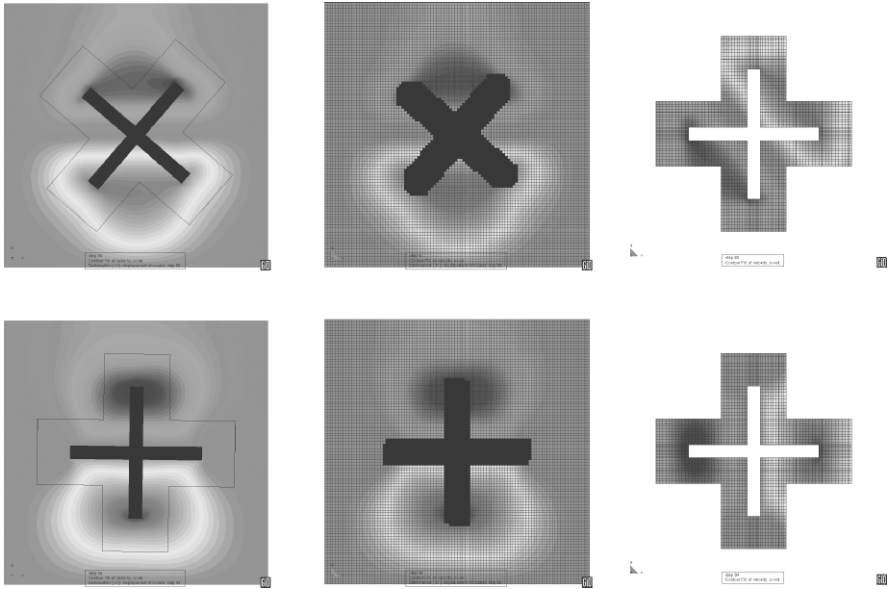


Fig. 15. Solutions for the horizontal velocity of the Chimera example problem (The difference in angle of rotation from top to bottom corresponds to 14 timesteps): Combined solution from patch and background (left), solution on the background subdomain (in the middle; the deactivated nodes are displayed in black) and solution on the patch domain before transforming to the global coordinate system (right).

very small changes in the boundary values over one iteration step, hence, the nonlinear iteration solved for the subproblems converges very rapidly in these cases.

Figure 15 displays the results for the horizontal velocity on all subdomains for two time steps. The horizontal velocity was favored in comparison to the absolute velocity to emphasize the different frames of reference on which the solution is determined on the different subdomains.

6.2 Interaction of a Flexible Beam with a Cylinder Vortex Street

To test the presented methods and implementations for long time fluid structure simulations, the ‘benchmark problem’ depicted in Figure 16 is considered. The simulation was performed for several inflow velocities corresponding to the Reynolds numbers 20, 100 and 200. While the final benchmark results are obviously not yet published, the internal comparison within the DFG research group FG493 indicates the good quality for long term simulations.

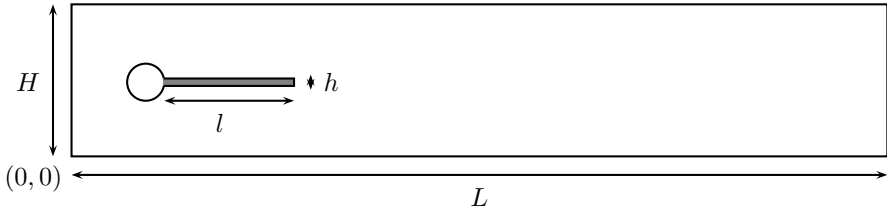


Fig. 16. Benchmark configuration: From the left, a constant fluid inflow generates a vortex street behind the rigid cylinder, which brings the attached flexible to oscillation.

For more details on the configuration and material properties, see [38]. A preliminary benchmark result for $RE = 200$ is shown in Figure 17. As expected the simulation ran stable over the complete simulation time (here 30s or 6000 time steps) and did not show any sign of instability.

6.3 Snap through of Rubber Gasket

The distinct interaction of incompressible flow and a shell structure is addressed with this three-dimensional numerical example of a machine part,

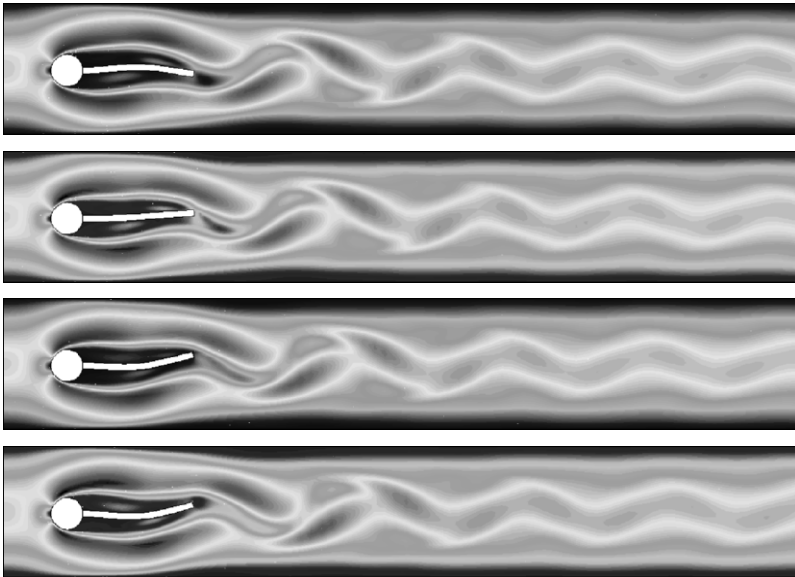


Fig. 17. Snapshots of solution of benchmark problem at $RE = 200$: the figures show the velocity magnitude at 4 different time steps. The beam oscillates with double the frequency of the generated vortex street.

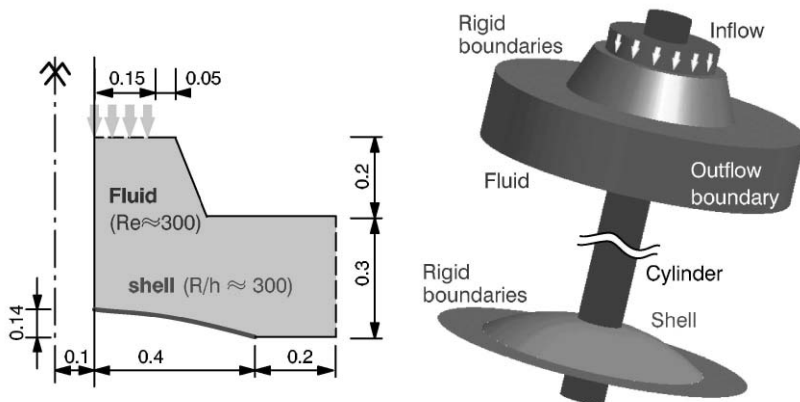


Fig. 18. Gasket with spherical shell (cross section and exploded isometric view)

Figure 18. The axisymmetric gasket has a rigid, maybe moving inner solid cylinder and a fixed outside casing allowing an inflow in the center at the top and an outflow to the sides of the structure. At the bottom the gasket is closed by a thin, elastic spherical shell cap which is for the purpose of visualization pulled out in the right picture of Figure 18.

It turned out that the shell snaps through by buckling under the impact of the fluid pressure. Although the three dimensionality of the simulation allows a non-symmetric buckling mode the final deformation mode was, except for a few perturbations, primarily axisymmetric; this was also the case when geometrical imperfections were superimposed to the initial geometry. This means that the fluid had a regularizing effect on the thin shell which is as a stand alone system known as extremely imperfection sensitive with a tendency to a non-axisymmetric failure mode. Overall the buckling process, shown in Figure 19, was regularized by the viscous damping effect and the inertia of the fluid which had to follow the snap-through deformation of the shell.

7 Concluding Remarks

In this paper we concentrated on specific issues when dealing with large deformation fluid structure interaction problems. Main subjects were advances on coupling issues and on ALE based approaches as well as the development of new fixed-grid schemes.

With respect to field coupling an analysis of the instability of sequential staggered schemes has been presented followed by an accelerated, robust strong coupling scheme. In addition important issues related to the transfer of variables at the interface and on specific modeling and solution requirements for surface coupled problems have been addressed.

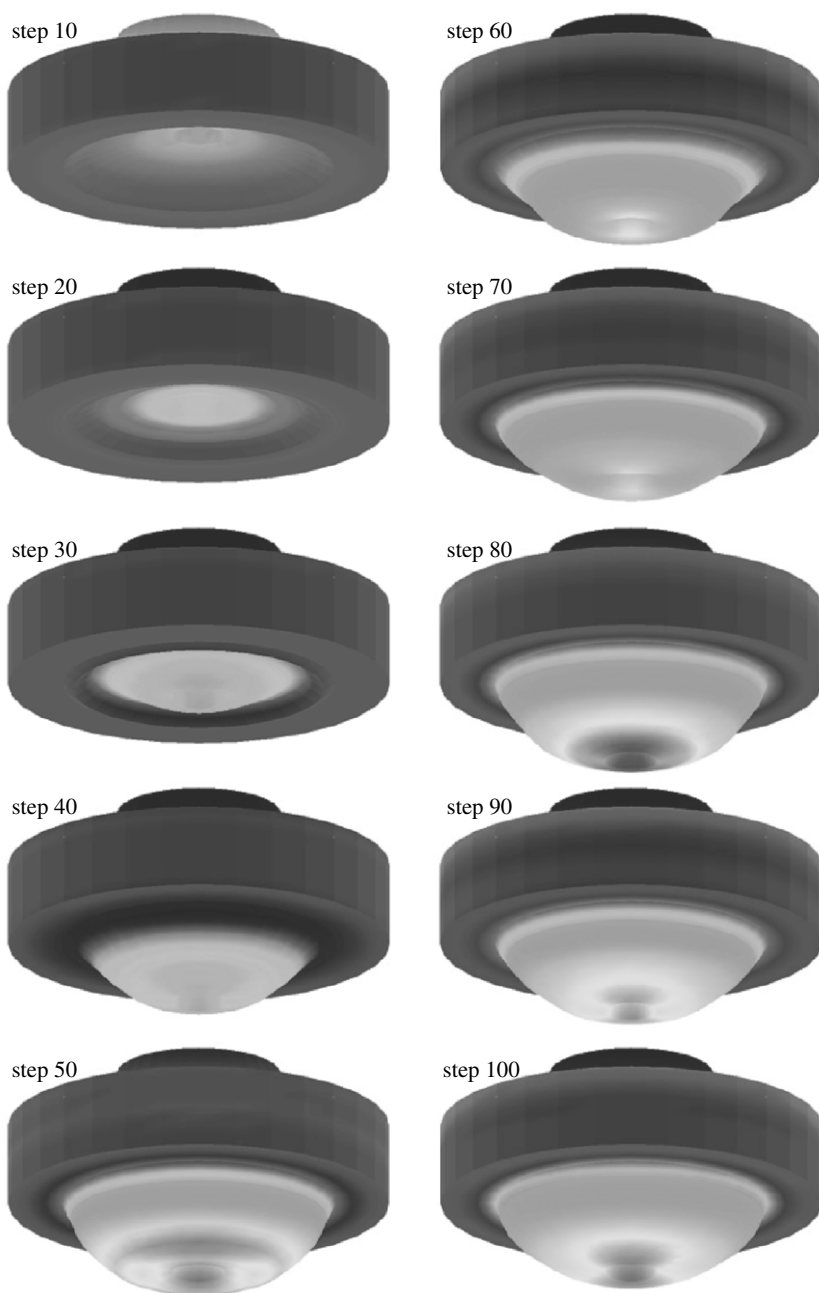


Fig. 19. Snap-through of gasket (view from below with pressure)

For ALE type approaches the question of geometric conservation has been revisited. In this context an advantageous formulation was presented that easily enables to transfer time discretization properties from the fixed-grid to the moving grid case for finite element formulations. In addition the achieved high accuracy of the flow solver on distorted meshes has been presented followed by short comments on mesh moving schemes.

Finally two fixed-grid schemes were presented, a Chimera-type scheme and a XFEM-based scheme. These approaches promise to overcome some shortcomings of currently available fixed-grid methods. Both schemes are still developed further and in addition we are working on a very promising new hybrid approach, which promises to combine the respective benefits of both methods.

8 Acknowledgement

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Fluid-Structure Interaction on Cartesian Grids: Flow Simulation and Coupling Environment

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Abstract. Despite their frequently supposed problems concerning the approximation of complicated and changing geometries, hierarchical Cartesian grids such as those defined by spacetrees have proven to be advantageous in many simulation scenarios. Probably their most important advantage is the simple, efficient, and flexible interface they offer and which allows for an elegant embedding of numerical simulations in some broader context, as it is encountered in a partitioned solution approach to coupled or multi-physics problems in general and to fluid-structure interaction in particular. For the latter, a flow solver, a structural solver, and a tool or library performing the data exchange and algorithmic interplay are required. Here, the main challenge still unsolved is to keep the balance between flexibility concerning the concrete codes used on the one hand and overall efficiency or performance on the other hand.

This contribution addresses two of the above-mentioned modules of a partitioned approach to FSI – the coupling environment and the flow solver, focusing on a strictly Cartesian, hierarchical grid. We report the design, implementation, and tuning of both and present first steps towards FSI simulations.

1 Introduction

One of the most important advantages of a partitioned approach to coupled problems in general or to fluid-structure interaction (FSI) in particular, see [35], e. g., is that existing and tried and tested codes can be utilised to simulate interaction. Typically, however, both simulation tools have to be modified to be able to directly exchange the respective physical quantities. Thus, the codes are closely connected by the underlying communication pattern. This entails the drawback that the codes must be modified if this pattern is changed due to a variation in the coupling scheme or due to the replacement of one of the simulation tools.

This unsatisfactory situation is caused by the established concept of integrating the coupling scheme into the source code of the applications. Even a sophisticated tool box that supports a lot of useful functionalities for coupled problems such as `MpCCI` [13], the successor of `GRISLI` [2], can not avoid this drawback entirely, since for particular cases such as higher order elements

in one of the solvers, the user has to re-implement significant parts of the coupling (the complete interpolation, for instance). In addition, control by the user is restricted due to the fact that the source code is not available.

In this paper, we present an alternative client-server approach that encapsulates the coupling scheme in a third independent module. Thereby, we avoid the undesirable changes in the source code of the simulation applications mentioned above. Each solver (server) only has to be enhanced with an interface to some central description of the geometry of the coupling surface the coupling client provides (a triangulation, e.g.). For the link-up of the solvers to the interface, we have to establish efficient algorithms to determine the geometric and topological relations, in particular the dependencies between surface data (supplied by the coupling environment) and spatial data (from the respective discretisations, supplied by the solvers). Therefore, we derived an embedded spacetree [1] model. Such a hierarchical volume-oriented model enables us to find the triangles in the neighbourhood of a given node of the grid of a solver very fast and efficiently. Examples of applications for our coupling environment can be found in [39,14].

Due to the importance of coupled and embedded simulations, the question of suitable software interfaces and embedding tools becomes more and more crucial. In this context, Cartesian grids in general and hierarchical structures such as spacetrees in particular have substantially gained attractivity in the last years due to the advantages mentioned above, see also [30]. But they are also perfectly suited for many simulation tasks such as the fluid application presented in this paper. The inherent spatial decomposition of the underlying geometry facilitates the implementation of numerical methods requiring hierarchical grid and data representations such as adaptive grid refinements and multigrid methods. In addition, using a Cartesian discretisation in our fluid solver enables us to directly create the computational domain for the flow part with the help of the embedded spacetree model and, in particular, to handle the moving structure in the fluid simulation very easily. We have developed two possible discretisation schemes for the Navier-Stokes equations on spacetree grids both ensuring momentum, energy, and mass conservation due to certain symmetry properties of the discrete operators corresponding to analogous symmetries of the continuous operators [43]. To efficiently implement and parallelise the corresponding multigrid solvers, we combine the spacetree concept with space-filling curves.

The remainder of this contribution is organised as follows. In Sect. 2, we present our coupling environment – from the underlying concept and design issues to its concrete implementation. Then, Sect. 3 deals with the second main part of our work, the Cartesian flow solver. Here, the essential topics addressed are the symmetry-preserving Navier-Stokes discretisation (finite volume and finite element), the realisation of multigrid and adaptivity abilities, and implementation aspects aiming at hardware efficiency and easy parallelisation properties. In Sect. 4, we discuss a crucial issue for FSI in a partitioned context, the problem of data exchange at the coupling surface.

Afterwards, Sect. 5 is dedicated to numerical experiments, dealing with both single modules and first steps for the simulation of FSI on Cartesian grids. Finally, some concluding remarks and a short outlook close the discussion.

2 Design of the Coupling Environment

It is our objective to enable the user to choose freely among simulation tools and coupling schemes to simulate his given fluid-structure interaction problem. Therefore, a software environment where, first, each simulation application can be replaced by another and, second, the coupling scheme is implemented outside the simulation programs must be developed. For that, we organise the data transfer between the simulation applications with a client-server approach based on an independent geometric representation of the coupling surface. In this section, we introduce this basic concept and present a first implementation of the corresponding programming interface. First, we describe the underlying client-server concept. Subsequently, we present the central surface model and discuss the embedded spacetree model bridging the gap between surface data of the coupling environment and spatial data of the respective fluid and structure solvers. Finally, some essential details of the programming interface are shown.

2.1 Client-Server Concept

In a partitioned fluid-structure interaction approach, the exchange of the physical quantities at the coupling surface is an essential part of the coupling process. This communication is strongly related to both the coupling scheme and the data representation of both solvers involved. Let us look at the simple explicit scheme. It starts at a time step with the calculation of the fluid flow around a structure, for instance. Subsequently, the resulting stresses acting on the surface of the structure have to be transferred to the structural dynamics simulation, which is now performed at the current time step. The structure is deflecting, and the movement is transferred back to the fluid simulation, where the next time step will be started.

Usually, the data are transmitted directly between the applications: one side is sending, while the other side is receiving the data. Such a direct communication has two major drawbacks. First, the codes have to “decide” themselves when to send and receive data and what to do with them. In the example above, the fluid solver has to stop after the first time step, send data to the structure solver, and wait for the results from the structure solver until it can continue with the next time step. Thus, it is not possible to switch from an explicit to an implicit coupling scheme (see [12], e.g.) without a modification of the source code of both applications. Second, neither the grid points nor the representation of functions (e.g. by coefficients of a certain basis, nodal or hierarchical, ...) of the two solvers will coincide in general. Thus, we have

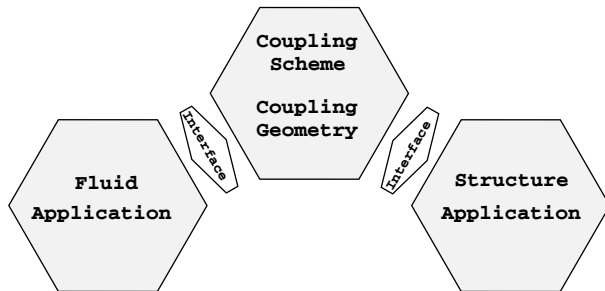


Fig. 1. Diagram showing the communication scheme based on a client-server approach. The simulation codes communicate with the central instance via a unique interface. The coupling scheme is encapsulated in the central instance. Hence, this communication architecture is much more flexible than a direct communication.

to find a communication pattern suitable for both applications for *each* combination of a fluid solver with a structural solver. This means that if we use another fluid application, for example, we will have to modify the source code of both the new fluid application and the existing structure application according to the respective communication pattern. The second problem can be avoided if we define a unique communication interface. Unfortunately, this does not help for the first problem, apart from the not very reasonable case that we include all possible coupling schemes. Even if we are able to do this, it will be necessary to modify the applications as soon as someone wants to slightly change an existing scheme.

Our way to overcome this problem is to change the basic communication architecture from direct communication to a client-server concept. For that, we use an additional application that controls the coupling procedure. This central instance sends requests to the simulation applications, while the latter act as servers. With each request, the servers receive all necessary information to compute a new step in their respective simulation. They return the calculated quantities to the central instance, and the central instance continues with the FSI simulation. In this approach, a server application communicates only with the central client (cf. Fig. 1).

Hence, the simulation applications do not know anything about each other and we have solved both problems mentioned above. On the one hand, we can replace each of the simulation codes without any effect on the other application. On the other hand, the coupling scheme is completely isolated in the central instance. With this encapsulation, a change of the coupling scheme can be realised without any impact on the simulation applications ¹.

¹ Of course, the used simulation codes have to provide the general functionalities needed for the respective coupling scheme such as iterations within one time step for implicit coupling schemes.

The user has to change his source codes only *once* to adapt them to the described client-server concept and the respective central geometric description of the coupling surface. Afterwards, he fully profits from a flexible coupling environment. He can freely choose the tools most suitable for his applications or which he is most familiar with, and he can define – or test – appropriate coupling schemes without touching the source code of the simulation tools again. The user is not restricted to a certain Computational Fluid Dynamics (CFD) tool being coupled with a certain Computational Structure Dynamics (CSD) tool. Together with our partners [5,14,39], we use this flexibility to study and compare different simulation applications with different coupling schemes in a partitioned coupling approach for FSI.

Obviously, the concept described needs a central discrete representation of the coupling surface that is independent of the simulation tools. This unique representation defines points in the computational domain. The simulation codes must return the physical state of the coupling surface at these positions. The next subsection introduces this boundary representation model used in our concept.

2.2 Geometric Representation of the Coupling Surface

For the representation of the fluid-structure interface, a triangular mesh is used which describes the geometry of the interface on the one hand, and which allows for the nodal-based definition of coupling data such as displacements and stresses on the other hand. Although we work in a Cartesian setting, for this task, a classical triangular mesh is the straightforward and the simplest choice since typical problems such as the need for healing distorted meshes resulting from changes of the structure's geometry by local mesh refinement or coarsening are solvable at tolerable cost. In our approach, the triangulated surface mesh has to meet two demands. First, the surface must be closed, which means that each edge has to be shared by two triangles, and second, it must be oriented, which means in our case that the normal vectors of the triangles point outwards with respect to the structure's domain. Figure 2 shows a part of a triangulated cylinder surface as an example for such a surface description.

The extraction of this surface description lies in the user's responsibility and can be done cheaply in most cases, for example using the nodes on the surface of the structure as used in the CSD tool. It must be sufficiently fine both for the coupling data to be interpolated accurately enough and for a precise geometry representation. Thus, it is also possible – and quite easy with the help of the embedded spacetree model described in Sect. 2.3 – for solvers to derive the geometry information for their individual space discretisations from this central surface mesh in the initial phase of a coupled fluid-structure interaction computation, if necessary.

In our current setting, where a finite volume code based on a Cartesian grid for the fluid simulation is combined with a p-FEM structural code [40],

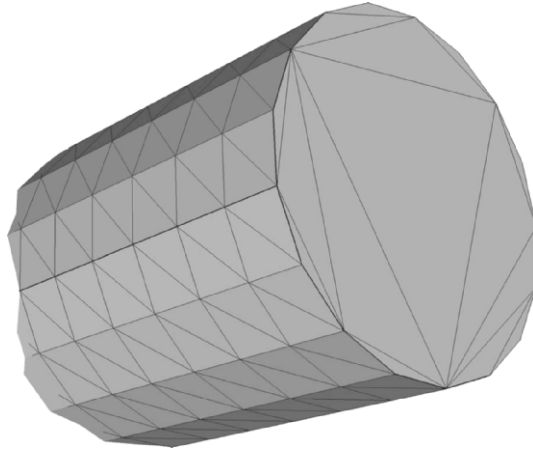


Fig. 2. An example for the central geometric representation of the coupling surface: a closed triangulated surface mesh, where the normal vectors of the triangles point outwards with respect to the structure.

the central triangular mesh is derived from the surfaces of high-order hexahedral finite elements given by the structural code. The mesh is constructed in such a way that the Gauss points of the structural surfaces are nodes of the mesh. The structural displacements can be computed at all nodes of the central surface mesh with arbitrarily high polynomial degree.

For the link-up of spatial grid data of the solver to the central surface mesh, we have to spatially localise the nodes of the central surface mesh. For this, we have established and implemented spacetree methods introduced in the next subsection.

2.3 The Embedded Spacetree Model

The triangular surface mesh as mentioned in the previous subsection is the reference for each data exchange between the flow solver and the structural solver. Therefore, the users of our coupling environment must be able to identify the triangles of the central surface mesh in the neighbourhood of the boundary nodes representing the coupling surface in the grids of their solvers. We address this crucial point with an embedded spacetree model. A big advantage of spacetrees and their representatives in 3D, octrees, is their inherent location awareness: any question related to finding neighbours for a specific cell or the spatial positioning of objects can be easily answered. Thus, determining the voxel including a specific node of the triangulated surface mesh and, therewith, the supply of all informations needed for the exchange of data between the central instance and the fluid or structural domain is cheap as soon as a suitable spacetree is available.

At first glance, the derivation of such a spacetree-based volume-oriented model from a surface-oriented one requires a lot of computations since in each refinement step during the generation of the spacetree the newly created voxels have to be tested for intersection with the boundary of the surface-oriented model again and again. However, in [31], a new approach to tackle this problem, allowing a fast and efficient spacetree generation even in real time and *on-the-fly*, was presented. Since we have to derive a new spacetree representation from the altered representation of the coupling surface after each coupling step, the integration of this approach into our coupling environment is decisive for its overall efficiency.

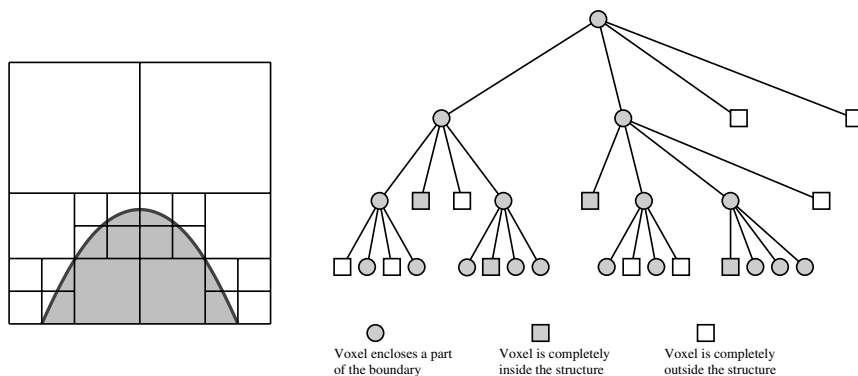


Fig. 3. Surface model together with the derived spacetree in 2D (left) and the corresponding quadtree data structure (right).

So far, we have explained our idea to establish the communication between the fluid and the structure simulation tool in a flexible but still efficient and user-friendly way. We pointed out that the user profits from both the free choice of the coupling strategy as soon as he has *once* linked up his solvers to the coupling environment and, second, from the flexibility to exchange one solver without having to change the source code of the solver of the other application since both codes are hidden from each other by the coupling environment. Concretely, our concept is implemented in a coupling library that has to be linked to the simulation tools. To demonstrate the features of this library, we turn to a more technical description of the corresponding programming interface in the next subsection.

2.4 Programming Interface

Above, we described our client-server communication model, which is the basis for the corresponding coupling library. Now, we give a short description of the programming interface related to this library. The interface follows two

principles. It should be easy to be integrated into an existing application, and it should be as simple as possible. Up to now, the library supports explicit coupling only and is implemented in the programming language C. It includes five interface functions, and, additionally, the user has the possibility to write a function that controls the progress of simulation time.

For reasons of convenience, we base our library on a more general communication library such as MPI [29]. All communication processes are encapsulated and neither visible nor accessible to the user. Hence, we can change the communication library without changing the interface. This gives us the possibility to extend the library in future releases using other communication concepts.

At the beginning of the coupling procedure, the application must initialise the coupling process. For that, the function `Fsi_Init()` has to be called. The declaration of the function is

```
int Fsi_Init(Mesh* M,
              int* put_get_mesh,
              double* dt,
              int* put_get_dt,
              int* start_calc_recv,
              int* argc, char** argv);
```

where the function parameter `M` is a pointer to the mesh structure, `put_get_mesh` is the information whether the application sets or gets the mesh, `dt` is the length of the time step, `put_get_dt` is the information whether the application sets or gets the length of the time step, `start_calc_recv` returns the information whether the application starts with an calculation or waits first for a request, and `argc` and `argv` pass the runtime arguments to the application.

At the end of the coupling procedure, the application must finalise the coupling process. For that, the function `Fsi_Finalize()` with the declaration

```
int Fsi_Finalize();
```

is used. After calculating the physical state at the interface, the application must provide its result with a call of the function

```
int Fsi_Send_quantity(Mesh* M, char* provide);
```

where the tag `provide` classifies the provided data. Currently, stresses and displacements are supported as data. The corresponding function to wait for a request is defined as

```
int Fsi_Recv_quantity(Mesh* M, char* require);
```

where the tag `require` classifies the required data. The declaration of the user function that controls the advancement of the time (called in the client) looks like

```
int Fsi_do_advance_coupling(Mesh* M, int time_step);
```

The return value of this function decides whether the coupling process will be continued. It is passed to the applications via the return value of the function `Fsi_Recv_quantity()`.

Finally, we look at a snippet of a fluid simulation code to illustrate the usage of the library:

```
do {
    /* -- some computations -- */
    ...

    Fsi_Send_quantity(M, provide);
    fsi_status = Fsi_Recv_quantity(M, require);

    step++;
} while(fsi_status==FSII_STATUS_GO);
```

We assume an explicit coupling scheme and a successful initialisation. The fluid application enters the time loop and computes the flow around the structure. By calling the function `Fsi_Send_quantity()`, the stresses acting on the structure are sent to the central instance. Afterwards, the application waits for the next request by calling the function `Fsi_Recv_quantity()`. The function returns the information whether there is a request or not. If there is a request, the CFD program will advance and start the calculation of the next time step. Otherwise, the program leaves the time loop, calls the function `Fsi_Finalize`, and stops.

3 Discretisation and Implementation of the Navier-Stokes Solver

As mentioned above, the fluid solver uses a spacetree representation for both the domain and the computational grid. Thus, our grid consists of square cells in two dimensions and cubic cells in three dimensions, resp. Spacetree grids offer both a high degree of structure and a high flexibility, that is the possibility for arbitrary (isotropic) adaptive refinement. We associate all velocity values to the vertices of the cells and the pressure to the midpoint of the cells. In contrast to the standard fully staggered grid as used for example in [17], this so-called semi-staggered grid allows for a continuous interpolation of velocity values, which is decisive in particular in connection with dynamic adaptivity and multigrid schemes.

We will describe two variants of a symmetry-preserving discretisation: a finite volume method and a finite element method. For the finite volume method, the implementation is completed both in 2D and in 3D. Our finite element method was developed to remedy deficiencies of this method but – at the same time – to maintain all advantages induced by the operator symmetry. In contrast to the finite volume method, the generalisation of the

finite element method to adaptive spacetree grids is straightforward and can be realised very efficiently using cell-based operator evaluation. To achieve a high hardware efficiency also for the adaptive grids and to provide a cheap and balanced parallelisation, we use space-filling curves to develop a particular algorithm working on strictly linear data structures.

3.1 Symmetry-Preserving Operator Discretisation

Our discretisation of spatial derivatives guarantees the fulfilment of symmetry properties of the continuous operators also for the discrete operators. According to [43], symmetry-preserving operator discretisations lead to a correct discretisation with respect to physical properties of the system, which means in our case conservation of energy and momentum. Using the duality

$$\nabla = -\operatorname{div}^*$$

of the continuous gradient and divergence operator, we get the general form of the semi-discrete Navier-Stokes equations

$$\Omega \frac{d\mathbf{u}_h}{dt} = C(\mathbf{u}_h)\mathbf{u}_h - D\mathbf{u}_h + M^T \mathbf{p}_h, \quad (1)$$

$$M\mathbf{u}_h = 0 \quad (2)$$

with the mass matrix Ω , the (non-constant!) convection matrix $C(\mathbf{u}_h)$, the diffusion matrix D , and the divergence matrix M . In the following, we will shortly describe both the finite volume and the finite element discretisation for which the symmetry and consistency requirements

$$\Omega \quad \text{symmetric positive definite}, \quad (3)$$

$$D + D^T \quad \text{positive semidefinite}, \quad D\mathbf{e} = \mathbf{0}, \quad (4)$$

$$C(\mathbf{u}) = -C^T(\mathbf{u}), \quad C(\mathbf{u})\mathbf{e} = \mathbf{0} \text{ for all } \mathbf{u} \quad (5)$$

hold. $\mathbf{e}$ is the vector with all entries equal to 1. To display the basics of our methods without complicating things too much, we restrict ourselves to the two-dimensional case. For a generalisation of the operators to 3D, see [11,4]. For time discretisation, we use Chorin's projection method [9].

The Continuity Equation The discrete continuity equation operates cell-wise on the grid. That is, in terms of the finite volume method, each cell of the spacetree grid defines a control volume at the same time (see Fig. 4). To achieve volume streams over the faces of a cell, we assume linear interpolation of velocities on the edges of the cell and get

$$M_{cell}\mathbf{u}_h := \frac{h}{2} \left(\underbrace{u_{NE} + u_{SE}}_{\text{right face}} - \underbrace{u_{NW} - u_{SW}}_{\text{left face}} + \underbrace{v_{NW} + v_{NE}}_{\text{upper face}} - \underbrace{v_{SW} - v_{SE}}_{\text{lower face}} \right),$$

where M_{cell} denotes the cell part of the overall operator matrix M .

The Momentum Equation

Finite Volume Discretisation For the momentum equation, we use a control volume of the same size as for the continuity equation, but with a vertex of a cell as the centre (see Fig. 4). Thus, Ω is h^2 times the identity matrix if h is the mesh width of the spatial grid. The diffusive term is discretised with the help of the well-known five-point stencil which corresponds to the approximation of the gradients of the velocities at the volume faces by a central difference of the velocities at the respective two neighbouring cell vertices. It is easy to see that both Ω and D fulfil the symmetry requirements.

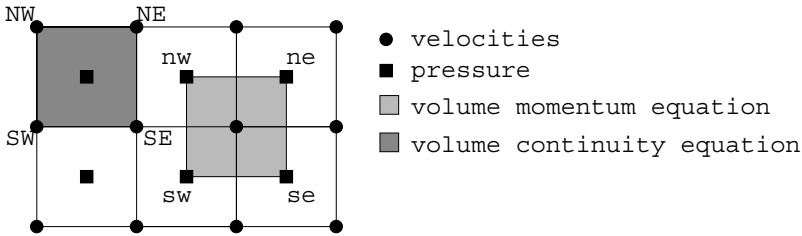


Fig. 4. Allocation of degrees of freedom for velocities and pressure and control volumes for the continuity and the momentum equation.

For the convective term, we need velocity values at the vertices of the control volume. Thus, before we can compute the interpolant on the volume's edges, we have to interpolate these values with the help of neighbouring degrees of freedom. To fulfil all symmetry and consistency requirements for the convective operator matrix, we use a very particular interpolation of the values at the volumes' vertices and edges: First, the interpolation has to be symmetric, that is, all neighbouring cell vertices must have the same weight. This ensures antisymmetry of the discrete operator. Second, the interpolation has to ensure that an analogue to the discrete divergence holds for the four subquadrants of the control volume if it holds for all four cells that share a subquadrant with the control volume. This ensures consistency of the discrete convection operator. Figure 5 shows the two steps of the evaluation of the resulting convection term: interpolation of velocities to vertices and edges of the control volume and approximation of the convective flow over the volume faces. For a detailed description, see [11].

Finite Element Discretisation Our finite element discretisation uses particular ansatz functions for the velocity space in combination with cell-wise operator evaluation to overcome two deficiencies of the finite volume method: First, due to the possibility of a cell-wise evaluation, the generalisation of

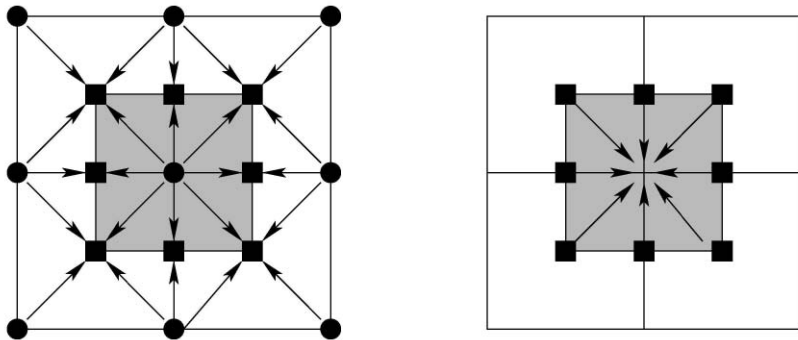


Fig. 5. Visualisation of the two steps of the evaluation of the discrete convection: interpolation to vertices and faces of the control volume (left) and approximation of the integral over the faces (right).

the discretisation to adaptive spacetree grids is straightforward² and gets along without storing specialised operators at boundaries between different refinement depths. Second, in contrast to the non-trivial construction of the symmetric convection operator in the finite volume context, things become very easy in the finite element context. The symmetry requirements can be shown to be equivalent to the pointwise fulfilment of the continuity equation.

For the flow velocities u (x -direction) and v (y -direction), we assume a general ansatz space. The associated basis shall be denoted as $\{f_1, \dots, f_{2N}\}$ with the number N of grid points. The basis functions $f_1, \dots, f_N$ correspond to the first component (u) of the velocity vector $\mathbf{u}$, $f_{N+1}, \dots, f_{2N}$ to the second component (v). As we concentrate on a nodal basis, the functions f_i and f_{N+i} have the values $(1, 0)$ and $(0, 1)$, respectively, at the i th grid point and zero at all other grid points. Thus, the approximative solution is represented by

$$u_h = \sum_{i=1}^N (u_i f_i + v_i f_{N+i})$$

² Since each cell has four vertices independent of the refinement depth of surrounding cells, the evaluation of the cell-parts of the operators also does not depend on the adaptive structure of the grid. At boundaries between different refinement depths, we only have to interpolate values to hanging nodes before the cell-wise operator evaluation and perform the corresponding restrictions afterwards to ensure a correct transport of fine-grid operator parts to vertices at refinement boundaries.

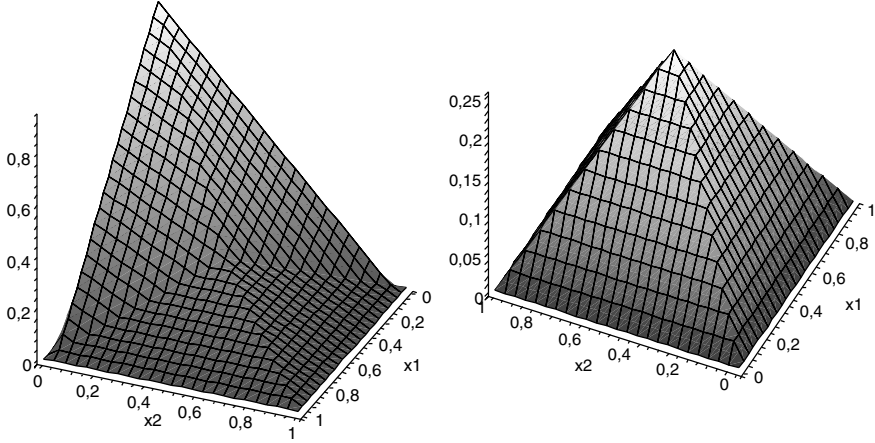


Fig. 6. Shape of the components of our basis functions $f_i = (\Phi_i, \phi_i)^T$ and $f_{N+i} = (\phi_i, \Phi_i)^T$ (left: Φ_i , right: ϕ_i).

with the solution vector $\mathbf{u}_h = (u_1, \dots, u_N, v_1, \dots, v_N)^T$. With the requirement $f_i|_{\partial V} = 0$, we get

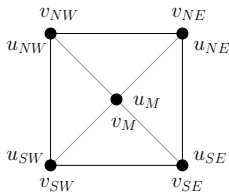
$$\begin{aligned} (C(\mathbf{u}_h))_{i,j} &= \int_V f_i \cdot (u_h \cdot \nabla) f_j dV = \\ &= \int_V f_i \cdot \nabla (u_h f_j) dV - \int_V f_i \cdot \nabla^T u_h \cdot f_j dV = \\ &= - \int_V f_j \cdot (u_h \cdot \nabla) f_i dV - \int_V f_i \cdot \nabla^T u_h \cdot f_j dV = \\ &= -(C(\mathbf{u}_h))_{j,i} - \int_V f_i \cdot \nabla^T u_h \cdot f_j dV. \end{aligned}$$

Thus, the requirement $C(\mathbf{u}_h)^T = -C(\mathbf{u}_h)$ is equivalent to the pointwise fulfilment of the continuity equation $\nabla^T u_h = 0$. To ensure this, we use particular basis functions

$$f_i = \begin{pmatrix} \Phi_i \\ \phi_i \end{pmatrix} \text{ and } f_{N+i} = \begin{pmatrix} \phi_i \\ \Phi_i \end{pmatrix}$$

with piecewise linear components shown in Fig. 6. These basis functions result in a piecewise (on the four subtriangles of a quadratic cell) linear interpolation of the velocities with values at the midpoint of the cells computed as shown in Fig. 7.

It is easy to show that all other conditions from (3), (4), and (5) and, thus, conservation of energy and momentum are also fulfilled. As the basis functions are piecewise linear, we achieve a second order discretisation as in the case of bilinear ansatz functions.



$$u_M = \frac{1}{4} (u_{SW} + u_{SE} + u_{NW} + u_{NE} + v_{SW} - v_{SE} - v_{NW} + v_{NE}),$$

$$v_M = \frac{1}{4} (u_{SW} - u_{SE} - u_{NW} + u_{NE} + v_{SW} + v_{SE} + v_{NW} + v_{NE})$$

Fig. 7. Continuity-preserving interpolation of the velocity values at a cell's vertices to the midpoint of the cell.

In both bases, the finite volume and the finite element method, we end up with a discretisation that conserves energy and momentum with respect to space for *any* spatial mesh width [43,11]. To ensure the overall stability, we only have to take into account some stability conditions for the time step [11].

3.2 Multigrid and Adaptivity

Continuity-Preserving Interpolation In connection with multigrid schemes and/or dynamic adaptivity, many discretisations of the Navier-Stokes equations suffer from interpolation routines violating the continuity equation. This leads – after a refinement or a coarse grid correction – to velocity values that are not mass-conserving. Based on the continuity-preserving finite element basis functions described above, we can easily define a piecewise linear interpolation coupling all components of the velocity vector within a cell and leading to an interpolant that fulfils the continuity equation pointwise if only the discrete continuity equation is fulfilled. Thus, we can ensure a divergence-free interpolant for every dynamic refinement and after every transport of coarse grid data to the fine grid in terms of multigrid methods.

Cell-Oriented Operator Evaluation For the finite element method, we apply a cell-oriented processing of the grid in combination with the so-called cell-oriented operator evaluation, which is a variant for the computation of operator values at cells' vertices. Instead of processing the grid vertex by vertex and computing the respective operator values as a whole at the respective vertex with the help of the values at neighbouring vertices, we process the grid cell by cell, compute the parts of the operators for which we only need data owned by the current cell, that is data located at the four vertices of the cell, and obtain the complete operator values at the vertices by an accumulation process over all participating cells [6,4,20,36,44,32], see Fig. 8 for an example. The cell-oriented approach results in two advantages for simulations on adaptively refined grids. First, since we do not need any informations about neighbouring cells for the evaluation of the cell-part of an operator, there is no need to derive and store specialised difference stencils at boundaries between

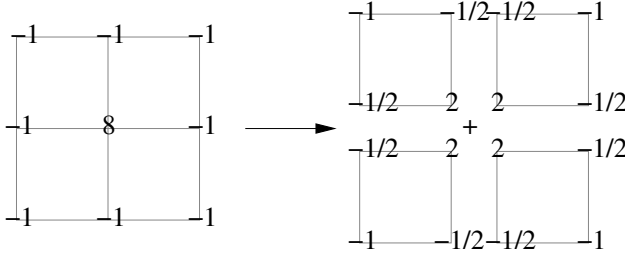


Fig. 8. Cell-oriented operator evaluation for the example of the finite element nine-point stencil for the Laplacian

different refinement depths. Second, together with suitable data structures (see Sect. 3.3), the locality of data access is substantially improved.

3.3 Data Structures, Algorithms, and Implementation

Space-Filling Curves and Data Structures For adaptively refined space-tree grids, there is no obvious traversal order such as row-wise or column-wise for regular grids. In addition, we don't have data on the vertices of the locally finest grid level only but, instead, we work with a generating system for our multigrid method (see [16], e. g.). In combination with our cell-wise operator evaluation, this means that we have to process a whole spacetree cell by cell. The discrete iterates of self-similar, recursively defined space-filling curves [38] are one possibility to find a uniquely defined top-down depth-first traversal of such a cell-tree. In the literature, space-filling curves are widely used as an efficient tool for finding a balanced partitioning of adaptively refined grids [18,19,33,34,37,45] with quasi-minimal communication costs [45]. We go one step further and exploit the quasi-optimal locality properties of space-filling curves together with particular properties of the Peano curve – which we use in our codes – to define strictly linear data structures for the storage of vertex data during one sweep over the whole grid, the so-called stacks. The construction of these stacks is very easy to understand for nodal data on regular two-dimensional grids, where we can do with only two stacks – one handling all points at the left-hand side of the Peano curve and the other handling all points at the right-hand side of the curve. As can be seen from Fig. 9, within these two groups of vertices, data are processed in one direction during the first pass of the curve and in the opposite direction during the second pass of the curve which perfectly fits with the idea of stacks, particular data structures for which only two operations are allowed: put a datum on top of the stack and pop a datum from the top of the stack. During the first pass of the curve, points are put on the stack, and during the second pass, they are popped from the stack in the reverse order. In the three-dimensional case, this essential property of a forward and backward processing of grid

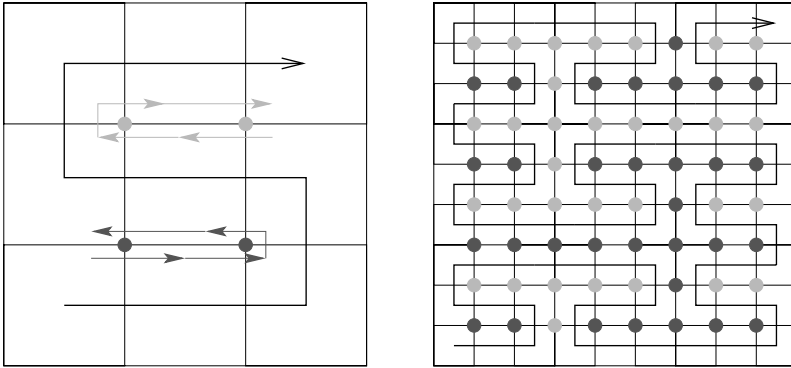


Fig. 9. Iterates of the Peano curve for two-dimensional regular grids and colouring of vertices with respect to the associated stacks. Vertices at the left-hand side of the Peano curve are assigned to the dark grey stack, vertices at the right-hand side to the light grey stack.

points on lower dimensional substructures of the grid could only be shown for the Peano curve so far. Therefore, we exclusively use this curve although it implicates a partitioning into three in each coordinate direction in each refinement step of a cell. For detailed descriptions of the three-dimensional case, adaptive grids, and hierarchical generating systems see [20,36,26,22,23].

Our concept based on structured spacetree grids, cell-oriented operator evaluation, and stacks as data structures results in very low storage requirements of about 5 Byte [36,28] per degree of freedom and high spatial (due to the use of stacks) and time (due to the locality properties of the Peano curve) locality of data and, thus, a very good cache-performance [20,36,22,23], which results in only about 10% more cache misses than the theoretical minimum [20,36]. Starting from this basis, we implemented an additive multigrid algorithm [20,36], dynamic adaptivity [10], higher order approximations [26], and a two-dimensional Navier-Stokes solver with an additive multigrid solver for the pressure Poisson equation [44,32].

Parallelisation For the parallelisation of our algorithm, we use the order of the grid cells given by the Peano curve and cut the resulting stream of cells into equal pieces to achieve a balanced partitioning (Fig. 10). The data structures for vertex data can be realised as stacks for each process in a completely analogous way to the sequential case. For vertices located on the boundary of the domain of two or more processes, the only difference in operator evaluation compared to the sequential algorithm is that the cell parts of the operator are accumulated from several processes via inter-process communication. The same is valid for coarse grid cells which – in contrast to many other approaches – do not cause any complications: coarse grid cells covering parts of the domains of several processes are assigned to each

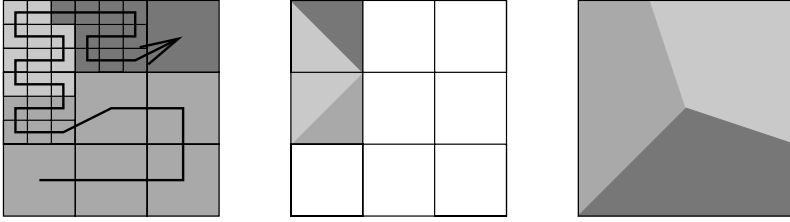


Fig. 10. Partitioning of grid cells according to their order along the Peano curve in a two-dimensional example. Coarse grid cells associated to different processes are marked with different colours.

of these processes and all operator values at their vertices are accumulated from all processes involved during the communication phase. For details of the parallelisation see [27,24,21].

4 Data Exchange: Interpolation and Approximation

Every exchange of quantities between the applications is based on the central surface description. Hence, both the fluid and the structure solver has to map their values to the nodes of this central surface mesh and – in the other direction – map the required values from these nodes to its own grid. In this section, we describe our concept of calculating and transferring the stresses to and the displacements from the central surface mesh to the Cartesian computational grid of our fluid solver.

4.1 Calculating and Transferring the Load

The fluid flow induces forces acting on the surface of the structure, resulting in movements and deformations of the structure. Thus, we have to compute these forces with the help of the information computed by the fluid solver and transfer them to the central surface description. In the following, we present two different methods, the first one for our finite volume discretisation and the second one for our finite element discretisation.

Method One: Using the Finite Volume Discretisation In this method, we compute the force F on a surface by integrating the momentum flux density over the area of the surface Γ_S . If there is no flow through the surface, the momentum flux density on the surface of the structure will be equal to the stress σ and, thus, the force is calculated by

$$F = \int_{\Gamma_S} \sigma \cdot \mathbf{n} \, ds. \quad (6)$$

In an incompressible Newtonian fluid, the stress tensor is given by

$$\sigma_{ij} = -p\delta_{ij} + \eta \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right),$$

where p denotes the pressure, η the dynamic viscosity, u the velocity, and δ_{ij} the Kronecker- δ . Thus, we have to establish a suitable interpolation to compute the pressure and the spatial derivatives of the velocity at the nodes of the central surface mesh. We will demonstrate the type of interpolation we use with a two-dimensional example shown in Fig. 11, where we see a couple of cells and a part of the central mesh. (The dashed lines represent the edges and the black points the nodes.) In principle, the interpolation follows the same rules as described in Sect. 3.1. We use a linear interpolant on the four subtriangles of each volume for the velocity. Hence, the spatial derivatives of the velocities are constant on each triangle. The pressure is interpolated as a constant in each cell. Thus, since all cells intersecting the surface are counted as fluid cells, we know the stress tensor at a node of the central surface mesh when we know its surrounding triangle and its surrounding cell. This location problem can be solved fast and efficiently with the spacetree representation of the central mesh, as mentioned above.

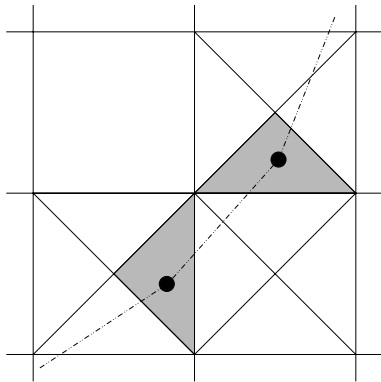


Fig. 11. Cutaway of a two-dimensional Cartesian grid with a surface triangulation marked by dashed lines (grey: surrounding triangles of the surface nodes).

Method Two: Using Consistent Forces In the context of our finite element approach, we use the method of consistent forces proposed in [15] for the calculation of forces acting on the structure. The main idea is to compute these forces with the help of the equilibrium between the contribution of the momentum equation at the boundary Γ_S of the structure and the sustaining forces F_s on the fluid. For this purpose, virtual degrees of freedom with ansatz functions $\tilde{f}_j$ and corresponding coefficients $\tilde{F}_j$ are introduced on the Dirichlet

boundary nodes $N + 1, \dots, N_S$ of the structure³. With these ansatz functions we represent the sustaining forces as

$$F_s(x) = \sum_{j=N+1}^{N_S} \tilde{F}_j \tilde{f}_j(x) \quad \forall x \in \Gamma_S. \quad (7)$$

In this formulation, the coefficients $\tilde{F}_j$ denote the local forces on the boundary nodes j . The equilibrium condition with the contributions of the momentum equation at the boundary node i ($i \in \{N + 1, \dots, N_S\}$) yields the equations

$$\sum_{j=N+1}^{N_S} \tilde{F}_j \int_{\Gamma_S} f_i(x) \cdot \tilde{f}_j(x) \, ds = \left(\Omega \frac{d\mathbf{u}_h}{dt} + D\mathbf{u}_h + C(\mathbf{u}_h)\mathbf{u}_h - M^T p_h \right)_i \quad (8)$$

for the local forces $\tilde{F}_j$ with the notation for the semi-discrete momentum equations taken from equation (1) used in the finite element sense and, thus, representing an integral form, too. To avoid the solution of the whole linear system of equations (8) for $\tilde{F}_i$ in each time step, we use mass lumping for the boundary mass matrix. Thus, the left-hand side of (8) turns into

$$\int_{\Gamma_S} f_i(x) \cdot F_s(x) \, ds = \tilde{F}_i \int_{\Gamma_S} f_i(x) \, ds,$$

and we get the explicit formula

$$\tilde{F}_i = \frac{1}{\int_{\Gamma_S} f_i(x) \, ds} \left(\Omega \frac{d\mathbf{u}_h}{dt} + D\mathbf{u}_h + C(\mathbf{u}_h)\mathbf{u}_h - M^T p_h \right)_i. \quad (9)$$

Finally, we calculate the forces exerted by the fluid on the structure by integration of the negative local contributions (7) of the sustaining force over the boundary Γ_S ,

$$F = - \int_{\Gamma_S} F_s(x) \, ds = - \int_{\Gamma_S} \sum_{j=N+1}^{N_S} \tilde{F}_j \tilde{f}_j(x) \, ds, \quad (10)$$

which can be done efficiently and exactly by a trapezoidal sum for our choice of ansatz functions.

This approach of consistent forces has two drawbacks and two major advantages. It is not possible to separate the contributions of the pressure and the viscous part when calculating the force on the structure and the supplementary boundary data have to be stored. However, the storage requirements for these data are rather small compared to the requirements for the rest of the fluid data. So the price we pay for this method is reasonable compared

³ Since the velocity has two or three components in 2D and 3D, respectively, $\tilde{f}_j$ and $\tilde{F}_j$ are two- or three-dimensional vectors.

to the gain of getting force values with nearly no additional computation (all components in (9) are already calculated with our cell-oriented finite element approach) and that we achieve an accuracy of the order of the primary variables $\mathbf{u}_h$ and p_h (see [15]). For a more detailed description of the ansatz functions used and the algorithmic details see [32].

4.2 Transferring the Movement

In the previous subsection, we discussed the transfer of stresses from the fluid grid to the central surface description. Now, we have to define methods for the way back, namely the transfer of the velocity of the surface to the boundary nodes of our Cartesian fluid grid. In our CFD approach, we define all cells as obstacle (non-fluid) cells that are completely enclosed by the surface of the structure. All other cells are defined as fluid cells. Thus, the triangular surface intersects only fluid cells directly neighboured to obstacle cells as displayed in Fig. 12.

To transfer the velocity of the structure to the boundary nodes of the Cartesian grid (black squares in Fig. 12), we look at the planes (edges in 2D, resp.) defined by the triangles of the central mesh. For each boundary node representing the structure in the fluid grid we search for the nearest plane and perform the orthogonal projection of the boundary node to the

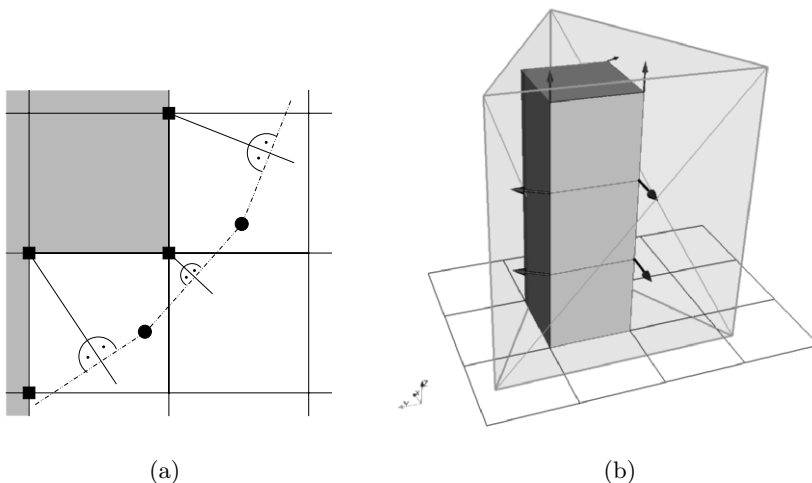


Fig. 12. Normal projection of the velocities at the coupling surface to the boundary nodes of our Cartesian fluid grid: (a) grey cells denote the obstacle, the dashed line the surface triangulation; (b) the light grey structure denotes a triangulated surface, the dark grey cubes obstacle cells of the Cartesian fluid grid, the projections from the vertices of the Cartesian grid cells to the surface triangulations are marked as arrows.

respective plane. Then, we use the linear interpolant of the velocity of the structure at the projection point as boundary value for the Cartesian grid node. This idea is illustrated in Fig. 12.

In general, this projection violates the conservation of mass. Hence, a correction of the boundary values which guarantees that the net flow over the Cartesian and the net flow over the central triangular representation of the coupling surface are equal, is needed in addition.

5 Numerical Experiments

In this section, we present numerical experiments as well as simulation results concerning the two main topics of this paper: a coupling environment including a central surface representation for the fluid-structure interface, as well as a Cartesian grid-based flow solver, especially designed for FSI applications and for being used in a partitioned approach. The collection of results is organized as follows. First, we illustrate single features for the CFD part, from the discretization up to the efficient implementation – some being already integrated into the flow solver and some being still in a stand-alone version. We conclude this part with benchmark computations comparing our results to those of a well-known CFD benchmark [42], which clearly shows that our approach is competitive from the fluid flow point of view. Next, we discuss important issues regarding the coupling interface and the geometry representation. Finally, first results with the set of benchmark configurations defined in the DFG Research Unit 493 on FSI [41] are presented.

5.1 Flow Simulations

Studies on an Efficient Implementation To show the potential of our approach in terms of hardware efficiency of the implementation, we present some results achieved for the two-dimensional Stokes equation on a unit square Ω with Driven Cavity boundary conditions:

$$\Delta \mathbf{u} - \nabla p = 0 \quad (11)$$

$$\nabla^T \mathbf{u} = 0 \quad (12)$$

$$\mathbf{u}(x, y) = (1, 0)^T \text{ for } y = 1, \quad (13)$$

$$\mathbf{u} = (0, 0) \text{ for } (x, y)^T \text{ in } \partial\Omega \setminus \{(x, y)^T : y = 1\}, \quad (14)$$

where $\partial\Omega$ denotes the boundary of a domain Ω . Figure 13 shows the exemplary adaptive grid we used for the test simulations.

Cache-Efficiency and Runtime Table 1 shows that we achieved a very high cache-hit-rate for the level 2 cache of more than 99% with our strictly linear data structures and the good time locality induced by the locality properties of the Peano curve. The runtimes per degree of freedom and iteration are a

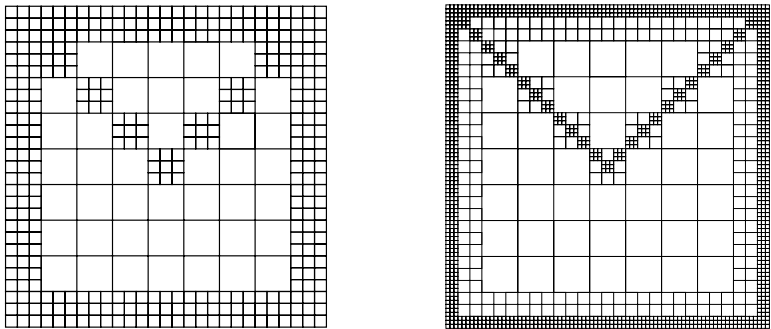


Fig. 13. Adaptive grid used for the Driven Cavity on the unit square at a resolution of 27×27 (left) and 81×81 (right).

factor of 5 to 10 slower than optimised solvers for the Poisson equation (for example DiME [25,28]), which is quite good if we take into account that we work on adaptive instead of regular grids and, in addition, solve the Stokes equation.

Table 1. Cache-performance and runtimes per degree of freedom and iteration of the Stokes solver for the Driven Cavity on the unit square computed on a Dual Xeon 2.4 GHz with 4 GByte main memory.

grid resol.	#dof	% of regular grid	L2-Hit-Rate	runtime/(dof*it)
27×27	298	40.05	99.74%	$4.04 \cdot 10^{-6}$ sec
81×81	1,140	15.95	99.42%	$3.83 \cdot 10^{-6}$ sec
243×243	3,800	5.78	99.20%	$3.83 \cdot 10^{-6}$ sec
729×729	13,840	2.30	99.12%	$3.63 \cdot 10^{-6}$ sec

Storage Requirements For the storage requirements, we will concentrate on the three-dimensional case as they become crucial only in this case. As described in [20,36], we have to store only two bits in addition to the values of the degrees of freedom per grid cell: the geometric information (obstacle or fluid) and the refinement information (refined or not). This results in an overall storage requirement of only about 5 Bytes per degree of freedom in the case of the Poisson equation [36]. In addition, the strictly linear data processing of our algorithm allows us to load only the currently needed data into a buffer and to store the rest on the hard disk. With this method, we solved a Poisson equation with 10^{10} degrees of freedom with only 470 MByte main memory (and 55 GByte on the hard disk) without a loss of efficiency in terms of runtime, see [36] and Table 2.

Table 2. Storage requirements in terms of main memory and hard disk space for our implementation loading only currently needed data via buffers from the hard disk to the main memory.

resolution	degrees of freedom	main memory	hard disk
243	14, 702, 584	69 MByte	83 MByte
729	400, 530, 936	157 MByte	2, 000 MByte
2187	10, 846, 541, 792	470 MByte	55, 000 MByte

Parallelisation We tested the performance of our parallel code on a cluster with *Myrinet*[®] (*Myricom*[®]) network connection (2Gbit/s bidirectional), consisting of 8 dual Pentium III processors with 2 GByte RAM each. As a test equation, we used the three-dimensional Poisson equation on the unit sphere. The results are shown in Table 3.

Table 3. Parallel speedup achieved for the solution of the three-dimensional Poisson equation on spherical domain on an adaptive grid with 23, 118, 848 degrees of freedom. The computations were performed on a myrinet cluster consisting of 8 dual Pentium III processors with 2 GByte RAM per node.

processes	1	2	4	8	16
speedup	1.00	1.95	3.73	6.85	12.93

Numerical Features Due to the still experimental character of our new cache-efficient implementation, we show results for the most sophisticated numerical methods for the example of the Poisson equation only:

$$\Delta u = 3\pi^2 \sin(\pi x_1) \sin(\pi x_2) \sin(\pi x_3) \text{ in }]0; 1[^3 \tag{15}$$

$$u = 0 \text{ on } \partial \left(]0; 1[^3 \right). \tag{16}$$

Dynamic Adaptivity and Multigrid PDEs can be solved most efficiently by a combination of dynamically adaptively refined grids and multigrid methods. We implemented several strategies and criteria to derive an optimal adaptive grid for the particular geometry and the particular accuracy requirements of the solution [10]. A very important issue for fluid-structure-interactions is the correct representation of the geometry. Figure 14 shows two grids successively generated in a refinement process detecting also steep tips of a structure, which is non-trivial for common methods.

To achieve an efficient overall solver, the dynamic adaptivity is combined with a full multigrid method. That is, we start with a coarse regular or a priori adaptively refined grid, apply one or a few multigrid cycles on this grid, perform a dynamic refinement step, interpolate onto the new grid, and perform a multigrid cycle again, etc. Table 4 shows the resulting numbers of

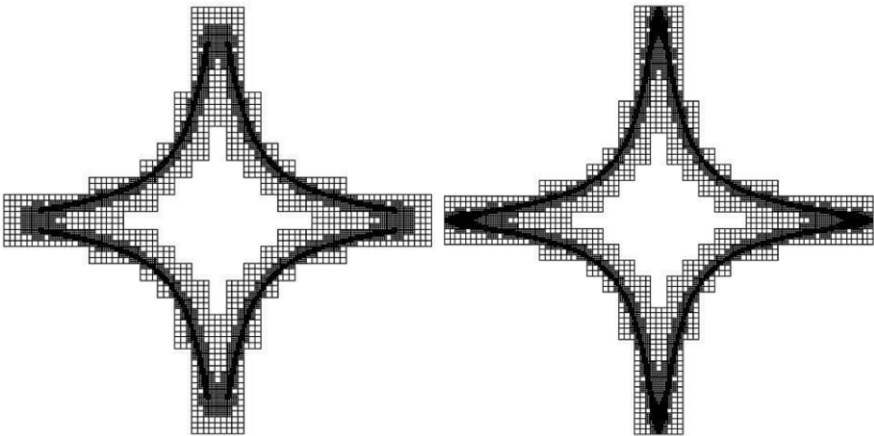


Fig. 14. Successive adaptive grids correctly detecting steep tips of a structure (from [10]).

iterations for equations (15) and (16) with the linear surplus as adaptivity criterion [10].

Table 4. Number of iterations needed to reduce the residual by a factor of 10^{-2} on each grid level within an F-cycle with dynamically adaptive grid refinement [10].

grid refinement	0th	1st	2nd	3rd
iterations	9	10	9	9
residual	$2.615 \cdot 10^{-3}$	$3.576 \cdot 10^{-4}$	$3.437 \cdot 10^{-5}$	$3.261 \cdot 10^{-6}$

High Order Discretisation We solve the three-dimensional Poisson equation (15) and (16) on an adaptive grid with a spherical region at the centre of the unit cube refined one level further than the rest of the domain (see Fig. 15). To achieve a fourth order discretisation, we apply the τ -extrapolation, which can be generalised to the Navier-Stokes equations [3]. Particularly, in the context of turbulent flow, a higher order discretisation is a very important feature if we want to achieve a reasonable accuracy with a realisable grid resolution. Figure 15 shows an almost perfect fourth order convergence for (15) and (16).

Adaptive FEM Solution of the 2D-Navier-Stokes Equations In order to show the general potential of our cell-oriented finite element algorithm to solve the Navier-Stokes Equations not only on regular but also on adaptively

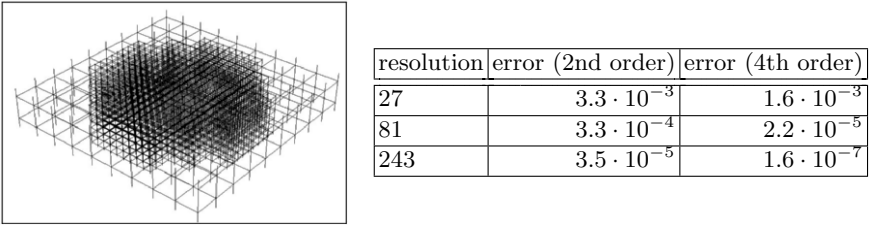


Fig. 15. Error in the L2-norm achieved with our second and fourth order discretisation for the Poisson equation (15) and (16).

refined grids, we present some numerical results for two two-dimensional scenarios.

Figure 16 shows the a priori adaptive grid for the driven cavity and the resulting velocities for laminar flow at $Re = 1$.

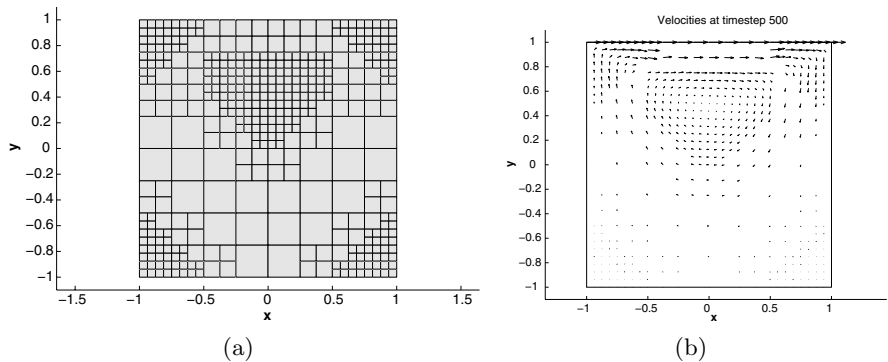


Fig. 16. Adaptive 2D driven cavity scenario for $Re = 1$: The grid (a) is based on a quadtree. In (b), the velocity results are shown for the calculation with our finite element approach.

The second scenario is the FSI-CFD1 benchmark described in detail in [41]. Simulations have been carried out on two rather coarse levels of a priori adaptive grids (see Figure 17). Level 1 has 2312 fluid cells leading to a total number of 4574 degrees of freedom (dof), whereas on level 2 there are 26297 dof on 9007 cells. The results for the drag and lift forces calculated with the method of consistent forces are shown in Table 5. For these coarse grids, the lift is in the right order of magnitude but shows oscillations around the reference values of [41], whereas the drag seems to converge slowly to the desired values.

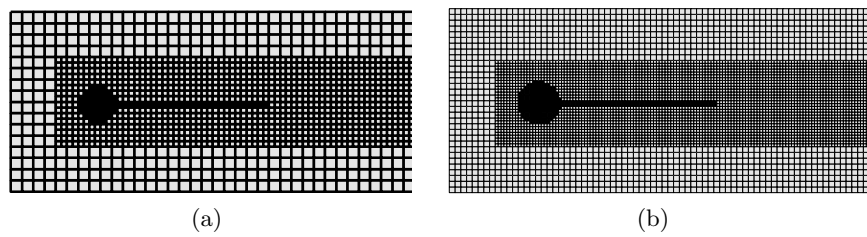


Fig. 17. FSI-CFD1 benchmark scenario for $Re = 20$: Two adaptive example grids with 2312 fluid cells (a) and 9007 cells (b), respectively. The second part of the channel has been cut off for the visualisation.

Table 5. Forces acting on the surface of the cylinder and on the beam for two different grid levels.

CFD1 ($Re = 20$)	drag [$\frac{kg}{s^2} m$]	lift [$\frac{kg}{s^2} m$]
level 1 (2312 cells)	14.48	0.89
level 2 (9007 cells)	14.39	1.53
Ref. Values [41]	14.29	1.119

Benchmark: Laminar Flow Around a Cylinder In this section, we demonstrate the correctness and applicability of both our finite volume and our finite element Navier-Stokes solvers with the help of the established benchmark cases *Laminar Flow around a Cylinder* [42]. In particular, we validated the computation of force coefficients (drag, lift, and pressure difference), which play a decisive role in the context of fluid-structure interactions.

Finite Volumes 2D For the two-dimensional finite volume solver, we perform a detailed analysis of all coefficients by means of the benchmark **2D-1** from [42] with a stationary flow at Reynolds number 20. Table 6 shows the resulting values for the drag and lift coefficients as well as for the pressure difference between a position before and a position behind the cylinder for different grid resolutions. For the computation of the forces we use a similar method based on consistent forces as described for the finite element method in Sect. 4.1. The convergence to the reference values from [42] is obvious.

Table 6. Force coefficients of laminar fluid flow around a cylinder and difference of the pressure before and behind the cylinder measured in the 2D scenario **2D-1** (Reynolds number = 20) introduced in [42]. All quantities are given for different resolutions of the grid.

2D-1 ($Re = 20$)		drag coefficient c_d	lift coefficient c_l	pressure [Pa]
number of cells along the diameter of the cylinder	10	4.74	0.0068	0.1017
	20	5.26	0.0134	0.1101
	40	5.44	0.0116	0.1136
	80	5.51	0.0111	0.1155
	160	5.55	0.0108	0.1164
	320	5.565	0.0107	0.1170
Ref. Values [42]		5.58 ± 0.01	0.0107 ± 0.0003	0.1174 ± 0.0002

In addition, the Figs. 18–20 show the development of the respective coefficients over the simulation time. Again, we can see the good convergence of the values to the reference values for growing grid resolutions. In the first time steps, we do not get correct results yet. This effect can be ascribed to the fact that our time-dependent simulation needs some time to “arrive” at the stationary solution for which the reference values hold.

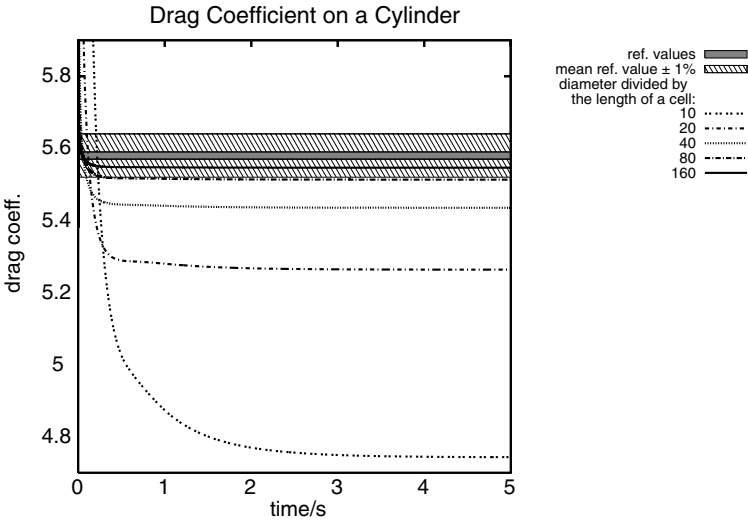


Fig. 18. Drag coefficient c_d of the 2D cylinder scenario **2D-1** (Reynolds number = 20) introduced in [42]. The coefficient is plotted over the simulation time for different resolutions of the grid.

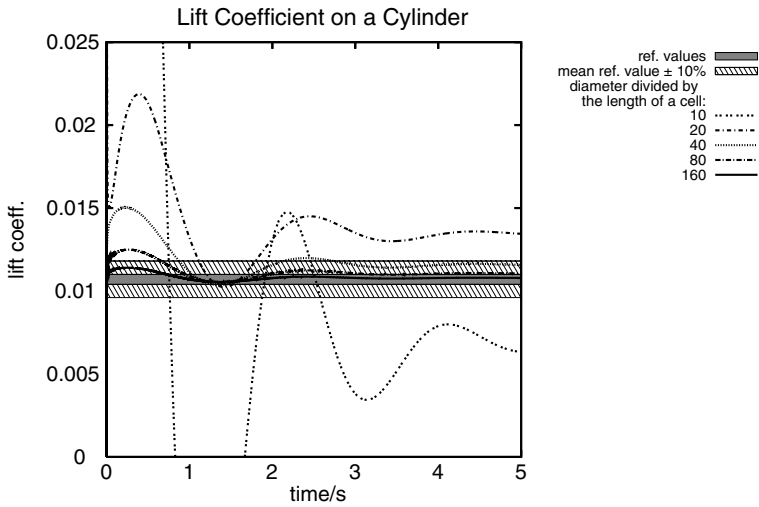


Fig. 19. Lift coefficient c_l of the 2D cylinder scenario **2D-1** (Reynolds number = 20) introduced in [42]. The coefficient is plotted over the simulation time for different resolutions of the grid.

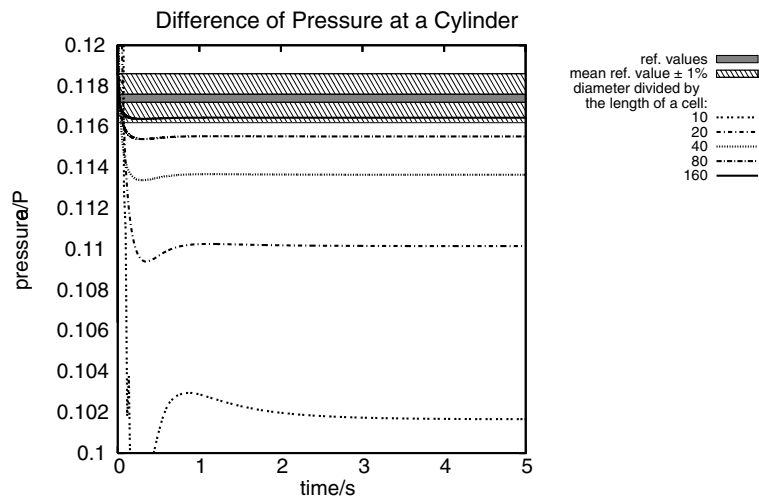


Fig. 20. Difference of the pressure in front of and behind the cylinder for the 2D scenario **2D-1** (Reynolds number = 20) introduced in [42]. The pressure difference is plotted over the simulation time for different resolutions of the grid.

Finite Volumes 3D In the three-dimensional case, we achieve similar results as for the three-dimensional benchmarks. We compute a benchmark case with stationary flow at Reynolds number 20, again (**3D-1Z** from [42]). Figure 21 shows the good convergence of the pressure difference also for this case.

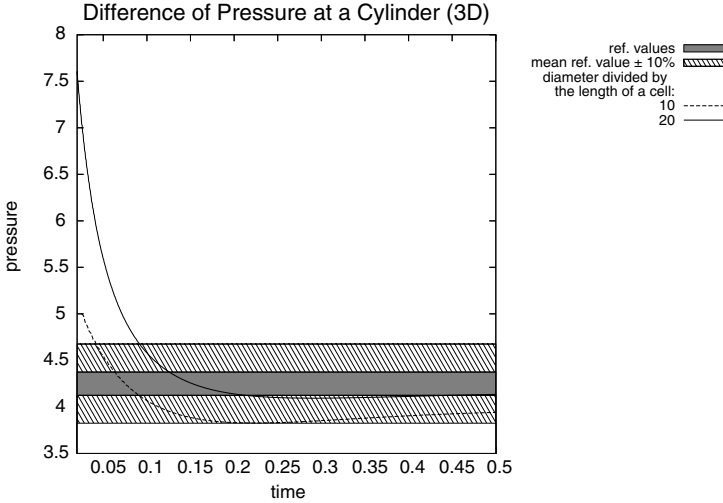


Fig. 21. Difference of the pressure in front of and behind the cylinder for the 3D scenario **3D-1Z** (Reynolds number = 20) introduced in [42]. The pressure difference is given in terms of the internal dimensionless unit and plotted over the simulation time for two resolutions of the grid.

Finite Elements 2D Our future work will mainly rely on our new finite element method based on cell-oriented operator evaluations on adaptively refined spacetree grids. Thus, we present the numerical results of the laminar flow around a cylinder obtained by our finite element solver in this section. The steady case **2D-1** at $Re = 20$ as well as the unsteady case **2D-2** at $Re = 100$ [42] resulting in the typical vortex-shedding are considered. The force values are calculated with the approach of the consistent forces described in Sect. 4.

Table 7 shows the values of the drag and lift coefficient for three different grid resolutions in the case **2D-1**. We can observe an even slightly better convergence to the reference values from [42] than achieved with the finite volume code. For the time dependent scenario **2D-2**, we consider as reference values the maximum drag coefficient $c_{d,max}$, the maximum lift coefficient $c_{l,max}$ and the Strouhal number St presented in [42]. The results of our simulations for two grids are shown in Table 8. The results of the second grid with 20 cells along the diameter of the cylinder fit well to the benchmark values. Figure 22 shows the oscillatory development of the drag coefficient c_d and the lift coefficient c_l in time for the second grid with 20 cells along the diameter of the cylinder. As the Strouhal number already indicates, the frequency of the oscillations is equal to that of the benchmark results.

Table 7. Drag and lift coefficients of laminar fluid flow around a cylinder in the 2D scenario **2D-1** ($Re = 20$) introduced in [42].

2D-1 ($Re = 20$)		drag coefficient c_d	lift coefficient c_l
number of cells along the diameter of the cylinder	10	5.39	0.0080
	20	5.42	0.0105
	40	5.48	0.0105
Ref. Values [42]		5.58 ± 0.01	0.0107 ± 0.0003

Table 8. Drag and lift coefficients of laminar fluid flow around a cylinder in the 2D scenario **2D-2** ($Re = 100$) introduced in [42].

2D-2 ($Re = 100$)		drag coeff. $c_{d,max}$	lift coeff. $c_{l,max}$	Strouhal number
number of cells along the diameter of the cylinder	10	3.58	1.58	0.312
	20	3.16	0.98	0.303
Ref. Values [42]		3.23 ± 0.01	1.00 ± 0.01	0.300 ± 0.005

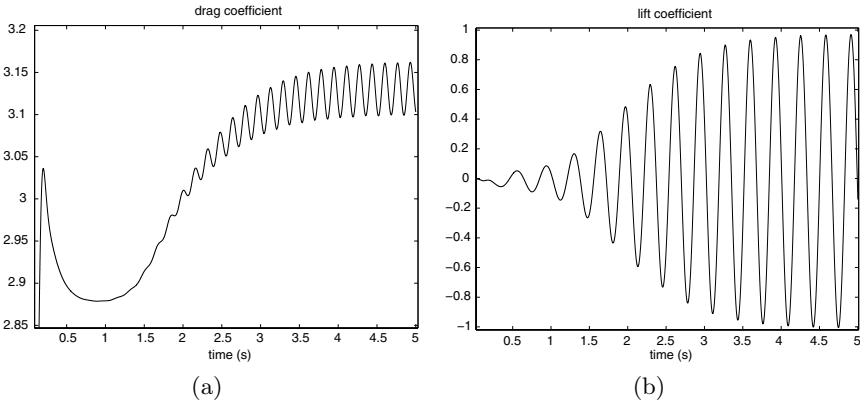


Fig. 22. Drag coefficient (a) and lift coefficient (b) in the **2D-2** benchmark ($Re = 100$) with 20 cells along the diameter of the cylinder.

5.2 Experiments with Complex Geometries

Example: Spacetree Model As described in Sect. 2.3, we embed the central surface mesh of our coupling environment into a spatial description using a spacetree to link surface and spatial data in our coupled simulations. Figure 23 shows the two corresponding representations of a car: the surface triangulation describing and transporting coupling data on the one hand and the octree representation supplying the spatial context of the object and, thus, enabling us to establish correct connections between surface data from the coupling interface and spatial data from the structural and the fluid solver, on the other hand.

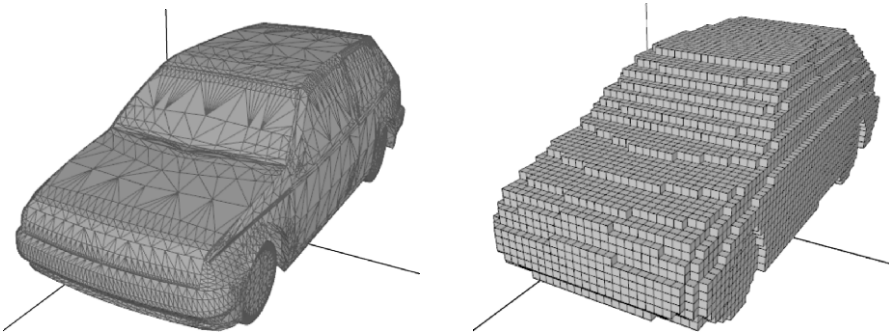


Fig. 23. Different boundary representations of a car: the triangulated surface model – created by the 3D modelling tool *Maya*[®] (Alias|Wavefront) – on the left-hand and the corresponding spacetree model on the right-hand side.

Example: Large Movement To demonstrate the ability of our spacetree concept to handle large movements of objects, we present an example focusing on the fluid simulation and using a virtual structure side that provides a triangulated sphere moving with a constant velocity through the fluid domain. Figure 24 shows a snapshot of this configuration. At the beginning of the simulation, all velocities in the fluid are zero. The moving sphere pushes the fluid and induces a flow field. The current state of the fluid flow around the sphere is also given in Fig. 24. This test shows two properties of our fluid

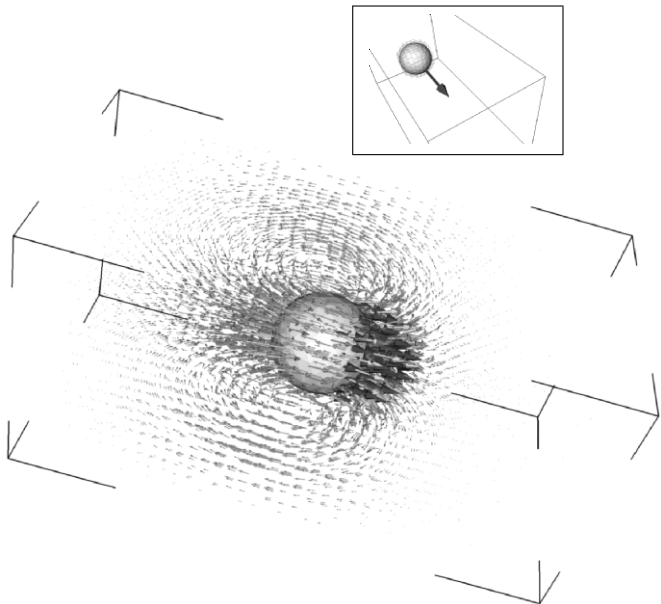


Fig. 24. A spherical structure moving with constant velocity through a fluid domain and inducing a flow field.

simulation application that are crucial for the simulation of FSI. First, we can alter our computational domain corresponding to the boundary representation of the structure. Second, we can handle arbitrary displacements of the structure.

5.3 Benchmark: FSI (CFD1–CFD3)

The first part of the FSI Benchmark includes non-coupled two-dimensional scenarios for the single solvers. We used the first part in order to verify our Cartesian discretisation and to prepare the planned FSI benchmark simulations. All computations were performed with our finite volume solver, again combined with a computation of forces at the surface of the structure based on the method of consistent forces. The pure CFD benchmark covers three scenarios with Reynolds numbers 20, 100, and 200 (**CFD1–CFD3** from [41]). In all scenarios, the structure surrounded by the fluid is given as a circle with a beam attached downstream. The scenarios **CFD1** and **CFD2** have a stationary solution. Our results for the drag force and the lift force were computed after a simulation time of five seconds for different resolutions of the grid (see Table 9 and Table 10). The scenario CFD3 has a transient solution. Our results for the drag force and the lift force were taken after three seconds for a period of 17 seconds. The simulated quantities were given for different resolutions of the grid in Table 11. In Fig. 25 the forces are plotted over the simulation time, the corresponding Fourier-spectra are shown in Fig. 26. As for the laminar flow around a cylinder, we observe convergence to the correct values for both drag and lift force in all three cases.

Table 9. Forces acting on the surface of a cylinder and the beam for different grid spacing.

CFD1 ($Re = 20$)		drag [$\frac{kg \cdot m}{s^2}$]	lift [$\frac{kg \cdot m}{s^2}$]
diameter of the cylinder divided by the length of a cell	20	14.67	1.160
	40	14.49	1.140
	80	14.38	1.128
Ref. Values [41]		14.29	1.119

Table 10. Forces acting on the surface of a cylinder and the beam for different grid spacing.

CFD2 ($Re = 100$)		drag [$\frac{kg \cdot m}{s^2}$]	lift [$\frac{kg \cdot m}{s^2}$]
diameter of the cylinder divided by the length of a cell	20	139.8	10.97
	40	138.2	10.67
	80	137.5	10.59
Ref. Values [41]		136.7	10.53

Table 11. Forces acting on the surface of a cylinder and the beam for different grid spacing.

CFD3 ($Re = 200$)		drag [$\frac{kg \cdot m}{s^2}$]	lift [$\frac{kg \cdot m}{s^2}$]
diameter of the cylinder	20	$466 \pm 10.5[4.35]$	$35 \pm 570[4.35]$
divided by the length of	40	$445 \pm 6.6[4.40]$	$-5.2 \pm 463[4.40]$
a cell	80	$442 \pm 5.8[4.45]$	$-11.3 \pm 447.5[4.45]$
Ref. Values [41]		$439.5 \pm 5.62[4.40]$	$-11.89 \pm 437.8[4.40]$

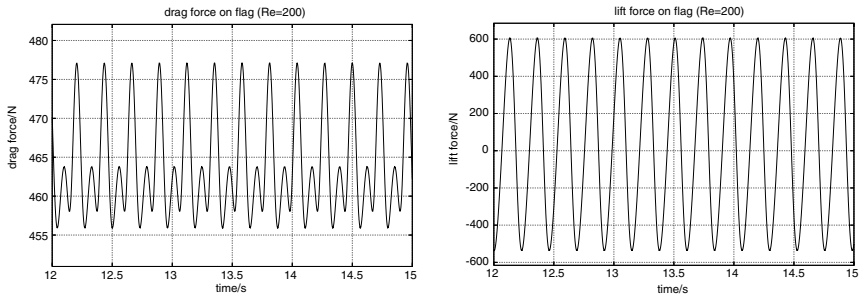


Fig. 25. Drag force (left) and lift force (right) on the surface of the cylinder and the beam over the simulation time.

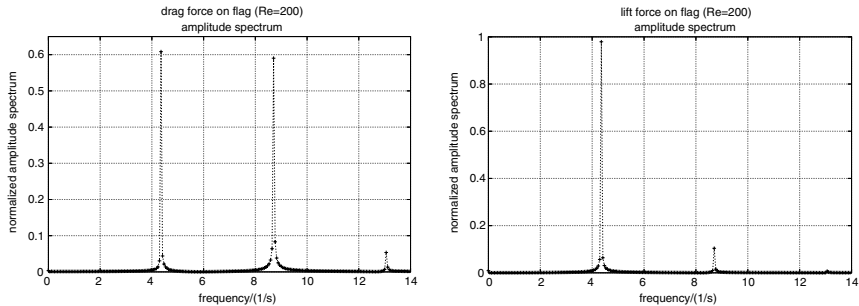


Fig. 26. Normalised amplitude spectrum of drag force (left) and lift force (right).

5.4 Evaluation

We have presented the results of numerous enhancements of the Cartesian fluid solver, the development of a client-server based coupling environment, and several extensions of our spacetime concept. With these three building blocks, we have established a good basis for the efficient simulation of FSI on Cartesian grids. As we could also show the general applicability of Cartesian discretisations in this context, it is to be assumed that first coupling applications can be simulated in the very near future.

6 Conclusions and Future Work

In this paper, we have presented our concepts for two main components of a partitioned approach for simulating fluid-structure interactions: A client-server based coupling environment on the one hand and a Cartesian fluid solver on the other hand. The integrating moment of the whole work is given by the spacetree concept, which is used as a tool to link surface and spatial data in the coupling interface and, second, as a structuring concept of the adaptively refined Cartesian grids used in the fluid solver. Both the coupling environment and the fluid solver were optimised in terms of robustness and numerical and hardware efficiency, and both have been tested for several test examples.

In the future, we will further enhance both components with the task of providing ‘ready to use’ modules for easy and efficient three-dimensional coupled simulations. For the coupling environment, the focus is on the implementation of further coupling strategies, the link-up to commercial codes, and the validation of our approach for ‘real-world’ scenarios. For the fluid solver, we will concentrate on the implementation of the new finite-element discretisation for the three-dimensional case based on an efficient adaptive parallel multigrid solver optimised in particular for this purpose. Altogether, the focus will be on doing FSI simulations using the p-FEM code [39], which has already been the structural component during the design of the coupling environment.

7 Acknowledgement

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Lattice-Boltzmann Method on Quadtree-Type Grids for Fluid-Structure Interaction

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Abstract. In this work a Lattice Boltzmann (LB) fluid flow solver based on unstructured quadtree/octree type Eulerian grids is coupled with a spectral Finite Element (p-FEM) structural mechanics solver based on a Lagrangian description to predict bidirectional fluid-structure interaction (FSI). The methods and algorithms are described in detail. Benchmark computations of a coupled transient problem of a 2D beam-like structure in a channel as defined by the DFG-Research Unit 493 are presented.

1 Introduction

In computational fluid dynamics grids can usually be classified as either body-conformed or Cartesian (embedded approach) [21]. The most common approach for the body-conformed description in the FSI simulation is the arbitrary Lagrangian Eulerian (ALE) formulation. In this case, an ALE-based FV/FE discretization for the Navier-Stokes equations is combined with a Lagrangian FE-solver for the structure. There are partitioned as well as monolithic approaches [29]. The advantage of the ALE method is the handling of the interface and the prevention of the activation/deactivation of fluid nodes due to the moving structure mesh. The most important disadvantage of this approach is the remeshing procedure if large displacements occur.

The Lattice Boltzmann approach [27] has to be categorized as an embedded approach, as it is based on fixed Cartesian grids. Local refinement [6,9,33] can be achieved e.g. by using quad-/octree type grids. The extra computational cost for handling moving geometries over the fixed grid is due to the computation of link distances to the moving boundaries and restricted to a region close to these boundaries.

This paper is organized as follows: In Section 2 the Lattice Boltzmann method and its extensions for FSI are discussed. In Section 2.2 an introduction into the Lattice Boltzmann method using the multiple relaxation time (MRT) model is given, in Section 2.4 a detailed discussion of the boundary condition and the implementation for moving walls is given, in Section 2.5 the procedure for the activation/deactivation of fluid nodes and in Section 2.6 the evaluation of forces on solid boundaries is described. Section 3 briefly comments on the efficiency of LB methods in terms of computational efficiency. After a short description of the solvers used for the Computational Structural Mechanics (CSD) problems in Section 4, the coupling algorithm for

the Fluid-Structure interaction is presented in Section 5. Finally preliminary results for the FSI benchmark are given in Section 6.

2 Lattice Boltzmann Method

2.1 Introduction

The Boltzmann equation is the basic equation of kinetic gas theory describing the dynamics of a particle distribution function $f(t, \mathbf{x}, \boldsymbol{\xi})$ in time t and phase space $\mathbf{x}, \boldsymbol{\xi}$, where $\mathbf{x}$ is the position and $\boldsymbol{\xi}$ the microscopic velocity:

$$\partial_t f(t, \mathbf{x}, \boldsymbol{\xi}) + (\boldsymbol{\xi} \cdot \nabla_{\mathbf{x}}) f(t, \mathbf{x}, \boldsymbol{\xi}) = \Omega(f, f') \quad (1)$$

The left hand side of Eq. (1) describes the advection of the distributions with velocity $\boldsymbol{\xi}$ and the collision term on the right hand side the interaction of the particle distributions. If one is not interested in the details of the collision operator, it can be approximated by a simplified model, e.g. the Bhatnagar - Gross and Krook model (BGK) [1]:

$$\Omega = -\frac{1}{\tau} (f(t, \mathbf{x}, \boldsymbol{\xi}) - f^{eq}(t, \mathbf{x}, \boldsymbol{\xi})) \quad (2)$$

Here $f^{eq}(t, \mathbf{x}, \boldsymbol{\xi})$ is a Maxwellian equilibrium distribution and τ a relaxation time. This collision term drives the distributions to a local equilibrium state. If one is interested in continuum flows at low Mach numbers, the velocity space can be discretized by a small set of $b+1$ fixed velocities $\{\mathbf{e}_i, i = 0, \dots, b\}$. Here we use the $d2q9$ -model [27] with 9 velocities ($b = 8$) for two-dimensional problems:

$$\{\mathbf{e}_i, i = 0, \dots, b\} = c \cdot \left\{ \begin{array}{ccccccccc} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{array} \right\}. \quad (3)$$

After projecting the collision operator into this space, we obtain the discrete velocity model (DVM) [14], consisting of $b + 1$ coupled equations:

$$\partial_t f_i(t, \mathbf{x}) + (\mathbf{e}_i \cdot \nabla_{\mathbf{x}}) f_i(t, \mathbf{x}) = \Omega_i, \quad i = 0, \dots, b, \quad (4)$$

where $\mathbf{f} = (f_i(\mathbf{x}, t), i = 0 \dots, b)$ are distribution functions (unit: $\frac{kg}{m^3}$) propagating with speed $\mathbf{e}_i$. The equilibrium distribution functions f_i^{eq} are usually chosen as quadratic polynomials of the first two hydrodynamic moments [27], i.e. the density fluctuation ρ and the momentum $\rho_0 \mathbf{u}$,

$$\rho = \sum_{i=0}^b f_i \quad \rho_0 \mathbf{u} = \sum_{i=0}^b \mathbf{e}_i f_i. \quad (5)$$

The equilibrium distribution tuned for incompressible flows [13] are

$$f_i^{eq} = w_i \left(\rho + \rho_0 \left(3 \frac{\mathbf{e}_i \cdot \mathbf{u}}{c^2} + \frac{9}{2} \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{c^4} - \frac{3}{2} \frac{u^2}{c^2} \right) \right), \quad (6)$$

where ρ_0 is a constant reference density and w_i are weighting coefficients which can be conveniently represented in vector form as

$$\mathbf{w} = \left\{ \frac{4}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36} \right\}. \quad (7)$$

The constant velocity c is usually related to the speed of sound by $c_s^2 = c^2/3$.

Discretizing Eq. (4) with an upwind finite difference approach coupled in space and time by setting $\Delta x = c\Delta t$, the evolution equation called Lattice Boltzmann equation [14]

$$f_i(t + \Delta t, \mathbf{x} + \mathbf{e}_i \Delta t) - f_i(t, \mathbf{x}) = \Omega_i, \quad i = 0, \dots, b \quad (8)$$

on a space-filling lattice with a nodal distance $\Delta x = c\Delta t$ results. The collision operator is now

$$\Omega_i = -\omega (f_i - f_i^{eq}), \quad (9)$$

where $\omega = \frac{\Delta t}{\tau}$. Using a Chapman-Enskog analysis [4,10] it can be shown that the moments (5) of the solution of Eq. (8) are solutions up to errors $\mathcal{O}(\Delta x^2), \mathcal{O}(Ma^2)$ of the incompressible Navier-Stokes equations. The collision rate ω is related to the kinematic viscosity ν by

$$\nu = c^2 \Delta t \left(\frac{1}{3\omega} - \frac{1}{6} \right). \quad (10)$$

A typical scheme for implementation is to split equation (8) into a local collision step,

$$\tilde{f}_i(t + \Delta t, \mathbf{x}) = f_i(t, \mathbf{x}) + \Omega_i(t, \mathbf{x}), \quad (11)$$

where $\tilde{f}_i(t + \Delta t, \mathbf{x})$ are the so-called post-collision values and a propagation step, where the distributions are moved (propagated) to their neighboring nodes.

2.2 Multiple Relaxation Time (MRT) Model

In the MRT model the distributions are transformed to the moment space by a transformation matrix $\mathbf{M}$ consisting of orthogonal eigenvectors [15,16,19]

$$\mathbf{m} = \mathbf{M} \mathbf{f} \quad \mathbf{f} = \mathbf{M}^{-1} \mathbf{m} \quad (12)$$

The moments $\mathbf{m}$ are labeled as

$$\mathbf{m} = (\rho, e, \epsilon, \rho_0 u_x, q_x, \rho_0 u_y, q_y, p_{xx}, p_{xy}). \quad (13)$$

The second order moments e, p_{xx}, p_{xy} are related to the stress tensor $\sigma_{\alpha\beta}$ by

$$\sigma_{11} = - \left(1 - \frac{\omega}{2} \right) \left(\frac{1}{3} \rho c^2 + \frac{1}{2} p_{xx} + \frac{1}{6} e - \rho_0 u_x^2 \right) \quad (14)$$

$$\sigma_{22} = - \left(1 - \frac{\omega}{2} \right) \left(\frac{1}{3} \rho c^2 - \frac{1}{2} p_{xx} + \frac{1}{6} e - \rho_0 u_y^2 \right) \quad (15)$$

$$\sigma_{12} = - \left(1 - \frac{\omega}{2} \right) (p_{xy} - \rho_0 u_x u_y), \quad (16)$$

where ω is a relaxation rate. The other quantities like higher order dissipation ϵ and heat fluxes q_x, q_y are irrelevant for the simulation of incompressible flows. The transformation matrix is given with

$$\mathbf{M} = \begin{bmatrix} 1 \cdot & (1 & 1 & 1 & 1 & 1 & 1 & 1 & 1) \\ c^2 \cdot & (-4 & -1 & -1 & -1 & -1 & 2 & 2 & 2) \\ c^4 \cdot & (4 & -2 & -2 & -2 & -2 & 1 & 1 & 1) \\ c \cdot & (0 & 1 & 0 & -1 & 0 & 1 & -1 & -1) \\ c^3 \cdot & (0 & -2 & 0 & 2 & 0 & 1 & -1 & -1) \\ c \cdot & (0 & 0 & 1 & 0 & -1 & 1 & 1 & -1) \\ c^3 \cdot & (0 & 0 & -2 & 0 & 2 & 1 & 1 & -1) \\ c^2 \cdot & (0 & 1 & -1 & 1 & -1 & 0 & 0 & 0) \\ c^2 \cdot & (0 & 0 & 0 & 0 & 0 & 1 & -1 & 1) \end{bmatrix} \quad (17)$$

In the MRT model the collision operator acts in moment space and is given by

$$\boldsymbol{\Omega} = -\mathbf{M}^{-1} \mathbf{S} [(\mathbf{M} \mathbf{f}) - \mathbf{m}^{eq}], \quad (18)$$

where $\mathbf{m}^{eq}$ are equilibrium moments defined as

$$\begin{aligned} m_0^{eq} &= \rho \\ m_1^{eq} &= e^{eq} = -2c^2 \rho + 3\rho_0 u^2 \\ m_2^{eq} &= \epsilon^{eq} = c^4 \rho - 3c^2 \rho_0 u^2 \\ m_3^{eq} &= \rho_0 u_x \\ m_4^{eq} &= q_x^{eq} = -c^2 \rho_0 u_x \\ m_5^{eq} &= \rho_0 u_y \\ m_6^{eq} &= q_y^{eq} = -c^2 \rho_0 u_y \\ m_7^{eq} &= p_{xx}^{eq} = \rho_0 (u_x^2 - u_y^2) \\ m_8^{eq} &= p_{xy}^{eq} = \rho_0 u_x u_y. \end{aligned} \quad (19)$$

$\mathbf{S}$ is a diagonal matrix composed of relaxation rates s_i . The collision rates s_0, s_3, s_5 are not relevant, since they are related to the conserved moments. Rates s_7 and s_8 have to be equal $s_7 = s_8 = \omega$ to obtain a consistent dynamic viscosity, see equation (10).

The other relaxation rates have no physical meaning for incompressible flows and can be freely chosen in the range $[0, 2]$ and can be utilized to improve stability [19]. The optimal values depend on the specific system under consideration (geometry, initial and boundary conditions) and can not be computed in advance. Some reasonable values for these parameters are given in [16,19]. Here we choose $s_1 = s_2 = s_4 = s_6 = 1$ implying an instantaneous relaxation of the corresponding moments to their equilibrium values.

Using Chapman-Enskog expansion techniques [4,10] or asymptotic analysis [17,18], it can be shown that the order of convergence with respect to the corresponding incompressible Navier-Stokes solution is first order in time and second order in space.

Using the MRT model instead of BGK improves efficiency substantially, the maximum cell Reynolds number is up to two orders of magnitude larger while the scheme still has no numerical viscosity.

2.3 Grid Refinement

The use of locally refined meshes for the direct discretization of the Navier-Stokes equations is mandatory for an efficient solution of CFD problems. The LB method requires a square in 2 dimensions and a cube in 3 dimensions as a basis cell. Therefore we use recursively refined quadtree/octree-type grids in 2D/3D [6]. Due to the coupling of space and time discretization $\Delta x = c\Delta t$, this leads to a nested time stepping scheme explained below, i.e. for each grid level l we have $\Delta t_l = c\Delta x_l$. If the speed of sound and the physical viscosity are assumed to be equal on all grid levels (Mach and Reynolds number equal on all grids), then two time steps on the finer grid have to be done during one time step on the coarser mesh [9]. An approach where also the Mach number is scaled down simultaneously on finer grid levels to ensure a faster convergence to the *incompressible* Navier-Stokes equations can be found in [28]. This approach requires four time steps on the fine grid during one time step on the coarse mesh.

Theory The analysis of the LB equations on different grid levels reveals, that it is not sufficient to glue two Cartesian grids of different resolution together by interpolating the distributions during the propagation from the coarse to the fine grid. It was first shown in [9], that in order to obtain smooth transitions for pressure, velocity and stresses, one has to rescale the distributions in an appropriate manner. The derivation of the rescaling procedure is presented here following [33,34], since we use a slightly different procedure due to improvements in the grid coupling procedure sketched later in the subsection *Implementation*. We present the derivation using the BGK method for simplicity.

Across the grid interface the density and velocity must be equal on the fine (f) and coarse (c) grid, resulting in

$$f^{c,eq} = f^{f,eq}, \quad (20)$$

where the index i is omitted for the sake of clarity. The non-equilibrium distributions $f^{neq} = f - f^{eq}$ are scaled by equating the deviatoric stress tensor formula [6,7,9,33,34]:

$$\left(1 - \frac{\Delta t^c}{2\tau^c}\right) \cdot f^{c,neq} = \left(1 - \frac{\Delta t^f}{2\tau^f}\right) \cdot f^{f,neq} \quad (21)$$

Introducing the scale factor $sF^{f \rightarrow c}$ and rearranging (21) leads to

$$f^{c,neq} = sF^{f \rightarrow c} \cdot f^{f,neq} = sF^{f \rightarrow c} \cdot (f^f - f^{f,eq}). \quad (22)$$

Substituting (10) into (21) gives:

$$sF^{f \rightarrow c} = \frac{6\nu + \Delta t^c}{6\nu + \Delta t^f}. \quad (23)$$

Substituting equation (22) into $f^{neq} = f - f^{eq}$ leads to:

$$f^c = f^{f,eq} + sF^{f \rightarrow c} \cdot (f^f - f^{f,eq}). \quad (24)$$

The post-collision state $\tilde{f}$ is given by

$$\tilde{f}^c = f^c - \frac{\Delta t^c}{\tau^c} (f^c - f^{c,eq}). \quad (25)$$

Substituting (24) into (25)

$$\tilde{f}^c = f^{f,eq} + sF^{f \rightarrow c} \cdot (f^f - f^{f,eq}) - \frac{\Delta t^c}{\tau^c} \cdot sF^{f \rightarrow c} \cdot (f^f - f^{f,eq}), \quad (26)$$

results in

$$\tilde{f}^c = f^{f,eq} + \left(1 - \frac{\Delta t^c}{\tau^c}\right) \cdot sF^{f \rightarrow c} \cdot (f^f - f^{f,eq}). \quad (27)$$

Equation (27) computes the collision step on the fine node and scales the distributions to a coarse domain post collision state. This approach avoids the singularity at viscosity $\nu = \frac{1}{6} \frac{m^2}{s}$ as described in [9]. For the scaling from coarse to fine grid the scale factor is $sF^{c \rightarrow f} = \frac{1}{sF^{f \rightarrow c}}$.

Using the MRT model for grid refinement gives more freedom, since only the non-equilibrium moments $p_{xx}^{neq}, p_{xy}^{neq}$ have to be rescaled in such a manner that the deviatoric stresses are continuous across the interface. The relaxation rates for s_7, s_8 are fixed by Eq. (10) on all grid levels, but the other relaxation rates can be chosen arbitrarily in the range $[0, 2]$ on all grid levels. A good choice is to use $s_1 = s_2 = s_4 = s_6 = 1$ on all grid levels. The relation for rescaling the moments on different grid levels after the propagation step is

$$m_c^{neq} = \frac{s_f}{s_c} \frac{\Delta t_c}{\Delta t_f} m_f^{neq} \quad (28)$$

This rescaling is required to obtain smooth transitions of the moments across the grid levels.

Implementation In contrast to the standard approach with an overlapping zone on the interface [6,33] we used a grid layout with no overlap and without hanging nodes as shown in Fig. 1. The nodes are stored in hash-maps and have additional attributes like position and different flags (boundary, scaleCoarseToFine, scaleFineToCoarse) indicating if distributions have to be scaled or if a boundary condition has to be set. Via simple bit casts it is possible to decide where the distributions should propagate. If a node has

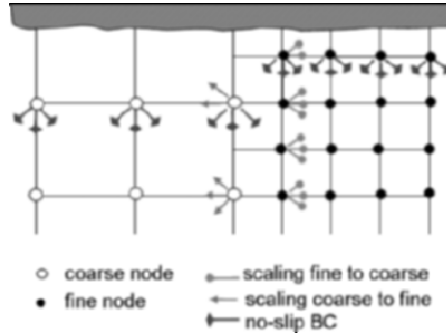


Fig. 1. Grid layout and attributes of the nodes

no neighbor it is either a boundary node or a node which receives a scaled coarse to fine distribution. For such a scaling node the distributions have to be interpolated in space with at least quadratic accuracy. We use cubic interpolations in space to improve symmetry and linear interpolations in time. The evolution in time is based on a nested time stepping scheme. This implies that for a fine level node one has to compute 2^n fine timesteps for each coarse node timestep, where n is the difference from the fine and coarse level index.

2.4 Boundary Conditions

Macroscopic flow quantities can only be set implicitly via the particle distribution functions. A well known and simple way to introduce no-slip walls is the so-called bounce back scheme, which allows spatial second order accuracy if the boundary is aligned with one of the lattice vectors $\hat{\mathbf{e}}_i = \mathbf{e}_i/c$ and first order otherwise. As we have arbitrarily shaped moving objects, we use the modified bounce back scheme developed in [2,20] for velocity boundary conditions, which is second order accurate for arbitrarily shaped boundaries. In Fig. 2 we identify two cases along a link i :

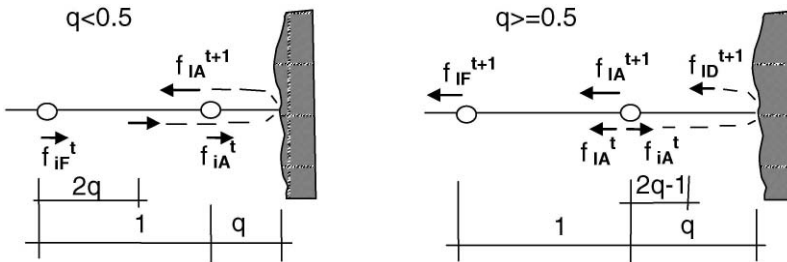


Fig. 2. Interpolations for second order bounce back scheme

- (a) wall-node distance $q_i < 0.5 |\mathbf{e}_i \Delta t|$ and
- (b) wall-node distance $q_i \geq 0.5 |\mathbf{e}_i \Delta t|$.

The modified bounce back scheme is

$$f_{IA}^{t+1} = (1 - 2q) \cdot f_{iF}^t + 2q \cdot f_{iA}^t - 2\rho w_i \frac{e_i u_w}{c_s^2}, \quad 0.0 < q < 0.5 \quad (29)$$

$$f_{IA}^{t+1} = \frac{2q-1}{2q} \cdot f_{iA}^t + \frac{1}{2q} \cdot f_{iA}^t - \rho w_i \frac{e_i u_w}{qc_s^2}, \quad 0.5 \leq q \leq 1.0, \quad (30)$$

With this scheme second order accurate results in space even for curved geometries are obtained [11]. For a detailed discussion of LBE boundary conditions we refer to [12]. In contrast to the simple bounce-back scheme the use of these interpolation based no-slip boundary conditions results in a notable mass loss across the no-slip lines. Yet, the results obtained with bounce-back were inferior which highlights the importance of a proper geometric resolution of the flow domain. Pressure boundary conditions are implemented by setting the incoming distributions to [30]

$$f_I(t + \Delta t, \mathbf{x}) = -f_i(t, \mathbf{x}) + f_I^{eq}(p_0, \mathbf{u}(t_B, \mathbf{x}_B)) + f_i^{eq}(p_0, \mathbf{u}(t_B, \mathbf{x}_B)) \quad (31)$$

where p_0 is the given pressure, $t_B = t + \frac{1}{2}\Delta t$, $\mathbf{x}_B = \mathbf{x} + \frac{1}{2}\hat{\mathbf{e}}_i$, (f_I, f_i) are an anti-parallel pair of distributions with velocities $\mathbf{e}_i = -\mathbf{e}_I$, and f_I is the incoming and f_i the outgoing distribution function value. $\mathbf{u}$ is obtained by extrapolation.

2.5 Activation/Deactivation of New Fluid Nodes

If a new fluid node is created due to the moving structure, we use linear inter- or extrapolation (depending on the geometrical configuration) to compute the local velocity at that node. A local Poisson type iteration described in [23] is used for this node to compute the pressure and the higher order moments.

2.6 Force Evaluation

There are two possibilities to evaluate forces on boundaries using the LB method. The first one is the momentum-exchange based method and the second one is the pressure/stress integration based method. A detailed description and comparison of both methods can be found in [25]. The boundary of the structure is given by the interface mesh consisting of flat triangles in 3D or line elements in 2D as discussed in Section 5.

Momentum-exchange The force acting along a link i on a boundary (see Fig. 3) results from the momentum-exchange between the particle distributions hitting the moving object and the moving object itself [26]:

$$\Delta \mathbf{F} = \frac{\Delta x^3}{\Delta t} (\mathbf{e}_i - \mathbf{u}_B) (f_i^t + f_I^{t+1}). \quad (32)$$

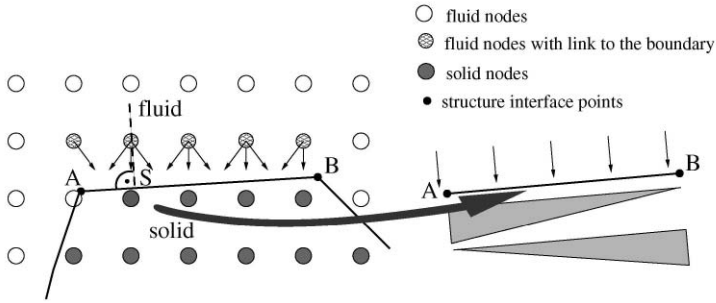


Fig. 3. Interpolation scheme to compute forces acting on the structure

Here $\mathbf{u}_B$ is the velocity of the moving boundary and $\mathbf{e}_I = -\mathbf{e}_i$. Figure 3 shows the links of the LB grid which contribute to the force on the structure. The forces $\Delta \mathbf{F}$ acting on the boundary between interface points A and B are weighted by a linear interpolation scheme and summed up to obtain the forces acting on A and B. The weight for the force integration for point A is the normalized distance between the intersection point S and the structure interface point A. All fluid nodes with a link cut by the boundary contribute to the resulting force. Note that this method is conserving the total momentum transferred to the structure.

Stress Integration The momentum exchange works well for integral forces on large structural elements, but it is not feasible for the calculation of forces when the extension of the boundary element is in the order of the fluid mesh resolution. Incorrect local forces are introduced if boundary elements have too few or even no matching links. For this case the stress integration method is preferable. One advantage compared to conventional CFD solvers is the locally available stress tensor in the Lattice Boltzmann method. The stress tensor $S_{\alpha\beta}$, including the scalar pressure, can be computed from

$$S_{\alpha\beta} = -c_s^2 \rho \delta_{\alpha\beta} + \sigma_{\alpha\beta}, \quad (33)$$

where $\sigma_{\alpha\beta}$ is given by equation (14–16) and δ is the Kronecker-Delta. The stresses at the nodes of the boundary are computed by extrapolation. In Fig. 4 the 16 different cases for the intersection of a structure boundary including a node and a fluid cell are shown. The different cases correspond to the Marching Squares (2D) and Marching Cubes (3D) algorithms [22], generating isolines and isosurfaces on fixed Cartesian grids. The index in Fig. 4 is the sum of the bits of the fluid nodes. As the extension of the structure is usually larger than the lattice distance of two neighboring LB-nodes, we exclude cases 10 and 5. For case 15 the stress interpolation is bilinear. For the rare case 0 we impose the stresses of the last timestep. In all other cases we extrapolate the stresses to the imaginary nodes in the solid domain before

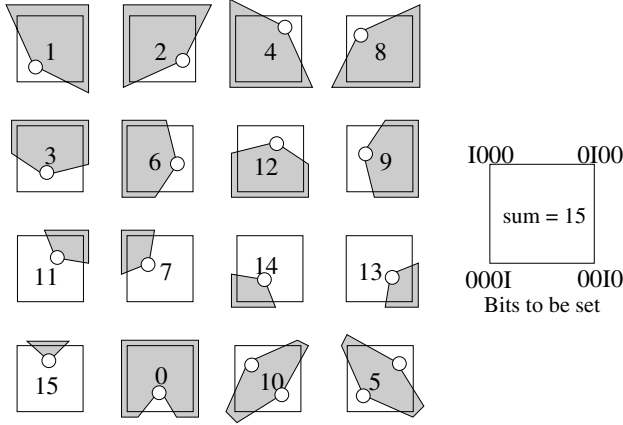


Fig. 4. Interpolation scheme to compute forces acting on the structure

using bilinear interpolation as shown in Fig. 5. Exemplarily

case (a) in Fig. 5

$$\begin{aligned} S_{SW} &= 2 \cdot S_{NW} - S_{NNW} \\ S_{SE} &= 2 \cdot S_{NE} - S_{NNE} \end{aligned} \quad (34)$$

case (b) in Fig. 5

$$S_{SW} = 2 \cdot S_{NE} - S_{NENE} \quad (35)$$

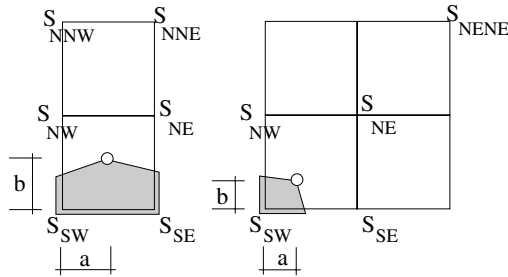


Fig. 5. Specified extrapolation cases (a) and (b)

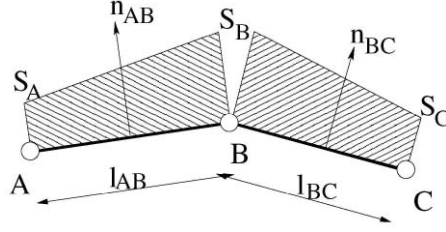


Fig. 6. Force evaluation from stress integration

After these extrapolations we use bilinear interpolation for the stresses S :

$$S_{point} = (1-a)(1-b)S_{SW} + (a)(1-b)S_{SE} + (a)(b)S_{NE} + (1-a)(b)S_{NW} \quad (36)$$

where a and b are normalized weighting factors $0 \leq \{a, b\} \leq 1$ at the point located inside the square. To obtain the load vector $\mathbf{F}_{load}$ for the structure, we have to multiply the stress tensor with the normal of the boundary and to integrate along the boundary Γ

$$\mathbf{F}_{load} = \int_{\Gamma} S \cdot \mathbf{n}_{boundary} d\Gamma. \quad (37)$$

We assume that the stress components are linear between two boundary nodes (see Fig. 6). The load vector $\mathbf{F}_B$ at point B is computed by extending the integration (37) from point $\frac{1}{2}(B + C)$ to point $\frac{1}{2}(A + B)$ in a counter clockwise manner with the structure on the left hand side. The result is

$$\mathbf{F}_B = \left(\frac{1}{4}S_A + \frac{3}{4}S_B \right) \mathbf{n}_{AB} \frac{1}{2}l_{AB} + \left(\frac{3}{4}S_B + \frac{1}{4}S_C \right) \mathbf{n}_{BC} \frac{1}{2}l_{BC} \quad (38)$$

Note that this method does not conserve the momentum but is profile preserving.

3 Computational Efficiency of LB-methods

Before starting investigations in the coupled fluid structure simulation we set up a benchmark problem published in [11]. The goal was to contribute to the discussion whether Lattice-Boltzmann (LB) methods can be regarded as efficient CFD solvers. We compared the accuracy and computational efficiency of two research simulation codes based on the LB and the Finite-Element method (FEM) and Finite Volume method (FV) for incompressible laminar two-dimensional flow problems in complex geometries. The *Featflow* software [35] is a finite element (in-)stationary CFD solver using multigrid methods. The commercial code *CFX* [36] is a Finite Volume based CFD package that

is second order accurate in space and time and contains an algebraic multigrid solver to provide optimal solutions even for complex geometries. The LB Solver is the research prototype *Virtual Fluids* [11,31].

Our results indicate that for the quantities studied (lift, drag, pressure drop) the LB solver is competitive for incompressible transient problems but asymptotically slower for steady-state Stokes flow as the asymptotic algorithmic complexity of the classical LB-method is not optimal compared to the multigrid solvers incorporated in the FEM and CFX code. For the weakly compressible case, the LB approach has a significant wall clock time advantage as compared to CFX, indicating that the LB solver is extremely efficient in the area of weakly compressible flows.

We believe that it is misleading to ask whether LB methods in general are better suited for CFD simulations than ‘conventional’ methods, because both terms are too general and one has to differentiate carefully between modeling errors (especially for complex fluids), discretization errors and implementational issues (which in the case of unstructured grids can have substantial influence on computational efficiency).

4 Structural Solver

The LB solver has been coupled with two structural solvers. The first one is a transient structural solver based on one-dimensional beam elements, implemented by the authors for the sake of developing, implementing, testing and benchmarking the FSI. The model is a Euler-Bernoulli beam and the finite element discretization is based on isoparametric Hermite elements with nodal displacements and slopes as independent degrees of freedom [24]. For the time integration a Newmark scheme and a modal analysis has been used. The second one is the CSD solver *AdhoC* [8] which is a fully three-dimensional transient structural solver based on high-order finite elements (p-version) for the spatial discretization. The time domain is discretized using the generalized- α -method [5], also a time integration scheme based on modal analysis is available.

5 Description of the Coupling Approach

Since we have different grids for the structure (coarse and curved p-elements) and the fluid (in 3D relatively small cubes and a locally refined octree type grid), we introduce an interface mesh to couple both meshes. The interface mesh is a moving surface mesh consisting of flat triangles. On each node the values for the velocity, the load vector and other physical quantities required for the exchange are stored. The mesh is constructed as follows. The nodes are defined by the Gaussian integration points of the p-FEM solver, then a triangulation to obtain the triangles is carried out. The interface mesh can be adapted by the p-FEM solver as well as by the LB solver. An interface

mesh and the coupling is shown in Fig. 7. For problems in two dimensions the interface mesh is reduced to a polyline. In order to increase flexibility con-

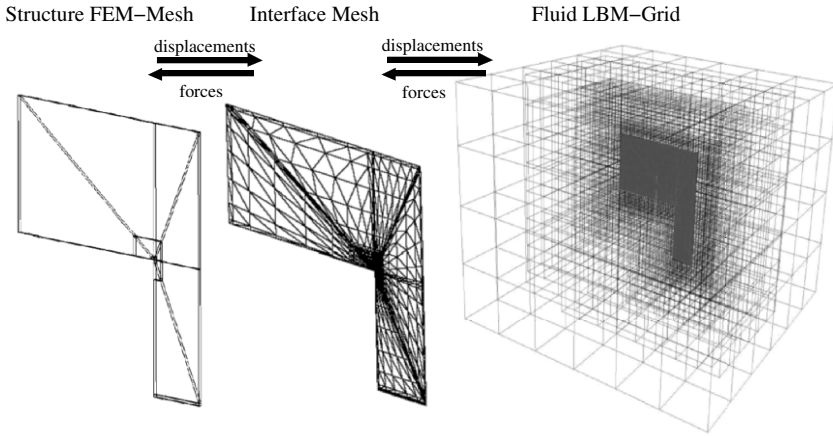


Fig. 7. Coupling using interface mesh

cerning the coupling to different structure codes we use the general coupling software framework described in [3].

5.1 Adaptive Grid Generation

In the grid generation scheme two aspects have to be considered: first the generation process of the quadtree/octree-type grid itself and second the automatic calculation of distances q_i as sketched in Fig. 2 for moving structures.

Specified nodes of the quad-/octree type grid can be modified adaptively by refinement and coarsening due to different criteria, e.g. divergence of velocity, vorticity as well as geometric criteria. Fast algorithms are used for the manipulation of the grid. The mapping of a complex moving geometric object inside the fluid domain requires the calculation of the normalized lattice distances (q -values) of the nodes near the boundary of the solid body. Therefore efficient point in polyhedron tests combined with boundary fill algorithms also known as polyfill algorithms have been implemented. For distance calculations intersection algorithms are performed.

5.2 Transformation of the Physical Quantities

A reasonable set of dimensionless quantities which uniquely define an FSI problem are given e.g. by

- the Aerolastic number $Ae = \frac{E_s}{\rho_f U_{0f}^2}$,

- the Reynolds number $Re = \frac{U_{0f}H}{\nu_f}$,
- the Density ratio $\beta = \frac{\rho_s}{\rho_f}$,
- and the Poisson ratio of the elastic structure ν_s .

Here E_s is the Young's modulus, ρ_s and ρ_f the density of structure and fluid respectively, U_{0f} a reference velocity, H a reference length and ν_f the kinematic viscosity.

The fluid solver rescales the system in such a way, that numerical errors are reduced (avoiding very small and/or large numerical values). Since this rescaling is inherent to most LB implementations, the procedure for rescaling two different systems labeled as *LB* and *real* is presented below. The scaling of the forces and the time from the fluid solver system (LB) to the real system is done by setting the dimensionless drag coefficient equal in both systems. The result is

$$\mathbf{F}_{real} = \mathbf{F}_{LB} \cdot \frac{H_{real} \cdot \rho_{real} \cdot \mathbf{u}_{real}^2}{H_{LB} \cdot \rho_{LB} \cdot \mathbf{u}_{LB}^2}, \quad (39)$$

with reference height H , reference density ρ and reference velocity u . By equating the square of the Reynolds number we obtain the following force scaling formula:

$$\mathbf{F}_{real} = \mathbf{F}_{LB} \cdot \frac{\rho_{real} \cdot \nu_{real}^2 \cdot H_{LB}}{\rho_{LB} \cdot \nu_{LB}^2 \cdot H_{real}} \quad (40)$$

This force transformation is necessary, if no reference velocity as in Section 6.3 is given. By equating the Reynolds number in both systems and using $u = \frac{H}{\Delta T}$ we obtain the following time scaling:

$$\Delta T_{real} = \Delta T_{LB} \cdot \frac{\nu_{LB} \cdot H_{real}^2}{\nu_{real} \cdot H_{LB}^2} \quad (41)$$

Computing the displacements as well as the velocity of the structure the wall velocity (boundary condition) for the LB system is

$$\mathbf{u}_{LB} = \frac{\mathbf{u}_{real}}{\Delta t_{real}} \cdot \frac{\nu_{LB} \cdot H_{real}}{\nu_{real} \cdot H_{LB}}. \quad (42)$$

5.3 FSI Coupling Algorithm

The explicit time coupling is sketched in Fig. 8 and can be summarized as follows:

- (1) The fluid solver computes the load vector on the interface mesh points.
- (2) Exchange of loads through interface mesh.
- (3) The structure solver computes the new displacements due to the loads.
- (4) Exchange of the new displacements through interface mesh.
- (5) Fluid solver: Time interpolation of the position of the interface surface geometry and dynamic adaptation of the quad-/octree.
- (6) Repeat step 5 for all internal iterations (nested time stepping).
- (7) Repeat step 5 and 6 for the number n of fluid time steps.

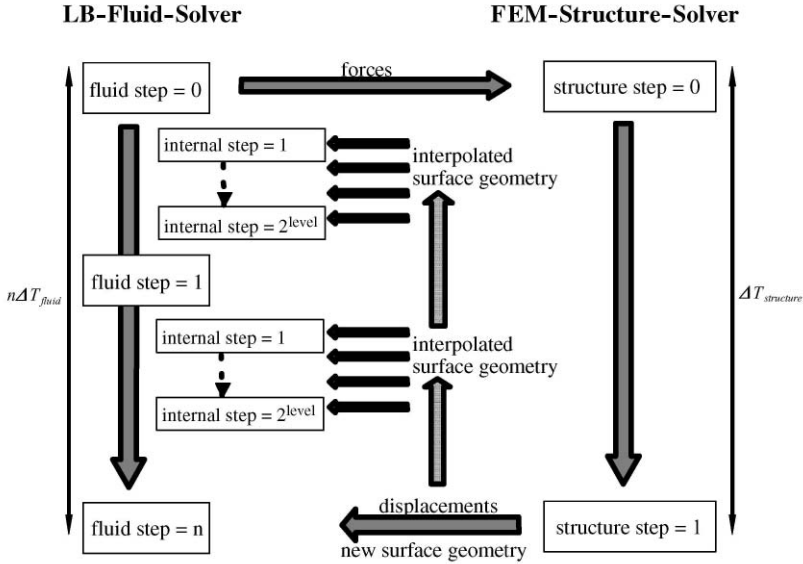


Fig. 8. Coupling algorithm.

6 FSI-Benchmark of the DFG-Research Unit 493

The detailed description of the benchmark defined in the DFG-Research Unit 493 can be found in a separate paper [32]. As depicted in Fig. 9 a cylinder with an elastic bar attached to it is asymmetrically placed in a channel. The benchmark consists of three different parts,

- a pure fluid benchmark (CFD),
- a pure structure benchmark (CSM) and
- a coupled benchmark (FSI).

Furthermore the benchmark covers Reynolds numbers 20, 100 and 200. The CFD and CSM test cases are for validating the individual solvers. The geometric definition parameters are the length $L = 2.5\text{m}$, height $H = 0.41\text{m}$ of the channel, the circle center position $C = (0.2\text{m}, 0.2\text{m})$ and circle radius $r = 0.05\text{m}$, length of the elastic bar $l = 0.35\text{m}$ and height $h = 0.02\text{m}$. The

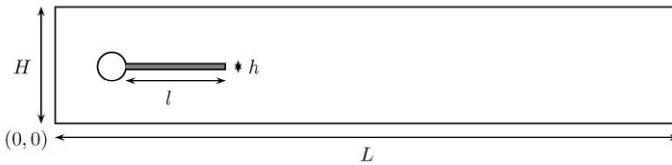


Fig. 9. Computational domain

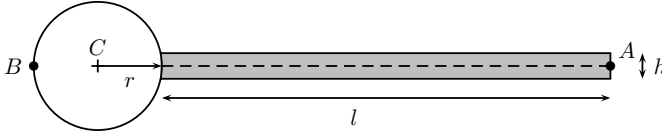


Fig. 10. Detail of the structure part

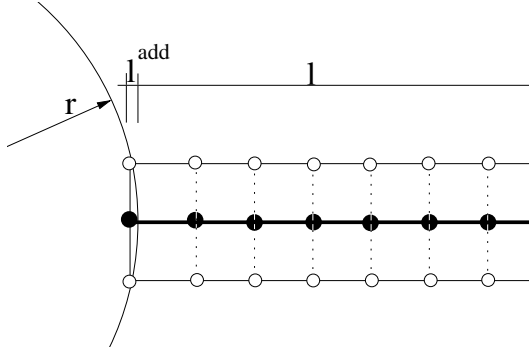


Fig. 11. Detail of the 1-D beam configuration

control point $A = (0.6\text{m}, 0.2\text{m})$ is defined for measurement of the displacements. The physical parameters are given later in the subsections.

For the one-dimensional beam-elements the geometry of the bar is set up according to Fig. 11. The beam elements are positioned in the center of the bar. The loads are transferred from the nodes at the boundary (marked as un-filled circles) to the nodes of the beam elements (marked as filled circles) by adding the loads, the displacement of the bar is calculated for the filled nodes. Finally the displacements for the boundary nodes are computed by simple geometric relations. The length l_{bar} of the bar is $l_{bar} = l + 0.00102\text{m}$.

6.1 Benchmarks Part 1: CFD

A parabolic velocity profile at the inlet and a constant reference pressure at the outlet is imposed. The Reynolds number is defined as $Re = \frac{\bar{U}D}{\nu_f}$, where $\bar{U}$ the average inflow velocity and D the diameter of the cylinder. Table 1 shows a good agreement with the test case $Re = 20$ and $Re = 100$. The error with respect to the reference values in [32] is below one percent for drag and lift of the obstacle. The test case $Re = 200$ is an instationary flow yielding periodic oscillations. For the test case $Re = 200$ we have a variance below 3% for the average value of the drag and a variance of ca. 15% for the amplitude of the oscillation of the drag value. The variance is 30% for the average value of the lift and below 10% for the amplitude of the oscillation of the lift value. For the stationary problems we used error indicators e.g.

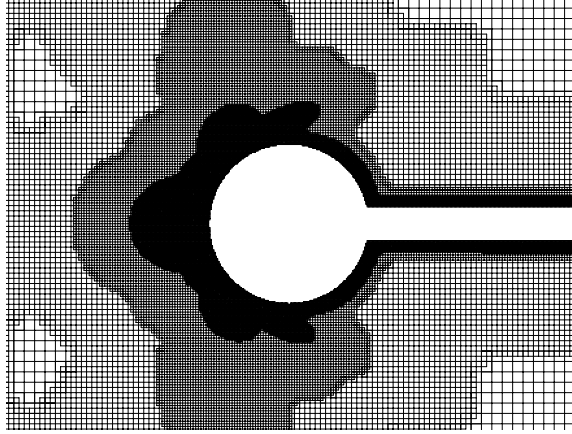


Fig. 12. Adaptively refined grid for the Re 20 case

based on velocity divergence to refine the grid adaptively (see Fig. 12). It is mentioned that the results for the test case CFD3 (Re = 200) are very sensitive to the setup of the mesh and this behavior is also experienced by the other groups participating at the benchmark. The simulations have been conducted with a Mach number of $Ma = 0.02$.

6.2 Benchmarks Part 2: CSM

CSM1 and CSM2 are steady state problems for a constant body force $g = 2 \frac{\text{m}}{\text{s}^2}$. The dynamics of the beam in CSM3 is driven by an instantaneously imposed body force ($g = 2 \frac{\text{m}}{\text{s}^2}$). The Young's modulus for CSM1 and CSM3 cases is $1.4 \times 10^6 \frac{\text{kg}}{\text{m s}^2}$ and $5.6 \times 10^6 \frac{\text{kg}}{\text{m s}^2}$ for CSM2. For the one-dimensional beam the Young's modulus $E_{xx} = \frac{E}{1-\nu^2}$ has been used due to the reduction to a one-dimensional system. The structural density is $\rho_s = 1000.0 \frac{\text{kg}}{\text{m}^3}$. The values in Table 2 show a good agreement with the reference values, especially if one takes into account the model error with respect to approximating the 2D-structure as a beam. The number of eigenmodes considered for the case *1D-FE-Beam modal* was 4.

6.3 FSI Example 1: Oscillating Beam in a Box Filled with Fluid

Before the investigation of the main FSI benchmark another variant of the configuration was studied. Here we consider a beam in a box filled with fluid and an instantaneously imposed constant load of $p = 40 \frac{\text{N}}{\text{m}}$. The fluid viscosity is $\nu_f = 0.001 \frac{\text{m}^2}{\text{s}}$ and the fluid as well as the structural density are $\rho_{f,s} = 1000.0 \frac{\text{kg}}{\text{m}^3}$. The steady state solution is known and corresponds to the

Table 1. CFD benchmark results

benchmark	Featflow	LB, Level 6–7	LB, Level 6–8	LB, Level 6–9
CFD1, Re 20				
– drag in [N]	14.29	14.224	14.264	14.267
– lift in [N]	1.119	1.098	1.108	1.116
– #dof	11225600	406458	431082	544311
CFD2, Re 100				
– drag in [N]	136.7	135.983	138.576	136.745
– lift in [N]	-10.53	-11.905	-10.997	-10.587
– #dof	11225600	771525	889524	1338660
CFD3, Re 200				
– average drag in [N]	439.38	459.03	456.78	450.345
– oscill. drag in [N]	± 5.6183	± 6.97	± 4.4	± 6.295
– average lift in [N]	-11.893	-15.795	-6.15	-15.47
– oscill. lift in [N]	± 437.81	± 401.035	± 361.61	± 466.64
– frequency in [Hz]	4.39	4.35	4.35	4.47
– #dof	705152	1470978	1835811	2368944

Table 2. CSM benchmark results

benchmark	Featflow	1D-FE-Beam Newmark	1D-FE-Beam Modal	AdhoC generalized- α
CSM1				
– y-dipl. in [m]	-0.06610	-0.06832	-0.06832	-0.06610
– #dof	435776	52	52	684
CSM2				
– y-dipl. in [m]	-0.01697	-0.01708	-0.01708	-0.01697
– #dof	435776	52	52	684
CSM3				
– ΔT	0.005	0.00125	0.00125	0.005
– average value in [m]	-0.06361	-0.06865	-0.06768	-0.06321
– oscill. amplitude in [m]	± 0.06516	± 0.06989	± 0.06918	± 0.06523
– frequency in [Hz]	1.0995	1.0702	1.0677	1.0742
– #dof	98820	52	52	684

CSM 1 test case. The additional effect is the damping of the structure motion by the viscous fluid.

Using a Newmark time discretization approach for the structure solver (1d beam elements) the initialization/deactivation of fluid nodes yields small pressure disturbances exciting the high frequency modes of the structure, which interact with the fluid, resulting in a self-excitation and an unstable scheme.

If we perform a modal analysis by computing eigenvalues and eigenvectors of the structural system, and use only the lowest 4 eigenmodes for the time integration, a stable scheme results.

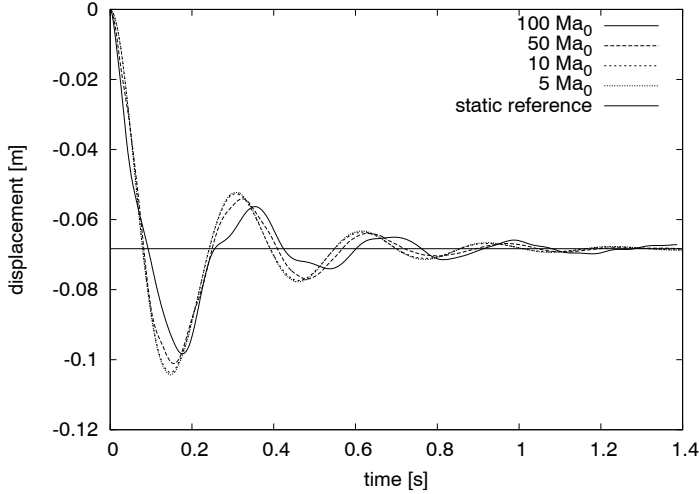


Fig. 13. Damped oscillation of a beam (point A, see Fig. 10) in a box for different Mach numbers

Figure 13 shows the displacement of the point A for different time step resolutions, i.e. for different Mach numbers and indicates a convergence to the incompressible solution. The Mach number scales as $Ma = \frac{\nu_{LB}}{\nu_f} \cdot Ma_0$, where Ma_0 is a constant reference value. In Fig. 13 the results for $\nu_{LB} = 0.1, 0.05, 0.01, 0.005$ resulting in timesteps $\Delta T_{real} = 4.307 \times 10^{-3}, 2.153 \times 10^{-3}, 4.307 \times 10^{-4}, 2.153 \times 10^{-4}$ [s] (according to Eq. (41)) are shown. To compute the actual Mach number, we compute the average velocity of point A in the interval $[0, T_P]$, where T_P is the time, where the first minimum of the displacement is reached, and divide it by the speed of sound. The Mach number for the largest timestep was quite high and has a value of approximately 0.6, so that Ma_0 can be set to $Ma_0 = 0.006$. This test case also shows that large displacements (one quarter of the channel height) can be handled easily.

6.4 FSI Example 2: FSI-Benchmark of the DFG-Research Unit 493

A parabolic velocity profile at the inlet and a constant reference pressure at the outlet is given. We investigated the setups for the FSI1 and FSI3 benchmark listed in Table 3. FSI2 has not been investigated yet.

The y -displacement of point A is shown in Table 4. FSI1 results in a steady state solution, whereas FSI3 has a periodic solution.

For all FSI computations we coupled *Virtual Fluids* with the 1D-beam FEM code and the structural solver *AdhoC*. Regarding the time integration scheme for the structure neither the Newmark integration scheme for the

Table 3. Parameters for the FSI benchmark

parameter	FSI1	FSI3
Density ratio β	1.0	1.0
Aerolastic number Ae	35000.0	1400.0
Reynolds number Re	20.0	200.0
Poisson ratio ν_s	0.4	0.4
$\rho_s = \rho_f$ in $\frac{\text{kg}}{\text{m}^3}$	1000	1000
E in Pa	1.4×10^6	5.6×10^6
$\bar{U}$ in $\frac{\text{m}}{\text{s}}$	0.2	2
ν_f in $\frac{\text{m}^2}{\text{s}}$	0.001	0.001

Table 4. FSI benchmark: y-displacements of point A.

benchmark	Featflow	LB1 (*)	LB2(*)	LB3(*)
FSI1, Re 20				
– #dof	4835328	972054	972054	773334
– ΔT [s]	–	1.263×10^{-3}	1.263×10^{-3}	0.2526×10^{-3}
– displ. [m]	0.00082	0.002026	0.00076	0.00083
FSI3, Re 200				
– #dof	304128	2318904	1835863	1836495
– ΔT [s]	5.0×10^{-4}	1.263×10^{-4}	1.263×10^{-4}	0.2526×10^{-4}
– average [m]	0.00143	0.00112	0.00123	0.001386
– amplitude [m]	± 0.03437	± 0.0295	± 0.03988	± 0.033951
– frequency [Hz]	5.4	5.82	5.41	5.32

(*) Legend:

- case LB1: Level 6-8, $Ma=0.1$ coupled to 1D Beam
- case LB2: Level 6-8, $Ma=0.1$ coupled to *AdhoC*
- case LB3: Level 6-8, $Ma=0.02$ coupled to *AdhoC*

1D-Beam elements nor the generalized α -method of *AdhoC* was stable. Only the method of modal analysis and integrating the lowest eigenmodes yielded a stable scheme for the coupled problem. The timestep for the fluid solver *Virtual Fluids* on the coarsest grid level was the same as the timestep for the structural solver. The number of considered eigenmodes was 4 for case *LB1* and 9 for the cases *LB2, LB3* in Table 4.

The results for FSI1 indicate that the approximation with the 1D beam elements does not yield a correct solution, the model error is very large since only forces normal to the axis of the bar are taken into account. Here the coupling with the structural solver *AdhoC* gives good results, the error for the simulation with Mach number $Ma = 0.02$ is approximately 1% with respect to the reference solution of *FeatFlow*.

For the periodic FSI3 benchmark the results of simulation *LB3* have an error of ca. 1 % in the amplitude of the oscillation, approximately 1.5 %

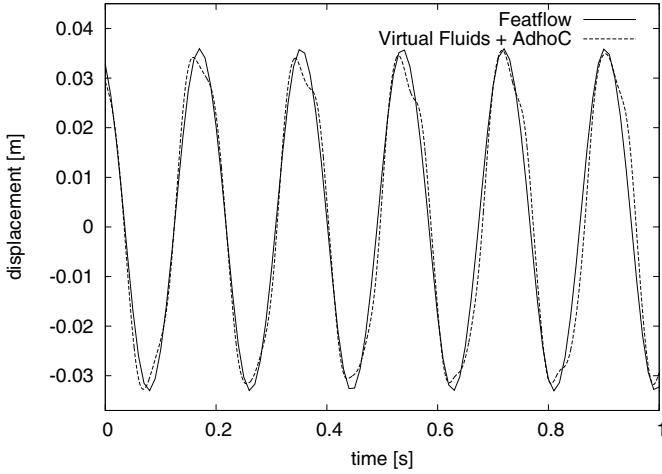


Fig. 14. Oscillation of point A for $Re = 200$

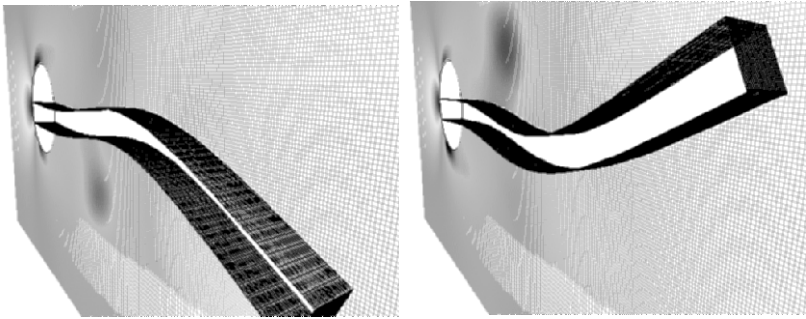


Fig. 15. Snapshots of moving beam dynamics for $Re = 200$

for the frequency and 3 % for the average value, which is approximately 30 times smaller than the oscillation amplitude. The computation time on a 2GHz AMD OPTERON for this case was approximately 5 hours per period, for the cases *LB1*, *LB2* it was about 1 hour per period. In Fig. 14 the results for the oscillation of point A is shown for the *FeatFlow* and our solution. In Fig. 15 snapshots of the moving beam dynamics are shown.

7 Conclusions

A partitioned coupling approach for two highly efficient solvers for CFD and CSD respectively has been described in detail. The preliminary results presented above show that the basic coupling approach is feasible and yields

consistent quantitative results. Further improvements of the coupling algorithm with respect to robustness are expected from the following measures:

- the surface mesh responsible for the load transfer is presently defined by the Gauss points of the p-FEM solver. This implies that the local size of the triangles is proportional to the inverse square of the degree of the Ansatz functions resulting in very small sizes for high order elements in the vicinity of the p-element edges resulting in localized reduced accuracy with respect to load transfer. The introduction of a coupling mesh independent of the Gauss points in combination with a suitable projection of the loads and a global error minimization with respect to the transfer of the displacements is expected to substantially reduce these effects leading to improved stability for cases where higher modes become essential for the coupled problem.
- the accuracy of the initialization of new fluid nodes is obviously important for the overall quality of the coupled solution. Presently the combination of the non-conservative second order no-slip boundary condition and the initialization approach based on local extrapolations and smoothing tends to destabilize the coupling scheme. Recently we have developed a new locally mass conservative second order boundary condition scheme which will be tested in combination with improved local Poisson type smoothing algorithms.

In addition, an implicit coupling approach will be tested which will allow a quantitative equilibration of loads and displacements on the fluid and structural side respectively. The extension of the coupling strategy to three dimensions is theoretically straightforward but technically elaborate. In the near future we are planning to extend the approach for FSI under special consideration of turbulent flow fields.

8 Acknowledgements

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Thin Solids for Fluid-Structure Interaction

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Abstract. In this contribution the use of hexahedral elements for the structural simulation in a fluid structure interaction framework is presented, resulting in a consistent kinematic and geometric description of the solid. In order to compensate the additional numerical effort of the three-dimensional approach, an anisotropic p -adaptive method for linear elastodynamic problems is proposed, resulting in a clearly higher efficiency and higher convergence rates than uniform p -extensions. Special emphasis is placed on the accurate transfer of loads considering the fluid discretization for computation of the surface load integrals. For a coupling with a cartesian grid based Lattice Boltzmann code it was found that oscillations in the interface tractions may excite higher structural modes possibly leading to a non-stable coupling behavior. A first remedy to this problem was a linear modal analysis of the structure, thus allowing to control the number of modes to be considered without disregarding bidirectional fluid structure interactions. Preliminary results are presented for the FSI benchmark configuration proposed in this book.

1 Introduction

Our approach is based on the application of high order solid elements to describe the behavior of structures interacting with fluids — e.g. thin-walled wind-loaded plates and shells. High order solid elements can provide a fully three-dimensional solution also including arbitrary three-dimensional stress states, and, nevertheless, can cope with high aspect ratios of thin-walled structures being less prone to locking effects. Thus, only one element type for thin- as well as thick-walled structures is sufficient, without making transition elements between thin-walled and massive parts of the structure necessary, see e.g. Figures 10, 11 and 12 on pages 305 and 306, respectively, showing a thin hemispherical shell connected to a compact stiffener ring. A detailed discussion of the advantages of high order solid elements for thin-walled (non-linear) continua can be found in [1] and [2].

For fluid-structure interaction problems, another advantage of three-dimensional models is obvious, i.e. the inherently correct representation of the *wet surface* being in contact with the fluid, without the necessity of reconstructing this skin surface from the middle surface and certain kinematic assumptions, as it is the usual approach for dimensionally reduced plate and shell models.

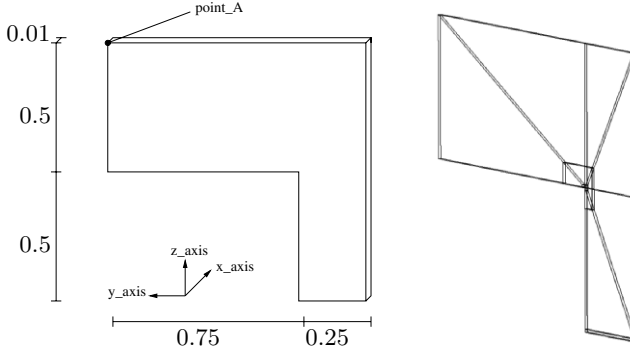


Fig. 1. System of a plate with re-entrant corner (left), hexahedral mesh (right)

An example by way of illustration is the plate with a re-entrant corner depicted in Figure 1 (left), being discretized by 16 high-order hexahedral elements, see Figure 1 (right). The interaction of the plate with a fluid flow in x -direction perpendicular to the plate ($Re = 50$) is computed by a partitioned coupling to a finite volume based CFD (*Computational Fluid Dynamics*) code provided by a project partner, see e.g. [3,4]. The transient response of the plate is plotted in Figure 2, showing the displacement of point A, see Figure 1 (left), in x -direction. The amplitudes of the plate's vibration decrease over time due to the damping influence of the surrounding fluid. Considering geometric nonlinearities for large displacements, the structure is computed with *anisotropic* high order hexahedral elements, since our implementation *AdhoC* [5] allows the polynomial degree of the hexahedra to be chosen independently for the three local directions of the elements. In this fluid-structure interaction ex-

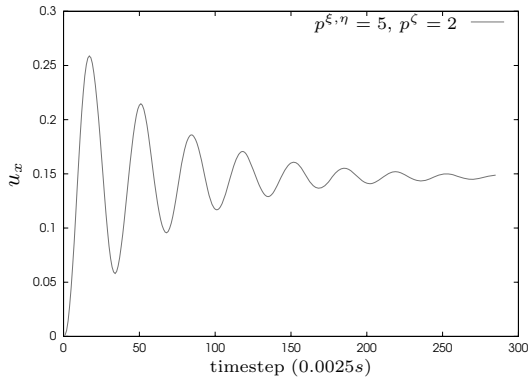


Fig. 2. Displacement u_x of point A over time (FSI)

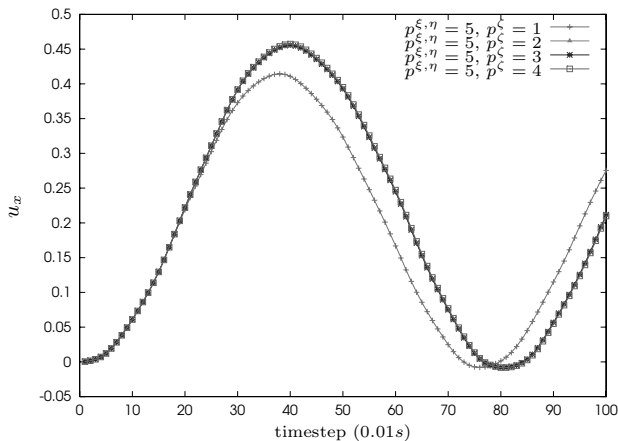


Fig. 3. Displacements of point A over time for varying p^ζ

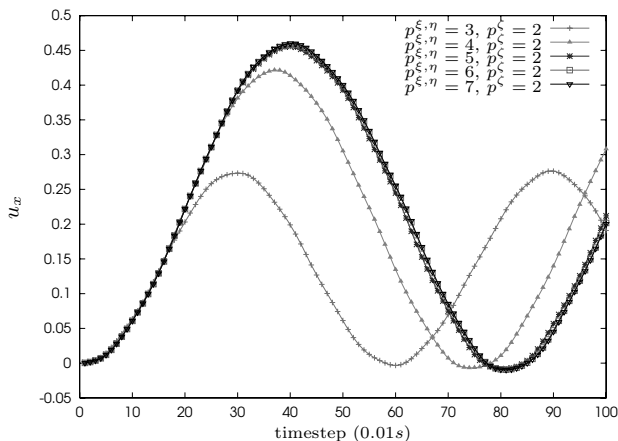


Fig. 4. Displacements of point A over time for varying $p^{\xi,\eta}$

ample, the *in-plane* polynomial degrees are chosen to be $p^{\xi,\eta} = 5$, whereas the *transversal* polynomial degree corresponding to the thickness direction is set to $p^\zeta = 2$ for *all* elements.

However, this choice is by far not arbitrary. The following sequence of computations for a simplified problem was performed in order to determine a reasonable choice for $p^{\xi,\eta}$ and p^ζ . Instead of computing the fluid-structure interaction problem, the fluid load was substituted by a similar uniform surface traction being yet constant over time.

In a first sequence of runs, the in-plane polynomial degrees were fixed to $p^{\xi,\eta} = 5$ and the transversal polynomial degree was varied, $p^\zeta = 1, \dots, 4$. The corresponding displacements in x -direction of point A for approximately one period of vibration is plotted in Figure 3. It is obvious from these results, that $p^\zeta = 1$ led to strong deviations from the reference solution. However, with $p^\zeta = 2$, the converged deflection was already matched satisfactorily. In a second sequence of runs, the transversal polynomial degree was fixed, $p^\zeta = 2$ and the in-plane polynomial degrees were varied, $p^{\xi,\eta} = 3, \dots, 7$, see Figure 4. Here, $p^{\xi,\eta} = 5$ was necessary to sufficiently approximate the reference solution. From these results, a choice of $p^{\xi,\eta} = 5$ and $p^\zeta = 2$ was considered reasonable for this system.

Although this heuristic approach yields accurate results for the investigated example, it has many disadvantages. Two sequences of computations are necessary to only adjust two global discretization parameters, $p^{\xi,\eta}$ and p^ζ for *all* elements. The approach only makes sense for strictly thin-walled structures, where orientations like “in-plane directions” and “transversal direction” are reasonable. On the other hand, no use is made of local, elementwise selection of polynomial degrees. Starting from these preliminary results we will discuss in this paper several steps which we consider crucial for an efficient use of high order structural elements for fluid structure interaction. Important are anisotropic solid elements with a precise description of structural surfaces, a verification of direct time integration schemes for the p -FEM, an adaptive choice of local polynomial degrees and, last but not least, a framework for load interpolation from (low order) fluid discretizations to high order structural elements.

The outline of this paper is as follows: In Section 2 the spatial discretization of thin-walled and massive structures with anisotropic high order hexahedral elements is described and the necessity of appropriate geometric mapping using the blending function method is emphasized.

The generalized α -method recalled in Section 3 is used for the discretization of the time domain. The section concludes with a numerical example, the elastodynamic computation of a thin-walled clamped plate. The results for geometric linear and geometric nonlinear computations using hexahedral elements are compared to solutions from literature based on a two-dimensional spectral element approach in order to verify the code.

A local anisotropic p -adaptive method based on a hierarchic anisotropic error estimator is proposed for linear elastodynamic problems in Section 4. The efficiency of the approach is investigated with a numerical example.

In Section 5 the anisotropic hexahedral elements are used within a flexible fluid structure interaction framework [6] based on a database for the geometrical description of the interface (wet surface) and for the coupling data defined thereon. Findings and first results are given for the coupling to a Lattice Boltzmann code [7,8] for the benchmark configuration proposed in [9]. Section 6 summarizes this article and contains conclusions for future work.

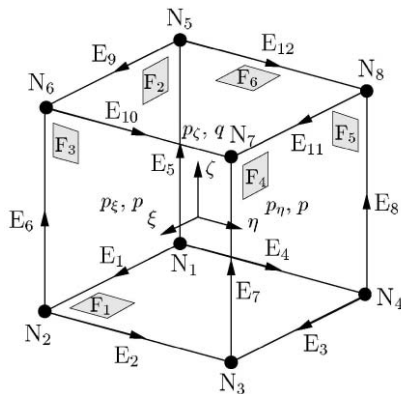


Fig. 5. Standard hexahedral element Ω_{st}^h : definition of nodes, edges, faces and polynomial degree

2 High Order FEM for Thin Solids

2.1 Hierarchical Shape Functions

Our implementation of high order finite elements in three dimensions is based on a hexahedral element formulation (see Figure 5), using the hierarchic shape functions introduced in [10]. High order hexahedral elements are suited for solid, 'thick' structures and for thin-walled structures alike. In the case of plate- or shell-like structures one local coordinate can be identified to correspond with the thickness direction and it is possible to choose the polynomial degree in thickness direction differently from those in in-plane direction (see [2]). Three-dimensional shape functions can be classified into four groups:

1. **Nodal or vertex modes:** The nodal modes

$$N_{1,i,1}^{N_i}(\xi, \eta, \zeta) = \frac{1}{8} (1 + \xi_i \xi)(1 + \eta_i \eta)(1 + \zeta_i \zeta), \quad i = 1, \dots, 8 \quad (1)$$

are the standard trilinear shape functions, well known from the isoparametric eight-noded brick element. (ξ_i, η_i, ζ_i) are the local coordinates of node N_i .

2. **Edge modes:** These modes are defined separately for each individual edge. If we consider, for example, edge E_1 (see Figure 5), the corresponding edge modes read:

$$N_{i,1,1}^{E_1}(\xi, \eta, \zeta) = \frac{1}{4} (1 - \eta)(1 - \zeta)\phi_i(\xi), \quad i \geq 2 \quad (2)$$

3. **Face modes:** These modes are defined separately for each individual face. If we consider, for example, face F_1 , the corresponding face modes read:

$$N_{i,j,1}^{F_1}(\xi, \eta, \zeta) = \frac{1}{2} (1 - \zeta)\phi_i(\xi)\phi_j(\eta), \quad i, j \geq 2 \quad (3)$$

4. Internal modes: The internal modes

$$N_{i,j,k}^{\text{int}}(\xi, \eta, \zeta) = \phi_i(\xi)\phi_j(\eta)\phi_k(\zeta), \quad i, j, k \geq 2 \quad (4)$$

are purely local and vanish at the faces of the hexahedral element.

The indices i, j, k of the shape functions denote the polynomial degrees in the local directions ξ, η, ζ .

Three different types of trial spaces can readily be defined: the *trunk space* $\mathcal{S}_{\text{ts}}^{p_\xi, p_\eta, p_\zeta}(\Omega_{\text{st}}^h)$, the *tensor product space* $\mathcal{S}_{\text{ps}}^{p_\xi, p_\eta, p_\zeta}(\Omega_{\text{st}}^h)$ and an *anisotropic tensor product space* $\mathcal{S}^{p, p, q}(\Omega_{\text{st}}^h)$. A detailed description of these trial spaces can be found in [1, 2, 10, 11]. The polynomial degree for the trial spaces $\mathcal{S}_{\text{ts}}^{p_\xi, p_\eta, p_\zeta}(\Omega_{\text{st}}^h)$ and $\mathcal{S}_{\text{ps}}^{p_\xi, p_\eta, p_\zeta}(\Omega_{\text{st}}^h)$ can be varied separately in each local direction (see Figure 5).

Owing to the built-in anisotropic behaviour of the trial space $\mathcal{S}^{p, p, q}(\Omega_{\text{st}}^h)$ it is important to consider the orientation of the local coordinates of a hexahedral element. Figure 6 shows how hexahedral elements should be oriented when three-dimensional, thin-walled structures are discretized. The local coordinate ζ of the hexahedral element corresponds to the thickness direction. If the orientation of all elements is identical, it is possible to construct discretizations where the polynomial degree for the in-plane and thickness direction of thin-walled structures can be treated differently.

2.2 Geometric Model

In low order finite element analysis, the most frequently used mapping concept for the geometric description of the domain of computation is the application of isoparametric elements. The same shape functions are used for the approximation of the unknown solution and for the geometry of the elements. Using elements of order $p = 1$ or $p = 2$ the boundary of the domain is therefore approximated by a polygonal or by a piecewise parabolic curve, respectively. As the mesh is refined, the boundary of the domain is approximated more and more accurately. When using the p -version, on the other

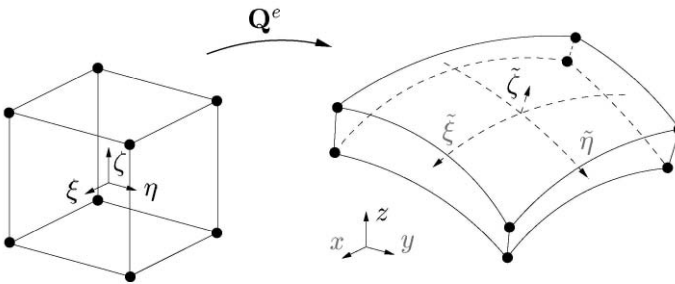


Fig. 6. Modelling thin-walled structures with hexahedral elements

hand, the mesh remains fixed. It is therefore important to model the geometry of the structure accurately with the fixed number of elements in use. This calls for a method which is able to describe complex geometries using only a few elements. GORDON and HALL[12] and [13] proposed the so-called *blending function method*, which is usually applied when describing curved boundaries of p -version finite elements (see, for example, [10] and [2]).

The Blending Function Method Our formulation of the blending function method follows closely the work of [14]. Consider a hexahedral element, as pictured in Figure 7. $\mathbf{X}_i = (X_i, Y_i, Z_i)^T$, $i = 1, \dots, 8$ denote the global coordinates of the nodes. $\mathbf{E}_i = (E_{ix}, E_{iy}, E_{iz})^T$, $i = 1, \dots, 12$ are functions which depend on local coordinates (ξ, η, ζ) and describe the shape of each edge. $\mathbf{F}_i = (F_{ix}, F_{iy}, F_{iz})^T$, $i = 1, \dots, 6$ denote the functions describing the shape of each face. The mapping function $\mathbf{Q}^e(\xi, \eta, \zeta)$ from local $\boldsymbol{\xi} = (\xi, \eta, \zeta)^T$ to global coordinates $\mathbf{x} = (x, y, z)^T$ is obtained by

$$\mathbf{x} = \mathbf{Q}^e(\xi, \eta, \zeta) = \sum_{i=1}^8 N_{1,1,1}^{N_i}(\xi, \eta, \zeta) \mathbf{X}_i + \sum_{i=1}^6 \mathbf{f}_i(\xi, \eta, \zeta) - \sum_{i=1}^{12} \mathbf{e}_i(\xi, \eta, \zeta). \quad (5)$$

The first term is the standard mapping of isoparametric eight-noded hexahedral elements. The second term is referred to as face blending (see [14] and [2]). Consider e.g. face 6 of the hexahedral element shown in Figure 7, where $\mathbf{F}_6(\xi, \eta)$ describes the parametric mapping of the local (ξ, η) -plane to the surface of the element:

$$\mathbf{f}_6(\xi, \eta, \zeta) := \left(\mathbf{F}_6(\xi, \eta) - \frac{1}{4}((1-\xi)(1-\eta)\mathbf{X}_5 + (1+\xi)(1-\eta)\mathbf{X}_6 + (1+\xi)(1+\eta)\mathbf{X}_7 + (1-\xi)(1+\eta)\mathbf{X}_8) \right) \left(\frac{1+\zeta}{2} \right)$$

At face 6, where $\zeta = 1$ the blending term $\frac{1+\zeta}{2}$ equals 1 and therefore $\mathbf{f}_6(\xi, \eta, \zeta) = 1$ describes the difference between the curved and the bilinear face coordinates (ξ, η) . Due to the blending term it is guaranteed, on the other hand, that this difference (i.e. the function $\mathbf{f}_6(\xi, \eta, \zeta)$) decreases

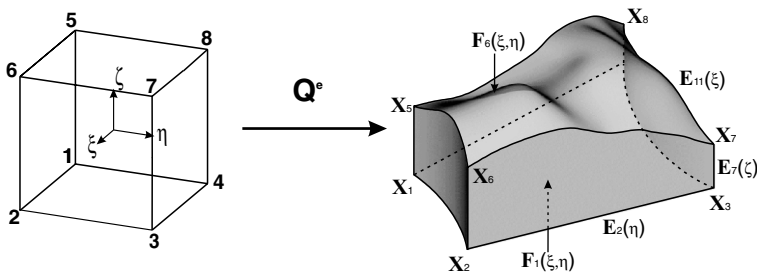


Fig. 7. Mapping by the blending function method

linearly to the opposite face 1, where $\zeta = -1$ such that $\mathbf{f}_6(\xi, \eta, -1) = 0$.

The third term in Equation (5) corresponds to the edge blending. Considering e.g. edge 1 we have

$$\mathbf{e}_1(\xi, \eta, \zeta) := \left(\mathbf{E}_1(\xi) - \frac{(1-\xi)\mathbf{X}_1 + (1+\xi)\mathbf{X}_2}{2} \right) \left(\frac{1-\eta}{2} \right) \left(\frac{1-\zeta}{2} \right). \quad (7)$$

$\mathbf{e}_i(\xi, \eta, \zeta)$, $i = 1, \dots, 12$ denotes the difference between the curved edge and the linear connection of the two end points, multiplied by a blending term (see [14] and [2]). The structure of the edge blending is similar to the face mapping, now with a blending term being linear in two variables. Because each edge belongs to two faces of a hexahedral element, a correction with respect to a straight line edge appears twice in the surface blending term (second sum) in (5). Therefore, the corresponding edge blending term has to be subtracted.

Substituting $\mathbf{f}_i(\xi, \eta, \zeta)$, $i = 1, \dots, 6$ and $\mathbf{e}_i(\xi, \eta, \zeta)$, $i = 1, \dots, 12$ in Equation (5), by inserting relations similar to the ones of (6) and (7) and rearranging terms we finally obtain

$$\begin{aligned} \mathbf{x} &= \mathbf{Q}^e(\xi, \eta, \zeta) \quad (8) \\ &= \frac{1}{2} \left((1-\zeta) \mathbf{F}_1(\xi, \eta) + (1-\eta) \mathbf{F}_2(\xi, \zeta) + (1+\xi) \mathbf{F}_3(\eta, \zeta) + \right. \\ &\quad \left. + (1+\eta) \mathbf{F}_4(\xi, \zeta) + (1-\xi) \mathbf{F}_5(\eta, \zeta) + (1+\zeta) \mathbf{F}_6(\xi, \eta) \right) \\ &\quad - \frac{1}{4} \left((1-\zeta)(1-\eta) \mathbf{E}_1(\xi) + (1-\zeta)(1+\xi) \mathbf{E}_2(\eta) + (1-\zeta)(1+\eta) \mathbf{E}_3(\xi) + \right. \\ &\quad \left. + (1-\zeta)(1-\xi) \mathbf{E}_4(\eta) + (1-\xi)(1-\eta) \mathbf{E}_5(\zeta) + (1+\xi)(1-\eta) \mathbf{E}_6(\zeta) + \right. \\ &\quad \left. + (1+\xi)(1+\eta) \mathbf{E}_7(\zeta) + (1-\xi)(1+\eta) \mathbf{E}_8(\zeta) + (1+\zeta)(1-\eta) \mathbf{E}_9(\xi) + \right. \\ &\quad \left. + (1+\zeta)(1+\xi) \mathbf{E}_{10}(\eta) + (1+\zeta)(1+\eta) \mathbf{E}_{11}(\xi) + (1+\zeta)(1-\xi) \mathbf{E}_{12}(\eta) \right) \\ &\quad + N_{1,1,1}^{N_1}(\xi, \eta, \zeta) \mathbf{X}_1 + N_{1,1,1}^{N_2}(\xi, \eta, \zeta) \mathbf{X}_2 + N_{1,1,1}^{N_3}(\xi, \eta, \zeta) \mathbf{X}_3 + \\ &\quad + N_{1,1,1}^{N_4}(\xi, \eta, \zeta) \mathbf{X}_4 + N_{1,1,1}^{N_5}(\xi, \eta, \zeta) \mathbf{X}_5 + N_{1,1,1}^{N_6}(\xi, \eta, \zeta) \mathbf{X}_6 + \\ &\quad + N_{1,1,1}^{N_7}(\xi, \eta, \zeta) \mathbf{X}_7 + N_{1,1,1}^{N_8}(\xi, \eta, \zeta) \mathbf{X}_8 \end{aligned}$$

Applying the mapping function (8) allows to model almost arbitrary shapes where the parametric functions $\mathbf{E}_i$ and $\mathbf{F}_i$ describing the geometry of the edges and faces may be, for example, polynomials, NURBS [15] or transcendental functions. However, when using a superparametric concept, where the mapping functions can not be represented by the shape functions interpolating the displacement field, rigid body modes may be not represented precisely. In [1,16] this drawback was investigated in detail. Summarizing these investigations, one can conclude that if the geometry is 'smooth' and

can be represented satisfactorily by means of polynomials, the error due to an inaccurate representation of rigid body modes decreases very rapidly.

To control the accuracy of the geometric description, we apply the quasi-regional mapping which has been proposed by KIRÁLYFALVI and SZABÓ [14]. The idea is to combine the blending function method with a polynomial interpolation of geometry, sampled at an optimized set of points. Lagrange polynomials are used to interpolate the geometry at the BABUŠKA-CHEN points [17,18], leading to a smooth description of geometry.

To be more precise let us consider a hexahedral element with an interpolation of geometry based on a polynomial degree of p_g . The mapping (8) defined by the blending function method involves the position of the 8 nodes and the parametric description of the 12 edges and 6 faces. Concerning the edges, the exact parametric description $\mathbf{E}_i^{ex}(r)$ is interpolated by

$$\mathbf{E}_i(r) = \sum_{k=1}^{p_g+1} \mathbf{E}_i^{ex}(r_k) N_k^{p_g}(r). \quad (9)$$

$N_k^{p_g}(r), k = 1, \dots, p_g + 1$ denote the Lagrangian polynomials of order p_g . The position of the interpolation points $r_k \in [-1, +1]$ has been determined by BABUŠKA and CHEN [17,18] such that the accuracy of the interpolation is strongly improved when being compared to an equally distributed set of nodes. This has been realized by adjusting the interpolation points such that the Lebesgue constant is minimized. Applying a tensor product of Lagrange polynomials this concept can be extended to the interpolation of the exact description of the faces $\mathbf{F}_i^{ex}(r, s)$

$$\mathbf{F}_i(r, s) = \sum_{k=1}^{p_g+1} \sum_{l=1}^{p_g+1} \mathbf{F}_i^{ex}(r_k, s_l) N_k^{p_g}(r) N_l^{p_g}(s). \quad (10)$$

For fluid structure interaction problems the geometric model error has to be controlled in order to accurately represent the fluid structure interface. In the following the accuracy of the quasi-regional mapping will be addressed in more detail. For further investigations the reader is also referred to [16,19,20,11].

2.3 Numerical Examples

Scordelis-Lo Shell Figure 8 shows the classical Scordelis and Lo shell [21] used as a benchmark problem for shell structures by many authors. The structure is loaded by a vertical shell weight. The shell is supported at both ends as indicated in the figure. For an investigation of the accuracy of the mapping concept presented in the preceding section, linear elasticity is assumed.

Due to symmetry, only a quarter of the shell has to be discretized. A three-dimensional reference solution is obtained using a refined mesh with a

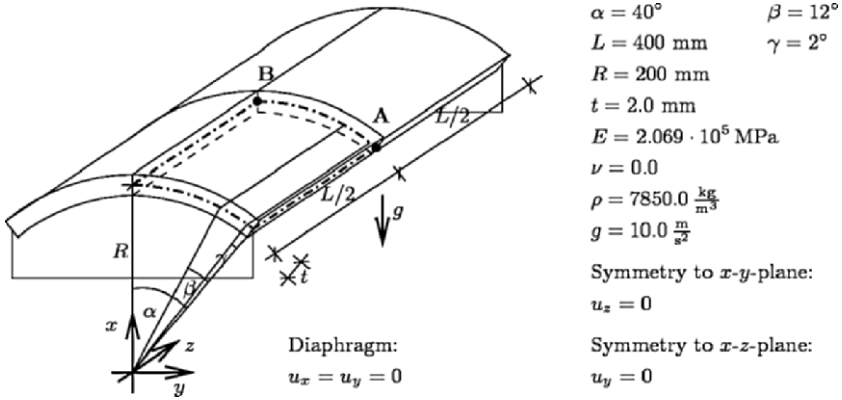


Fig. 8. Scordelis & Lo roof with dead load

high polynomial degree for all elements. To investigate the accuracy of the mapping concept, computations are performed on a mesh consisting of three elements, which are refined towards the free edge to resolve boundary layers (see Figure 8). The geometry is interpolated by Lagrangian polynomials of degrees $p_g = 3, 4, 8$. For each degree p_g a p -extension based on the trunk space $\mathcal{S}_{\text{ts}}^{p,p}(\Omega_{\text{st}}^h)$ is performed and the relative error in energy norm is computed. Figure 9 shows that $p_g = 3$ is not sufficient, but there is hardly any difference between an interpolation with $p_g = 4$ and $p_g = 8$. In general the interpolation error due to a finite degree $p_g = \text{const.}$ will, of course, not be known in advance. Therefore, we recommend to apply an “isoparametric” approach, where the polynomial degree of the geometric description p_g is chosen to be equal to that of the shape functions interpolating the displacement field. This approach ensures that a p -extension also reduces the interpolation error of geometry. Concerning the computational cost, the interpolation of the geometry amounts to a small fraction of the overall computational work involved.

A Hemispherical Shell with Stiffener In the previous example the accuracy of the geometric description, i.e. the mapping functions was investigated. As a further example we consider a hemispherical shell including a stiffener to demonstrate that the high order solid finite element formulation allows to simultaneously discretize thin and thick-walled structures. A transition from thin to thick-walled constructions is thus possible without the necessity of coupling models of differing dimensions and without imposing any restrictions on the (three-dimensional) kinematics of the structure. Geometry, boundary conditions and material parameters of linear elasticity are depicted in Figure 10. The cylindrical stiffener with a square cross-section is hard simply supported at the bottom surface. The structure is subjected to self-weight and

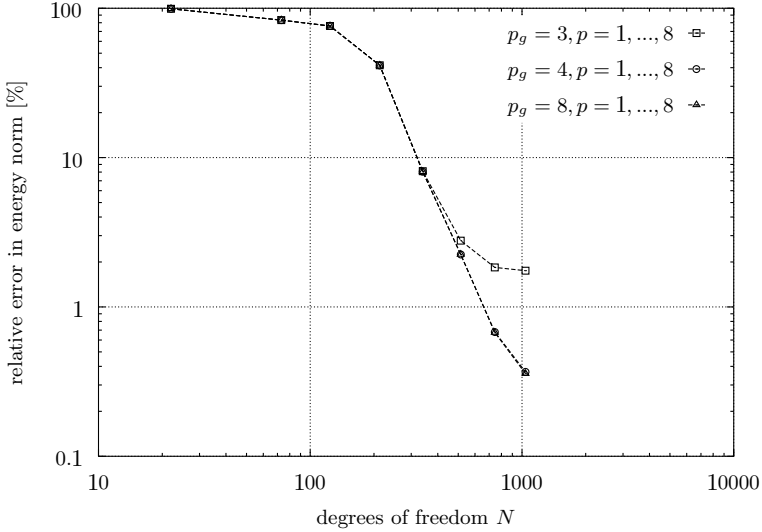


Fig. 9. Convergence study: p_g represents the polynomial degree of geometry representation, p corresponds to the polynomial degree of the Ansatz space $\mathcal{S}_{\text{ts}}^{p,p}(\Omega_{\text{st}}^{\text{h}})$

a pressure q is acting on the outer surface of the shell and stiffener. Due to symmetry only a quarter of the system has to be considered. This example has been investigated in [20] in order to compare a uniform p -extension for two different high order shell formulations.

The finite element mesh consists of 117 hexahedral elements, which are geometrically refined towards the re-entrant corners where stress singularities have to be expected, see Figures 11, 12. The aim of the computation based on an adaptive scheme is to reduce the error in energy norm down to a level of approximately 1%. The adaptive scheme, developed for elastostatic problems has been presented in detail in [22]. Its extension to elastodynamic problems is described in Section 4. In a first step we set the in-plane polynomial degree for all elements to $p_k = 4$ and automatically compute a hierarchic sequence where q_k is increased adaptively. After 6 levels of refinement we find the corresponding elementwise distribution of the transversal polynomial degree q_k which is plotted in Figure 11. From this we observe that the highest polynomial degree in thickness direction of order 6 is assigned to the shell elements which are close to the stiffener. The needle elements located at the re-entrant corners do not need to have a high degree which is in good accordance to an hp -refinement.

Having adapted the polynomial degree in thickness direction we compute a sequence of hierarchic refinements where the in-plane polynomial degree is automatically adapted. The adaptive refinement is carried out until the estimated error in energy norm is approximately 1%. In this case 5 adaptive

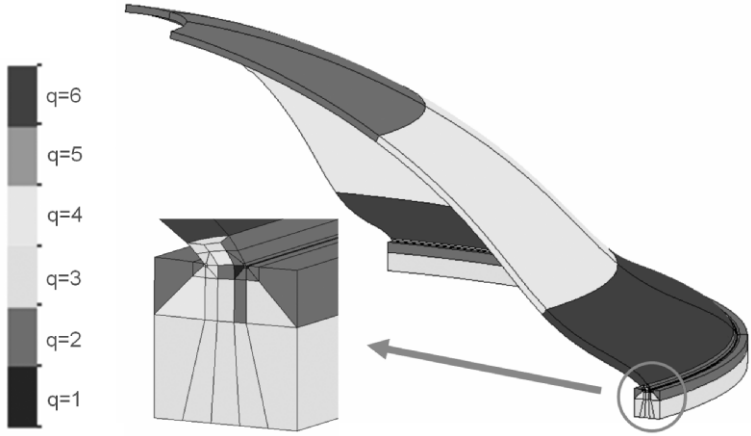


Fig. 11. Distribution of polynomial degree q_k (deformation is scaled by a factor of 500)

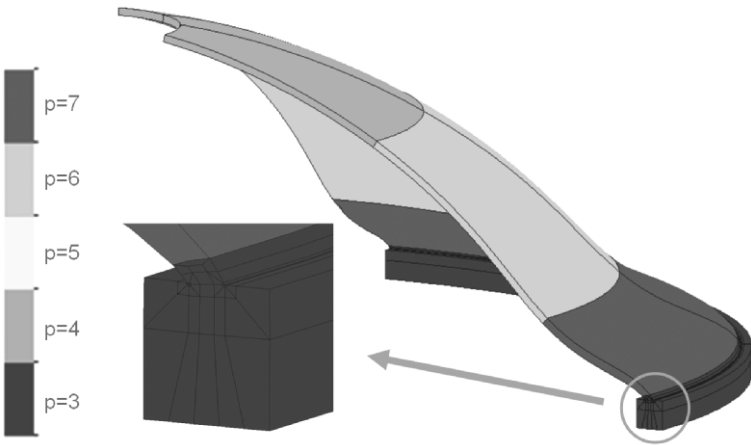


Fig. 12. Distribution of polynomial degree p_k (deformation is scaled by a factor of 500)

(P4, 3 GHz), yielding an approximation with less than 1.3% error in energy norm.

3 CSD for Geometrically Nonlinear Thin Solids

Since only very few approaches for structural dynamic computations using *high order solid elements* are known to the authors, this section briefly recalls the generalized α -method used for time integration and presents a numerical

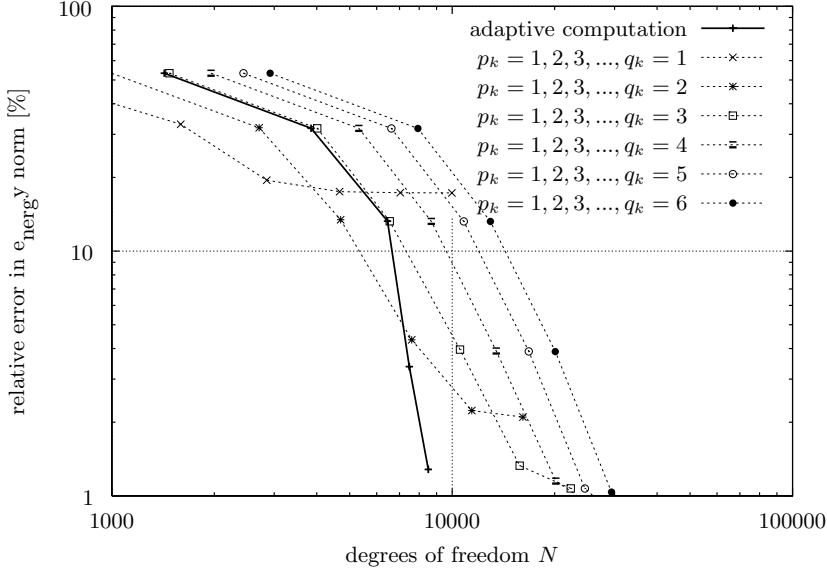


Fig. 13. Error in energy norm

example for verification of its suitability and implementation for hexahedral p -elements.

The semidiscrete equations of elastodynamics read

$$\mathbf{R}(\mathbf{u}) + \mathbf{M}\ddot{\mathbf{d}} - \mathbf{f} = \mathbf{0}, \quad (11)$$

with the (continuous) displacements $\mathbf{u}$, the finite element internal force vector $\mathbf{R}$ (generally depending on $\mathbf{u}$), the mass matrix $\mathbf{M}$, the finite element displacement vector $\mathbf{d}$ and its second time derivative $\ddot{\mathbf{d}}$ and the finite element external force vector $\mathbf{f}$. From experience it is known that vibrations are damped due to friction or viscous behavior. Often, damping effects are modeled by an additional damping term $\mathbf{C}\dot{\mathbf{d}}$ with the constant damping matrix $\mathbf{C}$. In many applications the damping matrix is a linear combination of the mass and the stiffness matrix, $\mathbf{C} = a\mathbf{K} + b\mathbf{M}$, which is referred to as RAYLEIGH damping having advantages in modal analysis of linear elastodynamic calculations, see e.g. BATHE [23]. With the additional damping term the semidiscrete equations of motion are given as

$$\mathbf{R}(\mathbf{u}) + \mathbf{C}\dot{\mathbf{d}} + \mathbf{M}\ddot{\mathbf{d}} - \mathbf{f} = \mathbf{0}. \quad (12)$$

These equations are to be fulfilled at every time t . To complete the definition of the initial value problem, initial conditions have to be prescribed at

$t_0 = 0$:

$$\mathbf{u}(t_0) = \bar{\mathbf{u}}_0, \quad (13)$$

$$\dot{\mathbf{u}}(t_0) = \bar{\dot{\mathbf{u}}}_0. \quad (14)$$

Since in general it is not possible to solve (12) analytically, numerical *time integration methods* have to be used. A very good overview over existing approaches can be found in HULBERT [24].

In this work an implicit finite difference based method, the generalized α -method was implemented, being a generalization of the NEWMARK method [25], which in turn is probably the most popular time integration scheme for finite element systems of type (12) of the last 30 years. The NEWMARK algorithm is based on the following approximations at time t_{n+1} :

$$\mathbf{d}_{n+1} = \mathbf{d}_n + \Delta t \dot{\mathbf{d}}_n + \frac{\Delta t^2}{2} \left[(1 - 2\beta) \ddot{\mathbf{d}}_n + 2\beta \ddot{\mathbf{d}}_{n+1} \right], \quad (15)$$

$$\dot{\mathbf{d}}_{n+1} = \dot{\mathbf{d}}_n + \Delta t \left[(1 - \gamma) \ddot{\mathbf{d}}_n + \gamma \ddot{\mathbf{d}}_{n+1} \right], \quad (16)$$

with the timestep size Δt and the NEWMARK parameters β and γ . HUGHES [26] postulates the following attributes of a time integration scheme to be competitive:

1. unconditionally stable when applied to linear problems,
2. no more than one set of implicit equations to be solved at each timestep,
3. second order accuracy,
4. controllable algorithmic dissipation of higher modes,
5. self-starting.

The NEWMARK algorithm is self-starting and only one implicit set of equations has to be solved, therefore requirements 2 and 5 are fulfilled. The attributes 1, 3 and 4 depend on the NEWMARK parameters β , γ , see [23,26–29] for details.

For linear problems and for $2\beta \geq \gamma \geq \frac{1}{2}$, NEWMARK's algorithm is unconditionally stable. For $\gamma > \frac{1}{2}$ algorithmic dissipation is introduced, i.e. the system is damped numerically which may be desired to damp artificial higher vibration modes. However, the only combination of parameters guaranteeing second order accuracy of the integration method is $\beta = \frac{1}{4}$ and $\gamma = \frac{1}{2}$, i.e. attributes 3 and 4 are mutually exclusive.

Many methods have been developed to overcome these difficulties. A very popular approach is the generalized α -method, using the NEWMARK formulae (15) and (16) as well, but differently weighting the force terms in the equation of motion (12) by introducing the additional parameters α_f , α_m

$$\mathbf{M} \ddot{\mathbf{d}}_{n+1-\alpha_m} + \mathbf{C} \dot{\mathbf{d}}_{n+1-\alpha_f} + \mathbf{R}_{n+1-\alpha_f} = \mathbf{f}(t_{n+1-\alpha_f}). \quad (17)$$

with

$$t_{n+1-\alpha_f} = (1 - \alpha_f)t_{n+1-\alpha_f} + \alpha_f t_n, \quad (18)$$

$$\mathbf{R}_{n+1-\alpha_f} = (1 - \alpha_f)\mathbf{R}(\mathbf{u}_{n+1}) + \alpha_f \mathbf{R}(\mathbf{u}_n), \quad (19)$$

$$\dot{\mathbf{d}}_{n+1-\alpha_f} = (1 - \alpha_f)\dot{\mathbf{d}}_{n+1} + \alpha_f \dot{\mathbf{d}}_n, \quad (20)$$

$$\ddot{\mathbf{d}}_{n+1-\alpha_m} = (1 - \alpha_m)\ddot{\mathbf{d}}_{n+1} + \alpha_m \ddot{\mathbf{d}}_n. \quad (21)$$

$\mathbf{R}_{n+1-\alpha_f}$ in (19) is defined analogously to CRISFIELD [30], see e.g. HEISERER [31]. Alternatively, it can be chosen to be $\mathbf{R}_{n+1-\alpha_f} = \mathbf{R}(\mathbf{u}_{n+1-\alpha_f})$, see e.g. WRIGGERS [29].

The generalized α -method eliminates the mutual exclusion of second order accuracy and algorithmic damping for linear problems. Analysis of the method is e.g. given in CHUNG and HULBERT [32], concluding that second order accuracy, unconditional stability and maximized algorithmic damping is achieved with

$$\gamma = \frac{1}{2} - \alpha_m - \alpha_f, \quad (22)$$

$$\beta = \frac{1}{4}(1 - \alpha_m + \alpha_f)^2 \quad (23)$$

$$\alpha_m = 2\rho_\infty \quad (24)$$

$$\alpha_f = \frac{\rho_\infty}{\rho_\infty + 1}. \quad (25)$$

ρ_∞ is the user-specified value for high frequency damping. $\rho_\infty = 1$ corresponds to no algorithmic damping, $\rho_\infty = 0$ results in maximum algorithmic high frequency dissipation.

The generalized α -method reduces for $\alpha_m = 0$ to the HHT- α method proposed by HILBER, HUGHES and TAYLOR [33] and with $\alpha_f = 0$ the WBZ- α method proposed by WOOD, BOSSAK and ZIENKIEWICZ [34] is obtained.

3.1 Numerical Example

Although the generalized α -method has been investigated extensively for low-order finite element approximations, only little is known about its suitability to high-order elements, especially for the case of thin solid elements as applied in our fluid structure interaction approach. For verification of our p -version code, and to illustrate the differences between linear and nonlinear vibrations, the transient forced response of the clamped plate depicted in Figure 14 is computed. The length of the quadratic plate is $L = 12$, the thickness equals to $t = 0.06$, YOUNG's modulus is $E = 2.1 \cdot 10^{11}$, POISSON's ratio is $\nu = 0.3$ and the density is $\rho = 7850$. The uniform face load starting to act in negative z -direction on the upper surface at $t_0 = 0$ is constant over time, $t_z = -5692$.

The plate is discretized into 49 high order hexahedral elements with uniform polynomial degree $p = 5$ in all directions for all elements. For the

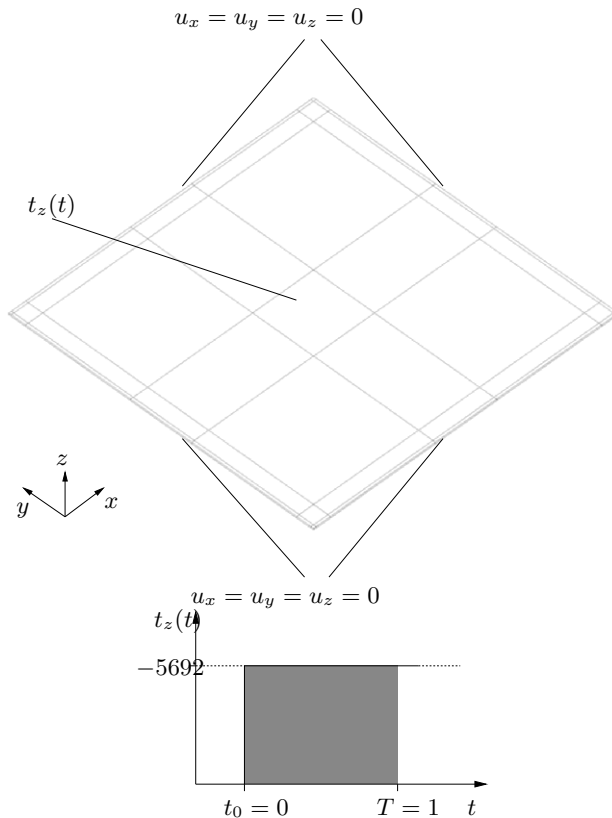


Fig. 14. System and hexahedral mesh of clamped plate (top) and load function over time $t_z(t)$ (bottom)

time discretization, the generalized α -method was used with a timestep size of $\Delta t = 0.0003375$ and $\rho_\infty = 0.8$. The problem was computed considering both geometric linear and geometric nonlinear behavior and the results were compared to corresponding results obtained by YOSIBASH and KIRBY [35] and KIRBY and YOSIBASH [36] using a spectral element approach for solving the two-dimensional geometric linear KIRCHHOFF plate model and the two-dimensional geometric nonlinear von-Kármán plate model. In Figure 15 the centerpoint displacements are plotted over time. The accordance of the three-dimensional and two-dimensional approaches is obvious for this thin plate computed geometrically linear and nonlinear. The necessity of considering the nonlinear effects is evident for this plate clamped on all four sides.

However, use of anisotropic Ansatz spaces has not been made, yet. An anisotropic p -adaptive method is proposed in Section 4 for linear elastodynamic problems.

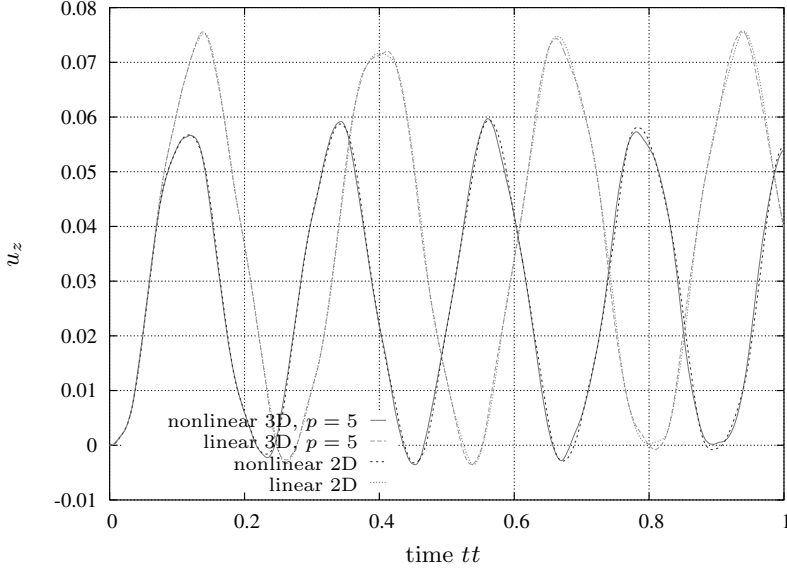


Fig. 15. Centerpoint displacements of plate over time

4 Control of Model and Spatial Discretization Error

In order to control the spatial discretization error, a p -adaptive method for linear elastodynamic problems is proposed in this section. The *temporal* domain is assumed to be discretized accurately enough. However, time step adaptivity as suggested e.g. in [37–40] could be applied in a straight forward manner.

Time integration methods for linear systems can be categorized into *direct* and *indirect* schemes. When using *direct* time integration, the time integration scheme is applied directly to the semidiscrete coupled system of ordinary differential equations

$$\mathcal{B}(\mathbf{u}, \mathbf{v}) + \mathcal{C}(\dot{\mathbf{u}}, \mathbf{v}) + \mathcal{D}(\ddot{\mathbf{u}}, \mathbf{v}) = \mathcal{F}(\mathbf{v}), \quad \forall \mathbf{v} \in \mathcal{S} \quad (26)$$

with $\mathcal{S}$ being the finite elements space, or in matrix notation

$$\mathbf{K}\mathbf{d}(t) + \mathbf{C}\dot{\mathbf{d}}(t) + \mathbf{M}\ddot{\mathbf{d}}(t) = \mathbf{F}(t). \quad (27)$$

Usually, direct implicit time integration requires the solution of linear equation systems. For time integration schemes of the NEWMARK family or related schemes, direct time integration demands for the solution of systems with the number of spatial degrees of freedom N in size in every timestep.

When the eigenvectors ϕ_k associated with the eigenfrequencies ω_k of the system are not only $\mathbf{K}$ - and $\mathbf{M}$ -orthogonal but also $\mathbf{C}$ -orthogonal, e.g. in

case of RAYLEIGH damping, or $\mathbf{C} = \mathbf{0}$, the system (27) can be completely decoupled by modal analysis. Integrating these decoupled ordinary differential equations is referred to as *indirect* time integration. This requires the solution of the eigenvalue problem corresponding to (27) with $\mathbf{F} = \mathbf{0}$

$$\mathbf{K}\phi_k = \omega_k^2 \mathbf{M}\phi_k. \quad (28)$$

The main computational effort in direct time integration is the repeated solution of the linear equation system depending on the integration scheme, in the case of indirect time integration the dominating task is the one-time solution of the eigenvalue problem (28).

In most applications, also in fluid structure interaction, specific frequencies and corresponding eigenvectors dominate the solution of the time dependent problem, e.g. resulting from a specific spatial distribution of the load, a specific load frequency, resonance effects, etc. The dominating frequencies are usually only few compared to the number of spatial degrees of freedom of the system. Furthermore, for many practical problems only the lowest frequencies are of interest. In Figure 16 (top) the internal energy — the sum of kinetic and strain energy $E_{kin} + E_{str}$ — of a clamped square plate subjected to a sudden uniform surface load is plotted, whereas Figure 16 (bottom) shows the decomposition of $E_{kin} + E_{str}$ into the proportions associated to the lowest five eigenfrequencies. It is obvious that in this case the first eigenfrequency clearly dominates the dynamic behavior of the plate. The plate example will be investigated in further detail in Section 4.3. These facts can be exploited by adapting indirect time integration schemes to only those eigenfrequencies and eigenvectors dominating the problem. In the case of direct integration schemes, spatial discretizations are sought, which are able to *represent* these dominating eigenfrequencies and eigenvectors *efficiently*.

Independent from which time integration method is used, direct or indirect, an efficient spatial discretization for the solution of (28) is desired, which can be found by an adaptive eigensolver representing the important eigenfrequencies and the corresponding eigenvectors accurately with adaptive refinements saving computational resources.

4.1 An Anisotropic p -Adaptive Hierarchic Eigensolver

In order to efficiently represent specific eigenmodes, an anisotropic p -adaptive eigensolver adapts the finite element discretization determined by the polynomial degrees of the shape functions $\mathcal{P}$, the mesh Δ and the mapping $\mathcal{Q}$

$$\mathcal{S} = \mathcal{S}(\mathcal{P}, \Delta, \mathcal{Q}), \quad (29)$$

only by varying the set of polynomial degree templates

$$\mathcal{P} = \{\mathbf{p}_1, \dots, \mathbf{p}_{n_{el}}\} \quad (30)$$

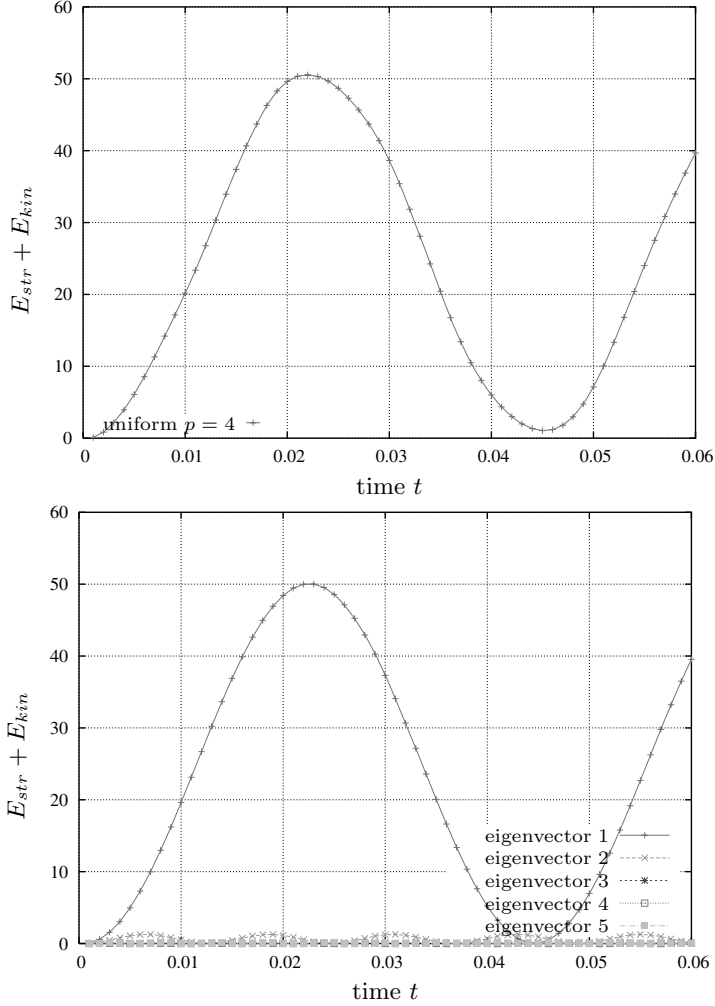


Fig. 16. Internal energy $E_{str} + E_{kin}$ of clamped plate subjected to sudden surface load over time (top) and decomposition of $E_{kin} + E_{str}$ into the proportions associated to the lowest five eigenfrequencies (bottom)

with

$$\mathbf{p}_e = \begin{matrix} & \begin{matrix} u_x & u_y & u_z \end{matrix} \\ \begin{matrix} \xi \\ \eta \\ \zeta \end{matrix} & \begin{pmatrix} p_e^{1,1} & p_e^{1,2} & p_e^{1,3} \\ p_e^{2,1} & p_e^{2,2} & p_e^{2,3} \\ p_e^{3,1} & p_e^{3,2} & p_e^{3,3} \end{pmatrix} \end{matrix}. \quad (31)$$

Therefore, for a fixed mesh and fixed mapping, the finite element discretization can be written as

$$\mathcal{S} = \mathcal{S}^{\mathcal{P}}. \quad (32)$$

The sensitivity of the solution of the eigenvalue problem (28) has to be analyzed with respect to changes of the discretization $\mathcal{S}^{\mathcal{P}}$ in order to be in the position to set up a p -adaptive procedure.

An Anisotropic Hierarchic Error Indicator For the dynamic eigenvalue problem, a reasonable measure for quantifying the sensitivity of the solution to the change of a specific $p_{\bar{e}}^{\bar{i},\bar{j}}$ in (31) has turned out to be the RAYLEIGH quotient ρ , which is related to the lowest circular frequency ω_1 as follows:

$$\omega_1^2 = \min_{\mathbf{u} \in \mathcal{V}} \rho(\mathbf{u}) = \min_{\mathbf{u} \in \mathcal{V}} \frac{\mathcal{B}(\mathbf{u}, \mathbf{u})}{\mathcal{D}(\mathbf{u}, \mathbf{u})} = \frac{\mathcal{B}(\mathbf{u}_1, \mathbf{u}_1)}{\mathcal{D}(\mathbf{u}_1, \mathbf{u}_1)}, \quad (33)$$

i.e., the eigenfunction $\mathbf{u}_1$ corresponding to ω_1 is that function $\mathbf{u} \in \mathcal{V}$ minimizing the RAYLEIGH quotient ρ . The higher frequencies ω_k are obtained by searching the eigenfunctions $\mathbf{u}_k$ in the space $\mathcal{V}_{k\perp}$ restricted to being orthogonal to the already obtained eigenfunctions $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$

$$\omega_k^2 = \min_{\mathbf{u} \in \mathcal{V}_{k\perp}} \rho(\mathbf{u}) = \min_{\mathbf{u} \in \mathcal{V}_{k\perp}} \frac{\mathcal{B}(\mathbf{u}, \mathbf{u})}{\mathcal{D}(\mathbf{u}, \mathbf{u})} = \frac{\mathcal{B}(\mathbf{u}_k, \mathbf{u}_k)}{\mathcal{D}(\mathbf{u}_k, \mathbf{u}_k)}. \quad (34)$$

Since only finite dimensional Ansatz spaces $\mathcal{S}^{\mathcal{P}}$, $\mathcal{S}_{k\perp}^{\mathcal{P}}$ are used instead of $\mathcal{V}$, $\mathcal{V}_{k\perp}$, respectively, we obtain the approximations $\omega_1^{\mathcal{P}}$ with

$$\omega_1^2 \approx \min_{\mathbf{u} \in \mathcal{S}^{\mathcal{P}}} \rho(\mathbf{u}) = \frac{\mathcal{B}(\mathbf{u}_1^{\mathcal{P}}, \mathbf{u}_1^{\mathcal{P}})}{\mathcal{D}(\mathbf{u}_1^{\mathcal{P}}, \mathbf{u}_1^{\mathcal{P}})} = \frac{\phi_1^T \mathbf{K} \phi_1}{\phi_1^T \mathbf{M} \phi_1} = (\omega_1^{\mathcal{P}})^2 \quad (35)$$

and $\omega_k^{\mathcal{P}}$ with

$$\omega_k^2 \approx \min_{\mathbf{u} \in \mathcal{S}_{k\perp}^{\mathcal{P}}} \rho(\mathbf{u}) = \frac{\mathcal{B}(\mathbf{u}_k^{\mathcal{P}}, \mathbf{u}_k^{\mathcal{P}})}{\mathcal{D}(\mathbf{u}_k^{\mathcal{P}}, \mathbf{u}_k^{\mathcal{P}})} = \frac{\phi_k^T \mathbf{K} \phi_k}{\phi_k^T \mathbf{M} \phi_k} = (\omega_k^{\mathcal{P}})^2. \quad (36)$$

Applying a sequence of hierarchically nested finite element spaces

$$\mathcal{S}^{\mathcal{P}_1} \subset \mathcal{S}^{\mathcal{P}_2} \subset \mathcal{S}^{\mathcal{P}_3} \subset \dots, \quad (37)$$

the sequence of approximations

$$\omega_k^{\mathcal{P}_1} \geq \omega_k^{\mathcal{P}_2} \geq \omega_k^{\mathcal{P}_3} \geq \dots \quad (38)$$

is obtained. Two hierarchic spaces $\mathcal{S}^{\mathcal{P}}$ and $\mathcal{S}^{\mathcal{P}^-}$ are constructed by varying one specific entry $p_{\bar{e}}^{\bar{i},\bar{j}}$ in the set of polynomial degree templates

$$\mathcal{S}^{\mathcal{P}} \supset \mathcal{S}^{\mathcal{P}^-} = \mathcal{S}^{\mathcal{P} - \Delta \mathcal{P}_{\bar{e}}^{\bar{i},\bar{j}}} \quad (39)$$

with

$$\Delta \mathcal{P}_{\bar{e}}^{\bar{i},\bar{j}} = \begin{cases} \Delta p_{\bar{e}}^{\bar{i},\bar{j}} > 0 \\ \Delta p_{\bar{e}}^{\bar{i},\bar{j}} = 0 \end{cases} \quad \text{for } \{e, i, j\} \neq \{\bar{e}, \bar{i}, \bar{j}\} \quad (40)$$

in order to analyze the sensitivity of the solution with respect to $p_{\bar{e}}^{\bar{i},\bar{j}}$. An anisotropic hierarchic error indicator for the entry $p_{\bar{e}}^{\bar{i},\bar{j}}$ based on reduction can be written as

$$\lambda_{\bar{e},k}^{\bar{i},\bar{j}} = \rho(\mathbf{u}^{\mathcal{P}^-}) - \rho(\mathbf{u}^{\mathcal{P}}) \geq 0 \quad (41)$$

An Anisotropic Hierarchic Error Indicator Computed on Element Level Since the global computation of $\rho(\mathbf{u}^{\mathcal{P}^-})$ for every combination of $\{e, i, j\}$ is prohibitively expensive, the cost is reduced by an approximation, which can be computed locally on element level and explicitly, i.e. without solving an equation system.

The two assumptions for computing the local, explicit approximation of (41) are

- that the influence of varying $p_{\bar{e}}^{\bar{i},\bar{j}}$ may be reduced to the domain of element $\bar{e}$, and
- that $\phi_{\bar{e},k}^{\mathcal{P}^-}$ can be approximated by simply cutting off those entries of $\phi_{\bar{e},k}^{\mathcal{P}}$ corresponding to the difference template $\Delta\mathcal{P}_{\bar{e}}^{\bar{i},\bar{j}}$

$$\tilde{\phi}_{\bar{e}}^{\mathcal{P}^-} = \pi \phi_{\bar{e}}^{\mathcal{P}}, \quad (42)$$

with π being the cut-off operator.

Using these assumptions, the anisotropic hierarchic error indicator computed on element level can be written as

$$\begin{aligned} \lambda_{\bar{e}}^{\bar{i},\bar{j}} &= \rho(u^{\mathcal{P}^-}) - \rho(u^{\mathcal{P}}) \\ &= \frac{\phi_{1,k}^{\mathcal{P}^{-T}} \mathbf{K}_1 \phi_{1,k}^{\mathcal{P}^-} + \dots + \phi_{\bar{e},k}^{\mathcal{P}^{-T}} \mathbf{K}_{\bar{e}} \phi_{\bar{e},k}^{\mathcal{P}^-} + \dots + \phi_{n_{el},k}^{\mathcal{P}^{-T}} \mathbf{K}_{n_{el}} \phi_{n_{el},k}^{\mathcal{P}^-}}{\phi_{1,k}^{\mathcal{P}^{-T}} \mathbf{M}_1 \phi_{1,k}^{\mathcal{P}^-} + \dots + \phi_{\bar{e},k}^{\mathcal{P}^{-T}} \mathbf{M}_{\bar{e}} \phi_{\bar{e},k}^{\mathcal{P}^-} + \dots + \phi_{n_{el},k}^{\mathcal{P}^{-T}} \mathbf{M}_{n_{el}} \phi_{n_{el},k}^{\mathcal{P}^-}} - \rho(u^{\mathcal{P}}) \\ &\approx \left| \frac{\phi_{1,k}^{\mathcal{P}^T} \mathbf{K}_1 \phi_{1,k}^{\mathcal{P}} + \dots + \tilde{\phi}_{\bar{e},k}^{\mathcal{P}^{-T}} \mathbf{K}_{\bar{e}} \tilde{\phi}_{\bar{e},k}^{\mathcal{P}^-} + \dots + \phi_{n_{el},k}^{\mathcal{P}^T} \mathbf{K}_{n_{el}} \phi_{n_{el},k}^{\mathcal{P}}}{\phi_{1,k}^{\mathcal{P}^T} \mathbf{M}_1 \phi_{1,k}^{\mathcal{P}} + \dots + \tilde{\phi}_{\bar{e},k}^{\mathcal{P}^{-T}} \mathbf{M}_{\bar{e}} \tilde{\phi}_{\bar{e},k}^{\mathcal{P}^-} + \dots + \phi_{n_{el},k}^{\mathcal{P}^T} \mathbf{M}_{n_{el}} \phi_{n_{el},k}^{\mathcal{P}}} - \rho(u^{\mathcal{P}}) \right|. \end{aligned} \quad (43)$$

In (43) the terms $\phi_{e,k}^{\mathcal{P}^T} \mathbf{K}_e \phi_{e,k}^{\mathcal{P}}$ and $\phi_{e,k}^{\mathcal{P}^T} \mathbf{M}_e \phi_{e,k}^{\mathcal{P}}$ are required for all elements $e = 1, \dots, n_{el}$ such that computations on all elements seem to be necessary for this approximation of $\lambda_{\bar{e}}^{\bar{i},\bar{j}}$. However, these element contributions have to be computed only once and may be stored temporarily. Therefore, for all subsequent computations, only element level calculations are required.

A p -Adaptive Strategy for the Eigenvalue Problem In the following the strategy for the p -adaptive approach is outlined. As a stopping criterion a hierarchic extrapolation may be used to compute a reference solution ω_k^{ref}

to be compared to a prescribed tolerance TOL . Alternatively, a maximum number of iterations may be prescribed, or a maximum number of degrees of freedom N_{max} . The prescription of N_{max} may be particularly desirable when an efficient discretization for a time stepping scheme is sought, where the transient problem is dominated by specific frequencies, see Section 4.2, and available computational resources are limited. The number of degrees of freedom in a direct time integration scheme is, of course, even more crucial than in the static case, since a system of order N equations has to be solved in every time step.

The error indicator used for obtaining the sensitivity of the frequency ω_k to a variation of a specific entry $p_e^{i,j}$ is the explicit error indicator computed on element level (43), where a ‘look back’ difference polynomial degree $\Delta p_e^{i,j} = 2$ is used. The error indicators $\lambda_{e,k}^{i,j}$ corresponding to the entries $p_e^{i,j}$ are sorted and fixed proportions are refined according to a staggered scheme, see algorithm A-1.

Since adaption to more than one eigenfrequency ω_k , $k = k_1, \dots, k_n$ may be desired, where $k_1, \dots, k_n$ do not necessarily have to be sequential integers, it is recommended to choose the new polynomial degree template to be the maximum hull of the templates $\mathcal{P}_{iter+1,k}$ suggested for adaption to all frequencies ω_k

$$\mathcal{P}_{iter+1} = \max_{k=k_1, \dots, k_n} p_{e,iter+1,k}^{i,j}, \quad (44)$$

see the inner loop over k in the following algorithm. Alternatively, an outer loop over the number of frequencies to be adapted to, n , could be used, to call the eigensolver for every frequency ω_k , separately. However, this might be less efficient depending on the eigensolver used, e.g. the LANCZOS based eigensolver used here computes all frequencies in one run much more efficiently than in separate runs for every ω_k , see SCHWARZ [41,42] and RABOLD [43].

4.2 A p -Adaptive Strategy for Time Dependent Problems

The p -adaptive strategy for solving time dependent elastodynamic problems suggested here consists of the following main steps:

1. Find frequencies dominating the problem by initial computation of the transient problem with coarse spatial discretization,
2. find a discretization adapted to dominating frequencies and
3. recompute problem of interest using adaptively found discretization.

The task of identifying the dominating frequencies will be described in the following. For an adaptive computation of the governing frequencies, the p -adaptive eigensolver of Section 4.1 is used. The initial time dependent computation described in the following and the recomputation with the adaptively

Algorithm A-1 p -adaptive strategy for eigenvalue problem

```

1: set  $iter = 0$ ,  $\mathcal{P}_{iter}$  uniformly to  $p_{e,iter}^{i,j} = 4$ 
2: repeat
3:   compute  $\omega_k^{\mathcal{P}_{iter}}$ ,  $k = k_1, \dots, k_n$ 
4:   for  $k = k_1, \dots, k_n$  do
5:     for all  $\{e, i, j\}$  compute  $\lambda_{e,k}^{i,j}$  using (43)
6:     sort all  $\lambda_{e,k}^{i,j}$  and find  $\lambda_{70\%}$ ,  $\lambda_{90\%}$ 
7:     set

$$\Delta \mathcal{P}_{iter+1,k} = \Delta p_{e,iter+1,k}^{i,j} = \begin{cases} 0 & \text{for } \check{\lambda}_e^{i,j} < \lambda_{70\%}(1 - \varepsilon) \\ 1 & \text{for } \lambda_{70\%}(1 - \varepsilon) \leq \check{\lambda}_e^{i,j} < \lambda_{90\%}(1 - \varepsilon) \\ 2 & \text{for } \lambda_{90\%}(1 - \varepsilon) \leq \check{\lambda}_e^{i,j} \end{cases}$$

8:     set  $\mathcal{P}_{iter+1,k} = \mathcal{P}_{iter} + \Delta \mathcal{P}_{iter+1,k}$ 
9:     if ( $iter \geq 2$ ) then
10:       compute  $\eta_k = \omega_k^{\mathcal{P}_{iter}} - \omega_k^{ref}$ 
11:     else
12:       set  $\eta_k = TOL + \varepsilon$  (stopping criterion not fulfilled yet)
13:     end if
14:   end for
15:   set

$$\mathcal{P}_{iter+1} = \max_{k=k_1, \dots, k_n} p_{e,iter+1,k}^{i,j},$$


$$\eta = \max_{k=k_1, \dots, k_n} \eta_k,$$


$$iter = iter + 1$$

16: until ( $\eta < TOL$ ) OR ( $iter > maxiter$ ) OR ( $N_{iter} > N_{max}$ )

```

chosen discretization may be computed using an arbitrary time integration scheme. The numerical example discussed in Section 4.3 was computed applying members of the generalized- α -family of algorithms, see Section 3.

To identify the dominating frequencies, an initial solution of the structural dynamic problem (27) is computed with an initial uniform polynomial degree distribution $p_e^{i,j} = 4$, where $\mathbf{d}(t), \dot{\mathbf{d}}(t)$ are required in the following to extract the prevailing frequencies. Further, the eigenvalue problem (28) has to be solved, yielding the eigenfrequencies $\omega_1, \dots, \omega_{n_{eigen}}$ with $n_{eigen} \ll N$, again using an initial uniform p -distribution $p_e^{i,j} = 4$.

Knowing the approximations of eigenvectors ϕ_k , the approximations $\mathbf{d}(t), \dot{\mathbf{d}}(t)$ can be projected onto the corresponding eigenvector space spanned by ϕ_k . If all N eigenvectors were known, the amplitudes $\mathbf{x}(t)$ associated with the eigenvectors ϕ_k would have to fulfill the relation

$$\mathbf{d}(t) = \Phi \mathbf{x}(t) \quad (45)$$

in order to fully describe the displacement vector $\mathbf{d}(t)$ with

$$\Phi = [\phi_1, \dots, \phi_N] \quad (46)$$

and

$$\mathbf{x}(t) = [x_1(t), \dots, x_N(t)]^T. \quad (47)$$

Multiplying (45) with $\Phi^T \mathbf{M}$ from the left we obtain

$$\Phi^T \mathbf{M} \mathbf{d}(t) = \Phi^T \mathbf{M} \Phi \mathbf{x}(t) = \mathbf{x}(t), \quad (48)$$

using the $\mathbf{M}$ -orthogonality

$$\phi_i^T \mathbf{M} \phi_j = \delta_{ij} \quad (49)$$

of the eigenvectors, or

$$x_k(t) = \phi_k^T \mathbf{M} \mathbf{d}(t). \quad (50)$$

Accordingly, in order to project the velocity vector $\dot{\mathbf{d}}(t)$ onto the eigenvector space, we can differentiate (50) with respect to time to obtain

$$\dot{x}_k(t) = \phi_k^T \mathbf{M} \dot{\mathbf{d}}(t). \quad (51)$$

It is obvious that for computation of (50) and (51) only the eigenvector ϕ_k is required, not the complete set of N eigenvectors as required first in Equation (45). Knowing $x_k, \dot{x}_k, k = 1, \dots, n_{eigen}$, the strain energy associated with eigenvector ϕ_k can be easily computed as

$$E_{str,k}(t) = \frac{1}{2} (\phi_k x_k(t))^T \mathbf{K} \phi_k x_k(t) \quad (52)$$

$$= \frac{1}{2} x_k^2(t) \phi_k^T \mathbf{K} \phi_k \quad (53)$$

$$= \frac{1}{2} x_k^2(t) \omega_k^2 \quad (54)$$

using the $\mathbf{K}$ -orthogonality of eigenvectors

$$\phi_i^T \mathbf{K} \phi_j = \delta_{ij} \omega_i^2, \quad (55)$$

and, using again (49), the kinetic energy associated to ϕ_k as

$$E_{kin,k}(t) = \frac{1}{2} (\phi_k \dot{x}_k(t))^T \mathbf{M} \phi_k \dot{x}_k(t) \quad (56)$$

$$= \frac{1}{2} \dot{x}_k^2(t) \phi_k^T \mathbf{M} \phi_k \quad (57)$$

$$= \frac{1}{2} \dot{x}_k^2(t). \quad (58)$$

Finally, a reasonable measure is required to decide which eigenfrequencies dominate the solution of the transient problem and, since only n_{eigen} eigenfrequencies are considered instead of N eigenfrequencies existing for the spatially discretized system, whether this reduction is justified, i.e. n_{eigen} is sufficiently large.

For identifying the prevailing frequencies, the *mean internal energy* of the system

$$\bar{E}_{int} = \frac{1}{T} \int_T E_{str}(t) + E_{kin}(t) dt \quad (59)$$

can be used to compare the proportions

$$\bar{E}_{int,k} = \frac{1}{T} \int_T E_{str,k}(t) + E_{kin,k}(t) dt \quad (60)$$

contributed by the different eigenfrequencies ω_k . A frequency ω_k can be regarded as dominant, when its proportion of the overall mean internal energy exceeds a certain limit

$$\frac{\bar{E}_{int,k}}{\bar{E}_{int}} > r_1, \quad (61)$$

where r_1 can be chosen according to the accuracy desired, e.g. $r_1=0.05$.

Furthermore, it should be ensured that energies of the chosen frequencies represent a sufficient proportion of the overall mean internal energy, i.e.

$$\frac{\sum_{k=k_1}^{k_n} \bar{E}_{int,k}}{\bar{E}_{int}} > r_2, \quad (62)$$

with r_2 again to be chosen to yield a desired accuracy, e.g. $r_2 = 0.95$. If this criterion is not fulfilled, the number of computed eigenfrequencies n_{eigen} has to be increased.

Thus, the n dominant eigenfrequencies ω_k , $k = k_1, \dots, k_n$, can be selected, where the $k_1, \dots, k_n$ in increasing order do not necessarily have to be consecutive integers, and the transient problem can be recomputed using the adapted p -discretization.

The proposed p -adaptive strategy is summarized in algorithm A-2.

Algorithm A-2 *p*-Adaptive strategy for transient problems

1: **set** $p_e^{i,j} = 4$ 2: **solve** transient problem

$$\mathbf{K}\mathbf{d}(t) + \mathbf{C}\dot{\mathbf{d}}(t) + \mathbf{M}\ddot{\mathbf{d}}(t) = \mathbf{F}(t) \quad \text{for } t \in [0, T]$$

3: **compute**

$$\bar{E}_{int} = \frac{1}{T} \int_T E_{str}(t) + E_{kin}(t) dt$$

4: **solve** eigenvalue problem

$$\mathbf{K}\phi_k = \omega_k^2 \mathbf{M}\phi_k \quad \text{with } k = 1, \dots, n_{eigen}, n_{eigen} \ll N$$

5: **project** $\mathbf{d}, \dot{\mathbf{d}}$ on eigenvectors $\phi_1, \dots, \phi_{n_{eigen}}$:

$$x_k(t) = \phi_k^T \mathbf{M}\mathbf{d}(t)$$

$$\dot{x}_k(t) = \phi_k^T \mathbf{M}\dot{\mathbf{d}}(t)$$

6: **compute** proportions of $\bar{E}_{int}$ associated with ω_k

$$\bar{E}_{int,k} = \frac{1}{T} \int_T \frac{1}{2} x_k^2(t) \omega_k^2 + \frac{1}{2} \dot{x}_k^2(t) dt$$

7: **select** dominant frequencies $\omega_k, k = k_1, \dots, k_n$ fulfilling

$$\frac{\bar{E}_{int,k}}{\bar{E}_{int}} > r_1$$

8: **if** $(\frac{\sum_{k=k_1}^{k_n} \bar{E}_{int,k}}{\bar{E}_{int}} \leq r_2)$ **then**9: **increase** n_{eigen} 10: **goto** step 411: **end if**12: **apply** *p*-adaptive eigensolver (Section 4.1) for $\omega_k, k = k_1, \dots, k_n$ 13: **recompute** transient problem (step 2) using adaptively selected discretization

4.3 Numerical Example

In this section, the transient behavior of a clamped quadratic plate being subjected to a sudden surface load is investigated. The uniform surface load will be applied at time $t = t_0 = 0$, acting until $t = T = 0.05$, the end of the time period of interest, which is slightly larger than the maximum eigenperiod of the system. The system, mesh and load function over time are depicted in Figure 17. The material of the plate is assumed to be linear elastic described by YOUNG's modulus of elasticity $E = 30000000$, POISSON's ratio $\nu = 0.2$ and density $\rho = 1.0$.

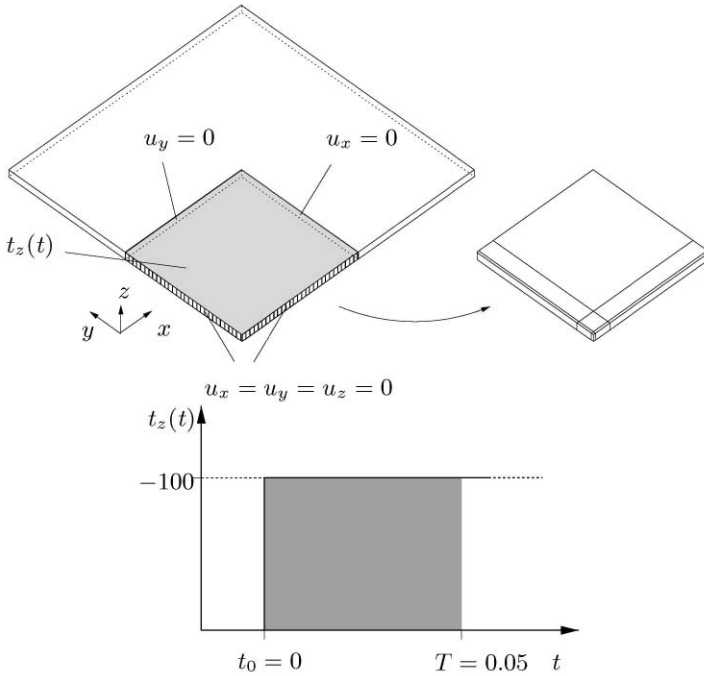


Fig. 17. System of clamped plate (top left) and hexahedral mesh (top right) and load function over time $t_z(t)$ (bottom)

An initial solution of the transient problem (27) is computed using a uniform polynomial degree $p_e^{i,j} = 4$ (according to steps 1, 2 of algorithm A-2). As a time integration scheme, the *trapezoidal rule* is used ($\alpha_m = \alpha_f = 0, \beta = 0.25, \gamma = 0.5$) and a timestep size $\Delta t = 0.001$.

The strain and kinetic energies are evaluated at every timestep (necessary for step 3) as depicted in Figure 18.

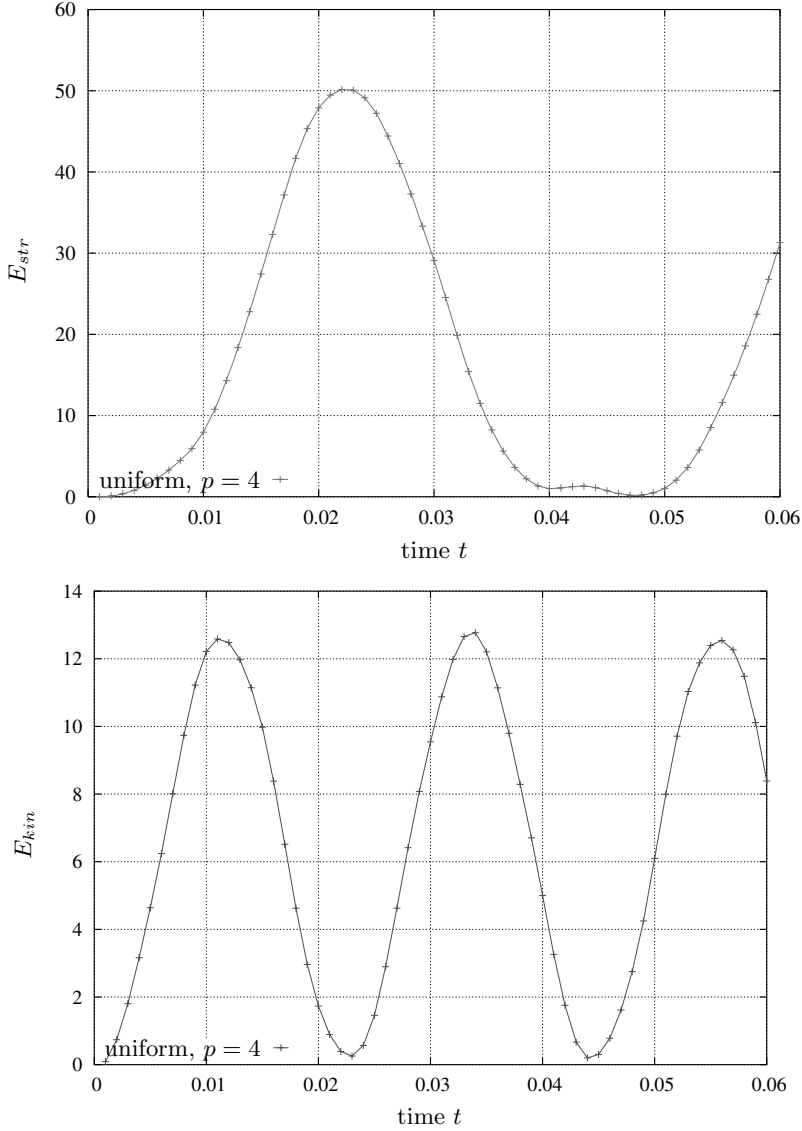


Fig. 18. Strain energy E_{str} and kinetic energy E_{kin} of clamped plate subjected to sudden surface load over time

In order to find the dominant frequencies of the problem, the eigenvalue problem (28) is solved, and $n_{eigen} = 5$ eigenfrequencies and eigenvectors are computed (step 4). Both here and in the following adaptive iterations for specific eigenfrequencies, the LANCZOS eigensolver is applied. The displacement and velocity vectors $\mathbf{d}$, $\dot{\mathbf{d}}$ are projected on the eigenvector space (step 5) and

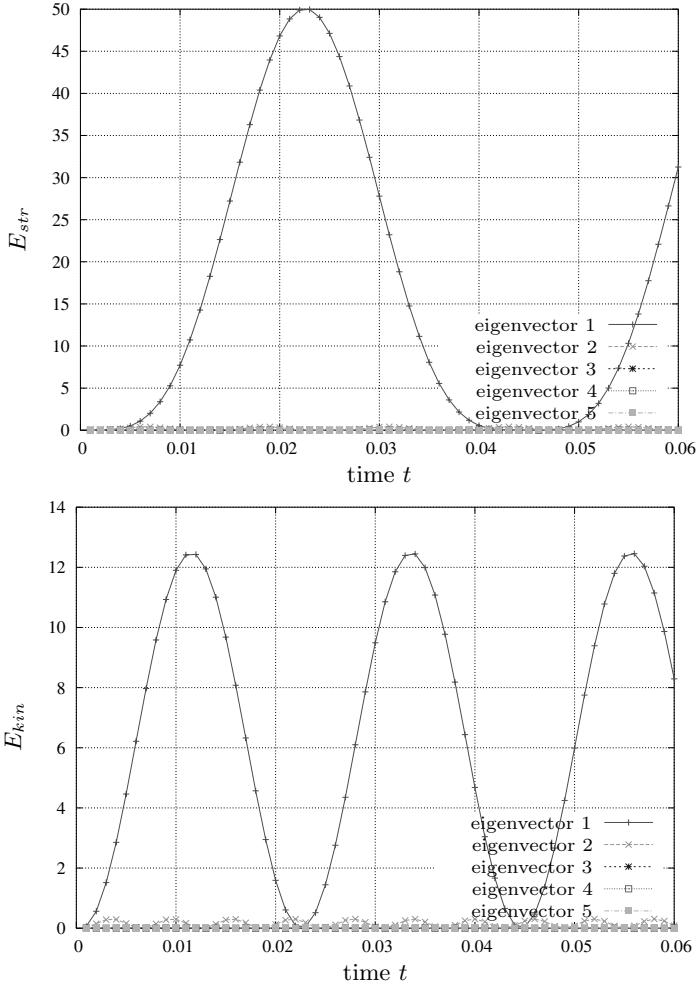


Fig. 19. Decomposition of strain energy E_{str} and E_{kin} into the proportions associated to the lowest five eigenfrequencies

the strain and kinetic energies associated with the eigenvectors $\phi_1, \dots, \phi_5$ are evaluated (necessary for step 6). The decomposition is plotted in Figure 19 from which it is evident that the dominant frequency is ω_1 . For ω_1 we obtain $\frac{\bar{E}_{int,1}}{\bar{E}_{int}} = 0.970$, thus, using $r_1 = 0.05$ and $r_2 = 0.95$ as suggested in Section 4.2, both criteria (61) and (62) are fulfilled (step 7 and 8).

For ω_1 the adaptive eigensolver is applied (step 12) with a stopping criterion $N_{max} = 2500$ resulting in five adaptive runs, see Figure 20. As a reference solution a hierarchically refined higher order solution was used. The adaptive convergence curve is always below the curve corresponding to uniform p -refinement, i.e. is more efficient, and, interestingly, also shows a

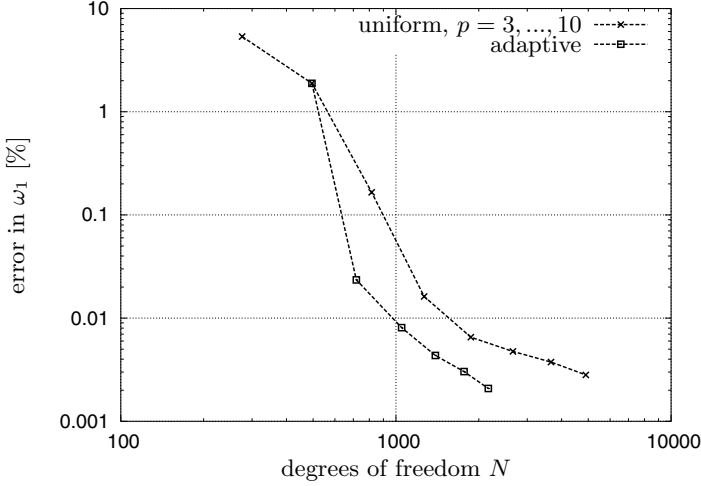


Fig. 20. Convergence of eigenfrequency ω_1 for plate

clearly higher rate of convergence in the asymptotic range, where uniform p -refinement shows poor behavior.

The set of polynomial degree templates $\mathcal{P}_5$ corresponding to the iteration $iter = 5$ is visualized in Figure 21, revealing the following characteristics: large polynomial degrees at the largest element, low polynomial degrees at small elements at corner, largest polynomial degrees at long boundary layer elements at the clamped support indicating that further mesh refinement would be reasonable for uniform p -version.

For analysis purposes the discretizations of all iterative steps of the p -adaptive eigensolver are used for recomputing the transient problem. In order to compare the efficiency of the adaptive and uniform p -discretizations the error measure

$$\eta(t) = \left(\frac{|\Delta E_{str}(t)| + |\Delta E_{kin}(t)|}{E_{max}} \right)^{\frac{1}{2}} \quad (63)$$

is introduced. In Figure 22 the maximum error and mean error over time

$$\eta_{max} = \max_t \eta(t) \quad (64)$$

$$\bar{\eta} = \frac{\int_{\Delta t} \eta(t) dt}{\Delta t} \quad (65)$$

are plotted. From both plots it is obvious, that the adaptive refinements show clearly higher efficiency than uniform p -refinements.

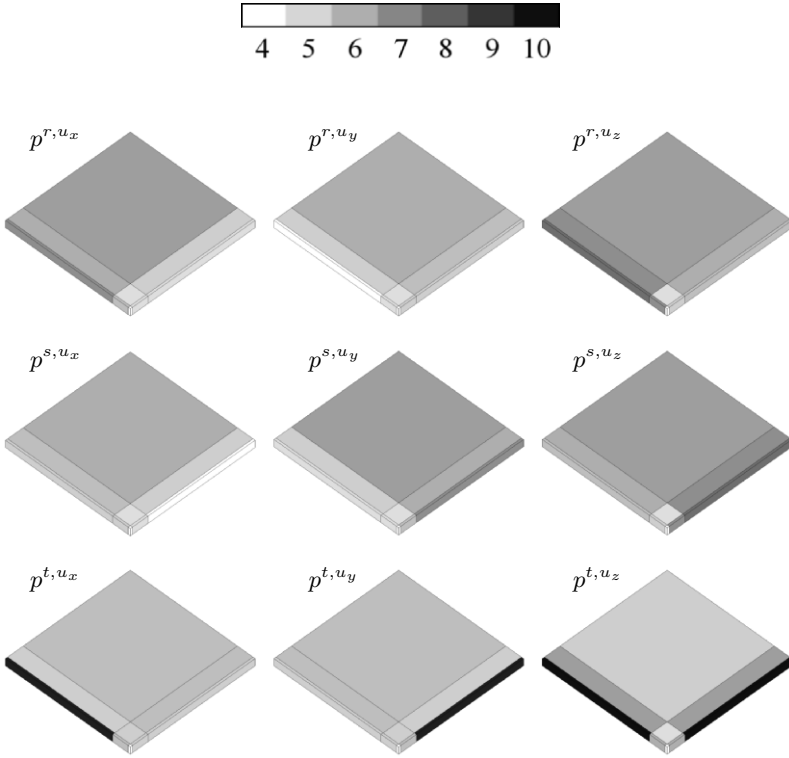


Fig. 21. Graphical representation of polynomial degree templates corresponding to iteration 5 of adaptive eigensolver for a 3D plate

5 Fluid Structure Interaction with High Order Thin Solids

5.1 Partitioned Solution Approach

Since in our approach different specialized codes for the fluid and structural fields — also based on different discretization techniques — are applied, a partitioned solution strategy is used. In order to increase flexibility concerning the coupling to different fluid codes the p -FEM code *AdhoC* [5] is connected to a general coupling software framework [6].

The idea of this framework is to provide coupling functionality for the partitioned solution approach, see Figure 23, and to allow for arbitrary explicit and implicit coupling strategies. The interface between fluid and structure (wet surface) is represented by an appropriate fine triangular or quadrilateral mesh on which coupling data — like stresses and displacements — can be defined. The interface mesh and coupling data are contained in a geometric

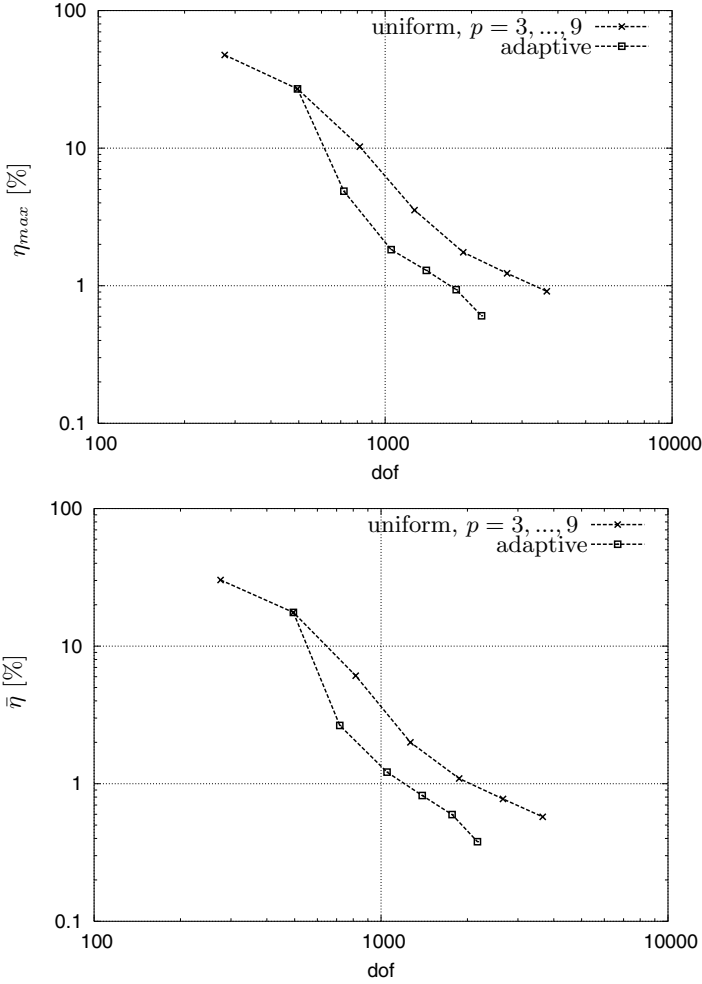


Fig. 22. Maximum error η_{max} and mean error $\bar{\eta}$ for transient computation of a plate

database playing a central role in the coupling framework. Inter-process communication of meshes and related data is provided on this basis.

5.2 Transfer of Loads onto the Structure

An important aspect of a coupled fluid structure interaction computation is the consistent and accurate transfer of the fluid loads and their integration on the structural surface.

The subsequent discussion will focus on FSI applications where the fluid load is represented on an appropriate discretization not matching the (high

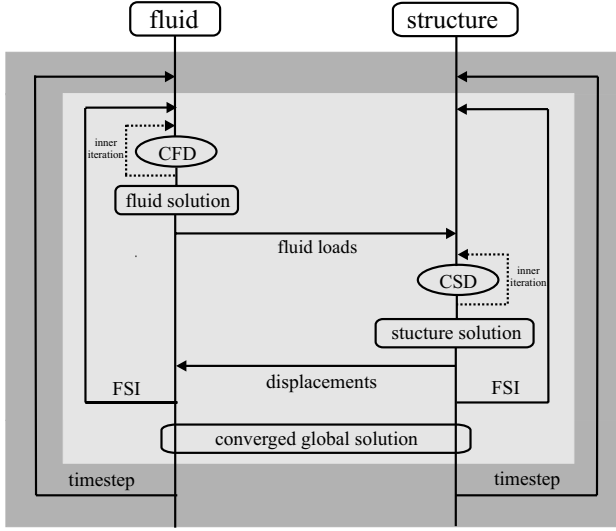


Fig. 23. Coupling scheme of partitioned solution approach.

order) structural discretization, e.g. the surface discretization of an ALE-based fluid code or a representation of the stresses computed with a Cartesian-based fluid code.

A first approach to this problem could be a simple interpolation technique of the fluid stress function onto the Gauss points of the structure, e.g. linear or bilinear interpolation in the case of triangles or quadrilaterals, respectively, discretizing the fluid interface. A major drawback of this method is that the integrand is a piecewise constant or piecewise linear or bilinear function over the integration domain due to the non-matching grids, i.e. the integrand contains kinks or jumps on the structural element surface, and using a Gauss quadrature (which is optimal for smooth polynomials) can lead to inaccurate results. To illustrate this deficiency, the load transfer between two non-matching plane quadrilateral bilinear grids is considered, where the integral of the bilinear load function t_z in z -direction defined on mesh 2 (e.g. the fluid) is evaluated on mesh 1 (e.g. the structure), see Figure 24. From Figure 25 it is evident that even with a Gauss order of 10, i.e. 100 Gauss points on one element surface, the resulting relative error η in the load integral over the interface

$$\eta = 100 \cdot \left| \frac{\sum_{i_e=1}^{n_e^S} \sum_{i_g=1}^{n_{g,i_e}^S} (w_{i_g,i_e} t_z(x_{i_g,i_e}) \det J(x_{i_g,i_e})) - \int_{\partial\Omega^F} t_z d\partial\Omega^F}{\int_{\partial\Omega^F} t_z d\partial\Omega^F} \right| \quad (66)$$

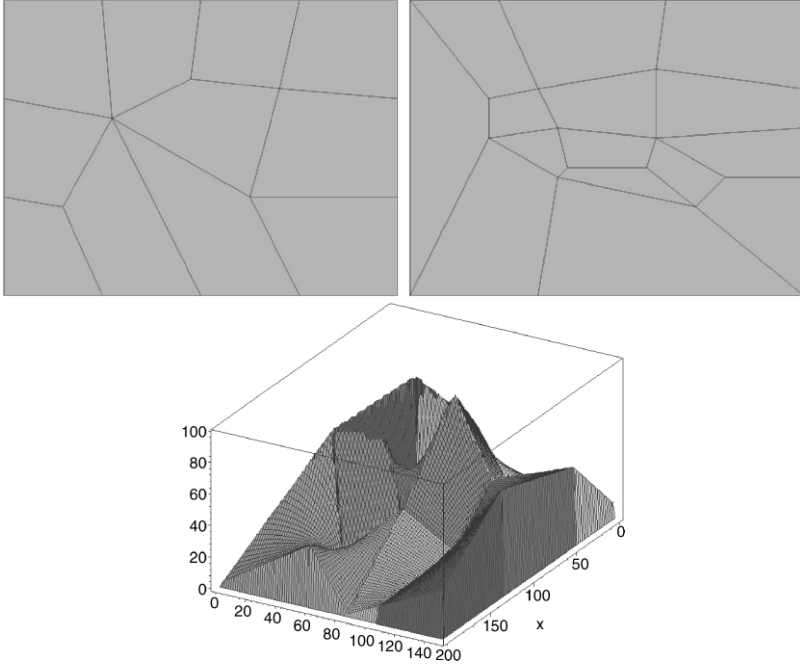


Fig. 24. Mesh 1 (e.g. structure, top left), mesh 2 (e.g. fluid, top right), bilinear load function on mesh 2 (bottom)

is still greater than 0.5%. In Equation (66), n_e^S is the number of structural element surfaces, n_{g,i_e}^S is the number of Gauss points on element surface i_e , w_{i_g,i_e} are the weights of the Gauss points, x_{i_g,i_e} are the global coordinates of the Gauss points, J is the Jacobian and $\partial\Omega^F$ is the fluid interface. The convergence for higher Gauss orders is extremely poor.

Another possibility to compute the load interpolation would be a ‘sum-conservative’ approach as proposed in [44]. Unfortunately, this approach implicitly assumes the structural shape functions to sum up to unity in every point of the interface, which is not the case for the p -version with hierarchic shape functions. Since an adaption to hierarchic shape functions is not trivial, we adopt another strategy: In order to integrate the piecewise defined fluid load function on the structural surface, we split the integration domain, i.e. the structural elements, into subdomains according to the fluid discretization. On these subdomains, where the fluid load function is smooth, the integration can be performed accurately. This approach is similar to the load transfer in [45] proposing Mortar-based methods implemented for triangles. In the following the extension of this technique for the transfer of loads to quadrilaterals is proposed.

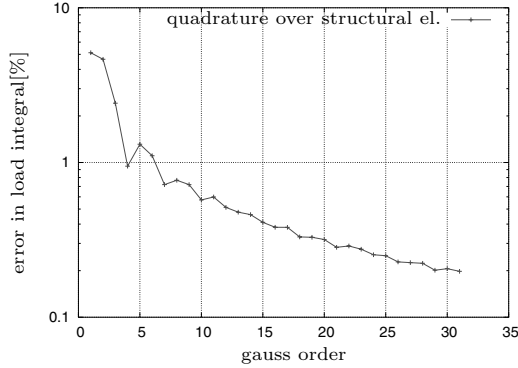


Fig. 25. Relative error of load integral η using bilinear interpolation from mesh 1 to mesh 2 and integration over structural elements

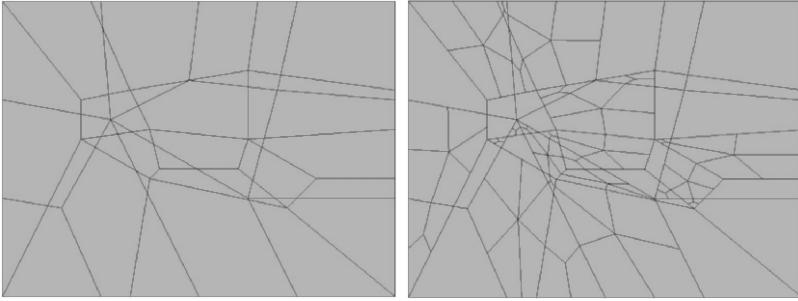


Fig. 26. Intersection of mesh 1 and 2 (left), intersection of mesh 1 and 2 divided into quadrilaterals (right)

In order to create the integration subdomains of the structural elements, we intersect the elements of both meshes in a first step. The resulting polygons have 3, 4, 5, 6, ... edges (see Figure 26, left).

These polygons can be divided for integration into e.g. quadrilaterals using an automatic mesh generator, where only the polygons with a number of edges not equal to 4 must be processed. When both the fluid and the structural elements have straight edges and no re-entrant corners, the resulting intersection polygons are convex. In this case it is possible to create a quadrilateral mesh from every polygon in a very simple manner by inserting a node in the interior of the polygon and a node on every midpoint of each edge (see Figure 26, right). The hanging nodes seem to define an inconsistent quadrilateral mesh, but they cause no problems, since this subdomain mesh is only used for the load integration.

Having performed these geometric operations, it is now possible to evaluate the load integral over the quadrilaterals resulting from the intersection. In Figure 27 the convergence of the load integral is plotted versus the

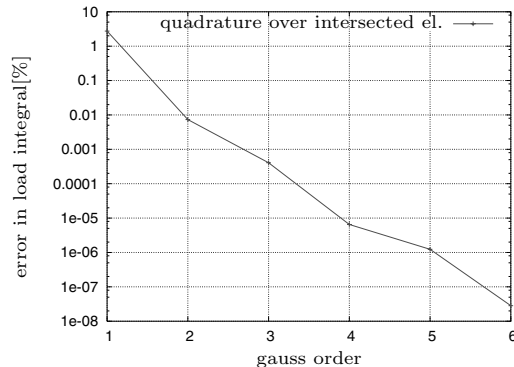


Fig. 27. Relative error of load integral η using piecewise integration over structural elements

corresponding Gauss order. The relative error in the load integral for a Gauss order of 2 is already less than 0.01%, since here the Gauss quadrature can do what this method is well suited for: integrating smooth functions.

5.3 Coupling to a Lattice-Boltzmann CFD Code

By coupling the structural p -FEM code with the Lattice-Boltzmann method (LBM) fluid code [7,8] using the coupling framework described in Section 5.1 (explicit coupling strategy) it has been found that the initialization and deletion of nodes of the Cartesian fluid discretization, necessary when their state changes from solid to fluid and vice versa, causes oscillations in the interface tractions. These artificial oscillations depend on the discretization and excite higher structural modes possibly leading to a non-stable coupling behavior. A first remedy to this problem was a linear modal analysis of the structure, thus allowing to control the number of modes to be considered without disregarding bidirectional fluid structure interactions. For the timestep-wise computation of the generalized forces with respect to every mode taken into account, we utilized the geometric database mentioned in Section 5.1. The integration of the forces was performed using a simple midpoint quadrature rule. Time integration of the decoupled system was accomplished by analytically integrating the single degree of freedom oscillators assuming a constant force during every timestep, which is justified due to the small timestep size necessary for the explicit coupling scheme.

As an example, the benchmark configuration FSI3 described in [9] was computed using the modal analysis approach, where the first nine eigenmodes were considered, computed with three hexahedral elements. To account for the plane strain conditions prescribed by the *two-dimensional* benchmark, motion in z -direction perpendicular to the plane (Figure 28) is suppressed and anisotropic Ansatz spaces were applied, $p^{x,y} = 10$ in in-plane and $p^z = 1$

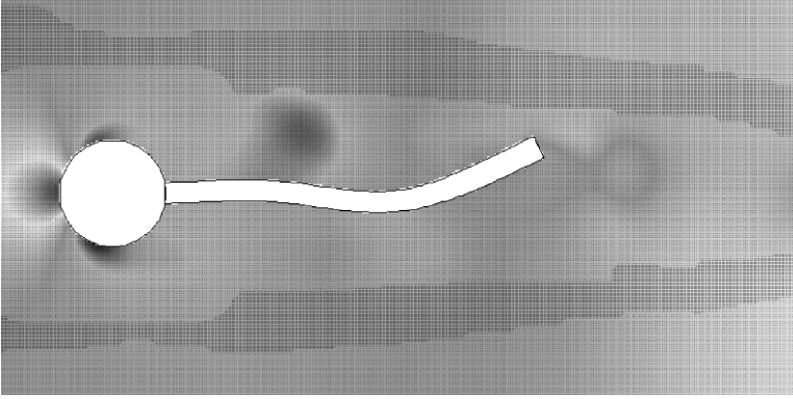


Fig. 28. Deflected cantilever

perpendicular to the plane. A timestep size of $\Delta t = 0.0002526$ was used. In Figures 28 and 29 the deflected beam during the fluid structure interaction and the (in this case triangular) surface mesh are depicted. In Figure 30, the quasi steady state displacements at the end of the beam are plotted over a time period of $T = 1$ s and, for comparison, corresponding results obtained by [9] are added, showing very good accordance.

6 Summary

In this contribution the use of hexahedral elements for the structural simulation in a fluid structure interaction framework is presented, resulting in a consistent kinematic and geometric description of the structural behavior. In order to compensate the additional effort of the three-dimensional approach, an anisotropic p -adaptive method for elastodynamic problems is proposed, resulting in clearly higher efficiency and higher convergence rates than uniform p -version. The method seems to be promising to improve the discretization for nonlinear problems as well.

Special emphasis is placed on the accurate transfer of loads considering the fluid discretization for computation of the surface load integrals. In future work also the transfer of displacements will be improved by projection-based, Mortar-like methods.

For a coupling with a Cartesian based Lattice Boltzmann code it was found that oscillations in the interface tractions may excite higher structural modes possibly leading to a non-stable coupling behavior. A first remedy to this problem was a linear modal analysis of the structure, thus allowing for controlling the number of modes to be considered without disregarding bidirectional fluid structure interactions. Results were presented for the FSI

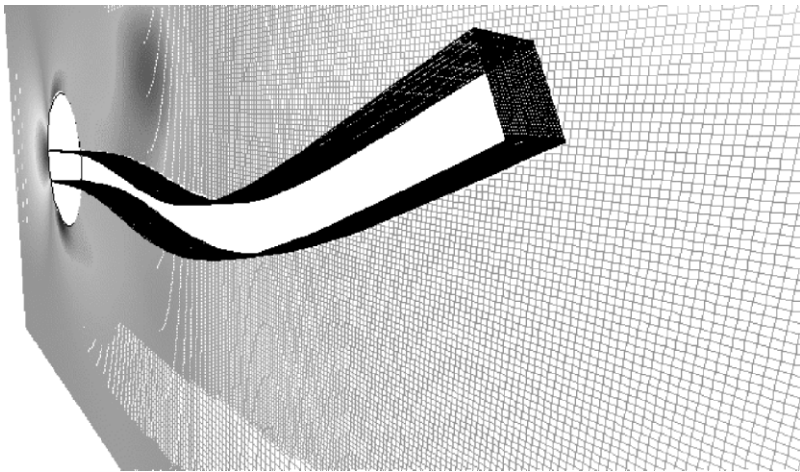


Fig. 29. Deflected surface mesh

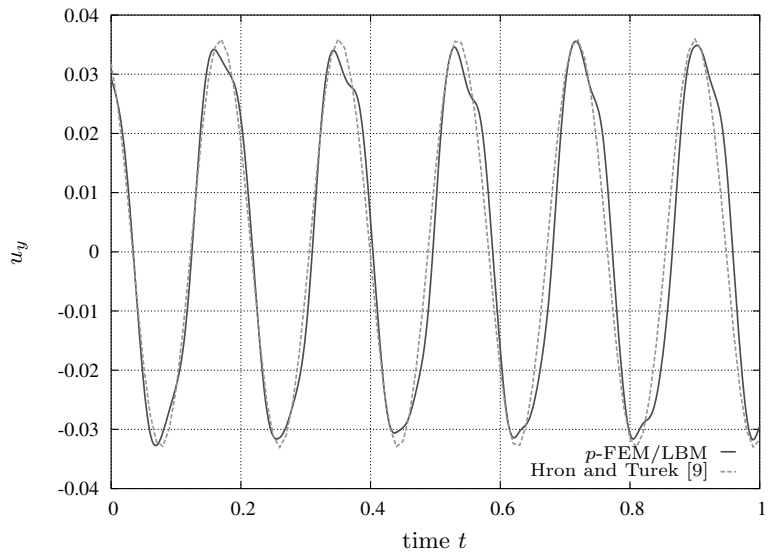


Fig. 30. Quasi steady state displacements u_y of tip of cantilever (point A) over time

benchmark configuration proposed in this book. Improving the physical representation of the system by getting rid of the (linear) modal reduction is subject of current work.

Acknowledgement

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Algorithmic Treatment of Shells and Free Form-Membranes in FSI

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Abstract. The aim of this contribution is to propose a methodology for the analysis and improvement of light, thin-walled structures with reference to aeroelastic effects. Those kind of problems demand for the appropriate combination of different physical and numerical disciplines to account for the relevant factors inherent to the simulation of light, thin-walled structures undergoing large displacements as well as highly turbulent air flows. To fulfill these requirements the occurring wind-structure interaction is accessed by a surface-coupled fluid-structure interaction (FSI) method. This is realized in a modular and flexible software environment with the use of a partitioned coupling approach: the structural field is solved by the in-house finite element program CARAT using several finite element types and advanced solution techniques for form finding, nonlinear and dynamical problems. The flow field is solved by the CFD software package CFX-5 of ANSYS Inc. A prerequisite to allow for the assessment of aeroelastic problems, beyond the mere exchange of data between the two physical fields, is the utilization of stable as well as efficient coupling strategies. In particular, it is shown that in the case of light-weight structures interacting with incompressible fluid flows the coupling strategy plays an important role regarding the feasibility of the simulations. This contribution will present theory and realization of a corresponding implementation enhanced by illustrative examples. Moreover, the comprehensiveness of this approach opens the possibility for multiphysics optimization.

1 Introduction

Applying methods of structural optimization aims at defining light and efficient structures. Typically, the resulting shape is a free-form shell or membrane structure which closes itself to any analytical description while fully utilizing the material parameters. However, in the case of structures subject to loads by surrounding fluid flows, increasing lightness and slenderness bring along a higher susceptibility to flow-induced deformations and vibrations. In the application to the civil engineering context this exemplifies in wind-induced effects on thin shells and membrane structures.

In general, an assessment of wind effects with deterministic tools is complicated by the stochastic nature of wind phenomena. Nevertheless these wind effects can define the decisive design loads and therefore require an in-depth analysis. This analysis is both interesting and challenging in cases of an interaction between structure and wind. This phenomenon is called aeroelasticity

and can occur on constructions such as towers, high-rise buildings, bridges, cable and membrane roofs, etc. In these cases the usual approach in structural engineering of reducing the complex problem of structures subject to wind to simpler models by finding appropriate assumptions involves the risk of neglecting essential effects which result from the strong coupling of the two different physical fields.

Applying numerical simulations to aeroelastic effects in wind engineering is a promising complement to and enhancement of experimental approaches. The analysis of aeroelasticity requires a multiphysics approach which can be assessed by a surface-coupled fluid-structure interaction (FSI) method [5],[8].

2 Approach

For a partitioned FSI simulation of light-weight structures subjected to wind the following requirements are identified:

- Correct treatment of geometrical nonlinearities in stationary and transient structural analysis.
- Integration of form finding to determine the proper initial geometry in the case of membrane structures due to strong coherence between shape and load carrying behavior.
- Adequate solution of highly turbulent flows and modeling of fluid boundary conditions to represent the characteristics of physical wind.
- Treatment of moving boundaries in the fluid solver.
- Accurate transfer of coupling quantities, especially in the case of non-matching grids.
- Respect of strong coherences between the physical fields by appropriate coupling schemes

In order to meet these requirements, a partitioned approach is chosen [7],[10]. Therefore, both fields are solved in specialized, well-suited, and tested single field solvers exchanging the necessary boundary conditions. In the current approach, the form finding and the structural part are solved by the research code CARAT, the fluid part by the general purpose Computational Fluid Dynamics Code CFX-5 of Ansys Inc., and for the coupling the MpCCI-library is used. Figure 1 gives a detailed view of the procedure of a coupled simulation and the used software. The integration of an optimization loop in the analysis framework, provides the possibility to compute structures less susceptible to wind loads. In Section 2.3 a more detailed view at the coupled simulation itself is provided.

2.1 Structural Part

The structural problem is characterized by the governing equations for an elastic body with large deformations (and the corresponding boundary and

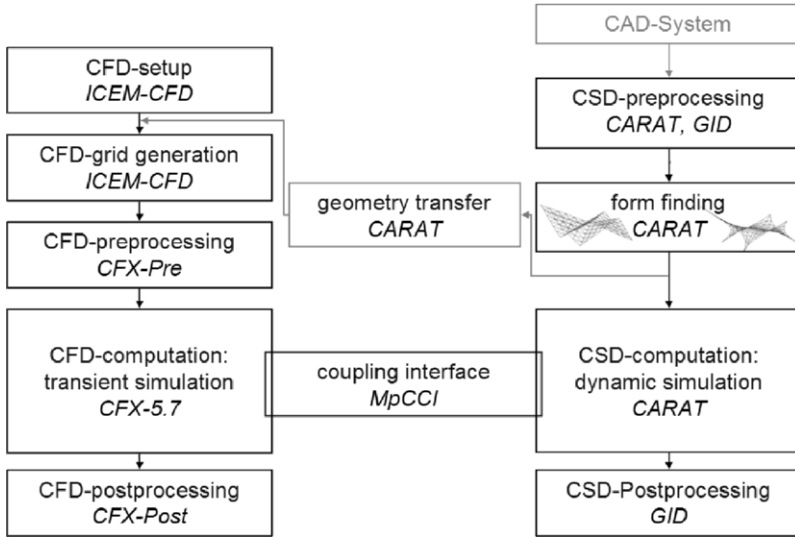


Fig. 1. Software environment and workflow of a coupled simulation.

initial conditions): equilibrium (1), material law (3), and kinematics (4). The acceleration vector vanishes and the first equation reduces to the static equilibrium condition (2), if only stationary problems are considered.

$$\rho_0 \cdot \ddot{\mathbf{u}} = \text{DIV}(\mathbf{F} \cdot \mathbf{S}) + \rho_0 \cdot \mathbf{b} \quad (1)$$

$$\mathbf{0} = \text{DIV}(\mathbf{F} \cdot \mathbf{S}) + \rho_0 \cdot \mathbf{b} \quad (2)$$

$$\mathbf{S} = \mathbf{C} : \mathbf{E} \quad (3)$$

$$\mathbf{E} = \frac{1}{2} (\mathbf{F}^T \cdot \mathbf{F} - \mathbf{I}) \quad (4)$$

Where ρ_0 is the density in the reference configuration, $\mathbf{S}$ the second Piola-Kirchhoff stress tensor, $\mathbf{b}$ the body forces, $\mathbf{F}$ the deformation gradient, $\mathbf{E}$ the Green-Lagrange strain tensor, and $\mathbf{C}$ the elasticity tensor. These equations are solved with the finite element method. For this purpose the Lagrangian description is used, i.e. the nodes of the corresponding mesh are fixed to the material points [1]. This leads to the semi-discrete equations of motion

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{D}\dot{\mathbf{u}}(t) + \mathbf{F}_{int}(\mathbf{u}(t)) = \mathbf{F}_{ext}(t) \quad (5)$$

where $\mathbf{M}$ is the mass matrix, $\mathbf{D}$ the damping matrix representing the structural damping (if considered in the simulation), $\mathbf{F}_{int}$ the vector of internal forces which leads via linearization to the tangential stiffness matrix, and $\mathbf{F}_{ext}$ the vector of external loads acting on the structure.

This nonlinear time-dependent problem with properly defined boundary and initial conditions is solved by a time integration algorithm. In the presented example the (implicit) generalized- α method [4] was used due to its

advantageous properties: With a certain choice of parameters the procedure is second order accurate and the user has full control over the numerical dissipation of the spurious high frequencies by minimized errors in the lower modes of interest. In the case of stationary response analysis the inertial and damping terms vanish and the equation of motion reduces to the discretised equilibrium condition. The nonlinear equations which are evolving in static, dynamic, and form finding computations are solved by the Newton-Raphson method.

In order to attain the desired flexibility and modularity in the proposed partitioned approach the fluid-structure coupling only affects the right-hand side of (5). Thereby the displacement dependence of the fluid load which would lead to additional contributions in the linearization of the internal forces is neglected. For cases in which this contribution is important, it is resolved by iterations between the fluid and the structural solution (see Section 2.3).

The structures under consideration are thin-walled and flexible and, therefore, are described by either shell or membrane theory. In the following the used elements and algorithms for those kind of structures are shortly displayed with regard to their peculiarities in FSI simulations.

The computation of the shell example in the present paper was done by so-called 7-parameter shell elements, which assume a linear strain distribution over the thickness. These models are able to approximately represent three-dimensional effects, while pertaining the efficiency of a two dimensional formulation due to pre-integration of the structural stiffness matrix across the thickness [2]. In Fig. 2, the reference and the current configurations of a shell structure are shown with the position vectors $\mathbf{X}$ and $\mathbf{x}$ of an arbitrary material point, the corresponding displacement vector $\mathbf{u}$ and the covariant base vectors $\mathbf{A}_i$ of the midsurface of the shell. Quantities of the undeformed (reference) and deformed (current) configurations are identified with upper case and lower case letters, respectively.

The position vector $\mathbf{X}$ of an arbitrary point of the shell can be expressed with the help of $\mathbf{R}$ and $\mathbf{A}_3$:

$$\mathbf{X} = \mathbf{R} + \theta^3 \cdot \mathbf{A}_3, \quad \mathbf{x} = \mathbf{r} + \theta^3 \cdot \mathbf{a}_3 \quad (6)$$

The covariant base vectors of the midsurface of the shell $\theta_3 = 0$ are obtained from the partial derivatives of the position vector $\mathbf{R} = \mathbf{X}(\theta_3 = 0)$ and $\mathbf{r} = \mathbf{x}(\theta_3 = 0)$, respectively.

$$\mathbf{A}_\alpha = \frac{\partial \mathbf{R}}{\partial \theta^\alpha} = \mathbf{R}_{,\alpha}, \quad \mathbf{a}_\alpha = \frac{\partial \mathbf{r}}{\partial \theta^\alpha} = \mathbf{r}_{,\alpha} \quad (7)$$

The director vector $\mathbf{A}_3$ is (by definition) perpendicular to the midsurface and has an initial length of $h/2$, where h is the shell thickness.

$$\mathbf{A}_3 = \frac{h}{2} \cdot \frac{\mathbf{A}_1 \times \mathbf{A}_2}{|\mathbf{A}_1 \times \mathbf{A}_2|} \quad (8)$$

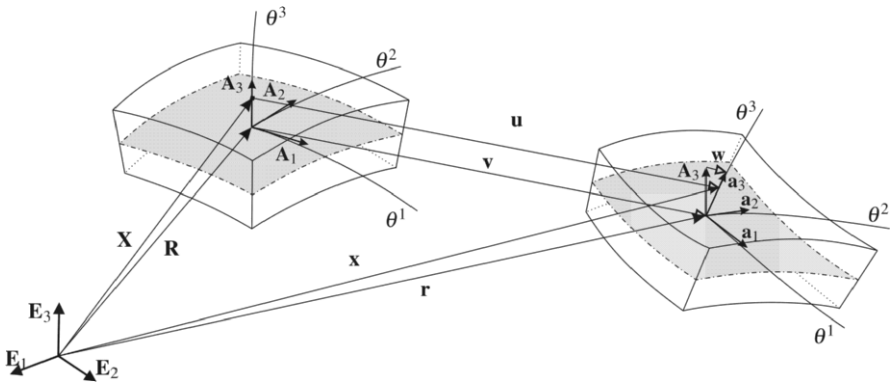


Fig. 2. Geometry and kinematics of 7-parameter shell model.

The covariant base vectors of an arbitrary point in the shell body are given by

$$\mathbf{G}_\alpha = \mathbf{X}_{,\alpha} = \mathbf{A}_\alpha + \theta^3 \cdot \mathbf{A}_{3,\alpha}, \quad \mathbf{G}_3 = \mathbf{X}_{,3} = \mathbf{A}_3 \quad (9)$$

The displacement $\mathbf{v}$ of a point of the midsurface together with the update of the director via the vector of the difference displacements $\mathbf{w}$

$$\mathbf{r} = \mathbf{R} + \mathbf{v}, \quad \mathbf{a}_3 = \mathbf{A}_3 + \mathbf{w} \quad (10)$$

render an expression for the displacement $\mathbf{u}$ of an arbitrary point in the shell body:

$$\mathbf{u} = \mathbf{x} - \mathbf{X} = \mathbf{R} + \mathbf{v} + \theta^3 \cdot (\mathbf{A}_3 + \mathbf{w}) - \mathbf{R} - \theta^3 \cdot \mathbf{A}_3 = \mathbf{v} + \theta^3 \cdot \mathbf{w} \quad (11)$$

As can be seen in (11), the 7-parameter shell formulation has first of all six degrees of freedom, namely the three displacements of the midsurface $\mathbf{v}_x$, $\mathbf{v}_y$ and $\mathbf{v}_z$ and three difference displacements $\mathbf{w}_x$, $\mathbf{w}_y$ and $\mathbf{w}_z$ for the update of the director. These are evolving directly from the linear displacement assumption in thickness direction. A seventh degree of freedom is added with the help of the enhanced assumed strain technique to establish a linear distribution of transverse normal strains in order to avoid Poisson thickness locking.

The general advantage of this description is the possibility to use unmodified, complete three-dimensional material laws within shell analysis. In the special case of FSI problems the deformation of the coupling interface incorporates also information about the thickness change of the shell (see Section 2.3). The specified shell model is implemented in the in-house research finite element code CARAT under consideration of well-known methods (like ANS and EAS) to avoid locking effects.

In contrast to shells, the thickness h of membranes is comparatively small and assumed to be constant during deformation, i.e. the Poisson effect in thickness direction is neglected. This is in accordance with the behavior of available membrane materials. This assumption and the negligible thickness reduce the kinematic relations stated above to the midplane. Therefore, the behavior of the membrane is completely described by three degrees of freedom, namely the three displacement components of the membrane surface. This has an effect on the formulation of the coupling interface for membranes with surrounding fluid flow (see Section 2.3).

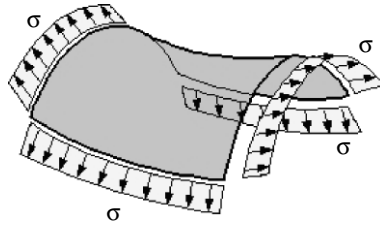


Fig. 3. Tangential surface stress field.

Further complications are caused by the special load carrying behavior of membranes. Membranes are pure tension structures (see Fig. 3). Therefore, they have no bending resistance and in the case of compression wrinkling patterns emerge. On the one hand, this enables e.g. the construction of very light, elegant and wide-spanned roofs since the material is optimally used. On the other hand, it is the reason for their extreme sensitivity to wind loads which makes a certain prestress necessary to avoid the occurrence of flutter. This prestress can be introduced by additional dead load, pressure differences or simply by pulling the membrane into the previously planned positions of the supports.

Accordingly finding the right initial shape is naturally the first step for a computation concerning membrane structures (see Fig. 1). The initial shape is defined by the equilibrium of surface stresses and edge cable forces, found as the result of a form finding computation [16].

The reason for special form finding procedures is given by the singularity of the inverse problem to find the corresponding shape to a given stress distribution. This difficulty is overcome by the updated reference strategy [3], a regularization by a homotopy mapping which is implemented in CARAT. The modified virtual work expression which is solved by the finite element

method is given by:

$$\delta w_\lambda = \lambda h \int_A \det \mathbf{F} (\boldsymbol{\sigma} \cdot \mathbf{F}^{-T}) : \delta \mathbf{F} dA + (1 - \lambda) h \int_A (\mathbf{F} \cdot \mathbf{S}) : \delta \mathbf{F} dA = 0 \quad (12)$$

The stabilizing modification (in terms of *PK2* stresses $\mathbf{S}$) fades out as the solution is approached. Even pure tangential movements which are very challenging for form finding algorithms are possible, which is demonstrated at the example of a four-point structure with flexible, cable-reinforced edges forcing a tangential adjustment of the surface finite element mesh during the form finding procedure. Figure 4 shows the preliminary computation for the wind simulation given in the second example (Section 5).



Fig. 4. Determination of membrane shape for FSI simulation by means of form finding (left: before, right: after form finding).

After the initial shape is found, further analysis respecting wind or snow loads can be performed. For the coupled computation, naturally this initial shape defines the FSI interface. However, if a certain shape shows shortcomings under wind action the prestress distribution must be modified such that the improved shape prevents flutter. This gives the perspective of a sequential application of form finding and FSI computation to find an ideal initial shape least susceptible to wind and the according prestress distribution. This sequential analysis can be realized by using the software environment proposed in this paper.

With the implementation of MpCCI-functions in CARAT, it is possible to apply CARAT together with different research and commercial codes for coupled field simulations.

2.2 Fluid Part

The viscous fluid flow is described by the governing Navier-Stokes equations which state the conservation of mass and momentum. In the scope of this paper, an incompressible fluid with constant properties is assumed. The governing equations are:

$$\rho \left(\frac{\partial U_j}{\partial t} + U_i \frac{\partial U_j}{\partial x_i} \right) = - \frac{\partial P}{\partial x_j} + \mu \frac{\partial^2 U_j}{\partial x_i \partial x_i} \quad (13)$$

$$\frac{\partial U_i}{\partial x_i} = 0 \quad (14)$$

with U_j as the velocity component in the j -direction, x_i as the Cartesian coordinate in the i -direction, P as the pressure, μ as the dynamic viscosity, and ρ as the density of the fluid. Adequate boundary conditions are describing wall, inflow, outflow, and symmetry type boundaries.

In the scope of this work, the finite volume approach is used to solve the Navier-Stokes equations. Therefore the governing equations are written in integral form. The equations of mass conservation and momentum conservation are applied to a control volume whose boundaries move with time.

$$\frac{d}{dt} \int_v \rho \Phi dV + \int_S \rho U_i \Phi dS_i - \int_S \rho U_{gi} \Phi dS_i - \int_S \Gamma_\Phi \frac{\partial \Phi}{\partial x_i} dS_i = \int_V q_\Phi dV \quad (15)$$

Equation (15) is used for both, the continuity equation and the momentum equation. For the *momentum equation* Φ equals U_j , Γ_Φ equals μ , and q_Φ equals $-\frac{\partial p}{\partial x_j}$; for the *continuity equation* Φ equals 1, Γ_Φ equals 0, and q_Φ equals 0. For cases of moving grids with the grid velocity U_{gi} the arbitrary Lagrangian Eulerian (ALE) approach is applied [12].

In the present work, the CFD calculations are performed by the commercial Computational Fluid Dynamics Package CFX-5. CFX-5 solves the 3D Navier-Stokes equations on structured and unstructured grids for compressible and incompressible flows. For the simulation of turbulent flows several advanced turbulence models are available, including Reynolds Averaged Navier-Stokes Models (RANS), Large Eddy Simulations (LES), and Detached Eddy Simulations (DES). For the following examples the Shear-Stress-Transport (SST) model was applied [11]. It is appropriate for separated flows as it combines the advantages of a $k-\omega$ turbulence model in the near wall region with those of the $k-\epsilon$ turbulence model outside the boundary layer.

2.3 Realization of the Fluid-Structure Coupling

In the procedure of a FSI simulation, the fluid and the structural simulation work together in a staggered algorithm, shown in Fig. 5. Stationary and transient simulations are possible. With a simple once-per-time-step exchange of coupling quantities, an explicit coupling scheme is possible. To ensure strong coupling a fully implicit coupling scheme is realized through the implicit iteration loop shown in Fig. 5: several coupled computations are conducted until an equilibrium state within one time-step is reached. In the current approach, the structural field is computed by the research code CARAT, the fluid part by the general purpose Computational Fluid Dynamics Code CFX-5 of Ansys Inc., and for the coupling the MpCCI-library is used.

For the stabilization of simulations with strong deformations between implicit iteration steps, an under-relaxation technique for the transferred loads

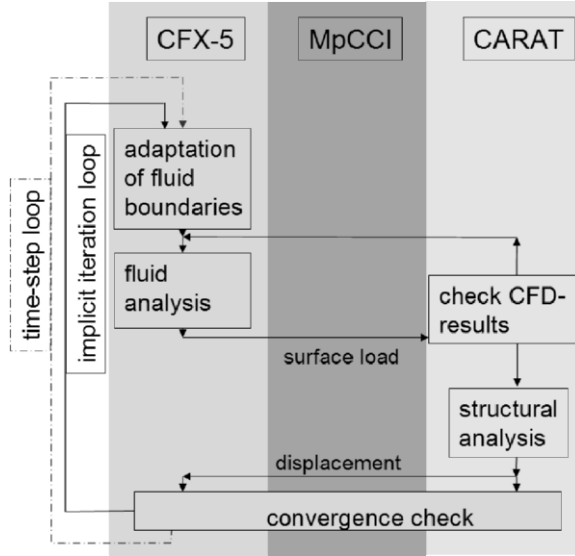


Fig. 5. Flow chart of the coupling algorithm.

within one time-step can be applied:

$$\mathbf{V}_{i+1} = \mathbf{V}_i + r_{i+1} \cdot \Delta \mathbf{V}_{i+1} \quad \text{with} \quad \Delta \mathbf{V}_{i+1} = \tilde{\mathbf{V}}_{i+1} - \mathbf{V}_i \quad (16)$$

$\mathbf{V}$ is a variable the under-relaxation is applied to, $\tilde{\mathbf{V}}$ is the non-under-relaxed variable, r is the under-relaxation-factor, and i is the number of the coupled iteration step within one time-step. However, small under-relaxation factors provide a good behavior of the coupled simulation towards stability for the price of a high computational effort. Therefore a reasonable choice for the parameter r is required. Following the work of Mok and Wall [13] in this approach the Aitken method is applied:

$$\mu_{i+1} = \mu_i + (\mu_i - 1) \frac{(\Delta \mathbf{V}_i - \Delta \mathbf{V}_{i+1})^T \cdot \Delta \mathbf{V}_{i+1}}{(\Delta \mathbf{V}_i - \Delta \mathbf{V}_{i+1})^2} \quad (17)$$

$$r_{i+1} = 1 - \mu_{i+1} \quad \text{with} \quad \mu_0 = 0 \quad (18)$$

In the current coupling approach for non-matching grids [6], surface pressure and displacement are interpolated in a bilinear manner using the coupling library MpCCI. The Mesh-based parallel Code Coupling Interface MpCCI was developed as a standardized tool for the coupling of multiple codes. The main benefit of MpCCI is its ability to support different discretizations of the common interface. This possibility to transfer and interpolate mesh based quantities between non-matching grids of the structure and fluid part on the coupling interface ensures that each simulation code can work on an optimal mesh for the specific numerical simulation method.

Interface Treatment for Shell Elements

As mentioned above, for the description of the thin-walled structure in CARAT, shell elements are applied. This implies that for structural analysis a 3-dimensional thin-walled structure is reduced to its middle surface. However, for correct modeling of the geometry of the structure in the fluid simulation, a 3-dimensional structure is required. Therefore the exchange of data is performed on an 'inflated' interface on the structural side: the middle surface is divided into a front and a back surface, both with identical meshes. The front (*back*) surface is moved $\frac{1}{2}h$ ($-\frac{1}{2}h$) in the direction of the normal vectors of the finite elements, with h as the shell thickness. On the inflated interface the data is exchanged (Fig. 6): the received loads are treated consistently with respect to the underlying continuum shell theory [2]. The results of the structural calculation are the displacements of the middle surface in the three directions in space ($\mathbf{v}$) and the difference vector ($\mathbf{w}$). The latter describes the change of the director of the undeformed ($\mathbf{A}_3$) to the deformed ($\mathbf{a}_3$) structure (see Fig. 2). Hence, the transferred displacements of the interface are (according to (11)) for the front surface ($\theta^3 = 1$) : $\mathbf{u} = \mathbf{v} + \mathbf{w}$ and for the back surface ($\theta^3 = -1$) : $\mathbf{u} = \mathbf{v} - \mathbf{w}$. With the application of both, the deformation of the middle surface and the difference vector, it is possible to describe the correct deformation of the three-dimensional structure which is transferred to the fluid-simulation, while preserving the efficiency of a two-dimensional structural computation.

Interface treatment for membrane elements

For an interface concerning a membrane structure, the thickness of the structure is neglected and accordingly the structure is treated as an infinitesimally thin plane in both the fluid and the structural computation. In the fluid simulation this infinite thin plane possesses two interface surfaces, one on each side, whose meshes don't need to coincide, while in the structural analysis the meshes on both sides coincide. In order to ensure the correct coupling of the upper (lower) side of the fluid interface plane with the upper (lower) side of the structure interface plane, searchtags are introduced.

3 Application to the Ovalling Phenomenon

In this section a well known phenomenon in wind engineering should be simulated to test the presented software environment's abilities. Under certain conditions on circular cylindrical shells, such as tall chimney structures, periodic vortex shedding occurs on the sides of the structure. This leads to a difference in the surface pressure and thus to a periodic load on the cylinder transversal to the flow direction. In wind engineering this aeroelastic effect is known as the "ovalling effect" and was subject to extensive experimental studies [9].

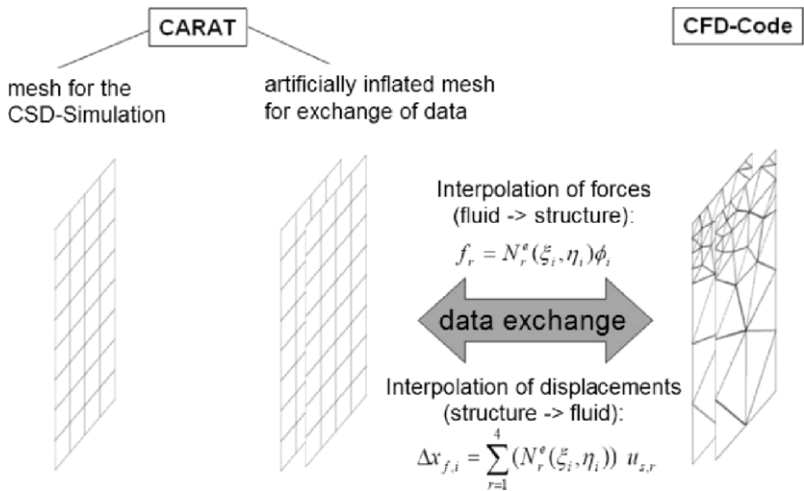


Fig. 6. Inflation of structural surface for CSD shell elements.

Since this phenomenon includes strong interaction between air flow and structure, it can be numerically accessed exclusively by methods which take this interaction into account, e.g. the approach proposed in this paper.

This experimental data provides a possibility to validate the results from the numerical coupled computations. However, these experiments are naturally in a 3D setting, which makes exact numerical simulations very elaborate. Therefore, as a first approach, a 2D setting is chosen (Fig. 7).

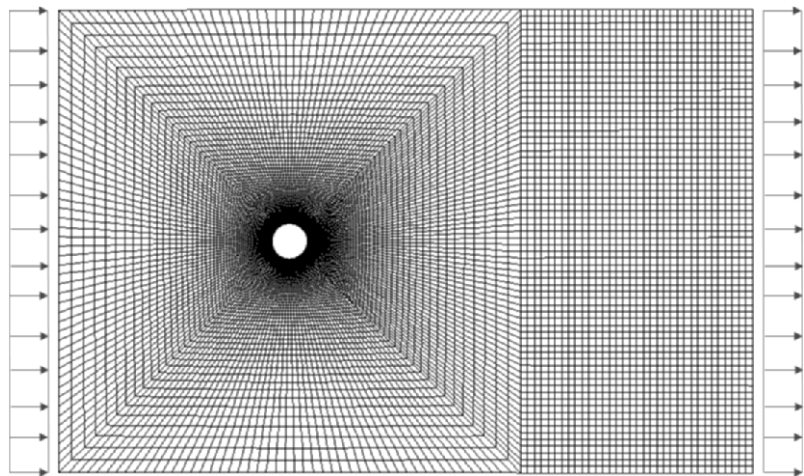


Fig. 7. Setup and CFD-mesh.

The structural problem is solved with the shell elements described in Section 2.1 and the generalized- α time integration scheme. The properties of the shell are a Young's modulus $E = 5.6 \cdot 10^8 \frac{N}{m^2}$, a density of $\rho = 1,8 \cdot 10^4 \frac{kg}{m^3}$, a thickness of $t = 0.5mm$ and a diameter of $7.62cm$. The inflow condition is defined with uniform velocity $U = 16.6 \frac{m}{s}$ and the outflow condition is an average static pressure of $0Pa$. The fluid is air at $25^\circ C$. A SST-turbulence model and a second order backward euler time integration scheme are applied in the fluid domain. With a considerably low time step of $5 \cdot 10^{-4}s$, this simulation shows stable behavior, even so only an explicit coupling scheme was applied. Figure 8 shows the time-deformation curve of point A.

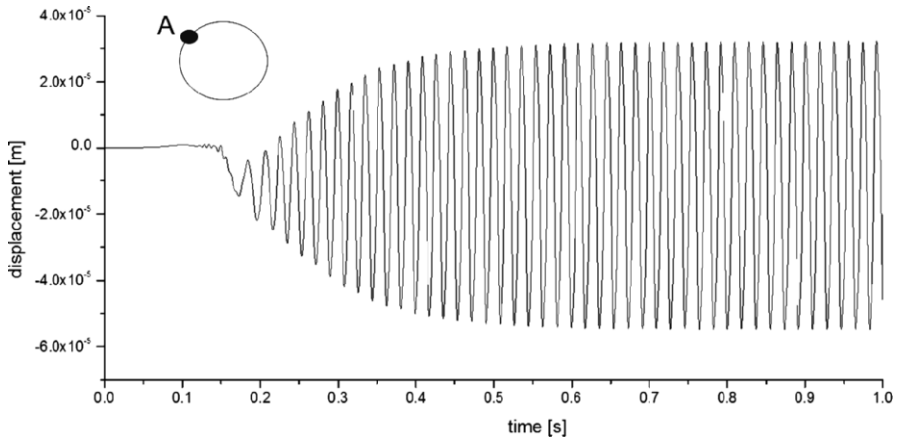


Fig. 8. Time-deformation curve for point A on the cylinder.

It is interesting to note that in accordance with literature the oscillation of the Cylinder occurs in a direction rotated by 45° (135°) degrees to the flow direction. Figure 9 presents a series of five deformation states of the structure.

4 Application to Instationary Simulation of a Hanging Roof

In the following example a hanging membrane roof under wind loading in a narrow 3D channel is presented. The parabolic velocity profile at the beginning of the channel has a maximum inflow velocity of $26 \frac{m}{s}$. Figure 10 shows the simulation setup and a detail view of the roof including the CFD-grid: the structure consists of a membrane roof with a span of $10m$ pre-stressed by its own weight and bordered by rigid walls. The membrane material is a



Fig. 9. Deformation state of the cylinder at different timesteps.

polyester fabric with PVC coating of type I and a thickness of 1 mm . The structural analysis is performed with 4-node membrane elements using the generalized- α method to solve the transient problem. In the fluid analysis a SST-turbulence model with wall functions is applied, time integration scheme in second order backward euler, and the fluid properties are those of air at 25°C .

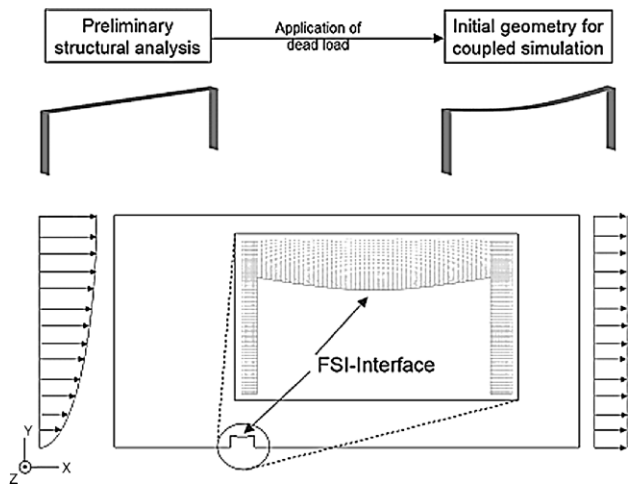


Fig. 10. Setup for the simulation of hanging roof under wind loading.

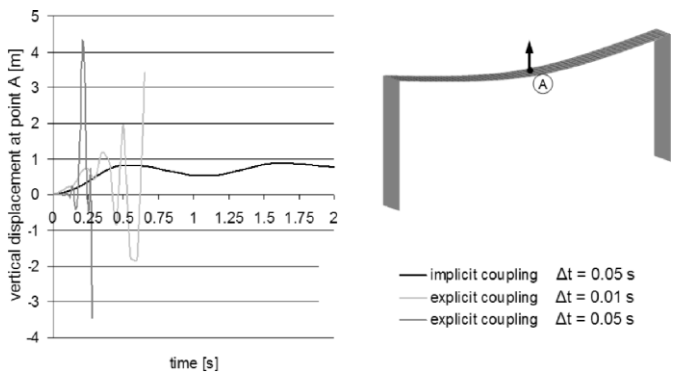


Fig. 11. Comparison of explicit and implicit coupling procedures.

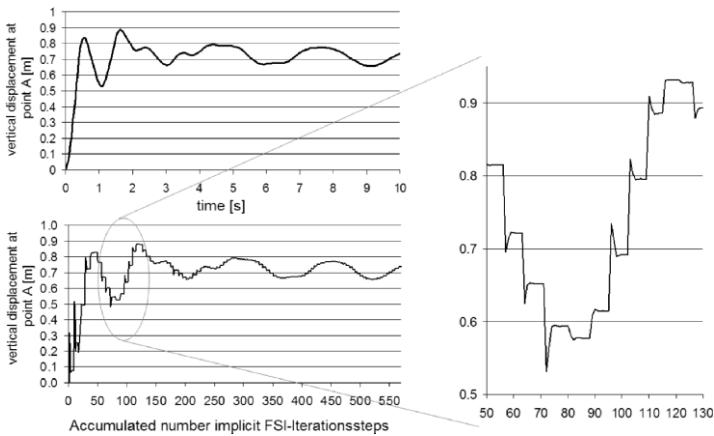


Fig. 12. Implicit coupling procedures: Iterations and results.

This simulation comprises of an interaction between incompressible fluid and flexible, nearly massless structure. In those cases iterative coupling is necessary due to the “artificial added mass” effect [13],[14]. Figure 11 gives a comparison of displacement-time diagram for the explicit coupling scheme with timestep $\Delta t = 0.05s$ and $\Delta t = 0.01s$ as well as an implicit (i.e. iterative) coupling scheme with $\Delta t = 0.05s$. Using the explicit iteration scheme leads to a instable simulation and decreasing the timestep size even worsens this behavior. The iteratively coupled algorithms show an unstable behavior as well if no under-relaxation is applied. For stabilisation here the adaptive under-relaxation technique presented above is applied.

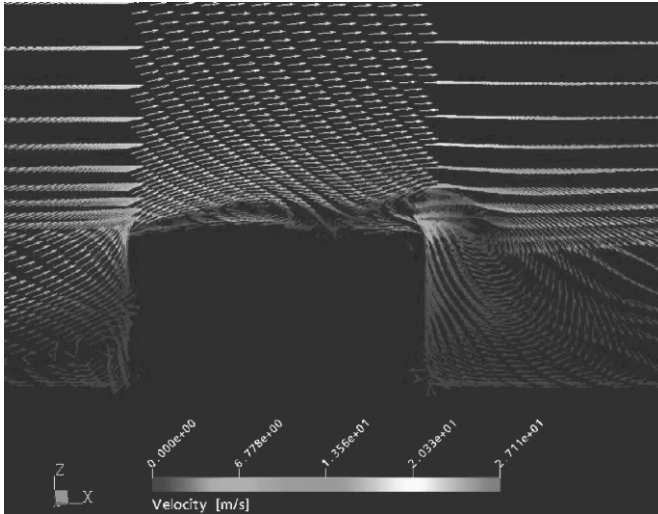


Fig. 13. Deformation of membrane roof and velocity vectors.

However, the required stability of the iteratively coupled computation is achieved by a higher computational effort. Figure 12 shows the time-displacement curve in the middle of the roof together with the time- “number of coupled iterations”-curve below. From the second curve and its detailed view it can be observed, that the necessary number of iterations to reach a kinematic equilibrium within one timestep increases with the gradient of the deformation of the structure and thereby with higher velocities of the FSI interface.

The simulation results show the expected upward movement, “snap-through” of the membrane roof and following oscillations. Figure 13 shows the deformation and the flow around the roof after the “snap-through”-point.

5 Application to a Tent Structure

In the following example a four point tent structure is analyzed under wind loading. Therefore a strongly coupled simulation in a full 3D-setup is performed. The influences of wind directions in the two planes of symmetry are analyzed. Besides testing the described FSI approach, the aim of this simulation is the assessment of the occurring effects and their magnitude in a qualitative way.

The inflow profile is logarithmic according to the requirements of German building code DIN 1055-2 but respecting an overstated maximum velocity $U_{max} = 40 \frac{m}{s}$. The four point tent structure resembles a saddle surface of a membrane with a uniform prestress of $2.5 \frac{kN}{m}$ stabilized by four cables at

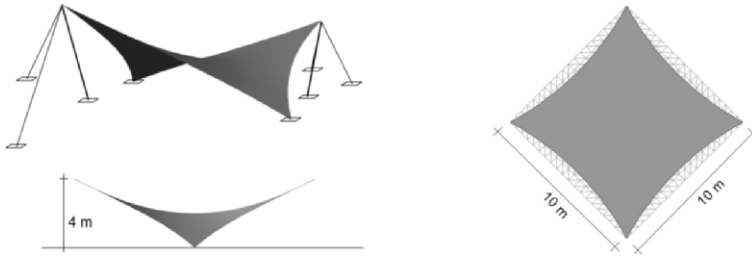


Fig. 14. Geometry and dimension of the four point tent structure.

the edges prestressed with $50kN$. Hence the result of the form finding is a minimal surface. The membrane material is a polyester fabric with PVC coating of type I and a thickness of $1mm$, the edge cables are spiral strands with a diameter of $16mm$. The dimensions are given in Fig. 14. Here in the top view the initial configuration before the form finding operation can be seen as an underlying mesh.

The setup of the analysis follows the flow chart given in Fig. 1, including the form finding procedure described in Section 2.1 to acquire the proper initial geometry. The interface is treated as a two sided infinite thin surface in both the structure and flow analysis.

The simulation setup is 3D (Fig. 15) using a tetrahedral mesh for the CFX simulation with refinement by prism layers nearby the membrane and the bottom. A SST-turbulence model using wall functions is applied. In the structural analysis 3-node membrane elements are used for a nonlinear analysis. The fluid is air at $25^{\circ}C$.

A steady state result for the most susceptible configuration with respect to the deformations as well as the pressure load on the membrane is presented: the higher tip of the structure heads towards the wind. As Fig. 16 shows, an upwards deformation occurs in the front part while the rear part deforms downwards. This also corresponds to the pressure distribution, here exemplarily given for a midplane in y - z direction in the domain.

From a civil engineer's point of view, the transient behavior of the structure is of high interest in order to avoid any oscillations that could shorten its lifetime due to wear and tear. Therefore, in addition to steady-state analysis, the sensitivity to fluttering and the reaction towards wind gusts are important to adjust the prestress and initial geometry.

This dynamic behavior of the coupled system was analyzed in a transient FSI simulation. With regard to the low fluid density and the large deformations of the structure, the computation was carried out in a strongly coupled, implicit manner. The time integration used on the fluid side was a second order backward euler scheme, on the structural side the generalized- α methode was applied.

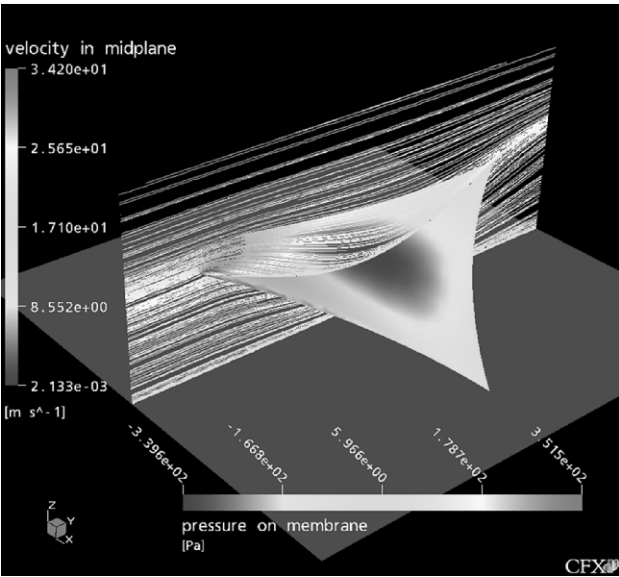


Fig. 15. Visualisation of flow and pressure at the membrane.

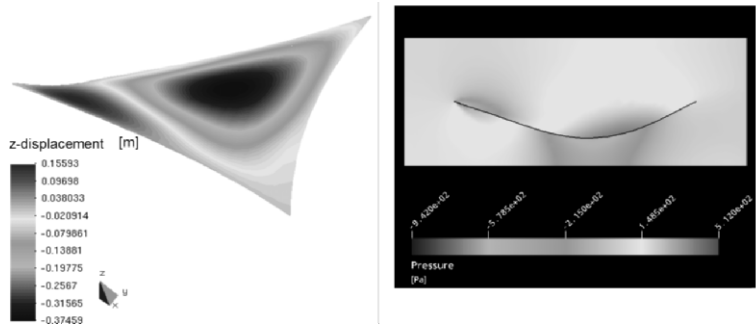


Fig. 16. Results for a wind flow in the y-direction.

Starting from a steady-state solution of the deformed membrane at $20 \frac{m}{s}$, the maximum windspeed was varied between $10 \frac{m}{s}$ and $30 \frac{m}{s}$ in a gust-like behavior (Fig. 17). The structural deformation followed the variation of the wind speed without any delay due to its prestress and its little mass. This supports earlier coupled simulation results including membranes which showed very little influence of the mass of the membrane on the inertia of the system. Figure 18 presents the state of deformation at different time steps.

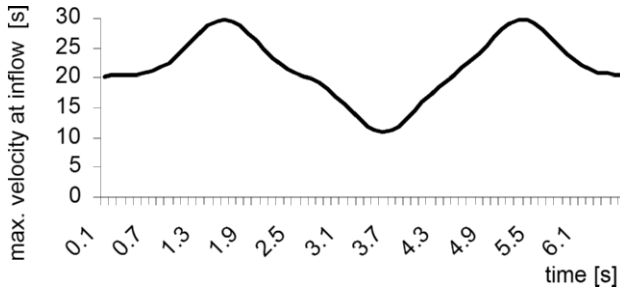


Fig. 17. Time-inflow velocity diagram for a height of 5m over the ground.

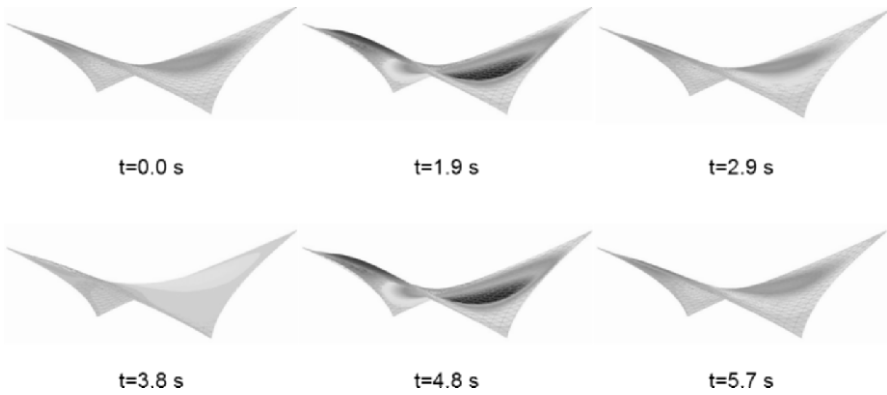


Fig. 18. Response of the structure at different timesteps.

These results provide a first assessment to the structure's behavior under wind loading, as well in magnitude of the deformation as in the occurring frequencies. Thereby, a possibility to quantify the effects, which are expected due to the geometry and the problem setting, is given.

6 Conclusion

In this paper, a modular software environment using methods of FSI specialized for the simulation of thin shells or membrane structures was presented. To take into account all demands arising from the physical problem, a partitioned approach using highly developed codes for the single field computations, namely CARAT and CFX-5, is applied.

Strongly and weakly coupled iteration schemes are possible. For the stabilization of strongly coupled iteration schemes, an adaptive under-relaxation technique is applied.

Special emphasis is laid on the discussion of data exchange on thin-walled structures as the structural model consists only of a spatially curved surface, while for the fluid part a 3D model of the structure is advantageous. Therefore an ‘inflated’ interface is proposed which, in combination with the solid shell model, can represent the three-dimensional solution of the structural problem by conserving the advantages and efficiency of two-dimensional formulations for nonlinear, thin-walled structures.

In the case of membrane structures, the coupling surface consists of an infinitely thin structure which is represented by two interface meshes on both wetted sides, respectively. It is demonstrated that the special load carrying behavior of membranes requires additional considerations concerning prestressing and form finding.

The application of the proposed software environment and the frame algorithm is shown in two examples from the civil engineering background: the coupled simulations of a hanging roof and a four-point sail subject to wind loading. Considering the requirements towards the occurring turbulence and the boundary conditions, it is obvious that further improvements are necessary in order to precisely approach the effects of wind loads. However they provide an idea about the occurring effects and their magnitude, showing the possibilities that the application of methods of FSI opens in wind engineering.

The possibility of modeling fluid-structure interactions numerically, opens the range to methods of optimization for coupled systems. Up to now, most optimizations techniques focus on the separate fields of the structural analysis or the analysis of the flow around or in a structure. On the basis of the presented approach, the extension of the separate optimization strategies to a multidisciplinary optimization, where the whole coupled system is subject to optimization, is topic of current research.

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Experimental Study on a Fluid-Structure Interaction Reference Test Case

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Abstract. Experimental studies on reference test cases are of capital importance to support the development of models and coupling strategies for numerical simulations on fluid-structure interaction problems. From the experimental view point, the study of the coupled unsteady fluid flow and structure motion requires specially adapted test rigs and measurement techniques to obtain accurate time-resolved results. This demand has triggered the present contribution to design and to study a two-dimensional reference test case on laminar fluid-structure interaction. A new experimental facility to be operated with high viscous fluids was build exclusively for the present study and a Particle Image Velocimetry system was adapted to measure both the periodical flow velocity field and the structure deflection modes. Finally the reference experiment was defined and performed. An extended investigation was conducted on the two-dimensional reference structure model at a Reynolds number of 170 and lead to very reproducible and accurate results compiled in a data base.

1 Introduction

Flow-structure interaction problems, involving the coupling of unsteady fluid flow and structure motion, arise in many fields of engineering [3,4] as well as in many other sciences, e.g. medicine. With the continuous increase of computer power, these problems have attracted more and more interest of the computational mechanics. However, in spite of the practical relevance of the prediction of coupled fluid and structural dynamics in many technical problems, this type of simulation is not yet considered a validated tool. Increased efforts in numerical research and development are presently being observed to develop models and coupling strategies for numerical simulations and to create coupling algorithms between computational fluid dynamics (CFD) and computational structural dynamics (CSD) solvers.

A group of University institutes has decided to form a research unit to work on this subject. This common effort shall result in new tools which permit a realistic simulation of such complex, nonlinear coupled problems. However, to ensure success, comparisons with experimental studies are needed. This demand has triggered the present contribution to perform a reference experiment specially dedicated toward the diagnostic and validation of the

numerical models for fluid-structure interaction simulations. The main objective was to establish a common experimental test case on fluid-structure interaction and to provide a reliable data base to serve the validation and comparison purposes of the different numerical methods and codes implementations.

The first reference experiment was focused on the two-dimensional periodical swiveling motion of a relatively simple flexible structure driven by a constant velocity incoming fluid flow. The structure had a linear mechanical behavior and the fluid was considered incompressible and in the laminar regime.

2 Experiment Definition

To provide the details needed for numerical validation and accuracy verification purposes the test case had to be conducted under very precise and well controlled working conditions. The requirements of the project imposed stringent restrictions to the selection of the reference structure while the necessity of very low Reynolds numbers determined the construction of a new facility specially dedicated to fluid-structure interaction studies.

The definition of the structure geometry took into account four principal aspects: (i) reproducibility of the resulting motion, (ii) two-dimensionality of the structure deflection, (iii) moderate structure motion frequency and (iv) significant excursion of the structure. The structure found to serve the requirements consisted of a 0,04 mm thick stainless steel membrane attached to an aluminum cylindrical front body. At the trailing edge of the flexible sheet a rectangular stainless steel mass was located. Both the rear mass and the front body were considered rigid. All the structure was free to rotate around an axle located in the center point of the cylindrical front body. The detailed dimensions of the structure are represented in Fig. 1. The density of the different materials used in the construction of the model are resumed in Table 1.

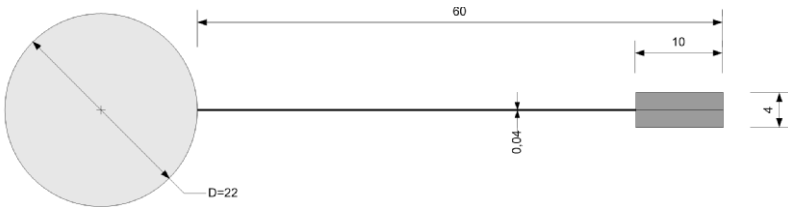


Fig. 1. Structure geometrical definition (all dimensions in millimeters).

Table 1. Density values of the structure components.

Front body	(aluminum)	2828 kg/m ³
Membrane	(stainless steel)	7855 kg/m ³
Rear mass	(stainless steel)	7800 kg/m ³

For the range of forces acting on the structure during the tests the mechanical behavior of the flexible stainless steel membrane has proven to be linear. The Young modulus was measured to be 200 kN/mm².

According to the specified requirements the first test case was in the laminar regime, with a Reynolds number, based on the diameter of the cylindrical front body, not bigger than 500. In order to achieve such small Reynolds number, the tests had to be performed in a fluid of high kinematic viscosity. The adopted test fluid was a mixture of Polyglycol and water and can be considered incompressible. It had a kinematic viscosity of 0,00019 m²/s and a density of 1050 kg/m³.

3 Experimental Apparatus

3.1 Flustruc Experimental Facility

Because of the need to perform the tests with hight viscous fluids a dedicated closed circuit tunnel was designed and constructed. The present investigation shall be expanded to turbulent flows in the near future, therefore, the facility was designed to be operated under very precise and controllable working conditions with both water and high viscous fluids. A sketch of the facility is shown in Fig. 2.

It is a vertical closed circuit tunnel powered by a 24 kW axial propeller pump. It was designed to be operated with liquids with kinematic viscosity ranging from 1×10^{-6} m²/s up to 5×10^{-4} m²/s. Emphasis was placed on the design of the 180 mm $\times$ 240 mm cross section, 338 mm long test section to allow flow investigations using laser optical measurement techniques. Therefore, it was entirely built out of glass to provide full optical access to all sides of the test model.

The model was located 55 mm downstream of the beginning of the test section and mounted on supports of negligible friction (see Fig. 3). The gravity force was aligned with the x-axis and so it does not introduce any asymmetry. As the working fluid polyethylene glycol PG-12000 (polyglycol) syrups were used. They permitted an accurate control of the kinematic viscosity of the tests within the design range of the facility. For the same concentration range the density of the syrups presented a minor variation from 1000 kg/m³ to 1100 kg/m³ (see Fig. 4).

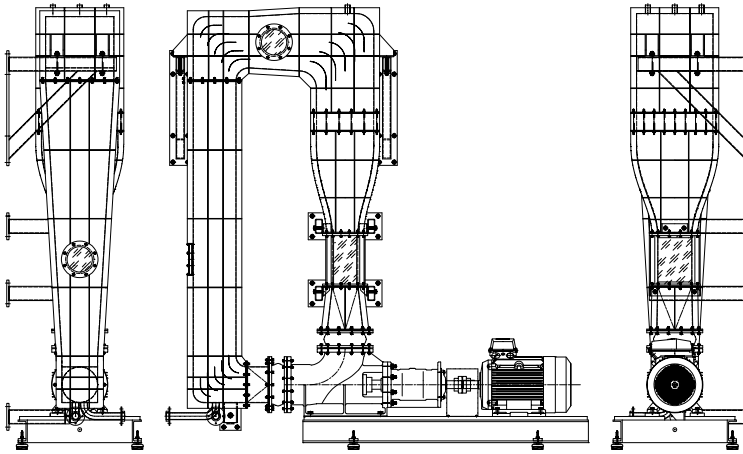


Fig. 2. FLUSTRUC tunnel layout.

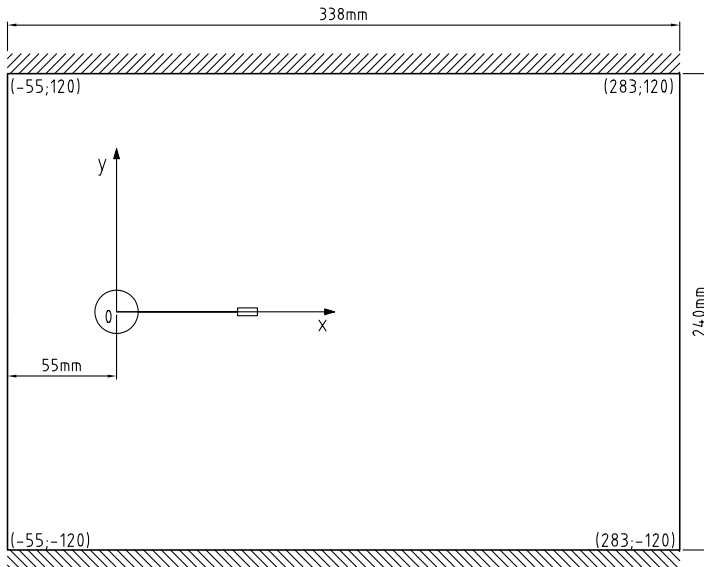


Fig. 3. Test section dimensions.

3.2 Time-phase Resolved Measurements

When investigating fluid-structure interaction problems two additional difficulties appear when it comes to resolve the measured data in time-phase space. Firstly, the periodicity of the structure motion is quite sensitive and, therefore, there are cycle-to-cycle variations of the period time. Secondly,

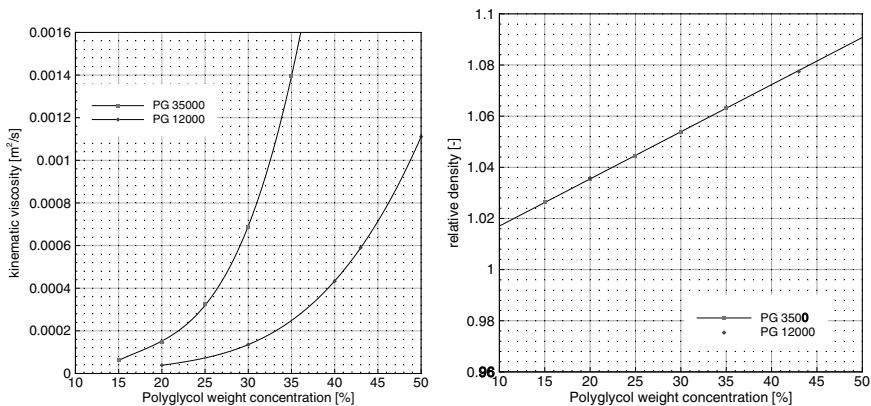


Fig. 4. Viscosity (left) and density (right) of Polyethylene glycol solutions versus Polyethylene glycol weight concentration.

the velocity of the structure motion is not predefined (as it is for example in crank shaft driven set-ups) which makes it impossible to reconstruct the time-phase resolved data from position resolved measurements. Because of these reasons, a different approach was adopted. Instead of triggering the data acquisition by the experiment the measuring system was operated at constant acquisition rate and both events the acquisition of a measurement and the start of a new cycle of the structure motion were recorded based upon an absolute clock. Using this time information the data was reorganized in a post-processing program in order to provide the time resolved data. Besides solving the trigger problem, this solution also resulted in a minimum acquisition time needed, because the measuring system could be operated at its optimum measuring rate independent of the structure swiveling frequency. The hardware module designed to perform the event monitoring was based on a FPGA (Field Programmable Gate Array) and a 1 MHz internal clock. In this way it was able to record up to 250 events per second with an accuracy of $\pm 2 \mu\text{s}$.

During the present tests two different kinds of events were recorded; the measurements (t_{ij}) and the beginning of the flexible structure motion period (t_i). Depending on the type of measurements performed (PIV or DMI) the measurement events were detected using the first laser pulse trigger signal or the camera first frame trigger signal, respectively. After deciding which angular position of the structure front body correspond to the beginning of the motion period this position was detected by a magnetic angular position sensor and a small magnet attached to the model. This sensor was elected to perform the task based on two criteria; non contacting position angular sensor and direction resolved output signal. An additional reason justified the employment of this sensor; the high sensitivity around a predefined an-

gular position. From the recorded events time information a specific software computed the period time of each individual period (T_j) and the measurement time phase angle within the structure motion period (t_j). Finally the time resolved measurements were reorganized in a reference structure motion period equal to the mean period of all acquired periods.

$$t_j = \frac{t_{ij} - t_i}{T_i} \times 360 \quad (1)$$

$$T_i = t_{i+1} - t_i \quad (2)$$

A typical timing diagram is given in Fig. 5.

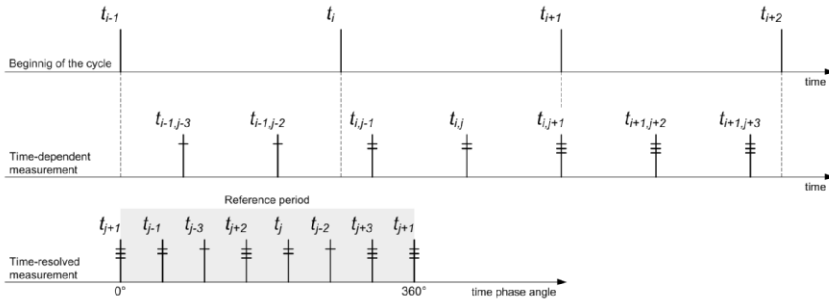


Fig. 5. Time phase resolved measurement reconstruction scheme.

3.3 Flow Velocity Measurements

The task of sampling the flow velocity of the fluid surrounding the flexible structure was addressed to a two-component Particle Image Velocimetry system. The PIV system adopted for the measurements consisted of two 1280 pixel $\times$ 1024 pixel synchronized cameras and a pulsed double-head 120 mJ laser.

The cameras were used to acquire two simultaneous time-dependent pair of PIV images from an extended flow field area per measurement. For that purpose they were mounted parallel to the rotating axle of the flexible structure to visualize the flow in a plane perpendicular to it. To assure a correct position of the two adjacent images a special support was designed to hold both cameras and to permit the adjustment of each individual camera in the 6 axis or of both cameras simultaneously. The PIV images acquired by the two cameras were then imported into a MatLab-based post-processing software to stitch the correspondent pairs of images before being cross-correlated. Opting for the solution of two synchronized cameras it was possible to achieve a large 272 mm $\times$ 170 mm flow field measuring area while keeping the spatial resolution as low as 133 μm $\times$ 133 μm per CCD pixel.

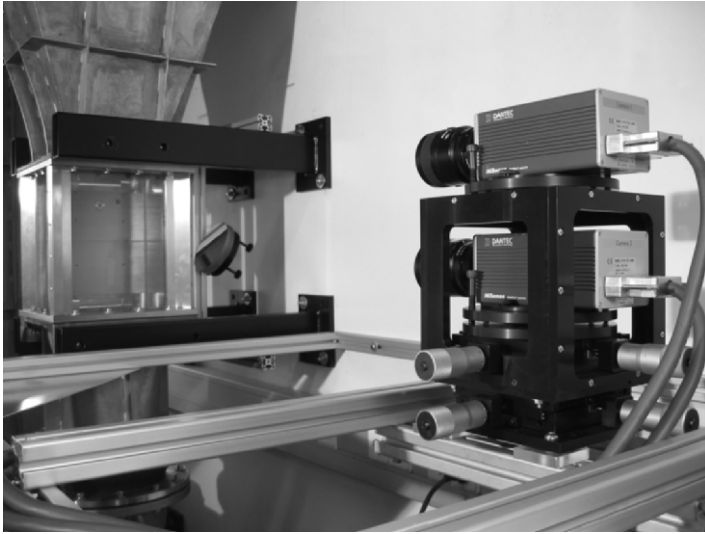


Fig. 6. Two-dimensional PIV cameras arrangement.

Preliminary tests had shown that a reliable and continuous data acquisition from two synchronized cameras was possible for long periods of time using a maximum frequency of 1 Hz.

Another improvement of the measuring technique to the specific task was related to the illumination of the flow field. The laser sheet was positioned perpendicular to the flexible structure rotating axis at the center of the structure in the spanwise direction. However the presence of the swiveling flexible structure imposed extra problems. The most important of them was related to the fact that the flexible structure was a swiveling opaque body which created an unsteady dark shadow region in the fluid. This behavior not only reduced the measuring area to almost only one side of the flexible structure but also made the masking of the PIV images in post-processing difficult to be preformed. To cope with this problem two different solutions were tried. First, a mirror was mounted on the opposite side of the test section to take advantage from the laser light passing the test section with the objective to deflect the laser sheet back and to illuminate the dark flow region. The second solution tried was to make use of multiple light sources to illuminate the structure from both sides to cover the entire flow field avoiding the shade behind the model. Both solutions presented in general the same advantages and disadvantages; the dark region behind the structure was extinguished and all the flow field surrounding the structure was accessible to perform PIV measurements at once. As a disadvantage it turned out that there were regions of significant different light intensity in the flow. However, this prob-

lem could be minimized with a proper adjustment of the laser light focus and the cameras optics. Because with the first approach the differences in light intensity in the images were more pronounced the solution of multiple light sources was finally adopted.

Different types of seeding particles were employed depending on the test fluid used. For tests in water $10\text{ }\mu\text{m}$ mean diameter hollow glass spheres were used while for Polyglycol syrups $10\text{ }\mu\text{m}$ mean diameter silver coated hollow glass spheres were adopted. Although hollow glass spheres appear to be the most suitable choice to be used in polyglycol syrups as far as density is concerned (the relative density of those spheres are about 1, 1) they introduce an additional problem because of their refractive index. The refractive index of the syrups is higher than the one of water and closer to the refractive index of the non coated spheres. On the other hand, silver coated hollow glass spheres are non-transparent particles and produce higher signal levels. One major drawback could be expected by using silver coated glass spheres; the relative density of this kind of particles is about 1, 4. Nevertheless, this drawback was neglectable because of the high viscosity and the velocity of the flow during the tests.

3.4 Structure Deflection Measurements

For structure deflection modes identification (DMI) measurements the PIV system was modified to provide it with structure deflection analysis capabilities. The idea behind this setup was to use the 3D stereoscopic PIV module to acquire and organize images from the swiveling structure using an especially developed software to analyze and reconstruct the time dependent deflection of the structure. The major advantage of this approach was that the same measuring system used for the velocity field measurements could be employed.

In order to generate coherent results, the relative position of the laser source was maintained to illuminate the time-dependent deflection of the flexible structure at the same plane as for the velocity field measurements. Just one camera was positioned in a skewed position to acquire the time dependent image of the light sheet reflected by the structure. The camera optics were improved with special filters correspondent to the wave length of the laser to increase the contrast between the reflection of the laser light on the structure and the ambient light.

After image acquisition, 3D PIV software including camera calibration routines that measure and account for perspective distortion were implemented to correct parallax errors and to reconstruct the real structure deflection image. The quantitative analysis of the time-dependent structure deflection was performed in MatLab workspace by a script developed for the specific task. It mapped the grayscale value of each pixel of the entire image and detected the line resulting from the intersection of the laser sheet and the structure. Together with the image calibration information the algorithm

was able to reconstruct the two-dimensional deflection of the flexible structure. Finally the deflection modes present in the structure were identified and characterized based on the results of the two-dimensional deflection data.

4 Experimental Results

The structure model was tested at different incoming velocities up to 2 m/s. It stayed steady and aligned with the approaching flow up to a velocity slightly bigger than 0,9 m/s. From this lower velocity limit the structure showed a stable symmetric and cyclic swiveling motion. The transition from rest to a stable cyclic motion was abrupt and it was not possible to define any evolution of the structure from one stage to the other.

Figure 7 shows the frequency response of the structure as a function of the incoming velocity. In the range of velocities were there was an interaction between the fluid flow and the structure the swiveling frequency varied linearly with the flow velocity and two distinct structure swiveling modes could be observed. The first swiveling mode appeared up to 1,25 m/s while the second started for flow velocities above 1,32 m/s. In this interval the mode shift displayed a behavior similar to hysteresis. Figure 8 refers to visualizations performed during the tests with the structure at 1,13 m/s and 1,44 m/s, respectively, and puts in evidence the difference between the two swiveling motion modes.

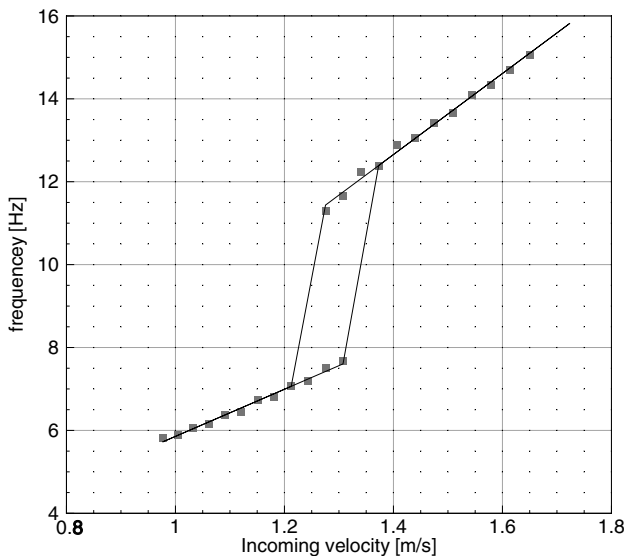


Fig. 7. Structure swiveling frequency versus incoming flow velocity.

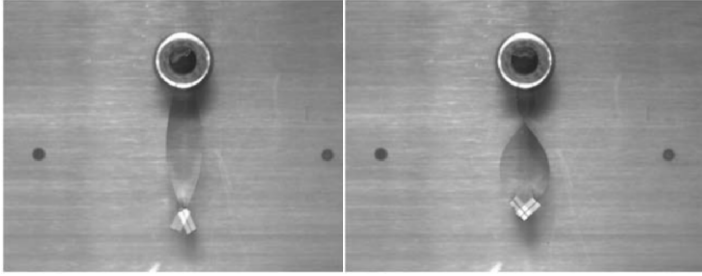


Fig. 8. Superimposition of images at a time-phase angle of 90° and 270° for an incoming flow velocity of 1,13 m/s (left) and 1,44 m/s (right).

In order to fully characterize the second swiveling motion mode of the structure an extended investigation was conducted at 1,44 m/s. The Reynolds number at this velocity, based on the diameter of the cylindrical front body, was 170. The tunnel was capable to maintain a uniform velocity profile across the test section except in the regions of the of the laminar boundary layer. The inlet velocity profile was measured in the absence of any model at $x = -55$ mm (see Fig. 9). Under those conditions the flow angularity was better than $0,5^\circ$ and the RMS of the velocity magnitude variance smaller than 1%.

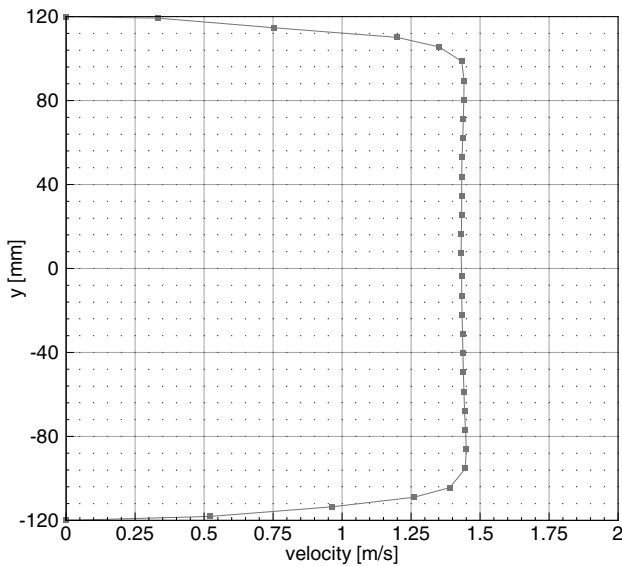


Fig. 9. Inlet velocity profile.

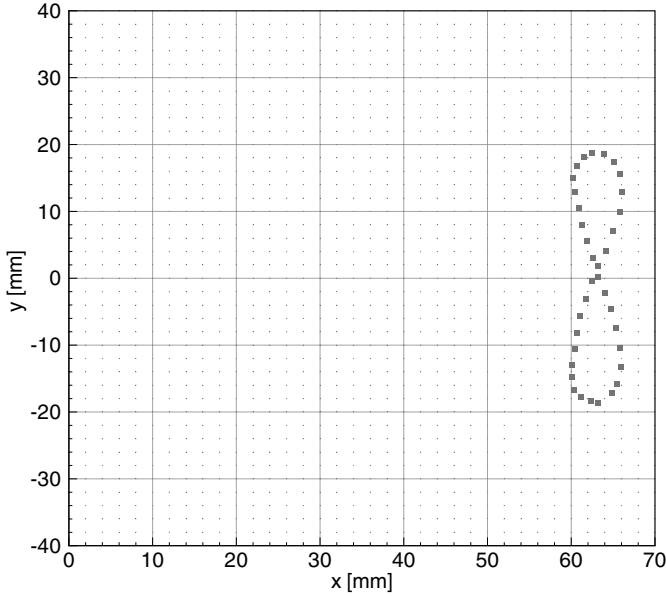


Fig. 10. Coordinates of the structure trailing edge within a period of motion.

Respecting the response of the structure two characteristic parameters were interesting to monitor: the front body angle of attack and the position of the trailing edge. For the velocity of 1,44 m/s the structure swiveled at a frequency of 13 Hz, and in the present test a maximum fluctuation on the structure motion frequency from cycle-to-cycle of $\pm 2\%$ was measured. In Fig. 10 the location of the trailing edge within a period of the structure movement is plotted whereas Fig. 11 shows the angle of attack of the cylindrical front body versus time.

As far as the flow field is concerned time-phase resolved flow field measurements were performed. The flow surrounding the two-dimensional flexible structure was registered with a time-phase angle resolution of 2.5° within an uncertainty of $\pm 0,2^\circ$. Figure 12 compiles the flow velocity field at different instants of the reference swiveling period. Each map corresponds to the average of 10 realizations. The velocity field measurements has covered a flow area from $(x, y) = (-36 \text{ mm}, -85 \text{ mm})$ to $(x, y) = (236 \text{ mm}, 85 \text{ mm})$. In the first three maps the front body is rotating clockwise while in the last one it is rotating in the opposite direction.

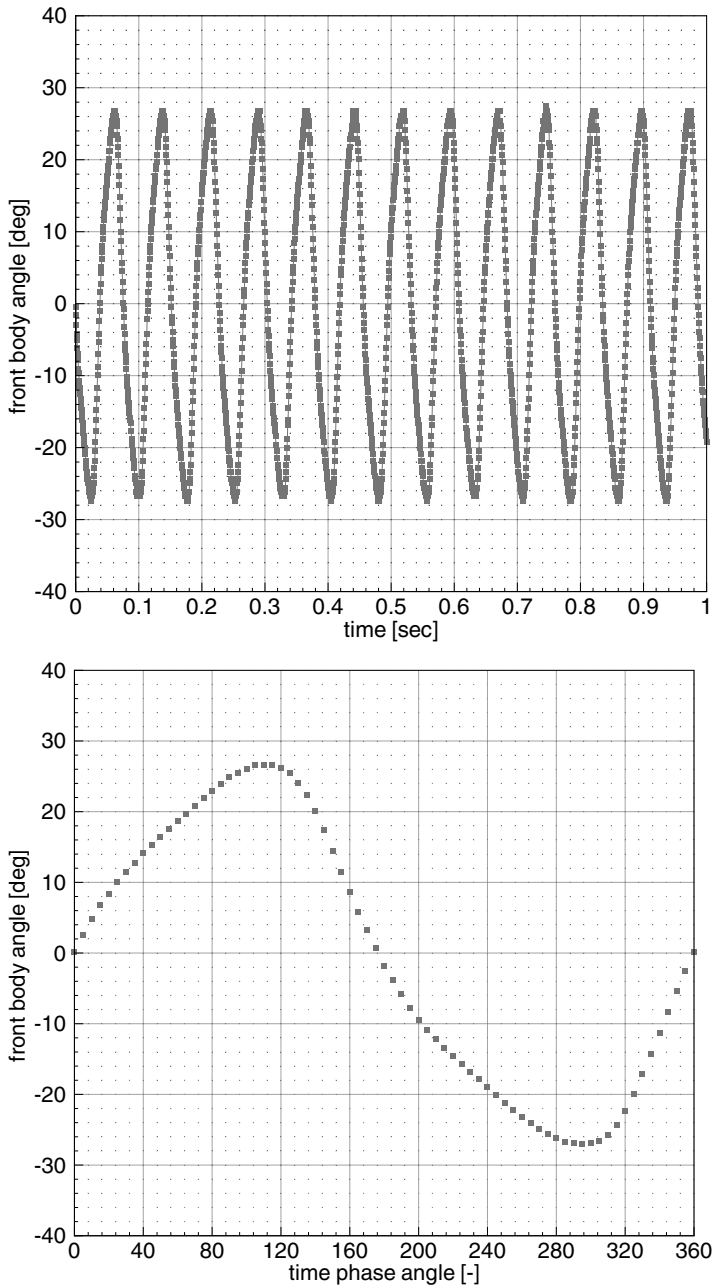


Fig. 11. Front body angle of attack over 1 second of test (top) and within the reference period of motion (bottom).

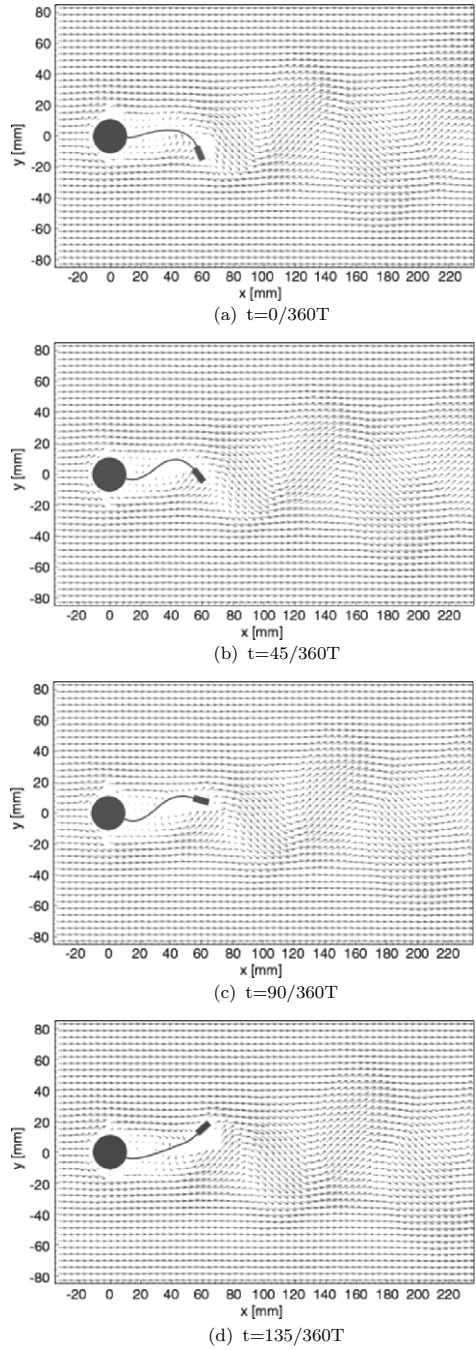


Fig. 12. Time phase resolved combined velocity flow field/structure deflection results for $t_j = 0^\circ$, $t_j = 45^\circ$, $t_j = 90^\circ$ and $t_j = 135^\circ$.

5 Conclusions

The increased efforts in numerical research to develop coupling strategies for the simulation of fluid-structure interaction problems have triggered the development of a reference experiment on the same research field. Within this work, a new test facility was constructed to form tests in laminar and turbulent flows under well defined boundary conditions. The dedicated tunnel was designed to be operated with fluids of different viscosities and has satisfied all the design objectives. It has shown to be capable to maintain a maximum constant velocity of 2 m/s in the test section when operated with a fluid of $0,00019\text{ m}^2/\text{s}$ of kinematic viscosity. The facility permits to perform the tests under very precise boundary and working conditions.

As far as the measurement techniques are concerned Particle Image Velocimetry was successfully applied to measure the velocities in the flow surrounding the flexible structure and, with some modifications, to measure the deflection of the structure. The time-phase detector module made the reconstruction of the unsteady periodic time phase resolved measurements reliable and extremely accurate. Because of its architecture, this approach has turned out to be more memory consuming compared to other solutions but yielded accurate time-phase resolved measurements reconstruction independently from the nature of the structure motion.

The reference structure model has proven to fulfill all the requirements needed. It showed a very reproducible two-dimensional swiveling motion. The excursion of the membrane was measured to be significant: at $1,44\text{ m/s}$, in particular, the front body angle of attack varies within $\pm 23^\circ$ and the maximum displacement of the trailing edge is about $\pm 19\text{ mm}$. All the movement of the flexible structure is very well symmetrical in relation to the x-axis. The oscillation response of the body is similar to the galloping response of square cross-section prisms as observed by Parkinson [1] and Novak [2].

The resulting 13 Hz periodic motion of the structure driven by a uniform incoming flow at $1,44\text{ m/s}$ was characterized in detail. A data base was compiled on the flow velocity field around the structure as well as on the deflection of the structure within the reference swiveling period. The experimental results are now available and constitute a useful tool for validation and comparison of different numerical simulations.

6 Acknowledgements

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Proposal for Numerical Benchmarking of Fluid-Structure Interaction between an Elastic Object and Laminar Incompressible Flow

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Abstract. We describe new benchmark settings for the rigorous evaluation of different methods for fluid-structure interaction problems. The configurations consist of laminar incompressible channel flow around an elastic object which results in self-induced oscillations of the structure. Moreover, characteristic flow quantities and corresponding plots are provided for a quantitative comparison.

1 Introduction

The main purpose of this benchmark proposal is to describe specific configurations which shall help in future to test and to compare different numerical methods and code implementations for the fluid-structure interaction (FSI) problem. In particular, the various coupling mechanisms, ranging from partitioned, weakly coupled approaches to fully coupled, monolithic schemes are of high interest. Moreover, it shall be possible to examine the quality of different discretization schemes (FEM, FV, FD, LBM, resp., beam, shell, volume elements), and the robustness and numerical efficiency of the integrated solver components shall be a further aspect. This new benchmark is based on the older successful *flow around cylinder* setting developed in [1] for incompressible laminar flow and on the setup in [2]. Similar to these older configurations we consider the fluid to be incompressible and in the laminar regime. The structure is allowed to be compressible, and the deformations of the structure should be significant. The overall setup of the interaction problem is such that the solid object with elastic part is submerged in a channel flow. Then, self induced oscillations in the fluid and the deformable part of the structure are obtained so that characteristic physical quantities and plots for the time-dependent results can be provided.

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2 Definitions

We consider the flow of an **incompressible Newtonian fluid** interacting with an **elastic solid**. We denote by Ω_t^f the domain occupied by the fluid and Ω_t^s by the solid at the time $t \in [0, T]$. Let $\Gamma_t^0 = \bar{\Omega}_t^f \cap \bar{\Omega}_t^s$ be the part of the boundary where the elastic solid interacts with the fluid.

2.1 Fluid Properties

The fluid is considered to be **Newtonian, incompressible** and its state is described by the velocity and pressure fields $\mathbf{v}^f, p^f$. The balance equations are

$$\begin{aligned} \varrho^f \frac{\partial \mathbf{v}^f}{\partial t} + \varrho^f (\nabla \mathbf{v}^f) \mathbf{v}^f &= \operatorname{div} \boldsymbol{\sigma}^f & \text{in } \Omega_t^f, \\ \operatorname{div} \mathbf{v}^f &= 0 \end{aligned} \quad (1)$$

The material constitutive equation is

$$\boldsymbol{\sigma}^f = -p^f \mathbf{I} + \varrho^f \nu^f (\nabla \mathbf{v}^f + \nabla \mathbf{v}^{fT}). \quad (2)$$

The constant density of the fluid is ϱ^f and the viscosity is denoted by ν^f . The Reynolds number is defined by $\operatorname{Re} = \frac{2r\bar{V}}{\nu^f}$, with the mean velocity $\bar{V} = \frac{2}{3}v(0, \frac{H}{2}, t)$, r radius of the cylinder and H height of the channel (see Fig. 1).

2.2 Structure Properties

The structure is assumed to be **elastic** and **compressible**. Its configuration is described by the displacement $\mathbf{u}^s$, with velocity field $\mathbf{v}^s = \frac{\partial \mathbf{u}^s}{\partial t}$. The balance equations are

$$\varrho^s \frac{\partial \mathbf{v}^s}{\partial t} + \varrho^s (\nabla \mathbf{v}^s) \mathbf{v}^s = \operatorname{div}(\boldsymbol{\sigma}^s) + \varrho^s \mathbf{g} \quad \text{in } \Omega_t^s. \quad (3)$$

Written in the more common Lagrangian description, i.e. with respect to some fixed reference (initial) state Ω^s , we have

$$\varrho^s \frac{\partial^2 \mathbf{u}^s}{\partial t^2} = \operatorname{div}(J \boldsymbol{\sigma}^s \mathbf{F}^{-T}) + \varrho^s \mathbf{g} \quad \text{in } \Omega^s \quad (4)$$

where $\mathbf{F} = \mathbf{I} + \nabla \mathbf{u}^s$ is the deformation gradient tensor. For further details see for example [3].

The material is specified by giving the Cauchy stress tensor $\boldsymbol{\sigma}^s$ (the 2nd Piola-Kirchhoff stress tensor is then given by $\mathbf{S}^s = J \mathbf{F}^{-1} \boldsymbol{\sigma}^s \mathbf{F}^{-T}$) by the

following constitutive law for the **St. Venant-Kirchhoff** material ($\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$)

$$\boldsymbol{\sigma}^s = \frac{1}{J} \mathbf{F} (\lambda^s (\text{tr } \mathbf{E}) \mathbf{I} + 2\mu^s \mathbf{E}) \mathbf{F}^T \quad (5)$$

$$\mathbf{S}^s = \lambda^s (\text{tr } \mathbf{E}) \mathbf{I} + 2\mu^s \mathbf{E} \quad (6)$$

The density of the structure in the undeformed configuration is ϱ^s . The elasticity of the material is characterized by the Poisson ratio ν^s ($\nu^s < \frac{1}{2}$ for a compressible structure) and by the Young modulus E . The alternative characterization is described by the Lamé coefficients λ^s and μ^s (the shear modulus):

$$\nu^s = \frac{\lambda^s}{2(\lambda^s + \mu^s)} \quad E = \frac{\mu^s(3\lambda^s + 2\mu^s)}{(\lambda^s + \mu^s)} \quad (7)$$

$$\mu^s = \frac{E}{2(1 + \nu^s)} \quad \lambda^s = \frac{\nu^s E}{(1 + \nu^s)(1 - 2\nu^s)} \quad (8)$$

2.3 Interaction Conditions

The boundary conditions on the fluid solid interface are assumed to be

$$\begin{aligned} \boldsymbol{\sigma}^f \mathbf{n} &= \boldsymbol{\sigma}^s \mathbf{n} \\ \mathbf{v}^f &= \mathbf{v}^s \end{aligned} \quad \text{on } \Gamma_t^0, \quad (9)$$

where $\mathbf{n}$ is a unit normal vector to the interface Γ_t^0 . This implies the no-slip condition for the flow, and that the forces on the interface are in balance.

2.4 Domain Definition

The domain is based on the 2D version of the well-known CFD benchmark in [1] and shown here in Figure 1. By omitting the elastic bar behind the cylinder one can exactly recover the setup of the *flow around cylinder* configuration which allows for validation of the flow part by comparing the results with the older flow benchmark.

- The domain dimensions are: length $L = 2.5$, height $H = 0.41$.
- The circle center is positioned at $C = (0.2, 0.2)$ (measured from the left bottom corner of the channel) and the radius is $r = 0.05$.
- The elastic structure bar has length $l = 0.35$ and height $h = 0.02$, the right bottom corner is positioned at $(0.6, 0.19)$, and the left end is fully attached to the fixed cylinder.
- The control points are $A(t)$, fixed with the structure with $A(0) = (0.6, 0.2)$, and $B = (0.15, 0.2)$.

The setting is intentionally non-symmetric (see [1]) to prevent the dependence of the onset of any possible oscillation on the precision of the computation.

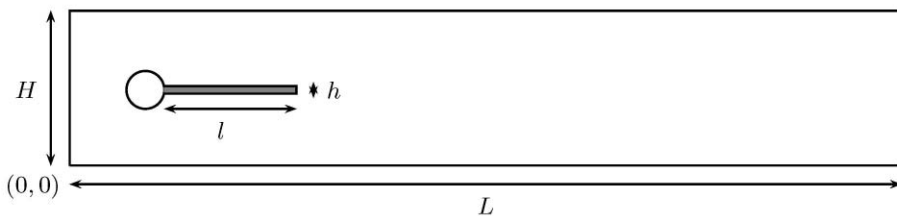


Fig. 1. Computational domain

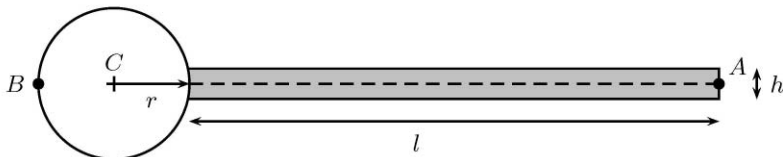


Fig. 2. Detail of the structure part

2.5 Boundary Conditions

- A parabolic velocity profile is prescribed at the left channel inflow

$$v^f(0, y) = 1.5\bar{U} \frac{y(H-y)}{\left(\frac{H}{2}\right)^2} = 1.5\bar{U} \frac{4.0}{0.1681} y(0.41 - y), \quad (10)$$

such that the mean inflow velocity is $\bar{U}$ and the maximum of the inflow velocity profile is $1.5\bar{U}$.

- The outflow condition can be chosen by the user, for example *stress free* or *do nothing* conditions. The outflow condition effectively prescribes some reference value for the pressure variable p . While this value could be arbitrarily set in the incompressible case, in the case of compressible structure this will have influence on the stress and consequently the deformation of the solid. In this proposal, we set the reference pressure at the outflow to have *zero mean value*.
- The *no-slip* condition is prescribed for the fluid on the other boundary parts. i.e. top and bottom wall, circle and fluid-structure interface Γ_t^0 .

2.6 Initial Conditions

Suggested starting procedure for the non-steady tests is to use a smooth increase of the velocity profile in time as

$$v^f(t, 0, y) = \begin{cases} v^f(0, y) \frac{1 - \cos(\frac{\pi}{2}t)}{2} & \text{if } t < 2.0 \\ v^f(0, y) & \text{otherwise} \end{cases} \quad (11)$$

where $v^f(0, y)$ is the velocity profile given in (10).

Table 1. Overview of the geometry parameters

geometry parameters		value [m]
channel length	L	2.5
channel width	H	0.41
cylinder center position	C	(0.2, 0.2)
cylinder radius	r	0.05
elastic structure length	l	0.35
elastic structure thickness	h	0.02
reference point (at $t = 0$)	A	(0.6, 0.2)
reference point	B	(0.2, 0.2)

2.7 Material Parameters

An overview of certain material properties for some relevant fluids and elastic materials is shown in the Table 2. The choice of the parameters for the benchmark is guided by several requirements:

First, we would like the flow to be in the laminar regime, which implies “small” Reynolds numbers. On the other hand, the flow should be capable of deforming the elastic structure. A typical fluid candidate for such experiments is glycerine.

In order not to introduce additional numerical complications connected with high aspect ratios in the geometry, the deformable structure has a

Table 2. Overview of some solid and fluid material parameters (densities ϱ^f , ϱ^s , Poisson ratio ν^s , Young modulus E , shear modulus μ^s , dynamic viscosity μ^f and kinematic viscosity ν^f)

material	ϱ^s [$\frac{\text{kg}}{\text{m}^3}$]	ν^s	E [$10^6 \frac{\text{kg}}{\text{ms}^2}$]	μ^s [$10^6 \frac{\text{kg}}{\text{ms}^2}$]
polybutadiene	910	0.50	1.6	0.53
polyurethane	1200	0.50	25	8.3
polypropylene	1100	0.42	900	317
PVC	1400	0.42	1500	528
steel	7800	0.29	210000	81400
cork	180	0.25	32	12.8

material	ϱ^f [$\frac{\text{kg}}{\text{m}^3}$]	ν^f [$10^{-6} \frac{\text{m}^2}{\text{s}}$]	μ^f [$10^{-3} \frac{\text{kg}}{\text{ms}}$]
air	1.23	0.015	0.018
acetone	790	0.405	0.32
ethyl alcohol	790	1.4	1.1
oil, vegetable	920	76.1	70
water	1000	1.14	1.14
blood	1035	3 – 4	3 – 4
glycerine	1260	1127	1420
honey	1420	7042	10000
mercury	13594	0.0114	1.55

Table 3. Proposed material combination

parameter	polybutadiene & glycerine	polypropylene & glycerine
ϱ^s [$10^3 \frac{\text{kg}}{\text{m}^3}$]	0.91	1.1
ν^s	0.5	0.42
μ^s [$10^6 \frac{\text{kg}}{\text{ms}^2}$]	0.53	317
ϱ^f [$10^3 \frac{\text{kg}}{\text{m}^3}$]	1.26	1.26
ν^f [$10^{-3} \frac{\text{m}^2}{\text{s}}$]	1.13	1.13

certain thickness which requires that the stiffness of the material should be low enough to allow significant deformations. Certain rubber-like materials fit into such a setting, namely polybutadiene (for a future incompressible configuration) and polypropylene.

In Table 3 the material parameters are presented for 2 combinations of glycerine and selected rubber-like material.

3 Quantities for Comparison

According to our preliminary calculations, self induced periodic oscillations develop in the flow and structure. The comparison will be done for *fully developed flow*, and particularly for *one full period of the oscillation* with respect to the position of the point $A(t)$. The suggested quantities of interest are:

- 1. The y -coordinate $y(t)$ of the end of the beam structure at point $A(t)$ (see the Figure 2).
- 2. Forces exerted by the fluid on the *whole* submerged body, i.e. lift and drag forces acting on the cylinder and the beam structure together

$$(F_D, F_L) = \int_S \boldsymbol{\sigma} \boldsymbol{n} dS,$$

where $S = S_1 \cup S_2$ (see Fig. 3) denotes the part of the circle being in contact with the fluid (i.e. S_1) plus part of the boundary of the beam structure being in contact with the fluid (i.e. S_2), and $\boldsymbol{n}$ is the outer unit normal vector to the integration path with respect to the fluid domain.

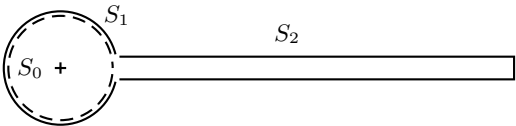


Fig. 3. Integration path $S = S_1 \cup S_2$ for the force calculation

Remark 1. The forces can be calculated in several different ways, i.e.

$$\begin{aligned}
 (F_D, F_L) &= \int_S \boldsymbol{\sigma} \mathbf{n} dS = \int_{S_1} \boldsymbol{\sigma}^f \mathbf{n} dS + \int_{S_2} \boldsymbol{\sigma}^f \mathbf{n} dS \\
 &= \int_{S_1} \boldsymbol{\sigma}^f \mathbf{n} dS + \int_{S_2} \boldsymbol{\sigma}^s \mathbf{n} dS \\
 &= \int_{S_1} \boldsymbol{\sigma}^f \mathbf{n} dS + \int_{S_2} \frac{1}{2} (\boldsymbol{\sigma}^s + \boldsymbol{\sigma}^f) \mathbf{n} dS \\
 &= \int_{S_0} \boldsymbol{\sigma} \mathbf{n} dS.
 \end{aligned}$$

That means that, up to numerical effects, all proposed evaluations will lead (asymptotically) to the same results.

3. Pressure difference between the points $A(t)$ and B

$$\Delta p^{AB} = p^B - p^{A(t)}.$$

The position of the point $A(t)$ is time dependent.

The time dependent values are represented by the mean value, amplitude and frequency. The mean value and amplitude are computed from the last period of the oscillations by taking the maximum and minimum values, then the mean value is taken as average of the min/max values, and the amplitude is the difference of the max/min from the mean:

$$\begin{aligned}
 \text{mean} &= \frac{1}{2}(\max + \min) \\
 \text{amplitude} &= \frac{1}{2}(\max - \min)
 \end{aligned}$$

The frequency of the oscillations can be computed either from the period time T as

$$\text{frequency} = \frac{1}{T}$$

or by using fourier analysis on the periodic data and taking the lowest significant frequency present in the spectrum. Additionally, a plot of the quantities over the period should be presented.

4 Partial Computational Tests

For the validation of the employed fluid and solid solvers, we first describe partial tests which are performed on different levels of mesh refinement (see Fig. and Table 4) with various time steps. We provide the results for the different discretization levels in the following since these sequences of results indicate that our given "reference results" are almost grid-independent. All simulations have been performed with a fully implicit monolithic ALE-FEM method with a fully coupled multigrid solver as described in [4].

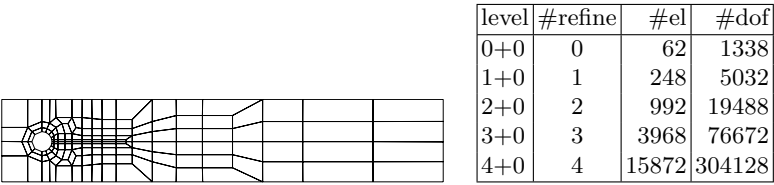


Fig. 4. Example of a coarse mesh and the number of degrees of freedom for refined levels

4.1 CFD Tests

Taking the flag as a rigid object, we perform 3 subtests focusing on the fluid dynamics part of the problem. The flag can be made almost rigid by setting the structural parameters to large values ($\varrho^s = 10^6 \frac{\text{kg}}{\text{m}^3}, \mu^s = 10^{12} \frac{\text{kg}}{\text{ms}^2}$) or completely rigid by considering the flow domain only with fixed boundary conditions on the flag interface.

4.2 CSM Tests

The structural tests are computed only for the elastic beam (without the surrounding fluid) adding the gravitational force *only*(!) on the structural part, $\mathbf{g} = (0, g) [\frac{\text{m}}{\text{s}^2}]$. The CSM3 test is computed as a time dependent case starting from the undeformed configuration while the tests CSM1 and CSM2 are the steady state solutions.

Table 4. Parameter settings for the CFD tests

dimensional parameter	CFD1	CFD2	CFD3
ϱ^f [$10^3 \frac{\text{kg}}{\text{m}^3}$]	1	1	1
ν^f [$10^{-3} \frac{\text{m}^2}{\text{s}}$]	1	1	1
$\bar{U}$ [$\frac{\text{m}}{\text{s}}$]	0.2	1	2
non-dimensional parameter	CFD1	CFD2	CFD3
$\text{Re} = \frac{\bar{U}d}{\nu^f}$	20	100	200
$\bar{U}$	0.2	1	2

Table 5. Results for **CFD1**

level	nel	ndof	drag	lift
0 + 0	144	3032	$1.41635 \times 10^{+01}$	$1.15592 \times 10^{+00}$
1 + 0	576	11536	$1.42236 \times 10^{+01}$	$1.11747 \times 10^{+00}$
2 + 0	2304	44960	$1.42730 \times 10^{+01}$	$1.11692 \times 10^{+00}$
3 + 0	9216	177472	$1.42880 \times 10^{+01}$	$1.11852 \times 10^{+00}$
4 + 0	36864	705152	$1.42919 \times 10^{+01}$	$1.11896 \times 10^{+00}$
5 + 0	147456	2811136	$1.42927 \times 10^{+01}$	$1.11904 \times 10^{+00}$
5 + 1	150528	2869504	$1.42929 \times 10^{+01}$	$1.11906 \times 10^{+00}$
5 + 2	156672	2986240	$1.42929 \times 10^{+01}$	$1.11905 \times 10^{+00}$
5 + 3	168960	3219712	$1.42929 \times 10^{+01}$	$1.11905 \times 10^{+00}$
6 + 0	589824	11225600	$1.42929 \times 10^{+01}$	$1.11905 \times 10^{+00}$
ref.			14.29	1.119

Table 6. Results for **CFD2**

level	nel	ndof	drag	lift
0 + 0	144	3032	$1.33188 \times 10^{+02}$	$1.18522 \times 10^{+01}$
1 + 0	576	11536	$1.34996 \times 10^{+02}$	$1.10739 \times 10^{+01}$
2 + 0	2304	44960	$1.36355 \times 10^{+02}$	$1.05337 \times 10^{+01}$
3 + 0	9216	177472	$1.36610 \times 10^{+02}$	$1.05303 \times 10^{+01}$
4 + 0	36864	705152	$1.36678 \times 10^{+02}$	$1.05347 \times 10^{+01}$
5 + 0	147456	2811136	$1.36696 \times 10^{+02}$	$1.05349 \times 10^{+01}$
5 + 1	150528	2869504	$1.36700 \times 10^{+02}$	$1.05346 \times 10^{+01}$
5 + 2	156672	2986240	$1.36701 \times 10^{+02}$	$1.05343 \times 10^{+01}$
5 + 3	168960	3219712	$1.36701 \times 10^{+02}$	$1.05340 \times 10^{+01}$
6 + 0	589824	11225600	$1.36700 \times 10^{+02}$	$1.05343 \times 10^{+01}$
ref.			136.7	10.53

CFD3: lift and drag forces on the cylinder+flag

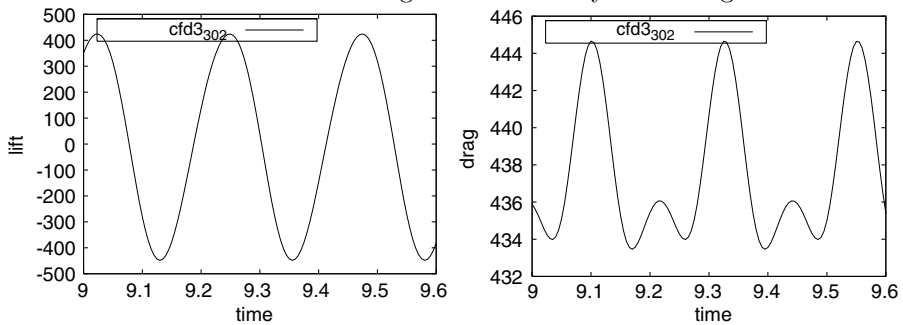


Table 7. Results for **CFD3** with $\Delta t = 0.01$ and 0.005

level	nel	ndof	drag	lift
1 + 0	576	11536	$416.8 \pm 3.3578[4.3825]$	$-24.702 \pm 342.38[4.3825]$
2 + 0	2304	44960	$437.29 \pm 5.3462[4.3825]$	$-11.085 \pm 429.88[4.3825]$
3 + 0	9216	177472	$438.99 \pm 5.4419[4.3825]$	$-10.289 \pm 433.09[4.3825]$
4 + 0	36864	705152	$439.38 \pm 5.4639[4.3825]$	$-9.9868 \pm 434.79[4.3825]$
level	nel	ndof	drag	lift
1 + 0	576	11536	$416.83 \pm 3.4023[4.3956]$	$-23.897 \pm 346.72[4.3956]$
2 + 0	2304	44960	$437.41 \pm 5.5856[4.3956]$	$-12.673 \pm 434.74[4.3956]$
3 + 0	9216	177472	$439.05 \pm 5.5804[4.3956]$	$-11.837 \pm 436.17[4.3956]$
4 + 0	36864	705152	$439.45 \pm 5.6183[4.3956]$	$-11.893 \pm 437.81[4.3956]$
ref.			$439.45 \pm 5.6183[4.3956]$	$-11.893 \pm 437.81[4.3956]$

Table 8. Parameter settings for the CSM tests

par. dim.	CSM1	CSM2	CSM3
$\varrho^s [10^3 \frac{\text{kg}}{\text{m}^3}]$	1	1	1
ν^s	0.4	0.4	0.4
$\mu^s [10^6 \frac{\text{kg}}{\text{ms}^2}]$	0.5	2.0	0.5
$\varrho^f [10^3 \frac{\text{kg}}{\text{m}^3}]$	1	1	1
$\nu^f [10^{-3} \frac{\text{m}^2}{\text{s}}]$	1	1	1
$\bar{U} [\frac{\text{m}}{\text{s}}]$	0	0	0
$g [\frac{\text{m}}{\text{s}^2}]$	2	2	2
par. non-dim.	CSM1	CSM2	CSM3
$\beta = \frac{\varrho^s}{\varrho^f}$	1	1	1
ν^s	0.4	0.4	0.4
$\text{E}^s [\frac{\text{kg}}{\text{ms}^2}]$	1.4×10^6	5.6×10^6	1.4×10^6
$\text{Re} = \frac{\bar{U} d}{\nu^f}$	0	0	0
$\bar{U}$	0	0	0
g	2	2	2

4.3 FSI Tests

The following FSI tests are performed for two different inflow speeds. FSI1 is resulting in a steady state solution, while the other two tests (FSI2, FSI3) result in periodic solutions and correspond to the final benchmark settings.

Table 9. Results for **CSM1**

level	nel	ndof	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]
2 + 0	320	6468	-7.17301	-66.0263
3 + 0	1280	25092	-7.18372	-66.0817
4 + 0	5120	98820	-7.18656	-66.0965
4 + 1	6260	120512	-7.18738	-66.1008
4 + 2	8552	164092	-7.18766	-66.1023
4 + 3	13148	251448	-7.18777	-66.1029
5 + 0	20480	392196	-7.18739	-66.1009
5 + 1	22772	435776	-7.18767	-66.1023
ref.			-7.187	-66.10

Table 10. Results for **CSM2**

level	nel	ndof	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]
2 + 0	320	6468	-0.468011	-16.9536
3 + 0	1280	25092	-0.468734	-16.9684
4 + 0	5120	98820	-0.468925	-16.9723
4 + 1	6260	120512	-0.468980	-16.9735
4 + 2	8552	164092	-0.468999	-16.9739
4 + 3	13148	251448	-0.469006	-16.9740
5 + 0	20480	392196	-0.468981	-16.9735
5 + 1	22772	435776	-0.469000	-16.9739
ref			-0.4690	-16.97

Table 11. Results for **CSM3** with timesteps $\Delta t = 0.02, 0.01, 0.005$

level	nel	ndof	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]
2 + 0	320	6468	$-14.384 \pm 14.389[1.0956]$	$-64.271 \pm 64.595[1.0956]$
3 + 0	1280	25092	$-14.402 \pm 14.406[1.0956]$	$-64.352 \pm 64.679[1.0956]$
4 + 0	5120	98820	$-14.404 \pm 14.408[1.0956]$	$-64.371 \pm 64.695[1.0956]$
level	nel	ndof	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]
2 + 0	320	6468	$-14.632 \pm 14.636[1.0978]$	$-64.744 \pm 64.907[1.0978]$
3 + 0	1280	25092	$-14.645 \pm 14.650[1.0978]$	$-64.765 \pm 64.946[1.0978]$
4 + 0	5120	98820	$-14.645 \pm 14.650[1.0978]$	$-64.766 \pm 64.948[1.0978]$
level	nel	ndof	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]
2 + 0	320	6468	$-14.279 \pm 14.280[1.0995]$	$-63.541 \pm 65.094[1.0995]$
3 + 0	1280	25092	$-14.299 \pm 14.299[1.0995]$	$-63.594 \pm 65.154[1.0995]$
4 + 0	5120	98820	$-14.305 \pm 14.305[1.0995]$	$-63.607 \pm 65.160[1.0995]$
ref			$-14.305 \pm 14.305[1.0995]$	$-63.607 \pm 65.160[1.0995]$

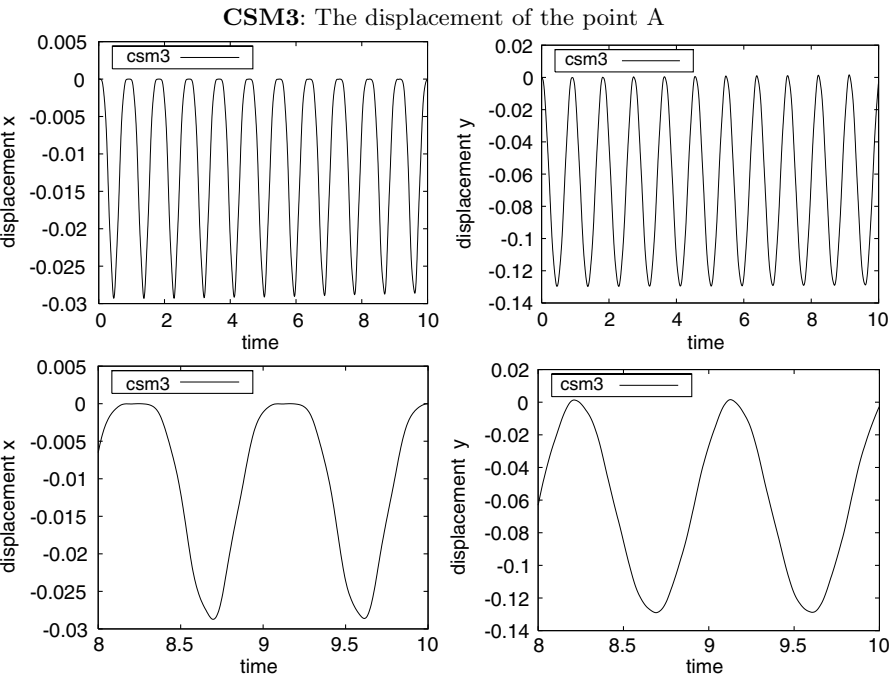


Table 12. Parameter settings for the full FSI benchmarks

parameter	FSI1	FSI2	FSI3
$\varrho^s [10^3 \frac{\text{kg}}{\text{m}^3}]$	1	10	1
ν^s	0.4	0.4	0.4
$\mu^s [10^6 \frac{\text{kg}}{\text{ms}^2}]$	0.5	0.5	2.0
$\varrho^f [10^3 \frac{\text{kg}}{\text{m}^3}]$	1	1	1
$\nu^f [10^{-3} \frac{\text{m}^2}{\text{s}}]$	1	1	1
$\bar{U} [\frac{\text{m}}{\text{s}}]$	0.2	1	2
parameter	FSI1	FSI2	FSI3
$\beta = \frac{\varrho^s}{\varrho^f}$	1	10	1
ν^s	0.4	0.4	0.4
$\text{Ae} = \frac{E^s}{\varrho^f \bar{U}^2}$	3.5×10^4	1.4×10^3	1.4×10^3
$\text{Re} = \frac{\bar{U} d}{\nu^f}$	20	100	200
$\bar{U}$	0.2	1	2

Table 13. Results for **FSI1**

level	nel	ndof	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]	drag	lift
2 + 0	992	19488	0.022871	0.81930	14.27360	0.76178
3 + 0	3968	76672	0.022775	0.82043	14.29177	0.76305
4 + 0	15872	304128	0.022732	0.82071	14.29484	0.76356
5 + 0	63488	1211392	0.022716	0.82081	14.29486	0.76370
6 + 0	253952	4835328	0.022708	0.82086	14.29451	0.76374
ref.			0.0227	0.8209	14.295	0.7638

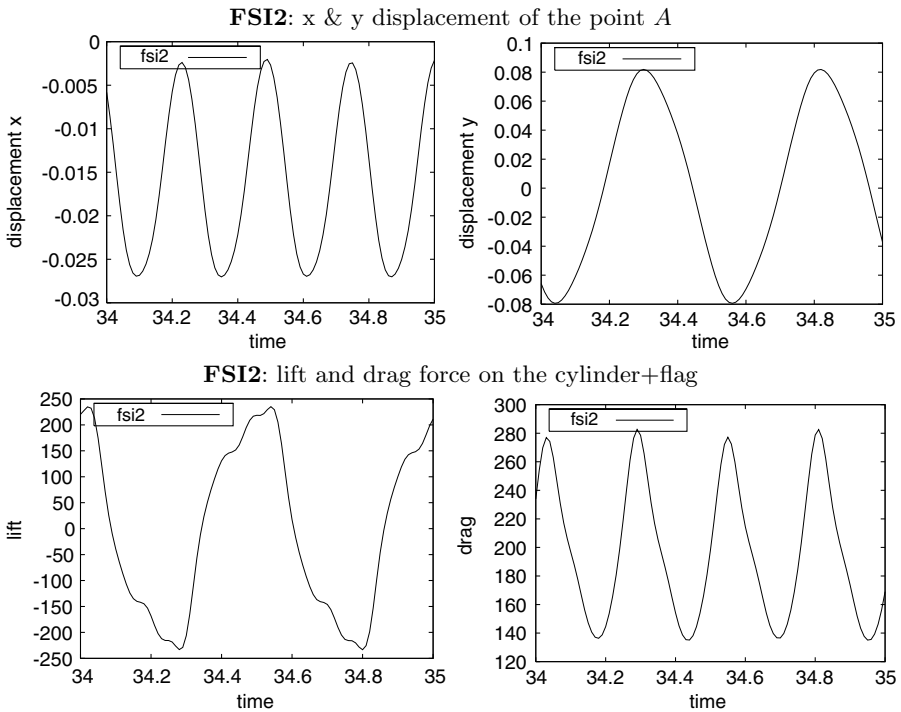


Table 14. Results for **FSI2** with timestep $\Delta t = 0.002, \Delta t = 0.001$

lev.	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]	drag	lift
2	$-14.00 \pm 12.03[3.8]$	$1.18 \pm 78.7[2.0]$	$209.46 \pm 72.30[3.8]$	$-1.18 \pm 269.6[2.0]$
3	$-14.25 \pm 12.03[3.8]$	$1.20 \pm 79.2[2.0]$	$202.55 \pm 67.02[3.8]$	$0.71 \pm 227.1[2.0]$
4	$-14.58 \pm 12.37[3.8]$	$1.25 \pm 80.7[2.0]$	$201.29 \pm 67.61[3.8]$	$0.97 \pm 233.2[2.0]$
lev.	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]	drag	lift
2	$-14.15 \pm 12.23[3.7]$	$1.18 \pm 78.8[1.9]$	$210.36 \pm 70.28[3.7]$	$0.80 \pm 286.0[1.9]$
3	$-13.97 \pm 12.01[3.8]$	$1.25 \pm 79.3[2.0]$	$203.54 \pm 68.43[3.8]$	$0.41 \pm 229.3[2.0]$
4	$-14.58 \pm 12.44[3.8]$	$1.23 \pm 80.6[2.0]$	$208.83 \pm 73.75[3.8]$	$0.88 \pm 234.2[2.0]$
ref.	$-14.58 \pm 12.44[3.8]$	$1.23 \pm 80.6[2.0]$	$208.83 \pm 73.75[3.8]$	$0.88 \pm 234.2[2.0]$

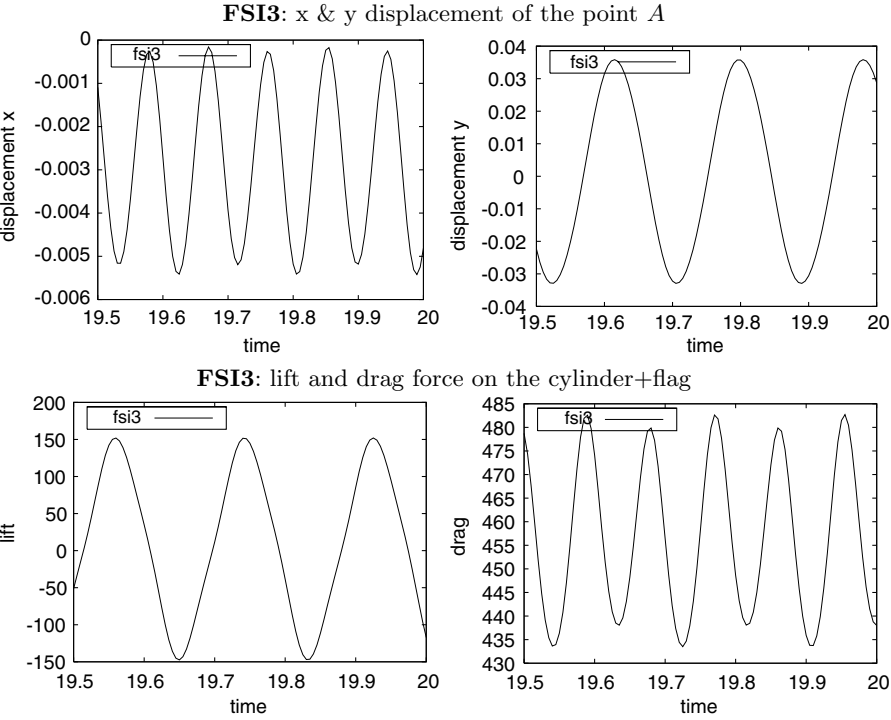


Table 15. Results for **FSI3** with timestep $\Delta t = 0.001, \Delta t = 0.0005$

lev.	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]	drag	lift
2	$-3.02 \pm 2.78[10.6]$	$0.99 \pm 35.70[5.3]$	$444.6 \pm 31.69[10.6]$	$9.48 \pm 151.55[5.3]$
3	$-3.02 \pm 2.83[10.6]$	$1.43 \pm 35.43[5.3]$	$457.1 \pm 20.05[10.6]$	$1.23 \pm 146.04[5.3]$
4	$-2.85 \pm 2.56[10.9]$	$1.53 \pm 34.35[5.3]$	$459.8 \pm 20.00[10.9]$	$1.51 \pm 148.76[5.3]$
lev.	ux of A [$\times 10^{-3}$]	uy of A [$\times 10^{-3}$]	drag	lift
2	$-3.00 \pm 2.79[10.7]$	$1.19 \pm 35.72[5.3]$	$445.0 \pm 35.09[10.7]$	$8.26 \pm 163.72[5.3]$
3	$-2.86 \pm 2.68[10.7]$	$1.45 \pm 35.34[5.3]$	$455.7 \pm 24.69[10.7]$	$1.42 \pm 146.43[5.3]$
4	$-2.69 \pm 2.53[10.9]$	$1.48 \pm 34.38[5.3]$	$457.3 \pm 22.66[10.9]$	$2.22 \pm 149.78[5.3]$
ref.	$-2.69 \pm 2.53[10.9]$	$1.48 \pm 34.38[5.3]$	$457.3 \pm 22.66[10.9]$	$2.22 \pm 149.78[5.3]$

5 Summary

The next step will be the specification of how to submit and to collect the results, and the publication of the test configurations in an international journal. Moreover, it is planned to prepare a webpage for collecting and presenting the FSI results. As we have learned from [1], a very important aspect will be the submission of the results on (at least) 3 different meshes and time steps. Then, based on the collected results, quantitative ratings regarding the main questions, particularly w.r.t. the coupling mechanisms and monolithic vs. partitioned approaches, might get possible.

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